

mielke-geometroynamics — formula report

1794 inline formulas (1 shown: the ones that did not render), 1149 display equations, 5 tables, 18 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometroynamics_313 EQ0966 (14.2.7) [conf 0.024]	313	■ 0.024 ● C +3	$\begin{aligned} d\hat{s}^2 &= \frac{1}{4+K} \operatorname{Tr} \left(\hat{\vartheta} \otimes_s \hat{\vartheta}^T \right) = \hat{g}_{AB} \hat{\vartheta}^A \otimes_s \hat{\vartheta}^B; \quad A, B, = 0, \dots, 3+K \\ &= \frac{1}{4+K} \operatorname{Tr} \left[\begin{array}{c c} \vartheta \otimes \vartheta + \hat{\ell}^2 \omega \otimes \omega & \hat{\ell} \omega \otimes \overset{\circ}{\vartheta} \\ \hline \hat{\ell} \vartheta \otimes \omega & \ddots \\ \hline \circ & \vartheta \otimes \end{array} \right]. \end{aligned}$	$d\hat{s}^2 = \frac{1}{4+K} \operatorname{Tr} \left(\hat{\vartheta} \otimes_s \hat{\vartheta}^T \right) = \hat{g}_{AB} \hat{\vartheta}^A \otimes_s \hat{\vartheta}^B; \quad A, B, = 0, \dots, 3+K$ $= \frac{1}{4+K} \operatorname{Tr} \left[\frac{\vartheta \otimes \vartheta + \hat{\ell}^2 \omega \otimes \omega}{\hat{\ell} \overset{\circ}{\vartheta} \otimes \omega} \middle \frac{\hat{\ell} \omega \otimes \overset{\circ}{\vartheta}}{\overset{\circ}{\vartheta} \otimes \overset{\circ}{\vartheta}} \right]. \quad (1.$	
mielke-geometroynamics_134 EQ0403 (6.4.23) [conf 0.028]	134	■ 0.028 ● C +4	$\Psi_{\parallel}(\vartheta) = \exp \left(- \int_{M_3} \overset{(\pm)}{C}_{\text{TT}} \right) = \exp \left(\frac{-1}{2\ell^2} \int_{M_3} \left[\overset{*}{+}_{\vartheta_B} \wedge \overset{*}{T}^B \right] \right)$	$\Psi_{\parallel}(\vartheta) = \exp \left(- \int_{M_3} \overset{(\pm)}{C}_{\text{TT}} \right) = \exp \left(\frac{-1}{2\ell^2} \int_{M_3} \left[\overset{*}{+}_{\vartheta_B} \wedge \overset{*}{T}^B \right] \right)$	
mielke-geometroynamics_275 EQ0850 (12.3.3) [conf 0.032]	275	■ 0.032 ● C +3	$\begin{aligned} \not{D}^2 &= -\frac{1}{2} \gamma^\alpha \gamma^\beta \left(\{ D_\alpha^{\{\}}, D_\beta^{\}\} + [D_\alpha^{\{\}}, D_\beta^{\}\}] \right) - 2im \not{D}^{\{\}} \\ &\quad - \frac{i}{4} \gamma_5 \left(D^{\{\} \not{A}} \right) + \frac{1}{2} \gamma_5 \sigma^{\alpha\beta} \not{A}_\alpha D_\beta^{\{\}} + m^2 - \frac{1}{2} m \gamma_5 \not{A} - \frac{1}{16} \not{K} \not{A} \\ &\cong -\square - \frac{1}{8} \sigma^{\alpha\beta} R_{\alpha\beta\mu\nu}^{\{\} } \sigma^{\mu\nu} \\ &\quad - \frac{i}{4} \gamma_5 \left(D^{\{\} \not{b}} \right) + \frac{1}{2} \gamma_5 \sigma^{\alpha\beta} \not{A}_\alpha D_\beta^{\{\}} - \frac{1}{16} \not{A}_\alpha \not{A}^\alpha - m^2 \end{aligned}$	$\not{D}^2 = -\frac{1}{2} \gamma^\alpha \gamma^\beta \left(\{ D_\alpha^{\{\}}, D_\beta^{\}\} + [D_\alpha^{\{\}}, D_\beta^{\}\}] \right) - 2im \not{D}^{\{\}} \\ - \frac{i}{4} \gamma_5 \left(D^{\{\} \not{A}} \right) + \frac{1}{2} \gamma_5 \sigma^{\alpha\beta} \not{A}_\alpha D_\beta^{\{\}} + m^2 - \frac{1}{2} m \gamma_5 \not{A} - \frac{1}{16} \not{K} \not{A} \\ \cong -\square - \frac{1}{8} \sigma^{\alpha\beta} R_{\alpha\beta\mu\nu}^{\{\} } \sigma^{\mu\nu} \\ - \frac{i}{4} \gamma_5 \left(D^{\{\} \not{b}} \right) + \frac{1}{2} \gamma_5 \sigma^{\alpha\beta} \not{A}_\alpha D_\beta^{\{\}} - \frac{1}{16} \not{A}_\alpha \not{A}^\alpha - m^2$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_134 EQ0402 (6.4.22) [conf 0.042]	134	■ 0.042 ●	$\begin{array}{ll} \text{q: } & \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta) = \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta), \\ \underline{p}: & \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta) = -i\ell^2 \frac{\delta}{\delta \underline{*}^{\vartheta}_{\alpha}} \Psi_{\parallel}(\vartheta), \end{array}$	$q: \quad \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta) = \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta),$ $\underline{p}: \quad \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta) = -i\ell^2 \frac{\delta}{\delta \underline{*}^{\vartheta}_{\alpha}} \Psi_{\parallel}(\vartheta),$	$q: \quad \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta) = \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta),$ $\underline{p}: \quad \underline{*}^{\vartheta\beta} \Psi_{\parallel}(\vartheta) = -i\ell^2 \frac{\delta}{\delta \underline{*}^{\vartheta}_{\alpha}} \Psi_{\parallel}(\vartheta),$
mielke-geometrodynamics_101 EQ0293 (4.5.4) [conf 0.052]	101	■ 0.052 ● N +0	$\xi \downarrow L \equiv (\xi \downarrow \vartheta^{\alpha}) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + (\xi \downarrow \Psi) \wedge \frac{\partial L}{\partial \Psi} + (\xi \downarrow D\Psi) \wedge \frac{\partial L}{\partial D\Psi}.$	$\xi \downarrow L \equiv (\xi \downarrow \vartheta^{\alpha}) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + (\xi \downarrow \Psi) \wedge \frac{\partial L}{\partial \Psi} + (\xi \downarrow D\Psi) \wedge \frac{\partial L}{\partial D\Psi}.$	$\xi \downarrow L \equiv (\xi \downarrow \vartheta^{\alpha}) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + (\xi \downarrow \Psi) \wedge \frac{\partial L}{\partial \Psi} + (\xi \downarrow D\Psi) \wedge \frac{\partial L}{\partial D\Psi}.$
mielke-geometrodynamics_047 EQ0064 (2.8.6) [conf 0.054]	047	■ 0.054 ● N -2	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \text{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \text{Tr}(\Omega \wedge \Omega) - \text{Tr}(\Omega) \wedge \text{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}^2(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$	$\stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \text{Tr} \Omega \quad (=0)$ $\stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \text{Tr}(\Omega \wedge \Omega) - \text{Tr}(\Omega) \wedge \text{Tr}(\Omega) \}$ $\left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right)$ $\stackrel{*}{\pi}^2(\gamma_k) = 0 \text{ for } k \geq 3.$	$\stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \text{Tr} \Omega \quad (=0)$ $\stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \text{Tr}(\Omega \wedge \Omega) - \text{Tr}(\Omega) \wedge \text{Tr}(\Omega) \}$ $\left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right)$ $\stackrel{*}{\pi}(\gamma_k) = 0 \text{ for } k \geq 3.$
mielke-geometrodynamics_133 EQ0400 (6.4.19) [conf 0.060]	133	■ 0.060 ● S +1	$\stackrel{(\pm)}{\mathcal{G}}_{\alpha} := \underline{D} \stackrel{(\pm)}{\Pi}_{\alpha} = \underline{D} \stackrel{(\pm)}{\underline{A}}_{\alpha} = 0,$	$\stackrel{(\pm)}{\mathcal{G}}_{\alpha} := \underline{D} \stackrel{(\pm)}{\Pi}_{\alpha} = \underline{D} \stackrel{(\pm)}{\underline{A}}_{\alpha} = 0,$	$\stackrel{(\pm)}{\mathcal{G}}_{\alpha} := \underline{D} \stackrel{(\pm)}{\Pi}_{\alpha} = \underline{D} \stackrel{(\pm)}{\underline{A}}_{\alpha} = 0,$
mielke-geometrodynamics_229 EQ0718 (10.9.4) [conf 0.066]	229	■ 0.066 ● N +0	$\Sigma_{\alpha} := e_{\alpha} \downarrow L_{\Psi} - (e_{\alpha} \downarrow \Psi) \wedge \frac{\partial L_{\Psi}}{\partial \Psi} - (e_{\alpha} \downarrow D\Psi) \wedge \frac{\partial L_{\Psi}}{\partial D\Psi} - (e_{\alpha} \downarrow D\Psi) \wedge \frac{\partial L_{\Psi}}{\partial D\Psi};$	$\Sigma_{\alpha} := e_{\alpha} \downarrow L_{\Psi} - (e_{\alpha} \downarrow \Psi) \wedge \frac{\partial L_{\Psi}}{\partial \Psi} - (e_{\alpha} \downarrow D\Psi) \wedge \frac{\partial L_{\Psi}}{\partial D\Psi} - (e_{\alpha} \downarrow D\Psi) \wedge \frac{\partial L_{\Psi}}{\partial D\Psi};$	$\Sigma_{\alpha} := e_{\alpha} \downarrow L_{\Psi} - (e_{\alpha} \downarrow \Psi) \wedge \frac{\partial L_{\Psi}}{\partial \Psi} - (e_{\alpha} \downarrow D\Psi) \wedge \frac{\partial L_{\Psi}}{\partial D\Psi} - (e_{\alpha} \downarrow D\Psi) \wedge \frac{\partial L_{\Psi}}{\partial D\Psi};$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					
[refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_131 EQ0391 (6.4.6) [conf 0.066]	131	■ 0.066 ●	$\begin{aligned} \Sigma_{\alpha} &:= \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ &\quad - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}. \end{aligned}$	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}.$	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}.$
mielke-geometrodynamics_168 EQ0528 (7.7.48) [conf 0.078]	168	■ 0.078 ● C +3	$\begin{aligned} 2 \ell^2 d C_{T \ T^*} &:= 2 d (\vartheta^{\alpha} \wedge {}^* T_{\alpha}) \\ &= R^{f\alpha\beta} \wedge \eta_{\alpha\beta} - R^{\alpha\beta} \wedge \eta_{\alpha\beta} \\ &\quad - T^{\alpha} * \left[T_{\alpha} - \vartheta_{\alpha} \wedge (e_{\beta} \rfloor T^{\beta}) - \frac{1}{2} e_{\alpha} \rfloor (T^{\beta} \wedge \vartheta_{\beta}) \right] \\ &= R^{f\alpha\beta} \wedge \eta_{\alpha\beta} - R^{\alpha\beta} \wedge \eta_{\alpha\beta} - T^{\alpha} \wedge * \left(- {}^{(1)}T_{\alpha} + 2 {}^{(2)}T_{\alpha} + \frac{1}{2} {}^{(3)}T_{\alpha} \right), \end{aligned}$	$2\ell^2 dC_{T \ T^*} := 2d(\vartheta^{\alpha} \wedge {}^* T_{\alpha}) \\ = R^{f\alpha\beta} \wedge \eta_{\alpha\beta} - R^{\alpha\beta} \wedge \eta_{\alpha\beta} \\ - T^{\alpha} * \left[T_{\alpha} - \vartheta_{\alpha} \wedge (e_{\beta} \rfloor T^{\beta}) - \frac{1}{2} e_{\alpha} \rfloor (T^{\beta} \wedge \vartheta_{\beta}) \right] \\ = R^{f\alpha\beta} \wedge \eta_{\alpha\beta} - R^{\alpha\beta} \wedge \eta_{\alpha\beta} - T^{\alpha} \wedge * \left(- {}^{(1)}T_{\alpha} + 2 {}^{(2)}T_{\alpha} + \frac{1}{2} {}^{(3)}T_{\alpha} \right),$	$2\ell^2 dC_{T T^*} := 2d(\vartheta^{\alpha} \wedge {}^* T_{\alpha}) \tag{7.7.4} \\ = R^{f\alpha\beta} \wedge \eta_{\alpha\beta} - R^{\alpha\beta} \wedge \eta_{\alpha\beta} \\ - T^{\alpha} * \left[T_{\alpha} - \vartheta_{\alpha} \wedge (e_{\beta} \rfloor T^{\beta}) - \frac{1}{2} e_{\alpha} \rfloor (T^{\beta} \wedge \vartheta_{\beta}) \right] \\ = R^{f\alpha\beta} \wedge \eta_{\alpha\beta} - R^{\alpha\beta} \wedge \eta_{\alpha\beta} - T^{\alpha} \wedge * \left(- {}^{(1)}T_{\alpha} + 2 {}^{(2)}T_{\alpha} + \frac{1}{2} {}^{(3)}T_{\alpha} \right),$
mielke-geometrodynamics_129 EQ0382 (6.2.33) [conf 0.087]	129	■ 0.087 ● W +1	$\begin{aligned} \stackrel{\circ}{\Gamma}_{\alpha}^{\beta} &:= \Gamma_{\alpha}^{\beta} + i \stackrel{\circ}{A}^{\beta}, \quad \stackrel{\circ}{A}^{\beta} := \pm \frac{1}{2} \ell^2 e_{\alpha} \rfloor \Pi^{\beta}, \\ \stackrel{\circ}{\Gamma}_{\alpha}^{\beta} &:= \Gamma_{\alpha}^{\beta} + i \stackrel{\circ}{A}^{\beta}, \quad \stackrel{\circ}{A}^{\beta} := \pm \frac{1}{2} \ell^2 e_{\alpha} \rfloor \Pi^{\beta}, \end{aligned}$	$\stackrel{(\pm)}{\Gamma}_{\alpha}^{\beta} := \Gamma_{\alpha}^{\beta} + i \stackrel{(\mp)}{A}_{\alpha}^{\beta}, \quad \stackrel{(\pm)}{A}_{\alpha}^{\beta} := \pm \frac{1}{2} \ell^2 e_{\alpha} \rfloor \Pi^{\beta},$	$\stackrel{(\pm)}{\Gamma}_{\alpha}^{\beta} := \Gamma_{\alpha}^{\beta} + i \stackrel{(\mp)}{A}_{\alpha}^{\beta}, \quad \stackrel{(\pm)}{A}_{\alpha}^{\beta} := \pm \frac{1}{2} \ell^2 e_{\alpha} \rfloor \Pi^{\beta},$
mielke-geometrodynamics_135 EQ0485 (6.4.25) [conf 0.098]	135	■ 0.098 ● C +8	$-\ell^2 \underline{D}^{(\pm)*} \left(\frac{\delta}{\delta^* \underline{\vartheta}_B} \right) = \underline{D}^{(\pm)} \underline{T}^B \equiv \underline{R}_C^{(\pm)B} \underline{\vartheta}^{(\pm)C} = 0,$	$-\ell^2 \underline{D}^{(\pm)*} \left(\frac{\delta}{\delta^* \underline{\vartheta}_B} \right) = \underline{D}^{(\pm)} \underline{T}^B \equiv \underline{R}_C^{(\pm)B} \underline{\vartheta}^{(\pm)C} = 0,$	$-\ell^2 \underline{D}^{(\pm)*} \left(\frac{\delta}{\delta^* \underline{\vartheta}_B} \right) = \underline{D}^{(\pm)} \underline{T}^B \equiv \underline{R}_C^{(\pm)B} \underline{\vartheta}^{(\pm)C} = 0,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_168 EQ0529 (7.7.49) [conf 0.099]	168	■ 0.099	$\begin{aligned} L_{\text{Euler}} &:= (-1)^{\text{sig}+1} \text{Tr} \left\{ \Omega \wedge \Omega^{(*)} \right\} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ &= dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}{}^{\beta(*)} \wedge \Gamma_{\beta}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\alpha} \right) \\ &\equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^* R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}{}^{\alpha} \wedge {}^* \text{Ric}_{\beta}{}^{\beta} \\ &\equiv -L_{\text{SKY}} \\ &\quad - 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}{}^{\alpha} \wedge {}^* \text{Ric}_{\beta}{}^{\beta} \right) \end{aligned}$	$L_{\text{Euler}} := (-1)^{\text{sig}+1} \text{Tr} \left\{ \Omega \wedge \Omega^{(*)} \right\} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)}$ $= dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}{}^{\beta(*)} \wedge \Gamma_{\beta}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\alpha} \right)$ $\equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^* R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}{}^{\alpha} \wedge {}^* \text{Ric}_{\beta}{}^{\beta}$ $\equiv -L_{\text{SKY}}$ $- 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}{}^{\alpha} \wedge {}^* \text{Ric}_{\beta}{}^{\beta} \right)$	$L_{\text{Euler}} := (-1)^{\text{sig}+1} \text{Tr} \{ \Omega \wedge \Omega^{(*)} \} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)}$ $= dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}{}^{\beta(*)} \wedge \Gamma_{\beta}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\alpha} \right)$ $\equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^* R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}{}^{\alpha} \wedge {}^* \text{Ric}_{\beta}{}^{\beta}$ $\equiv -L_{\text{SKY}}$ $- 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^* \text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}{}^{\alpha} \wedge {}^* \text{Ric}_{\beta}{}^{\beta} \right)$
mielke-geometrodynamics_315 EQ0974 (14.2.15) [conf 0.099]	315	■ 0.099 ● N +0	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\circ\Theta} = 0 \right],$	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\circ\Theta} = 0 \right],$	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\pi\overset{\circ}{\Theta} = 0} \right],$
mielke-geometrodynamics_180 EQ0553 (8.4.7)	180	■ 0.100 ● W +1	$\begin{aligned} L_{\text{SKY}}^{(*)} &= \frac{1}{2} \text{Tr} \left(\stackrel{\pm}{\Omega}^g \wedge {}^* \stackrel{\pm}{\Omega}^g \right) \\ &= -\frac{1}{2} R^{\alpha\beta} \wedge {}^* R_{\alpha\beta} \mp \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ &= -\frac{1}{4} \left(R_{\alpha\beta} \pm {}^* R_{\alpha\beta}^{(*)} \right) \wedge {}^* \left(R^{\alpha\beta} \pm {}^* R^{\alpha\beta(*)} \right). \end{aligned}$	$L_{\text{SKY}}^{(*)} = \frac{1}{2} \text{Tr} \left(\stackrel{\pm}{\Omega}^g \wedge {}^* \stackrel{\pm}{\Omega}^g \right)$ $= -\frac{1}{2} R^{\alpha\beta} \wedge {}^* R_{\alpha\beta} \mp \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)}$ $= -\frac{1}{4} \left(R_{\alpha\beta} \pm {}^* R_{\alpha\beta}^{(*)} \right) \wedge {}^* \left(R^{\alpha\beta} \pm {}^* R^{\alpha\beta(*)} \right).$	$L_{\text{SKY}}^{(*)} = \frac{1}{2} \text{Tr} \left(\stackrel{\pm}{\Omega}^g \wedge {}^* \stackrel{\pm}{\Omega}^g \right)$ $= -\frac{1}{2} R^{\alpha\beta} \wedge {}^* R_{\alpha\beta} \mp \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)}$ $= -\frac{1}{4} \left(R_{\alpha\beta} \pm {}^* R_{\alpha\beta}^{(*)} \right) \wedge {}^* \left(R^{\alpha\beta} \pm {}^* R^{\alpha\beta(*)} \right).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_104 EQ0311 (4.6.3)	104	■ 0.111 ● C +6	$\begin{aligned} & \widetilde{\leftarrow}_{\widetilde{y}} \widetilde{D} \widetilde{\xi}^{\alpha} = \widetilde{y} \left(\widetilde{D} \widetilde{D} \widetilde{\xi}^{\alpha} \right) + \widetilde{D} \left(\widetilde{y} \right) \widetilde{D} \widetilde{\xi}^{\alpha} \\ & = y J \left(D D \xi^{\alpha} + R^{(T)\alpha} \right) = 0, \end{aligned}$	$\widetilde{\mathbf{L}}_{\widetilde{y}} \widetilde{D} \widetilde{\xi}^{\alpha} = \widetilde{y} \left(\widetilde{D} \widetilde{D} \widetilde{\xi}^{\alpha} \right) + \widetilde{D} \left(\widetilde{y} \right) \widetilde{D} \widetilde{\xi}^{\alpha} = y \left[D D \xi^{\alpha} + R^{(T)\alpha} \right] = 0,$	
mielke-geometrodynamics_128 EQ0378 (6.2.29)	128	■ 0.136 ●	$D \, D \, \vartheta^{\alpha} = R_{\beta}^{\alpha} \wedge \vartheta^{\beta} = 0, \, D \, D \, \Pi_{\alpha} = - \, R_{\alpha}^{\beta} \wedge \Pi_{\beta} = \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \lrcorner \Pi^{\beta} \right) \wedge \Sigma_{\beta} \cong 0.$	$D D \vartheta^{\alpha} = R_{\beta}^{\alpha} \wedge \vartheta^{\beta} = 0, \, D \, D \, \Pi_{\alpha} = - \, R_{\alpha}^{\beta} \wedge \Pi_{\beta} = \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \lrcorner \Pi^{\beta} \right) \wedge \Sigma_{\beta} \cong 0.$	
mielke-geometrodynamics_346 EQ1059 (15.2.16)	346	■ 0.147 ● C +12	$\begin{aligned} & \begin{array}{c} \psi = \\ \left[\begin{array}{ccc c} \mathcal{P}_a & \mathcal{P}_b & \mathcal{P}_c & \nu_e \\ \eta_a & \eta_b & \eta_c & e^- \\ \lambda_a & \lambda_b & \lambda_c & \mu^- \\ \hline c_a & c_b & c_c & \nu_{\mu} \end{array} \right] \end{array} \\ & \begin{array}{c} \text{up} \\ \text{down} \\ \text{strange} \\ \text{charm} \end{array} \quad \begin{array}{c} \uparrow \\ \text{flavors} \\ \downarrow \end{array} \\ & \begin{array}{c} \text{red} \quad \text{yellow} \quad \text{blue} \quad \text{lilac} \end{array} \end{aligned}$	$\psi = \left[\begin{array}{ccc c} \mathcal{P}_a & \mathcal{P}_b & \mathcal{P}_c & \nu_e \\ \eta_a & \eta_b & \eta_c & e^- \\ \lambda_a & \lambda_b & \lambda_c & \mu^- \\ \hline c_a & c_b & c_c & \nu_{\mu} \end{array} \right] \begin{array}{c} \text{up} \\ \text{down} \\ \text{strange} \\ \text{charm} \end{array} \quad \begin{array}{c} \uparrow \\ \text{flavors} \\ \downarrow \end{array}$	
mielke-geometrodynamics_314 EQ0972 (14.2.13)	314	■ 0.153 ● K +0	$\stackrel{\circ}{D} \alpha := \stackrel{\circ}{d} \alpha - \omega^g \wedge \alpha - \alpha \wedge \stackrel{\circ}{\omega}$	$\stackrel{\circ}{D} \alpha := \stackrel{\circ}{d} \alpha - \omega^g \wedge \alpha - \alpha \wedge \stackrel{\circ}{\omega}$	
mielke-geometrodynamics_095 EQ0264 (4.3.29)	095	■ 0.154 ● W +2	$\Gamma_{ij}^{' } = \left\{ \begin{array}{c} \kappa \\ ij \end{array} \right\} - K_{ij}^{' \kappa},$	$\Gamma_{ij}^{',\kappa} = \left\{ \begin{array}{c} \kappa \\ ij \end{array} \right\} - K_{ij}^{',\kappa},$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_E00292 (4.5.3)	100	■0.165	$\begin{aligned} \mathfrak{f}_{\xi} L &= (\mathfrak{f}_{\xi} g_{\alpha\beta}) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + (\mathfrak{f}_{\xi} \vartheta^{\alpha}) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ (\mathfrak{f}_{\xi} \Psi) \wedge \frac{\partial L}{\partial \Psi} + (\mathfrak{f}_{\xi} D\Psi) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[(\xi \lrcorner \vartheta^{\alpha}) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + (\xi \lrcorner \Psi) \wedge \frac{\partial L}{\partial \Psi} + (\xi \lrcorner D\Psi) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- (\xi \lrcorner \vartheta^{\alpha}) D \frac{\partial L}{\partial \vartheta^{\alpha}} + (\xi \lrcorner T^{\alpha}) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + (\xi \lrcorner R_{\beta}^{\gamma}) \wedge I^{\beta}{}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ (\xi \lrcorner D\Psi) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p (\xi \lrcorner \Psi) \wedge D \frac{\delta L}{\delta \Psi} \end{aligned}$	$\begin{aligned} \mathfrak{f}_{\xi} L &= (\mathfrak{f}_{\xi} g_{\alpha\beta}) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + (\mathfrak{f}_{\xi} \vartheta^{\alpha}) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ (\mathfrak{f}_{\xi} \Psi) \wedge \frac{\partial L}{\partial \Psi} + (\mathfrak{f}_{\xi} D\Psi) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[(\xi \lrcorner \vartheta^{\alpha}) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + (\xi \lrcorner \Psi) \wedge \frac{\partial L}{\partial \Psi} + (\xi \lrcorner D\Psi) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- (\xi \lrcorner \vartheta^{\alpha}) D \frac{\partial L}{\partial \vartheta^{\alpha}} + (\xi \lrcorner T^{\alpha}) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + (\xi \lrcorner R_{\beta}^{\gamma}) \wedge I^{\beta}{}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ (\xi \lrcorner D\Psi) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p (\xi \lrcorner \Psi) \wedge D \frac{\delta L}{\delta \Psi} \end{aligned}$	
mielke-geometrodynamics_E00734 (11.1.1)	237	■0.220 W +1	$\pi_1 \left(SO_{\circ}(s, n-s), e \right) = \begin{cases} \mathbb{Z}_2 & n = 2k \\ \mathbb{Z}_2 \oplus \mathbb{Z}_2 & n = 4k \\ \mathbb{Z}_4 & n = 2(2k+1) \end{cases}$	$\pi_1(SO_{\circ}(s, n-s), e) = \begin{cases} \mathbb{Z}_2 & n = 2k \\ \mathbb{Z}_2 \oplus \mathbb{Z}_2 & n = 4k \\ \mathbb{Z}_4 & n = 2(2k+1) \end{cases}$	
mielke-geometrodynamics_E00961 (14.2.1)	312	■0.220 C +3	$\hat{\vartheta} = \hat{\vartheta}(m, g) = \underbrace{\left[\begin{array}{c c} \vartheta & \hat{\ell}\omega \\ \hline 0 & \vartheta \end{array} \right]}_4 \}^4 K$	$\hat{\vartheta} = \hat{\vartheta}(m, g) = \underbrace{\left[\begin{array}{c c} \vartheta & \hat{\ell}\omega \\ \hline 0 & \vartheta \end{array} \right]}_4 \}^4 K$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_085 E00211 (4.2.3)		■ 0.223 ● C +3	$\begin{aligned} & * \widetilde{\omega} \stackrel{*}{=} \left(d\widetilde{\omega} - \frac{1}{2} [\widetilde{\omega}, \widetilde{\omega}] \right) \\ & = d\omega^{\mathrm{L}} - \omega^{\mathrm{L}} \wedge \omega^{\mathrm{L}} + d\omega^{\mathrm{T}} - \omega^{\mathrm{T}} \wedge \omega^{\mathrm{L}} - \omega^{\mathrm{L}} \wedge \omega^{\mathrm{T}} + \frac{1}{2} [\omega^{\mathrm{T}}, \omega^{\mathrm{T}}] \\ & = \Omega^{\mathrm{L}} + \Omega^{\mathrm{T}} \end{aligned}$	$\begin{aligned} * \widetilde{\Omega} &= \chi \left(d\widetilde{\omega} - \frac{1}{2} [\widetilde{\omega}, \widetilde{\omega}] \right) \\ & = d\omega^{\mathrm{L}} - \omega^{\mathrm{L}} \wedge \omega^{\mathrm{L}} + d\omega^{\mathrm{T}} - \omega^{\mathrm{T}} \wedge \omega^{\mathrm{L}} - \omega^{\mathrm{L}} \wedge \omega^{\mathrm{T}} + \frac{1}{2} [\omega^{\mathrm{T}}, \omega^{\mathrm{T}}] \\ & = \Omega^{\mathrm{L}} + \Omega^{\mathrm{T}} \end{aligned}$	$\begin{aligned} * \widetilde{\chi} \widetilde{\Omega} &= * \chi (d\widetilde{\omega} - \frac{1}{2} [\widetilde{\omega}, \widetilde{\omega}]) \\ & = d\omega^{\mathrm{L}} - \omega^{\mathrm{L}} \wedge \omega^{\mathrm{L}} + d\omega^{\mathrm{T}} - \omega^{\mathrm{T}} \wedge \omega^{\mathrm{L}} - \omega^{\mathrm{L}} \wedge \omega^{\mathrm{T}} + \frac{1}{2} [\omega^{\mathrm{T}}, \omega^{\mathrm{T}}] \\ & = \Omega^{\mathrm{L}} + \Omega^{\mathrm{T}} \end{aligned}$
mielke-geometrodynamics_095 E00266 (4.3.31)		■ 0.227 ●	$R_{ijc}^{::d} = R_{ijc}^{\{!...^d} - 2D_{[i}K_{j]c}^{..d} + 2K_{[i c}^{..h}K_{j]h}^{..d}.$	$R_{ijc}^{::d} = R_{ijc}^{\{!...^d} - 2D_{[i}K_{j]c}^{..d} + 2K_{[i c}^{..h}K_{j]h}^{..d}.$	$R_{ijc}^{...d} = R_{ijc}^{\{!...^d} - 2D_{[i}K_{j]c}^{..d} + 2K_{[i c}^{..h}K_{j]h}^{..d}.$
mielke-geometrodynamics_099 E00286 (4.4.15)		■ 0.229 ● K +0	$D \Pi_{\square}^{\mathrm{T}} - E_{\square} = \Sigma_{\square}$	$D \Pi_{\square}^{\mathrm{T}} - E_{\square} = \Sigma_{\square}$	$D \Pi_{\square}^{\mathrm{T}} - E_{\square} = \Sigma_{\square}$
mielke-geometrodynamics_114 E00337 (5.3.14)		■ 0.239 ● C +3	$\begin{aligned} & R^{\{\alpha\beta\}} \wedge \eta_{\alpha\beta} \equiv R^{\alpha\beta} \wedge \eta_{\alpha\beta} - K^{\alpha\mu} \wedge K_{\mu}^{\beta} \wedge \eta_{\alpha\beta} + K^{\alpha\beta} \wedge T^{\gamma} \wedge \eta_{\alpha\beta\gamma} \\ & \quad + d(K^{\alpha\beta} \wedge \eta_{\alpha\beta}) \\ & = R^{\alpha\beta} \wedge \eta_{\alpha\beta} + T^{\alpha} \wedge * \left(-^{(1)}T_{\alpha} + 2^{(2)}T_{\alpha} + \frac{1}{2}^{(3)}T_{\alpha} \right) + 2dC_{\mathrm{TT}}^* \end{aligned}$	$\begin{aligned} R^{\{\alpha\beta\}} \wedge \eta_{\alpha\beta} &\equiv R^{\alpha\beta} \wedge \eta_{\alpha\beta} - K^{\alpha\mu} \wedge K_{\mu}^{\beta} \wedge \eta_{\alpha\beta} + K^{\alpha\beta} \wedge T^{\gamma} \wedge \eta_{\alpha\beta\gamma} \\ & \quad + d(K^{\alpha\beta} \wedge \eta_{\alpha\beta}) \\ & = R^{\alpha\beta} \wedge \eta_{\alpha\beta} + T^{\alpha} \wedge * \left(-^{(1)}T_{\alpha} + 2^{(2)}T_{\alpha} + \frac{1}{2}^{(3)}T_{\alpha} \right) + 2dC_{\mathrm{TT}}^* \end{aligned}$	$\begin{aligned} R^{\{\alpha\beta\}} \wedge \eta_{\alpha\beta} &\equiv R^{\alpha\beta} \wedge \eta_{\alpha\beta} - K^{\alpha\mu} \wedge K_{\mu}^{\beta} \wedge \eta_{\alpha\beta} + K^{\alpha\beta} \wedge T^{\gamma} \wedge \eta_{\alpha\beta\gamma} \\ & \quad + d(K^{\alpha\beta} \wedge \eta_{\alpha\beta}) \\ & = R^{\alpha\beta} \wedge \eta_{\alpha\beta} + T^{\alpha} \wedge * \left(-^{(1)}T_{\alpha} + 2^{(2)}T_{\alpha} + \frac{1}{2}^{(3)}T_{\alpha} \right) + 2dC_{\mathrm{TT}}^* \end{aligned} \tag{5.3}$
mielke-geometrodynamics_271 E00835 (12.2.3)		■ 0.247 ● C -3	$\begin{aligned} & L_{\mathrm{D}} \stackrel{*}{=} L \left(\gamma, \psi, D^{\dagger} \psi \right) + \frac{1}{4} \mathscr{A} \wedge j_5 + A \wedge j \\ & = L \left(\gamma, \psi, D^{(i)} \psi \right) - T^{\alpha} \wedge \mu_{\alpha} + A \wedge j. \end{aligned}$	$\begin{aligned} L_{\mathrm{D}} &= L \left(\gamma, \psi, D^{(f)} \psi \right) - \frac{i}{2} \bar{\psi} (*\gamma \wedge K - K \wedge *\gamma) \psi + A \wedge j \\ & = L \left(\gamma, \psi, D^{(\dagger)} \psi \right) + \frac{1}{4} \mathscr{A} \wedge j_5 + A \wedge j \\ & = L \left(\gamma, \psi, D^{(i)} \psi \right) - T^{\alpha} \wedge \mu_{\alpha} + A \wedge j. \end{aligned}$	$\begin{aligned} L_{\mathrm{D}} &= L(\gamma, \psi, D^{\dagger} \psi) - \frac{i}{2} \bar{\psi} (*\gamma \wedge K - K \wedge *\gamma) \psi + A \wedge j \\ & = L(\gamma, \psi, D^{\dagger} \psi) + \frac{1}{4} \mathscr{A} \wedge j_5 + A \wedge j \\ & = L(\gamma, \psi, D^{\dagger} \psi) - T^{\alpha} \wedge \mu_{\alpha} + A \wedge j. \end{aligned}$
mielke-geometrodynamics_248 E00787 (11.3.6)		■ 0.258 ● C +3	$i\gamma \wedge *D^{[1}\psi - 6\ell^{*2}\gamma^{n+1}\gamma \wedge *(\bar{\psi}\gamma^{n+1}\gamma\psi)\psi - m\psi\eta = 0$	$i\gamma \wedge *D^{\{\}\psi - 6\ell^{*2}\gamma^{n+1}\gamma \wedge *(\bar{\psi}\gamma^{n+1}\gamma\psi)\psi - m\psi\eta = 0 \Big ,$	$i\gamma \wedge *D^{\{\}\psi - 6\ell^{*2}\gamma^{n+1}\gamma \wedge *(\bar{\psi}\gamma^{n+1}\gamma\psi)\psi - m\psi\eta = 0 \Big ,$
Conf. MathPix confidence: ■ ■ ■ — ≥0.90.5–0.9<0.5no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● ● ● ● ● ● C componentW weakS stableN noiseK cleannot measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_037 EQ1078 (A.3)	365	0.258 W +1	$\begin{aligned} & \left(\sigma^a \right)^T := \left(\begin{array}{l} \sigma^0 \\ \sigma^1 \\ \sigma^2 \end{array} \right) \quad \sigma^0 := \left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right) \equiv \mathbb{1}, \quad \sigma^1 := \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) \equiv \sigma^2 := \left(\begin{array}{cc} 0 & -i \\ i & 0 \end{array} \right), \quad \sigma^3 := \left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right). \end{aligned}$	$(\sigma^a)^T := \begin{pmatrix} \sigma^0, \sigma^k & k = 1, 2, 3 \\ \sigma^0 := \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \equiv \mathbb{1}, & \sigma^1 := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \dots \\ \sigma^2 := \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, & \sigma^3 := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \end{pmatrix}.$	$(\sigma^a)^T := (\sigma^0, \sigma^k) \quad k = 1, 2, 3.$ $\sigma^0 := \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \equiv \mathbb{1}, \quad \sigma^1 := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$ $\sigma^2 := \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$
mielke-geometrodynamics_037 EQ0018 (2.3.4)	037	0.261 S +0	$\delta : \left\{ \begin{array}{ccc} G \times P & \rightarrow & P \\ w & \psi & \psi \\ g_p \cdot & p_o = p & \end{array} \middle e \cdot p_o = p_o \in P \right\}.$	$\delta : \left\{ \begin{array}{ccc} G \times P & \rightarrow & P \\ \psi & \psi & \psi \\ g_p \cdot & p_o = p & \end{array} \middle e \cdot p_o = p_o \in P \right\}.$	
mielke-geometrodynamics_177 EQ0544 (8.3.5)	177	0.261	$\left(\exp(\zeta) \right) := \mathbf{1} + \zeta + \frac{1}{2!} \zeta^2 + \frac{1}{3!} \zeta^3 + \dots$	$\exp(\zeta) := \mathbf{1} + \zeta + \frac{1}{2!} \zeta^2 + \frac{1}{3!} \zeta^3 + \dots$	
mielke-geometrodynamics_199 EQ0605 (9.2.4)	199	0.263 N +0	$\chi(M) := \sum_{p=0}^n (-1)^p b_p = \int_M \pi(\gamma_2^g)$	$\chi(M) := \sum_{p=0}^n (-1)^p b_p = \int_M \pi(\gamma_2^g)$	
mielke-geometrodynamics_211 EQ0627 (10.1.15)	211	0.271 N +0	$E_\alpha := \frac{\partial L}{\partial \vartheta^\alpha} = e_\alpha \lrcorner L + (e_\alpha \lrcorner T^\beta) \wedge H_\beta + 2(-1)^s (e_\alpha \lrcorner R^{\star\beta}) \wedge H_\beta^*,$	$E_\alpha := \frac{\partial L}{\partial \vartheta^\alpha} = e_\alpha \lrcorner L + (e_\alpha \lrcorner T^\beta) \wedge H_\beta + 2(-1)^s (e_\alpha \lrcorner R^{\star\beta}) \wedge H_\beta^*,$	
mielke-geometrodynamics_316 EQ0979 (14.2.20)	316	0.272 N +0	$\hat{\ell} = \left(\begin{array}{c} + \\ - \end{array} \right) \ell^* / \sqrt{\alpha_g} = \left(\begin{array}{c} + \\ - \end{array} \right) \ell^* \frac{\sqrt{\hbar c}}{g}$	$\hat{\ell} = \left(\begin{array}{c} + \\ - \end{array} \right) \ell^* / \sqrt{\alpha_g} = \left(\begin{array}{c} + \\ - \end{array} \right) \ell^* \frac{\sqrt{\hbar c}}{g}$	
mielke-geometrodynamics_216 EQ0654 (10.3.11)	216	0.274 N +0	$E_\alpha = e_\alpha \lrcorner L + [(e_\alpha \lrcorner T^\beta) + (-1)^s 2\delta\ell(e_\alpha \lrcorner R^{\star\beta})] \wedge H_\beta + (-1)^s \frac{2\gamma}{\ell} (e_\alpha \lrcorner R^{\star\beta}) \wedge \vartheta_\beta,$	$E_\alpha = e_\alpha \lrcorner L + [(e_\alpha \lrcorner T^\beta) + (-1)^s 2\delta\ell(e_\alpha \lrcorner R^{\star\beta})] \wedge H_\beta + (-1)^s \frac{2\gamma}{\ell} (e_\alpha \lrcorner R^{\star\beta}) \wedge \vartheta_\beta,$	
mielke-geometrodynamics_095 EQ0260 (4.3.33)	095	0.276 W -2	$R_{i[j\kappa l]}^{(l)} \equiv 0$	$R_{i[j\kappa l]}^{(l)} \equiv 0$	
mielke-geometrodynamics_213 EQ0639 (10.2.10)	213	0.278 N -1	$R_\alpha^{i\star} = -\Lambda_{\text{eff}} \eta_\alpha,$	$R_\alpha^{i\star} = -\Lambda_{\text{eff}} \eta_\alpha,$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_168 EQ0527 (7.7.47)		■ 0.282 ● N -1	$\begin{aligned} L_{\mathrm{Pontr}} := dC_{\mathrm{RR}} = & \frac{1}{2} R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha} - \frac{1}{2} R_{\alpha\beta}^{\{\} } \wedge R^{\{\} \alpha\beta} \\ & - \frac{1}{12} d \left[{}^* \mathscr{A} \wedge R^{\{\} } - \frac{1}{3} \mathscr{A} \wedge d\mathscr{A} + \frac{1}{9} {}^* \mathscr{A} \wedge {}^* (\mathscr{A} \wedge {}^* \mathscr{A}) \right]. \end{aligned}$	$L_{\mathrm{Pontr}} := dC_{\mathrm{RR}} = \frac{1}{2} R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha} - \frac{1}{2} R_{\alpha\beta}^{\{\} } \wedge R^{\{\} \alpha\beta} - \frac{1}{12} d \left[{}^* \mathscr{A} \wedge R^{\{\} } - \frac{1}{3} \mathscr{A} \wedge d\mathscr{A} + \frac{1}{9} {}^* \mathscr{A} \wedge {}^* (\mathscr{A} \wedge {}^* \mathscr{A}) \right].$	$L_{\mathrm{Pontr}} := dC_{\mathrm{RR}} = \frac{1}{2} R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha} - \frac{1}{2} R_{\alpha\beta}^{\{\} } \wedge R^{\{\} \alpha\beta} - \frac{1}{12} d \left[{}^* \mathscr{A} \wedge R^{\{\} } - \frac{1}{3} \mathscr{A} \wedge d\mathscr{A} + \frac{1}{9} {}^* \mathscr{A} \wedge {}^* (\mathscr{A} \wedge {}^* \mathscr{A}) \right].$
mielke-geometrodynamics_306 EQ0942 (14.1.23)		■ 0.287 ● N +0	$\xi \wedge {}^* \xi = \pm \frac{\Omega \wedge {}^* \Omega \operatorname{Tr}^*(\Omega \wedge {}^* \Omega) - (-1)^{2z} \Omega \wedge \Omega \operatorname{Tr}^*(\Omega \wedge \Omega)}{\sqrt{(\operatorname{Tr}^*(\Omega \wedge {}^* \Omega))^2 + (-1)^{2z} (\operatorname{Tr}^*(\Omega \wedge \Omega))^2}}.$	$\xi \wedge {}^* \xi = \pm \frac{\Omega \wedge {}^* \Omega \operatorname{Tr}^*(\Omega \wedge {}^* \Omega) - (-1)^{2z} \Omega \wedge \Omega \operatorname{Tr}^*(\Omega \wedge \Omega)}{\sqrt{(\operatorname{Tr}^*(\Omega \wedge {}^* \Omega))^2 + (-1)^{2z} (\operatorname{Tr}^*(\Omega \wedge \Omega))^2}}.$	$\xi \wedge {}^* \xi = \pm \frac{\Omega \wedge {}^* \Omega \operatorname{Tr}^*(\Omega \wedge {}^* \Omega) - (-1)^{2z} \Omega \wedge \Omega \operatorname{Tr}^*(\Omega \wedge \Omega)}{\sqrt{(\operatorname{Tr}^*(\Omega \wedge {}^* \Omega))^2 + (-1)^{2z} (\operatorname{Tr}^*(\Omega \wedge \Omega))^2}}.$
mielke-geometrodynamics_143 EQ0440 (6.8.11)		■ 0.288 ● N +0	$\left. D \Sigma_{\alpha} \cong (e_{\alpha} \rfloor T^{\gamma}) \wedge \Sigma_{\gamma} + (e_{\alpha} \rfloor R^{\gamma\delta}) \wedge \tau_{\gamma\delta} \right. \wedge \Sigma_{\alpha} \cong (e_{\alpha} \rfloor T^{\gamma}) \wedge \Sigma_{\gamma} + (e_{\alpha} \rfloor R^{\gamma\delta}) \wedge \tau_{\gamma\delta}$	$D\Sigma_{\alpha} \cong (e_{\alpha} \rfloor T^{\gamma}) \wedge \Sigma_{\gamma} + (e_{\alpha} \rfloor R^{\gamma\delta}) \wedge \tau_{\gamma\delta}$	$D\Sigma_{\alpha} \cong (e_{\alpha} \rfloor T^{\gamma}) \wedge \Sigma_{\gamma} + (e_{\alpha} \rfloor R^{\gamma\delta}) \wedge \tau_{\gamma\delta}$
mielke-geometrodynamics_105 EQ0315 (4.6.7)		■ 0.292 ●	$\Delta\Phi(\widetilde{\Gamma},\gamma)\simeq\frac{i}{\hbar}\int_S\Big(R^{(T)\alpha}P_{\alpha}+R_{\alpha}^{(L)\beta}L^{\alpha}_{\beta}\Big)$	$\Delta\Phi(\widetilde{\Gamma},\gamma)\simeq\frac{i}{\hbar}\int_S\Big(R^{(T)\alpha}P_{\alpha}+R_{\alpha}^{(L)\beta}L^{\alpha}_{\beta}\Big)$	$\Delta\Phi(\widetilde{\Gamma},\gamma)\simeq\frac{i}{\hbar}\int_S(R^{(T)\alpha}P_{\alpha}+R_{\alpha}^{(L)\beta}L^{\alpha}_{\beta}).$
mielke-geometrodynamics_313 EQ0965 (14.2.5)		■ 0.299 ● N +0	$\begin{aligned} & \stackrel{\circ}{\Theta} := \dot{\stackrel{\circ}{\vartheta}} - [\stackrel{\circ}{\omega}, \stackrel{\circ}{\vartheta}] = (1 - 2b) \stackrel{\circ}{\vartheta} \wedge \stackrel{\circ}{\vartheta}, \\ & \stackrel{\circ}{\Omega} := \dot{\stackrel{\circ}{\omega}} - \stackrel{\circ}{\omega} \wedge \stackrel{\circ}{\omega} = b(1 - b) \stackrel{\circ}{\vartheta} \wedge \stackrel{\circ}{\vartheta}. \end{aligned}$	$\begin{aligned} \stackrel{\circ}{\Theta} &:= \dot{\stackrel{\circ}{\vartheta}} - [\stackrel{\circ}{\omega}, \stackrel{\circ}{\vartheta}] = (1 - 2b) \stackrel{\circ}{\vartheta} \wedge \stackrel{\circ}{\vartheta}, \\ \stackrel{\circ}{\Omega} &:= \dot{\stackrel{\circ}{\omega}} - \stackrel{\circ}{\omega} \wedge \stackrel{\circ}{\omega} = b(1 - b) \stackrel{\circ}{\vartheta} \wedge \stackrel{\circ}{\vartheta}. \end{aligned}$	$\begin{aligned} \stackrel{\circ}{\Theta} &:= \dot{\stackrel{\circ}{\vartheta}} - [\stackrel{\circ}{\omega}, \stackrel{\circ}{\vartheta}] = (1 - 2b) \stackrel{\circ}{\vartheta} \wedge \stackrel{\circ}{\vartheta}, \\ \stackrel{\circ}{\Omega} &:= \dot{\stackrel{\circ}{\omega}} - \stackrel{\circ}{\omega} \wedge \stackrel{\circ}{\omega} = b(1 - b) \stackrel{\circ}{\vartheta} \wedge \stackrel{\circ}{\vartheta}. \end{aligned}$
mielke-geometrodynamics_087 EQ0226 (4.2.19)		■ 0.299 ● C +4	$\begin{aligned} A^{-1} \omega^{\mathrm{T}} = A^{-1} \widetilde{\vartheta} - G^{-1} D G A^{-1} \widetilde{\varphi} \\ = G^{-1} \widetilde{\vartheta} - G^{-1} D \widetilde{\varphi}, \end{aligned}$	$\begin{aligned} A^{-1} \omega^{\mathrm{T}} = A^{-1} \widetilde{\vartheta} - G^{-1} D G A^{-1} \widetilde{\varphi} \\ = G^{-1} \widetilde{\vartheta} - G^{-1} D \widetilde{\varphi}, \end{aligned}$	$\begin{aligned} A^{-1} \omega^{\mathrm{T}} &= A^{-1} \widetilde{\vartheta} - G^{-1} D G A^{-1} \widetilde{\varphi} \\ &= G^{-1} \widetilde{\vartheta} - G^{-1} D \widetilde{\varphi}, \end{aligned}$
mielke-geometrodynamics_135 EQ0407 (6.4.27)		■ 0.315 ● N +0	$\Psi_{\Lambda}(A)=\exp\left(\frac{3}{\Lambda}\int_{M_3}C_{\mathrm{RR}}^{\left(\pm\right)}\left(\right)\right),$	$\Psi_{\Lambda}(A)=\exp\left(\frac{3}{\Lambda}\int_{M_3}C_{\mathrm{RR}}^{\left(\pm\right)}\left(\right)\right),$	$\Psi_{\Lambda}(A)=\exp\left(\frac{3}{\Lambda}\int_{M_3}\underline{C}_{\mathrm{RR}}^{\left(\pm\right)}\left(\right)\right),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_082 EQ0196 (4.1.8)		0.322 N +1	$\begin{aligned} & \left[P_{\alpha}, P_{\beta} = 0 \right] \quad (P_{\alpha} := \partial / \partial x^{\alpha}), \\ & \left[P_{\alpha}, L_{\gamma \delta} \right] = g_{\alpha \delta} P_{\gamma} - g_{\alpha \gamma} P_{\delta}, \\ & \left[L_{\alpha \beta}, L_{\gamma \delta} \right] = c_{\alpha \beta \gamma \delta}^{\varepsilon} L_{\varepsilon \gamma}, \\ & c_{\alpha \beta \delta \gamma}^{\varepsilon \zeta} := \delta_{(\alpha}^{\varepsilon} \delta_{\delta}^{\zeta]} \eta_{\beta \gamma)}. \end{aligned}$	$\begin{aligned} & [P_{\alpha}, P_{\beta} = 0] \quad (P_{\alpha} := \partial / \partial x^{\alpha}), \\ & [P_{\alpha}, L_{\gamma \delta}] = g_{\alpha \delta} P_{\gamma} - g_{\alpha \gamma} P_{\delta}, \\ & [L_{\alpha \beta}, L_{\gamma \delta}] = c_{\alpha \beta \gamma \delta}^{\varepsilon} L_{\varepsilon \gamma}, \\ & c_{\alpha \beta \delta \gamma}^{\varepsilon \zeta} := \delta_{(\alpha}^{\varepsilon} \delta_{\delta}^{\zeta]} \eta_{\beta \gamma)}. \end{aligned}$	$\begin{aligned} & [P_{\alpha}, P_{\beta} = 0] \quad (P_{\alpha} := \partial / \partial x^{\alpha}), \\ & [P_{\alpha}, L_{\gamma \delta}] = g_{\alpha \delta} P_{\gamma} - g_{\alpha \gamma} P_{\delta}, \\ & [L_{\alpha \beta}, L_{\gamma \delta}] = c_{\alpha \beta \gamma \delta}^{\varepsilon} L_{\varepsilon \gamma}, \\ & c_{\alpha \beta \delta \gamma}^{\varepsilon \zeta} := \delta_{(\alpha}^{\varepsilon} \delta_{\delta}^{\zeta]} \eta_{\beta \gamma)}. \end{aligned}$
mielke-geometrodynamics_196 EQ0597 (9.1.15)		0.323 W +2	$\left[\mathrm{T}_{\alpha \beta \gamma} \right] = \left[\begin{array}{cccc} f & -h & \cdot & \cdot \\ \cdot & \cdot & -k & \cdot \\ \cdot & \cdot & \cdot & -k \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & -g \\ \cdot & \cdot & g & \cdot \\ 0 & 1 & 2 & 3 \end{array} \right] \begin{array}{l} 01 \\ 02 \\ 03 \\ 23 \\ 31 \\ 12 \end{array}$	$[T_{\alpha \beta}{}^{\gamma}] = \left[\begin{array}{cccc} f & -h & \cdot & \cdot \\ \cdot & \cdot & -k & \cdot \\ \cdot & \cdot & \cdot & -k \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & -g \\ \cdot & \cdot & g & \cdot \\ 0 & 1 & 2 & 3 \end{array} \right] \begin{array}{l} 01 \\ 02 \\ 03 \\ 23 \\ 31 \\ 12 \end{array}$	$[T_{\alpha \beta}{}^{\gamma}] = \left[\begin{array}{cccc} f & -h & \cdot & \cdot \\ \cdot & \cdot & -k & \cdot \\ \cdot & \cdot & \cdot & -k \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & -g \\ \cdot & \cdot & g & \cdot \\ 0 & 1 & 2 & 3 \end{array} \right] \begin{array}{l} 01 \\ 02 \\ 03 \\ 23 \\ 31 \\ 12 \end{array}$
mielke-geometrodynamics_255 EQ0803 (11.4.15)		0.324	$\psi = \frac{4}{\ell} \left(\frac{2\pi m}{3} \right)^{1/2} e^{-\mu/2 - \nu/4 - i\omega m t} \left[\begin{array}{cc} iH(\rho) & x_{\kappa}^m \\ F(\rho) & \frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } x_{\kappa}^m \end{array} \right].$	$\psi = \frac{4}{\ell} \left(\frac{2\pi m}{3} \right)^{1/2} e^{-\mu/2 - \nu/4 - i\omega m t} \left[\begin{array}{cc} iH(\rho) & x_{\kappa}^m \\ F(\rho) & \frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } x_{\kappa}^m \end{array} \right].$	$\psi = \frac{4}{\ell} \left(\frac{2\pi m}{3} \right)^{1/2} e^{-\mu/2 - \nu/4 - i\omega m t} \left[\begin{array}{cc} iH(\rho) & x_{\kappa}^m \\ F(\rho) & \frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } x_{\kappa}^m \end{array} \right].$
mielke-geometrodynamics_365 EQ1082 (A.7)		0.326 W +1	$\begin{aligned} & \gamma^0 := \begin{pmatrix} \sigma^0 & 0 \\ 0 & \sigma^0 \end{pmatrix}, \quad \gamma^k := \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix}; \\ & \gamma^5 := i\gamma^0 \gamma^1 \gamma^2 \gamma^3 = \begin{pmatrix} 0 & \sigma^0 \\ \sigma^0 & 0 \end{pmatrix}. \end{aligned}$	$\begin{aligned} & \gamma^0 := \begin{pmatrix} \sigma^0 & 0 \\ 0 & \sigma^0 \end{pmatrix}, \quad \gamma^k := \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix}; \\ & \gamma^5 := i\gamma^0 \gamma^1 \gamma^2 \gamma^3 = \begin{pmatrix} 0 & \sigma^0 \\ \sigma^0 & 0 \end{pmatrix}. \end{aligned}$	$\begin{aligned} & \gamma^0 := \begin{pmatrix} \sigma^0 & 0 \\ 0 & \sigma^0 \end{pmatrix}, \quad \gamma^k := \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix}; \\ & \gamma^5 := i\gamma^0 \gamma^1 \gamma^2 \gamma^3 = \begin{pmatrix} 0 & \sigma^0 \\ \sigma^0 & 0 \end{pmatrix}. \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_316 EQ0978 (14.2.19)		0.331 W -2	$\begin{aligned} \pi \, \hat{L}_{ } &= -\frac{\hat{\ell}^2}{\ell^{*2}} \operatorname{Tr}(\Omega' \wedge {}^*\Omega' \wedge \underbrace{\vartheta \wedge \cdots \wedge \vartheta}_{\text{K terms}}) \\ &= -\frac{1}{\alpha_g} \operatorname{Tr}(\Omega' \wedge {}^*\Omega') \sqrt{\det \overset{\circ}{g}} = L'_{\text{YM}} d_{\mu}(G) \end{aligned}$	$\pi \hat{L}_{ } = -\frac{\hat{\ell}^2}{\ell^{*2}} \operatorname{Tr}(\Omega' \wedge {}^*\Omega' \wedge \underbrace{\vartheta \wedge \cdots \wedge \vartheta}_{\text{K terms}})$ $= -\frac{1}{\alpha_g} \operatorname{Tr}(\Omega' \wedge {}^*\Omega') \sqrt{\det \overset{\circ}{g}} = L'_{\text{YM}} d_{\mu}(G)$	$\pi \hat{L}_{ } = -\frac{\hat{\ell}^2}{\ell^{*2}} \operatorname{Tr}(\Omega' \wedge {}^*\Omega' \wedge \underbrace{\vartheta \wedge \cdots \wedge \vartheta}_{\text{K terms}})$ $= -\frac{1}{\alpha_g} \operatorname{Tr}(\Omega' \wedge {}^*\Omega') \sqrt{\det \overset{\circ}{g}} = L'_{\text{YM}} d_{\mu}(G)$
mielke-geometrodynamics_186 EQ0979 (8.6.17)		0.351 K +0	$\left. \left. \left. 2 D \left(e^{\beta} \right] D H_{\alpha \beta} - \frac{1}{4} \vartheta_{\alpha} \wedge e^{\gamma} \right] e^{\delta} \right] D H_{\gamma \delta} \right) - E_{\alpha} = \Sigma_{\alpha} - D \mu_{\alpha},$	$2D\left(e^{\beta}\rfloor DH_{\alpha\beta}-\frac{1}{4}\vartheta_{\alpha}\wedge e^{\gamma}\rfloor e^{\delta}\rfloor DH_{\gamma\delta}\right)-E_{\alpha}=\Sigma_{\alpha}-D\mu_{\alpha},$	$2D\left(e^{\beta}\rfloor DH_{\alpha\beta}-\frac{1}{4}\vartheta_{\alpha}\wedge e^{\gamma}\rfloor e^{\delta}\rfloor DH_{\gamma\delta}\right)-E_{\alpha}=\Sigma_{\alpha}-D\mu_{\alpha},$
mielke-geometrodynamics_309 EQ0956 (14.1.38)		0.353 W -1	$\begin{aligned} d\delta &= (-1)^{-z*} \Sigma_{\text{YM}}(\Omega) \wedge D\Sigma_{\text{YM}}(\Omega) / (*\Sigma_{\text{YM}} \wedge \Sigma_{\text{YM}}) \\ &= (-1)^{-z*} (\vartheta \wedge {}^*\Omega^g) \wedge D(\vartheta \wedge {}^*\Omega^g) / (*\vartheta \wedge {}^*\Omega^g \wedge {}^*(\vartheta \wedge {}^*\Omega^g)). \end{aligned}$	$d\delta = (-1)^{-z*} \Sigma_{\text{YM}}(\Omega) \wedge D\Sigma_{\text{YM}}(\Omega) / (*\Sigma_{\text{YM}} \wedge \Sigma_{\text{YM}})$ $= (-1)^{-z*} (\vartheta \wedge {}^*\Omega^g) \wedge D(\vartheta \wedge {}^*\Omega^g) / (*\vartheta \wedge {}^*\Omega^g \wedge {}^*(\vartheta \wedge {}^*\Omega^g)).$	$d\delta = (-1)^{-z*} \Sigma_{\text{YM}}(\Omega) \wedge D\Sigma_{\text{YM}}(\Omega) / (*\Sigma_{\text{YM}} \wedge \Sigma_{\text{YM}})$ $= (-1)^{-z*} (\vartheta \wedge {}^*\Omega^g) \wedge D(\vartheta \wedge {}^*\Omega^g) / (*\vartheta \wedge {}^*\Omega^g \wedge {}^*(\vartheta \wedge {}^*\Omega^g)).$
mielke-geometrodynamics_180 EQ0955 (8.4.9)		0.371 N +0	$\left. \left. \left. E_{\alpha} = \frac{1}{2} \operatorname{Tr} \left(\overset{\pm}{\Omega}^g \wedge e_{\alpha} \right] {}^*\overset{\pm}{\Omega}^g - e_{\alpha} \right] {}^{\pm} \Omega^g \wedge {}^*\overset{\pm}{\Omega}^g \right) = 0,$	$E_{\alpha} = \frac{1}{2} \operatorname{Tr} \left(\overset{\pm}{\Omega}^g \wedge e_{\alpha} \rfloor {}^*\overset{\pm}{\Omega}^g - e_{\alpha} \rfloor \overset{\pm}{\Omega}^g \wedge {}^*\overset{\pm}{\Omega}^g \right) = 0,$	$E_{\alpha} = \frac{1}{2} \operatorname{Tr} \left(\overset{\pm}{\Omega}^g \wedge e_{\alpha} \rfloor {}^*\overset{\pm}{\Omega}^g - e_{\alpha} \rfloor \overset{\pm}{\Omega}^g \wedge {}^*\overset{\pm}{\Omega}^g \right) = 0,$
mielke-geometrodynamics_182 EQ0962 (8.5.5)		0.377	$\begin{aligned} \widetilde{L}_{\text{SKY}}^{(*)} &= \frac{1}{2} \operatorname{Tr} \left[\left(\otimes^g - i \frac{\theta_{\text{T}}^*}{4\ell^2} \sigma \right) \wedge {}^* \left(\overset{t}{\Omega}^g - i \frac{\theta_{\text{T}}^*}{4\ell^2} \sigma \right) \right] \\ &= -\frac{\theta_{\text{T}}^*}{2\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} + \theta_{\text{T}}^* \Lambda_{\text{eff}} \eta - \frac{1}{2} R^{\alpha\beta} \wedge {}^*R_{\alpha\beta} \mp \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)}. \end{aligned}$	$\widetilde{L}_{\text{SKY}}^{(*)} = \frac{1}{2} \operatorname{Tr} \left[\left(\otimes^g - i \frac{\theta_{\text{T}}^*}{4\ell^2} \sigma \right) \wedge {}^* \left(\overset{t}{\Omega}^g - i \frac{\theta_{\text{T}}^*}{4\ell^2} \sigma \right) \right]$ $= -\frac{\theta_{\text{T}}^*}{2\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} + \theta_{\text{T}}^* \Lambda_{\text{eff}} \eta - \frac{1}{2} R^{\alpha\beta} \wedge {}^*R_{\alpha\beta} \mp \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)}$	$\widetilde{L}_{\text{SKY}}^{(*)} = \frac{1}{2} \operatorname{Tr} \left[\left(\overset{\pm}{\Omega}^g - i \frac{\theta_{\text{T}}^*}{4\ell^2} \sigma \right) \wedge {}^* \left(\overset{\pm}{\Omega}^g - i \frac{\theta_{\text{T}}^*}{4\ell^2} \sigma \right) \right] \tag{8.5}$ $= -\frac{\theta_{\text{T}}^*}{2\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} + \theta_{\text{T}}^* \Lambda_{\text{eff}} \eta - \frac{1}{2} R^{\alpha\beta} \wedge {}^*R_{\alpha\beta} \mp \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)}.$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_132 EQ0393 (6.4.8)		0.382 W +1	$\begin{aligned} n^{\alpha} \Sigma_{\alpha} &= n \downarrow L - (\mathfrak{t}_n \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (n \downarrow \Psi) \wedge \frac{\delta L}{\delta \Psi} + D[(n \downarrow \Psi) \wedge \frac{\partial L}{\partial D \Psi}] \\ &\cong n \downarrow L - \left(\ell_n \Psi + \Gamma_{\perp}^{\alpha \beta} \rho(I_{\alpha \beta}) \Psi \right) \wedge \frac{\partial L}{\partial D \Psi} + D[(n \downarrow \Psi) \wedge \frac{\partial L}{\partial D \Psi}]. \end{aligned}$	$n^{\alpha} \Sigma_{\alpha} = n \downarrow L - (\mathfrak{t}_n \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (n \downarrow \Psi) \wedge \frac{\delta L}{\delta \Psi} + D[(n \downarrow \Psi) \wedge \frac{\partial L}{\partial D \Psi}]$ $\cong n \downarrow L - \left(\ell_n \Psi + \Gamma_{\perp}^{\alpha \beta} \rho(I_{\alpha \beta}) \Psi \right) \wedge \frac{\partial L}{\partial D \Psi} + D[(n \downarrow \Psi) \wedge \frac{\partial L}{\partial D \Psi}].$	$n^{\alpha} \Sigma_{\alpha} = n \downarrow L - (\mathfrak{t}_n \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (n \downarrow \Psi) \wedge \frac{\delta L}{\delta \Psi} + D[(n \downarrow \Psi) \wedge \frac{\partial L}{\partial D \Psi}]$ $\cong n \downarrow L - \left(\ell_n \Psi + \Gamma_{\perp}^{\alpha \beta} \rho(I_{\alpha \beta}) \Psi \right) \wedge \frac{\partial L}{\partial D \Psi} + D[(n \downarrow \Psi) \wedge \frac{\partial L}{\partial D \Psi}].$
mielke-geometrodynamics_094 EQ0261 (4.3.26)		0.394 N +0	$K_{\cdot \alpha \beta}^c = -\frac{1}{2} (T_{\cdot \alpha \beta}^c - T_{\alpha, \beta}^{\cdot c} + T_{\beta \cdot \alpha}^c) = -K_{\cdot \beta \alpha}^c.$	$K_{\cdot \alpha \beta}^c = -\frac{1}{2} (T_{\cdot \alpha \beta}^c - T_{\alpha, \beta}^{\cdot c} + T_{\beta \cdot \alpha}^c) = -K_{\cdot \beta \alpha}^c.$	$K_{\cdot \alpha \beta}^c = -\frac{1}{2} (T_{\cdot \alpha \beta}^c - T_{\alpha, \beta}^{\cdot c} + T_{\beta \cdot \alpha}^c) = -K_{\cdot \beta \alpha}^c.$
mielke-geometrodynamics_128 EQ0377 (6.2.28)		0.396 W +0	$\stackrel{\scriptscriptstyle\text{rel}}{\left(\begin{array}{c} \vartheta^{\alpha} \wedge \frac{\partial L}{\partial D} \end{array} \right)}_{\alpha} = -\frac{1}{2} \left[\vartheta^{\alpha} \wedge \frac{\partial L}{\partial D} \right]_{\alpha} + d \left(\vartheta^{\alpha} \wedge \frac{\partial L}{\partial D} \right)_{\alpha} = -\frac{1}{2} \left(\frac{\partial L}{\partial D} \vartheta^{\alpha} \wedge \frac{\partial L}{\partial D} \right)_{\alpha},$	$\stackrel{\scriptscriptstyle\text{rel}}{\left(\begin{array}{c} \vartheta^{\alpha} \wedge \frac{\partial L}{\partial D} \end{array} \right)}_{\alpha} = -\frac{1}{2} \left[\vartheta^{\alpha} \wedge \frac{\partial L}{\partial D} \right]_{\alpha} + d \left(\vartheta^{\alpha} \wedge \frac{\partial L}{\partial D} \right)_{\alpha} = -\frac{1}{2} \left(\frac{\partial L}{\partial D} \vartheta^{\alpha} \wedge \frac{\partial L}{\partial D} \right)_{\alpha},$	$\stackrel{\scriptscriptstyle\text{rel}}{\left(\begin{array}{c} \vartheta^{\alpha} \wedge \frac{\partial L}{\partial D} \end{array} \right)}_{\alpha} = -\frac{1}{2} \left[\vartheta^{\alpha} \wedge \frac{\partial L}{\partial D} \right]_{\alpha} + d \left(\vartheta^{\alpha} \wedge \frac{\partial L}{\partial D} \right)_{\alpha} = -\frac{1}{2} \left(\frac{\partial L}{\partial D} \vartheta^{\alpha} \wedge \frac{\partial L}{\partial D} \right)_{\alpha},$
mielke-geometrodynamics_094 EQ0262 (4.3.27)		0.399 C +3	$\begin{aligned} \Omega^g &= \Omega^{\{\}} - dK + K \wedge \omega^{\{\}} + \omega^{\{\}} \wedge K - K \wedge K \\ &= \Omega^{\{\}} - D^{\{\}} K - K \wedge K = \Omega^{\{\}} - DK + K \wedge K \end{aligned}$	$\Omega^g = \Omega^{\{\}} - dK + K \wedge \omega^{\{\}} + \omega^{\{\}} \wedge K - K \wedge K$ $= \Omega^{\{\}} - D^{\{\}} K - K \wedge K = \Omega^{\{\}} - DK + K \wedge K$	$\Omega^g = \Omega^{\{\}} - dK + K \wedge \omega^{\{\}} + \omega^{\{\}} \wedge K - K \wedge K$ $= \Omega^{\{\}} - D^{\{\}} K - K \wedge K = \Omega^{\{\}} - DK + K \wedge K$
mielke-geometrodynamics_184 EQ0570 (8.6.8)		0.403 K +0	$\left. E_{\alpha} \right = \frac{\partial L}{\partial \vartheta^{\alpha}} = e_{\alpha} \downarrow L + (e_{\alpha} \downarrow T^{\beta}) \wedge H_{\beta} + (e_{\alpha} \downarrow R^{\beta \gamma}) \wedge H_{\beta \gamma}$	$E_{\alpha} := \partial L / \partial \vartheta^{\alpha} = e_{\alpha} \downarrow L + (e_{\alpha} \downarrow T^{\beta}) \wedge H_{\beta} + (e_{\alpha} \downarrow R^{\beta \gamma}) \wedge H_{\beta \gamma}$	$E_{\alpha} := \partial L / \partial \vartheta^{\alpha} = e_{\alpha} \downarrow L + (e_{\alpha} \downarrow T^{\beta}) \wedge H_{\beta} + (e_{\alpha} \downarrow R^{\beta \gamma}) \wedge H_{\beta \gamma}$
mielke-geometrodynamics_103 EQ0306 (4.5.16)		0.406 N +0	$\left. \widetilde{\varepsilon}_{\text{MA}} \right := \varepsilon_{\text{RC}} + \xi \downarrow L \cong -(\mathcal{L}_{\xi} \Gamma_{\beta}^{\gamma}) \wedge H^{\beta}_{\gamma} + dH_{\infty}$	$\widetilde{\varepsilon}_{\text{MA}} := \varepsilon_{\text{RC}} + \xi \downarrow L \cong -(\mathcal{L}_{\xi} \Gamma_{\beta}^{\gamma}) \wedge H^{\beta}_{\gamma} + dH_{\infty}$	$\widetilde{\varepsilon}_{\text{MA}} := \varepsilon_{\text{RC}} + \xi \downarrow L \cong -(\mathcal{L}_{\xi} \Gamma_{\beta}^{\gamma}) \wedge H^{\beta}_{\gamma} + dH_{\infty}$
mielke-geometrodynamics_114 EQ0336 (5.3.13)		0.407 N +1	$R^{\alpha \beta} = R^{\{j \alpha \beta\}} - D^{\{\}} K^{\alpha \beta} - K^{\alpha}_{\mu} \wedge K^{\mu \beta}$	$R^{\alpha \beta} = R^{\{j \alpha \beta\}} - D^{\{\}} K^{\alpha \beta} - K^{\alpha}_{\mu} \wedge K^{\mu \beta}$	$R^{\alpha \beta} = R^{\{j \alpha \beta\}} - D^{\{\}} K^{\alpha \beta} - K^{\alpha}_{\mu} \wedge K^{\mu \beta}$
mielke-geometrodynamics_375 EQ1136 (C.4)		0.409 N +0	$c_{ij}^k = -c_{ji}^k$	$c_{ij}^k = -c_{ji}^k$	$c_{ij}^{} = -c_{ji}^{}$
Conf. MathPix confidence: 0.9 0.5-0.9 0.5 no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_279 EQ0861 (12.4.8)		■ 0.420	$\begin{aligned} d\,C_{\mathrm{RR}} &= -\operatorname{Tr}(\Omega\wedge\Omega) \\ &= \frac{1}{2}R_{\alpha\beta}^{\{\}}\wedge R^{\{\beta\alpha\}} \\ &+ \frac{1}{12}d\left[{}^*\mathcal{A}\wedge R^{\{\}}-\frac{1}{3}\mathcal{A}\wedge d\mathcal{A}+\frac{1}{9}{}^*\mathcal{A}\wedge{}^*(\mathcal{A}\wedge{}^*\mathcal{A})\right]. \end{aligned}$	$dC_{\mathrm{RR}}=-\operatorname{Tr}(\Omega\wedge\Omega)$ $=\frac{1}{2}R_{\alpha\beta}^{\{\}}\wedge R^{\{\beta\alpha\}}$ $+\frac{1}{12}d\left[{}^*\mathcal{A}\wedge R^{\{\}}-\frac{1}{3}\mathcal{A}\wedge d\mathcal{A}+\frac{1}{9}{}^*\mathcal{A}\wedge{}^*(\mathcal{A}\wedge{}^*\mathcal{A})\right].$	$dC_{\mathrm{RR}}=-\operatorname{Tr}(\Omega\wedge\Omega)$ $=\frac{1}{2}R_{\alpha\beta}^{\{\}}\wedge R^{\{\beta\alpha\}}$ $+\frac{1}{12}d\left[{}^*\mathcal{A}\wedge R^{\{\}}-\frac{1}{3}\mathcal{A}\wedge d\mathcal{A}+\frac{1}{9}{}^*\mathcal{A}\wedge{}^*(\mathcal{A}\wedge{}^*\mathcal{A})\right].$
mielke-geometrodynamics_151 EQ0451 (7.2.4)		■ 0.424 N +0	$\stackrel{+}{\Omega}:=\frac{1}{2}\left(\Omega^g\pm{}^*\Omega^{g(*)}\right),$	$\stackrel{+}{\Omega}:=\frac{1}{2}\left(\Omega^g\pm{}^*\Omega^{g(*)}\right),$	$\stackrel{\pm}{\Omega}^g:=\frac{1}{2}\left(\Omega^g\pm{}^*\Omega^{g(*)}\right),$
mielke-geometrodynamics_127 EQ0372 (6.2.23)		■ 0.437 C -3	$\left.\stackrel{(\pm)}{D}\stackrel{(\pm)}{\Pi}_\alpha\mp\frac{i}{4}\ell^2e_\alpha\right]\left(\stackrel{(+)}{\Pi}^\beta\wedge\stackrel{(-)}{\Pi}_\beta\right)=\Sigma_\alpha,$	$\left.\stackrel{(\pm)}{D}\stackrel{(\pm)}{\Pi}_\alpha\mp\frac{i}{4}\ell^2e_\alpha\right]\left(\stackrel{(+)}{\Pi}^\beta\wedge\stackrel{(-)}{\Pi}_\beta\right)=\Sigma_\alpha,$	$\stackrel{(\pm)}{D}\stackrel{(\pm)}{\Pi}_\alpha\mp\frac{i}{4}\ell^2e_\alpha\left(\stackrel{(+)}{\Pi}^\beta\wedge\stackrel{(-)}{\Pi}_\beta\right)=\Sigma_\alpha,$
mielke-geometrodynamics_322 EQ0997 (14.3.10)		■ 0.442 N -1	$D^{\mathrm{(int)}}=\stackrel{\circ}{\gamma}\wedge{}^*\stackrel{\circ}{D}=\stackrel{\circ}{\gamma}\wedge{}^*(\stackrel{\circ}{d}+i\stackrel{\circ}{\tilde{\omega}})$	$D^{\mathrm{(int)}}=\stackrel{\circ}{\gamma}\wedge{}^*\stackrel{\circ}{D}=\stackrel{\circ}{\gamma}\wedge{}^*(\stackrel{\circ}{d}+i\stackrel{\circ}{\tilde{\omega}})$	$\mathcal{D}^{(int)}=\stackrel{\circ}{\gamma}\wedge{}^*\stackrel{\circ}{D}=\stackrel{\circ}{\gamma}\wedge{}^*(\stackrel{\circ}{d}+i\stackrel{\circ}{\tilde{\omega}})$
mielke-geometrodynamics_156 EQ0474 (7.4.2)		■ 0.444 N -1	$R_{\alpha\beta}\equiv C_{\alpha\beta}+\frac{1}{2}Ri^\tau_{\alpha\beta}-\frac{R}{12}\vartheta_\alpha\wedge\vartheta_\beta,$	$R_{\alpha\beta}\equiv C_{\alpha\beta}+\frac{1}{2}Ri^\tau_{\alpha\beta}-\frac{R}{12}\vartheta_\alpha\wedge\vartheta_\beta,$	$R_{\alpha\beta}\equiv C_{\alpha\beta}+\frac{1}{2}Ri^\tau_{\alpha\beta}-\frac{R}{12}\vartheta_\alpha\wedge\vartheta_\beta,$
mielke-geometrodynamics_221 EQ0681 (10.3.38)		■ 0.452 C +4	$\begin{aligned} \chi &= \theta_{\mathrm{L}}\varepsilon - \tilde{\theta}_{\mathrm{L}}\tilde{\varepsilon}, \quad \Lambda = \frac{(-1)^s}{4\ell^2}\left(\theta_{\mathrm{L}}\varepsilon^3 - \tilde{\theta}_{\mathrm{L}}\tilde{\varepsilon}^3\right), \quad \theta_{\mathrm{T}} = \frac{(-1)^s}{2}\left(\theta_{\mathrm{L}}\varepsilon^2 - \tilde{\theta}_{\mathrm{L}}\tilde{\varepsilon}^2\right), \\ \Delta\theta_{\mathrm{L}} &= \theta_{\mathrm{L}} - \tilde{\theta}_{\mathrm{L}}, \quad \Delta\theta_{\mathrm{TL}} = \frac{1}{2}\left(\theta_{\mathrm{L}}\varepsilon - \tilde{\theta}_{\mathrm{L}}\tilde{\varepsilon}\right). \end{aligned}$	$\chi = \theta_{\mathrm{L}}\varepsilon - \tilde{\theta}_{\mathrm{L}}\tilde{\varepsilon}, \quad \Lambda = \frac{(-1)^s}{4\ell^2}\left(\theta_{\mathrm{L}}\varepsilon^3 - \tilde{\theta}_{\mathrm{L}}\tilde{\varepsilon}^3\right), \quad \theta_{\mathrm{T}} = \frac{(-1)^s}{2}\left(\theta_{\mathrm{L}}\varepsilon^2 - \tilde{\theta}_{\mathrm{L}}\tilde{\varepsilon}^2\right),$ $\Delta\theta_{\mathrm{L}} = \theta_{\mathrm{L}} - \tilde{\theta}_{\mathrm{L}}, \quad \Delta\theta_{\mathrm{TL}} = \frac{1}{2}\left(\theta_{\mathrm{L}}\varepsilon - \tilde{\theta}_{\mathrm{L}}\tilde{\varepsilon}\right).$	$\chi = \theta_{\mathrm{L}}\varepsilon - \tilde{\theta}_{\mathrm{L}}\tilde{\varepsilon}, \quad \Lambda = \frac{(-1)^s}{4\ell^2}\left(\theta_{\mathrm{L}}\varepsilon^3 - \tilde{\theta}_{\mathrm{L}}\tilde{\varepsilon}^3\right), \quad \theta_{\mathrm{T}} = \frac{(-1)^s}{2}\left(\theta_{\mathrm{L}}\varepsilon^2 - \tilde{\theta}_{\mathrm{L}}\tilde{\varepsilon}^2\right),$ $\Delta\theta_{\mathrm{L}} = \theta_{\mathrm{L}} - \tilde{\theta}_{\mathrm{L}}, \quad \Delta\theta_{\mathrm{TL}} = \frac{1}{2}\left(\theta_{\mathrm{L}}\varepsilon - \tilde{\theta}_{\mathrm{L}}\tilde{\varepsilon}\right).$
mielke-geometrodynamics_273 EQ0845 (12.2.13)		■ 0.457 N -2	$C_{\mathrm{TT}}\cong\frac{1}{4}j_5, \quad \mathcal{A}=2\ell^2{}^*C_{\mathrm{TT}}=\left(\ell^2/2\right)\bar{\psi}\gamma_5\gamma\psi.$	$C_{\mathrm{TT}}\cong\frac{1}{4}j_5, \quad \mathcal{A}=2\ell^2{}^*C_{\mathrm{TT}}=\left(\ell^2/2\right)\bar{\psi}\gamma_5\gamma\psi.$	$C_{\mathrm{TT}}\cong\frac{1}{4}j_5, \quad \mathcal{A}=2\ell^2{}^*C_{\mathrm{TT}}=\left(\ell^2/2\right)\bar{\psi}\gamma_5\gamma\psi.$
mielke-geometrodynamics_128 EQ0375 (6.2.26)		■ 0.466 S +2	$\stackrel{(\pm)}{L}_{\parallel}=\mp\frac{i}{4}\ell^2\stackrel{(\pm)}{\Pi}^\alpha\wedge\stackrel{(\pm)}{\Pi}_\alpha, \quad \stackrel{(+)}{L}_{\parallel}=\stackrel{(-)}{L}_{\parallel},$	$\stackrel{(\pm)}{L}_{\parallel}=\mp\frac{i}{4}\ell^2\stackrel{(\pm)}{\Pi}^\alpha\wedge\stackrel{(\pm)}{\Pi}_\alpha, \quad \stackrel{(+)}{L}_{\parallel}=\stackrel{(-)}{L}_{\parallel},$	$\stackrel{(\pm)}{L}_{\parallel}=\mp\frac{i}{4}\ell^2\stackrel{(\pm)}{\Pi}^\alpha\wedge\stackrel{(\pm)}{\Pi}_\alpha, \quad \stackrel{(+)}{L}_{\parallel}=\stackrel{(-)}{L}_{\parallel},$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E00222 (4.2.15)	087	■ 0.466 ● N +0	$\widetilde{\varphi}: A(M) \rightarrow C^\infty(\overset{\circ}{V})$	$\widetilde{\varphi}: A(M) \rightarrow C^\infty(\overset{\circ}{V})$	$\widetilde{\varphi}: A(M) \rightarrow C^\infty(\overset{\circ}{V})$
mielke-geometrodynamics_E00604 (9.2.3)	199	■ 0.474 ●	$\begin{aligned} L_{\text{Euler}}, &= \frac{1}{(4\pi)^2} \text{Tr} \left(\Omega^g \wedge \Omega^{g(*)} \right) \\ &= \frac{(-1)^{s/4}}{32\pi^2} R^{\alpha\beta\gamma\delta*} R^*_{\alpha\beta\gamma\delta} \sqrt{ g } d^4x. \end{aligned}$	$L_{\text{Euler}}, = {}^*(\gamma_2^g) = \frac{1}{(4\pi)^2} \text{Tr} \left(\Omega^g \wedge \Omega^{g(*)} \right) \\ = \frac{(-1)^{s/4}}{32\pi^2} R^{\alpha\beta\gamma\delta*} R^*_{\alpha\beta\gamma\delta} \sqrt{ g } d^4x.$	$L_{\text{Euler}}, = \pi^*(\gamma_2^g) = \frac{1}{(4\pi)^2} \text{Tr}(\Omega^g \wedge \Omega^{g(*)}) \\ = \frac{(-1)^{s/4}}{32\pi^2} R^{\alpha\beta\gamma\delta*} R^*_{\alpha\beta\gamma\delta} \sqrt{ g } d^4x.$
mielke-geometrodynamics_E00401 (6.4.20)	133	■ 0.482 ● W +1	$\mathscr{H}_{\parallel} = \underline{D} - \mathcal{A}_{\hat{0}}^{(\pm)*(\pm)} = 0,$	$\mathscr{H}_{\parallel} = \underline{D} - \mathcal{A}_{\hat{0}}^{(\pm)*(\pm)} = 0,$	$\mathscr{H}_{\parallel} = \underline{D} - \mathcal{A}_{\hat{0}}^{(\pm)*(\pm)} = 0,$
mielke-geometrodynamics_E00750 (11.1.17)	241	■ 0.484 ● N +0	$D_{\left[\frac{1}{2}\right]}^{\text{F}} = \underbrace{D^{\frac{1}{2}} \times \cdots \times D^{\frac{1}{2}}}_{(n-4)/4}.$	$D_{\left[\frac{1}{2}\right]}^{\text{F}} = \underbrace{D^{\frac{1}{2}} \times \cdots \times D^{\frac{1}{2}}}_{(n-4)/4}.$	$D_{\left[\frac{1}{2}\right]}^{\text{F}} = \underbrace{D^{\frac{1}{2}} \times \cdots \times D^{\frac{1}{2}}}_{(n-4)/4}.$
mielke-geometrodynamics_E00686 (10.3.44)	222	■ 0.497 ● N +0	$C_l = E_l^\alpha \left\{ D^{\{ \}} \left[2 \left(e^\beta \lrcorner D^{\{ \}} H_{[\alpha\beta]} \right) + \frac{1}{2} \vartheta_\alpha \wedge \left(e^\beta \lrcorner e^\gamma \lrcorner D^{\{ \}} H_{[\beta\gamma]} \right) \right] \right\}$	$C_l = E_l^\alpha \left\{ D^{\{ \}} \left[2 \left(e^\beta \lrcorner D^{\{ \}} H_{[\alpha\beta]} \right) + \frac{1}{2} \vartheta_\alpha \wedge \left(e^\beta \lrcorner e^\gamma \lrcorner D^{\{ \}} H_{[\beta\gamma]} \right) \right] \right\}$	$C_l = E_l^\alpha \left\{ D^{\{ \}} \left[2 \left(e^\beta \lrcorner D^{\{ \}} H_{[\alpha\beta]} \right) + \frac{1}{2} \vartheta_\alpha \wedge \left(e^\beta \lrcorner e^\gamma \lrcorner D^{\{ \}} H_{[\beta\gamma]} \right) \right] \right\}$
mielke-geometrodynamics_E00515 (7.7.33)	166	■ 0.497 ● N +0	$T := \frac{1}{4} \text{Tr}(\check{\gamma} \lrcorner \Theta) = e_\alpha \lrcorner T^\alpha, \quad \mathcal{A} := \frac{1}{4} {}^* \text{Tr}(\gamma \wedge \Theta) = {}^*(\vartheta_\alpha \wedge T^\alpha).$	$T := \frac{1}{4} \text{Tr}(\check{\gamma} \lrcorner \Theta) = e_\alpha \lrcorner T^\alpha, \quad \mathcal{A} := \frac{1}{4} {}^* \text{Tr}(\gamma \wedge \Theta) = {}^*(\vartheta_\alpha \wedge T^\alpha).$	$T := \frac{1}{4} \text{Tr}(\check{\gamma} \lrcorner \Theta) = e_\alpha \lrcorner T^\alpha, \quad \mathcal{A} := \frac{1}{4} {}^* \text{Tr}(\gamma \wedge \Theta) = {}^*(\vartheta_\alpha \wedge T^\alpha).$
mielke-geometrodynamics_E00799 (11.4.11)	254	■ 0.505 ● N +0	$ds^2 = \frac{1}{\lambda^2} \left(R_\circ^2 d\lambda^2 - \mathfrak{o}_{ij} dy^i \otimes_s dy^j \right).$	$ds^2 = \frac{1}{\lambda^2} \left(R_\circ^2 d\lambda^2 - \mathfrak{o}_{ij} dy^i \otimes_s dy^j \right).$	$ds^2 = \frac{1}{\lambda^2} \left(R_\circ^2 d\lambda^2 - \mathfrak{o}_{ij} dy^i \otimes_s dy^j \right).$
mielke-geometrodynamics_E00960 (14.1.42)	309	■ 0.508 ● W -1	$\xi_{\mu\nu} \xi_{\rho\sigma} = -\frac{1}{2} \bar{R}_{\mu\nu\rho\sigma}^{\{ \}} - \frac{1}{2} \frac{\bar{R}_{\mu\nu\gamma\delta}^{\{ \}} \bar{R}_{\rho\sigma}^{\{ \}}{}^{\gamma\delta}}{R_{\alpha\beta}^{\{ \}} R^{\{ \}}{}^{\alpha\beta}}.$	$\xi_{\mu\nu} \xi_{\rho\sigma} = -\frac{1}{2} \bar{R}_{\mu\nu\rho\sigma}^{\{ \}} - \frac{1}{2} \frac{\bar{R}_{\mu\nu\gamma\delta}^{\{ \}} \bar{R}_{\rho\sigma}^{\{ \}}{}^{\gamma\delta}}{R_{\alpha\beta}^{\{ \}} R^{\{ \}}{}^{\alpha\beta}}.$	$\xi_{\mu\nu} \xi_{\rho\sigma} = -\frac{1}{2} \bar{R}_{\mu\nu\rho\sigma}^{\{ \}} - \frac{1}{2} \frac{\bar{R}_{\mu\nu\gamma\delta}^{\{ \}} \bar{R}_{\rho\sigma}^{\{ \}}{}^{\gamma\delta}}{R_{\alpha\beta}^{\{ \}} R^{\{ \}}{}^{\alpha\beta}}.$
mielke-geometrodynamics_E01131 (B.43)	373	■ 0.509 ● W +2	$\langle \cdot, \cdot \rangle_M : \begin{cases} \bigwedge^p T^*(M) \times C_p(M, \mathbb{R}) \rightarrow \mathbb{R} \\ \langle \alpha^{(p)}, c_p \rangle_M \rightarrow \int_{c_p} \alpha^{(p)} \end{cases}$	$\langle \cdot, \cdot \rangle_M : \begin{cases} \bigwedge^p T^*(M) \times C_p(M, \mathbb{R}) \rightarrow \mathbb{R} \\ \langle \alpha^{(p)}, c_p \rangle_M \rightarrow \int_{c_p} \alpha^{(p)} \end{cases}$	$\langle \cdot, \cdot \rangle_M : \begin{cases} \bigwedge^p T^*(M) \times C_p(M, \mathbb{R}) \rightarrow \mathbb{R} \\ \langle \alpha^{(p)}, c_p \rangle_M \rightarrow \int_{c_p} \alpha^{(p)} \end{cases}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_101 EQ0294 (4.5.5)	101	■ 0.510 ● K +0	$\left.\Sigma_{\alpha} = e_{\alpha} \downarrow L - (e_{\alpha} \downarrow D \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (e_{\alpha} \downarrow \Psi) \wedge \frac{\partial L}{\partial \Psi} \right\} \wedge \frac{\partial L}{\partial D \Psi} - (e_{\alpha} \downarrow \Psi) \wedge \frac{\partial L}{\partial \Psi} .$	$\Sigma_{\alpha} = e_{\alpha} \downarrow L - (e_{\alpha} \downarrow D \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (e_{\alpha} \downarrow \Psi) \wedge \frac{\partial L}{\partial \Psi} .$	$\Sigma_{\alpha} = e_{\alpha} \downarrow L - (e_{\alpha} \downarrow D \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (e_{\alpha} \downarrow \Psi) \wedge \frac{\partial L}{\partial \Psi} .$
mielke-geometrodynamics_059 EQ0118 (3.3.19)	059	■ 0.513 ● N -1	$L_{\mathrm{Pontr}} := \frac{1}{2} F \wedge F = \mathbf{E} \cdot \mathbf{B} \eta .$	$L_{\mathrm{Pontr}} := -\frac{1}{2} F \wedge F = \mathbf{E} \cdot \mathbf{B} \eta .$	$L_{\mathrm{Pontr}} := -\frac{1}{2} F \wedge F = \mathbf{E} \cdot \mathbf{B} \eta .$
mielke-geometrodynamics_169 EQ0533 (7.7.53)	169	■ 0.513 ●	$\begin{aligned} G &:= \frac{1}{2} R^{\beta \gamma} \wedge \eta_{\alpha \beta \gamma} \gamma^{\alpha} \\ &= G^{(\beta)} + \frac{(-1)^{\mathrm{sig}}}{12} (e_{\alpha} \downarrow \mathcal{A} \wedge {}^* \mathcal{A} - \frac{1}{3} \mathcal{A} \wedge e_{\alpha} \downarrow {}^* \mathcal{A}) \gamma^{\alpha} + \frac{(-1)^{\mathrm{sig}}}{6} \gamma \wedge d \mathcal{A} \end{aligned}$	$G := \frac{1}{2} R^{\beta \gamma} \wedge \eta_{\alpha \beta \gamma} \gamma^{\alpha} = G^{(\beta)} + \frac{(-1)^{\mathrm{sig}}}{12} (e_{\alpha} \downarrow \mathcal{A} \wedge {}^* \mathcal{A} - \frac{1}{3} \mathcal{A} \wedge e_{\alpha} \downarrow {}^* \mathcal{A}) \gamma^{\alpha} + \frac{(-1)^{\mathrm{sig}}}{6} \gamma \wedge d \mathcal{A}$	$G := \frac{1}{2} R^{\beta \gamma} \wedge \eta_{\alpha \beta \gamma} \gamma^{\alpha} = G^{(\beta)} + \frac{(-1)^{\mathrm{sig}}}{12} (e_{\alpha} \downarrow \mathcal{A} \wedge {}^* \mathcal{A} - \frac{1}{3} \mathcal{A} \wedge e_{\alpha} \downarrow {}^* \mathcal{A}) \gamma^{\alpha} + \frac{(-1)^{\mathrm{sig}}}{6} \gamma \wedge d \mathcal{A} \quad (7.7.5)$
mielke-geometrodynamics_288 EQ0898 (13.5.3)	288	■ 0.515 ● N +0	$\hat{B} := B_A {}^B A_A B = (b_{\alpha} P_{*}^{\alpha} + \hat{b}^{\alpha} P_{\alpha}) \ell^2 + B_{\alpha} {}^{\beta} L^{\alpha} {}_{\beta}$	$\hat{B} := B_A {}^B A_A B = (b_{\alpha} P_{*}^{\alpha} + \hat{b}^{\alpha} P_{\alpha}) \ell^2 + B_{\alpha} {}^{\beta} L^{\alpha} {}_{\beta}$	$\hat{B} := B_A {}^B A_A B = (b_{\alpha} P_{*}^{\alpha} + \hat{b}^{\alpha} P_{\alpha}) \ell^2 + B_{\alpha} {}^{\beta} L^{\alpha} {}_{\beta}$
mielke-geometrodynamics_043 EQ0045 (2.6.8)	043	■ 0.516 ● N +0	$(d \, b)_{\mathrm{hor}} = D b = \omega \otimes b$	$(db)_{\mathrm{hor}} = D b = \omega \otimes b$	$(db)_{\mathrm{hor}} = D b = \omega \otimes b$
mielke-geometrodynamics_285 EQ0876 (13.3.5)	285	■ 0.518 ● W -1	$\begin{aligned} \{ \left[P_{\alpha}, P_{\beta} \right] = 0, \{ \left[P_{*}^{\alpha}, P_{*}^{\beta} \right] = 0, \{ \left[L_{\beta}^{\alpha}, P_{\gamma} \right] = \delta_{\gamma}^{\alpha} P_{\beta}, \{ \left[L_{\beta}^{\alpha}, P_{*}^{\gamma} \right] = -\delta_{\beta}^{\gamma} P_{*}^{\alpha} . \end{aligned}$	$[P_{\alpha}, P_{\beta}] = 0, [P_{*}^{\alpha}, P_{*}^{\beta}] = 0, [L_{\beta}^{\alpha}, P_{\gamma}] = \delta_{\gamma}^{\alpha} P_{\beta}, [L_{\beta}^{\alpha}, P_{*}^{\gamma}] = -\delta_{\beta}^{\gamma} P_{*}^{\alpha} .$	$[P_{\alpha}, P_{\beta}] = 0, [P_{*}^{\alpha}, P_{*}^{\beta}] = 0, [L_{\beta}^{\alpha}, P_{\gamma}] = \delta_{\gamma}^{\alpha} P_{\beta}, [L_{\beta}^{\alpha}, P_{*}^{\gamma}] = -\delta_{\beta}^{\gamma} P_{*}^{\alpha} .$
mielke-geometrodynamics_320 EQ0990 (14.3.3)	320	■ 0.526 ● W +0	$\begin{aligned} \hat{L}_{\mathrm{D}} &:= \frac{i}{2} \left(\overline{\hat{\psi}} \hat{\gamma} \wedge {}^* \hat{D}^{g+o} \hat{\psi} + \overline{(\hat{D}^{g+o} \hat{\psi})} \wedge {}^* \hat{\gamma} \hat{\psi} \right) - \hat{m} \overline{\hat{\psi}} \hat{\psi} \eta \\ &+ \frac{1}{2} \overline{\hat{\psi}} [\hat{\alpha}, {}^* \hat{\gamma}] \hat{\psi} . \end{aligned}$	$\hat{L}_{\mathrm{D}} = \frac{i}{2} \left(\overline{\hat{\psi}} \hat{\gamma} \wedge {}^* \hat{D}^{g+o} \hat{\psi} + \overline{(\hat{D}^{g+o} \hat{\psi})} \wedge {}^* \hat{\gamma} \hat{\psi} \right) - \hat{m} \overline{\hat{\psi}} \hat{\psi} \eta + \frac{1}{2} \overline{\hat{\psi}} [\hat{\alpha}, {}^* \hat{\gamma}] \hat{\psi} .$	$\hat{L}_{\mathrm{D}} = \frac{i}{2} \left(\overline{\hat{\psi}} \hat{\gamma} \wedge {}^* \hat{D}^{g+o} \hat{\psi} + \overline{(\hat{D}^{g+o} \hat{\psi})} \wedge {}^* \hat{\gamma} \hat{\psi} \right) - \hat{m} \overline{\hat{\psi}} \hat{\psi} \eta + \frac{1}{2} \overline{\hat{\psi}} [\hat{\alpha}, {}^* \hat{\gamma}] \hat{\psi} .$
mielke-geometrodynamics_152 EQ0456 (7.2.8)	152	■ 0.527 ● W +0	$\bar{R}_{..cd} {}^{ab} = 2 \mathcal{R}_{[c} {}^{[a} \delta_{d]} {}^{b]} ; \quad \bar{R}_{..cd} {}^{ab} = 2 \mathcal{R}_{[c} {}^{[a} \delta_{d]} {}^{b]} ; \quad \bar{R}_{..cd} {}^{ab} = 2 \mathcal{R}_{[c} {}^{[a} \delta_{d]} {}^{b]} ; \quad \bar{R}_{..cd} {}^{ab} = 2 \mathcal{R}_{[c} {}^{[a} \delta_{d]} {}^{b]} ;$	$\bar{R}_{..cd} {}^{ab} = 2 \mathcal{R}_{[c} {}^{[a} \delta_{d]} {}^{b]} ; \quad \bar{R}_{..cd} {}^{ab} = 2 \mathcal{R}_{[c} {}^{[a} \delta_{d]} {}^{b]} ; \quad \bar{R}_{..cd} {}^{ab} = 2 \mathcal{R}_{[c} {}^{[a} \delta_{d]} {}^{b]} ; \quad \bar{R}_{..cd} {}^{ab} = 2 \mathcal{R}_{[c} {}^{[a} \delta_{d]} {}^{b]} ;$	$\bar{R}_{..cd} {}^{ab} = 2 \mathcal{R}_{[c} {}^{[a} \delta_{d]} {}^{b]} ; \quad \bar{R}_{..cd} {}^{ab} = 2 \mathcal{R}_{[c} {}^{[a} \delta_{d]} {}^{b]} ; \quad \bar{R}_{..cd} {}^{ab} = 2 \mathcal{R}_{[c} {}^{[a} \delta_{d]} {}^{b]} ; \quad \bar{R}_{..cd} {}^{ab} = 2 \mathcal{R}_{[c} {}^{[a} \delta_{d]} {}^{b]} ;$
mielke-geometrodynamics_125 EQ0360 (6.2.11)	125	■ 0.531 ● N -1	$L_{\parallel} \equiv L_{\mathrm{HE}} + \frac{1}{2 \kappa} R_{\alpha \beta} \wedge {}^* (\vartheta^{\alpha} \wedge \vartheta^{\beta}) + \frac{2 \ell^2}{\kappa} d C_{\mathrm{TT}} {}^*,$	$L_{\parallel} \equiv L_{\mathrm{HE}} + \frac{1}{2 \kappa} R_{\alpha \beta} \wedge {}^* (\vartheta^{\alpha} \wedge \vartheta^{\beta}) + \frac{2 \ell^2}{\kappa} d C_{\mathrm{TT}} {}^*,$	$L_{\parallel} \equiv L_{\mathrm{HE}} + \frac{1}{2 \kappa} R_{\alpha \beta} \wedge {}^* (\vartheta^{\alpha} \wedge \vartheta^{\beta}) + \frac{2 \ell^2}{\kappa} d C_{\mathrm{TT}} {}^*,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_142 EQ0437 (6.8.8)		■ 0.532 ● N +0	$E_{\alpha}:=\frac{\partial L_g}{\partial \vartheta^\alpha}=e_\alpha\lrcorner L_g+(e_\alpha\lrcorner T^\beta)\wedge H_\beta+(e_\alpha\lrcorner R^{\beta\gamma})\wedge H_{\beta\gamma},$		
mielke-geometrodynamics_082 EQ0198 (4.1.11)		■ 0.533 ● W +0	$\begin{aligned} e_{\alpha}(m) &\rightarrow e_{\alpha}'(m)=g_{\alpha}^{\beta}e_{\beta}(m), \\ g &=\left[g_{\alpha}^{\beta}\right]\in GL(n,\mathbb{R}). \end{aligned}$		
mielke-geometrodynamics_196 EQ0599 (9.1.17)		■ 0.534 ●	$\begin{aligned} &f=-\frac{1}{r}\left(1+\frac{\kappa}{2\ell^{*2}}r^2\right)e^{-\nu/2},\,g=\frac{1}{r}e^{\nu/2} \\ &h=-\frac{1}{(+)\ell^{*}}\frac{\sqrt{-\kappa}}{3},\quad k=+\frac{1}{(-)2\ell^{*}}\sqrt{-3\kappa}. \end{aligned}$		
mielke-geometrodynamics_241 EQ0752 (11.1.19)		■ 0.535 ● K +0	$\Psi=\Psi\otimes\tilde{b}\in C^{\infty}\left(V^{\frac{1}{2}}\right).$		
mielke-geometrodynamics_344 EQ1047 (15.2.4)		■ 0.536 ● W +0	$\begin{aligned} \bar{E}_i &= \frac{1}{2} \left\{ E_{i\alpha}^{(f)j} \gamma^\alpha + i^* E_{i\alpha}^{(f)j} \gamma^\alpha \gamma^5 \right\} \lambda_j^{(f)} \\ &\oplus \frac{1}{2} \left\{ E_{i\alpha}^{(c)j} \gamma^\alpha + i^* E_{i\alpha}^{(c)j} \gamma^\alpha \gamma^5 \right\} \lambda_j^{(c)} \end{aligned}$		
mielke-geometrodynamics_317 EQ0982 (14.2.23)		■ 0.537 ● N -1	$\int_G \pi \hat{L}_{\mathrm{E.}} = \frac{1}{\ell^{*2}} L_{\mathrm{W.}} + L'_{\mathrm{YM}} + \frac{1}{2\ell^{*2}} \vartheta \wedge \vartheta \wedge \vartheta \wedge \vartheta \int_G \overset{\circ}{R} \, d\mu(G).$		
mielke-geometrodynamics_375 EQ1134 (C.2)		■ 0.539 ● K +0	$\exp:\mathfrak{g}\rightarrow G\ni g=\exp\left(\iota\xi^jI_j\right)$		
mielke-geometrodynamics_323 EQ1002 (14.3.15)		■ 0.541 ● N +0	$(i\not{D})^2=(-1)^{3n-1+s}\operatorname{Tr}\{\gamma\wedge\gamma\wedge^*(DD)-\gamma\wedge(D\gamma)\wedge^*D\}$		
mielke-geometrodynamics_365 EQ1085 (A.11)		■ 0.541 ● K +0	$\begin{aligned} \mathrm{C}(0,1) &= \mathbb{C} \quad (\text{field of complex numbers}), \\ \mathrm{C}(1,0) &= \mathbb{R} \oplus \mathbb{R}, \end{aligned}$		
mielke-geometrodynamics_033 EQ0004 (2.1.3)		■ 0.543 ● N +0	$e_{\alpha}(m)=e_{,\alpha}^i(m)\partial_i.$		

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_008558 (8.5.1)	181	■0.544 ●N +0	\Delta_{\mathrm{FP}}=s \operatorname{Tr}\left\{\frac{\theta_{\mathrm{T}}^*\mathrm{i} }{4\ell^2}\bar{\chi}\wedge{}^*\sigma-\bar{c}d^*\gamma\right\}	$\Delta L_\text{FP}=s\text{Tr}\left\{\frac{\theta_\text{T}^*i}{4\ell^2}\bar{\chi}\wedge{}^*\sigma-\bar{c}d^*\gamma\right\}$	$\Delta L_\text{FP}=s\text{Tr}\left\{\frac{\theta_\text{T}^*i}{4\ell^2}\bar{\chi}\wedge{}^*\sigma-\bar{c}d^*\gamma\right\}$
mielke-geometrodynamics_008304 (4.5.14)	102	■0.545 ●N +1	\begin{gathered} D\,\Sigma_{\alpha}\,\stackrel{*}{=}0\\[1ex]\left(\tau_{\alpha\beta}+x_{[\alpha}\wedge\Sigma_{\beta]}\right)\stackrel{*}{=}0.\end{gathered}	$D\Sigma_\alpha\simeq 0$ $D\left(\tau_{\alpha\beta}+x_{[\alpha}\wedge\Sigma_{\beta]}\right)\simeq 0.$	$D\Sigma_\alpha\simeq 0,$ $D(\tau_{\alpha\beta}+x_{[\alpha}\wedge\Sigma_{\beta]})\simeq 0.$
mielke-geometrodynamics_000270 (4.3.35)	095	■0.546 ●N +0	R:=R_{\mu}^\wedge\lrcorner\mu=R_{\alpha\mu}^\cdot\lrcorner\mu\lrcorner\alpha=\{\ }\wedge\left(0\omega^\wedge g\right)\wedge\vartheta\wedge\vartheta\right).	$R:=R_\mu^\mu=R_{\alpha\mu}^{\cdot\mu\alpha}={}^*(\Omega^g\wedge\vartheta\wedge\vartheta).$	$R:=R_\mu^\mu=R_{\alpha\mu}^{\cdot\mu\alpha}={}^*(\Omega^g\wedge\vartheta\wedge\vartheta).$
mielke-geometrodynamics_000174 (3.7.2)	071	■0.550 ●	\begin{aligned}D^*\Theta^{(1)}&=\frac{\lambda}{2}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)\varphi\eta\\ \Theta^{(1)}&=D\phi^{(0)}\end{aligned}	$D^*\Theta^{(1)}=\frac{\lambda}{2}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)\varphi\eta$ $\Theta^{(1)}=D\phi^{(0)}$	$D^*\Theta^{(1)}=\frac{\lambda}{2}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)\varphi\eta,$ $\Theta^{(1)}=D\phi^{(0)}.$
mielke-geometrodynamics_000404 (6.4.24)	134	■0.551 ●S +2	-ell^2\frac{\delta}{\delta^*\vartheta_B}\Psi_\parallel(\vartheta)=\overset{*}{T}^{(\pm)B}\Psi_\parallel(\vartheta).	$-\ell^2\frac{\delta}{\delta^*_\vartheta}\Psi_\parallel(\vartheta)=\overset{*}{T}^{(\pm)B}\Psi_\parallel(\vartheta).$	$-\ell^2\frac{\delta}{\delta^*\vartheta_B}\Psi_\parallel(\vartheta)=\overset{*}{T}^{(\pm)B}\Psi_\parallel(\vartheta).$
mielke-geometrodynamics_000392 (6.4.7)	132	■0.552 ●N +0	\left.\left.\left.n^\alpha\right \right.\right.\Sigma_\alpha=n\downarrow L-(n\downarrow D\Psi)\wedge\frac{\partial L}{\partial D\Psi}-(n\downarrow\Psi)\wedge\frac{\partial L}{\partial\Psi}.	$n^\alpha\Sigma_\alpha=n\downarrow L-(n\downarrow D\Psi)\wedge\frac{\partial L}{\partial D\Psi}-(n\downarrow\Psi)\wedge\frac{\partial L}{\partial\Psi}.$	$n^\alpha\Sigma_\alpha=n\downarrow L-(n\downarrow D\Psi)\wedge\frac{\partial L}{\partial D\Psi}-(n\downarrow\Psi)\wedge\frac{\partial L}{\partial\Psi}.$
mielke-geometrodynamics_000879 (13.3.4)	285	■0.553 ●N -1	\stackrel{\circ}{L}_{AB}:=\hat{g}_{[A C}\mathcal{E}^C{}_{ B]}=-\stackrel{\circ}{L}_{BA}	$\overset{\circ}{L}_{AB}:=\hat{g}_{[A C}\mathcal{E}^C{}_{ B]}=-\overset{\circ}{L}_{BA}$	$\overset{\circ}{L}_{AB}:=\hat{g}_{[A C}\mathcal{E}^C{}_{ B]}=-\overset{\circ}{L}_{BA}$
mielke-geometrodynamics_000013 (2.2.8)	034	■0.554 ●N +1	d s^{\scriptscriptstyle{2}}=g_{ij}dx^i\otimes_s dx^j\overset{(*)}{=}-\sum_{i=1}^s\left(\overset{\circ}{\vartheta}^i\right)^2+\sum_{j=s+1}^n\left(\overset{\circ}{\vartheta}^j\right)^2.	$ds^2=g_{ij}dx^i\otimes_s dx^j\overset{(*)}{=}-\sum_{i=1}^s\left(\overset{\circ}{\vartheta}^i\right)^2+\sum_{j=s+1}^n\left(\overset{\circ}{\vartheta}^j\right)^2.$	$ds^2=g_{ij}dx^i\otimes_s dx^j\overset{(*)}{=}-\sum_{i=1}^s(\overset{\circ}{\vartheta}^i)^2+\sum_{j=s+1}^n(\overset{\circ}{\vartheta}^j)^2.$
mielke-geometrodynamics_001011 (14.4.6)	325	■0.554 ●N +0	r_{(-)}^+=\frac{\hbar}{M^{*2}c}\left(M_{(-)}^+\sqrt{M^2-J(J+1)\frac{M^{*4}}{M^2}}-\alpha_e M^{*2}\operatorname{Tr}\overline{Q}^2\right)	$r_{(-)}^+=\frac{\hbar}{M^{*2}c}\left(M_{(-)}^+\sqrt{M^2-J(J+1)\frac{M^{*4}}{M^2}}-\alpha_e M^{*2}\operatorname{Tr}\overline{Q}^2\right)$	$r_{(-)}^+=\frac{\hbar}{M^{*2}c}\left(M_{(-)}^+\sqrt{M^2-J(J+1)\frac{M^{*4}}{M^2}}-\alpha_e M^{*2}\operatorname{Tr}\overline{Q}^2\right)$
mielke-geometrodynamics_000385 (6.3.3)	130	■0.561 ●W +2	\stackrel{\circ}{S}_\alpha:=\Gamma_\alpha^\beta\wedge\Pi_\beta+\Sigma_\alpha\cong d\Pi_\alpha,\quad\overset{(A)}{S}_\alpha=\overset{(\mp)}{S}_\alpha	$\overset{(\pm)}{S}_\alpha:=\overset{(\pm)}{\Gamma}_\alpha^\beta\wedge\overset{(\pm)}{\Pi}_\beta+\Sigma_\alpha\cong d\overset{(\pm)}{\Pi}_\alpha,\quad\overset{(\pm)}{S}_\alpha=\overset{(\mp)}{S}_\alpha$	$\overset{(\pm)}{S}_\alpha:=\overset{(\pm)}{\Gamma}_\alpha^\beta\wedge\overset{(\pm)}{\Pi}_\beta+\Sigma_\alpha\cong d\overset{(\pm)}{\Pi}_\alpha,\quad\overset{(\pm)}{S}_\alpha=\overset{(\mp)}{S}_\alpha$
mielke-geometrodynamics_000596 (9.1.14)	195	■0.564 ●N +0	(\stackrel{+}{D}K)-K\wedge K=\stackrel{+}{\Omega}_C\{\}.	$(\overset{+}{D}K)-K\wedge K=\overset{+}{\Omega}_C\{\}.$	$(\overset{+}{D}K)-K\wedge K=\overset{+}{\Omega}_C^{\{\}}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■0.5–0.9 ■ < 0.5 — no value					
LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ					
Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured					
[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_070 EQ0169 (3.6.4)		■ 0.570 ● W +1	$D_{\{ \pm \}} \left(F^{\pm} - \frac{1}{2} \bar{\psi} \sigma_{\pm} \psi \right) = \mp \frac{i}{2} \bar{\psi}^* D_{\pm} \psi = \mp \frac{1}{2} \bar{\psi}^* \gamma \wedge {}^*(i^* \gamma \wedge D_{\pm} \psi) .$	$D_{\pm} \left(F^{\pm} - \frac{1}{2} \bar{\psi} \sigma_{\pm} \psi \right) = \mp \frac{i}{2} \bar{\psi}^* D_{\pm} \psi = \mp \frac{1}{2} \bar{\psi}^* \gamma \wedge {}^*(i^* \gamma \wedge D_{\pm} \psi) .$	
mielke-geometrodynamics_286 EQ0889 (13.4.8)		■ 0.576 ● N +0	$\begin{aligned} & \Gamma^{\wedge(T)} \xrightarrow{A^{-1}(x)} \Gamma^{\wedge(T)'} = \Lambda^{-1}(x) \left[\Gamma^{\wedge(T)} + D\tau(x) \right] , \\ & \Gamma_*^{\wedge(T)} \xrightarrow{A^{-1}(x)} \Gamma_*^{\wedge(T)'} = \left[\Gamma_*^{\wedge(T)} + D\tau_*(x) \right] \Lambda(x); \end{aligned}$	$\Gamma^{(T)} \xrightarrow{A^{-1}(x)} \Gamma^{(T)'} = \Lambda^{-1}(x) \left[\Gamma^{(T)} + D\tau(x) \right] ,$ $\Gamma_*^{(T)} \xrightarrow{A^{-1}(x)} \Gamma_*^{(T)'} = \left[\Gamma_*^{(T)} + D\tau_*(x) \right] \Lambda(x);$	$\Gamma^{(T)} \xrightarrow{A^{-1}(x)} \Gamma^{(T)'} = \Lambda^{-1}(x) \left[\Gamma^{(T)} + D\tau(x) \right] ,$ $\Gamma_*^{(T)} \xrightarrow{A^{-1}(x)} \Gamma_*^{(T)'} = \left[\Gamma_*^{(T)} + D\tau_*(x) \right] \Lambda(x);$
mielke-geometrodynamics_059 EQ0116 (3.3.17)		■ 0.585 ●	$d\mathbf{H} = \mathbf{j} \times \mathbf{B} - \frac{\partial \mathbf{E}}{c^2 \partial t} = \mu \mathbf{J}$	$dH = j \left\{ \begin{array}{l} \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{c^2 \partial t} = \mu \mathbf{J} \\ \nabla \cdot \mathbf{E} = \rho / \epsilon . \end{array} \right.$	$dH = j \left\{ \begin{array}{l} \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{c^2 \partial t} = \mu \mathbf{J} \\ \nabla \cdot \mathbf{E} = \rho / \epsilon . \end{array} \right.$
mielke-geometrodynamics_129 EQ0381 (6.2.32)		■ 0.588 ● W +1	$\stackrel{\scriptscriptstyle\text{D}}{\Pi}_{\alpha} := \stackrel{\scriptscriptstyle\text{D}}{\Pi}_{\alpha} \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \downarrow \stackrel{\scriptscriptstyle\text{D}}{\Pi} \right) \wedge \stackrel{\scriptscriptstyle\text{D}}{\Pi}_{\beta} ,$	$\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} := D \stackrel{(\pm)}{\Pi}_{\alpha} \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \downarrow \stackrel{(\mp)}{\Pi} \right) \wedge \stackrel{(\pm)}{\Pi}_{\beta} ,$	$\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} := D \stackrel{(\pm)}{\Pi}_{\alpha} \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \downarrow \stackrel{(\mp)}{\Pi} \right) \wedge \stackrel{(\pm)}{\Pi}_{\beta} ,$
mielke-geometrodynamics_045 EQ0056 (2.7.6)		■ 0.592 ● N +0	$\phi = \varphi \otimes b = G^{-1} \varphi \otimes^G b \in C^{\infty}(V^{\rho}) .$	$\phi = \varphi \otimes b = G^{-1} \varphi \otimes^G b \in C^{\infty}(V^{\rho}) .$	$\phi = \varphi \otimes b = G^{-1} \varphi \otimes^G b \in C^{\infty}(V^{\rho}) .$
mielke-geometrodynamics_074 EQ0187 (3.7.17)		■ 0.593 ● N +0	$\begin{aligned} & \left(D^* \Omega \right) = \vec{D}^* \vec{\Omega} + [\omega_0, \Omega_0] \\ & = \vec{D}^* \vec{\Omega} + [\omega_0, \Theta^{\perp}] = 0. \end{aligned}$	$\begin{aligned} (D^* \Omega) &= \vec{D}^* \vec{\Omega} + [\omega_0, \Omega_0] \\ &= \vec{D}^* \vec{\Omega} + [\omega_0, \Theta^{\perp}] = 0. \end{aligned}$	$\begin{aligned} (D^* \Omega) &= \vec{D}^* \vec{\Omega} + [\omega_0, \Omega_0] \\ &= \vec{D}^* \vec{\Omega} + [\omega_0, \Theta^{\perp}] = 0. \end{aligned}$
mielke-geometrodynamics_186 EQ0581 (8.6.19)		■ 0.593 ● K +0	$\widehat{\Sigma}_{\alpha} := \Sigma_{\alpha} - D^{\{\}} \mu_{\alpha} ,$	$\widehat{\Sigma}_{\alpha} := \Sigma_{\alpha} - D^{\{\}} \mu_{\alpha} ,$	$\widehat{\Sigma}_{\alpha} := \Sigma_{\alpha} - D^{\{\}} \mu_{\alpha} ,$
mielke-geometrodynamics_307 EQ0948 (14.1.29)		■ 0.593 ● W +0	$D^* \Omega = (D^* \xi - (-1)^z \xi \wedge D \delta) \cos \delta - (-1)^z (D \xi + (-1)^{z*} \xi \wedge D \delta) \sin \delta = \tau_e$	$D^* \Omega = (D^* \xi - (-1)^z \xi \wedge D \delta) \cos \delta - (-1)^z (D \xi + (-1)^{z*} \xi \wedge D \delta) \sin \delta = \tau_e$	$D^* \Omega = (D^* \xi - (-1)^z \xi \wedge D \delta) \cos \delta - (-1)^z (D \xi + (-1)^{z*} \xi \wedge D \delta) \sin \delta = \tau_e$
mielke-geometrodynamics_125 EQ0357 (6.2.8)		■ 0.600 ● N +0	$L_{\mathrm{HE}} = -\frac{1}{2\kappa} R_{\alpha\beta}^{\{\}} \wedge {}^*(\vartheta^{\alpha} \wedge \vartheta^{\beta})$	$L_{\mathrm{HE}} = -\frac{1}{2\kappa} R_{\alpha\beta}^{\{\}} \wedge {}^*(\vartheta^{\alpha} \wedge \vartheta^{\beta})$	$L_{\mathrm{HE}} = -\frac{1}{2\kappa} R_{\alpha\beta}^{\{\}} \wedge {}^*(\vartheta^{\alpha} \wedge \vartheta^{\beta})$
mielke-geometrodynamics_225 EQ0704 (10.6.3)		■ 0.601 ● N +0	$\square F - m_{\mathrm{photon}}^2 F = 0,$	$\square F - m_{\mathrm{photon}}^2 F = 0,$	$\square F - m_{\mathrm{photon}}^2 F = 0,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_303 E09927 (14.1.8)	303	■ 0.604 ● W -1	$\begin{aligned} \Sigma_{\mathrm{YM}} &= \frac{(-1)^s}{2\alpha_g} \operatorname{Tr} \left\{ \Omega \wedge^* (\vartheta \wedge^* \Omega) +^* \vartheta \wedge^* (\Omega \wedge^* \Omega) \right\} \\ &= \frac{(-1)^s}{4\alpha_g} \operatorname{Tr} \left\{ \Omega \wedge^* (\vartheta \wedge^* \Omega) -^* \Omega \wedge^* (\vartheta \wedge \Omega) \right\}. \end{aligned}$	$\Sigma_{\mathrm{YM}} = \frac{(-1)^s}{2\alpha_g} \operatorname{Tr} \left\{ \Omega \wedge^* (\vartheta \wedge^* \Omega) +^* \vartheta \wedge^* (\Omega \wedge^* \Omega) \right\}$ $= \frac{(-1)^s}{4\alpha_g} \operatorname{Tr} \left\{ \Omega \wedge^* (\vartheta \wedge^* \Omega) -^* \Omega \wedge^* (\vartheta \wedge \Omega) \right\}.$	$\Sigma_{\mathrm{YM}} = \frac{(-1)^s}{2\alpha_g} \operatorname{Tr} \left\{ \Omega \wedge^* (\vartheta \wedge^* \Omega) +^* \vartheta \wedge^* (\Omega \wedge^* \Omega) \right\}$ $= \frac{(-1)^s}{4\alpha_g} \operatorname{Tr} \left\{ \Omega \wedge^* (\vartheta \wedge^* \Omega) -^* \Omega \wedge^* (\vartheta \wedge \Omega) \right\}.$
mielke-geometrodynamics_112 E09326 (5.3.3)	112	■ 0.613 ● N +0	$\left. \widehat{D}_G \right _{\alpha} \equiv \frac{1}{2} \left(e_{\alpha} \rfloor R^{\beta\gamma} \right) \wedge \eta_{\beta\gamma\mu} \wedge T^{\mu}$	$\widehat{D}G_{\alpha} \equiv \frac{1}{2} (e_{\alpha} \rfloor R^{\beta\gamma}) \wedge \eta_{\beta\gamma\mu} \wedge T^{\mu}$	$\widehat{D}G_{\alpha} \equiv \frac{1}{2} (e_{\alpha} \rfloor R^{\beta\gamma}) \wedge \eta_{\beta\gamma\mu} \wedge T^{\mu}$
mielke-geometrodynamics_242 E09758 (11.1.25)	242	■ 0.615 ● N +0	$\gamma = E_{\cdot j}^{\alpha}(m) \gamma_{\alpha} \otimes dx^j = \gamma_j dx^j \in C^{\infty}(T_{\mathbb{C}}^*(M) \times CV)$	$\gamma = E_{\cdot j}^{\alpha}(m) \gamma_{\alpha} \otimes dx^j = \gamma_j dx^j \in C^{\infty}(T_{\mathbb{C}}^*(M) \times CV)$	$\gamma = E_{\cdot j}^{\alpha}(m) \gamma_{\alpha} \otimes dx^j = \gamma_j dx^j \in C^{\infty}(T_{\mathbb{C}}^*(M) \times CV)$
mielke-geometrodynamics_308 E09953 (14.1.35)	308	■ 0.616 ●	$\begin{aligned} & (-1)^z \operatorname{Tr} \left\{ \vartheta \wedge^* \xi \wedge^* \xi - \vartheta \wedge \xi \wedge \xi \right\} \wedge d\delta = \operatorname{Tr} \left\{ \xi \wedge \vartheta \wedge D^* \xi +^* \xi \wedge \vartheta \wedge D\xi \right\} \\ &= \frac{1}{2} D \operatorname{Tr} \left\{^* (\xi \wedge^* (\vartheta \wedge^* \xi) -^* \xi \wedge^* (\vartheta \wedge \xi)) \right\} \\ &= (-1)^s 2\alpha_g D \Sigma_{\mathrm{YM}} = \frac{\alpha_g}{\ell^{*2}} D (\vartheta \wedge^* \Omega^g) \equiv 2 \frac{\alpha_g}{\ell^{*2}} \Theta \wedge^* \Omega^g. \end{aligned}$	$\begin{aligned} & (-1)^z \operatorname{Tr} \left\{ \vartheta \wedge^* \xi \wedge^* \xi - \vartheta \wedge \xi \wedge \xi \right\} \wedge d\delta = \operatorname{Tr} \left\{ \xi \wedge \vartheta \wedge D^* \xi +^* \xi \wedge \vartheta \wedge D\xi \right\} \\ &= \frac{1}{2} D \operatorname{Tr} \left\{^* (\xi \wedge^* (\vartheta \wedge^* \xi) -^* \xi \wedge^* (\vartheta \wedge \xi)) \right\} \\ &= (-1)^s 2\alpha_g D \Sigma_{\mathrm{YM}} = \frac{\alpha_g}{\ell^{*2}} D (\vartheta \wedge^* \Omega^g) \equiv 2 \frac{\alpha_g}{\ell^{*2}} \Theta \wedge^* \Omega^g. \end{aligned}$	$\begin{aligned} & (-1)^z \operatorname{Tr} \left\{ \vartheta \wedge^* \xi \wedge^* \xi - \vartheta \wedge \xi \wedge \xi \right\} \wedge d\delta = \operatorname{Tr} \left\{ \xi \wedge \vartheta \wedge D^* \xi +^* \xi \wedge \vartheta \wedge D\xi \right\} \\ &= \frac{1}{2} D \operatorname{Tr} \left\{^* (\xi \wedge^* (\vartheta \wedge^* \xi) -^* \xi \wedge^* (\vartheta \wedge \xi)) \right\} \\ &= (-1)^s 2\alpha_g D \Sigma_{\mathrm{YM}} = \frac{\alpha_g}{\ell^{*2}} D (\vartheta \wedge^* \Omega^g) \equiv 2 \frac{\alpha_g}{\ell^{*2}} \Theta \wedge^* \Omega^g. \end{aligned} \tag{14.1.3}$
mielke-geometrodynamics_322 E09998 (14.3.11)	322	■ 0.619 ● S +0	$\begin{array}{l} \psi_{\mathrm{L}}: \quad \gamma^5 = +1, \quad \overset{\circ}{\gamma}^{K+1} = +\widehat{\gamma}^{n+1} \\ \psi_{\mathrm{R}}: \quad \gamma^5 = -1, \quad \overset{\circ}{\gamma}^{K+1} = -\widehat{\gamma}^{n+1} \end{array}$	$\begin{array}{l} \psi_{\mathrm{L}}: \quad \gamma^5 = +1, \quad \overset{\circ}{\gamma}^{K+1} = +\widehat{\gamma}^{n+1} \\ \psi_{\mathrm{R}}: \quad \gamma^5 = -1, \quad \overset{\circ}{\gamma}^{K+1} = -\widehat{\gamma}^{n+1} \end{array}$	$\begin{array}{l} \psi_{\mathrm{L}}: \quad \gamma^5 = +1, \quad \overset{\circ}{\gamma}^{K+1} = +\widehat{\gamma}^{n+1} \\ \psi_{\mathrm{R}}: \quad \gamma^5 = -1, \quad \overset{\circ}{\gamma}^{K+1} = -\widehat{\gamma}^{n+1} \end{array}$
mielke-geometrodynamics_167 E08525 (7.7.44)	167	■ 0.623 ● N -1	$\begin{aligned} C_{\mathrm{TT}} &:= \frac{1}{8\ell^2} \operatorname{Tr}(\gamma \wedge \Theta) = \frac{1}{2\ell^2} \vartheta^{\alpha} \wedge T_{\alpha} = -\frac{(-1)^{\operatorname{sig}}}{2\ell^2} * \mathscr{A}, \\ C_{\mathrm{RR}} &:= \operatorname{Tr} \left(\Gamma \wedge \Omega^g - \frac{1}{3} \Gamma \wedge \Gamma \wedge \Gamma \right), \end{aligned}$	$\begin{aligned} C_{\mathrm{TT}} &:= \frac{1}{8\ell^2} \operatorname{Tr}(\gamma \wedge \Theta) = \frac{1}{2\ell^2} \vartheta^{\alpha} \wedge T_{\alpha} = -\frac{(-1)^{\operatorname{sig}}}{2\ell^2} * \mathscr{A}, \\ C_{\mathrm{RR}} &:= \operatorname{Tr} \left(\Gamma \wedge \Omega^g - \frac{1}{3} \Gamma \wedge \Gamma \wedge \Gamma \right), \end{aligned}$	$C_{\mathrm{TT}} := \frac{1}{8\ell^2} \operatorname{Tr}(\gamma \wedge \Theta) = \frac{1}{2\ell^2} \vartheta^{\alpha} \wedge T_{\alpha} = -\frac{(-1)^{\operatorname{sig}}}{2\ell^2} * \mathscr{A},$ $C_{\mathrm{RR}} := \operatorname{Tr} \left(\Gamma \wedge \Omega^g - \frac{1}{3} \Gamma \wedge \Gamma \wedge \Gamma \right),$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_044 EQ0050 (2.6.13)		■ 0.626 ● W +2	$\begin{aligned} D \, D \, D \, b = & \, D(\backslash\omega \, \otimes \, b) = (d \, \backslash\omega + [\backslash\Omega, \, \omega]) \, \otimes \, b \\ & = (d \, \backslash\omega + [\backslash\Omega, \, \omega]) \, \otimes \, b \\ & = (dd\omega - d\omega \wedge \omega + \omega \wedge d\omega \\ & \quad + d\omega \wedge \omega - \omega \wedge \omega \wedge \omega \\ & \quad + \omega \wedge \omega \wedge \omega - \omega \wedge d\omega) \otimes b \equiv 0. \end{aligned}$	$\begin{aligned} DDDb = & D(\Omega \otimes b) = (d\Omega + [\Omega, \omega]) \otimes b \\ & = (d\Omega + \Omega \wedge \omega - \omega \wedge \Omega) \otimes b \\ & = (dd\omega - d\omega \wedge \omega + \omega \wedge d\omega \\ & \quad + d\omega \wedge \omega - \omega \wedge \omega \wedge \omega \\ & \quad + \omega \wedge \omega \wedge \omega - \omega \wedge d\omega) \otimes b \equiv 0. \end{aligned}$	$DDDb = D(\Omega \otimes b) = (d\Omega + [\Omega, \omega]) \otimes b$ $= (d\Omega + \Omega \wedge \omega - \omega \wedge \Omega) \otimes b$ $= (dd\omega - d\omega \wedge \omega + \omega \wedge d\omega$ $+ d\omega \wedge \omega - \omega \wedge \omega \wedge \omega$ $+ \omega \wedge \omega \wedge \omega - \omega \wedge d\omega) \otimes b \equiv 0.$
mielke-geometrodynamics_059 EQ0115 (3.3.16)		■ 0.630 ● N +1	$d \, F \, \text{equiv} \, 0 \, \text{quad} \, \left\{ \begin{array}{l} \nabla \, \times \, \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} \equiv 0 \\ \nabla \cdot \mathbf{B} \equiv 0 \end{array} \right.$	$dF \equiv 0 \quad \left\{ \begin{array}{l} \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} \equiv 0 \\ \nabla \cdot \mathbf{B} \equiv 0 \end{array} \right.$	$dF \equiv 0 \quad \left\{ \begin{array}{l} \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} \equiv 0 \\ \nabla \cdot \mathbf{B} \equiv 0. \end{array} \right.$
mielke-geometrodynamics_070 EQ0171 (3.6.6)		■ 0.630 ● N +0	$i^{\wedge\{*\}} \, \gamma \, \wedge \, D_{\{ \, \backslash \text{pm} \}} \, \psi = 0 \, \text{quad} \, \text{\textit{and}} \, \, \, F^{\pm} = \frac{1}{2} \bar{\psi} \sigma_{\pm} \psi = m \mathscr{P}_{\pm},$	$i^* \gamma \wedge D_{\pm} \psi = 0 \quad \text{and} \quad F^{\pm} = \frac{1}{2} \bar{\psi} \sigma_{\pm} \psi = m \mathscr{P}_{\pm},$	$i^* \gamma \wedge D_{\pm} \psi = 0 \quad \text{and} \quad F^{\pm} = \frac{1}{2} \bar{\psi} \sigma_{\pm} \psi = m \mathscr{P}_{\pm},$
mielke-geometrodynamics_074 EQ0189 (3.7.19)		■ 0.631 ● N +0	$\{ \, \}^{\wedge\{*\}} \, \backslash\Omega = (\stackrel{\text{\tiny{rel}}}{+}\{-\}) \, \backslash\Omega \, \text{\textit{and}} \, \, \, \{ \, \}^{\wedge\{*\}} \, \backslash\Omega = (\stackrel{\text{\tiny{rel}}}{-}) \, \backslash\Omega.$	${}^* \Omega = (\overset{+}{-}) \Omega.$	${}^* \Omega = (\overset{+}{-}) \Omega.$
mielke-geometrodynamics_249 EQ0788 (11.3.7)		■ 0.635 ●	$\left\{ i \gamma^{\kappa} \nabla_{\kappa}^{()} - \frac{6}{n^2} \ell^{*2} \gamma^{n+1} \gamma^{\mu} \bar{\psi} \gamma^{n+1} \gamma_{\mu} \psi - m \right\} \psi = 0.$	$\left\{ i \gamma^{\kappa} \nabla_{\kappa}^{()} - \frac{6}{n^2} \ell^{*2} \gamma^{n+1} \gamma^{\mu} \bar{\psi} \gamma^{n+1} \gamma_{\mu} \psi - m \right\} \psi = 0.$	$\left\{ i \gamma^{\kappa} \nabla_{\kappa}^{()} - \frac{6}{n^2} \ell^{*2} \gamma^{n+1} \gamma^{\mu} \bar{\psi} \gamma^{n+1} \gamma_{\mu} \psi - m \right\} \psi = 0.$
mielke-geometrodynamics_068 EQ0164 (3.5.12)		■ 0.636 ● N +0	$L_{\mathrm{YM}} + L_{\infty} = (\stackrel{\text{\tiny{rel}}}{+}) \frac{1}{2\alpha_g} \, \mathrm{Tr} \left(\, {}^* \Omega \, (\stackrel{\text{\tiny{rel}}}{+}) \, \Omega \right)^2 \geq 0$	$L_{\mathrm{YM}} + L_{\infty} = (\overset{+}{-}) \frac{1}{2\alpha_g} \, \mathrm{Tr} \left({}^* \Omega \, (\overset{+}{-}) \, \Omega \right)^2 \geq 0$	$L_{\mathrm{YM}} + L_{\infty} = (\overset{+}{-}) \frac{1}{2\alpha_g} \mathrm{Tr} ({}^* \Omega (\overset{+}{-}) \Omega)^2 \geq 0$
mielke-geometrodynamics_179 EQ0549 (8.4.3)		■ 0.637 ● W +2	$\begin{aligned} L_{\mathrm{FP}} := & \, -s \, \mathrm{Tr} \left\{ \bar{\chi} \wedge {}^* \Omega^g + \bar{\Phi} D^* \Psi + \frac{1}{2} \rho \bar{\chi} \wedge {}^* \beta + \bar{C} d^* \Gamma + \frac{1}{2} \bar{C} \wedge {}^* \mathbb{B} \right\} \\ & = \frac{1}{2} s \left[\bar{\chi}^{\alpha\beta} \wedge {}^* R_{\alpha\beta}^{(\pm)} + \bar{\Phi}_{\alpha\beta} D^* \Psi^{\alpha\beta} + \frac{1}{2} \rho \bar{\chi}^{\alpha\beta} \wedge {}^* b_{\alpha\beta} + \tilde{\Psi}_{\mathrm{F}} \right] \end{aligned}$	$\begin{aligned} L_{\mathrm{FP}} := & \, -s \, \mathrm{Tr} \left\{ \bar{\chi} \wedge {}^* \Omega^g + \bar{\Phi} D^* \Psi + \frac{1}{2} \rho \bar{\chi} \wedge {}^* \beta + \bar{C} d^* \Gamma + \frac{1}{2} \bar{C} \wedge {}^* \mathbb{B} \right\} \\ & = \frac{1}{2} s \left[\bar{\chi}^{\alpha\beta} \wedge {}^* R_{\alpha\beta}^{(\pm)} + \bar{\Phi}_{\alpha\beta} D^* \Psi^{\alpha\beta} + \frac{1}{2} \rho \bar{\chi}^{\alpha\beta} \wedge {}^* b_{\alpha\beta} + \tilde{\Psi}_{\mathrm{F}} \right] \end{aligned}$	$\begin{aligned} L_{\mathrm{FP}} := & \, -s \mathrm{Tr} \left\{ \bar{\chi} \wedge {}^* \bar{\Omega}^g + \bar{\Phi} D^* \Psi + \frac{1}{2} \rho \bar{\chi} \wedge {}^* \beta + \bar{C} d^* \Gamma + \frac{1}{2} \bar{C} \wedge {}^* \mathbb{B} \right\} \\ & = \frac{1}{2} s \left[\bar{\chi}^{\alpha\beta} \wedge {}^* R_{\alpha\beta}^{(\pm)} + \bar{\Phi}_{\alpha\beta} D^* \Psi^{\alpha\beta} + \frac{1}{2} \rho \bar{\chi}^{\alpha\beta} \wedge {}^* b_{\alpha\beta} + \tilde{\Psi}_{\mathrm{F}} \right]. \end{aligned} \quad (8)$
mielke-geometrodynamics_346 EQ1058 (15.2.15)		■ 0.638 ● K +0	$\bar{D}^{\{\}} := d + i \, \bar{\omega}^{\{\}}$	$\bar{D}^{\{\}} := d + i \bar{\omega}^{\{\}}$	$\bar{D}^{\{\}} := d + i \bar{\omega}^{\{\}}$

Conf.

MathPix confidence:

■ ≥0.9

■ 0.5-0.9

■ <0.5

— no value

LaTeX source

MathPix’s reading, unchanged;

Rendered

the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual

render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*]

beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_091008 (14.4.3)	325	■0.640 ●N +1	$\begin{aligned} ds^2 &= -\frac{\Delta}{\rho^2} \left[cdt - J_3 \frac{\hbar}{M} \sin^2 \vartheta d\phi \right]^2 + \frac{\rho^2}{\Delta} dr + \rho^2 d\vartheta^2 \\ &+ \frac{\sin^2 \vartheta}{\rho^2} \left\{ \left[r^2 + J(J+1) \frac{\hbar^2}{M^2 c^2} \right] d\phi - J_3 \frac{\hbar}{M} dt \right\}^2. \end{aligned}$	$ds^2 = -\frac{\Delta}{\rho^2} \left[cdt - J_3 \frac{\hbar}{M} \sin^2 \vartheta d\phi \right]^2 + \frac{\rho^2}{\Delta} dr + \rho^2 d\vartheta^2 + \frac{\sin^2 \vartheta}{\rho^2} \left\{ \left[r^2 + J(J+1) \frac{\hbar^2}{M^2 c^2} \right] d\phi - J_3 \frac{\hbar}{M} dt \right\}^2$	$ds^2 = -\frac{\Delta}{\rho^2} \left[cdt - J_3 \frac{\hbar}{M} \sin^2 \vartheta d\phi \right]^2 + \frac{\rho^2}{\Delta} dr + \rho^2 d\vartheta^2 + \frac{\sin^2 \vartheta}{\rho^2} \left\{ \left[r^2 + J(J+1) \frac{\hbar^2}{M^2 c^2} \right] d\phi - J_3 \frac{\hbar}{M} dt \right\}^2.$
mielke-geometrodynamics_090255 (4.3.20)	093	■0.642 ●N +0	$\stackrel{*}{\Omega}^g = \frac{1}{4} R_{\alpha\beta}^{..cd} L_{cd} \otimes \vartheta^\alpha \wedge \vartheta^\beta.$	$\stackrel{*}{\Omega}^g = \frac{1}{4} R_{\alpha\beta}^{..cd} L_{cd} \otimes \vartheta^\alpha \wedge \vartheta^\beta.$	$\stackrel{*}{\Omega}^g = \frac{1}{4} R_{\alpha\beta}^{..cd} L_{cd} \otimes \vartheta^\alpha \wedge \vartheta^\beta.$
mielke-geometrodynamics_091015 (14.4.10)	327	■0.646 ●N +1	$A = {}^* \omega = A^{(\circ)} \lambda_3.$	$A = {}^* \omega = A^{(\circ)} \lambda_3.$	$A = \stackrel{*}{\sigma} \omega = A^{(\circ)} \lambda_3.$
mielke-geometrodynamics_090119 (3.3.20)	059	■0.646 ●N +0	$\begin{aligned} \Sigma_\alpha &:= e_\alpha \rfloor L - (e_\alpha \rfloor dA) \wedge \frac{\partial L}{\partial (dA)} \\ &= \frac{1}{2} [(e_\alpha \rfloor F) \wedge H - F \wedge (e_\alpha \rfloor H)] \end{aligned}$	$\begin{aligned} \Sigma_\alpha &:= e_\alpha \rfloor L - (e_\alpha \rfloor dA) \wedge \frac{\partial L}{\partial (dA)} \\ &= \frac{1}{2} [(e_\alpha \rfloor F) \wedge H - F \wedge (e_\alpha \rfloor H)] \end{aligned}$	$\begin{aligned} \Sigma_\alpha &:= e_\alpha \rfloor L - (e_\alpha \rfloor dA) \wedge \frac{\partial L}{\partial (dA)} \\ &= \frac{1}{2} [(e_\alpha \rfloor F) \wedge H - F \wedge (e_\alpha \rfloor H)] \end{aligned}$
mielke-geometrodynamics_090120 (3.3.21)	060	■0.651 ●K +0	$d\Sigma_\alpha \simeq (e_\alpha \rfloor F) \wedge j$	$d\Sigma_\alpha \simeq (e_\alpha \rfloor F) \wedge j$	$d\Sigma_\alpha \simeq (e_\alpha \rfloor F) \wedge j$
mielke-geometrodynamics_090497 (7.7.15)	163	■0.653 ●	$ds^2 = o_{\alpha\beta} \vartheta^\alpha \otimes \vartheta^\beta = g_{ij} dx^i \otimes dx^j, \quad o_{\alpha\beta} := \text{diag}\{1, \underbrace{-1, \dots, -1}_{\text{sig}}, \dots, -1\}_{\text{sig}},$	$ds^2 = o_{\alpha\beta} \vartheta^\alpha \otimes \vartheta^\beta = g_{ij} dx^i \otimes dx^j, \quad o_{\alpha\beta} := \text{diag}\{1, \underbrace{-1, \dots, -1}_{\text{sig}}, \dots, -1\},$	$ds^2 = o_{\alpha\beta} \vartheta^\alpha \otimes \vartheta^\beta = g_{ij} dx^i \otimes dx^j, \quad o_{\alpha\beta} := \text{diag}\{1, \underbrace{-1, \dots, -1}_{\text{sig}}, \dots, -1\},$
mielke-geometrodynamics_090800 (11.4.12)	254	■0.658 ●N +0	$\left\{ i\gamma^k \partial_k - \frac{3i}{2R_o \lambda} \gamma^0 - \frac{m}{\lambda} \right\} \psi = 0,$	$\left\{ i\gamma^k \partial_k - \frac{3i}{2R_o \lambda} \gamma^0 - \frac{m}{\lambda} \right\} \psi = 0,$	$\left\{ i\gamma^k \partial_k - \frac{3i}{2R_o \lambda} \gamma^0 - \frac{m}{\lambda} \right\} \psi = 0,$
mielke-geometrodynamics_090316 (4.6.8)	105	■0.664 ●N +0	$\Phi(\widetilde{\Gamma}, \gamma) = (2\pi \hbar \mathcal{M}_n G / \hbar \ell c^2) = 2\pi n$	$\Phi(\widetilde{\Gamma}, \gamma) = (2\pi \hbar \mathcal{M}_n G / \hbar \ell c^2) = 2\pi n$	$\Phi(\widetilde{\Gamma}, \gamma) = (2\pi \hbar \mathcal{M}_n G / \hbar \ell c^2) = 2\pi n,$
mielke-geometrodynamics_090335 (5.3.12)	113	■0.665 ●W +0	$\begin{aligned} \widetilde{T}_{\mu\nu} &= T_{\mu\nu} + \ell^{*2} \left\{ -4\tau_{\mu\cdot[\lambda]}^{\cdot\kappa} \tau_{\nu\cdot[\kappa]}^{\cdot\lambda} - 2\tau_{\mu}^{\cdot\kappa\lambda} \tau_{\nu\kappa\lambda} + \tau_{\cdot,\mu}^{\kappa\lambda} \tau_{\kappa\lambda\nu} \right. \\ &\quad \left. + \frac{1}{2} g_{\mu\nu} \left(4\tau_{\delta\cdot[\lambda]}^{\cdot\kappa} \tau_{\cdot,\kappa]}^{\delta\lambda} + \tau^{\delta\kappa\lambda} \tau_{\delta\kappa\lambda} \right) \right\}. \end{aligned}$	$\begin{aligned} \widetilde{T}_{\mu\nu} &= T_{\mu\nu} + \ell^{*2} \left\{ -4\tau_{\mu\cdot[\lambda]}^{\cdot\kappa} \tau_{\nu\cdot[\kappa]}^{\cdot\lambda} - 2\tau_{\mu}^{\cdot\kappa\lambda} \tau_{\nu\kappa\lambda} + \tau_{\cdot,\mu}^{\kappa\lambda} \tau_{\kappa\lambda\nu} \right. \\ &\quad \left. + \frac{1}{2} g_{\mu\nu} \left(4\tau_{\delta\cdot[\lambda]}^{\cdot\kappa} \tau_{\cdot,\kappa]}^{\delta\lambda} + \tau^{\delta\kappa\lambda} \tau_{\delta\kappa\lambda} \right) \right\}. \end{aligned}$	$\begin{aligned} \widetilde{T}_{\mu\nu} &= T_{\mu\nu} + \ell^{*2} \left\{ -4\tau_{\mu\cdot[\lambda]}^{\cdot\kappa} \tau_{\nu\cdot[\kappa]}^{\cdot\lambda} - 2\tau_{\mu}^{\cdot\kappa\lambda} \tau_{\nu\kappa\lambda} + \tau_{\cdot,\mu}^{\kappa\lambda} \tau_{\kappa\lambda\nu} \right. \\ &\quad \left. + \frac{1}{2} g_{\mu\nu} (4\tau_{\delta\cdot[\lambda]}^{\cdot\kappa} \tau_{\cdot,\kappa]}^{\delta\lambda} + \tau^{\delta\kappa\lambda} \tau_{\delta\kappa\lambda}) \right\}. \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_345 EQ1055 (15.2.12)		0.665 N +0	$\bar{\mathrm{L}}_{\mathrm{w}}(f)=\frac{i}{8}\operatorname{Tr}\left(\bar{\gamma}\wedge\bar{\gamma}\wedge^{*}\bar{\Omega}\right).$	$\bar{L}_{\mathrm{w}}(f)=\frac{i}{8}\operatorname{Tr}\left(\bar{\gamma}\wedge\bar{\gamma}\wedge^{*}\bar{\Omega}\right).$	
mielke-geometrodynamics_377 EQ1145 (C.13)		0.667 N +0	$\stackrel{\circ}{g}_{lm}:=c_{lk}{}^ic_{mi}{}^k$	$\stackrel{\circ}{g}_{lm}:=c_{lk}{}^ic_{mi}{}^k$	
mielke-geometrodynamics_151 EQ0450 (7.2.3)		0.676 N +0	$\Omega^g=\stackrel{+}{\Omega}^g+\bar{\Omega}^g$	$\Omega^g=\stackrel{+}{\Omega}^g+\bar{\Omega}^g$	
mielke-geometrodynamics_064 EQ0143 (3.4.14)		0.679 N +0	$\operatorname{div}\iota:=(-1)^{1+s}*d^*\iota=0$	$\operatorname{div}\iota:=(-1)^{1+s}*d^*\iota=0$	
mielke-geometrodynamics_155 EQ0472 (7.3.11)		0.682 C +23	$\stackrel{+}{\Omega}^{\{\}}=0\iff\mathcal{R}_{ij}^{\{\}}=0\iff R_{ij}^{\{\}}=\varLambda'g_{ij}.$	$\stackrel{+}{\Omega}^{\{\}}=0\iff\mathcal{R}_{ij}^{\{\}}=0\iff R_{ij}^{\{\}}=\varLambda'g_{ij}.$	
mielke-geometrodynamics_256 EQ0812 (11.4.24)		0.686 C +8	$\begin{aligned} &\partial_{\rho^*}H+\frac{1+\kappa}{\rho}e^{(\nu-\mu)/2}H=\frac{1}{\beta}\left[\omega+e^{\nu/2}-\varepsilon e^{-\mu}\left(H^2-F^2-\frac{2}{3}\mathrm{a}H^2\right)\right]F,\\ &\partial_{\rho^*}F+\frac{1-\kappa}{\rho}e^{(\nu-\mu)/2}F=\frac{1}{\beta}\left[-\omega+e^{\nu/2}-\varepsilon e^{-\mu}\left(H^2-F^2+\frac{2}{3}\mathrm{a}F^2\right)\right]H. \end{aligned}$	$\begin{aligned} &\partial_{\rho^*}H+\frac{1+\kappa}{\rho}e^{(\nu-\mu)/2}H=\frac{1}{\beta}\left[\omega+e^{\nu/2}-\varepsilon e^{-\mu}\left(H^2-F^2-\frac{2}{3}\mathrm{a}H^2\right)\right]F,\\ &\partial_{\rho^*}F+\frac{1-\kappa}{\rho}e^{(\nu-\mu)/2}F=\frac{1}{\beta}\left[-\omega+e^{\nu/2}-\varepsilon e^{-\mu}\left(H^2-F^2+\frac{2}{3}\mathrm{a}F^2\right)\right]H. \end{aligned}$	$\begin{aligned} &\partial_{\rho^*}H+\frac{1+\kappa}{\rho}e^{(\nu-\mu)/2}H=\frac{1}{\beta}\left[\omega+e^{\nu/2}-\varepsilon e^{-\mu}\left(H^2-F^2-\frac{2}{3}\mathrm{a}H^2\right)\right]F,\\ &\partial_{\rho^*}F+\frac{1-\kappa}{\rho}e^{(\nu-\mu)/2}F=\frac{1}{\beta}\left[-\omega+e^{\nu/2}-\varepsilon e^{-\mu}\left(H^2-F^2+\frac{2}{3}\mathrm{a}F^2\right)\right]H. \end{aligned}$
mielke-geometrodynamics_111 EQ0321 (5.2.5)		0.686	$\tau_{\alpha\beta}:=\frac{\partial L_{\mathrm{D}}}{\partial\Gamma^{\alpha\beta}}=\frac{1}{8}\bar{\Psi}\left(*\gamma\sigma_{\alpha\beta}+\sigma_{\alpha\beta}* \gamma\right)\Psi=\frac{1}{4}\eta_{\alpha\beta\gamma\delta}\bar{\Psi}\gamma^{\delta}\gamma_5\Psi\eta^{\gamma}=\tau_{\alpha\beta\gamma}\eta^{\gamma}$	$\tau_{\alpha\beta}:=\frac{\partial L_{\mathrm{D}}}{\partial\Gamma^{\alpha\beta}}=\frac{1}{8}\bar{\Psi}\left(*\gamma\sigma_{\alpha\beta}+\sigma_{\alpha\beta}* \gamma\right)\Psi=\frac{1}{4}\eta_{\alpha\beta\gamma\delta}\bar{\Psi}\gamma^{\delta}\gamma_5\Psi\eta^{\gamma}=\tau_{\alpha\beta\gamma}\eta^{\gamma}$	
mielke-geometrodynamics_131 EQ0309 (6.4.4)		0.688 W +2	$\mathscr{H}=\mathscr{K}-\dot{\Psi}\wedge\left(\frac{\partial\mathscr{K}}{\partial\dot{\Psi}}\right),$	$\mathscr{H}=\mathscr{K}-\dot{\Psi}\wedge\left(\frac{\partial\mathscr{K}}{\partial\dot{\Psi}}\right),$	
mielke-geometrodynamics_307 EQ0949 (14.1.30)		0.688 W +0	$\begin{aligned} D\Omega&=(D\xi+(-1)^z*\xi\wedge D\delta)\cos\delta\\ &\quad+(-1)^z(D^*\xi-(-1)^z\xi\wedge D\delta)\sin\delta=0. \end{aligned}$	$\begin{aligned} D\Omega&=(D\xi+(-1)^z*\xi\wedge D\delta)\cos\delta\\ &\quad+(-1)^z(D^*\xi-(-1)^z\xi\wedge D\delta)\sin\delta=0. \end{aligned}$	$\begin{aligned} D\Omega&=(D\xi+(-1)^z*\xi\wedge D\delta)\cos\delta\\ &\quad+(-1)^z(D^*\xi-(-1)^z\xi\wedge D\delta)\sin\delta=0. \end{aligned}$
mielke-geometrodynamics_224 EQ0696 (10.5.7)		0.694 N +0	$\left.D^{\wedge{*}}\gamma\right\}\gamma=\{\}^{\wedge{*}}\gamma\wedge T,\quad T:=e_{\alpha}\rfloor T^{\alpha},$	$D^*\gamma=*\gamma\wedge T,\quad T:=e_{\alpha}\rfloor T^{\alpha},$	$D^*\gamma=*\gamma\wedge T,\quad T:=e_{\alpha}\rfloor T^{\alpha},$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_175 EQ0534 (8.2.1)		0.699 C +5	$\begin{aligned} s \Gamma &= \Psi - DC \\ sC &= \Phi - \frac{1}{2}[C, C] \\ s\Omega^g &= -D\Psi - [C, \Omega^g] \\ s\Psi &= -D\Phi - [C, \Psi] \\ s\Phi &= -[C, \Phi] \end{aligned}$	$s\Gamma = \Psi - DC$ $sC = \Phi - \frac{1}{2}[C, C]$ $s\Omega^g = -D\Psi - [C, \Omega^g]$ $s\Psi = -D\Phi - [C, \Psi]$ $s\Phi = -[C, \Phi]$	$s\Gamma = \Psi - DC,$ $sC = \Phi - \frac{1}{2}[C, C],$ $s\Omega^g = -D\Psi - [C, \Omega^g],$ $s\Psi = -D\Phi - [C, \Psi],$ $s\Phi = -[C, \Phi].$
mielke-geometrodynamics_112 EQ0329 (5.3.6)		0.699 N +0	$D \Theta \equiv \left[\Omega^g, \gamma \right], \quad D \Omega^g \equiv 0,$	$D\Theta \equiv [\Omega^g, \gamma], \quad D\Omega^g \equiv 0,$	$D\Theta \equiv [\Omega^g, \gamma], \quad D\Omega^g \equiv 0,$
mielke-geometrodynamics_155 EQ0471 (7.3.10)		0.706 N +0	$\bar{\omega}^{\{\}} = 0 \Longleftrightarrow \omega_{\mathrm{C}}^{\{\}} = 0 \text{ and } R^{\{\}} = 0.$	$\bar{\Omega}^{\{\}} = 0 \iff \Omega_{\mathrm{C}}^{\{\}} = 0 \text{ and } R^{\{\}} = 0.$	$\overline{\Omega}^{\{\}} = 0 \iff \Omega_{\mathrm{C}}^{\{\}} = 0 \text{ and } R^{\{\}} = 0.$
mielke-geometrodynamics_240 EQ0747 (11.1.14)		0.706 N +0	$M = \mathbb{R} \times S^3 \#^v W^3.$	$M = \mathbb{R} \times S^3 \#^v W^3.$	$M = \mathbb{R} \times S^3 \#^v W^3.$
mielke-geometrodynamics_105 EQ0314 (4.6.6)		0.710 N +0	$\Phi(\tilde{\Gamma}, \gamma) = P \exp \left[(i/\hbar) \oint \left(\Gamma^{(T)\alpha} P_\alpha + \Gamma_\alpha^{(L)\beta} L^\alpha_\beta \right) \right]$	$\Phi(\tilde{\Gamma}, \gamma) = P \exp \left[(i/\hbar) \oint \left(\Gamma^{(T)\alpha} P_\alpha + \Gamma_\alpha^{(L)\beta} L^\alpha_\beta \right) \right]$	$\Phi(\tilde{\Gamma}, \gamma) = P \exp \left[(i/\hbar) \oint \left(\Gamma^{(T)\alpha} P_\alpha + \Gamma_\alpha^{(L)\beta} L^\alpha_\beta \right) \right]$
mielke-geometrodynamics_058 EQ0108 (3.3.9)		0.711 N -1	$L_{\mathrm{Pontr}} = -\frac{1}{2} F \wedge F = -\frac{1}{2} dC,$	$L_{\mathrm{Pontr}} = -\frac{1}{2} F \wedge F = -\frac{1}{2} dC,$	$L_{\mathrm{Pontr}} = -\frac{1}{2} F \wedge F = -\frac{1}{2} dC,$
mielke-geometrodynamics_177 EQ0545 (8.3.6)		0.711 N +0	$d \rightarrow \exp(\zeta) d \exp(-\zeta).$	$d \rightarrow \exp(\zeta) d \exp(-\zeta).$	$d \rightarrow \exp(\zeta) d \exp(-\zeta).$
mielke-geometrodynamics_192 EQ0583 (9.1.1)		0.712 N +0	$ds^2 = -e^v dt^2 + e^\lambda dr^2 + r^2 d\Omega^{n-2}$	$ds^2 = -e^v dt^2 + e^\lambda dr^2 + r^2 d\Omega^{n-2}$	$ds^2 = -e^v dt^2 + e^\lambda dr^2 + r^2 d\Omega^{n-2}$
mielke-geometrodynamics_094 EQ0260 (4.3.25)		0.718 N +0	$\stackrel{*}{K} = \frac{1}{2} K_\alpha^{\beta c} L_{\beta c} \otimes \vartheta^\alpha.$	$\stackrel{*}{K} = \frac{1}{2} K_\alpha^{\beta c} L_{\beta c} \otimes \vartheta^\alpha.$	$\stackrel{*}{K} = \frac{1}{2} K_\alpha^{\beta c} L_{\beta c} \otimes \vartheta^\alpha.$
mielke-geometrodynamics_285 EQ0878 (13.3.7)		0.722 N +0	$\mathrm{SL}(n+1, \mathbb{R}) \approx \mathrm{A}_*(n, \mathbb{R}) := \mathbb{R}^n \otimes \mathrm{GL}(n, \mathbb{R}) \otimes \mathbb{R}_*^n.$	$\mathrm{SL}(n+1, \mathbb{R}) \approx \mathrm{A}_*(n, \mathbb{R}) := \mathbb{R}^n \otimes \mathrm{GL}(n, \mathbb{R}) \otimes \mathbb{R}_*^n.$	$\mathrm{SL}(n+1, \mathbb{R}) \approx \mathrm{A}_*(n, \mathbb{R}) := \mathbb{R}^n \otimes \mathrm{GL}(n, \mathbb{R}) \otimes \mathbb{R}_*^n.$
mielke-geometrodynamics_042 EQ0041 (2.6.4)		0.725	$\stackrel{*}{\Omega} = \frac{1}{2} F_\alpha \beta^j L_j \otimes \vartheta^\alpha \wedge \vartheta^\beta$	$\stackrel{*}{\Omega} = \frac{1}{2} F_\alpha \beta^j L_j \otimes \vartheta^\alpha \wedge \vartheta^\beta$	$\stackrel{*}{\Omega} = \frac{1}{2} F_\alpha \beta^j L_j \otimes \vartheta^\alpha \wedge \vartheta^\beta$
mielke-geometrodynamics_185 EQ0574 (8.6.12)		0.725 N +0	$L_{\mathrm{eff}} := L_{\mathrm{qPG}}(**) - dC_\theta = -\frac{\theta_{\mathrm{T}}^*}{2\ell^2} R^{\{\} \alpha \beta} \wedge \eta_{\alpha \beta} + \frac{\theta_{\mathrm{T}}^* \Lambda_{\mathrm{eff}}}{\ell^2} \eta.$	$L_{\mathrm{eff}} := L_{\mathrm{qPG}}(**) - dC_\theta = -\frac{\theta_{\mathrm{T}}^*}{2\ell^2} R^{\{\} \alpha \beta} \wedge \eta_{\alpha \beta} + \frac{\theta_{\mathrm{T}}^* \Lambda_{\mathrm{eff}}}{\ell^2} \eta.$	$L_{\mathrm{eff}} := L_{\mathrm{qPG}}(**) - dC_\theta = -\frac{\theta_{\mathrm{T}}^*}{2\ell^2} R^{\{\} \alpha \beta} \wedge \eta_{\alpha \beta} + \frac{\theta_{\mathrm{T}}^* \Lambda_{\mathrm{eff}}}{\ell^2} \eta.$
mielke-geometrodynamics_126 EQ0366 (6.2.13)		0.725 N -2	$L_{\mathrm{EC}} := L_{\mathrm{EC}} \pm idC_{\mathrm{TT}} = -\frac{1}{2\ell^2} R_{\alpha \beta} \wedge *(\vartheta^\alpha \wedge \vartheta^\beta) + \frac{\Lambda}{\ell^2} \eta.$	$L_{\mathrm{EC}} := L_{\mathrm{EC}} \pm idC_{\mathrm{TT}} = -\frac{1}{2\ell^2} R_{\alpha \beta} \wedge *(\vartheta^\alpha \wedge \vartheta^\beta) + \frac{\Lambda}{\ell^2} \eta.$	$L_{\mathrm{EC}} := L_{\mathrm{EC}} \pm idC_{\mathrm{TT}} = -\frac{1}{2\ell^2} R_{\alpha \beta} \wedge *(\vartheta^\alpha \wedge \vartheta^\beta) + \frac{\Lambda}{\ell^2} \eta.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_ E01099 (B.10)	369	0.746 C +3	$\begin{aligned} & \sigma_* \left(\alpha^{(p)} + \beta^{(p)} \right) = \sigma_* \left(\alpha^{(p)} \right) + \sigma_* \left(\beta^{(p)} \right) \\ & \sigma_* \left(\alpha^{(p)} \wedge \beta^{(q)} \right) = \sigma_* \left(\alpha^{(p)} \right) \wedge \sigma_* \left(\beta^{(q)} \right) \\ & d \left(\sigma_* \alpha^{(p)} \right) = \sigma_* \left(d \alpha^{(p)} \right) \end{aligned}$	$\begin{aligned} \sigma_*(\alpha^{(p)} + \beta^{(p)}) &= \sigma_*(\alpha^{(p)}) + \sigma_*(\beta^{(p)}), \\ \sigma_*(\alpha^{(p)} \wedge \beta^{(q)}) &= \sigma_*(\alpha^{(p)}) \wedge \sigma_*(\beta^{(q)}), \\ d(\sigma_* \alpha^{(p)}) &= \sigma_*(d\alpha^{(p)}). \end{aligned}$	
mielke-geometrodynamics_ E00655 (10.3.12)	216	0.746 N +0	$\left. \left. E_{\alpha} \right L(\star) - \frac{2\gamma\varepsilon^2}{\ell^3(1+\delta\varepsilon)} \eta_{\alpha} + (-1)^s \frac{2\gamma}{\ell(1+\delta\varepsilon)} \left[(e_{\alpha}] R^{*\beta} \right) \wedge \vartheta_{\beta} \right],$	$E_{\alpha} = e_{\alpha}] L(\star) - \frac{2\gamma\varepsilon^2}{\ell^3(1+\delta\varepsilon)} \eta_{\alpha} + (-1)^s \frac{2\gamma}{\ell(1+\delta\varepsilon)} \left[(e_{\alpha}] R^{*\beta} \right) \wedge \vartheta_{\beta} \right],$	
mielke-geometrodynamics_ E00477 (7.4.5)	156	0.748 N +0	$g_{ij} = \Omega^2 o_{ij}^{\text{Mink}}$	$g_{ij} = \Omega^2 o_{ij}^{\text{Mink}},$	
mielke-geometrodynamics_ E00963 (14.2.3)	312	0.750 N +1	$\stackrel{\circ}{d}\mathcal{U}=\stackrel{\circ}{\vartheta} \wedge \stackrel{\circ}{\vartheta}.$	$\stackrel{\circ}{d}\mathcal{U}=\stackrel{\circ}{\vartheta} \wedge \stackrel{\circ}{\vartheta}.$	
mielke-geometrodynamics_ E00687 (10.3.45)	222	0.753 N +0	$-E_{\alpha} = -e_{\alpha}] L(\star) - (-1)^s \frac{2\gamma}{\ell} (e_{\alpha}] R^{\{\star\beta} \right) \wedge \vartheta_{\beta} = \ell \widetilde{\Sigma}_{\alpha}$	$-E_{\alpha} = -e_{\alpha}] L(\star) - (-1)^s \frac{2\gamma}{\ell} (e_{\alpha}] R^{\{\star\beta} \right) \wedge \vartheta_{\beta} = \ell \widetilde{\Sigma}_{\alpha}$	
mielke-geometrodynamics_ E00606 (9.2.5)	199	0.753 N +1	$\chi(M)=\int_M^* (\gamma_2^g)-\frac{1}{(4\pi)^2}\int_{\partial M}\left\{\alpha\wedge\Omega^{g(\star)}+\frac{2}{3}\alpha\wedge(\alpha\wedge\alpha)^{(\star)}\right\}.$	$\chi(M)=\int_M^* (\gamma_2^g)-\frac{1}{(4\pi)^2}\int_{\partial M}\left\{\alpha\wedge\Omega^{g(\star)}+\frac{2}{3}\alpha\wedge(\alpha\wedge\alpha)^{(\star)}\right\}.$	
mielke-geometrodynamics_ E00992 (14.3.5)	320	0.754 N +0	$\stackrel{\circ}{\gamma}^{K+1} \chi_{\rho F}^{\pm}(\xi) = \pm \chi_{\rho F}^{\pm}(\xi)$	$\stackrel{\circ}{\gamma}^{K+1} \chi_{\rho F}^{\pm}(\xi) = \pm \chi_{\rho F}^{\pm}(\xi)$	
mielke-geometrodynamics_ E00249 (4.3.14)	092	0.755 N +0	$D_k g_{ij} = E_k^{\alpha} \nabla_{\alpha} g_{ij} = 0$	$D_k g_{ij} = E_k^{\alpha} \nabla_{\alpha} g_{ij} = 0$	
mielke-geometrodynamics_ E00399 (6.4.17)	133	0.755 W +0	$\begin{aligned} & [D_{\alpha}, D_{\beta}] = -T_{\alpha\beta}{}^{\gamma}(x) D_{\gamma} + R_{\alpha\beta\gamma}{}^{\delta}(x) L^{\gamma}{}_{\delta}, \\ & [L^{\alpha}{}_{\beta}, D_{\gamma}] = \delta_{\gamma}^{\alpha} D_{\beta}, \\ & [L^{\alpha}{}_{\beta}, L^{\gamma}{}_{\delta}] = \delta_{\delta}^{\alpha} L^{\gamma}{}_{\beta} - \delta_{\beta}^{\gamma} L^{\alpha}{}_{\delta}, \end{aligned}$	$\begin{aligned} [D_{\alpha}, D_{\beta}] &= -T_{\alpha\beta}{}^{\gamma}(x) D_{\gamma} + R_{\alpha\beta\gamma}{}^{\delta}(x) L^{\gamma}{}_{\delta}, \\ [L^{\alpha}{}_{\beta}, D_{\gamma}] &= \delta_{\gamma}^{\alpha} D_{\beta}, \\ [L^{\alpha}{}_{\beta}, L^{\gamma}{}_{\delta}] &= \delta_{\delta}^{\alpha} L^{\gamma}{}_{\beta} - \delta_{\beta}^{\gamma} L^{\alpha}{}_{\delta}, \end{aligned}$	
mielke-geometrodynamics_ E00957 (14.1.39)	309	0.756	$\partial_{\kappa} \delta = \sqrt{ g } \varepsilon_{\kappa\lambda\rho\sigma} R^{\{\}^{\sigma}{}_{\nu}} \nabla^{\rho} R^{\{\}^{\nu\lambda}} / \left(R_{\alpha\beta}^{\{\}^{\lambda}} R^{\{\}^{\alpha\beta}} \right).$	$\partial_{\kappa} \delta = \sqrt{ g } \varepsilon_{\kappa\lambda\rho\sigma} R^{\{\}^{\sigma}{}_{\nu}} \nabla^{\rho} R^{\{\}^{\nu\lambda}} / (R_{\alpha\beta}^{\{\}^{\lambda}} R^{\{\}^{\alpha\beta}}).$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_314 EQ0970 (14.2.11)		■ 0.758 ● C +6	$\hat{\Omega}=\hat{d}\,\hat{\omega}=\hat{\omega}-\hat{\omega}\wedge\hat{\omega}=\left[\begin{array}{l l}\Omega^g+\alpha\wedge\alpha^{\mathrm{T}}&-D\alpha\\\hline\circ&\overset{\circ}{\Omega}+\alpha^{\mathrm{T}}\wedge\alpha\end{array}\right].$	$\hat{\Omega}=\hat{d}\hat{\omega}-\hat{\omega}-\hat{\omega}\wedge\hat{\omega}=\left[\begin{array}{l l}\Omega^g+\alpha\wedge\alpha^{\mathrm{T}}&-D\alpha\\\hline\circ&\overset{\circ}{\Omega}+\alpha^{\mathrm{T}}\wedge\alpha\end{array}\right].$	
mielke-geometrodynamics_327 EQ1016 (14.4.11)		■ 0.758 ● N +0	$ds^2=-\left(1-\left(\frac{r_{\circ}}{r}\right)^4\right)\frac{\Delta}{r^2}dt^2+\frac{r^2}{\Delta}dr^2+r^2d\Omega^2.$	$ds^2=-\left(1-\left(\frac{r_{\circ}}{r}\right)^4\right)\frac{\Delta}{r^2}dt^2+\frac{r^2}{\Delta}dr^2+r^2d\Omega^2.$	
mielke-geometrodynamics_104 EQ0310 (4.6.2)		■ 0.760 ● W +2	$\begin{aligned} \widetilde{D}\widetilde{\xi}^{\alpha} &= d\xi^{\alpha} + \Gamma_{\beta}^{(L)\alpha} \wedge \xi^{\beta} + \Gamma^{(T)\alpha} \\ &= D\xi^{\alpha} + \Gamma^{(T)\alpha} = 0 \end{aligned}$	$\begin{aligned} \widetilde{D}\widetilde{\xi}^{\alpha} &= d\xi^{\alpha} + \Gamma_{\beta}^{(L)\alpha} \wedge \xi^{\beta} + \Gamma^{(T)\alpha} \\ &= D\xi^{\alpha} + \Gamma^{(T)\alpha} = 0. \end{aligned}$	
mielke-geometrodynamics_270 EQ0832 (12.1.4)		■ 0.763 ● N +0	$\left\langle d\widetilde{j}_5\right\rangle =0,$	$\langle d\widetilde{j}_5\rangle =0,$	
mielke-geometrodynamics_088 EQ0231 (4.2.24)		■ 0.766 ● N +2	$T_{\alpha\beta}^{\cdot\cdot c}=2D_{[\alpha}E_{\cdot] \beta]}^c=2\left(\partial_{[\alpha}E_{\cdot] \beta]}^c+\Gamma_{[\alpha d}^cE_{\cdot \beta]}^d\right).$	$T_{\alpha\beta}^{\cdot\cdot c}=2D_{[\alpha}E_{\cdot] \beta]}^c=2\left(\partial_{[\alpha}E_{\cdot] \beta]}^c+\Gamma_{[\alpha d}^{\cdot\cdot c}E_{\cdot \beta]}^d\right).$	
mielke-geometrodynamics_365 EQ1079 (A.4)		■ 0.766 ● K +0	$\left[\sigma^i,\sigma^j\right]=\varepsilon^{ij}{}_k\sigma^k$	$\left[\sigma^i,\sigma^j\right]=\varepsilon^{ij}{}_k\sigma^k$	
mielke-geometrodynamics_097 EQ0276 (4.4.4)		■ 0.767 ● W +0	$\begin{aligned} \delta L_g &= \delta \vartheta \wedge E - \delta \vartheta \wedge D \Pi^{\mathrm{T}} + d \left(\delta \vartheta \wedge \Pi^{\mathrm{T}} \right) \\ &\quad - \left[\delta \omega^g, \vartheta \right] \wedge \left[\Pi_{\square}^{\mathrm{T}}, \vartheta \right] - \delta \omega^g \wedge D \Pi^{\mathrm{L}} + d \left(\delta \omega^g \wedge \Pi^{\mathrm{L}} \right), \end{aligned}$	$\begin{aligned} \delta L_g &= \delta \vartheta \wedge E - \delta \vartheta \wedge D \Pi^{\mathrm{T}} + d \left(\delta \vartheta \wedge \Pi^{\mathrm{T}} \right) \\ &\quad - \left[\delta \omega^g, \vartheta \right] \wedge \left[\Pi_{\square}^{\mathrm{T}}, \vartheta \right] - \delta \omega^g \wedge D \Pi^{\mathrm{L}} + d \left(\delta \omega^g \wedge \Pi^{\mathrm{L}} \right), \end{aligned}$	
mielke-geometrodynamics_209 EQ0613 (10.1.1)		■ 0.769 ● N +0	$\vartheta^\alpha=E_i^\alpha dx^i\quad\text{and}\quad\Gamma_\alpha^\star:=\frac{1}{2}\eta_{\alpha\beta\gamma}\Gamma^{\beta\gamma}.$	$\vartheta^\alpha=E_i^\alpha dx^i\quad\text{and}\quad\Gamma_\alpha^\star:=\frac{1}{2}\eta_{\alpha\beta\gamma}\Gamma^{\beta\gamma}.$	
mielke-geometrodynamics_355 EQ1066 (16.2.5)		■ 0.770 ● C -4	$\rho^*=\int_o^{\rho}\frac{dx}{1\pm x^2}=\left\{\begin{array}{l}\operatorname{artan}\rho,\kappa>0\\\operatorname{Arth}\rho,\kappa<0\end{array}\right\}$	$\rho^*=\int_o^{\rho}\frac{dx}{1\pm x^2}=\left\{\begin{array}{l}\operatorname{artan}\rho,\kappa>0\\\operatorname{Arth}\rho,\kappa<0\end{array}\right\}.$	
mielke-geometrodynamics_337 EQ1030 (15.1.6)		■ 0.771 ● W -1	$\begin{aligned} \frac{1}{2}\vartheta^g\wedge^*\Omega^g+\Lambda^*\vartheta_g &= \ell^{*2}\left\{\Sigma_{\mathrm{m}}(\mathrm{leptons})+\Sigma^{\mathrm{g}}\right\}, \\ \frac{1}{2}\vartheta^f\wedge^*\Omega^f &= \ell^2\left\{\Sigma_{\mathrm{m}}(\mathrm{hadrons})+\Sigma^{\mathrm{f}}\right\}. \end{aligned}$	$\begin{aligned} \frac{1}{2}\vartheta^g\wedge^*\Omega^g+\Lambda^*\vartheta_g &= \ell^{*2}\left\{\Sigma_{\mathrm{m}}(\mathrm{leptons})+\Sigma^{\mathrm{g}}\right\}, \\ \frac{1}{2}\vartheta^f\wedge^*\Omega^f &= \ell^2\left\{\Sigma_{\mathrm{m}}(\mathrm{hadrons})+\Sigma^{\mathrm{f}}\right\}. \end{aligned}$	
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_00112 (3.3.13)	059	0.772	$\begin{aligned} j:= & \mathscr{J} \wedge dt + \hat{\rho} \\ \backslash \backslash = & - \backslash \left(J_{\backslash x} d y \wedge d z + J_{\backslash y} d z \wedge d x + J_{\backslash z} d x \wedge d y \right) \wedge dt \\ & + \rho d x \wedge d y \wedge d z \end{aligned}$	$j := - \mathscr{J} \wedge dt + \hat{\rho} \\ = - \left(J_x dy \wedge dz + J_y dz \wedge dx + J_z dx \wedge dy \right) \wedge dt \\ + \rho dx \wedge dy \wedge dz$	$j := - \mathscr{J} \wedge dt + \hat{\rho} \\ = - (J_x dy \wedge dz + J_y dz \wedge dx + J_z dx \wedge dy) \wedge dt \\ + \rho dx \wedge dy \wedge dz .$
mielke-geometrodynamics_00130 (3.4.1)	062	0.780 N -1	$\mathscr{F} := \frac{1}{\alpha_g} \operatorname{Tr} \backslash \left\{ \right. \left. \right\} ^{\ast} \backslash \left(\Omega \wedge \right. \left. \right\} ^{\ast} \Omega \backslash \right) \backslash \right\} \\ \backslash = \frac{1}{\alpha_g} \left\{ 4 \alpha_g \right\} F_{\alpha \beta} \{ \} ^{\ast} F^{\alpha \beta} \{ \} _{\alpha \beta} \{ \} _{\beta j}$	$\mathscr{F} := \frac{1}{\alpha_g} \operatorname{Tr} \left\{ \ast \left(\Omega \wedge \ast \Omega \right) \right\} = \frac{1}{4 \alpha_g} F_{\alpha \beta}{}^j F^{\alpha \beta}{}_j$	$\mathscr{F} := \frac{1}{\alpha_g} \operatorname{Tr} \left\{ \ast (\Omega \wedge \ast \Omega) \right\} = \frac{1}{4 \alpha_g} F_{\alpha \beta}{}^j F^{\alpha \beta}{}_j$
mielke-geometrodynamics_00470 (7.3.9)	155	0.782 N +0	$\{ \} ^{\ast} \Omega ^g \{ \} = \zeta \Omega ^g \{ \} ,$	$\ast \Omega ^g \{ \} = \zeta \Omega ^g ,$	$\ast \Omega ^g \{ \} = \zeta \Omega ^g ,$
mielke-geometrodynamics_00846 (2.6.9)	043	0.783 N +0	$\begin{aligned} \Delta b &= \int_{\partial U} \omega \otimes b = \int_U d(\omega \otimes b)_{\mathrm{hor}} = \int_U D(\omega \otimes b) \\ &= \int_U \Omega \otimes b \simeq \Omega \otimes b \end{aligned}$	$\Delta b = \int_{\partial U} \omega \otimes b = \int_U d(\omega \otimes b)_{\mathrm{hor}} = \int_U D(\omega \otimes b) \\ = \int_U \Omega \otimes b \simeq \Omega \otimes b$	$\Delta b = \int_{\partial U} \omega \otimes b = \int_U d(\omega \otimes b)_{\mathrm{hor}} = \int_U D(\omega \otimes b) \\ = \int_U \Omega \otimes b \simeq \Omega \otimes b$
mielke-geometrodynamics_01049 (15.2.6)	344	0.785 N +1	$\{ \} ^{\ast} \backslash \operatorname{bar} \{ \omega \} = \operatorname{bar} \{ \omega \} _{\{ i \} ^{\ast} \{ j \} } \backslash \operatorname{bar} \{ L \} _{\{ j \} } \\ \otimes d x^{\{ i \} } = \operatorname{bar} \{ \omega \} _{\{ i \} } d x^{\{ i \} }$	$\ast \bar{\omega} = \bar{\omega}_i^j \bar{L}_j \otimes dx^i = \bar{\omega}_i dx^i$	$\ast \bar{\omega} = \bar{\omega}_i^j \bar{L}_j \otimes dx^i = \bar{\omega}_i dx^i$
mielke-geometrodynamics_00133 (3.4.4)	063	0.785 N -2	$L_{\Omega} = L \left(\mathscr{F}, \{ \} ^{\ast} \mathscr{F} \right) \eta .$	$L_{\omega} = L \left(\mathscr{F}, \ast \mathscr{F} \right) \eta .$	$L_{\omega} = L(\mathscr{F}, \ast \mathscr{F}) \eta .$
mielke-geometrodynamics_00312 (4.6.4)	104	0.786 W +0	$\begin{aligned} \Delta \xi^{\alpha} &= - \oint_C y \left[(DD \xi^{\alpha}) = \oint_C y \right] R^{(T) \alpha} = \int_S R^{(T) \alpha} \\ &\simeq \frac{1}{2} \left(T_{ij}^{\alpha} - R_{ij \beta}^{\alpha} \xi^{\beta} \right) \int_S dy^i \wedge dy^j \end{aligned}$	$\Delta \xi^{\alpha} = - \oint_C y \left[(DD \xi^{\alpha}) = \oint_C y \right] R^{(T) \alpha} = \int_S R^{(T) \alpha} \\ \simeq \frac{1}{2} \left(T_{ij}^{\alpha} - R_{ij \beta}^{\alpha} \xi^{\beta} \right) \int_S dy^i \wedge dy^j$	$\Delta \xi^{\alpha} = - \oint_C y \rfloor (DD \xi^{\alpha}) = \oint_C y \rfloor R^{(T) \alpha} = \int_S R^{(T) \alpha} \\ \simeq \frac{1}{2} (T_{ij}^{\alpha} - R_{ij \beta}^{\alpha} \xi^{\beta}) \int_S dy^i \wedge dy^j .$
mielke-geometrodynamics_00224 (4.2.17)	087	0.787 N +0	$\widetilde{\varphi} \rightarrow A^{-1} \widetilde{\varphi} = G^{-1} (\widetilde{\varphi} - T) .$	$\widetilde{\varphi} \rightarrow A^{-1} \widetilde{\varphi} = G^{-1} (\widetilde{\varphi} - T) .$	$\widetilde{\varphi} \rightarrow A^{-1} \widetilde{\varphi} = G^{-1} (\widetilde{\varphi} - T) .$
mielke-geometrodynamics_00941 (14.1.22)	306	0.788 N +0	$\tan 2 \delta = (-1)^z \frac{\operatorname{Tr}^{\ast} (\Omega \wedge \Omega)}{\operatorname{Tr}^{\ast} (\Omega \wedge \ast \Omega)} .$	$\tan 2 \delta = (-1)^z \frac{\operatorname{Tr}^{\ast} (\Omega \wedge \Omega)}{\operatorname{Tr}^{\ast} (\Omega \wedge \ast \Omega)} .$	$\tan 2 \delta = (-1)^z \frac{\operatorname{Tr}^{\ast} (\Omega \wedge \Omega)}{\operatorname{Tr}^{\ast} (\Omega \wedge \ast \Omega)} .$
mielke-geometrodynamics_00760 (11.1.27)	242	0.791 N +0	$\left(\gamma^{n+1} \right)^2 = \mp \mathbb{1} .$	$\left(\gamma^{n+1} \right)^2 = \mp \mathbb{1} .$	$\left(\gamma^{n+1} \right)^2 = \mp \mathbb{1} .$
mielke-geometrodynamics_00725 (10.10.5)	230	0.793 N +0	$\delta_{\mathrm{susy}} L \cong \bar{\sigma} \left(i \gamma^{\alpha} \Psi \wedge \frac{\delta L}{\delta \vartheta^{\alpha}} - 2 D \frac{\delta L}{\delta \bar{\Psi}} + c \gamma \wedge \frac{\delta L}{\delta \bar{\Psi}} \right) + 2 d \left(\bar{\sigma} \wedge \frac{\delta L}{\delta \bar{\Psi}} \right) .$	$\delta_{\mathrm{susy}} L \cong \bar{\sigma} \left(i \gamma^{\alpha} \Psi \wedge \frac{\delta L}{\delta \vartheta^{\alpha}} - 2 D \frac{\delta L}{\delta \bar{\Psi}} + c \gamma \wedge \frac{\delta L}{\delta \bar{\Psi}} \right) + 2 d \left(\bar{\sigma} \wedge \frac{\delta L}{\delta \bar{\Psi}} \right) .$	$\delta_{\mathrm{susy}} L \cong \bar{\sigma} \left(i \gamma^{\alpha} \Psi \wedge \frac{\delta L}{\delta \vartheta^{\alpha}} - 2 D \frac{\delta L}{\delta \bar{\Psi}} + c \gamma \wedge \frac{\delta L}{\delta \bar{\Psi}} \right) + 2 d \left(\bar{\sigma} \wedge \frac{\delta L}{\delta \bar{\Psi}} \right) .$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E0810 (11.4.22)	256	■0.793 ●N +0	$\frac{3}{8}\ell^2\bar{\psi}\psi=\frac{4\pi m}{e^{-\mu-\nu/2}}\left(H^2-F^2\right)\left Y_{ \kappa -1}^{ \kappa -1}(\vartheta,\phi)\right ^2$	$\frac{3}{8}\ell^2\bar{\psi}\psi=4\pi me^{-\mu-\nu/2}\left(H^2-F^2\right)\left Y_{ \kappa -1}^{ \kappa -1}(\vartheta,\phi)\right ^2$	
mielke-geometrodynamics_E01032 (15.1.9)	338	■0.793 ●	$\frac{1}{2}*(\vartheta^g\wedge^*\Omega^g)+\Lambda\vartheta^g=0,$	$\frac{1}{2}*(\vartheta^g\wedge^*\Omega^g)+\Lambda\vartheta^g=0,$	$\frac{1}{2}*(\vartheta^g\wedge^*\Omega^g)+\Lambda\vartheta^g=0,$
mielke-geometrodynamics_E08305 (4.5.15)	103	■0.793 ●N +0	$\mathrm{varepsilon}_{\mathrm{RC}}:=\mathrm{x}\mathrm{i}^{\mathrm{\alpha}}\mathrm{\Sigma}_{\mathrm{\alpha}}+\mathrm{\left(e_{\mathrm{\beta}}\downharpoonleft\overrightarrow{\hspace{0.1cm}}\right)\mathrm{x}\mathrm{i}^{\mathrm{\gamma}}\mathrm{\gamma}}\mathrm{\tau}_{\mathrm{\beta}}$	$\varepsilon_{\mathrm{RC}}:=\xi^{\alpha}\Sigma_{\alpha}+\left(e_{\beta}\downarrow\widehat{D}\xi^{\gamma}\right)\tau^{\beta}_{\gamma},$	$\varepsilon_{\mathrm{RC}}:=\xi^{\alpha}\Sigma_{\alpha}+(e_{\beta}\downarrow\widehat{D}\xi^{\gamma})\tau^{\beta}_{\gamma},$
mielke-geometrodynamics_E0848 (12.3.1)	274	■0.794 ●C +6	$\left\langle d j_5\right\rangle =2 i m\left\langle \bar{\psi} \gamma_5 \psi\right\rangle \eta-\frac{1}{4 \pi^2} \operatorname{Tr}(G \wedge G)-\frac{1}{96 \pi^2}\left[2 R_{\alpha \beta}^{\{\}} \wedge R^{\{\beta \alpha \beta\}}+d \mathscr{A} \wedge d \mathscr{A}\right]$	$\langle dj_5\rangle=2im\langle\bar{\psi}\gamma_5\psi\rangle\eta-\frac{1}{4\pi^2}\operatorname{Tr}(G\wedge G)-\frac{1}{96\pi^2}\left[2R_{\alpha\beta}^{\{\}}\wedge R^{\{\beta\alpha\beta\}}+d\mathscr{A}\wedge d\mathscr{A}\right]$	$\langle dj_5\rangle=2im\langle\bar{\psi}\gamma_5\psi\rangle\eta-\frac{1}{4\pi^2}\operatorname{Tr}(G\wedge G)-\frac{1}{96\pi^2}\left[2R_{\alpha\beta}^{\{\}}\wedge R^{\{\beta\alpha\beta\}}+d\mathscr{A}\wedge d\mathscr{A}\right]$
mielke-geometrodynamics_E08319 (5.2.3)	110	■0.795 ●N +1	$\begin{aligned} L_{\mathrm{D}} &= L\left(\gamma, \psi, D^{\{\}}\psi\right)-\frac{i}{2} \bar{\psi}\left(*\gamma \wedge K-K \wedge *\gamma\right) \psi \\ &= L\left(\gamma, \psi, D^{\{\}}\psi\right)+\frac{1}{4} \mathscr{A} \wedge j_5 . \end{aligned}$	$L_{\mathrm{D}}=L\left(\gamma, \psi, D^{\{\}}\psi\right)-\frac{i}{2} \bar{\psi}\left(*\gamma \wedge K-K \wedge *\gamma\right) \psi \\ =L\left(\gamma, \psi, D^{\{\}}\psi\right)+\frac{1}{4} \mathscr{A} \wedge j_5 .$	$L_{\mathrm{D}}=L(\gamma, \psi, D^{\{\}}\psi)-\frac{i}{2} \bar{\psi}\left(*\gamma \wedge K-K \wedge *\gamma\right) \psi \\ =L(\gamma, \psi, D^{\{\}}\psi)+\frac{1}{4} \mathscr{A} \wedge j_5 .$
mielke-geometrodynamics_E08552 (8.4.6)	179	■0.797 ●N +0	$\stackrel{+}{\Omega}:=\frac{1}{2}\left(\Omega^g\pm*\Omega^{g(*)}\right)=0$	$\stackrel{+}{\Omega}:=\frac{1}{2}\left(\Omega^g\pm*\Omega^{g(*)}\right)=0$	$\stackrel{\pm}{\Omega}^g:=\frac{1}{2}\left(\Omega^g\pm*\Omega^{g(*)}\right)=0$
mielke-geometrodynamics_E01001 (14.3.14)	323	■0.798 ●N -1	$(i\not{D})^2:=-\operatorname{Tr}\left\{\gamma\wedge^*\left(\gamma\wedge^*D\right)\right\}.$	$(i\not{D})^2:=-\operatorname{Tr}\left\{\gamma\wedge^*D^*(\gamma\wedge^*D)\right\}.$	$(i\not{D})^2:=-\operatorname{Tr}\left\{\gamma\wedge^*D^*(\gamma\wedge^*D)\right\}.$
mielke-geometrodynamics_E08065 (2.8.8)	047	■0.803 ●N +0	$c_2(M)=-\int_M\pi(\gamma_2)=\frac{1}{8\pi^2}\int_M\operatorname{Tr}(\Omega\wedge\Omega).$	$c_2(M)=-\int_M^*\pi(\gamma_2)=\frac{1}{8\pi^2}\int_M\operatorname{Tr}(\Omega\wedge\Omega).$	$c_2(M)=-\int_M^*\pi(\gamma_2)=\frac{1}{8\pi^2}\int_M\operatorname{Tr}(\Omega\wedge\Omega).$
mielke-geometrodynamics_E08652 (10.3.9)	216	■0.803 ●K +0	$H_{\alpha}=-\frac{\gamma\varepsilon}{\ell^2(1+\delta\varepsilon)}\vartheta_{\alpha}\quad\Leftrightarrow\quad T_{\alpha}=\frac{\varepsilon}{\ell}\eta_{\alpha},$	$H_{\alpha}=-\frac{\gamma\varepsilon}{\ell^2(1+\delta\varepsilon)}\vartheta_{\alpha}\quad\Leftrightarrow\quad T_{\alpha}=\frac{\varepsilon}{\ell}\eta_{\alpha},$	$H_{\alpha}=-\frac{\gamma\varepsilon}{\ell^2(1+\delta\varepsilon)}\vartheta_{\alpha}\quad\Leftrightarrow\quad T_{\alpha}=\frac{\varepsilon}{\ell}\eta_{\alpha},$
mielke-geometrodynamics_E01065 (16.2.4)	355	■0.803 ●N +0	$\rho:=\sqrt{\frac{1}{3}\Lambda_{\mathrm{eff}}}\,r,\quad r= \mathbf{x} $	$\rho:=\sqrt{\frac{1}{3}\Lambda_{\mathrm{eff}}}\,r,\quad r= \mathbf{x} $	$\rho:=\sqrt{\frac{1}{3}\Lambda_{\mathrm{eff}}}\,r,\quad r= \mathbf{x} $

Conf. MathPix confidence: ■ ≥0.9■ 0.5–0.9■ <0.5— no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component● W weak● S stable● N noise● K clean● not measured

[refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_069 EQ0166 (3.6.1)		■ 0.806 ● W +0	$\begin{aligned} L_{\mathrm{SW}} = & \frac{1}{2} \overline{D_{\pm} \psi} \wedge D_{\pm} \psi \mp i \left(F^{\pm} \wedge F^{\pm} \right) + \frac{1}{2} \overline{D_{\pm} \psi} \wedge D_{\pm} \psi \\ & \pm i \overline{\psi} \sigma_{\pm} \psi \wedge F^{\pm} \mp \frac{i}{4} \overline{\psi} \sigma_{\pm} \psi \wedge \psi \sigma_{\pm} \psi ; \end{aligned}$	$L_{\mathrm{SW}} = \frac{1}{2} \overline{D_{\pm} \psi} \wedge D_{\pm} \psi \mp i \left(F^{\pm} \wedge F^{\pm} \right) + \frac{1}{2} \overline{D_{\pm} \psi} \wedge D_{\pm} \psi \pm i \overline{\psi} \sigma_{\pm} \psi \wedge F^{\pm} \mp \frac{i}{4} \overline{\psi} \sigma_{\pm} \psi \wedge \psi \sigma_{\pm} \psi ;$	$L_{\mathrm{SW}} = \frac{1}{2} \overline{D_{\pm} \psi} \wedge D_{\pm} \psi \mp i \left(F^{\pm} \wedge F^{\pm} \right) + \frac{1}{2} \overline{D_{\pm} \psi} \wedge D_{\pm} \psi \pm i \overline{\psi} \sigma_{\pm} \psi \wedge F^{\pm} \mp \frac{i}{4} \overline{\psi} \sigma_{\pm} \psi \wedge \psi \sigma_{\pm} \psi ;$
mielke-geometrodynamics_356 EQ1070 (16.2.9)		■ 0.806 ● N +0	$V_{\mathrm{eff}}(\rho^*) = \left(\frac{1}{\beta^2} - 2 \right) [1 + \tan^2 \rho^*]$	$V_{\mathrm{eff}}(\rho^*) = \left(\frac{1}{\beta^2} - 2 \right) [1 + \tan^2 \rho^*]$	$V_{\mathrm{eff}}(\rho^*) = \left(\frac{1}{\beta^2} - 2 \right) [1 + \tan^2 \rho^*]$
mielke-geometrodynamics_144 EQ0446 (6.9.2)		■ 0.806 ●	$\begin{aligned} \stackrel{\theta}{L} = & \frac{1}{2\ell} \left[T^{\alpha} \wedge T_{\alpha} + \theta_{\mathrm{T}} (T^{\alpha} \wedge T_{\alpha} + R_{\alpha\beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta}) \right] \\ & + \frac{1}{2} \left[R^{\alpha\beta} \wedge R_{\alpha\beta} + \theta_{\mathrm{L}} R^{\alpha\beta} \wedge R_{\alpha\beta} \right] . \end{aligned}$	$\stackrel{\theta}{L} = \frac{1}{2\ell} [T^{\alpha} \wedge T_{\alpha} + \theta_{\mathrm{T}} (T^{\alpha} \wedge T_{\alpha} + R_{\alpha\beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta})] + \frac{1}{2} [R^{\alpha\beta} \wedge R_{\alpha\beta} + \theta_{\mathrm{L}} R^{\alpha\beta} \wedge R_{\alpha\beta}] .$	$\stackrel{\theta}{L} = \frac{1}{2\ell} [T^{\alpha} \wedge T_{\alpha} + \theta_{\mathrm{T}} (T^{\alpha} \wedge T_{\alpha} + R_{\alpha\beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta})] + \frac{1}{2} [R^{\alpha\beta} \wedge R_{\alpha\beta} + \theta_{\mathrm{L}} R^{\alpha\beta} \wedge R_{\alpha\beta}] .$
mielke-geometrodynamics_202 EQ0610 (9.3.1)		■ 0.806 ● N -1	$\begin{aligned} d C_{\mathrm{RR}}^{\alpha\beta(\star)} = & \frac{1}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(\star)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(\star)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ \equiv & -L_{\mathrm{SKY}} - 2 \mathrm{Ric}_{\alpha\beta} \wedge \mathrm{Ric}^{\alpha\beta} + \frac{1}{2} \mathrm{Ric}_{\alpha}^{\alpha} \wedge \mathrm{Ric}_{\beta}^{\beta} \end{aligned}$	$d C_{\mathrm{RR}}^{\alpha\beta(\star)} := \frac{1}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(\star)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(\star)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \equiv -L_{\mathrm{SKY}} - 2 \mathrm{Ric}_{\alpha\beta} \wedge \mathrm{Ric}^{\alpha\beta} + \frac{1}{2} \mathrm{Ric}_{\alpha}^{\alpha} \wedge \mathrm{Ric}_{\beta}^{\beta}$	$d C_{\mathrm{RR}}^{\alpha\beta(\star)} := \frac{1}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(\star)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(\star)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \equiv -L_{\mathrm{SKY}} - 2 \mathrm{Ric}_{\alpha\beta} \wedge \mathrm{Ric}^{\alpha\beta} + \frac{1}{2} \mathrm{Ric}_{\alpha}^{\alpha} \wedge \mathrm{Ric}_{\beta}^{\beta}$
mielke-geometrodynamics_244 EQ0767 (11.1.34)		■ 0.807 ● N +0	$\psi \longrightarrow S^{-1} \psi = S^{-1} \psi$	$\psi \longrightarrow S^{-1} \psi = S^{-1} \psi$	$\psi \longrightarrow S^{-1} \psi = S^{-1} \psi$
mielke-geometrodynamics_095 EQ0267 (4.3.32)		■ 0.808 ● W +2	$\begin{aligned} R_{(ij)\kappa l} = & R_{ij(\kappa l)} = 0 \\ R_{ij\kappa l}^{\} = & R_{\kappa l i j}^{\} . \end{aligned}$	$R_{(ij)\kappa l}^{\} = R_{ij(\kappa l)}^{\} = 0 \\ R_{ij\kappa l}^{\} = & R_{\kappa l i j}^{\} .$	$R_{(ij)\kappa l}^{\} = R_{ij(\kappa l)}^{\} = 0 \\ R_{ij\kappa l}^{\} = & R_{\kappa l i j}^{\} .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_101 EQ0298 (4.5.8)	101	■ 0.811 ● W +1	$\begin{aligned} \delta L &= \delta g_{\alpha\beta} \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \delta \vartheta^\alpha \wedge \frac{\partial L}{\partial \vartheta^\alpha} \\ &+ \delta \Gamma_\alpha{}^\beta \wedge I^\alpha{}_\beta \Psi \wedge \frac{\partial L}{\partial D\Psi} + \delta \Psi \wedge \frac{\delta L}{\delta \Psi} + D \left(\delta \Psi \wedge \frac{\partial L}{\partial D\Psi} \right) \end{aligned}$	$\delta L = \delta g_{\alpha\beta} \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \delta \vartheta^\alpha \wedge \frac{\partial L}{\partial \vartheta^\alpha} + \delta \Gamma_\alpha{}^\beta \wedge I^\alpha{}_\beta \Psi \wedge \frac{\partial L}{\partial D\Psi} + \delta \Psi \wedge \frac{\delta L}{\delta \Psi} + D \left(\delta \Psi \wedge \frac{\partial L}{\partial D\Psi} \right)$	$\delta L = \delta g_{\alpha\beta} \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \delta \vartheta^\alpha \wedge \frac{\partial L}{\partial \vartheta^\alpha} + \delta \Gamma_\alpha{}^\beta \wedge I^\alpha{}_\beta \Psi \wedge \frac{\partial L}{\partial D\Psi} + \delta \Psi \wedge \frac{\delta L}{\delta \Psi} + D \left(\delta \Psi \wedge \frac{\partial L}{\partial D\Psi} \right).$
mielke-geometrodynamics_295 EQ0919 (13.8.5)	295	■ 0.811 ● N +0	$D \, T^{\{\alpha\}} \equiv R_{\{\beta\}}^{\{\alpha\}} \wedge \vartheta^{\beta}$	$DT^\alpha \equiv R_\beta{}^\alpha \wedge \vartheta^\beta$	$DT^\alpha \equiv R_\beta{}^\alpha \wedge \vartheta^\beta$
mielke-geometrodynamics_254 EQ0802 (11.4.14)	254	■ 0.812 ● W +0	$\begin{aligned} &\left[i \gamma^0 \partial_0 - e^{(\nu-\mu)/2} \left[i \vec{\gamma} \cdot \vec{\partial} + i \gamma^1 \partial_r \left(\frac{\mu}{2} + \frac{\nu}{4} \right) \right] \right. \\ &\left. + \frac{3\varepsilon}{8} \ell^2 e^{\nu/2} (\bar{\psi} \psi - a \bar{\psi} \gamma^5 \psi \gamma^5) - e^{\nu/2} m \right] \psi = 0 \end{aligned}$	$\left\{ i \gamma^0 \partial_0 - e^{(\nu-\mu)/2} \left[i \vec{\gamma} \cdot \vec{\partial} + i \gamma^1 \partial_r \left(\frac{\mu}{2} + \frac{\nu}{4} \right) \right] + \frac{3\varepsilon}{8} \ell^2 e^{\nu/2} (\bar{\psi} \psi - a \bar{\psi} \gamma^5 \psi \gamma^5) - e^{\nu/2} m \right\} \psi = 0$	$\left\{ i \gamma^0 \partial_0 - e^{(v-\mu)/2} \left[i \vec{\gamma} \cdot \vec{\partial} + i \gamma^1 \partial_r \left(\frac{\mu}{2} + \frac{v}{4} \right) \right] + \frac{3\varepsilon}{8} \ell^2 e^{v/2} (\bar{\psi} \psi - a \bar{\psi} \gamma^5 \psi \gamma^5) - e^{v/2} m \right\} \psi = 0$
mielke-geometrodynamics_255 EQ0807 (11.4.19)	255	■ 0.812 ●	$\vec{\sigma} \cdot \vec{L} \chi_\kappa^m = -(\kappa + 1) \chi_\kappa^m.$	$\vec{\sigma} \cdot \vec{L} \chi_\kappa^m = -(\kappa + 1) \chi_\kappa^m.$	$\vec{\sigma} \cdot \vec{L} \chi_\kappa^m = -(\kappa + 1) \chi_\kappa^m.$
mielke-geometrodynamics_132 EQ0397 (6.4.12)	132	■ 0.813 ● N +0	$\mathscr{H} \cong n^\alpha \mathscr{G}_\alpha + \Gamma_\perp^{\alpha\beta} \mathscr{G}_{\alpha\beta},$	$\mathcal{H} \cong n^\alpha \mathcal{G}_\alpha + \Gamma_\perp^{\alpha\beta} \mathcal{G}_{\alpha\beta},$	$\mathcal{H} \cong n^\alpha \mathcal{G}_\alpha + \Gamma_\perp^{\alpha\beta} \mathcal{G}_{\alpha\beta},$
mielke-geometrodynamics_227 EQ0709 (10.7.5)	227	■ 0.813 ● N +0	$R_{\beta}{}^{\alpha} \wedge \vartheta^{\beta} = \frac{\ell^2}{2} \varepsilon (x^\alpha Y \wedge X \wedge Y - y^\alpha X \wedge X \wedge Y) = 0.$	$R_{\beta}{}^{\alpha} \wedge \vartheta^{\beta} = \frac{\ell^2}{2} \varepsilon (x^\alpha Y \wedge X \wedge Y - y^\alpha X \wedge X \wedge Y) = 0.$	$R_{\beta}{}^{\alpha} \wedge \vartheta^{\beta} = \frac{\ell^2}{2} \varepsilon (x^\alpha Y \wedge X \wedge Y - y^\alpha X \wedge X \wedge Y) = 0.$
mielke-geometrodynamics_085 EQ0210 (4.2.2)	085	■ 0.814 ● N +1	$\stackrel{*}{\chi} \approx \left[\begin{array}{c c} \omega^L & \omega^T \\ \hline 0 & 0 \end{array} \right].$	${}^*\tilde{\chi} = \left[\begin{array}{c c} \omega^L & \omega^T \\ \hline 0 & 0 \end{array} \right].$	${}^*\tilde{\chi} \tilde{\omega} = \left[\begin{array}{c c} \omega^L & \omega^T \\ \hline 0 & 0 \end{array} \right].$
mielke-geometrodynamics_244 EQ0769 (11.1.36)	244	■ 0.817 ● N +0	$\widetilde{\omega} \longrightarrow {}^S \widetilde{\omega} = S \widetilde{\omega} S^{-1} + (dS) S^{-1}$	$\widetilde{\omega} \longrightarrow {}^S \widetilde{\omega} = S \widetilde{\omega} S^{-1} + (dS) S^{-1}$	$\widetilde{\omega} \longrightarrow {}^S \widetilde{\omega} = S \widetilde{\omega} S^{-1} + (dS) S^{-1}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_008398 (6.4.14)	133	0.820 W +0	$\begin{aligned} & \left(\mathbf{G}_\alpha(t, \mathbf{x}), \mathbf{G}_\beta(t, \mathbf{y}) \right) = \left(-T_{\alpha\beta}{}^\gamma \mathcal{G}_\gamma + R_{\alpha\beta\gamma}{}^\delta \mathcal{G}^\gamma{}_\delta \right) \cdot \delta(\mathbf{x} - \mathbf{y}), \\ & \mathcal{G}^\alpha{}_\beta(t, \mathbf{x}), \mathcal{G}_\gamma(t, \mathbf{y}) = \delta_\beta^\gamma \mathcal{G}_\beta \delta(\mathbf{x} - \mathbf{y}), \\ & \mathcal{G}^\alpha{}_\beta(t, \mathbf{x}), \mathcal{G}^\gamma{}_\delta(t, \mathbf{y}) = (\delta_\delta^\alpha \mathcal{G}^\gamma{}_\beta - \delta_\beta^\gamma \mathcal{G}^\alpha{}_\delta) \cdot \delta(\mathbf{x} - \mathbf{y}). \end{aligned}$	$\begin{aligned} & \left(\mathcal{G}_\alpha(t, \mathbf{x}), \mathcal{G}_\beta(t, \mathbf{y}) \right) = \left(-T_{\alpha\beta}{}^\gamma \mathcal{G}_\gamma + R_{\alpha\beta\gamma}{}^\delta \mathcal{G}^\gamma{}_\delta \right) \cdot \delta(\mathbf{x} - \mathbf{y}), \\ & \mathcal{G}^\alpha{}_\beta(t, \mathbf{x}), \mathcal{G}_\gamma(t, \mathbf{y}) = \delta_\beta^\gamma \mathcal{G}_\beta \delta(\mathbf{x} - \mathbf{y}), \\ & \mathcal{G}^\alpha{}_\beta(t, \mathbf{x}), \mathcal{G}^\gamma{}_\delta(t, \mathbf{y}) = \left(\delta_\delta^\alpha \mathcal{G}^\gamma{}_\beta - \delta_\beta^\gamma \mathcal{G}^\alpha{}_\delta \right) \cdot \delta(\mathbf{x} - \mathbf{y}). \end{aligned}$	$\begin{aligned} & \left(\mathcal{G}_\alpha(t, \mathbf{x}), \mathcal{G}_\beta(t, \mathbf{y}) \right) = \left(-T_{\alpha\beta}{}^\gamma \mathcal{G}_\gamma + R_{\alpha\beta\gamma}{}^\delta \mathcal{G}^\gamma{}_\delta \right) \cdot \delta(\mathbf{x} - \mathbf{y}), \\ & \mathcal{G}^\alpha{}_\beta(t, \mathbf{x}), \mathcal{G}_\gamma(t, \mathbf{y}) = \delta_\beta^\gamma \mathcal{G}_\beta \delta(\mathbf{x} - \mathbf{y}), \\ & \mathcal{G}^\alpha{}_\beta(t, \mathbf{x}), \mathcal{G}^\gamma{}_\delta(t, \mathbf{y}) = (\delta_\delta^\alpha \mathcal{G}^\gamma{}_\beta - \delta_\beta^\gamma \mathcal{G}^\alpha{}_\delta) \cdot \delta(\mathbf{x} - \mathbf{y}). \end{aligned}$
mielke-geometrodynamics_008059 (2.7.9)	045	0.820 K +0	$\bar{\varphi} \rightarrow \bar{\varphi} G \quad \in C^\infty(V^{\bar{s}}).$	$\bar{\varphi} \rightarrow \bar{\varphi} G \quad \in C^\infty(V^{\bar{s}}).$	$\bar{\varphi} \rightarrow \bar{\varphi} G \quad \in C^\infty(V^{\bar{s}}).$
mielke-geometrodynamics_009993 (14.3.6)	320	0.822 W +0	$\begin{aligned} & \int_G \pi \hat{L}_D = \frac{i}{2} \left(\bar{\psi} \gamma \wedge^* D^{g+\omega} \psi + \overline{D^{g+\omega} \psi} \wedge^* \gamma \psi \right) \\ & + \frac{\hat{\ell}}{2} \bar{\psi} \gamma \wedge \gamma \wedge^* \Omega \psi - \hat{m} \bar{\psi} \psi \eta, \end{aligned}$	$\int_G \pi \hat{L}_D = \frac{i}{2} \left(\bar{\psi} \gamma \wedge^* D^{g+\omega} \psi + \overline{D^{g+\omega} \psi} \wedge^* \gamma \psi \right) + \frac{\hat{\ell}}{2} \bar{\psi} \gamma \wedge \gamma \wedge^* \Omega \psi - \hat{m} \bar{\psi} \psi \eta,$	$\int_G \pi \hat{L}_D = \frac{i}{2} \left(\bar{\psi} \gamma \wedge^* D^{g+\omega} \psi + \overline{D^{g+\omega} \psi} \wedge^* \gamma \psi \right) + \frac{\hat{\ell}}{2} \bar{\psi} \gamma \wedge \gamma \wedge^* \Omega \psi - \hat{m} \bar{\psi} \psi \eta,$
mielke-geometrodynamics_00844 (12.2.12)	273	0.831 N +0	$\mathrm{L} = \frac{1}{2\ell^2} \mathrm{Tr}(\Omega \wedge^* \sigma) + \mathrm{L}_D = \frac{1}{2\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} + \mathrm{L}_D,$	$\mathrm{L} = \frac{1}{2\ell^2} \mathrm{Tr}(\Omega \wedge^* \sigma) + \mathrm{L}_D = \frac{1}{2\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} + \mathrm{L}_D,$	$\mathrm{L} = \frac{1}{2\ell^2} \mathrm{Tr}(\Omega \wedge^* \sigma) + \mathrm{L}_D = \frac{1}{2\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} + \mathrm{L}_D,$
mielke-geometrodynamics_008218 (4.2.10)	086	0.836	$\begin{aligned} & \omega^L \rightarrow A^{-1} \omega^L = G^{-1} \omega^L G - G^{-1} dG, \\ & \omega^T \rightarrow A^{-1} \omega^T = G^{-1} \omega^T - G^{-1} DT. \end{aligned}$	$\omega^L \rightarrow A^{-1} \omega^L = G^{-1} \omega^L G - G^{-1} dG, \\ \omega^T \rightarrow A^{-1} \omega^T = G^{-1} \omega^T - G^{-1} DT.$	$\omega^L \rightarrow A^{-1} \omega^L = G^{-1} \omega^L G - G^{-1} dG, \\ \omega^T \rightarrow A^{-1} \omega^T = G^{-1} \omega^T - G^{-1} DT.$
mielke-geometrodynamics_008248 (4.3.5)	090	0.836 S -1	$\begin{aligned} & F(m, f\vartheta) = f F(m, \vartheta), \\ & F(m, \vartheta) > 0 \quad \text{if } \vartheta \neq 0. \end{aligned}$	$F(m, f\vartheta) = f F(m, \vartheta), \\ F(m, \vartheta) > 0 \quad \text{if } \vartheta \neq 0.$	$F(m, f\vartheta) = f F(m, \vartheta), \\ F(m, \vartheta) > 0 \quad \text{if } \vartheta \neq 0.$
mielke-geometrodynamics_008445 (6.9.1)	144	0.837 N +0	$ds^2 = - \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2} \right) dt^2 + \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2} \right)^{-1} dr^2 + r^2 d\Omega^2,$	$ds^2 = - \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2} \right) dt^2 + \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2} \right)^{-1} dr^2 + r^2 d\Omega^2,$	$ds^2 = - \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2} \right) dt^2 + \left(1 - \frac{2M}{r} + \frac{Q^2}{r^2} \right)^{-1} dr^2 + r^2 d\Omega^2,$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_EQ0227 (4.2.20)	088	■ 0.839 ● N +1	$\widetilde{\vartheta}\rightarrow^{A^{-1}}\widetilde{\vartheta}=G^{-1}\widetilde{\vartheta}.$ $\widetilde{\vartheta}\rightarrow^{A^{-1}}\widetilde{\vartheta}=G^{-1}\widetilde{\vartheta}.$		
mielke-geometrodynamics_EQ0773 (11.2.2)	245	■ 0.840 ● N +0	$i\,\gamma\wedge^*D\psi-\frac{i}{2}(D\wedge^*\gamma)\psi-m\psi\eta=0$		
mielke-geometrodynamics_EQ0811 (11.4.23)	256	■ 0.840 ● N +0	$\frac{3}{8}\ell^2\bar{\psi}\gamma^5\psi=4\pi me^{-\mu-v/2}\frac{2}{3}HF\left Y_{ \kappa -1}^{ \kappa -1}(\vartheta,\phi)\right ^2$		
mielke-geometrodynamics_EQ0584 (9.1.2)	192	■ 0.843 ● N +0	$\begin{aligned} e^{\{\nu\}}&=e^{\{-\lambda\}}\,\&=1-\left(\frac{2}{r}\right)^{n-3}-\frac{2\Lambda_{\text{eff.}}}{(n-1)(n-2)}r^2\\ \mu&=\frac{M}{m^{*2}};\quad d\Omega^2:=d\vartheta^2+\sin^2\vartheta d\phi^2 \end{aligned}$		
mielke-geometrodynamics_EQ0349 (5.4.7)	117	■ 0.845 ● N +0	$\left\langle d j_5\right\rangle=2 i m\left\langle\bar{\psi} \gamma_5 \psi\right\rangle \eta-\frac{1}{96 \pi^2}\left[2 R_{\alpha \beta}^{\{\}} \wedge R^{\{\beta \alpha \beta}+d \mathscr{A} \wedge d \mathscr{A}\right]$		
mielke-geometrodynamics_EQ0792 (11.4.4)	253	■ 0.845 ● N +0	$\begin{aligned} ds^2&=g_{kl}dx^k\otimes_sdx^l\\&=e^{\nu}dt^2-e^{\mu}\left(dr^2+r^2d\vartheta^2+r^2\sin^2\vartheta d\phi^2\right). \end{aligned}$		
mielke-geometrodynamics_EQ0742 (11.1.9)	238	■ 0.846 ● W +1	$\Lambda:\left\{\begin{array}{cl} \widetilde{SO}_{\circ}(s,n-s) & \longrightarrow SO_{\circ}(s,n-s), \\ \Psi & \longrightarrow g^{\alpha}_{\beta}=-\frac{i}{N}\text{Tr}\left(\gamma^{\alpha}\text{Si}\,\gamma_{\beta}S^{-1}\right), \end{array}\right.$		
mielke-geometrodynamics_EQ0897 (13.5.2)	288	■ 0.846 ●	$\begin{aligned} \hat{R} \,\&:=R_{\{A\}}^{\{ \}^{\{B\}}}\,E^{\{A\}}_{\{ \}_{\{B\}}}:=\left[d\,\Gamma_{\{A\}}^{\{ \}^{\{B\}}}-\Gamma_{\{A\}}^{\{ \}^{\{C\}}}\wedge\Gamma_{\{C\}}^{\{ \}^{\{B\}}}\right]E^{\{A\}}_{\{ \}_{\{B\}}}\,\&= \left[R_{\alpha}^{\beta}-\frac{\Lambda}{3}\theta_{\alpha}\wedge\vartheta^{\beta}\right]L^{\alpha}_{\beta}+T^{\beta}P_{\beta}+D\theta_{\alpha}P^{\alpha}_{*} \end{aligned}$		
mielke-geometrodynamics_EQ0296 (4.5.7)	101	■ 0.847 ● K +0	$F_{\alpha}=(e_{\alpha}\lrcorner D\Psi)\frac{\delta L}{\delta\Psi}+(-1)^p(e_{\alpha}\lrcorner\Psi)\wedge D\frac{\delta L}{\delta\Psi}$		

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometroynamics_091 EQ0241 (4.3.6)		0.848 N +0	$g_{\alpha\beta}(m,\vartheta):=\frac{1}{2}\frac{\partial^2F^2(m,\vartheta)}{\partial\vartheta^\alpha\otimes_s\partial\vartheta^\beta}.$	$g_{\alpha\beta}(m,\vartheta):=\frac{1}{2}\frac{\partial^2F^2(m,\vartheta)}{\partial\vartheta^\alpha\otimes_s\partial\vartheta^\beta}.$	
mielke-geometroynamics_157 EQ0478 (7.4.6)		0.849 N -1	$\operatorname{Ric}_{\alpha\beta}:=\operatorname{Ric}_{\alpha\beta}-\frac{1}{4}\operatorname{Rg}_{\alpha\beta}=0.$	$\operatorname{Ric}_{\alpha\beta}:=\operatorname{Ric}_{\alpha\beta}-\frac{1}{4}\operatorname{Rg}_{\alpha\beta}=0.$	$R\mathcal{I}_{\alpha\beta}:=\operatorname{Ric}_{\alpha\beta}-\frac{1}{4}Rg_{\alpha\beta}=0.$
mielke-geometroynamics_101 EQ0295 (4.5.6)		0.850 K +0	$\begin{aligned} D\,\Sigma_{\alpha}&\&\left.\left(e_{\alpha}\right\rfloor T^{\beta}\right)\wedge\Sigma_{\beta}+\left(e_{\alpha}\right\rfloor R^{\beta\gamma}\right)\wedge\tau_{\beta\gamma}+F_{\alpha}\\ &\simeq\left(e_{\alpha}\right\rfloor T^{\beta}\right)\wedge\Sigma_{\beta}+\left(e_{\alpha}\right\rfloor R^{\beta\gamma}\right)\wedge\tau_{\beta\gamma}. \end{aligned}$	$D\Sigma_{\alpha}=\left(e_{\alpha}\right\rfloor T^{\beta}\right)\wedge\Sigma_{\beta}+\left(e_{\alpha}\right\rfloor R^{\beta\gamma}\right)\wedge\tau_{\beta\gamma}+F_{\alpha}\\ \simeq\left(e_{\alpha}\right\rfloor T^{\beta}\right)\wedge\Sigma_{\beta}+\left(e_{\alpha}\right\rfloor R^{\beta\gamma}\right)\wedge\tau_{\beta\gamma}.$	$D\Sigma_{\alpha}=\left(e_{\alpha}\right\rfloor T^{\beta}\right)\wedge\Sigma_{\beta}+\left(e_{\alpha}\right\rfloor R^{\beta\gamma}\right)\wedge\tau_{\beta\gamma}+F_{\alpha}\\ \simeq\left(e_{\alpha}\right\rfloor T^{\beta}\right)\wedge\Sigma_{\beta}+\left(e_{\alpha}\right\rfloor R^{\beta\gamma}\right)\wedge\tau_{\beta\gamma}.$
mielke-geometroynamics_272 EQ0842 (12.2.10)		0.852 N +0	$\mu:=e_{\alpha}\downarrow\mu^{\alpha}=\frac{3}{4}^*j_5$	$\mu:=e_{\alpha}\downarrow\mu^{\alpha}=\frac{3}{4}^*j_5$	$\mu:=e_{\alpha}\downarrow\mu^{\alpha}=\frac{3}{4}^*j_5$
mielke-geometroynamics_164 EQ0506 (7.7.24)		0.852 K +0	$\gamma_5:=(i/4!)^*(\gamma\wedge\gamma\wedge\gamma\wedge\gamma)$	$\gamma_5:=(i/4!)^*(\gamma\wedge\gamma\wedge\gamma\wedge\gamma)$	$\gamma_5:=(i/4!)^*(\gamma\wedge\gamma\wedge\gamma\wedge\gamma)$
mielke-geometroynamics_326 EQ1014 (14.4.9)		0.856 N +0	$M^{\{2\}}=\left(M_{\mathrm{ir}}+\frac{\alpha_eM^{*2}}{4M_{\mathrm{ir}}}\mathrm{Tr}\,\bar{Q}^2\right)^2+\frac{1}{4}J(J+1)\frac{M^{*4}}{M_{\mathrm{ir}}^2},$	$M^2=\left(M_{\mathrm{ir}}+\frac{\alpha_eM^{*2}}{4M_{\mathrm{ir}}}\mathrm{Tr}\,\bar{Q}^2\right)^2+\frac{1}{4}J(J+1)\frac{M^{*4}}{M_{\mathrm{ir}}^2},$	$M^2=\left(M_{\mathrm{ir}}+\frac{\alpha_eM^{*2}}{4M_{\mathrm{ir}}}\mathrm{Tr}\,\bar{Q}^2\right)^2+\frac{1}{4}J(J+1)\frac{M^{*4}}{M_{\mathrm{ir}}^2},$
mielke-geometroynamics_241 EQ0753 (11.1.20)		0.859 N +0	$\widehat{\Psi}_{ \widehat{SO}(1,3)}=\left[\begin{array}{c}\Psi_1\\\Psi_2\\\vdots\\\Psi_{\left[\frac{n}{4}\right]}\end{array}\right].$	$\hat{\Psi}_{ \widehat{SO}(1,3)}=\left[\begin{array}{c}\Psi_1\\\Psi_2\\\vdots\\\Psi_{\left[\frac{n}{4}\right]}\end{array}\right].$	$\hat{\Psi}_{ \widehat{SO}(1,3)}=\left[\begin{array}{c}\Psi_1\\\Psi_2\\\vdots\\\Psi_{\left[\frac{n}{4}\right]}\end{array}\right].$
mielke-geometroynamics_247 EQ0783 (11.3.2)		0.862	$\begin{aligned} L_{\mathrm{D}}&\&\frac{i}{2}\left(\overline{\psi}\gamma\wedge^*D^{\{\}}\psi+\left(\overline{D^{\{\}}\psi}\right)\wedge^*\gamma\psi\right)-m\overline{\psi}\psi\eta\\ &+\frac{1}{2}\overline{\psi}\left[\widetilde{K},^*\gamma\right]\psi=:L_{\mathrm{D}}^{\{\}}+L_{\mathrm{NL}}. \end{aligned}$	$L_{\mathrm{D}}=\frac{i}{2}\left(\overline{\psi}\gamma\wedge^*D^{\{\}}\psi+\left(\overline{D^{\{\}}\psi}\right)\wedge^*\gamma\psi\right)-m\overline{\psi}\psi\eta\\ +\frac{1}{2}\overline{\psi}\left[\widetilde{K},^*\gamma\right]\psi=:L_{\mathrm{D}}^{\{\}}+L_{\mathrm{NL}}.$	$L_{\mathrm{D}}=\frac{i}{2}\left(\overline{\psi}\gamma\wedge^*D^{\{\}}\psi+\left(\overline{D^{\{\}}\psi}\right)\wedge^*\gamma\psi\right)-m\overline{\psi}\psi\eta\\ +\frac{1}{2}\overline{\psi}\left[\widetilde{K},^*\gamma\right]\psi=:L_{\mathrm{D}}^{\{\}}+L_{\mathrm{NL}}.$
mielke-geometroynamics_369 EQ1100 (B.12)		0.864 N +0	$\begin{aligned} \{ \}^{*} \alpha^{\{ (p) \}} &:= \frac{1}{(n-p)! \cdot (p)!} \sqrt{ \det g_{ij} } \varepsilon_{\alpha_1 \cdots \alpha_p \beta_1 \cdots \beta_{n-p}} \\ &\quad \times A^{\alpha_1 \cdots \alpha_p} \vartheta^{\beta_1} \wedge \cdots \wedge \vartheta^{\beta_{n-p}}. \end{aligned}$	$^*\alpha^{(p)}:=\frac{1}{(n-p)!\cdot(p!)}\sqrt{ \det g_{ij} }\varepsilon_{\alpha_1\cdots\alpha_p\beta_1\cdots\beta_{n-p}}\\ \times A^{\alpha_1\cdots\alpha_p}\vartheta^{\beta_1}\wedge\cdots\wedge\vartheta^{\beta_{n-p}}.$	$^*\alpha^{(p)}:=\frac{1}{(n-p)!\cdot(p!)}\sqrt{ \det g_{ij} }\varepsilon_{\alpha_1\cdots\alpha_p\beta_1\cdots\beta_{n-p}}\\ \times A^{\alpha_1\cdots\alpha_p}\vartheta^{\beta_1}\wedge\cdots\wedge\vartheta^{\beta_{n-p}}.$

Conf. MathPix confidence:

■

≥0.9

■

0.5–0.9

■

<0.5

—

no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E00184 (3.7.14)	074	■0.864 ●C +16	\vec{\Omega}=d\,\vec{\omega}-({\overrightarrow{\omega}}\wedge	$\vec{\Omega} = d\vec{\omega} - (\vec{\omega}\wedge$	$\vec{\Omega} = d\vec{\omega} - (\overrightarrow{\omega \wedge \vec{\omega}}) = d\vec{\omega} - \vec{\omega} \wedge \vec{\omega}.$
mielke-geometrodynamics_E00165 (3.5.13)	069	■0.865 ●N +0	{ }^{*} \Omega=(\stackrel{+}{-}) \Omega ;	$*\Omega = (\overset{+}{-})\Omega;$	$*\Omega = (\overset{+}{-})\Omega ;$
mielke-geometrodynamics_E00737 (11.1.4)	238	■0.867 ●N +0	x^{\alpha} \rightarrowtail x^{\prime \alpha}=g_{\beta}^{\alpha}\left[g_{\beta}^{\alpha}\right] \in S O(s, n-s), \quad x^{\alpha} \in \mathbf{E}^n .	$x^\alpha \rightarrow x'^\alpha = g_\beta^\alpha \quad x^\beta, [g_\beta^\alpha] \in SO(s,n-s), \quad x^\alpha \in \mathbf{E}^n.$	$x^\alpha \rightarrow x'^\alpha = g_\beta^\alpha \; x^\beta, [g_\beta^\alpha] \in SO(s,n-s), \; x^\alpha \in \mathbf{E}^n.$
mielke-geometrodynamics_E00853 (12.3.6)	276	■0.868 ●C +3	\left\langle\vartheta \wedge \sigma_{\alpha}\right\rangle=-\frac{1}{3 \pi^2}\left[\mathrm{Tr}(G \wedge * G)+\frac{1}{24}\left(2 R^{\alpha \beta} \wedge R_{\alpha \beta}^{(j *)}+d \mathcal{A} \wedge * d \mathcal{A}\right)\right],	$\langle\vartheta^\alpha \wedge \sigma_\alpha\rangle = -\frac{1}{3\pi^2} \left[\text{Tr}(G \wedge {}^*G) + \frac{1}{24} \left(2R^{\alpha\beta}{}_{\{\} \wedge} R_{\alpha\beta}^{\{j(*)} + d\mathscr{A} \wedge {}^*d\mathscr{A} \right) \right],$	$\langle\vartheta^\alpha \wedge \sigma_\alpha\rangle = -\frac{1}{3\pi^2} \left[Tr(G \wedge {}^*G) + \frac{1}{24} \left(2R^{\alpha\beta}{}_{\{\} \wedge} R_{\alpha\beta}^{\{j(*)} + d\mathscr{A} \wedge {}^*d\mathscr{A} \right) \right],$
mielke-geometrodynamics_E00774 (11.2.3)	245	■0.868 ●N +0	i(\overline{D \psi}) \wedge^* \gamma+\frac{i}{2} \bar{\psi} D \wedge^* \gamma+m \bar{\psi} \eta=0,	$i(\overline{D\psi}) \wedge^*\gamma + \frac{i}{2}\bar{\psi}D \wedge^*\gamma + m\bar{\psi}\eta = 0,$	$i(\overline{D\psi}) \wedge^*\gamma + \frac{i}{2}\bar{\psi}D \wedge^*\gamma + m\bar{\psi}\eta = 0,$
mielke-geometrodynamics_E00548 (8.4.2)	178	■0.871 ●N -1	s L_{\text {Pontr }}=-2 d \operatorname{Tr} \backslash \left.\Psi\right\rfloor \wedge \Omega^g\}= -d\left(\Psi_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}\right) .	$sL_{\text{Pontr}} = -2d\operatorname{Tr}\{\Psi \wedge \Omega^g\} = -d(\Psi_\alpha{}^\beta \wedge R_\beta{}^\alpha).$	$sL_{\text{Pontr}} = -2d\operatorname{Tr}\{\Psi \wedge \Omega^g\} = -d(\Psi_\alpha{}^\beta \wedge R_\beta{}^\alpha).$
mielke-geometrodynamics_E00244 (4.3.9)	091	■0.873 ●N +1	g_{ij} e_{\alpha}^i e_{\beta}^j=g_{\alpha \beta} .	$g_{ij}e_\alpha^ie_\beta^j = g_{\alpha\beta}.$	$g_{ij}e_\alpha^ie_\beta^j = g_{\alpha\beta}.$
mielke-geometrodynamics_E00890 (13.4.9)	287	■0.876 ●N +0	\hat{x}_i=\binom{x_i}{1}=\binom{x_i}{1} P_{\alpha} \\ \hat{x}_{i *}=\left(x_{i *}, 1\right)=\left(x_{\beta *}^* P_{*}^{\beta}, 1\right)	$\hat\xi = \begin{pmatrix} \xi \\ 1 \end{pmatrix} = \begin{pmatrix} \xi^\alpha P_\alpha \\ 1 \end{pmatrix}, \quad \hat\xi_* = (\xi_*, 1) = (\xi_\beta^* P_*^\beta, 1)$	$\hat\xi = \begin{pmatrix} \xi \\ 1 \end{pmatrix} = \begin{pmatrix} \xi^\alpha P_\alpha \\ 1 \end{pmatrix}, \quad \hat\xi_* = (\xi_*, 1) = (\xi_\beta^* P_*^\beta, 1)$
mielke-geometrodynamics_E01007 (14.4.2)	325	■0.877 ●N +1	\begin{aligned} A^{\left(\circ\right)} &= \sigma^{\omega^{\left(\circ\right)}} =-\frac{r^2}{\rho^2}\left[d t-J_3 \frac{\hbar}{M c} \sin ^2 \vartheta d \phi\right] Q^{\left(\circ\right)} \cos \delta \\ &\quad -\frac{\cos \vartheta}{\rho^2}\left[\left(r^2+J(J+1) \frac{\hbar^2}{M^2 c^2}\right) d \phi-J_3 \frac{\hbar}{M} d t\right] P^{\left(\circ\right)} \sin \delta . \end{aligned}	$A^{(\circ)} = \overset{*}{\sigma} \omega^{(\circ)} = -\frac{r^2}{\rho^2} \left[dt - J_3 \frac{\hbar}{Mc} \sin^2 \vartheta d\phi \right] Q^{(\circ)} \cos \delta$ $- \frac{\cos \vartheta}{\rho^2} \left[\left(r^2 + J(J+1) \frac{\hbar^2}{M^2c^2} \right) d\phi - J_3 \frac{\hbar}{M} dt \right] P^{(\circ)} \sin \delta$	$A^{(\circ)} = \overset{*}{\sigma} \omega^{(\circ)} = -\frac{r^2}{\rho^2} \left[dt - J_3 \frac{\hbar}{Mc} \sin^2 \vartheta d\phi \right] Q^{(\circ)} \cos \delta$ $- \frac{\cos \vartheta}{\rho^2} \left[\left(r^2 + J(J+1) \frac{\hbar^2}{M^2c^2} \right) d\phi - J_3 \frac{\hbar}{M} dt \right] P^{(\circ)} \sin \delta.$
mielke-geometrodynamics_E01062 (16.2.1)	355	■0.877 ●	\left[\square-(\mathrm{mc}/\hbar)^2\right] \varphi=0	$[\square - (mc/\hbar)^2] \varphi = 0$	$[\square - (mc/\hbar)^2] \varphi = 0$
mielke-geometrodynamics_E00051 (2.7.1)	044	■0.881 ●N -2	G(p): P \rightarrowtail P, \quad G(p) \in \mathscr{G}_p	$G(p) : P \rightarrow P, \quad G(p) \in \mathscr{G}_p$	$G(p) : P \rightarrow P, \quad G(p) \in \mathscr{G}_p$
mielke-geometrodynamics_E00178 (3.7.8)	072	■0.881 ●K +0	\left \left\langle O\right ^{\lambda} \varphi\right O\rangle=\mu / \sqrt{\lambda}	$\left\langle O\right ^{\lambda} \varphi\mid O\rangle = \mu/\sqrt{\lambda}$	$\langle O ^{\lambda} \varphi O \rangle = \mu/\sqrt{\lambda}$
mielke-geometrodynamics_E00028 (2.5.1)	040	■0.885 ●N +0	V_{p}^{*}=T_{p}^{*}\left(F_m(M)\right),	$V_p^* = T_p^*(F_m(M)),$	$V_p^* = T_p^*(F_m(M)),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					
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mielke-geometrodynamics_295 EQ0920 (13.8.6)		■ 0.888 ● C -3	$\begin{aligned} d\,C_{\mathrm{RR}(\star)} &:= \frac{1}{2} d\left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(\star)} - \frac{1}{3} \Gamma_{\alpha}{}^{\beta(\star)} \wedge \Gamma_{\beta}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\alpha}\right) \\ &\equiv -L_{\mathrm{SKY}} - 2\,\mathrm{Ric}_{\alpha\beta} \wedge {}^*\mathrm{Ric}^{\alpha\beta} + \frac{1}{2} \mathrm{Ric}_{\alpha}{}^{\alpha} \wedge {}^*\mathrm{Ric}_{\beta}{}^{\beta} \end{aligned}$	$dC_{\mathrm{RR}(\star)} := \frac{1}{2} d\left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(\star)} - \frac{1}{3} \Gamma_{\alpha}{}^{\beta(\star)} \wedge \Gamma_{\beta}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\alpha}\right) \\ \equiv -L_{\mathrm{SKY}} - 2\,\mathrm{Ric}_{\alpha\beta} \wedge {}^*\mathrm{Ric}^{\alpha\beta} + \frac{1}{2} \mathrm{Ric}_{\alpha}{}^{\alpha} \wedge {}^*\mathrm{Ric}_{\beta}{}^{\beta}$	$dC_{\mathrm{RR}(\star)} := \frac{1}{2} d\left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(\star)} - \frac{1}{3} \Gamma_{\alpha}{}^{\beta(\star)} \wedge \Gamma_{\beta}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\alpha}\right) \\ \equiv -L_{\mathrm{SKY}} - 2\,\mathrm{Ric}_{\alpha\beta} \wedge {}^*\mathrm{Ric}^{\alpha\beta} + \frac{1}{2} \mathrm{Ric}_{\alpha}{}^{\alpha} \wedge {}^*\mathrm{Ric}_{\beta}{}^{\beta}$
mielke-geometrodynamics_220 EQ0674 (10.3.31)		■ 0.890 ● C -5	$\check{\mathrm{E}}_{\alpha} = -\frac{\theta_{\mathrm{T}}}{\theta_{\mathrm{TL}}} \frac{\theta_{\mathrm{TL}}}{2\ell^2} \left(\frac{1}{\theta_{\mathrm{TL}}} d\vartheta_{\alpha} - \frac{4\ell}{\theta_{\mathrm{T}}\theta_{\mathrm{L}}} \check{\mathcal{J}}_{\alpha}^{\star} \right),$	$\check{\mathcal{E}}_{\alpha} = -\frac{\theta_{\mathrm{T}}\theta_{\mathrm{TL}}}{2\ell^2} \left(\frac{1}{\theta_{\mathrm{TL}}} d\vartheta_{\alpha} - \frac{4\ell}{\theta_{\mathrm{T}}\theta_{\mathrm{L}}} \check{\mathcal{J}}_{\alpha}^{\star} \right),$	$\check{\mathcal{E}}_{\alpha} = -\frac{\theta_{\mathrm{T}}\theta_{\mathrm{TL}}}{2\ell^2} \left(\frac{1}{\theta_{\mathrm{TL}}} d\vartheta_{\alpha} - \frac{4\ell}{\theta_{\mathrm{T}}\theta_{\mathrm{L}}} \check{\mathcal{J}}_{\alpha}^{\star} \right),$
mielke-geometrodynamics_169 EQ0531 (7.7.51)		■ 0.890 ● N +0	$\left. \left. \left. R := e_{\beta} \rfloor e_{\alpha} \rfloor R^{\alpha\beta}, \quad {}^*R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta} \equiv R^{\alpha\beta} \wedge \eta_{\alpha\beta} \equiv -R\eta, \right. \right. \quad \left. \left. R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta} \equiv R^{\alpha\beta} \wedge \eta_{\alpha\beta} \equiv -R\eta, \right. \right.$	$R := e_{\beta} \rfloor e_{\alpha} \rfloor R^{\alpha\beta}, \quad {}^*R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta} \equiv R^{\alpha\beta} \wedge \eta_{\alpha\beta} \equiv -R\eta,$	$R := e_{\beta} \rfloor e_{\alpha} \rfloor R^{\alpha\beta}, \quad {}^*R^{\alpha\beta} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta} \equiv R^{\alpha\beta} \wedge \eta_{\alpha\beta} \equiv -R\eta,$
mielke-geometrodynamics_176 EQ0540 (8.3.1)		■ 0.891 ● W +2	$\begin{aligned} (d \oplus s) \gamma + [\tilde{\Gamma}, \gamma] &= \tilde{\Theta} \\ (d \oplus s) \tilde{\Theta} + [\tilde{\Gamma}, \tilde{\Theta}] &\equiv [\tilde{\Omega}^g, \gamma] \end{aligned}$	$\begin{aligned} (d \oplus s) \gamma + [\tilde{\Gamma}, \gamma] &= \tilde{\Theta} \\ (d \oplus s) \tilde{\Theta} + [\tilde{\Gamma}, \tilde{\Theta}] &\equiv [\tilde{\Omega}^g, \gamma] \end{aligned}$	$\begin{aligned} (d \oplus s) \gamma + [\tilde{\Gamma}, \gamma] &= \tilde{\Theta}, \\ (d \oplus s) \tilde{\Theta} + [\tilde{\Gamma}, \tilde{\Theta}] &\equiv [\tilde{\Omega}^g, \gamma], \end{aligned}$
mielke-geometrodynamics_303 EQ0925 (14.1.5)		■ 0.892 ● N +0	$\begin{aligned} D^* \Omega &= \tau_e, \\ D\Omega &\equiv 0, \\ \frac{1}{2} * (\vartheta \wedge {}^* \Omega^g) &= \ell^{*2} (\Sigma_{\mathrm{YM}} + \Sigma_{\mathrm{mat}}), \end{aligned}$	$\begin{aligned} D^* \Omega &= \tau_e, \\ D\Omega &\equiv 0, \\ \frac{1}{2} * (\vartheta \wedge {}^* \Omega^g) &= \ell^{*2} (\Sigma_{\mathrm{YM}} + \Sigma_{\mathrm{mat}}), \end{aligned}$	$\begin{aligned} D^* \Omega &= \tau_e, \\ D\Omega &\equiv 0, \\ \frac{1}{2} * (\vartheta \wedge {}^* \Omega^g) &= \ell^{*2} (\Sigma_{\mathrm{YM}} + \Sigma_{\mathrm{mat}}), \end{aligned}$
mielke-geometrodynamics_283 EQ0867 (13.2.3)		■ 0.893 ● N +0	$B \cong dA, \quad dB \cong dF \equiv 0,$	$B \cong dA, \quad dB \cong dF \equiv 0,$	$B \cong dA, \quad dB \cong dF \equiv 0,$
mielke-geometrodynamics_201 EQ0609 (9.2.11)		■ 0.894 ● N +0	$\begin{aligned} &\left((1 + \zeta^2) \int_{M^4} L_{\mathrm{SKY}} - (-1)^s \zeta (4\pi)^2 \cdot \chi(M^4) \right) \\ &= \frac{1}{4} \int_{M^4} \left(\Omega^g - \zeta^* \Omega^{g(\star)} \right) \wedge {}^* \left(\Omega^g - \zeta^* \Omega^{g(\star)} \right) \geq 0, \end{aligned}$	$\begin{aligned} &\left((1 + \zeta^2) \int_{M^4} L_{\mathrm{SKY}} - (-1)^s \zeta (4\pi)^2 \cdot \chi(M^4) \right) \\ &= \frac{1}{4} \int_{M^4} \left(\Omega^g - \zeta^* \Omega^{g(\star)} \right) \wedge {}^* \left(\Omega^g - \zeta^* \Omega^{g(\star)} \right) \geq 0, \end{aligned}$	$\begin{aligned} &(1 + \zeta^2) \int_{M^4} L_{\mathrm{SKY}} - (-1)^s \zeta (4\pi)^2 \cdot \chi(M^4) \\ &= \frac{1}{4} \int_{M^4} (\Omega^g - \zeta^* \Omega^{g(\star)}) \wedge {}^*(\Omega^g - \zeta^* \Omega^{g(\star)}) \geq 0, \end{aligned}$
mielke-geometrodynamics_309 EQ0959 (14.1.41)		■ 0.894 ●	$\oint d\delta = 2\pi b_1 = 2\pi v$	$\oint d\delta = 2\pi b_1 = 2\pi v$	$\oint d\delta = 2\pi b_1 = 2\pi v$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_090299 (4.5.10)	102	0.896 N +1	$\begin{aligned} & \& \delta g_{\alpha\beta} = 2\varepsilon_{(\alpha}{}^{\gamma} g_{\beta)\gamma}, \quad \delta\vartheta^{\alpha} = -\vartheta^{\beta}\varepsilon_{\beta}{}^{\alpha}, \\ & \delta\Gamma_{\alpha}{}^{\beta} = D\varepsilon_{\alpha}{}^{\beta}, \quad \delta\Psi = -\varepsilon_{\alpha}{}^{\beta}I^{\alpha}{}_{\beta}\Psi. \end{aligned}$	$\delta g_{\alpha\beta} = 2\varepsilon_{(\alpha}{}^{\gamma} g_{\beta)\gamma}, \quad \delta\vartheta^{\alpha} = -\vartheta^{\beta}\varepsilon_{\beta}{}^{\alpha},$ $\delta\Gamma_{\alpha}{}^{\beta} = D\varepsilon_{\alpha}{}^{\beta}, \quad \delta\Psi = -\varepsilon_{\alpha}{}^{\beta}I^{\alpha}{}_{\beta}\Psi.$	$\delta g_{\alpha\beta} = 2\varepsilon_{(\alpha}{}^{\gamma} g_{\beta)\gamma}, \quad \delta\vartheta^{\alpha} = -\vartheta^{\beta}\varepsilon_{\beta}{}^{\alpha},$ $\delta\Gamma_{\alpha}{}^{\beta} = D\varepsilon_{\alpha}{}^{\beta}, \quad \delta\Psi = -\varepsilon_{\alpha}{}^{\beta}I^{\alpha}{}_{\beta}\Psi.$
mielke-geometrodynamics_090236 (4.3.1)	090	0.896 N +0	$\begin{aligned} & (\Delta s)^2 = -c^2(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 \\ & = g_{ij}\Delta x^i\Delta x^j. \end{aligned}$	$(\Delta s)^2 = -c^2(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$ $= g_{ij}\Delta x^i\Delta x^j.$	$(\Delta s)^2 = -c^2(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$ $= g_{ij}\Delta x^i\Delta x^j.$
mielke-geometrodynamics_0901005 (14.3.18)	323	0.896 N -1	$\begin{aligned} & \not{D}^2 \& = \operatorname{Tr} \left(\frac{1}{2} \gamma \wedge \gamma \wedge {}^* \{D, D\} + \frac{1}{2} \gamma \wedge \gamma \wedge {}^* \tilde{\Omega} - \gamma \wedge \tilde{\Theta} \wedge {}^* D \right) \\ & = \left(\Box^g + \frac{1}{4} R \right) \eta - \vartheta \wedge \Theta \wedge {}^* D \end{aligned}$	$D^2 = \operatorname{Tr} \left(\frac{1}{2} \gamma \wedge \gamma \wedge {}^* \{D, D\} + \frac{1}{2} \gamma \wedge \gamma \wedge {}^* \tilde{\Omega} - \gamma \wedge \tilde{\Theta} \wedge {}^* D \right)$ $= \left(\Box^g + \frac{1}{4} R \right) \eta - \vartheta \wedge \Theta \wedge {}^* D$	$D^2 = \operatorname{Tr} \left(\frac{1}{2} \gamma \wedge \gamma \wedge {}^* \{D, D\} + \frac{1}{2} \gamma \wedge \gamma \wedge {}^* \tilde{\Omega} - \gamma \wedge \tilde{\Theta} \wedge {}^* D \right)$ $= (\Box^g + \frac{1}{4} R) \eta - \vartheta \wedge \Theta \wedge {}^* D$
mielke-geometrodynamics_090230 (4.2.23)	088	0.899 N +0	$\stackrel{*}{\sigma} \Theta = \frac{1}{2} T_{\alpha\beta}^{\cdot c} P_c \otimes \vartheta^\alpha \wedge \vartheta^\beta = \frac{1}{2} T_{ij}^{\cdot c} P_c \otimes dx^i \wedge dx^j,$	$\overset{*}{\sigma} \Theta = \frac{1}{2} T_{\alpha\beta}^{\cdot c} P_c \otimes \vartheta^\alpha \wedge \vartheta^\beta = \frac{1}{2} T_{ij}^{\cdot c} P_c \otimes dx^i \wedge dx^j,$	$\overset{*}{\sigma} \Theta = \frac{1}{2} T_{\alpha\beta}^{\cdot c} P_c \otimes \vartheta^\alpha \wedge \vartheta^\beta = \frac{1}{2} T_{ij}^{\cdot c} P_c \otimes dx^i \wedge dx^j,$
mielke-geometrodynamics_090723 (10.10.3)	230	0.900 C +4	$\begin{aligned} & \delta_{\text{susy}} \vartheta^\alpha = i\bar{\sigma} \Psi \gamma^\alpha, \quad \delta_{\text{susy}} \Gamma_\alpha^* = i\bar{\sigma} \gamma_\alpha^* D\Psi + i c \bar{\sigma} (\gamma_\alpha \Psi + e_\alpha]{}^* \Psi) \\ & \delta_{\text{susy}} \Psi = 2D\sigma + c\gamma\sigma, \end{aligned}$	$\delta_{\text{susy}} \vartheta^\alpha = i\bar{\sigma} \Psi \gamma^\alpha, \quad \delta_{\text{susy}} \Gamma_\alpha^* = i\bar{\sigma} \gamma_\alpha^* D\Psi + i c \bar{\sigma} (\gamma_\alpha \Psi + e_\alpha]{}^* \Psi)$ $\delta_{\text{susy}} \Psi = 2D\sigma + c\gamma\sigma,$	$\delta_{\text{susy}} \vartheta^\alpha = i\bar{\sigma} \Psi \gamma^\alpha, \quad \delta_{\text{susy}} \Gamma_\alpha^* = i\bar{\sigma} \gamma_\alpha^* D\Psi + i c \bar{\sigma} (\gamma_\alpha \Psi + e_\alpha]{}^* \Psi),$ $\delta_{\text{susy}} \Psi = 2D\sigma + c\gamma\sigma, \tag{1}$
mielke-geometrodynamics_090216 (4.2.8)	086	0.902 N +0	$\widetilde{\omega} \rightarrow^{A^{-1}} \widetilde{\omega} = A^{-1} \widetilde{\omega} A - A^{-1} dA, \quad A \in \mathscr{A}_p.$	$\widetilde{\omega} \rightarrow^{A^{-1}} \widetilde{\omega} = A^{-1} \widetilde{\omega} A - A^{-1} dA, \quad A \in \mathscr{A}_p.$	$\widetilde{\omega} \rightarrow^{A^{-1}} \widetilde{\omega} = A^{-1} \widetilde{\omega} A - A^{-1} dA, \quad A \in \mathscr{A}_p.$
mielke-geometrodynamics_0901042 (15.1.19)	340	0.902 K +0	$\Gamma_{\mu\alpha\beta} : (2^+) \oplus 2(1^-) \oplus (0^+) \oplus (2^-) \oplus 2(1^+) \oplus (0^-),$	$\Gamma_{\mu\alpha\beta} : (2^+) \oplus 2(1^-) \oplus (0^+) \oplus (2^-) \oplus 2(1^+) \oplus (0^-),$	$\Gamma_{\mu\alpha\beta} : (2^+) \oplus 2(1^-) \oplus (0^+) \oplus (2^-) \oplus 2(1^+) \oplus (0^-),$
mielke-geometrodynamics_090481 (7.5.2)	157	0.903 W -1	$\square^c g_{ij} = \square^c \left(\Omega^2 o_{ij}^{\text{Mink}} \right) = \Omega^6 \square^c o_{ij}^{\text{Mink}} = 0$	$\square^c g_{ij} = \square^c \left(\Omega^2 o_{ij}^{\text{Mink}} \right) = \Omega^6 \square^c o_{ij}^{\text{Mink}} = 0$	$\square^c g_{ij} = \square^c (\Omega^2 o_{ij}^{\text{Mink}}) = \Omega^6 \square^c o_{ij}^{\text{Mink}} = 0$
mielke-geometrodynamics_090117 (3.3.18)	059	0.905 N -1	$L_{\mathrm{Max}} := -\frac{1}{2} F \wedge {}^* F = \frac{1}{2} (\mathbf{E}^2 - \mathbf{B}^2) \eta,$	$L_{\mathrm{Max}} := -\frac{1}{2} F \wedge {}^* F = \frac{1}{2} (\mathbf{E}^2 - \mathbf{B}^2) \eta,$	$L_{\mathrm{Max}} := -\frac{1}{2} F \wedge {}^* F = \frac{1}{2} (\mathbf{E}^2 - \mathbf{B}^2) \eta,$
mielke-geometrodynamics_0901110 (B.22)	371	0.905 K +0	$\square \alpha = 0,$	$\square \alpha = 0,$	$\square \alpha = 0,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_EQ0831 (12.1.3)	270	■ 0.907 ●	$\tilde{j}_5:=j_5+\left(1\,/\,96\,\pi^2\right)A\wedge dA$	$\tilde{j}_5:=j_5+\left(1/96\pi^2\right)A\wedge dA$	$\tilde{j}_5:=j_5+(1/96\pi^2)A\wedge dA$
mielke-geometrodynamics_EQ0409 (6.4.29)	136	■ 0.907 ● C +5	$T^{\mathrm{A}}=\frac{1}{f}\left(d f\wedge\vartheta^A-2\eta^{0 A \beta \gamma} \vartheta_{\beta} \wedge \vartheta_{\gamma}\right)=-\frac{2}{f} \eta^A{}_{\mu \nu} \vartheta^{\mu} \wedge \vartheta^{\nu}=\pm * T^A, \quad T^{\hat{0}}=0,$	$T^A=\frac{1}{f}\left(df\wedge\vartheta^A-2\eta^{0A\beta\gamma}\vartheta_\beta\wedge\vartheta_\gamma\right)=-\frac{2}{f}\eta^A{}_{\mu\nu}\vartheta^\mu\wedge\vartheta^\nu=\pm^*T^A,\quad T^{\hat{0}}=0,$	$T^A=\frac{1}{f}\left(df\wedge\vartheta^A-2\eta^{0A\beta\gamma}\vartheta_\beta\wedge\vartheta_\gamma\right)=-\frac{2}{f}\eta^A{}_{\mu\nu}\vartheta^\mu\wedge\vartheta^\nu=\pm^*T^A,\quad T^{\hat{0}}=0,$
mielke-geometrodynamics_EQ0657 (10.3.14)	217	■ 0.907 ● N +0	$\left.E_{\left[\alpha\right]}=e_{\left[\alpha\right]}\right\rfloor L(\star)-\frac{2}{\delta \ell^2} T_{\alpha},$	$E_{\alpha}=e_{\alpha}\rfloor L(\star)-\frac{2\gamma}{\delta\ell^2}T_{\alpha},$	$E_{\alpha}=e_{\alpha}\rfloor L(\star)-\frac{2\gamma}{\delta\ell^2}T_{\alpha},$
mielke-geometrodynamics_EQ0396 (6.4.11)	132	■ 0.909 ● W +1	$\begin{aligned} \mathscr{H} &\cong n^{\alpha} \frac{\delta L}{\delta \vartheta^{\alpha}} + \Gamma_{\perp}^{\alpha \beta} \frac{\delta L}{\delta \Gamma^{\alpha \beta}} \\ &\quad - \underline{d}[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi} + n^{\alpha} \frac{\partial L}{\partial T^{\alpha}} + \Gamma_{\perp}^{\alpha \beta} \frac{\partial L}{\partial R^{\alpha \beta}}] . \end{aligned}$	$\mathcal{H} \cong n^{\alpha} \frac{\delta L}{\delta \vartheta^{\alpha}} + \Gamma_{\perp}^{\alpha \beta} \frac{\delta L}{\delta \Gamma^{\alpha \beta}} \\ \quad - \underline{d}[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi} + n^{\alpha} \frac{\partial L}{\partial T^{\alpha}} + \Gamma_{\perp}^{\alpha \beta} \frac{\partial L}{\partial R^{\alpha \beta}}] .$	$\mathcal{H} \cong n^{\alpha} \frac{\delta L}{\delta \vartheta^{\alpha}} + \Gamma_{\perp}^{\alpha \beta} \frac{\delta L}{\delta \Gamma^{\alpha \beta}} \\ \quad - \underline{d}\Big[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi} + n^{\alpha} \frac{\partial L}{\partial T^{\alpha}} + \Gamma_{\perp}^{\alpha \beta} \frac{\partial L}{\partial R^{\alpha \beta}}\Big] .$
mielke-geometrodynamics_EQ0641 (10.2.12)	213	■ 0.909 ● N +0	$d s^2=-\left(1-\Lambda_{\mathrm{eff}} r^2\right) d t^2+\left(1-\Lambda_{\mathrm{eff}} r^2\right)^{-1} d r^2+r^2 d \phi^2$	$ds^2=-\left(1-\Lambda_{\mathrm{eff}}r^2\right)dt^2+\left(1-\Lambda_{\mathrm{eff}}r^2\right)^{-1}dr^2+r^2d\phi^2$	$ds^2=-(1-\Lambda_{\mathrm{eff}}r^2)dt^2+(1-\Lambda_{\mathrm{eff}}r^2)^{-1}dr^2+r^2d\phi^2$
mielke-geometrodynamics_EQ0263 (4.3.28)	094	■ 0.912 ● N +0	$\Gamma_{ij}^{\cdot k}=e_b^{\kappa}\Gamma_{i\alpha}^{\cdot b}E_j^{\alpha}+e_{\beta}^{\kappa}\partial_iE_j^{\beta}$	$\Gamma_{ij}^{\cdot k}=e_b^{\kappa}\Gamma_{i\alpha}^{\cdot b}E_j^{\alpha}+e_{\beta}^{\kappa}\partial_iE_j^{\beta}$	$\Gamma_{ij}^{\cdot\kappa}=e_b^{\kappa}\Gamma_{i\alpha}^{\cdot b}E_j^{\alpha}+e_{\beta}^{\kappa}\partial_iE_j^{\beta}$
mielke-geometrodynamics_EQ0364 (6.2.16)	126	■ 0.912 ● N +0	$D\,D\,\lambda_{\alpha\beta}=-2\,R_{[\alpha}\,\lambda_{\gamma \beta]}=0$	$DD\lambda_{\alpha\beta}=-2R_{[\alpha]}{}^{\gamma}\wedge\lambda_{\gamma \beta]}=0$	$DD\lambda_{\alpha\beta}=-2R_{[\alpha]}{}^{\gamma}\wedge\lambda_{\gamma \beta]}=0$
mielke-geometrodynamics_EQ0840 (12.2.8)	272	■ 0.916 ● N +1	$\begin{aligned} \tau_{\alpha\beta}&:=\frac{\partial L_{\mathrm{D}}}{\partial\Gamma^{\alpha\beta}}=\frac{1}{8}\bar{\psi}\left(*\gamma\sigma_{\alpha\beta}+\sigma_{\alpha\beta}*\gamma\right)\psi\\ &=\frac{1}{4}\eta_{\alpha\beta\gamma\delta}\bar{\psi}\gamma^{\delta}\gamma_5\psi\eta^{\gamma}=\tau_{\alpha\beta\gamma}\eta^{\gamma}=\vartheta_{[\alpha}\wedge\mu_{\beta]}. \end{aligned}$	$\tau_{\alpha\beta}:=\frac{\partial L_{\mathrm{D}}}{\partial\Gamma^{\alpha\beta}}=\frac{1}{8}\bar{\psi}\left(*\gamma\sigma_{\alpha\beta}+\sigma_{\alpha\beta}*\gamma\right)\psi\\ =\frac{1}{4}\eta_{\alpha\beta\gamma\delta}\bar{\psi}\gamma^{\delta}\gamma_5\psi\eta^{\gamma}=\tau_{\alpha\beta\gamma}\eta^{\gamma}=\vartheta_{[\alpha}\wedge\mu_{\beta]}$	$\tau_{\alpha\beta}:=\frac{\partial L_{\mathrm{D}}}{\partial\Gamma^{\alpha\beta}}=\frac{1}{8}\bar{\psi}\left(*\gamma\sigma_{\alpha\beta}+\sigma_{\alpha\beta}*\gamma\right)\psi\\ =\frac{1}{4}\eta_{\alpha\beta\gamma\delta}\bar{\psi}\gamma^{\delta}\gamma_5\psi\eta^{\gamma}=\tau_{\alpha\beta\gamma}\eta^{\gamma}=\vartheta_{[\alpha}\wedge\mu_{\beta]}.$
mielke-geometrodynamics_EQ0232 (4.2.25)	088	■ 0.917 ● K +0	$\Theta\rightarrow{}^{A^{-1}}\Theta=G^{-1}\Theta,$	$\Theta\rightarrow{}^{A^{-1}}\Theta=G^{-1}\Theta,$	$\Theta\rightarrow{}^{A^{-1}}\Theta=G^{-1}\Theta,$
mielke-geometrodynamics_EQ0577 (8.6.15)	185	■ 0.920 ● N +1	$-E_{\left[\alpha\right]}=\frac{\theta_{\mathrm{T}}^{\ast}}{2}R^{\left[\beta\gamma\right]\wedge}\eta_{\alpha\beta\gamma}-\theta_{\mathrm{T}}^{\ast}\Lambda_{\mathrm{eff}}\eta_{\alpha}=\ell^2\widehat{\Sigma}_{\alpha}.$	$-E_{\alpha}=\frac{\theta_{\mathrm{T}}^{\ast}}{2}R^{\left[\beta\gamma\right]\wedge}\eta_{\alpha\beta\gamma}-\theta_{\mathrm{T}}^{\ast}\Lambda_{\mathrm{eff}}\eta_{\alpha}=\ell^2\widehat{\Sigma}_{\alpha}.$	$-E_{\alpha}=\frac{\theta_{\mathrm{T}}^{\ast}}{2}R^{\left[\beta\gamma\right]\wedge}\eta_{\alpha\beta\gamma}-\theta_{\mathrm{T}}^{\ast}\Lambda_{\mathrm{eff}}\eta_{\alpha}=\ell^2\widehat{\Sigma}_{\alpha}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_337 EQ1031 (15.1.8)		■0.920 ●	$\begin{aligned} & \frac{1}{2} * \left(\vartheta^f \wedge * \Omega^f \right) - a \ell^2 \vartheta^f \wedge * \left\{ 2L_{fg} - \frac{1}{2b\ell^{*2}} (\Omega^g \wedge \vartheta^g \wedge \vartheta^g + 2\Lambda\eta) \right\} \\ & = \ell^2 * \left\{ \Sigma_{\mathrm{m}}(\text{ hadrons }) + \frac{a}{b} * (\vartheta^{\mathrm{g}} \wedge * \Sigma_{\mathrm{m}}(\text{ leptons })) \wedge \vartheta^{\mathrm{f}} \right\} \end{aligned}$	$\frac{1}{2} * \left(\vartheta^f \wedge * \Omega^f \right) - a \ell^2 \vartheta^f \wedge * \left\{ 2L_{fg} - \frac{1}{2b\ell^{*2}} (\Omega^g \wedge \vartheta^g \wedge \vartheta^g + 2\Lambda\eta) \right\} \\ = \ell^2 * \left\{ \Sigma_{\mathrm{m}}(\text{ hadrons }) + \frac{a}{b} * (\vartheta^{\mathrm{g}} \wedge * \Sigma_{\mathrm{m}}(\text{ leptons })) \wedge \vartheta^{\mathrm{f}} \right\}$	$\frac{1}{2} * (\vartheta^f \wedge * \Omega^f) - a \ell^2 \vartheta^f \wedge * \left\{ 2L_{fg} - \frac{1}{2b\ell^{*2}} (\Omega^g \wedge \vartheta^g \wedge \vartheta^g + 2\Lambda\eta) \right\} \\ = \ell^2 * \left\{ \Sigma_{\mathrm{m}}(\text{hadrons}) + \frac{a}{b} * (\vartheta^{\mathrm{g}} \wedge * \Sigma_{\mathrm{m}}(\text{leptons})) \wedge \vartheta^{\mathrm{f}} \right\}$
mielke-geometrodynamics_156 EQ0473 (7.4.1)		■0.921 ● N +0	$\begin{aligned} & \int \mathcal{D}\Gamma \exp \left[\int L_{\mathrm{SKY}}^{(*)} / \hbar \right] \\ & = \int \mathcal{D}\Gamma \exp \left[- \int \left(R_{\alpha\beta} + * R_{\alpha\beta}^{(*)} \right) \wedge * \left(R^{\alpha\beta} + * R^{\alpha\beta(*)} \right) / 4\hbar \right] \end{aligned}$	$\int \mathcal{D}\Gamma \exp \left[\int L_{\mathrm{SKY}}^{(*)} / \hbar \right] \\ = \int \mathcal{D}\Gamma \exp \left[- \int \left(R_{\alpha\beta} + * R_{\alpha\beta}^{(*)} \right) \wedge * \left(R^{\alpha\beta} + * R^{\alpha\beta(*)} \right) / 4\hbar \right]$	$\int \mathcal{D}\Gamma \exp \left[\int L_{\mathrm{SKY}}^{(*)} / \hbar \right] \\ = \int \mathcal{D}\Gamma \exp \left[- \int \left(R_{\alpha\beta} + * R_{\alpha\beta}^{(*)} \right) \wedge * \left(R^{\alpha\beta} + * R^{\alpha\beta(*)} \right) / 4\hbar \right]$
mielke-geometrodynamics_115 EQ0343 (5.4.1)		■0.922 ● N +0	$k \partial_k \Gamma_k = \frac{1}{2} \text{S Tr} \left\{ \left[\Gamma_k^{(2)} + R_k \right]^{-1} (k \partial_k R_k) \right\},$	$k \partial_k \Gamma_k = \frac{1}{2} \text{S Tr} \left\{ \left[\Gamma_k^{(2)} + R_k \right]^{-1} (k \partial_k R_k) \right\},$	$k \partial_k \Gamma_k = \frac{1}{2} \text{STr} \left\{ [\Gamma_k^{(2)} + R_k]^{-1} (k \partial_k R_k) \right\},$
mielke-geometrodynamics_269 EQ0829 (12.1.1)		■0.922 ● W +0	$\bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon) \rightarrow \bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon) \mathbf{P} \exp \left\{ i \int_x^{x+\varepsilon} A \right\}.$	$\bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon) \rightarrow \bar{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon) \mathbf{P} \exp \left\{ i \int_x^{x+\varepsilon} A \right\}.$	$\overline{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon) \rightarrow \overline{\psi}(x) \gamma_5^* \gamma \psi(x + \varepsilon) \mathbf{P} \exp \left\{ i \int_x^{x+\varepsilon} A \right\}.$
mielke-geometrodynamics_238 EQ0736 (11.1.3)		■0.922 ● N +0	$\gamma^0 := \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}, \quad \gamma^k := \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix}, \quad \sigma^k : \text{Pauli matrices.}$	$\gamma^0 := \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}, \quad \gamma^k := \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix}, \quad \sigma^k : \text{Pauli matrices.}$	$\gamma^0 := \begin{pmatrix} \mathbb{1} & 0 \\ 0 & -\mathbb{1} \end{pmatrix}, \quad \gamma^k := \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix}, \quad \sigma^k : \text{Pauli matrices.}$
mielke-geometrodynamics_217 EQ0661 (10.3.18)		■0.923 ● N +0	$\square H - (-1)^{s*} D^* D H = dt + \frac{\gamma}{\delta \ell^2} d\mathcal{A} + \frac{1}{\delta \ell} dh,$	$\square H - (-1)^{s*} D^* D H = dt + \frac{\gamma}{\delta \ell^2} d\mathcal{A} + \frac{1}{\delta \ell} dh,$	$\square H - (-1)^{s*} D^* D H = dt + \frac{\gamma}{\delta \ell^2} d\mathcal{A} + \frac{1}{\delta \ell} dh,$
mielke-geometrodynamics_222 EQ0685 (10.3.43)		■0.923 ● K +0	$C_\alpha - E_\alpha = \Sigma_\alpha - D^{\{\}} \mu_\alpha,$	$C_\alpha - E_\alpha = \Sigma_\alpha - D^{\{\}} \mu_\alpha,$	$C_\alpha - E_\alpha = \Sigma_\alpha - D^{\{\}} \mu_\alpha,$
mielke-geometrodynamics_225 EQ0702 (10.6.1)		■0.924 ● N -1	$L_A = \frac{1}{2} \left(F \wedge * F + m_{\text{photon}} F \wedge A \right).$	$L_A = \frac{1}{2} \left(F \wedge * F + m_{\text{photon}} F \wedge A \right).$	$L_A = \frac{1}{2} (F \wedge * F + m_{\text{photon}} F \wedge A).$
mielke-geometrodynamics_087 EQ0221 (4.2.14)		■0.924 ● K +0	$\Omega^{\mathrm{T}} \rightarrow {}^{A^{-1}} \Omega^{\mathrm{T}} = G^{-1} (\Omega^{\mathrm{T}} - D D T) = G^{-1} \Omega^{\mathrm{T}} - G^{-1} \Omega^{\mathrm{L}} \otimes T.$	$\Omega^{\mathrm{T}} \rightarrow {}^{A^{-1}} \Omega^{\mathrm{T}} = G^{-1} (\Omega^{\mathrm{T}} - D D T) = G^{-1} \Omega^{\mathrm{T}} - G^{-1} \Omega^{\mathrm{L}} \otimes T.$	$\Omega^{\mathrm{T}} \rightarrow {}^{A^{-1}} \Omega^{\mathrm{T}} = G^{-1} (\Omega^{\mathrm{T}} - D D T) = G^{-1} \Omega^{\mathrm{T}} - G^{-1} \Omega^{\mathrm{L}} \otimes T.$
mielke-geometrodynamics_143 EQ0443 (6.8.14)		■0.924 ● N -1	$H_\alpha = -\frac{1}{\ell^2} * \left(\sum_{i=1}^3 a_i^{(i)} T_\alpha \right),$	$H_\alpha = -\frac{1}{\ell^2} * \left(\sum_{i=1}^3 a_i^{(i)} T_\alpha \right),$	$H_\alpha = -\frac{1}{\ell^2} * \left(\sum_{i=1}^3 a_i^{(i)} T_\alpha \right),$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_081 EQ0192 (4.1.2)		■ 0.924 ●	a:=\left[\begin{array}{c c} g & \mathrm{a} \\ \hline 0 & 1 \end{array}\right] \mathrm{g} \, \& \, \mathrm{a} \, \, \, \hline 0 \, \& \, 1 \, \end{array}\right] \mathrm{g} \, \& \, 0 \, \, \, \hline 0 \, \& \, 1 \, \end{array}\right]=\mathrm{a}^{\mathrm{T}} \cdot \mathrm{a}^{\mathrm{L}} \in A(n,\mathbb{R}) \, \, \, \mathrm{in} \, A(n, \, \mathrm{mathbb{b}{R}}) \, \, .	$a := \left[\begin{array}{c c} g & \mathrm{a} \\ \hline 0 & 1 \end{array}\right] = \left[\begin{array}{c c} \mathbb{1}_n & \mathrm{a} \\ \hline 0 & 1 \end{array}\right] \left[\begin{array}{c c} g & 0 \\ \hline 0 & 1 \end{array}\right] = a^{\mathrm{T}} \cdot a^{\mathrm{L}} \in A(n,\mathbb{R}).$	$a := \left[\begin{array}{c c} g & \mathrm{a} \\ \hline 0 & 1 \end{array}\right] = \left[\begin{array}{c c} \mathbb{1}_n & \mathrm{a} \\ \hline 0 & 1 \end{array}\right] \left[\begin{array}{c c} g & 0 \\ \hline 0 & 1 \end{array}\right] = a^{\mathrm{T}} \cdot a^{\mathrm{L}} \in A(n,\mathbb{R}).$
mielke-geometrodynamics_278 EQ0857 (12.4.4)		■ 0.925 ● C +3	\left.\mathrm{mathbf{f}\{C\}}_n\, \mathrm{mathscr{G}\{A\}} \, \mathrm{cong} \, n\right\rfloor \mathrm{left}(D \, \mathrm{underline{\tau}}^{\mathrm{A}}\mathrm{right}),	(not rendered)	$\mathbb{L}_n\mathscr{G}^A \cong n\rfloor(D\underline{\tau}^A),$
mielke-geometrodynamics_336 EQ1025 (15.1.1)		■ 0.926 ● W +0	\begin{aligned} L_{\mathrm{f-g}}= & \, \mathrm{frac{1}{\ell^2}} L_{\mathrm{W}}(f) + \mathrm{frac{1}{\ell^{*2}}} L_{\mathrm{W}}(g) + L_{fg} + \mathrm{frac{\Lambda}{\ell^{*2}}} \vartheta^g \wedge \vartheta^g \wedge \vartheta^g \wedge \vartheta^g \\ & + L_{\mathrm{m}}(\mathrm{hadrons},\mathrm{f}) + L_{\mathrm{m}}(\mathrm{leptons},\mathrm{g}). \end{aligned}	$L_{f-g} = \frac{1}{\ell^2} L_{\mathrm{W}}(f) + \frac{1}{\ell^{*2}} L_{\mathrm{W}}(g) + L_{fg} + \frac{\Lambda}{\ell^{*2}} \vartheta^g \wedge \vartheta^g \wedge \vartheta^g \wedge \vartheta^g + L_{\mathrm{m}}(\mathrm{hadrons},\mathrm{f}) + L_{\mathrm{m}}(\mathrm{leptons},\mathrm{g}).$	$L_{f-g} = \frac{1}{\ell^2} L_{\mathrm{W}}(f) + \frac{1}{\ell^{*2}} L_{\mathrm{W}}(g) + L_{fg} + \frac{\Lambda}{\ell^{*2}} \vartheta^g \wedge \vartheta^g \wedge \vartheta^g \wedge \vartheta^g + L_{\mathrm{m}}(\mathrm{hadrons},\mathrm{f}) + L_{\mathrm{m}}(\mathrm{leptons},\mathrm{g}).$
mielke-geometrodynamics_058 EQ0109 (3.3.10)		■ 0.927 ● N -1	L_{\mathrm{Max}}=-\mathrm{frac{1}{2}} F \, \mathrm{wedge} \, \mathrm{^*} F,	$L_{\mathrm{Max}} = -\frac{1}{2} F \wedge \, ^* F,$	$L_{\mathrm{Max}} = -\frac{1}{2} F \wedge \, ^* F,$
mielke-geometrodynamics_194 EQ0587 (9.1.5)		■ 0.928 ● N +0	d \, r^{\mathrm{*}}:=\mathrm{e}^{\{(\lambda\!-\!v)/2\}} \, d \, r \, .	$dr^* := e^{(\lambda-v)/2} dr.$	$dr^* := e^{(\lambda-v)/2} dr.$
mielke-geometrodynamics_370 EQ1101 (B.13)		■ 0.929 ● W -1	*(\underbrace{\vartheta \wedge \cdots \wedge \vartheta}_{\mathrm{p\,products}})=\mathrm{frac{p!}{(n-p)!}} \underbrace{\vartheta \wedge \cdots \wedge \vartheta}_{\mathrm{n-p\,products}}(\star),	$*(\underbrace{\vartheta \wedge \cdots \wedge \vartheta}_{\mathrm{p\,products}}) = \frac{p!}{(n-p)!} \underbrace{\vartheta \wedge \cdots \wedge \vartheta}_{\mathrm{n-p\,products}}(\star),$	$*(\underbrace{\vartheta \wedge \cdots \wedge \vartheta}_{\mathrm{p\,products}}) = \frac{p!}{(n-p)!} \underbrace{\vartheta \wedge \cdots \wedge \vartheta}_{\mathrm{n-p\,products}}(\star),$
mielke-geometrodynamics_137 EQ0413 (6.5.1)		■ 0.929 ● N +0	W(A,\,\gamma)=\operatorname{Tr}\, \mathrm{bfbf}\{P\} \, \exp \, \oint A = \operatorname{Tr}\, \mathrm{bfbf}\{P\} \, \exp \, \oint d s \, \dot{\gamma}^i(s) \, A_{i\,}^J(\gamma(s)) \, \lambda_J,	$W(A,\gamma) = \mathrm{Tr} \, \mathbf{P} \exp \oint A = \mathrm{Tr} \, \mathbf{P} \exp \oint ds \dot{\gamma}^i(s) A_i^J(\gamma(s)) \lambda_J,$	$W(A,\gamma) = \mathrm{Tr} \mathbf{P} \exp \oint A = \mathrm{Tr} \mathbf{P} \exp \oint ds \dot{\gamma}^i(s) A_i^J(\gamma(s)) \lambda_J,$
mielke-geometrodynamics_153 EQ0465 (7.3.4)		■ 0.929 ● N +0	\left.\left.\mathrm{left.E}_{\alpha}:=\partial L_{\mathrm{SKY}}\right/\partial\vartheta^{\alpha}=\mathrm{frac{1}{2}}\left(\mathrm{e}_{\alpha}\rfloor R^{\mu\nu}\wedge^*R_{\mu\nu}-R^{\mu\nu}\wedge \mathrm{e}_{\alpha}\rfloor^*R_{\mu\nu}\right)=0\right.\right.	$E_{\alpha} := \partial L_{\mathrm{SKY}}/\partial\vartheta^{\alpha} = \frac{1}{2} \left(e_{\alpha}\rfloor R^{\mu\nu} \wedge ^*R_{\mu\nu} - R^{\mu\nu} \wedge e_{\alpha}\rfloor^*R_{\mu\nu} \right) = 0$	$E_{\alpha} := \partial L_{\mathrm{SKY}}/\partial\vartheta^{\alpha} = \frac{1}{2} \left(e_{\alpha}\rfloor R^{\mu\nu} \wedge ^*R_{\mu\nu} - R^{\mu\nu} \wedge e_{\alpha}\rfloor^*R_{\mu\nu} \right) = 0$
mielke-geometrodynamics_338 EQ1034 (15.1.11)		■ 0.930 ● N +0	\Lambda_{\mathrm{eff.}}^f=\mathrm{frac{2 \, a}{\kappa b}} \Lambda - 2 a \ell^{2*} L_{fg}	$\Lambda_{\mathrm{eff.}}^f = \frac{2a}{\kappa b} \Lambda - 2a \ell^{2*} L_{fg}$	$\Lambda_{\mathrm{eff.}}^f = \frac{2a}{\kappa b} \Lambda - 2a \ell^2 \, ^* L_{fg}$
mielke-geometrodynamics_165 EQ0512 (7.7.30)		■ 0.931 ● N -1	\begin{aligned} \Omega_{\mathrm{g}}^{}:=& \, d\Gamma+\Gamma\wedge\Gamma=\mathrm{frac{i}{4}}\left(d\Gamma_{\alpha}{}^{\beta}-\Gamma_{\alpha}{}^{\gamma}\wedge\Gamma_{\gamma}{}^{\beta}\right)\sigma^{\alpha}{}_{\beta} \\ & =\mathrm{frac{i}{4}}R^{\alpha\beta}\sigma_{\alpha\beta}=\mathrm{frac{i}{8}}R_{ij}{}^{\alpha\beta}\sigma_{\alpha\beta}dx^i\wedge dx^j, \end{aligned}	$\Omega^g := d\Gamma + \Gamma \wedge \Gamma = \frac{i}{4} \left(d\Gamma_{\alpha}{}^{\beta} - \Gamma_{\alpha}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\beta} \right) \sigma^{\alpha}{}_{\beta} = \frac{i}{4} R^{\alpha\beta} \sigma_{\alpha\beta} = \frac{i}{8} R_{ij}{}^{\alpha\beta} \sigma_{\alpha\beta} dx^i \wedge dx^j,$	$\Omega^g := d\Gamma + \Gamma \wedge \Gamma = \frac{i}{4} (d\Gamma_{\alpha}{}^{\beta} - \Gamma_{\alpha}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\beta}) \sigma^{\alpha}{}_{\beta} = \frac{i}{4} R^{\alpha\beta} \sigma_{\alpha\beta} = \frac{i}{8} R_{ij}{}^{\alpha\beta} \sigma_{\alpha\beta} dx^i \wedge dx^j,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_00452 (7.2.5)	151	■ 0.948 ● W +0	$C_{\{. \ . \ c \ d\}^{\{a \ b\}}=R_{\{. \ . \ c \ d\}^{\{a \ b\}}-2 \, R_{\{.[c]^\{[a \ \backslash cdot\} \ \backslash delta_{[d]\}^{\{b\}}\}+\frac{R\}{6} \ \backslash delta_{\{[c]^\{a\} \ \backslash delta_{[d]\}^{\{b\}} \} \ .$	$C_{..cd}^{ab}=R_{..cd}^{ab}-2R_{.[c}^{[a}\delta_{d]}^{b]}+\frac{R}{6}\delta_{[c}^a\delta_{d]}^b.$	$C_{..cd}^{ab}=R_{..cd}^{ab}-2R_{.[c}^{[a}\delta_{d]}^{b]}+\frac{R}{6}\delta_{[c}^a\delta_{d]}^b.$
mielke-geometrodynamics_00661 (2.8.2)	046	■ 0.949 ● N +1	$\{ \ }^{\{*\}}\backslashleft(\backslashgamma_{\{k\}}\backslashright)=\frac{((-1)^{\{k\}})\{(2 \ \backslash pi \ i)^{\{k\}} \ k!\} \ \operatorname{Tr}(\backslash0mega \ \backslashwedge \ \backslashcdots \ \backslashwedge \ \backslash0mega) \ .$	$^*(\gamma_k)=\frac{(-1)^k}{(2\pi i)^k k!}\operatorname{Tr}(\Omega\wedge\cdots\wedge\Omega).$	$\pi^*(\gamma_k)=\frac{(-1)^k}{(2\pi i)^k k!}\operatorname{Tr}(\varOmega\wedge\cdots\wedge\varOmega).$
mielke-geometrodynamics_00756 (11.1.23)	242	■ 0.950 ●	$\begin{aligned} S(p) \ \&= \exp \ \imath \mathit{\theta}_{\alpha\beta}(m) \ \Lambda^{-1} (L^{\alpha\beta}) = \exp \ \imath \theta_{\alpha\beta}(m) \sigma^{\alpha\beta} \\ \&\&= \exp \ \left(-\frac{1}{4} \theta_{\alpha\beta}(m) \left[\gamma^\alpha, \gamma^\beta \right] \right) . \end{aligned}$	$S(p)=\exp \imath \theta_{\alpha\beta}(m) \Lambda^{-1} \left(L^{\alpha\beta} \right) = \exp \imath \theta_{\alpha\beta}(m) \sigma^{\alpha\beta} \\ = \exp \left(-\frac{1}{4} \theta_{\alpha\beta}(m) \left[\gamma^\alpha, \gamma^\beta \right] \right) .$	$S(p)=\exp \imath \theta_{\alpha\beta}(m) \Lambda^{-1} (L^{\alpha\beta}) = \exp \imath \theta_{\alpha\beta}(m) \sigma^{\alpha\beta} \\ = \exp \left(-\frac{1}{4} \theta_{\alpha\beta}(m) [\gamma^\alpha, \gamma^\beta] \right) .$
mielke-geometrodynamics_00939 (14.1.20)	305	■ 0.950 ● N +0	$\backslash xi \ \backslashwedge \ \backslash xi=0 \ \backslash quad (=2 \ \backslash vec{E} \ \backslash cdot \ \backslash vec{B} \ \backslash text \{ \text{ in 3-dimensional vector notation } \})$	$\xi \wedge \xi = 0 \quad (=2\vec{E} \cdot \vec{B} \text{ in 3-dimensional vector notation })$	$\xi \wedge \xi = 0 \quad (=2\vec{E} \cdot \vec{B} \text{ in 3-dimensional vector notation})$
mielke-geometrodynamics_009568 (8.6.5)	183	■ 0.950 ● C -4	$L_{\{\backslashtheta\}}:=\backslashtheta_{\{\mathrm{T}\}} \ d \ C_{\{\mathrm{TT}\}}+\backslashtheta_{\{\mathrm{T}\}^{\{*\}} \ d \ C_{\{\mathrm{TT}\}^{\{*\}}\}+\backslashtheta_{\{\mathrm{L}\}} \ d \ C_{\{\mathrm{RR}\}}+\backslashtheta_{\{\mathrm{L}\}^{\{\backslashstar\}}(-1)^{\{\mathrm{sig}+1\}} \ d \ C_{\{\mathrm{RR}\}^{\{*\}}}=d \ C_{\{\backslashtheta\}} \ .$	$L_\theta:=\theta_\mathrm{T}dC_\mathrm{TT}+\theta_\mathrm{T}^*dC_\mathrm{TT}^*+\theta_\mathrm{L}dC_\mathrm{RR}+\theta_\mathrm{L}^*(-1)^{\mathrm{sig}+1}dC_\mathrm{RR}^*=dC_\theta.$	$L_\theta:=\theta_\mathrm{T}dC_\mathrm{TT}+\theta_\mathrm{T}^*dC_\mathrm{TT}^*+\theta_\mathrm{L}dC_\mathrm{RR}+\theta_\mathrm{L}^*(-1)^{\mathrm{sig}+1}dC_\mathrm{RR}^*=dC_\theta \ .$
mielke-geometrodynamics_009234 (4.2.27)	088	■ 0.950 ● N +0	$\backslashwidetilde{\backslashwidetilde{D}} \ \backslashwidetilde{\backslashwidetilde{\backslashvarphi}}=D \ \backslashwidetilde{\backslashwidetilde{\backslashwidetilde{\backslashvarphi}}}=0$	$\widetilde{\widetilde{D}}\widetilde{\widetilde{\varphi}}=D\widetilde{\widetilde{\varphi}}=0$	$\widetilde{\widetilde{D}}\widetilde{\widetilde{\varphi}}=D\widetilde{\widetilde{\varphi}}=0$
mielke-geometrodynamics_009392 (4.5.9)	102	■ 0.950 ● W +0	$\begin{aligned} \widehat{D} \ \backslash Sigma_{\{\alpha\}} \ \&\ \backslash left.\backcong \ \backslash left(e_{\{\alpha\}}\backslashright\rceil\backslash floor \ R_{\{\backslashbeta\}^{\{\backslashgamma\}}\backslashright) \ \backslashwedge \ \backslash Delta^{\{\backslashbeta\}} \ \backslashgamma \ \&\ \&\ \backslash left.-\backslashfrac{1\}{2} \ \backslash left(e_{\{\alpha\}}\backslashright\rceil\backslash floor \ Q_{\{\backslashbeta \ \backslashgamma\}}\backslashright) \ \backslashwedge \ \backslash sigma^{\{\backslashbeta \ \backslashgamma\}} \ \backslash end{aligned}$	$\widehat{D}\Sigma_\alpha\cong(e_\alpha\rfloor R_\beta^\gamma)\wedge\Delta^\beta\gamma \\ -\frac{1}{2}(e_\alpha\rfloor Q_{\beta\gamma})\wedge\sigma^{\beta\gamma}$	$\widehat{D}\ \varSigma_\alpha\cong(e_\alpha\rfloor R_{\beta}^\gamma)\wedge\Delta_{\gamma}^{\beta} \\ -\frac{1}{2}(e_\alpha\rfloor Q_{\beta\gamma})\wedge\sigma^{\beta\gamma}$
mielke-geometrodynamics_009151 (3.4.22)	066	■ 0.951 ● C -4	$L_{\{\text{Mills}\}}\backslashleft(\backslashmathscr{F},\{ \ }^{\{*\}} \ \backslashmathscr{F} \ \backslashright)=\backslashmathscr{F}\backslashleft[1\backslashleft(1+\backslashfrac{1\}{b^2}\}\backslashb^{\{2\}} \ \backslashmathscr{F}\backslashright)^{\{-1\}}\backslashright]$	$L_{\mathrm{Mills}}\left(\mathscr{F},{}^*\mathscr{F}\right)=\mathscr{F}\left[1-\left(1+\frac{1}{b^2}\mathscr{F}^2\right)^{-1}\right]$	$L_{\mathrm{Mills}}(\mathscr{F},{}^*\mathscr{F})=\mathscr{F}\left[1-\left(1+\frac{1}{b^2}\mathscr{F}^2\right)^{-1}\right]$
mielke-geometrodynamics_009560 (8.5.3)	181	■ 0.953 ● U +0	$\backslashstackrel{\rm l}{\backslash pm}\{\backslash0mega^{\{g\}}\}=i \ \backslashfrac{\{\backslashtheta_{\{\mathrm{T}\}^{\{*\}}\}\{4 \ \backslash ell^{\{2\}}\} \ \backslashsigma$	$\stackrel{\pm}{\Omega}^g=i\frac{\theta_{\mathrm{T}}^*}{4\ell^2}\sigma$	$\stackrel{\pm}{\Omega}^g=i\frac{\theta_{\mathrm{T}}^*}{4\ell^2}\sigma$
mielke-geometrodynamics_001106 (B.18)	370	■ 0.953 ● N +0	$\backslashdelta \ \backslashalpha^{\{p\}}:=(-1)^{\{p \ n+n+1+s \ * \} \ d^{\{*\}} \ \backslashalpha^{\{p\}} \} \ .$	$\delta\alpha^{(p)}:=(-1)^{pn+n+1+s*}d^*\alpha^{(p)}.$	$\delta\alpha^{(p)}:=(-1)^{pn+n+1+s*}d^*\alpha^{(p)}.$
mielke-geometrodynamics_00786 (11.3.5)	248	■ 0.954 ● W +1	$\begin{aligned} L_{\{\mathrm{NL}\}} \ \&= -3 \ \backslash ell^{\{*\} \ 2} \ \backslashtau_{\{s\}} \ \backslashwedge^{\{*\}} \ \backslashtau_{\{s\}}=3 \ \backslash ell^{\{*\} \ 2} \ \backslashbar{\backslashpsi}^{\{*\}} \ \backslashgamma \ \backslashpsi \ \backslashwedge^{\{*\}}\backslashleft(\backslashbar{\backslashpsi}^{\{*\}} \ \backslashgamma \ \backslashpsi\backslashright) \\ \&\&=3 \ \backslash ell^{\{*\} \ 2} \ \backslashbar{\backslashpsi} \ \backslashgamma^{\{n+1\}} \ \backslashgamma \ \backslashpsi \ \backslashwedge^{\{*\}}\backslashleft(\backslashbar{\backslashpsi} \ \backslashgamma^{\{n+1\}} \ \backslashgamma \ \backslashpsi \ \backslashright) \ . \ \backslash end{aligned}$	$L_{\mathrm{NL}}=-3\ell^{*2}\tau_s\wedge^*\tau_s=3\ell^{*2}\bar{\psi}^*\gamma\psi\wedge^*(\bar{\psi}^*\gamma\psi) \\ =3\ell^{*2}\bar{\psi}\gamma^{n+1}\gamma\psi\wedge^*(\bar{\psi}\gamma^{n+1}\gamma\psi).$	$L_{\mathrm{NL}}=-3\ell^{*2}\tau_s\wedge^*\tau_s=3\ell^{*2}\overline{\psi}^*\gamma\psi\wedge^*(\overline{\psi}^*\gamma\psi) \\ =3\ell^{*2}\overline{\psi}\gamma^{n+1}\gamma\psi\wedge^*(\overline{\psi}\gamma^{n+1}\gamma\psi).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_231 EQ0727 (10.10.7)	231	■ 0.955 ●	$\begin{aligned} & i \gamma^{\alpha} \Psi \wedge \frac{\delta L}{\delta \vartheta^{\alpha}} + c \gamma \wedge \frac{\delta L}{\delta \bar{\Psi}} - 2 D \frac{\delta L}{\delta \bar{\Psi}} \\ & \cong i \gamma^{\alpha} \Psi \wedge \left(\frac{\theta_{\mathrm{TL}}}{\ell} R_{\alpha}^{\star} + \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} + \Sigma_{\alpha} \right) \\ & + c \gamma \wedge \left(\frac{i}{2} D \Psi + \frac{i}{4} m \gamma \wedge \Psi \right) - D \left(i D \Psi + \frac{i}{2} m \gamma \wedge \Psi \right) \\ & = i \gamma^{\alpha} \Psi \wedge \left(\frac{\theta_{\mathrm{TL}}}{\ell} R_{\alpha}^{\star} + \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} \right) + \gamma^{\alpha} \Psi \wedge \left(\frac{1}{4} m \bar{\Psi} \gamma_{\alpha} \Psi \right) \\ & + c \gamma \wedge \left(\frac{i}{2} D \Psi + \frac{i}{4} m \gamma \wedge \Psi \right) - i R_{\alpha}^{\star} \gamma^{\alpha} \Psi - \frac{i}{2} m T_{\alpha} \gamma^{\alpha} \Psi \\ & + \frac{i}{2} m \gamma \wedge D \Psi. \end{aligned}$	$\begin{aligned} & i \gamma^{\alpha} \Psi \wedge \frac{\delta L}{\delta \vartheta^{\alpha}} + c \gamma \wedge \frac{\delta L}{\delta \bar{\Psi}} - 2 D \frac{\delta L}{\delta \bar{\Psi}} \\ & \cong i \gamma^{\alpha} \Psi \wedge \left(\frac{\theta_{\mathrm{TL}}}{\ell} R_{\alpha}^{\star} + \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} + \Sigma_{\alpha} \right) \\ & + c \gamma \wedge \left(\frac{i}{2} D \Psi + \frac{i}{4} m \gamma \wedge \Psi \right) - D \left(i D \Psi + \frac{i}{2} m \gamma \wedge \Psi \right) \\ & = i \gamma^{\alpha} \Psi \wedge \left(\frac{\theta_{\mathrm{TL}}}{\ell} R_{\alpha}^{\star} + \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} \right) + \gamma^{\alpha} \Psi \wedge \left(\frac{1}{4} m \bar{\Psi} \gamma_{\alpha} \Psi \right) \\ & + c \gamma \wedge \left(\frac{i}{2} D \Psi + \frac{i}{4} m \gamma \wedge \Psi \right) - i R_{\alpha}^{\star} \gamma^{\alpha} \Psi - \frac{i}{2} m T_{\alpha} \gamma^{\alpha} \Psi \\ & + \frac{i}{2} m \gamma \wedge D \Psi. \end{aligned}$	$\begin{aligned} & i \gamma^{\alpha} \Psi \wedge \frac{\delta L}{\delta \vartheta^{\alpha}} + c \gamma \wedge \frac{\delta L}{\delta \bar{\Psi}} - 2 D \frac{\delta L}{\delta \bar{\Psi}} \\ & \cong i \gamma^{\alpha} \Psi \wedge \left(\frac{\theta_{\mathrm{TL}}}{\ell} R_{\alpha}^{\star} + \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} + \Sigma_{\alpha} \right) \\ & + c \gamma \wedge \left(\frac{i}{2} D \Psi + \frac{i}{4} m \gamma \wedge \Psi \right) - D \left(i D \Psi + \frac{i}{2} m \gamma \wedge \Psi \right) \\ & = i \gamma^{\alpha} \Psi \wedge \left(\frac{\theta_{\mathrm{TL}}}{\ell} R_{\alpha}^{\star} + \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} \right) + \gamma^{\alpha} \Psi \wedge \left(\frac{1}{4} m \bar{\Psi} \gamma_{\alpha} \Psi \right) \\ & + c \gamma \wedge \left(\frac{i}{2} D \Psi + \frac{i}{4} m \gamma \wedge \Psi \right) - i R_{\alpha}^{\star} \gamma^{\alpha} \Psi - \frac{i}{2} m T_{\alpha} \gamma^{\alpha} \Psi \\ & + \frac{i}{2} m \gamma \wedge D \Psi. \end{aligned} \tag{10.1}$
mielke-geometrodynamics_244 EQ0768 (11.1.35)	244	■ 0.956 ● N +0	$\bar{\psi} \longrightarrow{ }^{\wedge\{S^{-1}\}} \bar{\psi} = \bar{\psi} S,$	$\bar{\psi} \longrightarrow^{S^{-1}} \bar{\psi} = \bar{\psi} S,$	
mielke-geometrodynamics_315 EQ0973 (14.2.14)	315	■ 0.956 ● C +3	$\begin{aligned} & \hat{\Theta} := \hat{D} \hat{\vartheta} = \hat{d} \hat{\vartheta} - [\hat{\omega}, \hat{\vartheta}] \\ & = \left[\begin{array}{c c} \Theta - \hat{\ell} \omega \wedge \alpha^{\mathrm{T}} & \hat{\ell} \left(d \omega - \omega^g \wedge \omega - \omega \wedge \mathring{\omega} \right) + \vartheta \wedge \alpha + \alpha \wedge \mathring{\vartheta} \\ \hline -\alpha^{\mathrm{T}} \wedge \vartheta - \vartheta \wedge \alpha^{\mathrm{T}} & \mathring{\Theta} - \hat{\ell} \alpha^{\mathrm{T}} \wedge \omega \end{array} \right]. \end{aligned}$	$\begin{aligned} & \hat{\Theta} := \hat{D} \hat{\vartheta} = \hat{d} \hat{\vartheta} - [\hat{\omega}, \hat{\vartheta}] \\ & = \left[\begin{array}{c c} \Theta - \hat{\ell} \omega \wedge \alpha^{\mathrm{T}} & \hat{\ell} \left(d \omega - \omega^g \wedge \omega - \omega \wedge \mathring{\omega} \right) + \vartheta \wedge \alpha + \alpha \wedge \mathring{\vartheta} \\ \hline -\alpha^{\mathrm{T}} \wedge \vartheta - \vartheta \wedge \alpha^{\mathrm{T}} & \mathring{\Theta} - \hat{\ell} \alpha^{\mathrm{T}} \wedge \omega \end{array} \right]. \end{aligned}$	$\begin{aligned} \hat{\Theta} &:= \hat{D} \hat{\vartheta} = \hat{d} \hat{\vartheta} - [\hat{\omega}, \hat{\vartheta}] \\ &= \left[\begin{array}{c c} \Theta - \hat{\ell} \omega \wedge \alpha^{\mathrm{T}} & \hat{\ell} \left(d \omega - \omega^g \wedge \omega - \omega \wedge \mathring{\omega} \right) + \vartheta \wedge \alpha + \alpha \wedge \mathring{\vartheta} \\ \hline -\alpha^{\mathrm{T}} \wedge \vartheta - \vartheta \wedge \alpha^{\mathrm{T}} & \mathring{\Theta} - \hat{\ell} \alpha^{\mathrm{T}} \wedge \omega \end{array} \right]. \end{aligned}$
mielke-geometrodynamics_243 EQ0761 (11.1.28)	243	■ 0.957 ● N +0	$\frac{1}{N} \operatorname{Tr}(\gamma \otimes_s \gamma) = ds^2 = g_{ij}(m) dx^i \otimes_s dx^j.$	$\frac{1}{N} \operatorname{Tr}(\gamma \otimes_s \gamma) = ds^2 = g_{ij}(m) dx^i \otimes_s dx^j.$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometroynamics_160 EQ0486 (7.7.2)		■ 0.957 ●	$\begin{aligned} & \vartheta^{\alpha} \rightarrow \widetilde{\vartheta}^{\alpha} := \vartheta^{\alpha} + \ell^2 e_{\beta} \Big]^{*} H^{\alpha\beta}, \\ & \Gamma_{\alpha}{}^{\beta} \rightarrow \widetilde{\Gamma}_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} + \ell^2 e_{\alpha} \Big]^{*} H^{\beta}. \end{aligned}$	$\vartheta^{\alpha} \rightarrow \widetilde{\vartheta}^{\alpha} := \vartheta^{\alpha} + \ell^2 e_{\beta} \Big]^{*} H^{\alpha\beta},$ $\Gamma_{\alpha}{}^{\beta} \rightarrow \widetilde{\Gamma}_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} + \ell^2 e_{\alpha} \Big]^{*} H^{\beta}.$	$\vartheta^{\alpha} \rightarrow \widetilde{\vartheta}^{\alpha} := \vartheta^{\alpha} + \ell^2 e_{\beta} \Big]^{*} H^{\alpha\beta},$ $\Gamma_{\alpha}{}^{\beta} \rightarrow \widetilde{\Gamma}_{\alpha}{}^{\beta} := \Gamma_{\alpha}{}^{\beta} + \ell^2 e_{\alpha} \Big]^{*} H^{\beta}.$
mielke-geometroynamics_370 EQ1104 (B.16)		■ 0.957 ● N +0	$\begin{aligned} & \left\langle \alpha^{(p)}, \beta^{(p)} \right\rangle := \alpha^{(p)} \wedge^{*} \beta^{(p)} = \beta^{(p)} \wedge^{*} \alpha^{(p)} \\ & = (-1)^{s*} \left(\alpha^{(p)} \wedge^{*} \beta^{(p)} \right) \eta \\ & = \frac{(-1)^s}{p!} A^{\alpha_1 \cdots \alpha_p} B_{\alpha_1 \cdots \alpha_p} d\mu(M) \in C^{\infty}(\wedge^n T^{*}(M)) \end{aligned}$	$\left\langle \alpha^{(p)}, \beta^{(p)} \right\rangle := \alpha^{(p)} \wedge^{*} \beta^{(p)} = \beta^{(p)} \wedge^{*} \alpha^{(p)}$ $= (-1)^{s*} \left(\alpha^{(p)} \wedge^{*} \beta^{(p)} \right) \eta$ $= \frac{(-1)^s}{p!} A^{\alpha_1 \cdots \alpha_p} B_{\alpha_1 \cdots \alpha_p} d\mu(M) \in C^{\infty}(\wedge^n T^{*}(M))$	$\langle \alpha^{(p)}, \beta^{(p)} \rangle := \alpha^{(p)} \wedge^{*} \beta^{(p)} = \beta^{(p)} \wedge^{*} \alpha^{(p)}$ $= (-1)^{s*} (\alpha^{(p)} \wedge^{*} \beta^{(p)}) \eta$ $= \frac{(-1)^s}{p!} A^{\alpha_1 \cdots \alpha_p} B_{\alpha_1 \cdots \alpha_p} d\mu(M) \in C^{\infty}(\wedge^n T^{*}(M))$
mielke-geometroynamics_176 EQ0537 (8.2.4)		■ 0.958 ● W +1	$\begin{array}{l} s\,\bar{C}=\mathbb{B},\quad s\mathbb{B}=0, \\ s\bar{\chi}=\beta,\quad s\beta=0, \\ s\bar{\Phi}=\bar{\eta},\quad s\bar{\eta}=0. \end{array}$	$s\bar{C}=\mathbb{B},\quad s\mathbb{B}=0,$ $s\bar{\chi}=\beta,\quad s\beta=0,$ $s\bar{\Phi}=\bar{\eta},\quad s\bar{\eta}=0.$	$s\bar{C}=\mathbb{B},\quad s\mathbb{B}=0,$ $s\bar{\chi}=\beta,\quad s\beta=0,$ $s\bar{\Phi}=\bar{\eta},\quad s\bar{\eta}=0.$
mielke-geometroynamics_095 EQ0269 (4.3.34)		■ 0.958 ● N +0	$R_{\{i\}j}:=R_{[\alpha}i\}j]^{\cdots\alpha}$	$R_{ij}:=R_{\alpha ij}^{\cdots\alpha}$	$R_{ij}:=R_{\alpha ij}^{\cdots\alpha}$
mielke-geometroynamics_132 EQ0394 (6.4.9)		■ 0.958 ● W +2	$\left.\mathop{\mathrm{H}}\nolimits\cong n^{\alpha}\Sigma_{\alpha}+\Gamma_{\perp}^{\alpha\beta}\rho(I_{\alpha\beta})\underline{\Psi}\wedge\frac{\partial L}{\partial D\Psi}-\underline{d}\bigl[(n]\Psi\bigr)\wedge\frac{\partial L}{\partial D\Psi}\Big].$	$\mathcal{H}\cong n^{\alpha}\underline{\Sigma}_{\alpha}+\Gamma_{\perp}^{\alpha\beta}\rho(I_{\alpha\beta})\underline{\Psi}\wedge\frac{\partial L}{\partial D\Psi}-\underline{d}\bigl[(n]\Psi\bigr)\wedge\frac{\partial L}{\partial D\Psi}\Big].$	$\mathcal{H}\cong n^{\alpha}\underline{\Sigma}_{\alpha}+\Gamma_{\perp}^{\alpha\beta}\rho(I_{\alpha\beta})\underline{\Psi}\wedge\frac{\partial L}{\partial D\Psi}-\underline{d}\bigl[(n]\Psi\bigr)\wedge\frac{\partial L}{\partial D\Psi}\Big].$
mielke-geometroynamics_135 EQ0406 (6.4.26)		■ 0.959 ● N -1	$\stackrel{\scriptscriptstyle{(\pm)}}{\underline{D}}\stackrel{\scriptscriptstyle{(\pm)}}{\underline{T}}^B=0$	$\stackrel{(\pm)}{\underline{D}}\stackrel{(\pm)}{\underline{T}}^B=0$	$\stackrel{(\pm)}{\underline{D}}\stackrel{(\pm)}{\underline{T}}^B=0$
mielke-geometroynamics_152 EQ0458 (7.2.10)		■ 0.959 ● K +0	$\mathfrak{s}\mathfrak{o}(4)\approx\mathfrak{s}\mathfrak{o}(3)\times\mathfrak{s}\mathfrak{u}(2)\approx\mathfrak{s}\mathfrak{u}(2)\times\mathfrak{s}\mathfrak{u}(2).$	$\mathfrak{s}\mathfrak{o}(4)\approx\mathfrak{s}\mathfrak{o}(3)\times\mathfrak{s}\mathfrak{o}(3)\approx\mathfrak{s}\mathfrak{u}(2)\times\mathfrak{s}\mathfrak{u}(2).$	$\mathfrak{s}\mathfrak{o}(4)\approx\mathfrak{s}\mathfrak{o}(3)\times\mathfrak{s}\mathfrak{o}(3)\approx\mathfrak{s}\mathfrak{u}(2)\times\mathfrak{s}\mathfrak{u}(2).$
mielke-geometroynamics_308 EQ0954 (14.1.36)		■ 0.959 ● W +0	$\begin{aligned} & (-1)^z \mathrm{Tr}_{\{ \, {}^* \xi \wedge \vartheta \wedge \xi - \xi \wedge \vartheta \wedge {}^* \xi \}} \wedge d\delta \\ & = \mathrm{Tr}_{\{ \, {}^* \xi \wedge \vartheta \wedge D {}^* \xi + \xi \wedge \vartheta \wedge D \xi \}} \\ & = -\frac{1}{2} D \mathrm{Tr}_{\{ \, {}^* \xi \wedge \vartheta \wedge {}^* \xi - \xi \wedge \vartheta \wedge \xi \}}. \end{aligned}$	$(-1)^z \mathrm{Tr}_{\{ \, {}^* \xi \wedge \vartheta \wedge \xi - \xi \wedge \vartheta \wedge {}^* \xi \}} \wedge d\delta$ $= \mathrm{Tr}_{\{ \, {}^* \xi \wedge \vartheta \wedge D {}^* \xi + \xi \wedge \vartheta \wedge D \xi \}}$ $= -\frac{1}{2} D \mathrm{Tr}_{\{ \, {}^* \xi \wedge \vartheta \wedge {}^* \xi - \xi \wedge \vartheta \wedge \xi \}}.$	$(-1)^z \mathrm{Tr}\{ {}^* \xi \wedge \vartheta \wedge \xi - \xi \wedge \vartheta \wedge {}^* \xi \} \wedge d\delta$ $= \mathrm{Tr}\{ {}^* \xi \wedge \vartheta \wedge D {}^* \xi + \xi \wedge \vartheta \wedge D \xi \}$ $= -\frac{1}{2} D \mathrm{Tr}\{ {}^* \xi \wedge \vartheta \wedge {}^* \xi - \xi \wedge \vartheta \wedge \xi \}.$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_008330 (5.3.7)	112	■ 0.960	$G:=G_{\alpha}\gamma^{\alpha}:=\frac{1}{2}R^{\mu\nu}\wedge\eta_{\mu\nu\lambda}\gamma^{\lambda}=-i\gamma_5(\Omega^g\wedge\gamma+\gamma\wedge\Omega^g)=i[\gamma,\gamma_5\Omega^g].$	$G:=G_{\alpha}\gamma^{\alpha}:=\frac{1}{2}R^{\mu\nu}\wedge\eta_{\mu\nu\lambda}\gamma^{\lambda}=-i\gamma_5(\Omega^g\wedge\gamma+\gamma\wedge\Omega^g)=i[\gamma,\gamma_5\Omega^g].$	$G:=G_{\alpha}\gamma^{\alpha}:=\frac{1}{2}R^{\mu\nu}\wedge\eta_{\mu\nu\lambda}\gamma^{\lambda}=-i\gamma_5(\Omega^g\wedge\gamma+\gamma\wedge\Omega^g)=i[\gamma,\gamma_5\Omega^g].$
mielke-geometrodynamics_00926 (14.1.7)	303	■ 0.960 N +0	$D\left(\vartheta\wedge\Omega^g\right)\equiv\Theta\wedge\Omega^g.$	$D\left(\vartheta\wedge\Omega^g\right)\equiv\Theta\wedge\Omega^g.$	$D(\vartheta\wedge\Omega^g)\equiv\Theta\wedge\Omega^g.$
mielke-geometrodynamics_00248 (4.3.13)	092	■ 0.961 N +0	$D\left(\vartheta\wedge\Omega^g\right)\equiv\Theta\wedge\Omega^g.$	$D\phi^{(\circ)}=\vartheta^{\alpha}\nabla_{\alpha}\phi^{(\circ)}.$	$D\phi^{(\circ)}=\vartheta^{\alpha}\nabla_{\alpha}\phi^{(\circ)}.$
mielke-geometrodynamics_00282 (4.4.11)	098	■ 0.961 N +1	$\begin{aligned} d\vartheta &= d\vartheta - d[\omega^L, \vartheta] \\ &= -[d\omega^L, \vartheta] + [\omega^L, d\vartheta] \\ &= -[\Omega^L, \vartheta] - [\omega^L \wedge \omega^L, \vartheta] + [\omega^L, \Theta] + [\omega^L, [\omega^L, \vartheta]] \\ &= [\omega^L, \Theta] + [\vartheta, \Omega^L] \end{aligned}$	$\begin{aligned} d\Theta &= dd\vartheta - d[\omega^L, \vartheta] \\ &= -[d\omega^L, \vartheta] + [\omega^L, d\vartheta] \\ &= -[\Omega^L, \vartheta] - [\omega^L \wedge \omega^L, \vartheta] + [\omega^L, \Theta] + [\omega^L, [\omega^L, \vartheta]] \\ &= [\omega^L, \Theta] + [\vartheta, \Omega^L] \end{aligned}$	$\begin{aligned} d\Theta &= dd\vartheta - d[\omega^L, \vartheta] \\ &= -[d\omega^L, \vartheta] + [\omega^L, d\vartheta] \\ &= -[\Omega^L, \vartheta] - [\omega^L \wedge \omega^L, \vartheta] + [\omega^L, \Theta] + [\omega^L, [\omega^L, \vartheta]] \\ &= [\omega^L, \Theta] + [\vartheta, \Omega^L]. \end{aligned}$
mielke-geometrodynamics_00910 (2.2.5)	034	■ 0.962 N +0	$T_p^q(M) := \bigcup_{m \in M} \otimes^p T_m^*(M) \otimes^q T_m(M).$	$T_p^q(M) := \bigcup_{m \in M} \otimes^p T_m^*(M) \otimes^q T_m(M).$	$T_p^q(M) := \bigcup_{m \in M} \otimes^p T_m^*(M) \otimes^q T_m(M).$
mielke-geometrodynamics_00340 (5.3.17)	115	■ 0.962 C +4	$\begin{aligned} G_{\alpha} &:= \frac{1}{2} R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \\ &= G_{\alpha}^{\{ \} } + \frac{(-1)^s}{12} (e_{\alpha} \rfloor \mathcal{A} \wedge {}^* \mathcal{A} - \frac{1}{3} \mathcal{A} \wedge e_{\alpha} \rfloor {}^* \mathcal{A}) + \frac{(-1)^s}{6} \vartheta_{\alpha} \wedge d\mathcal{A} \end{aligned}$	$\begin{aligned} G_{\alpha} &:= \frac{1}{2} R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \\ &= G_{\alpha}^{\{ \} } + \frac{(-1)^s}{12} (e_{\alpha} \rfloor \mathcal{A} \wedge {}^* \mathcal{A} - \frac{1}{3} \mathcal{A} \wedge e_{\alpha} \rfloor {}^* \mathcal{A}) + \frac{(-1)^s}{6} \vartheta_{\alpha} \wedge d\mathcal{A} \end{aligned}$	$\begin{aligned} G_{\alpha} &:= \frac{1}{2} R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \\ &= G_{\alpha}^{\{ \} } + \frac{(-1)^s}{12} \left(e_{\alpha} \rfloor \mathcal{A} \wedge {}^* \mathcal{A} - \frac{1}{3} \mathcal{A} \wedge e_{\alpha} \rfloor {}^* \mathcal{A} \right) + \frac{(-1)^s}{6} \vartheta_{\alpha} \wedge d\mathcal{A} \end{aligned} \tag{5.3}$
mielke-geometrodynamics_00424 (6.7.6)	140	■ 0.962 N +0	$\begin{aligned} \frac{1}{2\kappa} T^{\gamma} \wedge \eta_{\alpha\beta\gamma} - \frac{D\theta_{\mathrm{T}}}{2\ell^2} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta} + D\theta_{\mathrm{L}} \wedge R_{\alpha\beta} &= \vartheta_{[\alpha} \wedge \mu_{\beta]} \\ &= \frac{1}{4} \vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge {}^* j_5. \end{aligned}$	$\begin{aligned} \frac{1}{2\kappa} T^{\gamma} \wedge \eta_{\alpha\beta\gamma} - \frac{D\theta_{\mathrm{T}}}{2\ell^2} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta} + D\theta_{\mathrm{L}} \wedge R_{\alpha\beta} &= \vartheta_{[\alpha} \wedge \mu_{\beta]} \\ &= \frac{1}{4} \vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge {}^* j_5. \end{aligned}$	$\begin{aligned} \frac{1}{2\kappa} T^{\gamma} \wedge \eta_{\alpha\beta\gamma} - \frac{D\theta_{\mathrm{T}}}{2\ell^2} \wedge \vartheta_{\alpha} \wedge \vartheta_{\beta} + D\theta_{\mathrm{L}} \wedge R_{\alpha\beta} &= \vartheta_{[\alpha} \wedge \mu_{\beta]} \\ &= \frac{1}{4} \vartheta_{\alpha} \wedge \vartheta_{\beta} \wedge {}^* j_5. \end{aligned}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_255 EQ0809 (11.4.21)		■0.962 ●W +0	$\begin{array}{c} \text{i } \gamma^0 \vec{\gamma} \cdot \vec{\partial} \left[\begin{array}{cc} i H(\rho) & \chi_{\kappa}^m \\ -F(\rho) & \chi_{-\kappa}^m \end{array} \right] = \frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } \left(\partial_r - \frac{\vec{\sigma} \cdot \vec{L}}{r} \right) \left[\begin{array}{cc} i F(\rho) & \chi_{-\kappa}^m \\ H(\rho) & \chi_{\kappa}^m \end{array} \right] . \\ \begin{array}{c} \text{i } F(\rho) \text{ \& } \chi_{-\kappa}^m \\ \text{H}(\rho) \text{ \& } \chi_{\kappa}^m \end{array} \end{array} \right] .$	$i \gamma^0 \vec{\gamma} \cdot \vec{\partial} \left[\begin{array}{cc} i H(\rho) & \chi_{\kappa}^m \\ -F(\rho) & \chi_{-\kappa}^m \end{array} \right] = \frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } \left(\partial_r - \frac{\vec{\sigma} \cdot \vec{L}}{r} \right) \left[\begin{array}{cc} i F(\rho) & \chi_{-\kappa}^m \\ H(\rho) & \chi_{\kappa}^m \end{array} \right] .$	$i \gamma^0 \vec{\gamma} \cdot \vec{\partial} \left[\begin{array}{cc} i H(\rho) & \chi_{\kappa}^m \\ -F(\rho) & \chi_{-\kappa}^m \end{array} \right] = \frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } \left(\partial_r - \frac{\vec{\sigma} \cdot \vec{L}}{r} \right) \left[\begin{array}{cc} i F(\rho) & \chi_{-\kappa}^m \\ H(\rho) & \chi_{\kappa}^m \end{array} \right] .$
mielke-geometrodynamics_156 EQ0475 (7.4.3)		■0.963 ●	$\{ \}^{\ast} R_{\alpha \beta} = -C_{\alpha \beta} + \frac{1}{2} R i^{\tau}_{\alpha \beta} + \frac{R}{12} \vartheta_{\alpha} \wedge \vartheta_{\beta} .$	${}^* R_{\alpha \beta}^{(\star)} = -C_{\alpha \beta} + \frac{1}{2} R i^{\tau}_{\alpha \beta} + \frac{R}{12} \vartheta_{\alpha} \wedge \vartheta_{\beta} .$	${}^* R_{\alpha \beta}^{(\star)} = -C_{\alpha \beta} + \frac{1}{2} R j^{\tau}_{\alpha \beta} + \frac{R}{12} \vartheta_{\alpha} \wedge \vartheta_{\beta} .$
mielke-geometrodynamics_159 EQ0485 (7.7.1)		■0.963 ●W -1	$g_{ij} \rightarrow \widetilde{g}_{ij} = g_{ij} + a \operatorname{Ric}_{ij} + b g_{ij} \operatorname{Ric}_k^k$	$g_{ij} \rightarrow \widetilde{g}_{ij} = g_{ij} + a \operatorname{Ric}_{ij} + b g_{ij} \operatorname{Ric}_k^k$	$g_{ij} \rightarrow \widetilde{g}_{ij} = g_{ij} + a \operatorname{Ric}_{ij} + b g_{ij} \operatorname{Ric}_k^k$
mielke-geometrodynamics_376 EQ1143 (C.11)		■0.963 ●N +0	$(A, B) := \operatorname{Tr}(\operatorname{Ad} A \operatorname{Ad} B) = c_{ik}^i c_{si}^k A^i B^s =: \overset{\circ}{g}_{ts} A^t B^s$	$(A, B) := \operatorname{Tr}(\operatorname{Ad} A \operatorname{Ad} B) = c_{ik}^i c_{si}^k A^i B^s =: \overset{\circ}{g}_{ts} A^t B^s$	$(A, B) := \operatorname{Tr}(\operatorname{Ad} A \operatorname{Ad} B) = c_{ik}^i c_{si}^k A^i B^s =: \overset{\circ}{g}_{ts} A^t B^s$
mielke-geometrodynamics_372 EQ1122 (B.34)		■0.963 ●K +0	$H^p(S^n)=0 \quad \text{for} \quad 1 \leq p \leq n-1.$	$H^p(S^n)=0 \quad \text{for} \quad 1 \leq p \leq n-1.$	$H^p(S^n)=0 \quad \text{for} \quad 1 \leq p \leq n-1.$
mielke-geometrodynamics_252 EQ0791 (11.4.3)		■0.964 ●N +1	$\left\{ i \gamma^k \nabla_k^{(t)} + \frac{3\varepsilon}{8} \ell^2 (\bar{\psi} \psi - a \bar{\psi} \gamma^5 \psi \gamma^5) - m \right\} \psi = 0$	$\left\{ i \gamma^k \nabla_k^{(t)} + \frac{3\varepsilon}{8} \ell^2 (\bar{\psi} \psi - a \bar{\psi} \gamma^5 \psi \gamma^5) - m \right\} \psi = 0$	$\left\{ i \gamma^k \nabla_k^{(t)} + \frac{3\varepsilon}{8} \ell^2 (\bar{\psi} \psi - a \bar{\psi} \gamma^5 \psi \gamma^5) - m \right\} \psi = 0$
mielke-geometrodynamics_185 EQ0575 (8.6.13)		■0.964 ●N -1	$R := e_{\beta} \rfloor e_{\alpha} \rfloor R^{\alpha \beta} = \frac{6(\theta_{\mathrm{T}}^{\star} - a_0)}{\ell^2(\theta_{\mathrm{L}}^{\star} - b_6)}$	$R := e_{\beta} \rfloor e_{\alpha} \rfloor R^{\alpha \beta} = \frac{6(\theta_{\mathrm{T}}^{\star} - a_0)}{\ell^2(\theta_{\mathrm{L}}^{\star} - b_6)}$	$R := e_{\beta} \rfloor e_{\alpha} \rfloor R^{\alpha \beta} = \frac{6(\theta_{\mathrm{T}}^{\star} - a_0)}{\ell^2(\theta_{\mathrm{L}}^{\star} - b_6)}$
mielke-geometrodynamics_321 EQ0996 (14.3.9)		■0.965 ●S -2	$i \widehat{D} \widehat{\psi} := i \widehat{\gamma} \wedge^* \widehat{D} \widehat{\psi} = i \gamma \wedge^* D \widehat{\psi} + i \mathcal{D}^{(int)} \widehat{\psi} = 0,$	$i \widehat{D} \widehat{\psi} := i \widehat{\gamma} \wedge^* \widehat{D} \widehat{\psi} = i \gamma \wedge^* D \widehat{\psi} + i \mathcal{D}^{(int)} \widehat{\psi} = 0,$	$i \widehat{D} \widehat{\psi} := i \widehat{\gamma} \wedge^* \widehat{D} \widehat{\psi} = i \gamma \wedge^* D \widehat{\psi} + i \mathcal{D}^{(int)} \widehat{\psi} = 0,$
mielke-geometrodynamics_062 EQ0131 (3.4.2)		■0.966 ●N -1	${}^* \mathcal{F} := \frac{1}{\alpha_g} \operatorname{Tr} \{ {}^* (\Omega \wedge \Omega) \} = \frac{1}{4 \alpha_g} F_{\alpha \beta} {}^j {}^* F^{\alpha \beta}{}_j .$	${}^* \mathcal{F} := \frac{1}{\alpha_g} \operatorname{Tr} \{ {}^* (\Omega \wedge \Omega) \} = \frac{1}{4 \alpha_g} F_{\alpha \beta} {}^j {}^* F^{\alpha \beta}{}_j .$	${}^* \mathcal{F} := \frac{1}{\alpha_g} \operatorname{Tr} \{ {}^* (\Omega \wedge \Omega) \} = \frac{1}{4 \alpha_g} F_{\alpha \beta} {}^j {}^* F^{\alpha \beta}{}_j .$
Conf. MathPix confidence: ■ ■ ■ — ≥0.90.5–0.9<0.5no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● ● ● ● ● ● C componentW weakS stableN noiseK cleannot measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_366 EQ1089 (A.16)		■ 0.967 ● N +0	$\begin{aligned} & \left(\widehat{\gamma}^A\right)^{\mathrm{T}} := \left(\widehat{\gamma}^{\mu}, \widehat{\gamma}^{2p+1}, \widehat{\gamma}^{2p+2}\right), \\ & \widehat{\gamma}^{\mu} := \gamma^{\mu} \otimes \left[\bigotimes_{i=1}^{\left[\frac{n-2}{2}\right]} \sigma^1\right], \\ & \widehat{\gamma}^{2p+1} := i \mathbb{1} \otimes \left(\bigotimes_{i=1}^{p-2} \sigma^0\right) \otimes \sigma^3 \otimes \left(\bigotimes_{j=1}^{l-p+1} \sigma^1\right), \\ & \widehat{\gamma}^{2p+2} := i \mathbb{1} \otimes \left(\bigotimes_{i=1}^{p-2} \sigma^0\right) \otimes \sigma^2 \otimes \left(\bigotimes_{j=1}^{l-p+2} \sigma^1\right), \end{aligned}$	$\begin{aligned} & \left(\widehat{\gamma}^A\right)^{\mathrm{T}} := \left(\widehat{\gamma}^{\mu}, \widehat{\gamma}^{2p+1}, \widehat{\gamma}^{2p+2}\right), \\ & \widehat{\gamma}^{\mu} := \gamma^{\mu} \otimes \left[\bigotimes_{i=1}^{\left[\frac{n-2}{2}\right]} \sigma^1\right], \\ & \widehat{\gamma}^{2p+1} := i \mathbb{1} \otimes \left(\bigotimes_{i=1}^{p-2} \sigma^0\right) \otimes \sigma^3 \otimes \left(\bigotimes_{j=1}^{l-p+1} \sigma^1\right), \\ & \widehat{\gamma}^{2p+2} := i \mathbb{1} \otimes \left(\bigotimes_{i=1}^{p-2} \sigma^0\right) \otimes \sigma^2 \otimes \left(\bigotimes_{j=1}^{l-p+2} \sigma^1\right), \end{aligned}$	$\begin{aligned} & \left(\widehat{\gamma}^A\right)^{\mathrm{T}} := \left(\widehat{\gamma}^{\mu}, \widehat{\gamma}^{2p+1}, \widehat{\gamma}^{2p+2}\right), \\ & \widehat{\gamma}^{\mu} := \gamma^{\mu} \otimes \left[\bigotimes_{i=1}^{\left[\frac{n-2}{2}\right]} \sigma^1\right], \\ & \widehat{\gamma}^{2p+1} := i \mathbb{1} \otimes \left(\bigotimes_{i=1}^{p-2} \sigma^0\right) \otimes \sigma^3 \otimes \left(\bigotimes_{j=1}^{l-p+1} \sigma^1\right), \\ & \widehat{\gamma}^{2p+2} := i \mathbb{1} \otimes \left(\bigotimes_{i=1}^{p-2} \sigma^0\right) \otimes \sigma^2 \otimes \left(\bigotimes_{j=1}^{l-p+2} \sigma^1\right), \end{aligned}$
mielke-geometrodynamics_367 EQ1090 (A.17)		■ 0.967 ●	$\stackrel{*}{\alpha}\left(e_{\alpha_1}, \ldots, e_{\alpha_p}\right):=\alpha\left(\sigma_{*} e_{\alpha_1}, \ldots, \sigma_{*} e_{\alpha_p}\right) .$	$\overset{*}{\sigma}\alpha\left(e_{\alpha_1}, \ldots, e_{\alpha_p}\right):=\alpha\left(\sigma_{*} e_{\alpha_1}, \ldots, \sigma_{*} e_{\alpha_p}\right) .$	$\overset{*}{\sigma}\alpha\left(e_{\alpha_1}, \ldots, e_{\alpha_p}\right):=\alpha\left(\sigma_{*} e_{\alpha_1}, \ldots, \sigma_{*} e_{\alpha_p}\right) .$
mielke-geometrodynamics_195 EQ0592 (9.1.10)		■ 0.968 ● N +0	$\stackrel{+}{\Omega}=\frac{\gamma\kappa}{\ell^{*2}}\vartheta\wedge\vartheta.$	$\overset{+}{\Omega}^g=\frac{\gamma\kappa}{\ell^{*2}}\vartheta\wedge\vartheta.$	$\overset{+}{\Omega}^g=\frac{\gamma\kappa}{\ell^{*2}}\vartheta\wedge\vartheta.$
mielke-geometrodynamics_202 EQ0611 (9.3.2)		■ 0.968 ● K +0	$L_{\mathrm{SKY}}:=-\frac{1}{2}R_{\alpha\beta}\wedge{}^*R^{\alpha\beta},$	$L_{\mathrm{SKY}}:=-\frac{1}{2}R_{\alpha\beta}\wedge{}^*R^{\alpha\beta},$	$L_{\mathrm{SKY}}:=-\frac{1}{2}R_{\alpha\beta}\wedge{}^*R^{\alpha\beta},$
mielke-geometrodynamics_272 EQ0841 (12.2.9)		■ 0.968 ● K +0	$\mu_{\alpha}=\frac{1}{4}\vartheta_{\alpha}\wedge{}^*j_5.$	$\mu_{\alpha}=\frac{1}{4}\vartheta_{\alpha}\wedge{}^*j_5.$	$\mu_{\alpha}=\frac{1}{4}\vartheta_{\alpha}\wedge{}^*j_5.$
mielke-geometrodynamics_212 EQ0636 (10.2.7)		■ 0.970 ● K +0	$T_{\alpha}=\frac{\varepsilon}{\ell}\eta_{\alpha},\quad R_{\alpha}^{\star}=\frac{\rho}{\ell^2}\eta_{\alpha},$	$T_{\alpha}=\frac{\varepsilon}{\ell}\eta_{\alpha},\quad R_{\alpha}^{\star}=\frac{\rho}{\ell^2}\eta_{\alpha},$	$T_{\alpha}=\frac{\varepsilon}{\ell}\eta_{\alpha},\quad R_{\alpha}^{\star}=\frac{\rho}{\ell^2}\eta_{\alpha},$
mielke-geometrodynamics_376 EQ1137 (C.5)		■ 0.970 ● N +0	$3\,c_{[i s}{}^pc_{ jk]}{}^s:=c_{is}{}^pc_{jk}{}^s+c_{js}{}^pc_{ki}{}^s+c_{ks}{}^pc_{ij}{}^s\equiv0.$	$3c_{[i s}{}^pc_{ jk]}{}^s:=c_{is}{}^pc_{jk}{}^s+c_{js}{}^pc_{ki}{}^s+c_{ks}{}^pc_{ij}{}^s\equiv0.$	$3c_{[i s}{}^pc_{ jk]}{}^s:=c_{is}{}^pc_{jk}{}^s+c_{js}{}^pc_{ki}{}^s+c_{ks}{}^pc_{ij}{}^s\equiv0.$
mielke-geometrodynamics_137 EQ0415 (6.5.3)		■ 0.970 ● N +0	$\begin{aligned} & \langle 0 W(\gamma) 0\rangle^{(1)}=\int_{\mathcal{C} \mathcal{G}}\mathcal{D}Ae^{iC}W(A,\gamma) \\ & =\frac{1}{4\pi}\oint ds\oint dt\dot{\gamma}^{\mu}(s)\dot{\gamma}^{\nu}(t)\eta_{0\mu\nu\kappa}\frac{\gamma^{\kappa}(s)-\gamma^{\kappa}(t)}{ \gamma(s)-\gamma(t) ^3} \end{aligned}$	$\begin{aligned} & \langle 0 W(\gamma) 0\rangle^{(1)}=\int_{\mathcal{C} \mathcal{G}}\mathcal{D}Ae^{iC}W(A,\gamma) \\ & =\frac{1}{4\pi}\oint ds\oint dt\dot{\gamma}^{\mu}(s)\dot{\gamma}^{\nu}(t)\eta_{0\mu\nu\kappa}\frac{\gamma^{\kappa}(s)-\gamma^{\kappa}(t)}{ \gamma(s)-\gamma(t) ^3} \end{aligned}$	$\begin{aligned} & \langle 0 W(\gamma) 0\rangle^{(1)}=\int_{\mathcal{C} \mathcal{G}}\mathcal{D}Ae^{iC}W(A,\gamma) \\ & =\frac{1}{4\pi}\oint ds\oint dt\dot{\gamma}^{\mu}(s)\dot{\gamma}^{\nu}(t)\eta_{0\mu\nu\kappa}\frac{\gamma^{\kappa}(s)-\gamma^{\kappa}(t)}{ \gamma(s)-\gamma(t) ^3} \end{aligned}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured
[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_000495 (7.7.13)	162	0.971 N +0	$\{ \}^{\ast}\left(\backslash\vartheta^{\alpha}\backslash\wedge\backslash\vartheta^{\beta}\backslash\wedge\backslash\vartheta^{\gamma}\backslash\wedge\backslash\vartheta^{\delta}\right)=\eta^{\alpha\beta\gamma\delta},\backslash\text{\textit{ where }}\eta_{\alpha\beta\gamma\delta}:=+\delta_{\alpha\beta\gamma\delta}^{0123},$	$\left(\vartheta^{\alpha}\wedge\vartheta^{\beta}\wedge\vartheta^{\gamma}\wedge\vartheta^{\delta}\right)=\eta^{\alpha\beta\gamma\delta},\text{ where }\eta_{\alpha\beta\gamma\delta}:=+\delta_{\alpha\beta\gamma\delta}^{0123},$	$\left(\vartheta^{\alpha}\wedge\vartheta^{\beta}\wedge\vartheta^{\gamma}\wedge\vartheta^{\delta}\right)=\eta^{\alpha\beta\gamma\delta},\text{ where }\eta_{\alpha\beta\gamma\delta}:=+\delta_{\alpha\beta\gamma\delta}^{0123},$
mielke-geometrodynamics_001147 (C.16)	377	0.971 W +0	$\begin{aligned} &\left\{\left(\lambda_i,\lambda_j\right)\right\}_{i,j=1}^N=\left\{\left(c_{ij}^k\right)\right\}_{i,j=1}^N,\\ &\left\{\lambda_i,\lambda_j\right\}_{+}=\frac{4}{N}\delta_{ij}+2d_{ij}^k\lambda_k. \end{aligned}$	$\left[\lambda_i,\lambda_j\right]=2if_{ij}^k\lambda_k\quad\left(c_{ij}^k=2if_{ig}^k\right),$ $\left\{\lambda_i,\lambda_j\right\}_{+}=\frac{4}{N}\delta_{ij}+2d_{ij}^k\lambda_k.$	$\left[\lambda_i,\lambda_j\right]=2if_{ij}^k\lambda_k\quad\left(c_{ij}^k=2if_{ig}^k\right),$ $\left\{\lambda_i,\lambda_j\right\}_{+}=\frac{4}{N}\delta_{ij}+2d_{ij}^k\lambda_k.$
mielke-geometrodynamics_000129 (3.3.30)	061	0.972 C -6	$\begin{aligned} j:=\frac{\delta L_{\mathrm{matter}}}{\delta A}&=\frac{\partial L_{\mathrm{matter}}}{\partial A}+d\frac{\partial L_{\mathrm{matter}}}{\partial dA}\\ &=\Psi\wedge\frac{\partial L_{\mathrm{matter}}}{\partial D\Psi}+D\frac{\partial L_{\mathrm{matter}}}{\partial B}, \end{aligned}$	$j:=\frac{\delta L_{\mathrm{matter}}}{\delta A}=\frac{\partial L_{\mathrm{matter}}}{\partial A}+d\frac{\partial L_{\mathrm{matter}}}{\partial dA}$ $=\Psi\wedge\frac{\partial L_{\mathrm{matter}}}{\partial D\Psi}+D\frac{\partial L_{\mathrm{matter}}}{\partial B},$	$j:=\frac{\delta L_{\mathrm{matter}}}{\delta A}=\frac{\partial L_{\mathrm{matter}}}{\partial A}+d\frac{\partial L_{\mathrm{matter}}}{\partial dA}$ $=\Psi\wedge\frac{\partial L_{\mathrm{matter}}}{\partial D\Psi}+D\frac{\partial L_{\mathrm{matter}}}{\partial B},$
mielke-geometrodynamics_000784 (11.3.3)	248	0.972	$\begin{aligned} L_{\mathrm{NL}}&=\frac{1}{2}\bar{\psi}\left[\tilde{K},\gamma\right]\psi=-3i^{\ast}[\Theta,\vartheta]\wedge(\bar{\psi}^{\ast}\gamma\psi)\\ &=-3\tau_s\wedge[\Theta,\vartheta]. \end{aligned}$	$L_{\mathrm{NL}}=\frac{1}{2}\bar{\psi}\left[\tilde{K},\gamma\right]\psi=-3i^{\ast}[\Theta,\vartheta]\wedge(\bar{\psi}^{\ast}\gamma\psi)$ $=-3\tau_s\wedge[\Theta,\vartheta].$	$L_{\mathrm{NL}}=\frac{1}{2}\bar{\psi}[\tilde{K},\gamma]\psi=-3i^{\ast}[\Theta,\vartheta]\wedge(\bar{\psi}^{\ast}\gamma\psi)$ $=-3\tau_s\wedge[\Theta,\vartheta].$
mielke-geometrodynamics_000666 (10.3.23)	218	0.972 N +0	$\square\vartheta_{\alpha}+\frac{\ell}{\gamma}{}^{\ast}D^{\ast}(\delta\ell E_{\alpha}+S_{\alpha})=0,$	$\square\vartheta_{\alpha}+\frac{\ell}{\gamma}{}^{\ast}D^{\ast}(\delta\ell E_{\alpha}+S_{\alpha})=0,$	$\square\vartheta_{\alpha}+\frac{\ell}{\gamma}{}^{\ast}D^{\ast}(\delta\ell E_{\alpha}+S_{\alpha})=0,$
mielke-geometrodynamics_000945 (14.1.26)	306	0.972 W +0	$\begin{aligned} &\Sigma_{\mathrm{YM}}(\Omega)\wedge\Sigma_{\mathrm{YM}}(\Omega)=\frac{1}{4\alpha_g}\mathrm{Tr}\left\{-\xi\wedge\xi\wedge(\vartheta\wedge^{\ast}\xi)\wedge(\vartheta\wedge^{\ast}\xi)\right.\\ &\left.+\vartheta\wedge\vartheta\wedge^{\ast}(\xi\wedge^{\ast}\xi)^2-\vartheta\wedge^{\ast}\xi\wedge^{\ast}(\vartheta\wedge^{\ast}\xi)\wedge^{\ast}(\xi\wedge^{\ast}\xi)\right\}\\ &=\vartheta\wedge\vartheta\wedge^{\ast}(\Sigma_{\mathrm{YM}}(\xi)\wedge^{\ast}\Sigma_{\mathrm{YM}}(\xi)) \end{aligned}$	$\Sigma_{\mathrm{YM}}(\Omega)\wedge\Sigma_{\mathrm{YM}}(\Omega)=\frac{1}{4\alpha_g}\mathrm{Tr}\left\{-\xi\wedge\xi\wedge(\vartheta\wedge^{\ast}\xi)\wedge(\vartheta\wedge^{\ast}\xi)\right.$ $\left.+\vartheta\wedge\vartheta\wedge^{\ast}(\xi\wedge^{\ast}\xi)^2-\vartheta\wedge^{\ast}\xi\wedge^{\ast}(\vartheta\wedge^{\ast}\xi)\wedge^{\ast}(\xi\wedge^{\ast}\xi)\right\}$ $=\vartheta\wedge\vartheta\wedge^{\ast}(\Sigma_{\mathrm{YM}}(\xi)\wedge^{\ast}\Sigma_{\mathrm{YM}}(\xi))$	$\Sigma_{\mathrm{YM}}(\Omega)\wedge\Sigma_{\mathrm{YM}}(\Omega)=\frac{1}{4\alpha_g}\mathrm{Tr}\left\{-\xi\wedge\xi\wedge(\vartheta\wedge^{\ast}\xi)\wedge(\vartheta\wedge^{\ast}\xi)\right.$ $\left.+\vartheta\wedge\vartheta\wedge^{\ast}(\xi\wedge^{\ast}\xi)^2-\vartheta\wedge^{\ast}\xi\wedge^{\ast}(\vartheta\wedge^{\ast}\xi)\wedge^{\ast}(\xi\wedge^{\ast}\xi)\right\}$ $=\vartheta\wedge\vartheta\wedge^{\ast}(\Sigma_{\mathrm{YM}}(\xi)\wedge^{\ast}\Sigma_{\mathrm{YM}}(\xi))$
mielke-geometrodynamics_000462 (7.3.1)	153	0.973 N +0	$L_{\mathrm{SKY}}=\mathrm{Tr}\left(\Omega^g\wedge^{\ast}\Omega^g\right)=-\frac{1}{2}R^{\alpha\beta}\wedge^{\ast}R_{\alpha\beta},$	$L_{\mathrm{SKY}}=\mathrm{Tr}\left(\Omega^g\wedge^{\ast}\Omega^g\right)=-\frac{1}{2}R^{\alpha\beta}\wedge^{\ast}R_{\alpha\beta},$	$L_{\mathrm{SKY}}=\mathrm{Tr}(\Omega^g\wedge^{\ast}\Omega^g)=-\frac{1}{2}R^{\alpha\beta}\wedge^{\ast}R_{\alpha\beta},$
mielke-geometrodynamics_000356 (6.2.7)	124	0.974 N +0	$D\,T^{\alpha}\equiv R_{\beta}^{\alpha}\wedge\vartheta^{\beta}$	$DT^{\alpha}\equiv R_{\beta}^{\alpha}\wedge\vartheta^{\beta}$	$DT^{\alpha}\equiv R_{\beta}^{\alpha}\wedge\vartheta^{\beta}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_304 EQ0931 (14.1.12)		■ 0.981 ● N +0	$\Omega \rightarrow e^{*\delta} \Omega := \Omega \cos \delta + (-1)^z {}^* \Omega \sin \delta.$	$\Omega \rightarrow e^{*\delta} \Omega := \Omega \cos \delta + (-1)^z {}^* \Omega \sin \delta.$	
mielke-geometrodynamics_219 EQ0671 (10.3.28)		■ 0.982 ● N +0	$\check{\mathscr{J}}_{\alpha}^{\star} := dH_{\alpha}^{\star} \cong \tau_{\alpha}^{\star} + E_{\alpha}^{\star} - (-1)^s \eta_{\alpha\beta\gamma} \Gamma^{\star\beta} \wedge H^{\star\gamma},$	$\check{\mathscr{J}}_{\alpha}^{\star} := dH_{\alpha}^{\star} \cong \tau_{\alpha}^{\star} + E_{\alpha}^{\star} - (-1)^s \eta_{\alpha\beta\gamma} \Gamma^{\star\beta} \wedge H^{\star\gamma},$	
mielke-geometrodynamics_366 EQ1086 (A.12)		■ 0.982 ● W -1	$\begin{aligned} & \mathrm{C}(1,1)=\mathrm{C}(2,0)=\mathrm{GL}(2,\mathbb{R}), \\ & \mathrm{C}(0,2)=\mathbb{H} \quad (\text{field of quaternions}), \end{aligned}$	$\begin{aligned} & \mathrm{C}(1,1)=\mathrm{C}(2,0)=\mathrm{GL}(2,\mathbb{R}), \\ & \mathrm{C}(0,2)=\mathbb{H} \quad (\text{field of quaternions}), \end{aligned}$	
mielke-geometrodynamics_088 EQ0229 (4.2.22)		■ 0.982 ● K +0	$\Theta:=D\vartheta=d\vartheta-\left[\omega^L,\vartheta\right]$	$\Theta:=D\vartheta=d\vartheta-\left[\omega^L,\vartheta\right]$	
mielke-geometrodynamics_271 EQ0836 (12.2.4)		■ 0.983 ● N +0	$\mathscr{A}:=\frac{1}{4}\mathrm{Tr}(\gamma\wedge D\gamma)={}^*(\vartheta^{\alpha}\wedge T_{\alpha})=\frac{1}{2}T^{[\alpha\beta\gamma]}\eta_{\alpha\beta\gamma}=\mathscr{A}_idx^i,$	$\mathscr{A}:=\frac{1}{4}{}^*\mathrm{Tr}(\gamma\wedge D\gamma)={}^*(\vartheta^{\alpha}\wedge T_{\alpha})=\frac{1}{2}T^{[\alpha\beta\gamma]}\eta_{\alpha\beta\gamma}=\mathscr{A}_idx^i,$	
mielke-geometrodynamics_165 EQ0511 (7.7.29)		■ 0.983 ● W -1	$\begin{aligned} \Theta:=D\gamma=d\gamma+[\Gamma,\gamma]&=\left(d\vartheta^{\alpha}+\Gamma_{\beta}^{\alpha}\wedge\vartheta^{\beta}\right)\gamma_{\alpha} \\ &=T^{\alpha}\gamma_{\alpha}=\frac{1}{2}T_{ij}^{\alpha}\gamma_{\alpha}dx^i\wedge dx^j \end{aligned}$	$\begin{aligned} \Theta:=D\gamma=d\gamma+[\Gamma,\gamma]&=\left(d\vartheta^{\alpha}+\Gamma_{\beta}^{\alpha}\wedge\vartheta^{\beta}\right)\gamma_{\alpha} \\ &=T^{\alpha}\gamma_{\alpha}=\frac{1}{2}T_{ij}^{\alpha}\gamma_{\alpha}dx^i\wedge dx^j, \end{aligned}$	
mielke-geometrodynamics_212 EQ0637 (10.2.8)		■ 0.983 ● N +0	$\Gamma_{\alpha}^{\star}=\Gamma_{\alpha}^{\{\}\star}-K_{\alpha}^{\star}$	$\Gamma_{\alpha}^{\star}=\Gamma_{\alpha}^{\{\}\star}-K_{\alpha}^{\star}$	
mielke-geometrodynamics_096 EQ0271 (4.3.36)		■ 0.983 ●	$\left[D_{\alpha},D_{\beta}\right]=R_{\alpha\beta}^{..cd}L_{cd}-T_{\alpha\beta}^{..c}D_c,$	$\left[D_{\alpha},D_{\beta}\right]=R_{\alpha\beta}^{..cd}L_{cd}-T_{\alpha\beta}^{..c}D_c,$	
mielke-geometrodynamics_181 EQ0561 (8.5.4)		■ 0.984 ● N +0	$\{\}^{\{*\}}\sigma^{\{(\star)\}}=(-1)^{\mathrm{sig}}\sigma,$	${}^*\sigma^{(*)}=(-1)^{\mathrm{sig}}\sigma,$	
mielke-geometrodynamics_056 EQ0097 (3.2.7)		■ 0.985 ● K +0	$D\rightarrow^{G^{-1}}D=G^{-1}DG.$	$D\rightarrow^{G^{-1}}D=G^{-1}DG.$	
mielke-geometrodynamics_152 EQ0459 (7.2.11)		■ 0.985 ● N +0	$\Omega_{\mathrm{C}}=\stackrel{(+)}{\Omega}_{\mathrm{C}}+\stackrel{(-)}{\Omega}_{\mathrm{C}},\quad\stackrel{(\pm)}{\Omega}_{\mathrm{C}}:=\frac{1}{2}\left(\Omega_{\mathrm{C}}\pm\Omega_{\mathrm{C}}^{(*)}\right)$	$\varOmega_{\mathrm{C}}=\stackrel{(+)}{\varOmega}_{\mathrm{C}}+\stackrel{(-)}{\varOmega}_{\mathrm{C}},\quad\stackrel{(\pm)}{\varOmega}_{\mathrm{C}}:=\frac{1}{2}\left(\varOmega_{\mathrm{C}}\pm\varOmega_{\mathrm{C}}^{(*)}\right)$	
mielke-geometrodynamics_344 EQ1051 (15.2.8)		■ 0.985 ● N +0	$\begin{aligned} A_i&=\frac{1}{2}\left\{A_i^{(f)j}+{}^*A_i^{(f)j}\gamma^5\right\}\lambda_j^{(f)} \\ &\oplus\frac{1}{2}\left\{A_i^{(c)j}+{}^*A_i^{(c)j}\gamma^5\right\}\lambda_j^{(c)}, \end{aligned}$	$\begin{aligned} A_i&=\frac{1}{2}\left\{A_i^{(f)j}+{}^*A_i^{(f)j}\gamma^5\right\}\lambda_j^{(f)} \\ &\oplus\frac{1}{2}\left\{A_i^{(c)j}+{}^*A_i^{(c)j}\gamma^5\right\}\lambda_j^{(c)}, \end{aligned}$	
mielke-geometrodynamics_369 EQ1096 (B.6)		■ 0.985 ● N +0	$d\,d\,\alpha^{(p)}\equiv 0.$	$d\,d\alpha^{(p)}\equiv 0.$	
mielke-geometrodynamics_081 EQ0191 (4.1.1)		■ 0.985 ● N +0	$x\rightarrow x'=gx+a,\quad g\in GL(n,\mathbb{R}),\,a\in\mathbb{R}^n.$	$x\rightarrow x'=gx+a,\quad g\in GL(n,\mathbb{R}),\,a\in\mathbb{R}^n.$	

Conf. MathPix confidence:

■

≥0.9

■

0.5–0.9

■

<0.5

—

no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_224 EQ0694 (10.5.5)		■ 0.985 ● W +0	$\begin{aligned} L_{\mathrm{D}} &= \frac{i}{2} \left(\bar{\psi}^* \gamma \wedge D\psi + D\bar{\psi} \wedge^* \gamma \psi \right) - m \bar{\psi} \psi \eta \\ &= \bar{\psi} \left(i^* \gamma \wedge D - m \eta \right) \psi + d \left(\frac{i}{2} \bar{\psi}^* \gamma \psi \right), \end{aligned}$	$L_{\mathrm{D}} = \frac{i}{2} \left(\bar{\psi}^* \gamma \wedge D\psi + D\bar{\psi} \wedge^* \gamma \psi \right) - m \bar{\psi} \psi \eta$ $= \bar{\psi} \left(i^* \gamma \wedge D - m \eta \right) \psi + d \left(\frac{i}{2} \bar{\psi}^* \gamma \psi \right),$	$L_{\mathrm{D}} = \frac{i}{2} \left(\overline{\psi}^* \gamma \wedge D\psi + D\overline{\psi} \wedge^* \gamma \psi \right) - m \overline{\psi} \psi \eta$ $= \overline{\psi} \left(i^* \gamma \wedge D - m \eta \right) \psi + d \left(\frac{i}{2} \overline{\psi}^* \gamma \psi \right),$
mielke-geometrodynamics_180 EQ0554 (8.4.8)		■ 0.986 ● N +0	$D^* \stackrel{\pm}{\Omega} = 0$	$D^* \stackrel{\pm}{\Omega} = 0$	$D^* \stackrel{\pm}{\Omega} = 0$
mielke-geometrodynamics_045 EQ0058 (2.7.8)		■ 0.987 ● K +0	$b \rightarrow^G b := G b \quad \text{in } C^{\infty}(V^{\rho}).$	$b \rightarrow^G b := G b \quad \in C^{\infty}(V^{\rho}).$	$b \rightarrow^G b := G b \quad \in C^{\infty}(V^{\rho}).$
mielke-geometrodynamics_099 EQ0285 (4.4.14)		■ 0.987 ● N +0	$D \left(\vartheta \wedge^* \Omega^g \right) = \Theta \wedge^* \Omega^g$	$D \left(\vartheta \wedge^* \Omega^g \right) = \Theta \wedge^* \Omega^g$	$D \left(\vartheta \wedge^* \Omega^g \right) = \Theta \wedge^* \Omega^g$
mielke-geometrodynamics_271 EQ0834 (12.2.2)		■ 0.988 ● N +0	$i^* \gamma \wedge \left(\mathscr{D} + \frac{i}{4} m \gamma - \frac{1}{2} T \right) \psi = 0,$	$i^* \gamma \wedge \left(\mathscr{D} + \frac{i}{4} m \gamma - \frac{1}{2} T \right) \psi = 0,$	$i^* \gamma \wedge \left(\mathscr{D} + \frac{i}{4} m \gamma - \frac{1}{2} T \right) \psi = 0,$
mielke-geometrodynamics_031 EQ0002 (2.1.1)		■ 0.989 ● N +0	$x_2 \circ \stackrel{-1}{x}_1: \mathbb{R} \rightarrow \mathbb{R}$	$x_2 \circ \stackrel{-1}{x}_1: \mathbb{R} \rightarrow \mathbb{R}$	$x_2 \circ \stackrel{-1}{x}_1: \mathbb{R} \rightarrow \mathbb{R}$
mielke-geometrodynamics_114 EQ0339 (5.3.16)		■ 0.989 ● N +0	$L_{\mathrm{EC}} := -\frac{1}{2\kappa} R^{\alpha\beta} \wedge \eta_{\alpha\beta} = L_{\mathrm{HE}} + \frac{1}{12\kappa} \mathscr{A}^* \mathscr{A}$	$L_{\mathrm{EC}} := -\frac{1}{2\kappa} R^{\alpha\beta} \wedge \eta_{\alpha\beta} = L_{\mathrm{HE}} + \frac{1}{12\kappa} \mathscr{A}^* \mathscr{A}$	$L_{\mathrm{EC}} := -\frac{1}{2\kappa} R^{\alpha\beta} \wedge \eta_{\alpha\beta} = L_{\mathrm{HE}} + \frac{1}{12\kappa} \mathscr{A} \wedge^* \mathscr{A}$
mielke-geometrodynamics_039 EQ0024 (2.4.6)		■ 0.990 ● N +0	$[L_i, L_j] = c_{ij}^{k} L_k.$	$[L_i, L_j] = c_{ij}^{k} L_k.$	$[L_i, L_j] = c_{ij}^{k} L_k.$
mielke-geometrodynamics_164 EQ0507 (7.7.25)		■ 0.990 ● N +1	$\sigma_{\pm} := (\sigma \pm i^* \sigma) / 2 = \frac{1}{2} (1 \mp \gamma_5) \sigma, \quad \text{with} \quad i^* \sigma_{\pm} = \pm \sigma_{\pm}$	$\sigma_{\pm} := (\sigma \pm i^* \sigma) / 2 = \frac{1}{2} (1 \mp \gamma_5) \sigma, \quad \text{with} \quad i^* \sigma_{\pm} = \pm \sigma_{\pm}$	$\sigma_{\pm} := (\sigma \pm i^* \sigma) / 2 = \frac{1}{2} (1 \mp \gamma_5) \sigma, \quad \text{with} \quad i^* \sigma_{\pm} = \pm \sigma_{\pm}$
mielke-geometrodynamics_184 EQ0572 (8.6.10)		■ 0.990 ●	$\begin{aligned} H_{\alpha} &= \frac{\theta_{\mathrm{T}}^*}{2\ell^2} K^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} - \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} = \frac{1}{\ell^2} \left[\theta_{\mathrm{T}}^* K_{\alpha}^{(*)} - \theta_{\mathrm{T}} T_{\alpha} \right] \\ &= \frac{\theta_{\mathrm{T}}^*}{\ell^2} * \left[T_{\alpha} - \vartheta_{\alpha} \wedge (e_{\beta}] T^{\beta}) - \frac{1}{2} e_{\alpha} \right] (T^{\beta} \wedge \vartheta_{\beta}) \Big] - \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} \\ &\simeq -\partial L_{\theta} / \partial T^{\alpha}. \end{aligned}$	$\begin{aligned} H_{\alpha} &= \frac{\theta_{\mathrm{T}}^*}{2\ell^2} K^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} - \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} = \frac{1}{\ell^2} \left[\theta_{\mathrm{T}}^* K_{\alpha}^{(*)} - \theta_{\mathrm{T}} T_{\alpha} \right] \\ &= \frac{\theta_{\mathrm{T}}^*}{\ell^2} * \left[T_{\alpha} - \vartheta_{\alpha} \wedge (e_{\beta}] T^{\beta}) - \frac{1}{2} e_{\alpha} \right] (T^{\beta} \wedge \vartheta_{\beta}) \Big] - \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} \\ &\simeq -\partial L_{\theta} / \partial T^{\alpha}. \end{aligned}$	$\begin{aligned} H_{\alpha} &= \frac{\theta_{\mathrm{T}}^*}{2\ell^2} K^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} - \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} = \frac{1}{\ell^2} \left[\theta_{\mathrm{T}}^* K_{\alpha}^{(*)} - \theta_{\mathrm{T}} T_{\alpha} \right] \\ &= \frac{\theta_{\mathrm{T}}^*}{\ell^2} * \left[T_{\alpha} - \vartheta_{\alpha} \wedge (e_{\beta}] T^{\beta}) - \frac{1}{2} e_{\alpha} \right] (T^{\beta} \wedge \vartheta_{\beta}) \Big] - \frac{\theta_{\mathrm{T}}}{\ell^2} T_{\alpha} \\ &\simeq -\partial L_{\theta} / \partial T^{\alpha}. \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_058 E08110 (3.3.11)		■ 0.990 ● C -3	$\begin{aligned} F := & E \wedge dt + B \quad \wedge \quad \wedge \left(E_x dx + E_y dy + E_z dz \right) \wedge dt \\ & + B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy. \end{aligned}$	$F := E \wedge dt + B$ $= (E_x dx + E_y dy + E_z dz) \wedge dt$ $+ B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy.$	$F := E \wedge dt + B$ $= (E_x dx + E_y dy + E_z dz) \wedge dt$ $+ B_x dy \wedge dz + B_y dz \wedge dx + B_z dx \wedge dy.$
mielke-geometrodynamics_064 E08142 (3.4.13)		■ 0.990 ● W -2	$\begin{aligned} D \tau \wedge &= D \left(\frac{2}{\alpha_g} \left([\Omega, {}^* \Omega] \frac{\partial L}{\partial \mathcal{F}} + (-1)^s [\Omega, \Omega] \frac{\partial L}{\partial {}^* \mathcal{F}} \right) \right) \\ &= 0. \end{aligned}$	$D\tau = DD\Pi^\Omega = [\Omega, \Pi^\Omega]$ $= \frac{2}{\alpha_g} \left\{ [\Omega, {}^* \Omega] \frac{\partial L}{\partial \mathcal{F}} + (-1)^s [\Omega, \Omega] \frac{\partial L}{\partial {}^* \mathcal{F}} \right\} = 0.$	$D\tau = DD\Pi^\Omega = [\Omega, \Pi^\Omega]$ $= \frac{2}{\alpha_g} \left\{ [\Omega, {}^* \Omega] \frac{\partial L}{\partial \mathcal{F}} + (-1)^s [\Omega, \Omega] \frac{\partial L}{\partial {}^* \mathcal{F}} \right\} = 0.$
mielke-geometrodynamics_067 E08155 (3.5.3)		■ 0.990 ● K +0	$\stackrel{*}{\sigma} \Omega = \frac{1}{4} F_{\alpha\beta\mu\nu} \sigma^{\mu\nu} \otimes \vartheta^\alpha \wedge \vartheta^\beta$	$\stackrel{*}{\sigma} \Omega = \frac{1}{4} F_{\alpha\beta\mu\nu} \sigma^{\mu\nu} \otimes \vartheta^\alpha \wedge \vartheta^\beta$	$\stackrel{*}{\sigma} \Omega = \frac{1}{4} F_{\alpha\beta\mu\nu} \sigma^{\mu\nu} \otimes \vartheta^\alpha \wedge \vartheta^\beta$
mielke-geometrodynamics_221 E08682 (10.3.40)		■ 0.990 ● C -3	$\theta_L C_L - \tilde{\theta}_L \tilde{C}_L = L_{\text{MB}} + \frac{\Delta\theta_{\text{TL}}}{\ell} d(\Gamma^{\star\alpha} \wedge \vartheta_\alpha),$	$\theta_L C_L - \tilde{\theta}_L \tilde{C}_L = L_{\text{MB}} + \frac{\Delta\theta_{\text{TL}}}{\ell} d(\Gamma^{\star\alpha} \wedge \vartheta_\alpha),$	$\theta_L C_L - \tilde{\theta}_L \tilde{C}_L = L_{\text{MB}} + \frac{\Delta\theta_{\text{TL}}}{\ell} d(\Gamma^{\star\alpha} \wedge \vartheta_\alpha),$
mielke-geometrodynamics_284 E08874 (13.3.2)		■ 0.990 ● S -2	$\left[Z^A{}_B, Z^C{}_D \right] = \delta^A_D Z^C{}_B - \delta^C_B Z^A{}_D$	$[Z^A{}_B, Z^C{}_D] = \delta^A_D Z^C{}_B - \delta^C_B Z^A{}_D$	$[\mathcal{Z}^A{}_B, \mathcal{Z}^C{}_D] = \delta^A_D \mathcal{Z}^C{}_B - \delta^C_B \mathcal{Z}^A{}_D$
mielke-geometrodynamics_042 E08842 (2.6.5)		■ 0.991 ● C -3	$F_{\alpha\beta}{}^j = \partial_\alpha \Gamma_\beta{}^j - \partial_\beta \Gamma_\alpha{}^j - c_{kl}{}^j \Gamma_\alpha{}^k \Gamma_\beta{}^l,$	$F_{\alpha\beta}{}^j = \partial_\alpha \Gamma_\beta{}^j - \partial_\beta \Gamma_\alpha{}^j - c_{kl}{}^j \Gamma_\alpha{}^k \Gamma_\beta{}^l,$	$F_{\alpha\beta}{}^j = \partial_\alpha \Gamma_\beta{}^j - \partial_\beta \Gamma_\alpha{}^j - c_{kl}{}^j \Gamma_\alpha{}^k \Gamma_\beta{}^l,$
mielke-geometrodynamics_088 E08233 (4.2.26)		■ 0.991 ● N +0	$\Omega^T = f\Theta + df \wedge \vartheta - \Omega^L \otimes \tilde{\varphi}$	$\Omega^T = f\Theta + df \wedge \vartheta - \Omega^L \otimes \tilde{\varphi}$	$\Omega^T = f\Theta + df \wedge \vartheta - \Omega^L \otimes \tilde{\varphi}$
mielke-geometrodynamics_246 E08778		■ 0.991 ● N +0	$\langle -\infty D\tau_A \infty \rangle^{(+)} = \frac{1}{(4.)8\pi^2} \text{Tr} \left(\Omega^g \wedge \Omega^{g(*)} \right).$	$\langle -\infty D\tau_A \infty \rangle^{(+)} = \frac{1}{(4.)8\pi^2} \text{Tr} \left(\Omega^g \wedge \Omega^{g(*)} \right).$	$\langle -\infty D\tau_A \infty \rangle^{(+)} = \frac{1}{(4.)8\pi^2} \text{Tr} (\Omega^g \wedge \Omega^{g(*)}).$
mielke-geometrodynamics_214 E08644 (10.3.1)		■ 0.991 ● N +0	$\tilde{\Gamma}_\alpha^\star = \Gamma_\alpha^\star - (-1)^s \frac{\varepsilon}{2\ell} \vartheta_\alpha,$	$\tilde{\Gamma}_\alpha^\star = \Gamma_\alpha^\star - (-1)^s \frac{\varepsilon}{2\ell} \vartheta_\alpha,$	$\tilde{\Gamma}_\alpha^\star = \Gamma_\alpha^\star - (-1)^s \frac{\varepsilon}{2\ell} \vartheta_\alpha,$
mielke-geometrodynamics_065 E08149 (3.4.20)		■ 0.992 ●	$L(\mathcal{F}, {}^* \mathcal{F}) = -\mathcal{F}.$	$L(\mathcal{F}, {}^* \mathcal{F}) = -\mathcal{F}.$	$L(\mathcal{F}, {}^* \mathcal{F}) = -\mathcal{F}.$
mielke-geometrodynamics_127 E08368 (6.2.19)		■ 0.992 ● N -2	$\stackrel{(\pm)}{L}_\parallel := L_\parallel \pm i \frac{2\ell^2}{\kappa} dC_{\text{TT}} = L_{\text{HE}} - L_{\text{EC}} \pm i \frac{2\ell^2}{\kappa} d\tilde{C}_{\text{TT}}.$	$\stackrel{(\pm)}{L}_\parallel := L_\parallel \pm i \frac{2\ell^2}{\kappa} dC_{\text{TT}} = L_{\text{HE}} - L_{\text{EC}} \pm i \frac{2\ell^2}{\kappa} d\tilde{C}_{\text{TT}}.$	$\stackrel{(\pm)}{L}_\parallel := L_\parallel \pm i \frac{2\ell^2}{\kappa} dC_{\text{TT}} = L_{\text{HE}} - L_{\text{EC}} \pm i \frac{2\ell^2}{\kappa} d\tilde{C}_{\text{TT}}.$
mielke-geometrodynamics_151 E08449 (7.2.2)		■ 0.992 ● N +0	${}^* \Omega^{g(*)} := \frac{1}{4} {}^* R_{\alpha\beta}^{(*)cd} L_{cd} \otimes \vartheta^\alpha \wedge \vartheta^\beta,$	${}^* \Omega^{g(*)} := \frac{1}{4} {}^* R_{\alpha\beta}^{(*)cd} L_{cd} \otimes \vartheta^\alpha \wedge \vartheta^\beta,$	${}^* \Omega^{g(*)} := \frac{1}{4} {}^* R_{\alpha\beta}^{(*)cd} L_{cd} \otimes \vartheta^\alpha \wedge \vartheta^\beta,$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_EQ1050 (15.2.7)	344	■ 0.992 ● N +0	$\backslash\mathrm{left}\backslash\{\bar{L}_{\jmath}=\sigma^{\alpha\beta}\otimes\lambda_i\}\otimes\backslash\mathrm{lambda}_{\mathrm{i}}\},\backslash\mathrm{quad}\backslash\mathrm{lambda}_{\mathrm{i}}\},\backslash\mathrm{quad}\mathrm{i}\backslash\mathrm{gamma}^{\mathrm{5}}\backslash\mathrm{lambda}_{\mathrm{i}}\}\backslash\mathrm{mid}\mathrm{j}=1,\backslash\mathrm{ldots}8\mathrm{~N}^{\mathrm{2}}-1\backslash\mathrm{right}\backslash\},$	$\left\{\bar{L}_j=\sigma^{\alpha\beta}\otimes\lambda_i,\quad\lambda_i,\quad i\gamma^5\lambda_i\mid j=1,\dots 8N^2-1\right\},$	$\left\{\bar{L}_j=\sigma^{\alpha\beta}\otimes\lambda_i,\quad\lambda_i,\quad i\gamma^5\lambda_i\mid j=1,\dots 8N^2-1\right\},$
mielke-geometrodynamics_EQ0964 (14.2.4)	313	■ 0.992 ● N +0	$\backslash\mathrm{stackrel{rel}\{ \circ\}\{\omega\}}=\mathrm{b}\backslash\mathrm{stackrel{rel}\{ \circ\}\{\vartheta\}}\;.$	$\overset{\circ}{\omega}=b\overset{\circ}{\vartheta}\;.$	$\overset{\circ}{\omega}=b\overset{\circ}{\vartheta}\;.$
mielke-geometrodynamics_EQ0256 (4.3.21)	093	■ 0.992 ● W +0	$R_{\backslash\alpha\backslash\beta\mathrm{c}}^{\dots\mathrm{d}}=2\backslash\mathrm{left}\backslash\{\partial_{[\alpha}\Gamma_{\backslash\beta]\mathrm{c}}^{\dots\mathrm{d}}+\Gamma_{[\alpha \mathrm{h}}^{\dots\mathrm{d}}\Gamma_{ \backslash\beta]\mathrm{c}}^{\dots\mathrm{h}}\}\backslash\mathrm{Gamma}_{\backslash\beta\mathrm{c}}^{\mathrm{d}}\}\backslash\mathrm{Gamma}_{[\backslash\alpha\backslash\mathrm{mid}\mathrm{h}]^{\mathrm{d}}\backslash\beta\mathrm{c}}^{\dots\mathrm{h}}\}$	$R_{\alpha\beta c}^{\dots d}=2\left\{\partial_{[\alpha}\Gamma_{\beta]c}^{\dots d}+\Gamma_{[\alpha h}^{\dots d}\Gamma_{ \beta]c}^{\dots h}\right\}$	$R_{\alpha\beta c}^{\dots d}=2\left\{\partial_{[\alpha}\Gamma_{\beta]c}^{\dots d}+\Gamma_{[\alpha h}^{\dots d}\Gamma_{ \beta]c}^{\dots h}\right\}$
mielke-geometrodynamics_EQ0245 (4.3.10)	091	■ 0.992 ● N +0	$\backslash\mathrm{vartheta}\wedge^*\backslash\mathrm{vartheta}=\eta\;.$	$\vartheta\wedge^*\vartheta=\eta.$	$\vartheta\wedge^*\vartheta=\eta.$
mielke-geometrodynamics_EQ0633 (10.2.4)	212	■ 0.992 ● N +0	$-\left(-1\right)^{\mathrm{s}}\theta_{\mathrm{TL}}T_{\alpha}-\frac{1}{2\ell}\theta_{\mathrm{T}}\eta_{\alpha}-\theta_{\mathrm{L}}\ell R_{\alpha}^{\star}=\ell\tau_{\alpha}^{\star};$ $\ell\ell\theta_{\mathrm{T}}\eta_{\alpha}-\theta_{\mathrm{L}}\ell R_{\alpha}^{\star}=\ell\tau_{\alpha}^{\star};$	$-(-1)^s\theta_{\mathrm{TL}}T_{\alpha}-\frac{1}{2\ell}\theta_{\mathrm{T}}\eta_{\alpha}-\theta_{\mathrm{L}}\ell R_{\alpha}^{\star}=\ell\tau_{\alpha}^{\star};$	$-(-1)^s\theta_{\mathrm{TL}}T_{\alpha}-\frac{1}{2\ell}\theta_{\mathrm{T}}\eta_{\alpha}-\theta_{\mathrm{L}}\ell R_{\alpha}^{\star}=\ell\tau_{\alpha}^{\star};$
mielke-geometrodynamics_EQ0701 (10.5.12)	225	■ 0.992 ● N +0	$\hat{D}^*\hat{D}\psi-m^2\eta\psi+i^*\sigma\wedge\left(\hat{R}+T^{\alpha}\hat{D}_{\alpha}\right)\psi=0,$	$\hat{D}^*\hat{D}\psi-m^2\eta\psi+i^*\sigma\wedge\left(\hat{R}+T^{\alpha}\hat{D}_{\alpha}\right)\psi=0,$	$\hat{D}^*\hat{D}\psi-m^2\eta\psi+i^*\sigma\wedge\left(\hat{R}+T^{\alpha}\hat{D}_{\alpha}\right)\psi=0,$
mielke-geometrodynamics_EQ0847 (12.2.15)	273	■ 0.992 ● N +0	$\mathrm{\mathscr{A}}_{\backslash\alpha}\mathrm{\mathscr{A}}^{\backslash\alpha}\eta=\mathrm{\mathscr{A}}\wedge^*\mathrm{\mathscr{A}}\cong\left(\ell^4/4\right)^*j_5\wedge j_5=0.$	$\mathscr{A}_{\alpha}\mathscr{A}^{\alpha}\eta=\mathscr{A}\wedge^*\mathscr{A}\cong\left(\ell^4/4\right)^*j_5\wedge j_5=0.$	$\mathscr{A}_{\alpha}\mathscr{A}^{\alpha}\eta=\mathscr{A}\wedge^*\mathscr{A}\cong\left(\ell^4/4\right)^*j_5\wedge j_5=0.$
mielke-geometrodynamics_EQ0388 (6.4.3)	130	■ 0.993 ● N +0	$\backslash\mathrm{left}.\backslash\mathrm{underline}\{\Psi\}:=n\backslash\mathrm{rfloor}\left(\mathrm{d}\mathrm{t}\backslash\mathrm{wedge}\backslash\mathrm{Psi}\right)=\backslash\mathrm{left}\left(1-\backslash\mathrm{perp}\right)\backslash\mathrm{Psi},\backslash\mathrm{quad}n\backslash\mathrm{right}\backslash\mathrm{rfloor}\backslash\mathrm{underline}\{\Psi\}\backslash\mathrm{equiv}\mathrm{0}\;.$	$\underline{\Psi}:=n\rfloor(dt\wedge\Psi)=\left(1-\stackrel{\perp}{}\right)\Psi,\quad n\rfloor\underline{\Psi}\equiv 0.$	$\underline{\Psi}:=n\rfloor(dt\wedge\Psi)=\left(1-\stackrel{\perp}{}\right)\Psi,\quad n\rfloor\underline{\Psi}\equiv 0.$
mielke-geometrodynamics_EQ0228 (4.2.21)	088	■ 0.993 ● N +0	$\backslash\mathrm{widetilde}\{\backslash\mathrm{widetilde}\{\vartheta\}\}=f\backslash\mathrm{vartheta}\;.$	$\widetilde{\vartheta}=f\vartheta.$	$\widetilde{\vartheta}=f\vartheta.$
mielke-geometrodynamics_EQ0287 (4.4.16)	099	■ 0.993 ● N +0	$\mathrm{D}\mathrm{D}\backslash\mathrm{Pi}^{\backslash\mathrm{mathrm}\{\mathrm{L}\}}+\mathrm{D}\backslash\mathrm{Pi}_{\square}^{\backslash\mathrm{mathrm}\{\mathrm{T}\}}=\mathrm{D}\backslash\mathrm{tau}_{\mathrm{s}}\;.$	$DD\Pi^{\mathrm{L}}+D\Pi_{\square}^{\mathrm{T}}=D\tau_s.$	$D\,D\Pi^{\mathrm{L}}+D\Pi_{\square}^{\mathrm{T}}=D\tau_s.$
mielke-geometrodynamics_EQ0320 (5.2.4)	111	■ 0.993 ● N +0	$\mathrm{\mathscr{A}}:=\frac{1}{4}\mathrm{\backslash}\mathrm{Tr}\left(\gamma\wedge D\gamma\right)=\mathrm{\backslash}\left(\vartheta^{\alpha}\wedge T_{\alpha}\right)=\frac{1}{2}T^{[\alpha\beta\gamma]}{}_{\eta_{\alpha\beta\gamma}}=\mathscr{A}_i\mathrm{d}x^i,$	$\mathscr{A}:=\frac{1}{4}{}^*\mathrm{Tr}(\gamma\wedge D\gamma)={}^*(\vartheta^{\alpha}\wedge T_{\alpha})=\frac{1}{2}T^{[\alpha\beta\gamma]}{}_{\eta_{\alpha\beta\gamma}}=\mathscr{A}_i dx^i,$	$\mathscr{A}:=\frac{1}{4}{}^*Tr(\gamma\wedge D\gamma)={}^*(\vartheta^{\alpha}\wedge T_{\alpha})=\frac{1}{2}T^{[\alpha\beta\gamma]}{}_{\eta_{\alpha\beta\gamma}}=\mathscr{A}_i dx^i,$
mielke-geometrodynamics_EQ0100 (3.3.1)	056	■ 0.994 ● N -2	$F:=\mathrm{d}A=\frac{1}{2}F_{\mathrm{i}\mathrm{j}}\mathrm{d}x^{\mathrm{i}}\backslash\mathrm{wedge}\mathrm{d}x^{\mathrm{j}}\;.$	$F:=dA=\frac{1}{2}F_{ij}dx^i\wedge dx^j.$	$F:=dA=\frac{1}{2}F_{ij}dx^i\wedge dx^j.$
mielke-geometrodynamics_EQ1102 (B.14)	370	■ 0.994 ●	$\backslash\mathrm{begin}\{\mathrm{aligned}\}\mathrm{d}_{\backslash\mu}\{\mathrm{M}\}\&:=\backslash\mathrm{sqrt}\{\backslash\mathrm{left} \backslash\mathrm{operatorname}\{\mathrm{det}\}\mathrm{g}_{\backslash\alpha\backslash\beta}\backslash\mathrm{right} \}\backslash\mathrm{vartheta}^{\backslash\mathrm{1}}\backslash\mathrm{wedge}\backslash\mathrm{cdots}\backslash\mathrm{wedge}\backslash\mathrm{vartheta}^{\backslash\mathrm{n}}=\eta\backslash\mathrm{and}\backslash\mathrm{and}\backslash\mathrm{sqrt}\{\backslash\mathrm{left} \backslash\mathrm{operatorname}\{\mathrm{det}\}\mathrm{g}_{\mathrm{i}\mathrm{j}}\backslash\mathrm{right} \}\mathrm{d}x^{\backslash\mathrm{1}}\backslash\mathrm{wedge}\backslash\mathrm{cdots}\backslash\mathrm{wedge}\mathrm{d}x^{\backslash\mathrm{n}}=\frac{1}{n}\backslash\mathrm{vartheta}\wedge^*\backslash\mathrm{vartheta}\backslash\mathrm{end}\{\mathrm{aligned}\}$	$d_{\mu}(M):={}^*1=\sqrt{ \det g_{\alpha\beta} }\vartheta^1\wedge\cdots\wedge\vartheta^n=\eta$ $=\sqrt{ \det g_{ij} }dx^1\wedge\cdots\wedge dx^n=\frac{1}{n}\vartheta\wedge^*\vartheta.$	$d_{\mu}(M):={}^*1=\sqrt{ \det g_{\alpha\beta} }\vartheta^1\wedge\cdots\wedge\vartheta^n=\eta$ $=\sqrt{ \det g_{ij} }dx^1\wedge\cdots\wedge dx^n=\frac{1}{n}\vartheta\wedge^*\vartheta.$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_00467 (7.3.6)	154	0.994 N -1	$\begin{aligned} L_{\mathrm{SKY}}^{\left(\star\right)} &= -\frac{1}{2} R^{\alpha\beta} \wedge {}^{\star} R_{\alpha\beta} - \frac{\left(-1\right)^s}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{\left(\star\right)} \\ &= -\frac{1}{4} \left(R_{\alpha\beta} + {}^{\star} R_{\alpha\beta}^{\left(\star\right)} \right) \wedge {}^{\star} \left(R^{\alpha\beta} + {}^{\star} R^{\alpha\beta\left(\star\right)} \right) \\ &= \frac{1}{2} \mathrm{Tr} \left(\bar{\Omega}^g \wedge {}^{\star} \bar{\Omega}^g \right) . \end{aligned}$	$L_{\mathrm{SKY}}^{\left(\star\right)} = -\frac{1}{2} R^{\alpha\beta} \wedge {}^{\star} R_{\alpha\beta} - \frac{\left(-1\right)^s}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{\left(\star\right)}$ $= -\frac{1}{4} \left(R_{\alpha\beta} + {}^{\star} R_{\alpha\beta}^{\left(\star\right)} \right) \wedge {}^{\star} \left(R^{\alpha\beta} + {}^{\star} R^{\alpha\beta\left(\star\right)} \right)$ $= \frac{1}{2} \mathrm{Tr} \left(\bar{\Omega}^g \wedge {}^{\star} \bar{\Omega}^g \right) .$	$L_{\mathrm{SKY}}^{\left(\star\right)} = -\frac{1}{2} R^{\alpha\beta} \wedge {}^{\star} R_{\alpha\beta} - \frac{\left(-1\right)^s}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{\left(\star\right)}$ $= -\frac{1}{4} \left(R_{\alpha\beta} + {}^{\star} R_{\alpha\beta}^{\left(\star\right)} \right) \wedge {}^{\star} \left(R^{\alpha\beta} + {}^{\star} R^{\alpha\beta\left(\star\right)} \right)$ $= \frac{1}{2} \mathrm{Tr}(\bar{\Omega}^g \wedge {}^{\star} \bar{\Omega}^g) .$
mielke-geometrodynamics_00498 (7.7.16)	163	0.994 N +0	$\{ \}^{\star} \backslash \Phi := \frac{1}{\left(4-p\right)!} p! \sqrt{\det g_{ij}} \varepsilon^{i_1 \cdots i_p}_{j_1 \cdots j_{4-p}} \Phi_{i_1 \cdots i_p} dx^{j_1} \wedge \cdots \wedge dx^{j_{4-p}}$	${}^{\star} \Phi := \frac{1}{\left(4-p\right)! p!} \sqrt{\det g_{ij}} \varepsilon^{i_1 \cdots i_p}_{j_1 \cdots j_{4-p}} \Phi_{i_1 \cdots i_p} dx^{j_1} \wedge \cdots \wedge dx^{j_{4-p}}$	${}^{\star} \Phi := \frac{1}{\left(4-p\right)! p!} \sqrt{\det g_{ij}} \varepsilon^{i_1 \cdots i_p}_{j_1 \cdots j_{4-p}} \Phi_{i_1 \cdots i_p} dx^{j_1} \wedge \cdots \wedge dx^{j_{4-p}},$
mielke-geometrodynamics_00724 (10.10.4)	230	0.994 N +0	$\delta_{\mathrm{susy}} L = i \bar{\sigma} \Psi \gamma^{\alpha} \wedge \frac{\delta L}{\delta \vartheta^{\alpha}} + \delta_{\mathrm{susy}} \Gamma_{\alpha}^{\star} \wedge \frac{\delta L}{\delta \Gamma_{\alpha}^{\star}} + \left(2 \overline{D \sigma} + c \bar{\sigma} \gamma\right) \wedge \frac{\delta L}{\delta \overline{\Psi}},$	$\delta_{\mathrm{susy}} L = i \bar{\sigma} \Psi \gamma^{\alpha} \wedge \frac{\delta L}{\delta \vartheta^{\alpha}} + \delta_{\mathrm{susy}} \Gamma_{\alpha}^{\star} \wedge \frac{\delta L}{\delta \Gamma_{\alpha}^{\star}} + \left(2 \overline{D \sigma} + c \bar{\sigma} \gamma\right) \wedge \frac{\delta L}{\delta \overline{\Psi}},$	$\delta_{\mathrm{susy}} L = i \bar{\sigma} \Psi \gamma^{\alpha} \wedge \frac{\delta L}{\delta \vartheta^{\alpha}} + \delta_{\mathrm{susy}} \Gamma_{\alpha}^{\star} \wedge \frac{\delta L}{\delta \Gamma_{\alpha}^{\star}} + \left(2 \overline{D \sigma} + c \bar{\sigma} \gamma\right) \wedge \frac{\delta L}{\delta \overline{\Psi}},$
mielke-geometrodynamics_00881 (3.1.6)	053	0.994 N +0	$\begin{aligned} \delta L &= \delta \varphi \wedge {}^{\star} L_{\varphi} + \delta(d\varphi) \wedge \pi \\ &= \delta \varphi \wedge {}^{\star} L_{\varphi} + d(\delta \varphi) \wedge \pi \\ &= \delta \varphi \wedge \left({}^{\star} L_{\varphi} + d\pi \right) + d(\delta \varphi \wedge \pi), \end{aligned}$	$\delta L = \delta \varphi \wedge {}^{\star} L_{\varphi} + \delta(d\varphi) \wedge \pi$ $= \delta \varphi \wedge {}^{\star} L_{\varphi} + d(\delta \varphi) \wedge \pi$ $= \delta \varphi \wedge \left({}^{\star} L_{\varphi} + d\pi \right) + d(\delta \varphi \wedge \pi),$	$\delta L = \delta \varphi \wedge {}^{\star} L_{\varphi} + \delta(d\varphi) \wedge \pi$ $= \delta \varphi \wedge {}^{\star} L_{\varphi} + d(\delta \varphi) \wedge \pi$ $= \delta \varphi \wedge \left({}^{\star} L_{\varphi} + d\pi \right) + d(\delta \varphi \wedge \pi),$
mielke-geometrodynamics_00509 (7.7.27)	165	0.994 K +0	$D = d + [\Gamma, \quad]$	$D = d + [\Gamma, \quad]$	$D = d + [\Gamma, \quad]$
mielke-geometrodynamics_00387 (6.4.2)	130	0.994 K +0	$\left. \{ \}^{\perp} \backslash \Psi := d \, t \wedge \Psi_{\perp} \right\rfloor \Psi, \quad \Psi_{\perp} := n \rfloor \Psi,$	${}^{\perp} \Psi := dt \wedge \Psi_{\perp}; \quad \Psi_{\perp} := n \rfloor \Psi,$	${}^{\perp} \Psi := dt \wedge \Psi_{\perp}; \quad \Psi_{\perp} := n \rfloor \Psi,$
mielke-geometrodynamics_00646 (10.3.3)	214	0.995 N +0	$\Gamma_{\alpha}^{\star} = \left(-1\right)^s \frac{\varepsilon}{2 \ell} \vartheta_{\alpha},$	$\Gamma_{\alpha}^{\star} = \left(-1\right)^s \frac{\varepsilon}{2 \ell} \vartheta_{\alpha},$	$\Gamma_{\alpha}^{\star} = \left(-1\right)^s \frac{\varepsilon}{2 \ell} \vartheta_{\alpha},$
mielke-geometrodynamics_00806 (11.4.18)	255	0.995 W +1	$\overrightarrow{J^2} \chi_{\kappa}^m = j(j+1) \chi_{\kappa}^m$	$\overrightarrow{J^2} \chi_{\kappa}^m = j(j+1) \chi_{\kappa}^m$	$\overrightarrow{J^2} \chi_{\kappa}^m = j(j+1) \chi_{\kappa}^m$
mielke-geometrodynamics_00995 (14.3.8)	321	0.995	$L_{\mathrm{anom.}} = \frac{\widehat{\ell}}{2} \bar{\psi} \gamma \wedge \gamma \wedge {}^{\star} \Omega \psi = \frac{\ell^{\star}}{4 \sqrt{\alpha_g}} \bar{\psi} F_{\mu\nu} \gamma^5 [\gamma^{\mu}, \gamma^{\nu}] \psi \eta.$	$L_{\mathrm{anom.}} = \frac{\widehat{\ell}}{2} \bar{\psi} \gamma \wedge \gamma \wedge {}^{\star} \Omega \psi = \frac{\ell^{\star}}{4 \sqrt{\alpha_g}} \bar{\psi} F_{\mu\nu} \gamma^5 [\gamma^{\mu}, \gamma^{\nu}] \psi \eta.$	$L_{\mathrm{anom.}} = \frac{\widehat{\ell}}{2} \bar{\psi} \gamma \wedge \gamma \wedge {}^{\star} \Omega \psi = \frac{\ell^{\star}}{4 \sqrt{\alpha_g}} \bar{\psi} F_{\mu\nu} \gamma^5 [\gamma^{\mu}, \gamma^{\nu}] \psi \eta.$
mielke-geometrodynamics_00998 (3.2.8)	056	0.995 K +0	$\Omega \rightarrow {}^{G^{-1}} \Omega = G^{-1} \Omega G.$	$\Omega \rightarrow {}^{G^{-1}} \Omega = G^{-1} \Omega G.$	$\Omega \rightarrow {}^{G^{-1}} \Omega = G^{-1} \Omega G.$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

— no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_009930 (14.1.11)	304	■ 0.995 ● N +0	$\mathrm{R}=\{ \ }^*\!\!\left(\!\!\Omega^g\wedge\vartheta\right)\wedge\vartheta\wedge\vartheta\right)=0$	$R = {}^*(\Omega^g \wedge \vartheta \wedge \vartheta) = 0$	$R = {}^*(\Omega^g \wedge \vartheta \wedge \vartheta) = 0$
mielke-geometrodynamics_001033 (15.1.10)	338	■ 0.995 ● K +0	$\frac{1}{2}\{ \ }^*\!\!\left(\!\!\vartheta\wedge\Omega^f\right)+\Lambda_{\mathrm{eff}}^f\vartheta^f=0$	$\frac{1}{2}*(\vartheta^f \wedge^* \Omega^f) + \Lambda_{\mathrm{eff}}^f \vartheta^f = 0$	$\frac{1}{2}*(\vartheta^f \wedge^* \Omega^f) + \Lambda_{\mathrm{eff}}^f \vartheta^f = 0$
mielke-geometrodynamics_000207 (4.1.21)	083	■ 0.995 ● N +0	$\delta\!:\!\left[\begin{array}{l} A(n,\mathbb{R})\times A(M)\\ \downarrow\qquad\downarrow\\ (g,a)\qquad(e(m),x) \end{array}\right] \longrightarrow \begin{array}{c} A(M)\\ \downarrow\\ (\mathbf{g.e}(m),g^{-1}(x-a)). \end{array}$	$\delta:\left\{\begin{array}{ccc} A(n,\mathbb{R}) & \times A(M) & \longrightarrow \\ \downarrow & \downarrow & \\ (g,a) & (e(m),x) & = (\mathbf{g.e}(m),g^{-1}(x-a)). \end{array}\right.$	$\delta:\left\{\begin{array}{ccc} A(n,\mathbb{R}) & \times A(M) & \longrightarrow \\ \downarrow & \downarrow & \\ (g,a) & (e(m),x) & = (g.e(m),g^{-1}(x-a)). \end{array}\right.$
mielke-geometrodynamics_000658 (10.3.15)	217	■ 0.995 ● N +0	$\{ \ }^*\!\!\mathrm{D}\!\left(\!\!\vartheta^{\alpha}\wedge H_{\alpha}\right)-\{ \ }^*\!\!\left(\!\!T^{\alpha}\wedge H_{\alpha}\right)=\frac{\gamma}{\delta\ell^2}\{ \ }^*\!\!\left(\!\!\vartheta^{\alpha}\wedge T_{\alpha}\right)+\frac{1}{\delta\ell}\{ \ }^*\!\!\left(\!\!\eta^{\alpha}\wedge H_{\alpha}\right).$	$*D(\vartheta^\alpha \wedge H_\alpha) - {}^*(T^\alpha \wedge H_\alpha) = \frac{\gamma}{\delta \ell^2} {}^*(\vartheta^\alpha \wedge T_\alpha) + \frac{1}{\delta \ell} {}^*(\eta^\alpha \wedge H_\alpha).$	$*D(\vartheta^\alpha \wedge H_\alpha) - {}^*(T^\alpha \wedge H_\alpha) = \frac{\gamma}{\delta \ell^2} {}^*(\vartheta^\alpha \wedge T_\alpha) + \frac{1}{\delta \ell} {}^*(\eta^\alpha \wedge H_\alpha).$
mielke-geometrodynamics_000154 (3.5.2)	066	■ 0.995 ● N +0	$\stackrel{(-)}{\sigma}_{\mu\nu}:=\left\{\begin{array}{l} \sigma^{ij}=\frac{1}{2}\varepsilon^{ijk}\sigma_k\\ \sigma^{i0}=\binom{-}{-}\frac{1}{2}\sigma^i \end{array}\right \,i,j,k=1,2,3$	$\binom{-}{\sigma}_{\mu\nu}:=\left\{\begin{array}{l} \sigma^{ij}=\frac{1}{2}\varepsilon^{ijk}\sigma_k\\ \sigma^{i0}=\binom{-}{-}\frac{1}{2}\sigma^i \end{array}\right \,i,j,k=1,2,3$	$\binom{-}{\sigma}^{\mu\nu}:=\left\{\begin{array}{l} \sigma^{ij}=\frac{1}{2}\varepsilon^{ijk}\sigma_k\\ \sigma^{i0}=\binom{-}{-}\frac{1}{2}\sigma^i \end{array}\right \,i,j,k=1,2,3$
mielke-geometrodynamics_000928 (14.1.9)	304	■ 0.995 ● S -1	$\begin{aligned} \Sigma_{\mu\nu}^{\mathrm{YM}} &= \frac{1}{\alpha_g} \operatorname{Tr}\!\left(F_{\mu\alpha}F_{\nu}^{\alpha}-\frac{1}{4}g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right) \\ &= \frac{1}{2\alpha_g} \operatorname{Tr}\!\left(F_{\mu\alpha}F_{\nu}^{\alpha}-(-1)^s {}^*F_{\mu\alpha} {}^*F_{\nu}^{\alpha}\right), \end{aligned}$	$\Sigma_{\mu\nu}^{\mathrm{YM}} = \frac{1}{\alpha_g} \operatorname{Tr}\left(F_{\mu\alpha}F_{\nu}^{\alpha} - \frac{1}{4}g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right) \\ = \frac{1}{2\alpha_g} \operatorname{Tr}\left(F_{\mu\alpha}F_{\nu}^{\alpha} - (-1)^s {}^*F_{\mu\alpha} {}^*F_{\nu}^{\alpha}\right),$	$\Sigma_{\mu\nu}^{\mathrm{YM}} = \frac{1}{\alpha_g} \operatorname{Tr}(F_{\mu\alpha}F_{\nu}^{\alpha} - \frac{1}{4}g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}) \\ = \frac{1}{2\alpha_g} \operatorname{Tr}(F_{\mu\alpha}F_{\nu}^{\alpha} - (-1)^s {}^*F_{\mu\alpha} {}^*F_{\nu}^{\alpha}),$
mielke-geometrodynamics_000006 (2.2.1)	033	■ 0.996 ● K +0	$\left.e_{\beta}\right]\vartheta^{\alpha}=\delta^{\alpha}_{\beta}.$	$e_{\beta}]\vartheta^\alpha = \delta^\alpha_{\beta}.$	$e_{\beta}]\vartheta^\alpha = \delta^\alpha_{\beta}.$
mielke-geometrodynamics_000284 (4.4.13)	099	■ 0.996 ● W +2	$\begin{aligned} D[\Theta,\vartheta]&=[D\Theta,\vartheta]+[\Theta,D\vartheta] \\ &=[[\vartheta,\Omega^g],\vartheta]+[\Theta,\Theta] \\ &=\vartheta\wedge\Omega^g\wedge\vartheta-\Omega^g\wedge\vartheta\wedge\vartheta+\vartheta\wedge\vartheta\wedge\Omega^g-\vartheta\wedge\Omega^g\wedge\vartheta \\ &=[\vartheta\wedge\vartheta,\Omega^g]. \end{aligned}$	$D[\Theta,\vartheta] = [D\Theta,\vartheta] + [\Theta,D\vartheta] \\ = [[\vartheta,\Omega^g],\vartheta] + [\Theta,\Theta] \\ = \vartheta \wedge \Omega^g \wedge \vartheta - \Omega^g \wedge \vartheta \wedge \vartheta + \vartheta \wedge \vartheta \wedge \Omega^g - \vartheta \wedge \Omega^g \wedge \vartheta \\ = [\vartheta \wedge \vartheta, \Omega^g].$	$D[\Theta,\vartheta] = [D\Theta,\vartheta] + [\Theta,D\vartheta] \\ = [[\vartheta,\Omega^g],\vartheta] + [\Theta,\Theta] \\ = \vartheta \wedge \Omega^g \wedge \vartheta - \Omega^g \wedge \vartheta \wedge \vartheta + \vartheta \wedge \vartheta \wedge \Omega^g - \vartheta \wedge \Omega^g \wedge \vartheta \\ = [\vartheta \wedge \vartheta, \Omega^g].$
mielke-geometrodynamics_000308 (4.5.18)	103	■ 0.996 ●	$C_{\mathrm{AdS}}=\hat{M}^2+\frac{\Lambda}{3}\hat{J}^2$	$C_{\mathrm{AdS}} = \hat{M}^2 + \frac{\Lambda}{3} \hat{J}^2$	$C_{\mathrm{AdS}} = \hat{M}^2 + \frac{\Lambda}{3} \hat{J}^2$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_E08623 (10.1.11)	210	■0.996 ● S -2	$\frac{\delta C_{\mathrm{TL}}}{\delta \vartheta^\alpha} = \frac{1}{\ell} R_\alpha^\star \quad \text{and} \quad \frac{\delta C_{\mathrm{TL}}}{\delta \Gamma^{\star\alpha}} = \frac{1}{\ell} T_\alpha,$	$\frac{\delta C_{\mathrm{TL}}}{\delta \vartheta^\alpha} = \frac{1}{\ell} R_\alpha^\star \quad \text{and} \quad \frac{\delta C_{\mathrm{TL}}}{\delta \Gamma^{\star\alpha}} = \frac{1}{\ell} T_\alpha,$	
mielke-geometrodynamics_E08915 (13.8.1)	295	■0.996 ● K +0	$\hat{C} := -\frac{1}{2} \left(\Gamma_A{}^B \wedge d\Gamma_B{}^A - \frac{2}{3} \Gamma_A{}^B \wedge \Gamma_B{}^C \wedge \Gamma_C{}^A \right)$	$\hat{C} := -\frac{1}{2} \left(\Gamma_A{}^B \wedge d\Gamma_B{}^A - \frac{2}{3} \Gamma_A{}^B \wedge \Gamma_B{}^C \wedge \Gamma_C{}^A \right)$	
mielke-geometrodynamics_E01071 (16.2.10)	357	■0.996 ● N +0	$E_n = \beta mc^2 \left(2n + \iota + \frac{3}{2} + \sqrt{\frac{g}{4} + \frac{1}{\beta^2}} \right), n = 0, 1, 2 \dots$	$E_n = \beta mc^2 \left(2n + \iota + \frac{3}{2} + \sqrt{\frac{g}{4} + \frac{1}{\beta^2}} \right), n = 0, 1, 2 \dots$	
mielke-geometrodynamics_E08490 (7.7.7)	161	■0.996 ● K +0	$\begin{gathered} \left.\left.\widetilde{E}_{\alpha}\right \right\rfloor\left.\left.\widetilde{L}-\left(e_{\alpha}\right]\!R^{\beta\gamma}\right)\wedge\widetilde{H}_{\beta\gamma}=\Sigma_{\alpha}, \\ D\widetilde{H}_{\alpha\beta}=\tau_{\alpha\beta}. \end{gathered}$	$\begin{aligned} -\tilde{E}_{\alpha} &:= -e_{\alpha}\rfloor \tilde{L} - (e_{\alpha}\rfloor R^{\beta\gamma}) \wedge \tilde{H}_{\beta\gamma} = \Sigma_{\alpha}, \\ D\tilde{H}_{\alpha\beta} &= \tau_{\alpha\beta}. \end{aligned}$	
mielke-geometrodynamics_E08220 (4.2.13)	086	■0.996 ● K +0	$\Omega^{\mathrm{L}} \rightarrow^{A^{-1}} \Omega^{\mathrm{L}} = G^{-1} \Omega^{\mathrm{L}} G$	$\Omega^{\mathrm{L}} \rightarrow^{A^{-1}} \Omega^{\mathrm{L}} = G^{-1} \Omega^{\mathrm{L}} G$	
mielke-geometrodynamics_E08460 (7.2.12)	152	■0.996 ● W +0	$g_{ij} \rightarrow \bar{g}_{ij} = \phi^L g_{ij},$	$g_{ij} \rightarrow \bar{g}_{ij} = \phi^L g_{ij},$	
mielke-geometrodynamics_E08664 (10.3.21)	218	■0.996 ● N +0	$\square \vartheta_\alpha = \frac{\ell}{\gamma} {}^*D^* \left[\tau_\alpha^\star - S_\alpha - \delta \ell (\Sigma_\alpha + E_\alpha) \right] + \frac{\ell}{\gamma} D^* \left(D^* H_\alpha^\star - \delta \ell D^* H_\alpha \right).$	$\Box \vartheta_\alpha = \frac{\ell}{\gamma} {}^*D^* \left[\tau_\alpha^\star - S_\alpha - \delta \ell (\Sigma_\alpha + E_\alpha) \right] + \frac{\ell}{\gamma} D^* \left(D^* H_\alpha^\star - \delta \ell D^* H_\alpha \right).$	
mielke-geometrodynamics_E08977 (14.2.18)	315	■0.996 ● N +0	$\hat{L}_\parallel = -\frac{1}{2\ell^{*2}} \operatorname{Tr} [\hat{\alpha} \wedge \hat{\alpha}, \quad \hat{\vartheta} \wedge \cdots \wedge \hat{\vartheta}]$	$\hat{L}_\parallel = -\frac{1}{2\ell^{*2}} \operatorname{Tr} [\hat{\alpha} \wedge \hat{\alpha}, \quad \hat{\vartheta} \wedge \cdots \wedge \hat{\vartheta}]$	
mielke-geometrodynamics_E08152 (3.4.23)	066	■0.996 ● C -4	$L_{\mathrm{BI}}(\mathscr{F}, {}^*\mathscr{F}) = \sqrt{1 - 2\mathscr{F} - ({}^*\mathscr{F})^2} - 1.$	$L_{\mathrm{BI}}(\mathcal{F}, {}^*\mathcal{F}) = \sqrt{1 - 2\mathcal{F} - ({}^*\mathcal{F})^2} - 1.$	
mielke-geometrodynamics_E08250 (4.3.15)	092	■0.996 ● N +0	${}^*\omega^g = \frac{1}{2} \omega^{\alpha\beta} L_{\alpha\beta} = \frac{1}{2} \Gamma_i^{\cdot\alpha\beta} L_{\alpha\beta} \otimes dx^i,$	${}^*\omega^g = \frac{1}{2} \omega^{\alpha\beta} L_{\alpha\beta} = \frac{1}{2} \Gamma_i^{\cdot\alpha\beta} L_{\alpha\beta} \otimes dx^i,$	
mielke-geometrodynamics_E08376 (6.2.27)	128	■0.996 ● N +0	$\stackrel{\scriptscriptstyle{(+)}}{D}\vartheta^\alpha = \pm \frac{i}{2} \ell^2 g^{\alpha\beta} \stackrel{\scriptscriptstyle{(-)}}{\Pi}_\beta,$	$\overset{(\pm)}{D}\vartheta^\alpha = \pm \frac{i}{2} \ell^2 g^{\alpha\beta} \overset{(\pm)}{\Pi}_\beta,$	
mielke-geometrodynamics_E08367 (6.2.18)	126	■0.997 ● N +0	$R_{\alpha\beta} = \xi^* R_{\alpha\beta}^{(*)} + (\gamma/\ell^2) \left[\vartheta_\alpha \wedge \vartheta_\beta \pm i^*(\vartheta_\alpha \wedge \vartheta_\beta) \right],$	$R_{\alpha\beta} = \xi^* R_{\alpha\beta}^{(*)} + (\gamma/\ell^2) \left[\vartheta_\alpha \wedge \vartheta_\beta \pm i^*(\vartheta_\alpha \wedge \vartheta_\beta) \right],$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_127 EQ0371 (6.2.22)		■ 0.997 ● W +0	$\stackrel{+}{\Pi} \wedge \stackrel{-}{\Pi}_\beta = H^\beta \wedge H_\beta + \ell^{-4} T^\beta \wedge T_\beta = 0.$	$\stackrel{+}{\Pi}^\beta \wedge \stackrel{-}{\Pi}_\beta = H^\beta \wedge H_\beta + \ell^{-4} T^\beta \wedge T_\beta = 0.$	
mielke-geometrodynamics_198 EQ0602 (9.2.1)		■ 0.997 ●	$\begin{aligned} p_1(M^4) &= \frac{1}{2(2\pi)^2} \int_{M^4} \text{Tr}(\Omega^L \wedge \Omega^L) \\ &= \frac{(-1)^{s/2}}{(4\pi)^2} \int_{M^4} R_{\alpha\beta\gamma\delta}^* R^{\alpha\beta\gamma\delta} \sqrt{ g } d^4x \end{aligned}$	$p_1(M^4) = \frac{1}{2(2\pi)^2} \int_{M^4} \text{Tr}(\Omega^L \wedge \Omega^L) = \frac{(-1)^{s/2}}{(4\pi)^2} \int_{M^4} R_{\alpha\beta\gamma\delta}^* R^{\alpha\beta\gamma\delta} \sqrt{ g } d^4x$	
mielke-geometrodynamics_229 EQ0717 (10.9.3)		■ 0.997 ● N +0	$\Sigma_\alpha := \frac{\delta L_\Psi}{\delta \vartheta^\alpha} = \frac{\partial L_\Psi}{\partial \vartheta_\alpha} + D \frac{\partial L_\Psi}{\partial T^\alpha},$	$\Sigma_\alpha := \frac{\delta L_\Psi}{\delta \vartheta^\alpha} = \frac{\partial L_\Psi}{\partial \vartheta_\alpha} + D \frac{\partial L_\Psi}{\partial T^\alpha},$	
mielke-geometrodynamics_243 EQ0766 (11.1.33)		■ 0.997 ● N +0	$\sigma^* \tilde{\omega} = \frac{1}{2} \omega^{\alpha\beta} \sigma_{\alpha\beta} = \frac{1}{2} \Gamma_i^{\alpha\beta} \sigma_{\alpha\beta} \otimes dx^i =: \Gamma_i dx^i.$	$\sigma^* \tilde{\omega} = \frac{1}{2} \omega^{\alpha\beta} \sigma_{\alpha\beta} = \frac{1}{2} \Gamma_i^{\alpha\beta} \sigma_{\alpha\beta} \otimes dx^i =: \Gamma_i dx^i.$	
mielke-geometrodynamics_323 EQ1003 (14.3.16)		■ 0.997 ● K +0	$D D = \frac{1}{2} \{D, D\} + \frac{1}{2} [D, D] = \frac{1}{2} \{D, D\} + \frac{1}{2} \tilde{\Omega}$	$D D = \frac{1}{2} \{D, D\} + \frac{1}{2} [D, D] = \frac{1}{2} \{D, D\} + \frac{1}{2} \tilde{\Omega}$	
mielke-geometrodynamics_087 EQ0225 (4.2.18)		■ 0.997 ● N +0	$\omega^T = \tilde{\vartheta} - D \tilde{\varphi}$	$\omega^T = \tilde{\vartheta} - D \tilde{\varphi}$	
mielke-geometrodynamics_043 EQ0040 (2.6.11)		■ 0.997 ● N +0	$D D D b \equiv 0.$	$D D D b \equiv 0.$	
mielke-geometrodynamics_056 EQ0099 (3.2.9)		■ 0.997 ● K +0	$\Delta \omega := G^{-1} \omega - \omega = G^{-1} [\omega, G] - G^{-1} dG = -G^{-1} DG = -G^{-1} D.$	$\Delta \omega := G^{-1} \omega - \omega = G^{-1} [\omega, G] - G^{-1} dG = -G^{-1} DG = -G^{-1} D.$	
mielke-geometrodynamics_086 EQ0219 (4.2.12)		■ 0.997 ● N +0	$\tilde{\tilde{\Omega}} \rightarrow A^{-1} \tilde{\tilde{\Omega}} = A^{-1} \tilde{\tilde{\Omega}} A$	$\tilde{\tilde{\Omega}} \rightarrow A^{-1} \tilde{\tilde{\Omega}} = A^{-1} \tilde{\tilde{\Omega}} A$	
mielke-geometrodynamics_136 EQ0411 (6.4.31)		■ 0.997 ● W -1	$n_{\mathrm{NY}} := \int_{R^4} dC_{\mathrm{TT}} = \int_{S^3_\infty} \underline{C}_{\mathrm{TT}} = 3 \mathrm{Vol}(S^3) k = 6\pi^2 k.$	$n_{\mathrm{NY}} := \int_{R^4} dC_{\mathrm{TT}} = \int_{S^3_\infty} \underline{C}_{\mathrm{TT}} = 3 \mathrm{Vol}(S^3) k = 6\pi^2 k.$	
mielke-geometrodynamics_161 EQ0492 (7.7.10)		■ 0.997 ● N -1	$H_{\alpha\beta} := -\frac{\partial L}{\partial R^{\alpha\beta}} = \tilde{H}_{\alpha\beta} + \frac{1}{2} R^{\mu\nu} \wedge (\partial \tilde{H}_{\mu\nu} / \partial R^{\alpha\beta})$	$H_{\alpha\beta} := -\frac{\partial L}{\partial R^{\alpha\beta}} = \tilde{H}_{\alpha\beta} + \frac{1}{2} R^{\mu\nu} \wedge (\partial \tilde{H}_{\mu\nu} / \partial R^{\alpha\beta})$	
mielke-geometrodynamics_285 EQ0877 (13.3.6)		■ 0.997 ● N +0	$[P_\alpha, P_*^\beta] = \frac{1}{\ell^2} L^\beta{}_\alpha.$	$[P_\alpha, P_*^\beta] = \frac{1}{\ell^2} L^\beta{}_\alpha.$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_312 EQ0962 (14.2.2)	312	■ 0.997 ● N +0	$\stackrel{\circ}{\vartheta}=\varphi_j d\xi^j,$	$\overset{\circ}{\vartheta}=\varphi_j d\xi^j,$	$\overset{\circ}{\vartheta}=\varphi_j d\xi^j,$
mielke-geometrodynamics_354 EQ1060 (16.1.1)	354	■ 0.997 ● W +0	$R_{ij}^f-\frac{1}{2}f_{ij}\left(R^f-2\Lambda_{\mathrm{eff}}\right)=\ell^2T_{ij}$	$R_{ij}^f-\frac{1}{2}f_{ij}\left(R^f-2\Lambda_{\mathrm{eff}}\right)=\ell^2T_{ij}$	$R_{ij}^f-\frac{1}{2}f_{ij}\left(R^f-2\Lambda_{\mathrm{eff}}\right)=\ell^2T_{ij}$
mielke-geometrodynamics_315 EQ0976 (14.2.17)	315	■ 0.997 ●	$\begin{aligned} \hat{L}_{\mathrm{W}.} &= \frac{1}{2} \operatorname{Tr}(\hat{\Omega} \wedge \underbrace{\hat{\vartheta} \wedge \dots \wedge \hat{\vartheta}}_{2+\mathrm{K} \text{ termms}}) \\ &= \hat{L}_{\mathrm{W}.}^g + \hat{L}_{\mathrm{W}.}^G + \ell^{*2} \hat{L}_{\parallel} - \frac{1}{2} d \operatorname{Tr}(\hat{\alpha} \wedge \hat{\vartheta} \wedge \dots \wedge \hat{\vartheta}) \end{aligned}$	$\hat{L}_{\mathrm{W}.} = \frac{1}{2} \operatorname{Tr}(\hat{\Omega} \wedge \underbrace{\hat{\vartheta} \wedge \dots \wedge \hat{\vartheta}}_{2+\mathrm{K} \text{ termms}}) \\ = \hat{L}_{\mathrm{W}.}^g + \hat{L}_{\mathrm{W}.}^G + \ell^{*2} \hat{L}_{\parallel} - \frac{1}{2} d \operatorname{Tr}(\hat{\alpha} \wedge \hat{\vartheta} \wedge \dots \wedge \hat{\vartheta})$	$\hat{L}_{\mathrm{W}.} = \frac{1}{2} \operatorname{Tr}(\hat{\Omega} \wedge \underbrace{\hat{\vartheta} \wedge \dots \wedge \hat{\vartheta}}_{2+\mathrm{K} \text{ termms}}) \\ = \hat{L}_{\mathrm{W}.}^g + \hat{L}_{\mathrm{W}.}^G + \ell^{*2} \hat{L}_{\parallel} - \frac{1}{2} d \operatorname{Tr}(\hat{\alpha} \wedge \hat{\vartheta} \wedge \dots \wedge \hat{\vartheta})$
mielke-geometrodynamics_153 EQ0466 (7.3.5)	153	■ 0.997 ● W +0	$G_{\alpha}^{\{\}}:=\frac{1}{2}R^{\{\}\beta\gamma}\wedge\eta_{\alpha\beta\gamma},\quad G_{ij}:=\operatorname{Ric}_{ij}^{\{\}}-\frac{1}{2}g_{ij},$	$G_{\alpha}^{\{\}}:=\frac{1}{2}R^{\{\}\beta\gamma}\wedge\eta_{\alpha\beta\gamma},\quad G_{ij}:=\operatorname{Ric}_{ij}^{\{\}}-\frac{1}{2}g_{ij},$	$G_{\alpha}^{\{\}}:=\frac{1}{2}R^{\{\}\beta\gamma}\wedge\eta_{\alpha\beta\gamma},\quad G_{ij}:=\operatorname{Ric}_{ij}^{\{\}}-\frac{1}{2}g_{ij},$
mielke-geometrodynamics_058 EQ0105 (3.3.6)	058	■ 0.998 ● N -1	$H\!:=\!\frac{\partial L}{\partial dA}\!=\!\frac{\partial L}{\partial F}\quad\text{and}\quad j\!:=\!\frac{\delta L_{\mathrm{mat}}}{\delta A},$	$H:=-\frac{\partial L}{\partial dA}=-\frac{\partial L}{\partial F}\quad\text{and}\quad j:=\frac{\delta L_{\mathrm{mat}}}{\delta A},$	$H:=-\frac{\partial L}{\partial dA}=-\frac{\partial L}{\partial F}\quad\text{and}\quad j:=\frac{\delta L_{\mathrm{mat}}}{\delta A},$
mielke-geometrodynamics_063 EQ0136 (3.4.7)	063	■ 0.998 ● N +2	$\begin{aligned} \delta\Omega &= \delta(d\omega - \omega \wedge \omega) = \delta d\omega - \delta\omega \wedge \omega - \omega \wedge \delta\omega \\ &= d\delta\omega - [\omega, \delta\omega] = D\delta\omega. \end{aligned}$	$\delta\Omega = \delta(d\omega - \omega \wedge \omega) = \delta d\omega - \delta\omega \wedge \omega - \omega \wedge \delta\omega \\ = d\delta\omega - [\omega, \delta\omega] = D\delta\omega.$	$\delta\Omega = \delta(d\omega - \omega \wedge \omega) = \delta d\omega - \delta\omega \wedge \omega - \omega \wedge \delta\omega \\ = d\delta\omega - [\omega, \delta\omega] = D\delta\omega.$
mielke-geometrodynamics_131 EQ0390 (6.4.5)	131	■ 0.998 ● W +1	$\mathscr{H}=n\rfloor L-\ell_n\underline{\Psi}\wedge\left(\frac{\partial L}{\partial d\Psi}\right).$	$\mathscr{H}=n\rfloor L-\ell_n\underline{\Psi}\wedge\left(\frac{\partial L}{\partial d\Psi}\right).$	$\mathscr{H}=n\rfloor L-\ell_n\underline{\Psi}\wedge\left(\frac{\partial L}{\partial d\Psi}\right).$
mielke-geometrodynamics_195 EQ0594 (9.1.12)	195	■ 0.998 ● K +0	$\bar{\Omega}^{\{\}}=0.$	$\bar{\Omega}^{\{\}}=0.$	$\bar{\Omega}^{\{\}}=0.$
mielke-geometrodynamics_326 EQ1012 (14.4.7)	326	■ 0.998 ● N -1	$\begin{aligned} S_{\mathrm{A}} &:= 4\pi\left(r_+^2+\frac{J(J+1)\hbar^2}{M^2c^2}\right) \\ &= \frac{8\pi\hbar^2}{M^{*4}c^2}\left(M^2+\frac{\alpha_e}{2}M^{*2}\operatorname{Tr}\bar{Q}^2+M\sqrt{M^2-J(J+1)\frac{M^{*4}}{M^2}-\alpha_eM^{*2}\operatorname{Tr}\bar{Q}^2}\right) \end{aligned}$	$S_{\mathrm{A}} := 4\pi\left(r_+^2+\frac{J(J+1)\hbar^2}{M^2c^2}\right) \\ = \frac{8\pi\hbar^2}{M^{*4}c^2}\left(M^2+\frac{\alpha_e}{2}M^{*2}\operatorname{Tr}\bar{Q}^2+M\sqrt{M^2-J(J+1)\frac{M^{*4}}{M^2}-\alpha_eM^{*2}\operatorname{Tr}\bar{Q}^2}\right)$	$S_{\mathrm{A}} := 4\pi\left(r_+^2+\frac{J(J+1)\hbar^2}{M^2c^2}\right) \\ = \frac{8\pi\hbar^2}{M^{*4}c^2}\left(M^2+\frac{\alpha_e}{2}M^{*2}\operatorname{Tr}\bar{Q}^2+M\sqrt{M^2-J(J+1)\frac{M^{*4}}{M^2}-\alpha_eM^{*2}\operatorname{Tr}\bar{Q}^2}\right)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_049 E00073 (2.8.16)		■ 0.998 ● N +0	$\begin{aligned} & \mathrm{d} \stackrel{\infty}{\omega} - \stackrel{\infty}{\omega} \wedge \stackrel{\infty}{\omega} \\ &= -d \left(G^{-1} \right) \wedge dG - G^{-1} ddG - \left(G^{-1} dG \right) \wedge \left(G^{-1} dG \right) \\ &= -d \left(G^{-1} \right) \wedge dG + G^{-1} Gd \left(G^{-1} \right) \wedge dG = 0 \end{aligned}$	$\begin{aligned} & d \stackrel{\infty}{\omega} - \stackrel{\infty}{\omega} \wedge \stackrel{\infty}{\omega} \\ &= -d \left(G^{-1} \right) \wedge dG - G^{-1} ddG - \left(G^{-1} dG \right) \wedge \left(G^{-1} dG \right) \\ &= -d \left(G^{-1} \right) \wedge dG + G^{-1} Gd \left(G^{-1} \right) \wedge dG = 0 \end{aligned}$	$\begin{aligned} & d \stackrel{\infty}{\omega} - \stackrel{\infty}{\omega} \wedge \stackrel{\infty}{\omega} \\ &= -d \left(G^{-1} \right) \wedge dG - G^{-1} ddG - \left(G^{-1} dG \right) \wedge \left(G^{-1} dG \right) \\ &= -d \left(G^{-1} \right) \wedge dG + G^{-1} Gd \left(G^{-1} \right) \wedge dG = 0 \end{aligned}$
mielke-geometrodynamics_086 E00214 (4.2.6)		■ 0.998 ● N +0	$\mathrm{mathscr{G}}(n, \, \mathrm{mathbb{R}}) := C^{\infty} \left(A(M) \times_{\mathrm{Ad}} GL(n, \, \mathrm{mathbb{R}}) \right)$	$\mathscr{G}(n, \mathbb{R}) := C^{\infty} \left(A(M) \times_{\mathrm{Ad}} GL(n, \mathbb{R}) \right)$	$\mathscr{G}(n, \mathbb{R}) := C^{\infty} \left(A(M) \times_{\mathrm{Ad}} GL(n, \mathbb{R}) \right)$
mielke-geometrodynamics_216 E00656 (10.3.13)		■ 0.998 ● N +0	$\left. \left. E_{\alpha} = e_{\alpha} \right L(\star) + (-1)^s \frac{2\gamma}{\ell} \left[(e_{\alpha} \right R^{\star\beta}) \wedge \vartheta_{\beta} \right]$	$E_{\alpha} = e_{\alpha} \rfloor L(\star) + (-1)^s \frac{2\gamma}{\ell} \left[(e_{\alpha} \rfloor R^{\star\beta}) \wedge \vartheta_{\beta} \right]$	$E_{\alpha} = e_{\alpha} \rfloor L(\star) + (-1)^s \frac{2\gamma}{\ell} \left[(e_{\alpha} \rfloor R^{\star\beta}) \wedge \vartheta_{\beta} \right]$
mielke-geometrodynamics_256 E00813 (11.4.26)		■ 0.998 ●	$d \, \rho^* := e^{(\mu-\nu)/2} d \, \rho \, .$	$d\rho^* := e^{(\mu-\nu)/2} d\rho.$	$d\rho^* := e^{(\mu-\nu)/2} d\rho.$
mielke-geometrodynamics_046 E00062 (2.8.3)		■ 0.998 ● W +1	$\begin{aligned} & * \left(d \gamma_k \right) = d \pi^* \left(\gamma_k \right) = D \pi^* \left(\gamma_k \right) \\ &= \frac{(-1)^k}{(2\pi i)^k (k-1)!} \mathrm{Tr} (D \Omega \wedge \Omega \wedge \cdots \wedge \Omega) = 0. \end{aligned}$	$\begin{aligned} & * \left(d \gamma_k \right) = d \pi^* \left(\gamma_k \right) = D \pi^* \left(\gamma_k \right) \\ &= \frac{(-1)^k}{(2\pi i)^k (k-1)!} \mathrm{Tr} (D \Omega \wedge \Omega \wedge \cdots \wedge \Omega) = 0. \end{aligned}$	$\begin{aligned} & \pi^*(d\gamma_k) = d\pi^*(\gamma_k) = D\pi^*(\gamma_k) \\ &= \frac{(-1)^k}{(2\pi i)^k (k-1)!} \mathrm{Tr}(D\Omega \wedge \Omega \wedge \cdots \wedge \Omega) = 0. \end{aligned}$
mielke-geometrodynamics_141 E00426 (6.7.8)		■ 0.998 ● K +0	$\mu_{\alpha} = \frac{1}{4} \vartheta_{\alpha} \wedge {}^* j_5,$	$\mu_{\alpha} = \frac{1}{4} \vartheta_{\alpha} \wedge {}^* j_5,$	$\mu_{\alpha} = \frac{1}{4} \vartheta_{\alpha} \wedge {}^* j_5,$
mielke-geometrodynamics_048 E00069 (2.8.12)		■ 0.998 ● N +1	$\begin{aligned} & c_2(M) \, \&= \frac{1}{8\pi^2} \int_M d \, \mathrm{Tr} \left(\omega \wedge \Omega + \frac{1}{3} \omega \wedge \omega \wedge \omega \right) \\ &= \frac{1}{8\pi^2} \int_{\partial M} \mathrm{Tr} \left(\omega \wedge d\omega - \frac{2}{3} \omega \wedge \omega \wedge \omega \right) . \end{aligned}$	$\begin{aligned} c_2(M) &= \frac{1}{8\pi^2} \int_M d \, \mathrm{Tr} \left(\omega \wedge \Omega + \frac{1}{3} \omega \wedge \omega \wedge \omega \right) \\ &= \frac{1}{8\pi^2} \int_{\partial M} \mathrm{Tr} \left(\omega \wedge d\omega - \frac{2}{3} \omega \wedge \omega \wedge \omega \right) . \end{aligned}$	$\begin{aligned} c_2(M) &= \frac{1}{8\pi^2} \int_M d \mathrm{Tr} \left(\omega \wedge \Omega + \frac{1}{3} \omega \wedge \omega \wedge \omega \right) \\ &= \frac{1}{8\pi^2} \int_{\partial M} \mathrm{Tr} \left(\omega \wedge d\omega - \frac{2}{3} \omega \wedge \omega \wedge \omega \right) . \end{aligned}$
mielke-geometrodynamics_064 E00144 (3.4.15)		■ 0.998 ● N +0	$\iota := \tau + (-1)^{n+s} * \left[\omega, \Pi^{\Omega} \right]$	$\iota := \tau + (-1)^{n+s} * \left[\omega, \Pi^{\Omega} \right]$	$\iota := \tau + (-1)^{n+s} * \left[\omega, \Pi^{\Omega} \right]$
mielke-geometrodynamics_110 E00318 (5.2.2)		■ 0.998 ● N -1	$j := \bar{\psi}^* \gamma \psi = j^{\mu} \eta_{\mu} \quad \text{and} \quad j_5 := \bar{\psi}^* \gamma \gamma_5 \psi = j_5^{\mu} \eta_{\mu},$	$j := \bar{\psi}^* \gamma \psi = j^{\mu} \eta_{\mu} \quad \text{and} \quad j_5 := \bar{\psi}^* \gamma \gamma_5 \psi = j_5^{\mu} \eta_{\mu},$	$j := \overline{\psi}^* \gamma \psi = j^{\mu} \eta_{\mu} \quad \text{and} \quad j_5 := \overline{\psi}^* \gamma \gamma_5 \psi = j_5^{\mu} \eta_{\mu} ,$
mielke-geometrodynamics_116 E00347 (5.4.5)		■ 0.998 ● N +0	$\mu = \frac{4}{3} \kappa \Lambda = \frac{1}{3} (2\kappa)^2 \rho_{\Lambda} \leq \frac{4}{3} g_{\mathrm{N}*} \lambda_* \simeq 0.2$	$\mu = \frac{4}{3} \kappa \Lambda = \frac{1}{3} (2\kappa)^2 \rho_{\Lambda} \leq \frac{4}{3} g_{\mathrm{N}*} \lambda_* \simeq 0.2$	$\mu = \frac{4}{3} \kappa \Lambda = \frac{1}{3} (2\kappa)^2 \rho_{\Lambda} \leq \frac{4}{3} g_{\mathrm{N}*} \lambda_* \simeq 0.2$
mielke-geometrodynamics_163 E00500 (7.7.18)		■ 0.998 ● K +0	$\left\{ \right\}^{\star} \sigma = \frac{1}{2} \sigma_{\alpha\beta} \left(\vartheta^{\alpha} \wedge \vartheta^{\beta} \right) = \frac{1}{2} \sigma_{\alpha\beta} \eta^{\alpha\beta} =: \sigma^{(\star)} = i \gamma_5 \sigma.$	${}^* \sigma = \frac{1}{2} \sigma_{\alpha\beta} {}^* \left(\vartheta^{\alpha} \wedge \vartheta^{\beta} \right) = \frac{1}{2} \sigma_{\alpha\beta} \eta^{\alpha\beta} =: \sigma^{(*)} = i \gamma_5 \sigma.$	${}^* \sigma = \frac{1}{2} \sigma_{\alpha\beta} {}^* (\vartheta^{\alpha} \wedge \vartheta^{\beta}) = \frac{1}{2} \sigma_{\alpha\beta} \eta^{\alpha\beta} =: \sigma^{(*)} = i \gamma_5 \sigma .$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_00670 (10.3.27)	219	■ 0.998 ● N +0	$\check{\mathrm{H}}_{\alpha}:=\mathrm{H}_{\alpha}\cong\Sigma_{\alpha}+E_{\alpha}-(-1)^s\eta_{\alpha\beta\gamma}\vartheta_{\alpha}\wedge\Gamma^{\star\beta}\wedge H^{\gamma}$	$\check{\mathcal{E}}_{\alpha}:=dH_{\alpha}\cong\Sigma_{\alpha}+E_{\alpha}-(-1)^s\eta_{\alpha\beta\gamma}\Gamma^{\star\beta}\wedge H^{\gamma}$	$\check{\mathcal{E}}_{\alpha}:=dH_{\alpha}\cong\Sigma_{\alpha}+E_{\alpha}-(-1)^s\eta_{\alpha\beta\gamma}\Gamma^{\star\beta}\wedge H^{\gamma}$
mielke-geometrodynamics_00680 (10.3.37)	221	■ 0.998 ● W +1	$\theta_{\mathrm{L}}C_{\mathrm{L}}-\tilde{\theta}_{\mathrm{L}}\tilde{C}_{\mathrm{L}}=-\frac{\chi}{\ell}\vartheta^{\alpha}\wedge R_{\alpha}^{\star}-\frac{\Lambda}{\ell}\eta+\frac{\theta_{\mathrm{T}}}{2\ell}\vartheta^{\alpha}\wedge T_{\alpha}+\Delta\theta_{\mathrm{L}}C_{\mathrm{L}}+\frac{\Delta\theta_{\mathrm{TL}}}{\ell}d(\Gamma^{\star\alpha}\wedge\vartheta_{\alpha})$	$\theta_{\mathrm{L}}C_{\mathrm{L}}-\tilde{\theta}_{\mathrm{L}}\tilde{C}_{\mathrm{L}}=-\frac{\chi}{\ell}\vartheta^{\alpha}\wedge R_{\alpha}^{\star}-\frac{\Lambda}{\ell}\eta+\frac{\theta_{\mathrm{T}}}{2\ell}\vartheta^{\alpha}\wedge T_{\alpha}+\Delta\theta_{\mathrm{L}}C_{\mathrm{L}}+\frac{\Delta\theta_{\mathrm{TL}}}{\ell}d(\Gamma^{\star\alpha}\wedge\vartheta_{\alpha})$	$\theta_{\mathrm{L}}C_{\mathrm{L}}-\tilde{\theta}_{\mathrm{L}}\tilde{C}_{\mathrm{L}}=-\frac{\chi}{\ell}\vartheta^{\alpha}\wedge R_{\alpha}^{\star}-\frac{\Lambda}{\ell}\eta+\frac{\theta_{\mathrm{T}}}{2\ell}\vartheta^{\alpha}\wedge T_{\alpha}+\Delta\theta_{\mathrm{L}}C_{\mathrm{L}}+\frac{\Delta\theta_{\mathrm{TL}}}{\ell}d(\Gamma^{\star\alpha}\wedge\vartheta_{\alpha})$
mielke-geometrodynamics_00700 (10.5.11)	225	■ 0.998 ● K +0	$[\hat{D},\hat{D}]=\hat{R}+T^{\alpha}\hat{D}_{\alpha}$	$[\hat{D},\hat{D}]=\hat{R}+T^{\alpha}\hat{D}_{\alpha}$	$[\hat{D},\hat{D}]=\hat{R}+T^{\alpha}\hat{D}_{\alpha}$
mielke-geometrodynamics_00820 (11.4.34)	258	■ 0.998 ● N -1	$f^l(\rho)=\bar{\rho}^l\left(1+\bar{e}^{4l+2}\right)^{-1/2}$	$f^l(\rho)=\bar{\rho}^l\left(1+\bar{e}^{4l+2}\right)^{-1/2}$	$f^l(\rho)=\bar{\rho}^l(1+\bar{e}^{4l+2})^{-1/2}$
mielke-geometrodynamics_01126 (B.38)	373	■ 0.998 ● N +0	$\partial:C_p(M,\mathbb{R})\rightarrow C_{p-1}(M,\mathbb{R}),$	$\partial:C_p(M,\mathbb{R})\rightarrow C_{p-1}(M,\mathbb{R}),$	$\partial:C_p(M,\mathbb{R})\rightarrow C_{p-1}(M,\mathbb{R}),$
mielke-geometrodynamics_00516 (7.7.34)	166	■ 0.998 ●	$\widetilde{T}=T-3d\varphi,\quad\mathscr{A}^{\theta}=\mathscr{A}-id\theta.$	$\widetilde{T}=T-3d\varphi,\quad\mathscr{A}^{\theta}=\mathscr{A}-id\theta.$	$\widetilde{T}=T-3d\varphi,\quad\mathscr{A}^{\theta}=\mathscr{A}-id\theta.$
mielke-geometrodynamics_00967 (14.2.8)	313	■ 0.998 ● N +1	$d\stackrel{\circ}{s}=g_{ij}^{\circ}d\xi^i\otimes_sd\xi^j$	$d\stackrel{\circ}{s}=g_{ij}^{\circ}d\xi^i\otimes_sd\xi^j$	$d\stackrel{\circ}{s}=g_{ij}^{\circ}d\xi^i\otimes_sd\xi^j$
mielke-geometrodynamics_00060 (2.8.1)	046	■ 0.999 ● K +0	$\operatorname{det}\left(1+\frac{i}{2\pi}\Omega\right)=\pi^{\ast}\left(1+\gamma_1+\cdots+\gamma_m\right)$	$\det\left(1+\frac{i}{2\pi}\Omega\right)=\pi^{\ast}\left(1+\gamma_1+\cdots+\gamma_m\right)$	$\det\left(1+\frac{i}{2\pi}\Omega\right)=\pi^{\ast}\left(1+\gamma_1+\cdots+\gamma_m\right)$
mielke-geometrodynamics_00071 (2.8.14)	048	■ 0.999 ● K +0	$\stackrel{\infty}{\omega}:= -G^{-1}dG,$	$\stackrel{\infty}{\omega}:= -G^{-1}dG,$	$\stackrel{\infty}{\omega}:= -G^{-1}dG,$
mielke-geometrodynamics_00235 (4.2.28)	089	■ 0.999 ● K +0	$Q_{\kappa\mu\nu}:= -D_{\kappa}g_{\mu\nu}$	$Q_{\kappa\mu\nu}:= -D_{\kappa}g_{\mu\nu}$	$Q_{\kappa\mu\nu}:= -D_{\kappa}g_{\mu\nu}$
mielke-geometrodynamics_00278 (4.4.6)	097	■ 0.999 ● S +0	$D\Pi^{\mathrm{T}}-E=\Sigma$ $D\Pi^{\mathrm{L}}+\Pi_{\square}^{\mathrm{T}}=\tau_{\mathrm{s}}$	$D\Pi^{\mathrm{T}}-E=\Sigma$ $D\Pi^{\mathrm{L}}+\Pi_{\square}^{\mathrm{T}}=\tau_{\mathrm{s}}$	$D\Pi^{\mathrm{T}}-E=\Sigma$ $D\Pi^{\mathrm{L}}+\Pi_{\square}^{\mathrm{T}}=\tau_{\mathrm{s}}$
mielke-geometrodynamics_00313 (4.6.5)	105	■ 0.999 ● N +0	$\Phi(A,\gamma)=P\exp[(i/\hbar)\oint A],$	$\Phi(A,\gamma)=P\exp[(i/\hbar)\oint A],$	$\Phi(A,\gamma)=P\exp[(i/\hbar)\oint A],$
mielke-geometrodynamics_00358 (6.2.9)	125	■ 0.999 ● N +0	$L_{\parallel}:= -\frac{1}{2\kappa}T^{\alpha}\wedge^{\ast}\left({}^{(1)}T_{\alpha}-2{}^{(2)}T_{\alpha}-\frac{1}{2}{}^{(3)}T_{\alpha}\right),$	$L_{\parallel}:= -\frac{1}{2\kappa}T^{\alpha}\wedge^{\ast}\left({}^{(1)}T_{\alpha}-2{}^{(2)}T_{\alpha}-\frac{1}{2}{}^{(3)}T_{\alpha}\right),$	$L_{\parallel}:= -\frac{1}{2\kappa}T^{\alpha}\wedge^{\ast}\left({}^{(1)}T_{\alpha}-2{}^{(2)}T_{\alpha}-\frac{1}{2}{}^{(3)}T_{\alpha}\right),$
mielke-geometrodynamics_00461 (7.2.13)	152	■ 0.999 ● N +0	$\stackrel{(\mp)}{\Omega}_{\mathrm{C}}=0$	$\stackrel{(\mp)}{\Omega}_{\mathrm{C}}=0$	$\stackrel{(\mp)}{\Omega}_{\mathrm{C}}=0$
mielke-geometrodynamics_00620 (10.1.8)	209	■ 0.999 ● N +0	$\frac{\delta C_{\mathrm{T}}}{\delta\vartheta^{\alpha}}=\frac{1}{\ell^2}T_{\alpha}\quad\text{and}\quad\frac{\delta C_{\mathrm{T}}}{\delta\Gamma^{\star\alpha}}=\frac{(-1)^s}{\ell^2}\eta_{\alpha},$	$\frac{\delta C_{\mathrm{T}}}{\delta\vartheta^{\alpha}}=\frac{1}{\ell^2}T_{\alpha}\quad\text{and}\quad\frac{\delta C_{\mathrm{T}}}{\delta\Gamma^{\star\alpha}}=\frac{(-1)^s}{\ell^2}\eta_{\alpha},$	$\frac{\delta C_{\mathrm{T}}}{\delta\vartheta^{\alpha}}=\frac{1}{\ell^2}T_{\alpha}\quad\text{and}\quad\frac{\delta C_{\mathrm{T}}}{\delta\Gamma^{\star\alpha}}=\frac{(-1)^s}{\ell^2}\eta_{\alpha},$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_071 E08173 (3.7.1)		■ 0.999 ● N +0	$L_{\mathrm{H}}=-\bar{\mathrm{\Theta}}^{\mathrm{(1)}}\wedge^*\mathrm{D}\phi^0+\bar{\mathrm{\Theta}}^{\mathrm{(1)}}\wedge^*\Theta^{\mathrm{(1)}}+\frac{\lambda}{4}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)^2\eta.$	$L_{\mathrm{H}}=-\bar{\Theta}^{\mathrm{(1)}}\wedge^*D\phi^0+\bar{\Theta}^{\mathrm{(1)}}\wedge^*\Theta^{\mathrm{(1)}}+\frac{\lambda}{4}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)^2\eta.$	$L_{\mathrm{H}}=-\bar{\Theta}^{\mathrm{(1)}}\wedge^*D\phi^0+\bar{\Theta}^{\mathrm{(1)}}\wedge^*\Theta^{\mathrm{(1)}}+\frac{\lambda}{4}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)^2\eta.$
mielke-geometrodynamics_083 E08281 (4.1.15)		■ 0.999 ● N +0	$\delta\vartheta=g^{-1}\vartheta,\quad g\in GL(n,\mathbb{R}),$	$\delta_g^*\vartheta=g^{-1}\vartheta,\quad g\in GL(n,\mathbb{R}),$	$\delta_g^*\vartheta=g^{-1}\vartheta,\quad g\in GL(n,\mathbb{R}),$
mielke-geometrodynamics_097 E08275 (4.4.3)		■ 0.999 ● W +1	$\begin{aligned} & \mathrm{d}(\delta\vartheta)=\mathrm{d}(\delta\vartheta)-\mathrm{d}(\delta\vartheta)\\ & \mathrm{d}(\delta\vartheta)=\mathrm{d}(\delta\vartheta)-\mathrm{d}(\delta\vartheta)\\ & \mathrm{d}(\delta\vartheta)=\mathrm{d}(\delta\vartheta)-\mathrm{d}(\delta\vartheta) \end{aligned}$	$\begin{aligned} & =d(\delta\vartheta)-[\omega^{\mathrm{L}},\delta\vartheta]-[\delta\omega^{\mathrm{L}},\vartheta] \\ & =D(\delta\vartheta)-[\delta\omega^{\mathrm{L}},\vartheta]. \end{aligned}$	$\begin{aligned} & =d(\delta\vartheta)-[\omega^{\mathrm{L}},\delta\vartheta]-[\delta\omega^{\mathrm{L}},\vartheta] \\ & =D(\delta\vartheta)-[\delta\omega^{\mathrm{L}},\vartheta]. \end{aligned}$
mielke-geometrodynamics_115 E08341 (5.3.18)		■ 0.999 ● K +0	$\mathscr{A}=\frac{\kappa}{2}j_5,$	${}^*\mathscr{A}=\frac{\kappa}{2}j_5,$	${}^*\mathscr{A}=\frac{\kappa}{2}j_5,$
mielke-geometrodynamics_143 E08441 (6.8.12)		■ 0.999 ● N +0	$D\tau_{\alpha\beta}+\vartheta_{[\alpha}\wedge\Sigma_{\beta]}\cong0,$	$D\tau_{\alpha\beta}+\vartheta_{[\alpha}\wedge\Sigma_{\beta]}\cong0,$	$D\tau_{\alpha\beta}+\vartheta_{[\alpha}\wedge\Sigma_{\beta]}\cong0,$
mielke-geometrodynamics_156 E08476 (7.4.4)		■ 0.999 ● K +0	$C_{\alpha\beta}=0\quad\text{and}\quad R=0,$	$C_{\alpha\beta}=0\quad\text{and}\quad R=0,$	$C_{\alpha\beta}=0\quad\text{and}\quad R=0,$
mielke-geometrodynamics_166 E08519 (7.7.37)		■ 0.999 ● N +0	$D\Theta\equiv[\Omega^g,\gamma],\quad D\Omega^g\equiv0,$	$D\Theta\equiv[\Omega^g,\gamma],\quad D\Omega^g\equiv0,$	$D\Theta\equiv[\Omega^g,\gamma],\quad D\Omega^g\equiv0,$
mielke-geometrodynamics_183 E08567 (8.6.4)		■ 0.999 ●	$\begin{aligned} & H_{\alpha\beta}(\ast\ast)=\theta_{\mathrm{L}}R_{\alpha\beta}+\theta_{\mathrm{L}}^{\ast}R_{\alpha\beta}^{\ast}+\frac{\theta_{\mathrm{T}}^{\ast}}{2\ell^2}\eta_{\alpha\beta}-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}\wedge\vartheta_{\beta} \\ & \simeq-\partial L_{\theta}/\partial R^{\alpha\beta}, \end{aligned}$	$\begin{aligned} & H_{\alpha\beta}(\ast\ast)=\theta_{\mathrm{L}}R_{\alpha\beta}+\theta_{\mathrm{L}}^{\ast}R_{\alpha\beta}^{\ast}+\frac{\theta_{\mathrm{T}}^{\ast}}{2\ell^2}\eta_{\alpha\beta}-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}\wedge\vartheta_{\beta} \\ & \simeq-\partial L_{\theta}/\partial R^{\alpha\beta} \end{aligned}$	$\begin{aligned} & H_{\alpha\beta}(\ast\ast)=\theta_{\mathrm{L}}R_{\alpha\beta}+\theta_{\mathrm{L}}^{\ast}R_{\alpha\beta}^{\ast}+\frac{\theta_{\mathrm{T}}^{\ast}}{2\ell^2}\eta_{\alpha\beta}-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}\wedge\vartheta_{\beta} \\ & \simeq-\partial L_{\theta}/\partial R^{\alpha\beta}, \end{aligned}$
mielke-geometrodynamics_210 E08624 (10.1.12)		■ 0.999 ● K +0	$DH_{\alpha}-E_{\alpha}=\Sigma_{\alpha},$	$DH_{\alpha}-E_{\alpha}=\Sigma_{\alpha},$	$DH_{\alpha}-E_{\alpha}=\Sigma_{\alpha},$
mielke-geometrodynamics_220 E08673 (10.3.30)		■ 0.999 ● C -4	$H_{\alpha}=-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}-\frac{\theta_{\mathrm{TL}}}{\ell}\Gamma_{\alpha}^{\ast},\quad H_{\alpha}^{\ast}=-\frac{\theta_{\mathrm{L}}}{2}\Gamma_{\alpha}^{\ast},$	$H_{\alpha}=-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}-\frac{\theta_{\mathrm{TL}}}{\ell}\Gamma_{\alpha}^{\ast},\quad H_{\alpha}^{\ast}=-\frac{\theta_{\mathrm{L}}}{2}\Gamma_{\alpha}^{\ast},$	$H_{\alpha}=-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}-\frac{\theta_{\mathrm{TL}}}{\ell}\Gamma_{\alpha}^{\ast},\quad H_{\alpha}^{\ast}=-\frac{\theta_{\mathrm{L}}}{2}\Gamma_{\alpha}^{\ast},$
mielke-geometrodynamics_230 E08722 (10.10.2)		■ 0.999 ● N +0	$\delta L=\delta\vartheta^{\alpha}\wedge\frac{\delta L}{\delta\vartheta^{\alpha}}+\delta\Gamma_{\alpha}^{\ast}\wedge\frac{\delta L}{\delta\Gamma_{\alpha}^{\ast}}+\delta\bar{\Psi}\wedge\frac{\delta L}{\delta\bar{\Psi}},$	$\delta L=\delta\vartheta^{\alpha}\wedge\frac{\delta L}{\delta\vartheta^{\alpha}}+\delta\Gamma_{\alpha}^{\ast}\wedge\frac{\delta L}{\delta\Gamma_{\alpha}^{\ast}}+\delta\bar{\Psi}\wedge\frac{\delta L}{\delta\bar{\Psi}},$	$\delta L=\delta\vartheta^{\alpha}\wedge\frac{\delta L}{\delta\vartheta^{\alpha}}+\delta\Gamma_{\alpha}^{\ast}\wedge\frac{\delta L}{\delta\Gamma_{\alpha}^{\ast}}+\delta\bar{\Psi}\wedge\frac{\delta L}{\delta\bar{\Psi}},$
mielke-geometrodynamics_242 E08759 (11.1.26)		■ 0.999 ● N +0	$\underbrace{(\gamma\wedge\cdots\wedge\gamma)}_p=\frac{p!}{(n-p)!}\gamma^{n+1}\underbrace{\gamma\wedge\cdots\wedge\gamma}_{n-p}.$	$\underbrace{(\gamma\wedge\cdots\wedge\gamma)}_p=\frac{p!}{(n-p)!}\gamma^{n+1}\underbrace{\gamma\wedge\cdots\wedge\gamma}_{n-p}.$	$\underbrace{(\gamma\wedge\cdots\wedge\gamma)}_p=\frac{p!}{(n-p)!}\gamma^{n+1}\underbrace{\gamma\wedge\cdots\wedge\gamma}_{n-p}.$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamicsEQ0780 (11.2.8)	247	0.999 K +0	$(\psi,\chi):=\int_{H^{n-1}}\tau_s(\psi,\chi)=i\int_{H^{n-1}}\psi^*\gamma\chi$	$(\psi,\chi):=\int_{H^{n-1}}\tau_s(\psi,\chi)=i\int_{H^{n-1}}\psi^*\gamma\chi$	$(\psi,\chi):=\int_{H^{n-1}}\tau_s(\psi,\chi)=i\int_{H^{n-1}}\psi^*\gamma\chi$
mielke-geometrodynamicsEQ0789 (11.4.1)	251	0.999 N +0	$E_r=\sqrt{\alpha}r_{\circ}^2\left[1+\left(\frac{r}{r_{\circ}}\right)^4\right]^{-1/2}\sim\sqrt{\alpha}\frac{1}{r^2}$	$E_r=\sqrt{\alpha}r_{\circ}^2\left[1+\left(\frac{r}{r_{\circ}}\right)^4\right]^{-1/2}\sim\sqrt{\alpha}\frac{1}{r^2}$	$E_r=\sqrt{\alpha}r_{\circ}^2\left[1+\left(\frac{r}{r_{\circ}}\right)^4\right]^{-1/2}\sim\sqrt{\alpha}\frac{1}{r^2}$
mielke-geometrodynamicsEQ0825 (11.4.39)	259	0.999 N +0	$m_{\pi}=\sqrt{1-\omega^2}m$	$m_{\pi}=\sqrt{1-\omega^2}m$	$m_{\pi}=\sqrt{1-\omega^2}m$
mielke-geometrodynamicsEQ0892 (13.4.11)	287	0.999 K +0	$\vartheta\stackrel{A^{-1}(x)}{\longrightarrow}\vartheta'=\Lambda^{-1}(x)\vartheta,\quad\theta\stackrel{A^{-1}(x)}{\longrightarrow}\theta'=\theta\Lambda^{-1}(x).$	$\vartheta\stackrel{A^{-1}(x)}{\longrightarrow}\vartheta'=\Lambda^{-1}(x)\vartheta,\quad\theta\stackrel{A^{-1}(x)}{\longrightarrow}\theta'=\theta\Lambda^{-1}(x).$	$\vartheta\stackrel{A^{-1}(x)}{\longrightarrow}\vartheta'=\Lambda^{-1}(x)\vartheta,\quad\theta\stackrel{A^{-1}(x)}{\longrightarrow}\theta'=\theta\Lambda^{-1}(x).$
mielke-geometrodynamicsEQ0932 (14.1.13)	304	0.999 N +0	$e^{\circ\delta}\circ e^{\circ\varepsilon}=e^{\circ\varepsilon}\circ e^{\circ\delta}=e^{\circ(\delta+\varepsilon)},$	$e^{\circ\delta}\circ e^{\circ\varepsilon}=e^{\circ\varepsilon}\circ e^{\circ\delta}=e^{\circ(\delta+\varepsilon)},$	$e^{\circ\delta}\circ e^{\circ\varepsilon}=e^{\circ\varepsilon}\circ e^{\circ\delta}=e^{\circ(\delta+\varepsilon)},$
mielke-geometrodynamicsEQ0940 (14.1.21)	305	0.999 N +0	$\xi\wedge\xi=\xi\wedge^*\xi=0$	$\xi\wedge\xi=\xi\wedge^*\xi=0$	$\xi\wedge\xi=\xi\wedge^*\xi=0$
mielke-geometrodynamicsEQ0951 (14.1.33)	307	0.999 W +2	$\begin{aligned} & D^*\xi-(-1)^z\xi\wedge D\delta=\overline{\tau}_e\\ & D\xi+(-1)^z{}^*\xi\wedge D\delta=\overline{\tau}_p \end{aligned}$	$\begin{aligned} & D^*\xi-(-1)^z\xi\wedge D\delta=\overline{\tau}_e\\ & D\xi+(-1)^z{}^*\xi\wedge D\delta=\overline{\tau}_p \end{aligned}$	$\begin{aligned} & D^*\xi-(-1)^z\xi\wedge D\delta=\overline{\tau}_e,\\ & D\xi+(-1)^z{}^*\xi\wedge D\delta=\overline{\tau}_p. \end{aligned}$
mielke-geometrodynamicsEQ0968 (14.2.9)	314	0.999	$\hat{\omega}=\left[\begin{array}{c c}\omega^g&-\alpha\\\hline\alpha^T&\hat{\omega}\end{array}\right]=:\omega^g\oplus\overset{\circ}{\omega}-\hat{\alpha};\quad\hat{\alpha}=\left[\begin{array}{c c}0&\alpha\\\hline-\alpha^T&0\end{array}\right]$	$\hat{\omega}=\left[\begin{array}{c c}\omega^g&-\alpha\\\hline\alpha^T&\hat{\omega}\end{array}\right]=:\omega^g\oplus\overset{\circ}{\omega}-\hat{\alpha};\quad\hat{\alpha}=\left[\begin{array}{c c}0&\alpha\\\hline-\alpha^T&0\end{array}\right]$	$\hat{\omega}=\left[\begin{array}{c c}\omega^g&-\alpha\\\hline\alpha^T&\overset{\circ}{\omega}\end{array}\right]=:\omega^g\oplus\overset{\circ}{\omega}-\hat{\alpha};\quad\hat{\alpha}=\left[\begin{array}{c c}0&\alpha\\\hline-\alpha^T&0\end{array}\right]$
mielke-geometrodynamicsEQ1039 (15.1.16)	339	0.999 C +9	$\begin{aligned} & d\,s_{\scriptscriptstyle f}^2=f_{ij}dx^i\otimes_sdx^j\\ & =-\frac{3\gamma}{2}e^{-\lambda^f}dt^2+2\left[\gamma\left\{1-\left(1+\frac{9\gamma}{4}\right)e^{\lambda^f-\lambda^g}+\frac{9\gamma}{4}e^{2(\lambda^f-\lambda^g)}\right\}\right]^{1/2}dtdr\\ & \quad +e^{\lambda^g}\left\{\frac{2}{3}+\frac{3\gamma}{2}\left(1-e^{\lambda^g-\lambda^f}\right)\right\}dr^2+\frac{2}{3}r^2d\Omega^2 \end{aligned}$	$\begin{aligned} & ds_f^2=f_{ij}dx^i\otimes_sdx^j\\ & =-\frac{3\gamma}{2}e^{-\lambda^f}dt^2+2\left[\gamma\left\{1-\left(1+\frac{9\gamma}{4}\right)e^{\lambda^f-\lambda^g}+\frac{9\gamma}{4}e^{2(\lambda^f-\lambda^g)}\right\}\right]^{1/2}dtdr\\ & \quad +e^{\lambda^g}\left\{\frac{2}{3}+\frac{3\gamma}{2}\left(1-e^{\lambda^g-\lambda^f}\right)\right\}dr^2+\frac{2}{3}r^2d\Omega^2 \end{aligned}$	$\begin{aligned} ds_f^2&=f_{ij}dx^i\otimes_sdx^j\\ &=-\frac{3\gamma}{2}e^{-\lambda^f}dt^2+2\left[\gamma\left\{1-\left(1+\frac{9\gamma}{4}\right)e^{\lambda^f-\lambda^g}+\frac{9\gamma}{4}e^{2(\lambda^f-\lambda^g)}\right\}\right]^{1/2}dtdr\\ &\quad +e^{\lambda^g}\left\{\frac{2}{3}+\frac{3\gamma}{2}\left(1-e^{\lambda^g-\lambda^f}\right)\right\}dr^2+\frac{2}{3}r^2d\Omega^2 \end{aligned}\tag{15.1.16}$
mielke-geometrodynamicsEQ1088 (A.15)	366	0.999 N +0	$C(1,n-1)=\left[\bigotimes_{i=1}^{\left[\frac{n-2}{2}\right]}C(0,2)\right]\otimes C(1,1).$	$C(1,n-1)=\left[\bigotimes_{i=1}^{\left[\frac{n-2}{2}\right]}C(0,2)\right]\otimes C(1,1).$	$C(1,n-1)=\left[\bigotimes_{i=1}^{\left[\frac{n-2}{2}\right]}C(0,2)\right]\otimes C(1,1).$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometroynamics_00434 (6.8.5)	142	■ 0.999 ● N -1	$\begin{aligned} H_{\alpha} &:= -\frac{\partial L_g}{\partial d\vartheta^{\alpha}} = -\frac{\partial L_g}{\partial T^{\alpha}}, \quad \text{and} \quad H_{\alpha\beta} := -\frac{\partial L_g}{\partial d\Gamma^{\alpha\beta}} = -\frac{\partial L_g}{\partial R^{\alpha\beta}}. \\ H_{\alpha\beta} &:= -\partial L_{\mathrm{qPG}}/\partial R^{\alpha\beta} = \frac{a_0}{2\ell^2} \eta_{\alpha\beta} + \left(\sum_{N=1}^6 b_{(N)} {}^{(N)} R_{\alpha\beta} \right) \end{aligned}$	$H_{\alpha} := -\frac{\partial L_g}{\partial d\vartheta^{\alpha}} = -\frac{\partial L_g}{\partial T^{\alpha}}, \quad \text{and} \quad H_{\alpha\beta} := -\frac{\partial L_g}{\partial d\Gamma^{\alpha\beta}} = -\frac{\partial L_g}{\partial R^{\alpha\beta}}.$	$H_{\alpha} := -\frac{\partial L_g}{\partial d\vartheta^{\alpha}} = -\frac{\partial L_g}{\partial T^{\alpha}}, \quad \text{and} \quad H_{\alpha\beta} := -\frac{\partial L_g}{\partial d\Gamma^{\alpha\beta}} = -\frac{\partial L_g}{\partial R^{\alpha\beta}}.$
mielke-geometroynamics_00566 (8.6.3)	183	■ 0.999 ● N +0	$H_{\alpha\beta} := -\partial L_{\mathrm{qPG}}/\partial R^{\alpha\beta} = \frac{a_0}{2\ell^2} \eta_{\alpha\beta} + \left(\sum_{N=1}^6 b_{(N)} {}^{(N)} R_{\alpha\beta} \right)$	$H_{\alpha\beta} := -\partial L_{\mathrm{qPG}}/\partial R^{\alpha\beta} = \frac{a_0}{2\ell^2} \eta_{\alpha\beta} + \left(\sum_{N=1}^6 b_{(N)} {}^{(N)} R_{\alpha\beta} \right)$	$H_{\alpha\beta} := -\partial L_{\mathrm{qPG}}/\partial R^{\alpha\beta} = \frac{a_0}{2\ell^2} \eta_{\alpha\beta} + \left(\sum_{N=1}^6 b_{(N)} {}^{(N)} R_{\alpha\beta} \right)$
mielke-geometroynamics_00798 (11.4.10)	254	■ 0.999 ● W +0	$\begin{aligned} & \left(i\bar{\gamma}^k \left(\partial_k + i\bar{\Gamma}_k + \frac{3}{4}\partial_k\mu \right) \right. \\ & \left. + \frac{3\varepsilon}{8}\ell^2 e^{\mu/2} \left(\bar{\psi}\psi - \mathfrak{a}\bar{\psi}\gamma^5\psi\gamma^5 \right) - e^{\mu/2}m \right) \psi = 0. \end{aligned}$	$\left\{ i\bar{\gamma}^k \left(\partial_k + i\bar{\Gamma}_k + \frac{3}{4}\partial_k\mu \right) + \frac{3\varepsilon}{8}\ell^2 e^{\mu/2} \left(\bar{\psi}\psi - \mathfrak{a}\bar{\psi}\gamma^5\psi\gamma^5 \right) - e^{\mu/2}m \right\} \psi = 0.$	$\left\{ i\bar{\gamma}^k (\partial_k + i\bar{\Gamma}_k + \frac{3}{4}\partial_k\mu) + \frac{3\varepsilon}{8}\ell^2 e^{\mu/2} (\bar{\psi}\psi - \mathfrak{a}\bar{\psi}\gamma^5\psi\gamma^5) - e^{\mu/2}m \right\} \psi = 0.$
mielke-geometroynamics_00833 (12.2.1)	270	■ 0.999 ● N +0	$L_{\mathrm{D}} = L(\gamma, \psi, \mathscr{D}\psi) = \frac{i}{2} \left\{ \bar{\psi}^* \gamma \wedge \mathscr{D}\psi + \overline{\mathscr{D}\psi} \wedge^* \gamma \psi \right\} - m \bar{\psi} \psi \eta,$	$L_{\mathrm{D}} = L(\gamma, \psi, \mathscr{D}\psi) = \frac{i}{2} \left\{ \bar{\psi}^* \gamma \wedge \mathscr{D}\psi + \overline{\mathscr{D}\psi} \wedge^* \gamma \psi \right\} - m \bar{\psi} \psi \eta,$	$L_{\mathrm{D}} = L(\gamma, \psi, \mathscr{D}\psi) = \frac{i}{2} \left\{ \bar{\psi}^* \gamma \wedge \mathscr{D}\psi + \overline{\mathscr{D}\psi} \wedge^* \gamma \psi \right\} - m \bar{\psi} \psi \eta,$
mielke-geometroynamics_00920 (2.4.2)	038	■ 1.000 ● N +0	$V^{\rho} := V^{\rho} \left(M, \mathbb{C}^N, \rho(G) \subseteq GL(N, \mathbb{C}), P \right),$	$V^{\rho} := V^{\rho} \left(M, \mathbb{C}^N, \rho(G) \subseteq GL(N, \mathbb{C}), P \right),$	$V^{\rho} := V^{\rho} (M, \mathbb{C}^N, \rho(G) \subseteq GL(N, \mathbb{C}), P),$
mielke-geometroynamics_00991 (3.2.1)	055	■ 1.000 ●	$\mathscr{O} \rightarrow {}^{G^{-1}}\mathscr{O} = G^{-1}\mathscr{O}G.$	$\mathscr{O} \rightarrow {}^{G^{-1}}\mathscr{O} = G^{-1}\mathscr{O}G.$	$\mathscr{O} \rightarrow {}^{G^{-1}}\mathscr{O} = G^{-1}\mathscr{O}G.$
mielke-geometroynamics_00101 (3.3.2)	057	■ 1.000 ● N -1	$A \rightarrow A + d\theta(x) \quad \Rightarrow \quad F \rightarrow dA + dd\theta = dA = F.$	$A \rightarrow A + d\theta(x) \quad \Rightarrow \quad F \rightarrow dA + dd\theta = dA = F.$	$A \rightarrow A + d\theta(x) \quad \Rightarrow \quad F \rightarrow dA + dd\theta = dA = F.$
mielke-geometroynamics_00137 (3.4.8)	063	■ 1.000 ● N +0	$\begin{aligned} \delta L_{\omega} &= \delta\Omega \wedge \Pi^{\Omega} = D(\delta\omega) \wedge \Pi^{\Omega} \\ &= -\delta\omega \wedge D\Pi^{\Omega} + d\left(\delta\omega \wedge \Pi^{\Omega} \right). \end{aligned}$	$\delta L_{\omega} = \delta\Omega \wedge \Pi^{\Omega} = D(\delta\omega) \wedge \Pi^{\Omega} = -\delta\omega \wedge D\Pi^{\Omega} + d\left(\delta\omega \wedge \Pi^{\Omega} \right).$	$\delta L_{\omega} = \delta\Omega \wedge \Pi^{\Omega} = D(\delta\omega) \wedge \Pi^{\Omega} = -\delta\omega \wedge D\Pi^{\Omega} + d(\delta\omega \wedge \Pi^{\Omega}).$
mielke-geometroynamics_00146 (3.4.17)	065	■ 1.000 ● N +0	$Q = Q^{(i)} I_i := \int_{H^3} \iota = \int_{\partial H^3} \Pi^{\Omega},$	$Q = Q^{(i)} I_i := \int_{H^3} \iota = \int_{\partial H^3} \Pi^{\Omega},$	$Q = Q^{(i)} I_i := \int_{H^3} \iota = \int_{\partial H^3} \Pi^{\Omega},$
mielke-geometroynamics_00175 (3.7.4)	072	■ 1.000 ● N +0	$\tau_{\mathrm{H}} = - \left[\phi^{(0)}, {}^*\Theta^{(1)} \right]$	$\tau_{\mathrm{H}} = - \left[\phi^{(0)}, {}^*\Theta^{(1)} \right]$	$\tau_{\mathrm{H}} = -[\phi^{(0)}, {}^*\Theta^{(1)}]$
mielke-geometroynamics_00195 (4.1.5)	082	■ 1.000 ● N +0	$\mathfrak{a}(n, \mathbb{R}) = \mathbb{R}^n \otimes \mathfrak{gl}(n, \mathbb{R})$	$\mathfrak{a}(n, \mathbb{R}) = \mathbb{R}^n \otimes \mathfrak{gl}(n, \mathbb{R})$	$\mathfrak{a}(n, \mathbb{R}) = \mathbb{R}^n \otimes \mathfrak{gl}(n, \mathbb{R})$
mielke-geometroynamics_00208 (4.1.22)	084	■ 1.000 ● N +0	$T_{\mathrm{A}}(M) = \overset{\circ}{V}(M, \mathbb{R}^n, A(n, \mathbb{R}), A(M)) \subset V^{\mathrm{id}}$	$T_{\mathrm{A}}(M) = \overset{\circ}{V}(M, \mathbb{R}^n, A(n, \mathbb{R}), A(M)) \subset V^{\mathrm{id}}$	$T_{\mathrm{A}}(M) = \overset{\circ}{V}(M, \mathbb{R}^n, A(n, \mathbb{R}), A(M)) \subset V^{\mathrm{id}}$
mielke-geometroynamics_00212 (4.2.4)	085	■ 1.000 ● N +1	$\Omega^{\mathrm{T}} := D\omega^{\mathrm{T}} = d\omega^{\mathrm{T}} - [\omega^{\mathrm{T}}, \omega^{\mathrm{L}}],$	$\Omega^{\mathrm{T}} := D\omega^{\mathrm{T}} = d\omega^{\mathrm{T}} - [\omega^{\mathrm{T}}, \omega^{\mathrm{L}}],$	$\Omega^{\mathrm{T}} := D\omega^{\mathrm{T}} = d\omega^{\mathrm{T}} - [\omega^{\mathrm{T}}, \omega^{\mathrm{L}}],$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_086 EQ0215 (4.2.7)		■ 1.000 ● N +0	$\mathrm{T}(n,\mathbb{R}):=\hat{C}^{\infty}\left(A(M)\times_{\mathrm{Ad}}\mathbb{R}^n\right)$	$\mathcal{T}(n,\mathbb{R}):=C^{\infty}\left(A(M)\times_{\mathrm{Ad}}\mathbb{R}^n\right)$	$\mathcal{T}(n,\mathbb{R}):=C^{\infty}(A(M)\times_{\mathrm{Ad}}\mathbb{R}^n)$
mielke-geometrodynamics_086 EQ0217 (4.2.9)		■ 1.000 ● N +0	$A=\left[\begin{array}{c c} G&T\\\hline 0&1\end{array}\right]\in\mathscr{A}_p,\quad G\in\mathscr{G}_p,T\in\mathscr{T}_p$	$A=\left[\begin{array}{c c} G&T\\\hline 0&1\end{array}\right]\in\mathscr{A}_p,\quad G\in\mathscr{G}_p,T\in\mathscr{T}_p$	$A=\left[\begin{array}{c c} G&T\\\hline 0&1\end{array}\right]\in\mathscr{A}_p,\quad G\in\mathscr{G}_p,T\in\mathscr{T}_p$
mielke-geometrodynamics_091 EQ0243 (4.3.8)		■ 1.000 ● W +2	$\cos\phi=\frac{g_{\mu\nu}dV^{\mu}dW^{\nu}}{\sqrt{g_{\alpha\beta}dV^{\alpha}dV^{\beta}}\sqrt{g_{\alpha\beta}dW^{\alpha}dW^{\beta}}}$	$\cos\phi=\frac{g_{\mu\nu}dV^{\mu}dW^{\nu}}{\sqrt{g_{\alpha\beta}dV^{\alpha}dV^{\beta}}\sqrt{g_{\alpha\beta}dW^{\alpha}dW^{\beta}}}$	$\cos\phi=\frac{g_{\mu\nu}dV^{\mu}dW^{\nu}}{\sqrt{g_{\alpha\beta}dV^{\alpha}dV^{\beta}}\sqrt{g_{\alpha\beta}dW^{\alpha}dW^{\beta}}}.$
mielke-geometrodynamics_111 EQ0323 (5.2.7)		■ 1.000 ● N +0	$\begin{aligned}\Sigma_{\alpha}&\rightarrow\sigma_{\alpha}:=\Sigma_{\alpha}-D\mu_{\alpha}+e_{\beta}\rfloor\left(T^{\beta}\wedge\mu_{\alpha}\right),\\ \tau_{\alpha\beta}&\rightarrow\hat{\tau}_{\alpha\beta}:=\tau_{\alpha\beta}-\vartheta_{[\alpha}\wedge\mu_{\beta]}=0\end{aligned}$	$\begin{aligned}\Sigma_{\alpha}\rightarrow\sigma_{\alpha} &:=\Sigma_{\alpha}-D\mu_{\alpha}+e_{\beta}\rfloor\left(T^{\beta}\wedge\mu_{\alpha}\right),\\ \tau_{\alpha\beta}\rightarrow\hat{\tau}_{\alpha\beta} &:=\tau_{\alpha\beta}-\vartheta_{[\alpha}\wedge\mu_{\beta]}=0\end{aligned}$	$\begin{aligned}\Sigma_{\alpha}\rightarrow\sigma_{\alpha} &:=\Sigma_{\alpha}-D\mu_{\alpha}+e_{\beta}\rfloor\left(T^{\beta}\wedge\mu_{\alpha}\right),\\ \tau_{\alpha\beta}\rightarrow\hat{\tau}_{\alpha\beta} &:=\tau_{\alpha\beta}-\vartheta_{[\alpha}\wedge\mu_{\beta]}=0\end{aligned}$
mielke-geometrodynamics_112 EQ0328 (5.3.5)		■ 1.000 ● N +0	$\Theta:=D\gamma\quad\text{and}\quad\Omega^g:=d\Gamma+\Gamma\wedge\Gamma.$	$\Theta:=D\gamma\quad\text{and}\quad\Omega^g:=d\Gamma+\Gamma\wedge\Gamma.$	$\Theta:=D\gamma\quad\text{and}\quad\Omega^g:=d\Gamma+\Gamma\wedge\Gamma.$
mielke-geometrodynamics_127 EQ0370 (6.2.21)		■ 1.000 ● N +0	$\stackrel{(\pm)}{\Pi}_{\alpha}=-\partial^{\left(\pm\right)}L/\partial T^{\alpha}=H_{\alpha}^{\parallel}\mp\left(i/\kappa\right)T_{\alpha}$	$\stackrel{(\pm)}{\Pi}_{\alpha}=-\partial^{\left(\pm\right)}L/\partial T^{\alpha}=H_{\alpha}^{\parallel}\mp\left(i/\kappa\right)T_{\alpha}$	$\stackrel{(\pm)}{\Pi}_{\alpha}=-\partial^{\left(\pm\right)}L/\partial T^{\alpha}=H_{\alpha}^{\parallel}\mp\left(i/\kappa\right)T_{\alpha}$
mielke-geometrodynamics_129 EQ0380 (6.2.31)		■ 1.000 ● N +0	$\stackrel{(\pm)}{D}\stackrel{(\pm)}{\Pi}_{\alpha}=0,$	$\stackrel{(\pm)}{D}\stackrel{(\pm)}{\Pi}_{\alpha}=0,$	$\stackrel{(\pm)}{D}\stackrel{(\pm)}{\Pi}_{\alpha}=0,$
mielke-geometrodynamics_141 EQ0428 (6.7.10)		■ 1.000 ● N +0	$\kappa D\theta_{\mathrm{L}}\wedge DT_{\beta}=-T\wedge\eta_{\beta}=D\eta_{\beta},$	$\kappa D\theta_{\mathrm{L}}\wedge DT_{\beta}=-T\wedge\eta_{\beta}=D\eta_{\beta},$	$\kappa D\theta_{\mathrm{L}}\wedge DT_{\beta}=-T\wedge\eta_{\beta}=D\eta_{\beta},$
mielke-geometrodynamics_167 EQ0522 (7.7.41)		■ 1.000 ●	$\begin{gathered}K^{\left(\star\right)}:=\eta_{\alpha\beta\gamma}\wedge K^{\beta\gamma}\gamma^{\alpha},\\ \Omega^{g\left(\star\right)}:=\frac{i}{8}R_{\alpha\beta}\eta^{\alpha\beta\gamma\delta}\sigma_{\gamma\delta}=-\frac{1}{4}R^{\alpha\beta}\gamma_5\sigma_{\alpha\beta}=i\gamma_5\Omega^g.\end{gathered}$	$\begin{gathered}K^{\left(\star\right)}:=\eta_{\alpha\beta\gamma}\wedge K^{\beta\gamma}\gamma^{\alpha},\\ \Omega^{g\left(\star\right)}:=\frac{i}{8}R_{\alpha\beta}\eta^{\alpha\beta\gamma\delta}\sigma_{\gamma\delta}=-\frac{1}{4}R^{\alpha\beta}\gamma_5\sigma_{\alpha\beta}=i\gamma_5\Omega^g.\end{gathered}$	$\begin{gathered}K^{\left(\star\right)}:=\eta_{\alpha\beta\gamma}\wedge K^{\beta\gamma}\gamma^{\alpha},\\ \Omega^{g\left(\star\right)}:=\frac{i}{8}R_{\alpha\beta}\eta^{\alpha\beta\gamma\delta}\sigma_{\gamma\delta}=-\frac{1}{4}R^{\alpha\beta}\gamma_5\sigma_{\alpha\beta}=i\gamma_5\Omega^g.\end{gathered}$
mielke-geometrodynamics_168 EQ0526 (7.7.46)		■ 1.000 ● N -1	$L_{\mathrm{NY}}:=dC_{\mathrm{TT}}=\frac{1}{2\ell^2}\left(T^{\alpha}\wedge T_{\alpha}+R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}\right),$	$L_{\mathrm{NY}}:=dC_{\mathrm{TT}}=\frac{1}{2\ell^2}\left(T^{\alpha}\wedge T_{\alpha}+R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}\right),$	$L_{\mathrm{NY}}:=dC_{\mathrm{TT}}=\frac{1}{2\ell^2}\left(T^{\alpha}\wedge T_{\alpha}+R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}\right),$
mielke-geometrodynamics_175 EQ0535 (8.2.2)		■ 1.000 ● N +0	$\widetilde{\Gamma}:=\Gamma\oplus C,\quad\widetilde{\Omega}^g:=\Omega^g\oplus\Psi\oplus\Phi.$	$\widetilde{\Gamma}:=\Gamma\oplus C,\quad\widetilde{\Omega}^g:=\Omega^g\oplus\Psi\oplus\Phi.$	$\widetilde{\Gamma}:=\Gamma\oplus C,\quad\widetilde{\Omega}^g:=\Omega^g\oplus\Psi\oplus\Phi.$
mielke-geometrodynamics_176 EQ0538 (8.2.5)		■ 1.000 ● K +0	$s_{\mathrm{loc}}(\alpha+\lambda)=-d\lambda,\quad s_{\mathrm{loc}}\lambda=0$	$s_{\mathrm{loc}}(\alpha+\lambda)=-d\lambda,\quad s_{\mathrm{loc}}\lambda=0$	$s_{\mathrm{loc}}(\alpha+\lambda)=-d\lambda,\quad s_{\mathrm{loc}}\lambda=0$
mielke-geometrodynamics_176 EQ0539 (8.2.6)		■ 1.000 ● K +0	$(d\oplus s_{\mathrm{loc}})(\alpha+\lambda)=d\alpha.$	$(d\oplus s_{\mathrm{loc}})(\alpha+\lambda)=d\alpha.$	$(d\oplus s_{\mathrm{loc}})(\alpha+\lambda)=d\alpha.$

Conf. MathPix confidence:

■ ≥0.9

■ 0.5–0.9

■ <0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

● C component

● W weak

● S stable

● N noise

● K clean

● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_185 EQ0573 (8.6.11)	185	■1.000 ●W +2	$\begin{aligned} L_{\mathrm{qPG}}(* *) &= \frac{\Lambda}{\ell^2} \eta - \frac{a_0}{4\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} - \frac{1}{2} T^\alpha \wedge H_\alpha(* *) - \frac{1}{2} R^{\alpha\beta} \wedge H_{\alpha\beta}(* *) \\ &= \frac{\Lambda}{\ell^2} \eta - \frac{\theta_{\mathrm{T}}^* + a_0}{4\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} - \frac{1}{2\ell^2} T^\alpha \wedge \left[\theta_{\mathrm{T}}^* K_\alpha^{(*)} - \theta_{\mathrm{T}} T_\alpha \right] \\ &\quad - \frac{1}{2} R^{\alpha\beta} \wedge \left[\theta_{\mathrm{L}} R_{\alpha\beta} + \theta_{\mathrm{L}}^* R_{\alpha\beta}^{(*)} - \frac{\theta_{\mathrm{T}}}{2\ell^2} \vartheta_\alpha \wedge \vartheta_\beta \right]. \end{aligned}$	$L_{\mathrm{qPG}}(* *) = \frac{\Lambda}{\ell^2} \eta - \frac{a_0}{4\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} - \frac{1}{2} T^\alpha \wedge H_\alpha(* *) - \frac{1}{2} R^{\alpha\beta} \wedge H_{\alpha\beta}(* *)$ $= \frac{\Lambda}{\ell^2} \eta - \frac{\theta_{\mathrm{T}}^* + a_0}{4\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} - \frac{1}{2\ell^2} T^\alpha \wedge \left[\theta_{\mathrm{T}}^* K_\alpha^{(*)} - \theta_{\mathrm{T}} T_\alpha \right]$ $- \frac{1}{2} R^{\alpha\beta} \wedge \left[\theta_{\mathrm{L}} R_{\alpha\beta} + \theta_{\mathrm{L}}^* R_{\alpha\beta}^{(*)} - \frac{\theta_{\mathrm{T}}}{2\ell^2} \vartheta_\alpha \wedge \vartheta_\beta \right].$	$L_{\mathrm{qPG}}(* *) = \frac{\Lambda}{\ell^2} \eta - \frac{a_0}{4\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} - \frac{1}{2} T^\alpha \wedge H_\alpha(* *) - \frac{1}{2} R^{\alpha\beta} \wedge H_{\alpha\beta}(* *)$ $= \frac{\Lambda}{\ell^2} \eta - \frac{\theta_{\mathrm{T}}^* + a_0}{4\ell^2} R^{\alpha\beta} \wedge \eta_{\alpha\beta} - \frac{1}{2\ell^2} T^\alpha \wedge \left[\theta_{\mathrm{T}}^* K_\alpha^{(*)} - \theta_{\mathrm{T}} T_\alpha \right]$ $- \frac{1}{2} R^{\alpha\beta} \wedge \left[\theta_{\mathrm{L}} R_{\alpha\beta} + \theta_{\mathrm{L}}^* R_{\alpha\beta}^{(*)} - \frac{\theta_{\mathrm{T}}}{2\ell^2} \vartheta_\alpha \wedge \vartheta_\beta \right]. \quad (8.6)$
mielke-geometrodynamics_186 EQ0580 (8.6.18)	186	■1.000 ●N +0	$G_{\{\alpha\}^{\{\{\ \}\}} - \Lambda_{\mathrm{eff}} \eta_\alpha = \kappa_{\mathrm{eff}} \widehat{\Sigma}_\alpha.$	$G_\alpha^{\{\}} - \Lambda_{\mathrm{eff}} \eta_\alpha = \kappa_{\mathrm{eff}} \widehat{\Sigma}_\alpha.$	$G_\alpha^{\{\}} - \Lambda_{\mathrm{eff}} \eta_\alpha = \kappa_{\mathrm{eff}} \widehat{\Sigma}_\alpha.$
mielke-geometrodynamics_195 EQ0595 (9.1.13)	195	■1.000 ●N +0	$R^{\{\{\ \}\}} = -\frac{6}{\ell^{*2}} \kappa \gamma.$	$R^{\{\}} = -\frac{6\kappa\gamma}{\ell^{*2}}.$	$R^{\{\}} = -\frac{6\kappa\gamma}{\ell^{*2}}.$
mielke-geometrodynamics_214 EQ0645 (10.3.2)	214	■1.000 ●K +0	$\widetilde{T}_{\{\alpha\}} = T_{\{\alpha\}} - \frac{\varepsilon}{\ell} \eta_\alpha, \quad \widetilde{R}_{\alpha}^{\star} = R_{\alpha}^{\star} - (-1)^s \frac{\varepsilon}{2\ell} T_\alpha + (-1)^s \frac{\varepsilon^2}{4\ell^2} \eta_\alpha$	$\widetilde{T}_\alpha = T_\alpha - \frac{\varepsilon}{\ell} \eta_\alpha, \quad \widetilde{R}_\alpha^{\star} = R_\alpha^{\star} - (-1)^s \frac{\varepsilon}{2\ell} T_\alpha + (-1)^s \frac{\varepsilon^2}{4\ell^2} \eta_\alpha$	$\widetilde{T}_\alpha = T_\alpha - \frac{\varepsilon}{\ell} \eta_\alpha, \quad \widetilde{R}_\alpha^{\star} = R_\alpha^{\star} - (-1)^s \frac{\varepsilon}{2\ell} T_\alpha + (-1)^s \frac{\varepsilon^2}{4\ell^2} \eta_\alpha$
mielke-geometrodynamics_220 EQ0676 (10.3.33)	220	■1.000 ●W -2	$\check{\mathscr{E}}_{\{\alpha\}^{\{\{\ \}\}}_{\{\mathrm{MB}\}} = -(-1)^s \frac{\theta_{\mathrm{T}} \theta_{\mathrm{L}}}{2\chi \ell} \left(d\Gamma_\alpha^{\star} + \frac{2}{\theta_{\mathrm{L}}} \mathscr{J}_\alpha^{\mathrm{MB}\star} \right),$	$\check{\mathscr{E}}_{\alpha}^{\mathrm{MB}} = -(-1)^s \frac{\theta_{\mathrm{T}} \theta_{\mathrm{L}}}{2\chi \ell} \left(d\Gamma_\alpha^{\star} + \frac{2}{\theta_{\mathrm{L}}} \mathscr{J}_\alpha^{\mathrm{MB}\star} \right),$	$\check{\mathscr{E}}_{\alpha}^{\mathrm{MB}} = -(-1)^s \frac{\theta_{\mathrm{T}} \theta_{\mathrm{L}}}{2\chi \ell} \left(d\Gamma_\alpha^{\star} + \frac{2}{\theta_{\mathrm{L}}} \mathscr{J}_\alpha^{\mathrm{MB}\star} \right),$
mielke-geometrodynamics_222 EQ0688 (10.3.46)	222	■1.000 ●	$G_{\{\alpha\}^{\{\{\ \}\}} - \tilde{\Lambda} \eta_\alpha = \ell \tilde{\Sigma}_\alpha,$	$G_\alpha^{\{\}} - \tilde{\Lambda} \eta_\alpha = \ell \tilde{\Sigma}_\alpha,$	$G_\alpha^{\{\}} - \tilde{\Lambda} \eta_\alpha = \ell \tilde{\Sigma}_\alpha,$
mielke-geometrodynamics_223 EQ0690 (10.5.1)	223	■1.000 ●N +0	$\sigma^1 = \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right), \quad \sigma^2 = \left(\begin{array}{cc} 0 & -i \\ i & 0 \end{array} \right), \quad \sigma^3 = \left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right).$	$\sigma^1 = \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right), \quad \sigma^2 = \left(\begin{array}{cc} 0 & -i \\ i & 0 \end{array} \right), \quad \sigma^3 = \left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right).$	$\sigma^1 = \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right), \quad \sigma^2 = \left(\begin{array}{cc} 0 & -i \\ i & 0 \end{array} \right), \quad \sigma^3 = \left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right).$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 —no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_227 E00710 (10.7.6)	227	■ 1.000 ● W +2	$\begin{aligned} \vartheta^{\hat{0}} &= dt + \ell^2 \sigma \rho^{*2} [1 - \cos(\rho/\rho^*)] d\phi \\ \vartheta^{\hat{1}} &= d\rho, \quad \vartheta^{\hat{2}} = \rho^* \sin(\rho/\rho^*) d\phi \\ \Gamma^{\hat{1}\hat{2}} &= \cos(\rho/\rho^*) d\phi = -\Gamma^{\hat{2}\hat{1}} \end{aligned}$	$\vartheta^{\hat{0}} = dt + \ell^2 \sigma \rho^{*2} [1 - \cos(\rho/\rho^*)] d\phi$ $\vartheta^{\hat{1}} = d\rho, \quad \vartheta^{\hat{2}} = \rho^* \sin(\rho/\rho^*) d\phi$ $\Gamma^{\hat{1}\hat{2}} = \cos(\rho/\rho^*) d\phi = -\Gamma^{\hat{2}\hat{1}}$	$\vartheta^{\hat{0}} = dt + \ell^2 \sigma \rho^{*2} [1 - \cos(\rho/\rho^*)] d\phi$ $\vartheta^{\hat{1}} = d\rho, \quad \vartheta^{\hat{2}} = \rho^* \sin(\rho/\rho^*) d\phi,$ $\Gamma^{\hat{1}\hat{2}} = \cos(\rho/\rho^*) d\phi = -\Gamma^{\hat{2}\hat{1}}.$
mielke-geometrodynamics_229 E00716 (10.9.2)	229	■ 1.000 ● K +0	$D \Psi = d \Psi - \frac{1}{2} \gamma_{\alpha} \Gamma^{*\alpha} \wedge \Psi,$	$D\Psi = d\Psi - \frac{1}{2} \gamma_{\alpha} \Gamma^{*\alpha} \wedge \Psi,$	$D\Psi = d\Psi - \frac{1}{2} \gamma_{\alpha} \Gamma^{*\alpha} \wedge \Psi,$
mielke-geometrodynamics_229 E00720 (10.9.6)	229	■ 1.000 ● N +0	$\tau_{\alpha}^{\star} := \frac{1}{2} \eta_{\alpha\beta\gamma} \tau^{\beta\gamma} = \frac{(-1)^s}{2} \frac{\delta L_{\Psi}}{\delta \Gamma_{\alpha}^{\star}} = \frac{(-1)^s}{2} \frac{i}{4} \bar{\Psi} \wedge \gamma_{\alpha} \Psi.$	$\tau_{\alpha}^{\star} := \frac{1}{2} \eta_{\alpha\beta\gamma} \tau^{\beta\gamma} = \frac{(-1)^s}{2} \frac{\delta L_{\Psi}}{\delta \Gamma_{\alpha}^{\star}} = \frac{(-1)^s}{2} \frac{i}{4} \bar{\Psi} \wedge \gamma_{\alpha} \Psi.$	$\tau_{\alpha}^{\star} := \frac{1}{2} \eta_{\alpha\beta\gamma} \tau^{\beta\gamma} = \frac{(-1)^s}{2} \frac{\delta L_{\Psi}}{\delta \Gamma_{\alpha}^{\star}} = \frac{(-1)^s}{2} \frac{i}{4} \bar{\Psi} \wedge \gamma_{\alpha} \Psi.$
mielke-geometrodynamics_231 E00728 (10.10.8)	231	■ 1.000 ● K +0	$\gamma^{\alpha} \Psi \wedge \bar{\Psi} \gamma_{\alpha} \Psi = 0,$	$\gamma^{\alpha} \Psi \wedge \bar{\Psi} \gamma_{\alpha} \Psi = 0,$	$\gamma^{\alpha} \Psi \wedge \bar{\Psi} \gamma_{\alpha} \Psi = 0,$
mielke-geometrodynamics_244 E00770 (11.1.37)	244	■ 1.000 ● N +0	$D \longrightarrow {}^{S^{-1}} D = S^{-1} D S.$	$D \longrightarrow {}^{S^{-1}} D = S^{-1} D S.$	$D \longrightarrow {}^{S^{-1}} D = S^{-1} D S.$
mielke-geometrodynamics_272 E00839 (12.2.7)	272	■ 1.000 ● N +0	$j_{\pm} := \frac{1}{2} \bar{\psi} (1 \pm \gamma_5)^* \gamma \psi = \bar{\psi}_{\text{L,R}}^* \gamma \psi_{\text{L,R}}.$	$j_{\pm} := \frac{1}{2} \bar{\psi} (1 \pm \gamma_5)^* \gamma \psi = \bar{\psi}_{\text{L,R}}^* \gamma \psi_{\text{L,R}}.$	$j_{\pm} := \frac{1}{2} \bar{\psi} (1 \pm \gamma_5)^* \gamma \psi = \bar{\psi}_{\text{L,R}}^* \gamma \psi_{\text{L,R}}.$
mielke-geometrodynamics_284 E00873 (13.3.1)	284	■ 1.000 ● N +0	$\nvdash^{\{A\}\{ \}_B} := L^{\{A\}\{ \}_B} - \frac{1}{n} \delta_B^A L^C_C.$	$\not\vdash^A{}_B := L^A{}_B - \frac{1}{n} \delta_B^A L^C{}_C.$	$\not\vdash^A{}_B := L^A{}_B - \frac{1}{n} \delta_B^A L^C{}_C.$
mielke-geometrodynamics_287 E00893 (13.4.12)	287	■ 1.000 ● N +0	$\mathrm{A}_{\ast}(n,\mathbb{R})/\mathrm{GL}(n,\mathbb{R})\approx \mathbb{R}^n\otimes \mathbb{R}_{\ast}^n,$	$A_{\ast}(n,\mathbb{R})/\mathrm{GL}(n,\mathbb{R})\approx \mathbb{R}^n\otimes \mathbb{R}_{\ast}^n,$	$A_{\ast}(n,\mathbb{R})/\mathrm{GL}(n,\mathbb{R})\approx \mathbb{R}^n\otimes \mathbb{R}_{\ast}^n,$
mielke-geometrodynamics_290 E00907 (13.5.12)	290	■ 1.000 ● N +0	$\tilde{L}_{\mathrm{SSB}} + d\hat{C} \cong \frac{1}{2\mu} \eta_{\alpha\beta\gamma\delta} \left[R^{\alpha\beta} - \frac{\Lambda}{3} \vartheta^{\alpha} \wedge \vartheta^{\beta} \right] \wedge \left[R^{\gamma\delta} - \frac{\Lambda}{3} \vartheta^{\gamma} \wedge \vartheta^{\delta} \right].$	$\tilde{L}_{\mathrm{SSB}} + d\hat{C} \cong \frac{1}{2\mu} \eta_{\alpha\beta\gamma\delta} \left[R^{\alpha\beta} - \frac{\Lambda}{3} \vartheta^{\alpha} \wedge \vartheta^{\beta} \right] \wedge \left[R^{\gamma\delta} - \frac{\Lambda}{3} \vartheta^{\gamma} \wedge \vartheta^{\delta} \right].$	$\tilde{L}_{\mathrm{SSB}} + d\hat{C} \cong \frac{1}{2\mu} \eta_{\alpha\beta\gamma\delta} \left[R^{\alpha\beta} - \frac{\Lambda}{3} \vartheta^{\alpha} \wedge \vartheta^{\beta} \right] \wedge \left[R^{\gamma\delta} - \frac{\Lambda}{3} \vartheta^{\gamma} \wedge \vartheta^{\delta} \right].$
mielke-geometrodynamics_307 E00947 (14.1.28)	307	■ 1.000 ● N +0	$R_{i\alpha}^{\{ \} } R_{\cdot j}^{\{ \} \alpha} = \frac{1}{4} g_{ij} R_{\alpha\beta}^{\{ \} } R^{\{ \} \alpha\beta}$	$R_{i\alpha}^{\{ \} } R_{\cdot j}^{\{ \} \alpha} = \frac{1}{4} g_{ij} R_{\alpha\beta}^{\{ \} } R^{\{ \} \alpha\beta}$	$R_{i\alpha}^{\{ \} } R_{\cdot j}^{\{ \} \alpha} = \frac{1}{4} g_{ij} R_{\alpha\beta}^{\{ \} } R^{\{ \} \alpha\beta}$
mielke-geometrodynamics_323 E01004 (14.3.17)	323	■ 1.000 ● N +0	$\widetilde{\Theta} := D\gamma = \frac{1}{2} T_{\alpha\beta}^{\cdot\cdot c} \gamma_c \otimes \vartheta^{\alpha} \wedge \vartheta^{\beta}$	$\tilde{\Theta} := D\gamma = \frac{1}{2} T_{\alpha\beta}^{\cdot\cdot c} \gamma_c \otimes \vartheta^{\alpha} \wedge \vartheta^{\beta}$	$\tilde{\Theta} := D\gamma = \frac{1}{2} T_{\alpha\beta}^{\cdot\cdot c} \gamma_c \otimes \vartheta^{\alpha} \wedge \vartheta^{\beta}$
mielke-geometrodynamics_326 E01013 (14.4.8)	326	■ 1.000 ●	$M_{\mathrm{ir}}^2 := \frac{S_{\mathrm{A}}}{16\pi} \frac{M^{*4} c^2}{\hbar^2}$	$M_{\mathrm{ir}}^2 := \frac{S_{\mathrm{A}}}{16\pi} \frac{M^{*4} c^2}{\hbar^2}$	$M_{\mathrm{ir}}^2 := \frac{S_{\mathrm{A}}}{16\pi} \frac{M^{*4} c^2}{\hbar^2}$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_0375 EQ1135 (C.3)		■ 1.000 ● N +0	$\left[I_i, I_j \right] = c_{ij}^{k} I_k \; .$	$[I_i, I_j] = c_{ij}^{k} I_k.$	$[I_i, I_j] = c_{ij}^{k} I_k.$
mielke-geometrodynamics_024 EQ0001 (1.1)		■ 1.000 ● N +0	$\ell^* := \sqrt{8 \pi \hbar G_N / c^3} \simeq 8 \times 10^{-33} \; \mathrm{cm},$	$\ell^* := \sqrt{8 \pi \hbar G_N / c^3} \simeq 8 \times 10^{-33} \; \mathrm{cm},$	$\ell^* := \sqrt{8 \pi \hbar G_N / c^3} \simeq 8 \times 10^{-33} \mathrm{cm},$
mielke-geometrodynamics_032 EQ0003 (2.1.2)		■ 1.000 ● N -1	$e^{\mathrm{f}} := \left. \frac{df(s(t))}{dt} \right _{t_o = s^{-1}(m)} = \frac{\partial f}{\partial x^i} \cdot \frac{dx^i(s(t))}{dt}$	$e^{\mathrm{f}} := \left. \frac{df(s(t))}{dt} \right _{t_o = s^{-1}(m)} = \frac{\partial f}{\partial x^i} \cdot \frac{dx^i(s(t))}{dt}$	$e^{\mathrm{f}} := \left. \frac{df(s(t))}{dt} \right _{t_o = s^{-1}(m)} = \frac{\partial f}{\partial x^i} \cdot \frac{dx^i(s(t))}{dt}$
mielke-geometrodynamics_033 EQ0005 (2.1.4)		■ 1.000 ● N +1	$T(M) = \bigcup_{m \in M} T_m(M)$	$T(M) = \bigcup_{m \in M} T_m(M)$	$T(M) = \bigcup_{m \in M} T_m(M).$
mielke-geometrodynamics_034 EQ0008 (2.2.3)		■ 1.000 ● N -1	$\vartheta^{\alpha} = E_j^{\alpha}(m) \, d x^j \; .$	$\vartheta^{\alpha} = E_j^{\alpha}(m) d x^j.$	$\vartheta^{\alpha} = E_j^{\alpha}(m) \, d x^j.$
mielke-geometrodynamics_034 EQ0011 (2.2.6)		■ 1.000 ● W +2	$T = T_{\alpha_1 \cdots \alpha_p}^{\beta_1 \cdots \beta_q}(m) \, \vartheta^{\alpha_1} \cdots \vartheta^{\alpha_p} \otimes e_{\beta_1} \otimes \cdots \otimes e_{\beta_q} \in C^{\infty}\left(T_p^q(M)\right).$	$T = T_{\alpha_1 \cdots \alpha_p}^{\beta_1 \cdots \beta_q}(m) \vartheta^{\alpha_1} \otimes \cdots \otimes \vartheta^{\alpha_p} \otimes e_{\beta_1} \otimes \cdots \otimes e_{\beta_q} \in C^{\infty}\left(T_p^q(M)\right).$	$T = T_{\alpha_1 \cdots \alpha_p}^{\beta_1 \cdots \beta_q}(m) \vartheta^{\alpha_1} \otimes \cdots \otimes \vartheta^{\alpha_p} \otimes e_{\beta_1} \otimes \cdots \otimes e_{\beta_q} \in C^{\infty}(T_p^q(M)).$
mielke-geometrodynamics_034 EQ0012 (2.2.7)		■ 1.000 ● N -2	$ds^2 = g_{\alpha\beta} \vartheta^\alpha \otimes_s \vartheta^\beta =: g_{ij} dx^i \otimes_s dx^j.$	$ds^2 = g_{\alpha\beta} \vartheta^\alpha \otimes_s \vartheta^\beta =: g_{ij} dx^i \otimes_s dx^j.$	$ds^2 = g_{\alpha\beta} \vartheta^\alpha \otimes_s \vartheta^\beta =: g_{ij} dx^i \otimes_s dx^j.$
mielke-geometrodynamics_035 EQ0014 (2.2.9)		■ 1.000 ● N -1	$\alpha^{\{p\}} = \frac{1}{p!} A_{\alpha_1 \cdots \alpha_p} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \in C^{\infty}\left(\wedge^p T^*(M)\right).$	$\alpha^{(p)} = \frac{1}{p!} A_{\alpha_1 \cdots \alpha_p} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \in C^{\infty}\left(\wedge^p T^*(M)\right).$	$\alpha^{(p)} = \frac{1}{p!} A_{\alpha_1 \cdots \alpha_p} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \in C^{\infty}(\wedge^p T^*(M)).$
mielke-geometrodynamics_036 EQ0015 (2.3.1)		■ 1.000 ● N +0	$\pi\colon F\rightarrowtail M \; .$	$\pi:F\rightarrowtail M.$	$\pi:F\rightarrowtail M.$
mielke-geometrodynamics_037 EQ0016 (2.3.2)		■ 1.000 ● K +0	$\sigma\colon U\rightarrowtail F \quad \text{where} \quad \pi\circ\sigma=\mathrm{id}$	$\sigma:U\rightarrowtail F \quad \text{where} \quad \pi\circ\sigma=\mathrm{id}$	$\sigma:U\rightarrowtail F \quad \text{where} \quad \pi\circ\sigma=\mathrm{id}$
mielke-geometrodynamics_037 EQ0017 (2.3.3)		■ 1.000 ● S +0	$\iota\!\left(m,\,g_1g_2\right)=\iota\!\left(m,\,g_1\right)\cdot g_2,\quad m\in U,\,g\in G\; .$	$\iota\left(m,g_1g_2\right)=\iota\left(m,g_1\right)\cdot g_2,\quad m\in U,g\in G.$	$\iota(m,g_1g_2)=\iota(m,g_1)\cdot g_2,\quad m\in U,g\in G.$
mielke-geometrodynamics_038 EQ0019 (2.4.1)		■ 1.000 ● N -2	$\delta_g(p,\zeta) := \left(pg, g^{-1}\zeta \right) \subset P \times F, \quad g \in G.$	$\delta_g(p,\zeta) := \left(pg, g^{-1}\zeta \right) \subset P \times F, \quad g \in G.$	$\delta_g(p,\zeta) := \left(pg, g^{-1}\zeta \right) \subset P \times F, \quad g \in G.$
mielke-geometrodynamics_038 EQ0022 (2.4.4)		■ 1.000 ● N -2	$\varphi^{\{p\}} = \left\{ \varphi^{(p)A} \mid A = 1, \dots, N \right\}.$	$\varphi^{(p)} = \left\{ \varphi^{(p)A} \mid A = 1, \dots, N \right\}.$	$\varphi^{(p)} = \left\{ \varphi^{(p)A} A = 1, \dots, N \right\}.$
mielke-geometrodynamics_039 EQ0023 (2.4.5)		■ 1.000 ● N +0	$L_{\{j\}} := \left[\left. \frac{\partial \rho_A^B(g(\xi))}{\partial \xi^j} \right _{g=e} \right] \in T_e(\rho(G))$	$L_j := \left[\left. \frac{\partial \rho_A^B(g(\xi))}{\partial \xi^j} \right _{g=e} \right] \in T_e(\rho(G))$	$L_j := \left[\left. \frac{\partial \rho_A^B(g(\xi))}{\partial \xi^j} \right _{g=e} \right] \in T_e(\rho(G))$
mielke-geometrodynamics_039 EQ0025 (2.4.7)		■ 1.000 ● W -2	$\phi^{\{p\}} = \frac{1}{p!} \phi_{\alpha_1 \dots \alpha_p}^j L_j \otimes \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \in C^{\infty}\left(\wedge^p T_{\mathbb{C}}^*(M) \otimes V^{Ad}\right).$	$\phi^{(p)} = \frac{1}{p!} \phi_{\alpha_1 \dots \alpha_p}^j L_j \otimes \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \in C^{\infty}\left(\wedge^p T_{\mathbb{C}}^*(M) \otimes V^{Ad}\right).$	$\phi^{(p)} = \frac{1}{p!} \phi_{\alpha_1 \dots \alpha_p}^j L_j \otimes \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \in C^{\infty}\left(\wedge^p T_{\mathbb{C}}^*(M) \otimes V^{Ad}\right).$
mielke-geometrodynamics_039 EQ0026 (2.4.8)		■ 1.000 ● C -5	$\left[\phi^{\{p\}},\psi^{\{q\}}\right] := \phi^{\{p\}} \wedge \psi^{\{q\}} - (-1)^{pq} \psi^{\{q\}} \wedge \phi^{\{p\}} = (-1)^{pq+1} \left[\psi^{\{q\}},\phi^{\{p\}}\right]$	$\left[\phi^{(p)},\psi^{(q)}\right] := \phi^{(p)} \wedge \psi^{(q)} - (-1)^{pq} \psi^{(q)} \wedge \phi^{(p)} = (-1)^{pq+1} \left[\psi^{(q)},\phi^{(p)}\right]$	$\left[\phi^{(p)},\psi^{(q)}\right] := \phi^{(p)} \wedge \psi^{(q)} - (-1)^{pq} \psi^{(q)} \wedge \phi^{(p)} = (-1)^{pq+1} \left[\psi^{(q)},\phi^{(p)}\right]$

Conf. MathPix confidence:

■

≥0.9

■

0.5-0.9

■

<0.5

—

no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E00027 (2.4.9)	039	■ 1.000	$\left[\varphi^{\{2\,k\}},\varphi^{\{2\,k\}}\right]=-\left[\varphi^{\{2\,k\}},\varphi^{\{2\,k\}}\right]=0\;.$	$\left[\phi^{(2k)},\phi^{(2k)}\right]=-\left[\phi^{(2k)},\phi^{(2k)}\right]=0.$	$\left[\phi^{(2k)},\phi^{(2k)}\right]=-\left[\phi^{(2k)},\phi^{(2k)}\right]=0.$
mielke-geometrodynamics_E00029 (2.5.2)	040	■ 1.000	$\omega(e\,g)=g^{-1}\,\omega(e)\,g,\quad e\in T(P)\;.$	$\omega(eg)=g^{-1}\omega(e)g,\quad e\in T(P).$	$\omega(eg)=g^{-1}\omega(e)g,\quad e\in T(P).$
mielke-geometrodynamics_E00030 (2.5.3)	040	■ 1.000	$\omega(p\,d\,g)=g^{-1}\,d\,g\quad\text{in}\quad T_e(G)\approx\mathfrak{g}$	$\omega(pdg)=g^{-1}dg\quad\in\quad T_e(G)\approx\mathfrak{g}$	$\omega(p\,dg)=g^{-1}dg\quad\in\quad T_e(G)\approx\mathfrak{g}$
mielke-geometrodynamics_E00031 (2.5.4)	040	■ 1.000	$D:C^{\infty}(V^{\rho})\longrightarrow C^{\infty}(T_{\mathbb{C}}^*(M)\otimes V^{\rho})$	$D:C^{\infty}(V^{\rho})\longrightarrow C^{\infty}(T_{\mathbb{C}}^*(M)\otimes V^{\rho})$	$D:C^{\infty}(V^{\rho})\longrightarrow C^{\infty}(T_{\mathbb{C}}^*(M)\otimes V^{\rho})$
mielke-geometrodynamics_E00032 (2.5.6)	041	■ 1.000	$D\,b=\omega\otimes b,\quad b\in C^{\infty}(V^{\rho})\;.$	$D b=\omega\otimes b,\quad b\in C^{\infty}(V^{\rho}).$	$D b=\omega\otimes b,\quad b\in C^{\infty}(V^{\rho}).$
mielke-geometrodynamics_E00033 (2.5.7)	041	■ 1.000	$D\left(\varphi^{\left(p\right)}\otimes b\right)=d\varphi^{\left(p\right)}\otimes b+\left(-1\right)^p\varphi^{\left(p\right)}\otimes Db.$	$D\left(\varphi^{\left(p\right)}\otimes b\right)=d\varphi^{\left(p\right)}\otimes b+\left(-1\right)^p\varphi^{\left(p\right)}\otimes Db.$	$D(\varphi^{(p)}\otimes b)=d\varphi^{(p)}\otimes b+(-1)^p\varphi^{(p)}\otimes Db.$
mielke-geometrodynamics_E00034 (2.5.8)	041	■ 1.000	$D\left(\varphi^{\left(p\right)}\otimes b\right)=\left\{d\varphi^p+\left(-1\right)^p\left[\varphi^{\left(p\right)},\omega\right]\right\}\otimes b.$	$D\left(\varphi^{\left(p\right)}\otimes b\right)=\left\{d\varphi^p+\left(-1\right)^p\left[\varphi^{\left(p\right)},\omega\right]\right\}\otimes b.$	$D(\varphi^{(p)}\otimes b)=\left\{d\varphi^p+(-1)^p[\varphi^{(p)},\omega]\right\}\otimes b.$
mielke-geometrodynamics_E00035 (2.5.9)	041	■ 1.000	$D\,b_A=\omega_A{}^B\,b_B,\,A,B=1,\ldots,N\;.$	$D b_A=\omega_A{}^B b_B,A,B=1,\ldots,N.$	$D b_A=\omega_A{}^B b_B,A,B=1,\ldots,N.$
mielke-geometrodynamics_E00036 (2.5.10)	041	■ 1.000	$\Gamma:=\stackrel{*}{\sigma}\,\omega=\Gamma_{\alpha}{}^jL_j\otimes\vartheta^{\alpha},\quad j=1,\ldots,\dim G,$	$\Gamma:=\overset{*}{\sigma}\,\omega=\Gamma_{\alpha}{}^jL_j\otimes\vartheta^{\alpha},\quad j=1,\ldots,\dim G,$	$\Gamma:=\overset{*}{\sigma}\omega=\Gamma_{\alpha}{}^jL_j\otimes\vartheta^{\alpha},\quad j=1,\ldots,\dim G,$
mielke-geometrodynamics_E00037 (2.5.11)	041	■ 1.000	$A_{\alpha}{}^j(m):=\Gamma_{\alpha}{}^j(m)_{ G=U(f)}$	$A_{\alpha}{}^j(m):=\Gamma_{\alpha}{}^j(m)_{ G=U(f)}$	$A_{\alpha}{}^j(m):=\Gamma_{\alpha}{}^j(m)_{ G=U(f)}$
mielke-geometrodynamics_E00038 (2.6.1)	042	■ 1.000	$D\,D\,b=: \,\Omega\otimes b\;.$	$DDb=: \Omega\otimes b.$	$D\,D\,b=: \,\Omega\otimes b.$
mielke-geometrodynamics_E00039 (2.6.2)	042	■ 1.000	$\begin{aligned} D\,D\,b &= D(\omega\otimes b) \,\&= d\,\omega\otimes b - \omega\wedge D\,b \\ &\otimes b - \omega\wedge D\,b \,\&= (d\,\omega - \omega\wedge\omega)\otimes b \end{aligned}$	$\begin{aligned} D D b &= D(\omega\otimes b) = d\omega\otimes b - \omega\wedge Db \\ &= (d\omega - \omega\wedge\omega)\otimes b \end{aligned}$	$\begin{aligned} D D b &= D(\omega\otimes b) = d\omega\otimes b - \omega\wedge Db \\ &= (d\omega - \omega\wedge\omega)\otimes b \end{aligned}$
mielke-geometrodynamics_E00040 (2.6.3)	042	■ 1.000	$\Omega=d\,\omega-\omega\wedge\omega\quad\Omega=d\,\omega-\frac{1}{2}[\omega,\omega]$	$\Omega=d\omega-\omega\wedge\omega=d\omega-\frac{1}{2}[\omega,\omega]$	$\Omega=d\omega-\omega\wedge\omega=d\omega-\frac{1}{2}[\omega,\omega]\Big .$
mielke-geometrodynamics_E00043 (2.6.6)	042	■ 1.000	$F_{ij}=\partial_iA_j-\partial_jA_i$	$F_{ij}=\partial_iA_j-\partial_jA_i$	$F_{ij}=\partial_iA_j-\partial_jA_i$
mielke-geometrodynamics_E00044 (2.6.7)	042	■ 1.000	$\Delta b=b-b(\circ)=\oint (db)_{\mathrm{hor}}$	$\Delta b=b-b(\circ)=\oint (db)_{\mathrm{hor}}$	$\Delta b=b-b(\circ)=\oint (db)_{\mathrm{hor}},$
mielke-geometrodynamics_E00047 (2.6.10)	043	■ 1.000	$K=\lim_{U\rightarrow 0}\left(\int_U\Omega\otimes b\right)/\left(\int_U1\otimes b\right)$	$K=\lim_{U\rightarrow 0}\left(\int_U\Omega\otimes b\right)/\left(\int_U1\otimes b\right)$	$K=\lim_{U\rightarrow 0}\left(\int_U\Omega\otimes b\right)/\left(\int_U1\otimes b\right)$
mielke-geometrodynamics_E00049 (2.6.12)	043	■ 1.000	$D\,\Omega\equiv 0$	$D\Omega\equiv 0$	$D\Omega\equiv 0$
mielke-geometrodynamics_E00052 (2.7.2)	044	■ 1.000	$G(g\,p)=g\,G(p),\quad g\in G,\quad p\in P\;;$	$G(gp)=gG(p),\quad g\in G,\quad p\in P;$	$G(gp)=g\,G(p),\quad g\in G,\quad p\in P;$
mielke-geometrodynamics_E00053 (2.7.3)	044	■ 1.000	$\pi\circ G=\pi\;.$	$\pi\circ G=\pi.$	$\pi\circ G=\pi.$
mielke-geometrodynamics_E00054 (2.7.4)	045	■ 1.000	$\mathscr{G}_p\approx C^{\infty}(P\times_{\mathrm{Ad}}G)\;.$	$\mathscr{G}_p\approx C^{\infty}(P\times_{\mathrm{Ad}}G).$	$\mathscr{G}_p\approx C^{\infty}(P\times_{\mathrm{Ad}}G).$

Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_045 EQ0055 (2.7.5)	045	■ 1.000 ● N -1	$G(p)=\exp i \, \theta^{\wedge\{k\}(m)} \, L_{\{k\}} \, \in \, \mathscr{G}_{\{p\}} \, .$	$G(p) = \exp i \theta^k(m) L_k \in \mathscr{G}_p.$	$G(p) = \exp i \, \theta^k(m) L_k \, \in \, \mathscr{G}_p.$
mielke-geometrodynamics_045 EQ0057 (2.7.7)	045	■ 1.000 ●	$\varphi \rightarrow^{\wedge\{G^{-1}\}} \varphi := G^{\wedge\{-1\}} \varphi \quad \in \, C^{\infty}(M, \mathbb{C}), G \in \mathscr{G}_V,$ $\varphi \quad \in \, C^{\infty}(M, \mathbb{C}), G \in \mathscr{G}_V,$	$\varphi \rightarrow^{G^{-1}} \varphi := G^{-1} \varphi \quad \in \, C^{\infty}(M, \mathbb{C}), G \in \mathscr{G}_V,$	$\varphi \rightarrow^{G^{-1}} \varphi := G^{-1} \varphi \quad \in \, C^{\infty}(M, \mathbb{C}), G \in \mathscr{G}_V,$
mielke-geometrodynamics_046 EQ0063 (2.8.4)	046	■ 1.000 ● N +0	$p_{\{k\}} \backslash \left(V^{\wedge \{ \mathbb{R} \}} \backslash \right) = (-1)^{\wedge\{k\}} \, c_{2 \, k}(V) \, .$	$p_k \left(V^{\mathbb{R}} \right) = (-1)^k c_{2k}(V).$	$p_k(V^{\mathbb{R}}) = (-1)^k c_{2k}(V).$
mielke-geometrodynamics_047 EQ0066 (2.8.9)	047	■ 1.000 ● N +0	$\operatorname{Tr}(\backslash \omega \wedge \backslash \omega) = d \operatorname{Tr} \backslash \left(\omega \wedge \backslash \omega + \frac{1}{3} \omega \wedge \omega \wedge \omega \right)$	$\operatorname{Tr}(\Omega \wedge \Omega) = d \operatorname{Tr} \left(\omega \wedge \Omega + \frac{1}{3} \omega \wedge \omega \wedge \omega \right)$	$\operatorname{Tr}(\Omega \wedge \Omega) = d \operatorname{Tr} \left(\omega \wedge \Omega + \frac{1}{3} \omega \wedge \omega \wedge \omega \right)$
mielke-geometrodynamics_048 EQ0067 (2.8.10)	048	■ 1.000 ● C +7	$\begin{aligned} & \operatorname{Tr} \backslash \left\{ d \backslash \left(\omega \wedge d \, \omega - \frac{2}{3} \omega \wedge \omega \wedge \omega \right) \right\} = \operatorname{Tr} \{ d \, \omega \wedge d \, \omega - \omega \wedge d d \, \omega \\ & - \frac{2}{3} d \, \omega \wedge \omega \wedge \omega + \frac{2}{3} \omega \wedge d \, \omega \wedge \omega - \frac{2}{3} \omega \wedge \omega \wedge d \, \omega \} \\ & = \operatorname{Tr} \{ d \, \omega \wedge d \, \omega - d \, \omega \wedge \omega \wedge \omega - \omega \wedge \omega \wedge d \, \omega \} \\ & = \operatorname{Tr} \{ (d \, \omega - \omega \wedge \omega) \wedge (d \, \omega - \omega \wedge \omega) \} = \operatorname{Tr}(\Omega \wedge \Omega) \end{aligned}$	$\operatorname{Tr} \left\{ d \left(\omega \wedge d \omega - \frac{2}{3} \omega \wedge \omega \wedge \omega \right) \right\} = \operatorname{Tr} \{ d \omega \wedge d \omega - \omega \wedge d d \omega$ $- \frac{2}{3} d \omega \wedge \omega \wedge \omega + \frac{2}{3} \omega \wedge d \omega \wedge \omega - \frac{2}{3} \omega \wedge \omega \wedge d \omega \}$ $= \operatorname{Tr} \{ d \omega \wedge d \omega - d \omega \wedge \omega \wedge \omega - \omega \wedge \omega \wedge d \omega \}$ $= \operatorname{Tr} \{ (d \omega - \omega \wedge \omega) \wedge (d \omega - \omega \wedge \omega) \} = \operatorname{Tr}(\Omega \wedge \Omega)$	$\operatorname{Tr} \left\{ d \left(\omega \wedge d \omega - \frac{2}{3} \omega \wedge \omega \wedge \omega \right) \right\} = \operatorname{Tr} \{ d \omega \wedge d \omega - \omega \wedge d d \omega$ $- \frac{2}{3} d \omega \wedge \omega \wedge \omega + \frac{2}{3} \omega \wedge d \omega \wedge \omega - \frac{2}{3} \omega \wedge \omega \wedge d \omega \}$ $= \operatorname{Tr} \{ d \omega \wedge d \omega - d \omega \wedge \omega \wedge \omega - \omega \wedge \omega \wedge d \omega \}$ $= \operatorname{Tr} \{ (d \omega - \omega \wedge \omega) \wedge (d \omega - \omega \wedge \omega) \} = \operatorname{Tr}(\Omega \wedge \Omega).$
mielke-geometrodynamics_048 EQ0068 (2.8.11)	048	■ 1.000 ● K +0	$\operatorname{Tr} \backslash \{ \omega \wedge (\omega \wedge \omega \wedge \omega) \} = - \operatorname{Tr} \{ (\omega \wedge \omega \wedge \omega) \wedge \omega \} = 0.$	$\operatorname{Tr} \{ \omega \wedge (\omega \wedge \omega \wedge \omega) \} = - \operatorname{Tr} \{ (\omega \wedge \omega \wedge \omega) \wedge \omega \} = 0.$	$\operatorname{Tr} \{ \omega \wedge (\omega \wedge \omega \wedge \omega) \} = - \operatorname{Tr} \{ (\omega \wedge \omega \wedge \omega) \wedge \omega \} = 0.$
mielke-geometrodynamics_048 EQ0070 (2.8.13)	048	■ 1.000 ● N -1	$S_{\mathrm{YM}} = \int_M L_{\mathrm{YM}}$	$S_{\mathrm{YM}} = \int_M L_{\mathrm{YM}}$	$S_{\mathrm{YM}} = \int_M L_{\mathrm{YM}}$
mielke-geometrodynamics_048 EQ0072 (2.8.15)	048	■ 1.000 ● K +0	$\omega \sim \tilde{\omega}$	$\omega \sim \tilde{\omega}$	$\omega \sim \tilde{\omega}$
mielke-geometrodynamics_049 EQ0074 (2.8.17)	049	■ 1.000 ● K +0	$\Omega(\tilde{\omega}) = 0.$	$\Omega(\tilde{\omega}) = 0.$	$\Omega(\tilde{\omega}) = 0.$
mielke-geometrodynamics_049 EQ0075 (2.8.18)	049	■ 1.000 ● N +0	$c_2(M) = \frac{1}{24 \pi^2} \int_{\partial M} \operatorname{Tr}(\tilde{\omega} \wedge \tilde{\omega} \wedge \tilde{\omega}),$	$c_2(M) = \frac{1}{24 \pi^2} \int_{\partial M} \operatorname{Tr}(\tilde{\omega} \wedge \tilde{\omega} \wedge \tilde{\omega}),$	$c_2(M) = \frac{1}{24 \pi^2} \int_{\partial M} \operatorname{Tr}(\tilde{\omega} \wedge \tilde{\omega} \wedge \tilde{\omega}),$
mielke-geometrodynamics_052 EQ0076 (3.1.1)	052	■ 1.000 ● K +0	$L = L(\varphi, d \varphi) \eta$	$L = L(\varphi, d \varphi) \eta$	$L = L(\varphi, d \varphi) \eta$
mielke-geometrodynamics_052 EQ0077 (3.1.2)	052	■ 1.000 ● N +0	$\delta \int_M L = 0$	$\delta \int_M L = 0$	$\delta \int_M L = 0$
mielke-geometrodynamics_052 EQ0078 (3.1.3)	052	■ 1.000 ● N -1	$\delta \varphi = 0 \quad \text{at the boundary } \partial M \text{ of } M$	$\delta \varphi = 0 \quad \text{at the boundary } \partial M \text{ of } M$	$\delta \varphi = 0 \quad \text{at the boundary } \partial M \text{ of } M$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix's reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_052 EQ0079 (3.1.4)		■ 1.000 ● K +0	$\delta(d\varphi):=\delta(\varphi+\delta\varphi)-d\varphi=d\delta\varphi$ $\delta\varphi$	$\delta(d\varphi) := d(\varphi + \delta\varphi) - d\varphi = d\delta\varphi$	$\delta(d\varphi) := d(\varphi + \delta\varphi) - d\varphi = d\delta\varphi$
mielke-geometrodynamics_052 EQ0080 (3.1.5)		■ 1.000 ● K +0	$\pi:=\frac{\partial L}{\partial(\partial_\alpha\varphi)}\vartheta^\alpha$ ϑ^α	$\pi := \frac{\partial L}{\partial(\partial_\alpha\varphi)}\vartheta^\alpha$	$\pi := \frac{\partial L}{\partial(\partial_\alpha\varphi)}\vartheta^\alpha$
mielke-geometrodynamics_053 EQ0082 (3.1.7)		■ 1.000 ●	$\int_M d(\delta\varphi\wedge\pi)=\int_{\partial M}\delta\varphi\wedge\pi$ $\delta\varphi\wedge\pi$	$\int_M d(\delta\varphi\wedge\pi) = \int_{\partial M} \delta\varphi\wedge\pi$	$\int_M d(\delta\varphi\wedge\pi) = \int_{\partial M} \delta\varphi\wedge\pi.$
mielke-geometrodynamics_053 EQ0083 (3.1.9)		■ 1.000 ● N +0	$\lambda^\varphi:=(-1)^sL_\varphi+{}^*d\pi=0,$	$\lambda^\varphi := (-1)^s L_\varphi + {}^*d\pi = 0,$	$\lambda^\varphi := (-1)^s L_\varphi + {}^*d\pi = 0,$
mielke-geometrodynamics_053 EQ0084 (3.1.10)		■ 1.000 ● K +0	$\frac{\partial L}{\partial\varphi}-\partial_\alpha\frac{\partial L}{\partial(\partial_\alpha\varphi)}=0.$ $\frac{\partial L}{\partial\varphi}-\partial_\alpha\frac{\partial L}{\partial(\partial_\alpha\varphi)}=0$	$\frac{\partial L}{\partial\varphi} - \partial_\alpha \frac{\partial L}{\partial(\partial_\alpha\varphi)} = 0.$	$\frac{\partial L}{\partial\varphi} - \partial_\alpha \frac{\partial L}{\partial(\partial_\alpha\varphi)} = 0.$
mielke-geometrodynamics_053 EQ0085 (3.1.11)		■ 1.000 ● K +0	$\{ \}^{\ast}H:=L-d\varphi\wedge\pi$	${}^*H := L - d\varphi\wedge\pi$	${}^*H := L - d\varphi\wedge\pi$
mielke-geometrodynamics_053 EQ0086 (3.1.12)		■ 1.000 ● N +0	$d^{\ast}H=dL+d\varphi\wedge{}^*L_\varphi+d\varphi\wedge d\pi-dd\varphi\wedge\pi$	$d^*H = dL + d\varphi\wedge{}^*L_\varphi + d\varphi\wedge d\pi - dd\varphi\wedge\pi$	$d^*H = dL + d\varphi\wedge{}^*L_\varphi + d\varphi\wedge d\pi - d\,d\varphi\wedge\pi$
mielke-geometrodynamics_053 EQ0087 (3.1.13)		■ 1.000 ● K +0	$d^{\ast}H\simeq 0$	$d^*H \simeq 0$	$d^*H \simeq 0$
mielke-geometrodynamics_054 EQ0088 (3.1.14)		■ 1.000 ● N +0	$\beta\wedge{}^*D\psi=m\psi\eta.$	$\beta\wedge{}^*D\psi = m\psi\eta.$	$\beta\wedge{}^*D\psi = m\psi\eta.$
mielke-geometrodynamics_054 EQ0089 (3.1.15)		■ 1.000 ● N +0	$\square\varphi+m^2\varphi=0$	$\square\varphi + m^2\varphi = 0$	$\square\varphi + m^2\varphi = 0$
mielke-geometrodynamics_054 EQ0090 (3.1.16)		■ 1.000 ● S -1	$\overline{L}_{\mathrm{mat}}=\overline{L}_{\mathrm{mat}}(\bar{\psi}\psi,\bar{\psi}\beta\cdot d\psi)\eta=\bar{\psi}\beta\wedge{}^*d\psi-m\bar{\psi}\psi\eta.$ $\overline{L}_{\mathrm{mat}}(\bar{\psi}\psi,\bar{\psi}\beta\cdot d\psi)\eta=\bar{\psi}\beta\wedge{}^*d\psi-m\bar{\psi}\psi\eta$	$\overline{L}_{\mathrm{mat}} = \overline{L}_{\mathrm{mat}}(\bar{\psi}\psi, \bar{\psi}\beta \cdot d\psi)\eta = \bar{\psi}\beta \wedge {}^*d\psi - m\bar{\psi}\psi\eta.$	$\overline{L}_{\mathrm{mat}} = \overline{L}_{\mathrm{mat}}(\bar{\psi}\psi, \bar{\psi}\beta \cdot d\psi)\eta = \bar{\psi}\beta \wedge {}^*d\psi - m\bar{\psi}\psi\eta.$
mielke-geometrodynamics_055 EQ0092 (3.2.2)		■ 1.000 ● N -1	$L_{\mathrm{mat}}=L(\bar{\varphi}\varphi,\bar{\varphi}\beta\cdot(d+\omega')\varphi)\eta=L(\bar{\varphi}\varphi,\bar{\varphi}\beta\cdot D'\varphi)\eta$	$L_{\mathrm{mat}} = L(\bar{\varphi}\varphi, \bar{\varphi}\beta \cdot (d + \omega')\varphi)\eta = L(\bar{\varphi}\varphi, \bar{\varphi}\beta \cdot D'\varphi)\eta$	$L_{\mathrm{mat}} = L(\bar{\varphi}\varphi, \bar{\varphi}\beta \cdot (d + \omega')\varphi)\eta = L(\bar{\varphi}\varphi, \bar{\varphi}\beta \cdot D'\varphi)\eta$
mielke-geometrodynamics_055 EQ0094 (3.2.4)		■ 1.000 ● K +0	$D\left(\{ \}^G b\right)=\{ \}^G\omega\otimes b.$	$D\left({}^G b\right) = {}^G\omega \otimes {}^G b.$	$D({}^G b) = {}^G\omega \otimes {}^G b.$
mielke-geometrodynamics_055 EQ0095 (3.2.5)		■ 1.000 ● S +0	$D\left(\{ \}^G b\right)=dG\otimes b+G\omega\otimes b=\left\{dGG^{-1}+G\omega G^{-1}\right\}\otimes {}^G b.$	$D\left({}^G b\right) = dG \otimes b + G\omega \otimes b = \left\{dGG^{-1} + G\omega G^{-1}\right\} \otimes {}^G b.$	$D({}^G b) = dG \otimes b + G\omega \otimes b = \left\{dGG^{-1} + G\omega G^{-1}\right\} \otimes {}^G b.$
mielke-geometrodynamics_055 EQ0096 (3.2.6)		■ 1.000 ● N +0	$\omega\rightarrow {}^G\omega=G\omega G^{-1}+dGG^{-1}.$	$\omega \rightarrow {}^G\omega = G\omega G^{-1} + dGG^{-1}.$	$\omega \rightarrow {}^G\omega = G\omega G^{-1} + dG\,G^{-1}.$
mielke-geometrodynamics_057 EQ0102 (3.3.3)		■ 1.000 ● N +1	$\theta=-\frac{e}{\hbar}\int\mathbf{A}\cdot d\mathbf{l}$	$\theta = -\frac{e}{\hbar}\int\mathbf{A}\cdot d\mathbf{l}$	$\theta = -\frac{e}{\hbar}\int\mathbf{A}\cdot d\mathbf{l}.$
mielke-geometrodynamics_057 EQ0103 (3.3.4)		■ 1.000 ● N -2	$L=L(A,dA)+L_{\mathrm{mat}}(A,dA,\Psi,d\Psi).$	$L = L(A, dA) + L_{\mathrm{mat}}(A, dA, \Psi, d\Psi).$	$L = L(A, dA) + L_{\mathrm{mat}}(A, dA, \Psi, d\Psi).$
mielke-geometrodynamics_057 EQ0104 (3.3.5)		■ 1.000 ● N -1	$\frac{\delta L}{\delta A}:=\frac{\partial L}{\partial A}+d\frac{\partial L}{\partial dA}=0,$	$\frac{\delta L}{\delta A} := \frac{\partial L}{\partial A} + d\frac{\partial L}{\partial dA} = 0,$	$\frac{\delta L}{\delta A} := \frac{\partial L}{\partial A} + d\frac{\partial L}{\partial dA} = 0,$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_058 EQ0106 (3.3.7)		1.000 • N +1	$d\,F\equiv 0,\quad d\,H=j$	$dF\equiv 0,\quad dH=j$	$dF\equiv 0,\quad dH=j\,.$
mielke-geometrodynamics_058 EQ0107 (3.3.8)		1.000 • K +0	$d\,j\,\simeq 0\,.$	$dj\simeq 0.$	$dj\simeq 0\,.$
mielke-geometrodynamics_058 EQ0111 (3.3.12)		1.000 • N -1	$H\!:=\!\frac{H}{d\,t}\wedge d\,\frac{D}{t}\!=\!\frac{1}{2}\,H_{ij}\,H_{\{i\,j\}}\,d\,x^{\{i\}}\wedge d\,x^{\{j\}}$	$H:=-\mathfrak{H}\wedge dt+\mathfrak{D}=\frac{1}{2}H_{ij}dx^i\wedge dx^j$	$H:=-\mathfrak{H}\wedge dt+\mathfrak{D}=\frac{1}{2}H_{ij}dx^i\wedge dx^j\,,$
mielke-geometrodynamics_059 EQ0114 (3.3.15)		1.000 • W +1	$H_{mn}\!=\!\frac{\sqrt{ g }}{2}\,\varepsilon_{mnkl}\,g^{ki}\,g^{lj}\,F_{ij}\,.$	$H_{mn}=\frac{\sqrt{ g }}{2}\varepsilon_{mnkl}g^{ki}g^{lj}F_{ij}.$	$H_{mn}=\frac{\sqrt{ g }}{2}\varepsilon_{mnkl}\,g^{ki}\,g^{lj}\,F_{ij}\,.$
mielke-geometrodynamics_060 EQ0121 (3.3.22)		1.000	$L_{\mathrm{BF}}\!=\!-B\wedge F\!=\!-B\wedge d\,A\,.$	$L_{\mathrm{BF}}=-B\wedge F=-B\wedge dA.$	$L_{\mathrm{BF}}=-B\wedge F=-B\wedge dA\,.$
mielke-geometrodynamics_060 EQ0122 (3.3.23)		1.000 • N -1	$\begin{aligned} \tilde{L}_{\mathrm{BF}}&:= -B\wedge dA + \frac{1}{2}B\wedge B \\ &\cong -B\wedge dA + dC \end{aligned}$	$\begin{aligned} \tilde{L}_{\mathrm{BF}}&:= -B\wedge dA + \frac{1}{2}B\wedge B \\ &\cong -B\wedge dA + dC \end{aligned}$	$\begin{aligned} \tilde{L}_{\mathrm{BF}}&:= -B\wedge dA + \frac{1}{2}B\wedge B \\ &\cong -B\wedge dA + dC\,.\end{aligned}$
mielke-geometrodynamics_060 EQ0123 (3.3.24)		1.000 • N +0	$B\,\cong d\,A,\quad d\,B\,\cong d\,F\equiv 0$	$B\cong dA,\quad dB\cong dF\equiv 0$	$B\cong dA,\quad dB\cong dF\equiv 0\,,$
mielke-geometrodynamics_060 EQ0124 (3.3.25)		1.000 • K +0	$C\!=\!\frac{1}{2}\,A\wedge F,\quad d\,C\!=\!\frac{1}{2}\,F\wedge F,$	$C=\frac{1}{2}A\wedge F,\quad dC=\frac{1}{2}F\wedge F,$	$C=\frac{1}{2}A\wedge F,\quad dC=\frac{1}{2}F\wedge F\,,$
mielke-geometrodynamics_061 EQ0125 (3.3.26)		1.000 • K +0	$A\rightarrow A^{\prime}\!=\!A+d\,\theta(x)\,.$	$A\rightarrow A'=A+d\theta(x).$	$A\rightarrow A'=A+d\theta(x).$
mielke-geometrodynamics_061 EQ0126 (3.3.27)		1.000 • N +0	$A\rightarrow\widetilde{A}\!=\!A+\psi,\quad B\rightarrow\widetilde{B}\!=\!B+d\psi,$	$A\rightarrow\widetilde{A}=A+\psi,\quad B\rightarrow\widetilde{B}=B+d\psi,$	$A\rightarrow\widetilde{A}=A+\psi,\quad B\rightarrow\widetilde{B}=B+d\psi\,,$
mielke-geometrodynamics_061 EQ0127 (3.3.28)		1.000 • K +0	$L_{\mathrm{Max}}\!=\!-B\wedge d\,A\!+\!\frac{1}{2}\,B\wedge\,^*B\,+\,L_{\mathrm{matter}},$	$L_{\mathrm{Max}}=-B\wedge dA+\frac{1}{2}B\wedge{}^*B+L_{\mathrm{matter}},$	$L_{\mathrm{Max}}=-B\wedge dA+\frac{1}{2}B\wedge{}^*B+L_{\mathrm{matter}},$
mielke-geometrodynamics_061 EQ0128 (3.3.29)		1.000 • N +0	$-d\,B\!=\!d^*\,F\,\cong j,$	$-dB=d^*F\cong j,$	$-dB=d^*F\cong j,$
mielke-geometrodynamics_063 EQ0132 (3.4.3)		1.000 • N +0	$\alpha_g\!:=\!g^2/\hbar\,c$	$\alpha_g:=g^2/\hbar c$	$\alpha_g:=g^2/\hbar c$
mielke-geometrodynamics_063 EQ0134 (3.4.5)		1.000 • K +0	$\delta L_{\omega}\!:=\,\delta\Omega\wedge\Pi^{\Omega}\,.$	$\delta L_{\omega}:=\delta\Omega\wedge\Pi^{\Omega}.$	$\delta L_{\omega}:=\delta\Omega\wedge\Pi^{\Omega}.$
mielke-geometrodynamics_063 EQ0138 (3.4.9)		1.000 • N -1	$L\!=\!L_{\mathrm{mat}}\!+\!L_{\omega},$	$L=L_{\mathrm{mat}}+L_{\omega},$	$L=L_{\mathrm{mat}}+L_{\omega},$
mielke-geometrodynamics_063 EQ0139 (3.4.10)		1.000 • K +0	$\delta L_{\mathrm{mat}}\!:=\,\delta\omega\wedge\tau\,.$	$\delta L_{\mathrm{mat}}:=\delta\omega\wedge\tau.$	$\delta L_{\mathrm{mat}}:=\delta\omega\wedge\tau.$
mielke-geometrodynamics_064 EQ0140 (3.4.11)		1.000 • N +2	$D\,\Pi^{\{0\omega\}}\!=\!\tau$	$D\Pi^{\Omega}=\tau$	$\underline{D\Pi^{\Omega}}=\tau\,].$
mielke-geometrodynamics_064 EQ0141 (3.4.12)		1.000 • N +0	$D\,\omega\equiv 0\,.$	$D\Omega\equiv 0.$	$D\Omega\equiv 0.$
mielke-geometrodynamics_064 EQ0145 (3.4.16)		1.000 • N +0	$d\,\Pi^{\{0\omega\}}\!+\![\omega,\Pi^{\{0\omega\}}\!]\!=\!\tau\,.$	$d\Pi^{\Omega}+[\omega,\Pi^{\Omega}]=\tau.$	$d\,\Pi^{\Omega}+[\omega,\Pi^{\Omega}]=\tau.$
mielke-geometrodynamics_065 EQ0147 (3.4.18)		1.000 • N +0	$Q_{\mathrm{e}}\!=\!\frac{1}{2}\,Y\!+\!I_3\,.$	$Q_e=\frac{1}{2}Y+I_3.$	$Q_e=\frac{1}{2}Y+I_3.$

Conf. MathPix confidence:

≥0.9

0.5–0.9

<0.5

 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill):

C component

W weak

S stable

N noise

K clean

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_065 EQ0148 (3.4.19)		■ 1.000 ● N +0	$L_{\mathrm{YM}}=-\frac{1}{\alpha_g}\operatorname{Tr}\left(\Omega\wedge^*\Omega\right)$.	$L_{\mathrm{YM}}=-\frac{1}{\alpha_g}\operatorname{Tr}\left(\Omega\wedge^*\Omega\right).$	$L_{\mathrm{YM}}=-\frac{1}{\alpha_g}\operatorname{Tr}(\Omega\wedge^*\Omega).$
mielke-geometrodynamics_065 EQ0150 (3.4.21)		■ 1.000 ● N +1	$\Pi^{\Omega}=-\frac{2}{\alpha_g}\wedge^*\Omega$	$\Pi^{\Omega}=-\frac{2}{\alpha_g}\wedge^*\Omega$	$\Pi^{\Omega}=-\frac{2}{\alpha_g}\wedge^*\Omega.$
mielke-geometrodynamics_066 EQ0153 (3.5.1)		■ 1.000 ● N +1	$\Pi^{\Omega}=\zeta\Omega$.	$\Pi^{\Omega}=\zeta\Omega.$	$\Pi^{\Omega}=\zeta\Omega.$
mielke-geometrodynamics_067 EQ0156 (3.5.4)		■ 1.000 ● N +0	$\wedge^*\sigma^{\mu\nu}=\sigma^{\mu\nu},\quad\wedge^*\bar{\sigma}^{\mu\nu}=-\bar{\sigma}^{\mu\nu},$ $\bar{\sigma}^{\mu\nu}=-\bar{\sigma}^{\mu\nu},$	$\wedge^*\sigma^{\mu\nu}=\sigma^{\mu\nu},\quad\wedge^*\bar{\sigma}^{\mu\nu}=-\bar{\sigma}^{\mu\nu},$	$\wedge^*\sigma^{\mu\nu}=\sigma^{\mu\nu},\quad\wedge^*\bar{\sigma}^{\mu\nu}=-\bar{\sigma}^{\mu\nu},$
mielke-geometrodynamics_067 EQ0158 (3.5.6)		■ 1.000 ● N +0	$A_{\mu}=\mathrm{i}\sigma_{\mu i}\partial^i\ln h$.	$A_{\mu}=\mathrm{i}\sigma_{\mu i}\partial^i\ln h.$	$A_{\mu}=\mathrm{i}\sigma_{\mu i}\partial^i\ln h.$
mielke-geometrodynamics_067 EQ0159 (3.5.7)		■ 1.000 ● N +0	$\square h-h^{-1}\partial_i h\partial^i h=h^3$	$\square h-h^{-1}\partial_i h\partial^i h=h^3$	$\square h-h^{-1}\partial_i h\partial^i h=h^3$
mielke-geometrodynamics_067 EQ0160 (3.5.8)		■ 1.000 ●	$h=\sum_{i=1}^{k+1}\frac{l_{(i)}^2}{\left(x-x_{(i)}\right)^2},$	$h=\sum_{i=1}^{k+1}\frac{l_{(i)}^2}{\left(x-x_{(i)}\right)^2},$	$h=\sum_{i=1}^{k+1}\frac{l_{(i)}^2}{(x-x_{(i)})^2},$
mielke-geometrodynamics_067 EQ0161 (3.5.9)		■ 1.000 ● K +0	$p_1(M)=k$.	$p_1(M)=k.$	$p_1(M)=k.$
mielke-geometrodynamics_068 EQ0162 (3.5.10)		■ 1.000 ● N +2	$\langle\infty -\infty\rangle=Z\int_{\mathcal{M}_{\omega}}d\omega\mu[\omega]\exp i\int_M(L_{\omega}+L_{\infty})$	$\langle\infty -\infty\rangle=Z\int_{\mathcal{M}_{\omega}}d\omega\mu[\omega]\exp i\int_M(L_{\omega}+L_{\infty})$	$\langle\infty -\infty\rangle=Z\int_{\mathcal{M}_{\omega}}d\omega\mu[\omega]\exp i\int_M(L_{\omega}+L_{\infty}).$
mielke-geometrodynamics_068 EQ0163 (3.5.11)		■ 1.000 ● N +0	$L_{\infty}=(\stackrel{+}{-})\frac{1}{\alpha_g}\operatorname{Tr}(\Omega\wedge\Omega)$	$L_{\infty}=(\stackrel{+}{-})\frac{1}{\alpha_g}\operatorname{Tr}(\Omega\wedge\Omega)$	$L_{\infty}=(\stackrel{+}{-})\frac{1}{\alpha_g}\operatorname{Tr}(\Omega\wedge\Omega)$
mielke-geometrodynamics_070 EQ0167 (3.6.2)		■ 1.000 ● N +0	$\mathscr{P}_{\pm}:=\bar{\psi}\sigma_{\pm}\psi/2m,$	$\mathscr{P}_{\pm}:=\bar{\psi}\sigma_{\pm}\psi/2m,$	$\mathscr{P}_{\pm}:=\bar{\psi}\sigma_{\pm}\psi/2m,$
mielke-geometrodynamics_070 EQ0168 (3.6.3)		■ 1.000 ● N +0	$D_{\pm}\wedge^*\psi=\frac{1}{2}\bar{\psi}\sigma_{\pm}\psi\wedge\left(F^{\pm}-\frac{1}{2}\bar{\psi}\sigma_{\pm}\psi\right)\wedge\sigma_{\pm}\psi=0$	$D_{\pm}\wedge^*\psi=\frac{1}{2}\bar{\psi}\sigma_{\pm}\psi\wedge\left(F^{\pm}-\frac{1}{2}\bar{\psi}\sigma_{\pm}\psi\right)\wedge\sigma_{\pm}\psi=0$	$D_{\pm}\wedge^*\psi=\frac{1}{2}\bar{\psi}\sigma_{\pm}\psi\wedge\left(F^{\pm}-\frac{1}{2}\bar{\psi}\sigma_{\pm}\psi\right)\wedge\sigma_{\pm}\psi=0$
mielke-geometrodynamics_070 EQ0170 (3.6.5)		■ 1.000 ● N +0	$D^*\psi=\left(\nabla^*\nabla+\frac{1}{4}R+\frac{1}{2}F^+\right)\psi.$	$D^*D\psi=\left(\nabla^*\nabla+\frac{1}{4}R+\frac{1}{2}F^+\right)\psi.$	$D^*D\psi=\left(\nabla^*\nabla+\frac{1}{4}R+\frac{1}{2}F^+\right)\psi.$
mielke-geometrodynamics_070 EQ0172 (3.6.7)		■ 1.000 ● N +0	$\Omega_{\pm}^g=\frac{1}{2\ell^2}\sigma_{\pm},$	$\Omega_{\pm}^g=\frac{1}{2\ell^2}\sigma_{\pm},$	$\Omega_{\pm}^g=\frac{1}{2\ell^2}\sigma_{\pm},$
mielke-geometrodynamics_072 EQ0176 (3.7.5)		■ 1.000 ● N +0	$\begin{aligned}&\wedge^*\Theta^{\left(1\right)}-\frac{\lambda}{2}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)\varphi\eta=0,\\&D\Pi^{\Omega}+\left[\phi^{\left(0\right)},\wedge^*\Theta^{\left(1\right)}\right]=0.\end{aligned}$	$D^*\Theta^{\left(1\right)}-\frac{\lambda}{2}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)\varphi\eta=0,$ $D\Pi^{\Omega}+\left[\phi^{\left(0\right)},\wedge^*\Theta^{\left(1\right)}\right]=0.$	$D^*\Theta^{\left(1\right)}-\frac{\lambda}{2}\left(\frac{\mu^2}{\lambda}-\bar{\varphi}\cdot\varphi\right)\varphi\eta=0,$ $D\Pi^{\Omega}+\left[\phi^{\left(0\right)},\wedge^*\Theta^{\left(1\right)}\right]=0.$
mielke-geometrodynamics_072 EQ0177 (3.7.7)		■ 1.000 ● N +0	$\bar{\varphi}\cdot\varphi:=\varphi^{*j}\varphi_j=\frac{\mu^2}{\lambda}.$	$\bar{\varphi}\cdot\varphi:=\varphi^{*j}\varphi_j=\frac{\mu^2}{\lambda}.$	$\bar{\varphi}\cdot\varphi:=\varphi^{*j}\varphi_j=\frac{\mu^2}{\lambda}.$

Conf. MathPix confidence:

■

≥0.9

■

0.5–0.9

■

<0.5

—

no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_072 EQ0179 (3.7.9)		■ 1.000 ● N -1	$\varphi_{\min }: P \rightarrow C^{\infty }\left(\overset{\circ}{V}(M,\rho (G/H),\rho (G),P)\subset V^{\rho }\right).$	$\varphi _{\min }:P\rightarrow C^{\infty }\left(\overset{\circ}{V}(M,\rho (G/H),\rho (G),P)\subset V^{\rho }\right).$	$\varphi _{\min }:P\rightarrow C^{\infty }\left(\overset{\circ}{V}(M,\rho (G/H),\rho (G),P)\subset V^{\rho }\right).$
mielke-geometrodynamics_073 EQ0180 (3.7.10)		■ 1.000 ● N +0	$H:=\left\{h\in G\mid \rho (h)\mathrm{v}_0=\mathrm{v}_0\right\},$	$H:=\left\{h\in G\mid \rho (h)\mathrm{v}_0=\mathrm{v}_0\right\},$	$H:=\left\{h\in G\mid \rho (h)\mathrm{v}_0=\mathrm{v}_0\right\},$
mielke-geometrodynamics_073 EQ0181 (3.7.11)		■ 1.000 ● N -2	$Q=\left\{p\in P\mid \varphi (p)=\sigma (\mathrm{v}_0)\right\}$	$Q=\left\{p\in P\mid \varphi (p)=\sigma (\mathrm{v}_0)\right\}$	$Q=\left\{p\in P\mid \varphi (p)=\sigma (\mathrm{v}_0)\right\}$
mielke-geometrodynamics_073 EQ0182 (3.7.12)		■ 1.000 ● N +0	$D\,\phi ^{(0)}=0.$	$D\phi ^{(0)}=0.$	$D\phi ^{(0)}=0.$
mielke-geometrodynamics_073 EQ0183 (3.7.13)		■ 1.000 ● K +0	$\begin{aligned}\Theta ^{\perp }&:=\Omega _0=d\omega _0-(\omega \wedge \omega)_0=d\omega _0-\omega _0\wedge \omega -\omega \wedge \omega _0\\&=d\omega _0-[\omega _0,\omega]=D\omega _0\end{aligned}$	$\Theta ^{\perp }:=\Omega _0=d\omega _0-(\omega \wedge \omega)_0=d\omega _0-\omega _0\wedge \omega -\omega \wedge \omega _0$ $=d\omega _0-[\omega _0,\omega]=D\omega _0$	$\Theta ^{\perp }:=\Omega _0=d\omega _0-(\omega \wedge \omega)_0=d\omega _0-\omega _0\wedge \omega -\omega \wedge \omega _0$ $=d\omega _0-[\omega _0,\omega]=D\omega _0$
mielke-geometrodynamics_074 EQ0185 (3.7.15)		■ 1.000 ● K +0	$d_{\left\{ \right\} }^{\left\{ \right\} }\Omega _0=0.$	$d_0^*\Omega _0=0.$	$d_0^*\Omega _0=0.$
mielke-geometrodynamics_074 EQ0186 (3.7.16)		■ 1.000 ● N +0	$\left(D^*\Omega \right)_0=\vec{D}^*\Omega _0=\vec{D}^*\Theta ^{\perp }=0$	$(D^*\Omega)_0=\vec{D}^*\Omega _0=\vec{D}^*\Theta ^{\perp }=0$	$(D^*\Omega)_0=\vec{D}^*\Omega _0=\vec{D}^*\Theta ^{\perp }=0$
mielke-geometrodynamics_074 EQ0188 (3.7.18)		■ 1.000 ●	$\omega _0=\varphi ^jL_j\otimes \vartheta _0.$	$\omega _0=\varphi ^jL_j\otimes \vartheta _0.$	$\omega _0=\varphi ^jL_j\otimes \vartheta _0.$
mielke-geometrodynamics_080 EQ0190		■ 1.000 ● W +0	$\left[\begin{array}{l}T_{\{G\}}:\\quad T_{\{g\}}\,\psi (m)=D\left(g_{\{m\}}^{-1}\,g_{g_{\{g\}}^{-1}\,m},\,m_{\{0\}}\right)\,\psi \left(g^{-1}\,m\right)\\.\,\backslash\,\end{array}\right]$	$T_G:\quad T_g\psi (m)=D\left(g_m^{-1}gg_{g^{-1}m},m_0\right)\psi \left(g^{-1}m\right).$	$T_G:\quad T_g\psi (m)=D\left(g_m^{-1}gg_{g^{-1}m},m_0\right)\psi (g^{-1}m).$
mielke-geometrodynamics_081 EQ0193 (4.1.3)		■ 1.000 ● N +0	$A(n,\,\mathbb{R})=\mathbb{R}^n\otimes GL(n,\,\mathbb{R})$	$A(n,\mathbb{R})=\mathbb{R}^n\otimes GL(n,\mathbb{R})$	$A(n,\mathbb{R})=\mathbb{R}^n\otimes GL(n,\mathbb{R})$
mielke-geometrodynamics_081 EQ0194 (4.1.4)		■ 1.000 ● N +0	$P:=\mathbb{R}^n\otimes 0(1,\,n-1)\subset A(n,\mathbb{R}),$	$P:=\mathbb{R}^n\otimes 0(1,n-1)\subset A(n,\mathbb{R}),$	$P:=\mathbb{R}^n\otimes 0(1,n-1)\subset A(n,\mathbb{R}),$
mielke-geometrodynamics_082 EQ0197 (4.1.10)		■ 1.000 ● N +0	$e_{\alpha }(m)=e_{\alpha }^i\partial _i$	$e_{\alpha }(m)=e_{\alpha }^i\partial _i$	$e_{\alpha }(m)=e_{\alpha }^i\partial _i$
mielke-geometrodynamics_082 EQ0199 (4.1.13)		■ 1.000 ● N +0	$L(M)=\bigcup _{m\in M}L_m(M)$	$L(M)=\bigcup _{m\in M}L_m(M)$	$L(M)=\bigcup _{m\in M}L_m(M)$
mielke-geometrodynamics_083 EQ0200 (4.1.14)		■ 1.000 ● K +0	$L(M)=P(M, GL(n,\mathbb{R}),\pi,\delta)$	$L(M)=P(M, GL(n,\mathbb{R}),\pi,\delta)$	$L(M)=P(M, GL(n,\mathbb{R}),\pi,\delta)$
mielke-geometrodynamics_083 EQ0202 (4.1.16)		■ 1.000 ● K +0	$\vartheta (e)=0\Leftrightarrow \pi (e)=0,e\in T_p(P),$	$\vartheta (e)=0\Leftrightarrow \pi (e)=0,e\in T_p(P),$	$\vartheta (e)=0\Leftrightarrow \pi (e)=0,e\in T_p(P),$
mielke-geometrodynamics_083 EQ0204 (4.1.18)		■ 1.000 ● N +0	$T(M)=V\left(M,\mathbb{R}^n, GL(n,\mathbb{R}),L(M)\right)$	$T(M)=V\left(M,\mathbb{R}^n, GL(n,\mathbb{R}),L(M)\right)$	$T(M)=V(M,\mathbb{R}^n, GL(n,\mathbb{R}),L(M))$
mielke-geometrodynamics_083 EQ0205 (4.1.19)		■ 1.000 ● N +0	$A(M):=P(M,A(n,\mathbb{R}),\pi,\delta)=L(M)\times \mathbb{R}^n.$	$A(M):=P(M,A(n,\mathbb{R}),\pi,\delta)=L(M)\times \mathbb{R}^n.$	$A(M):=P(M,A(n,\mathbb{R}),\pi,\delta)=L(M)\times \mathbb{R}^n.$
mielke-geometrodynamics_083 EQ0206 (4.1.20)		■ 1.000 ● N +0	$\beta :A(n,\mathbb{R})\rightarrow GL(n,\mathbb{R})=A(n,\mathbb{R})/\mathbb{R}^n.$	$\beta :A(n,\mathbb{R})\rightarrow GL(n,\mathbb{R})=A(n,\mathbb{R})/\mathbb{R}^n.$	$\beta :A(n,\mathbb{R})\rightarrow GL(n,\mathbb{R})=A(n,\mathbb{R})/\mathbb{R}^n.$
mielke-geometrodynamics_086 EQ0213 (4.2.5)		■ 1.000 ● N +0	$\mathscr{A}(n,\mathbb{R}):=C^{\infty }\left(A(M)\times _{\mathrm{Ad}}A(n,\mathbb{R})\right).$	$\mathscr{A}(n,\mathbb{R}):=C^{\infty }\left(A(M)\times _{\mathrm{Ad}}A(n,\mathbb{R})\right).$	$\mathscr{A}(n,\mathbb{R}):=C^{\infty }\left(A(M)\times _{\mathrm{Ad}}A(n,\mathbb{R})\right).$
mielke-geometrodynamics_087 EQ0223 (4.2.16)		■ 1.000 ● N -2	$Q_{\{A\}}=\left\{p\in A(M)\mid \widetilde{\varphi }(p)=\sigma (0_m)\right\}=L(M)$	$Q_A=\left\{p\in A(M)\mid \widetilde{\varphi }(p)=\sigma (0_m)\right\}=L(M)$	$Q_A=\left\{p\in A(M)\mid \widetilde{\varphi }(p)=\sigma (0_m)\right\}=L(M)$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_090 EQ0237 (4.3.2)		■ 1.000 ● S +0	$\ell(\mathscr{C}) := \int_{t_0}^{t_1} ds(t)$	$\ell(\mathcal{C}) := \int_{t_0}^{t_1} ds(t)$	$\ell(\mathcal{C}) := \int_{t_0}^{t_1} ds(t)$
mielke-geometrodynamics_090 EQ0238 (4.3.3)		■ 1.000 ● K +0	$\ell(\mathscr{C}_1 + \mathscr{C}_2) = \ell(\mathscr{C}_1) + \ell(\mathscr{C}_2)$	$\ell(\mathcal{C}_1 + \mathcal{C}_2) = \ell(\mathcal{C}_1) + \ell(\mathcal{C}_2)$	$\ell(\mathcal{C}_1 + \mathcal{C}_2) = \ell(\mathcal{C}_1) + \ell(\mathcal{C}_2)$
mielke-geometrodynamics_090 EQ0239 (4.3.4)		■ 1.000 ● K +0	$ds = F(m, \vartheta) dt,$	$ds = F(m, \vartheta) dt,$	$ds = F(m, \vartheta) dt,$
mielke-geometrodynamics_091 EQ0242 (4.3.7)		■ 1.000 ● C -3	$ds^2 = (F(m, \vartheta) dt)^2 = g_{\alpha\beta} \vartheta^\alpha \otimes \vartheta^\beta = g_{ij}(m) dx^i \otimes_s dx^j;$	$ds^2 = (F(m, \vartheta) dt)^2 = g_{\alpha\beta} \vartheta^\alpha \otimes \vartheta^\beta = g_{ij}(m) dx^i \otimes_s dx^j;$	$ds^2 = (F(m, \vartheta) dt)^2 = g_{\alpha\beta} \vartheta^\alpha \otimes \vartheta^\beta = g_{ij}(m) dx^i \otimes_s dx^j;$
mielke-geometrodynamics_092 EQ0246 (4.3.11)		■ 1.000 ● N +0	$L^g(M) := P(M, O(s, n-s), \pi, \delta).$	$L^g(M) := P(M, O(s, n-s), \pi, \delta).$	$L^g(M) := P(M, O(s, n-s), \pi, \delta).$
mielke-geometrodynamics_092 EQ0247 (4.3.12)		■ 1.000 ● N +0	$De_{\alpha}^i = 0$	$De_{\alpha}^i = 0$	$De_{\alpha}^i = 0$
mielke-geometrodynamics_092 EQ0251 (4.3.16)		■ 1.000 ● N +0	$\Gamma_i^{\alpha\beta} = -\Gamma_i^{\beta\alpha}$	$\Gamma_i^{\alpha\beta} = -\Gamma_i^{\beta\alpha}$	$\Gamma_i^{\alpha\beta} = -\Gamma_i^{\beta\alpha}$
mielke-geometrodynamics_093 EQ0252 (4.3.17)		■ 1.000 ● N +0	$g_{ij} \stackrel{(*)}{=} o_{ij}$	$g_{ij} \stackrel{(*)}{=} o_{ij}$	$g_{ij} \stackrel{(*)}{=} o_{ij}$
mielke-geometrodynamics_093 EQ0253 (4.3.18)		■ 1.000 ● N +0	$\partial_k g_{ij} \stackrel{(*)}{=} 0.$	$\partial_k g_{ij} \stackrel{(*)}{=} 0.$	$\partial_k g_{ij} \stackrel{(*)}{=} 0.$
mielke-geometrodynamics_093 EQ0254 (4.3.19)		■ 1.000 ● N +1	$\Omega^g = d\omega^g - \omega^g \wedge \omega^g$	$\Omega^g = d\omega^g - \omega^g \wedge \omega^g$	$\Omega^g = d\omega^g - \omega^g \wedge \omega^g$
mielke-geometrodynamics_094 EQ0257 (4.3.22)		■ 1.000 ●	$\omega^g = \omega^{\{\}} - K.$	$\omega^g = \omega^{\{\}} - K.$	$\omega^g = \omega^{\{\}} - K.$
mielke-geometrodynamics_094 EQ0258 (4.3.23)		■ 1.000 ● K +0	$d\vartheta - [\omega^{\{\}}, \vartheta] = 0$	$d\vartheta - [\omega^{\{\}}, \vartheta] = 0$	$d\vartheta - [\omega^{\{\}}, \vartheta] = 0$
mielke-geometrodynamics_094 EQ0259 (4.3.24)		■ 1.000 ● N +0	$\Theta = [K, \vartheta]$	$\Theta = [K, \vartheta]$	$\Theta = [K, \vartheta]$
mielke-geometrodynamics_095 EQ0265 (4.3.30)		■ 1.000 ● W +0	$\left\{ \begin{array}{c} \kappa \\ ij \end{array} \right\} = \frac{1}{2} g^{\kappa l} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$	$\left\{ \begin{array}{c} \kappa \\ ij \end{array} \right\} = \frac{1}{2} g^{\kappa l} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$	$\left\{ \begin{array}{c} \kappa \\ ij \end{array} \right\} = \frac{1}{2} g^{\kappa l} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$
mielke-geometrodynamics_096 EQ0272 (4.4.1)		■ 1.000 ● N -1	$L_g = L(g_{\alpha\beta}; \vartheta, (\omega^g), \Theta, \Omega^g).$	$L_g = L(g_{\alpha\beta}; \vartheta, (\omega^g), \Theta, \Omega^g).$	$L_g = L(g_{\alpha\beta}; \vartheta, (\omega^g), \Theta, \Omega^g).$
mielke-geometrodynamics_097 EQ0273 (4.4.2)		■ 1.000 ● N +0	$\delta L_g =: \delta\vartheta \wedge E + \delta\Theta \wedge \Pi^T + \delta\Omega^g \wedge \Pi^L,$	$\delta L_g =: \delta\vartheta \wedge E + \delta\Theta \wedge \Pi^T + \delta\Omega^g \wedge \Pi^L,$	$\delta L_g =: \delta\vartheta \wedge E + \delta\Theta \wedge \Pi^T + \delta\Omega^g \wedge \Pi^L,$
mielke-geometrodynamics_097 EQ0274 (4.4.7)		■ 1.000 ● K +0	$\delta\Theta = \delta(d\vartheta - [\omega^L, \vartheta])$	$\delta\Theta = \delta(d\vartheta - [\omega^L, \vartheta])$	$\delta\Theta = \delta(d\vartheta - [\omega^L, \vartheta])$
mielke-geometrodynamics_097 EQ0277 (4.4.5)		■ 1.000 ● N +0	$S = \int_M (L_g + L_{\text{mat}})$	$S = \int_M (L_g + L_{\text{mat}})$	$S = \int_M (L_g + L_{\text{mat}})$
mielke-geometrodynamics_098 EQ0279 (4.4.8)		■ 1.000 ● N +0	$\delta L_{\text{matter}} =: \delta\vartheta \wedge \Sigma + \delta\omega^g \wedge \tau_s + \delta_{\varphi} \wedge \lambda^{\varphi}$	$\delta L_{\text{matter}} =: \delta\vartheta \wedge \Sigma + \delta\omega^g \wedge \tau_s + \delta_{\varphi} \wedge \lambda^{\varphi}$	$\delta L_{\text{matter}} =: \delta\vartheta \wedge \Sigma + \delta\omega^g \wedge \tau_s + \delta_{\varphi} \wedge \lambda^{\varphi}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_098 EQ0280 (4.4.9)		■ 1.000 ● N +0	$\mathrm{D}\,\vartheta\equiv\left[\vartheta,\,\Omega^{\mathrm{L}}\right]$	$D\Theta\equiv\left[\vartheta,\Omega^{\mathrm{L}}\right]$	$D\Theta\equiv\left[\vartheta,\,\Omega^{\mathrm{L}}\right]$
mielke-geometrodynamics_098 EQ0281 (4.4.10)		■ 1.000 ● N +0	$\mathrm{D}\,\Omega^{\mathrm{g}}\equiv\vartheta,$	$D\Omega^g\equiv 0,$	$D\Omega^g\equiv 0,$
mielke-geometrodynamics_099 EQ0283 (4.4.12)		■ 1.000 ● N +0	$\mathrm{D}[\vartheta,\vartheta]\equiv\left[\vartheta\wedge\vartheta,\Omega^g\right]$	$D[\Theta,\vartheta]\equiv\left[\vartheta\wedge\vartheta,\Omega^g\right]$	$D[\Theta,\vartheta]\equiv\left[\vartheta\wedge\vartheta,\,\Omega^g\right]$
mielke-geometrodynamics_100 EQ0289 (4.4.18)		■ 1.000 ● N +0	$\left[\Omega^{\mathrm{g}},\,\Pi^{\mathrm{L}}\right]+\mathrm{E}_{\varnothing}=\varnothing$	$\left[\Omega^g,\Pi^{\mathrm{L}}\right]+\mathrm{E}_{\varnothing}=0$	$\left[\Omega^g,\,\Pi^{\mathrm{L}}\right]+\mathrm{E}_{\varnothing}=0$
mielke-geometrodynamics_100 EQ0290 (4.5.1)		■ 1.000 ● N +0	$\mathrm{L}_{\mathrm{mat}}=\mathrm{L}\left(\mathrm{g}_{\alpha}\,\beta,\,\vartheta^{\alpha},\,\Psi,\,\mathrm{D}\,\Psi\right).$	$L_{\mathrm{mat}}=L\left(g_{\alpha\beta},\vartheta^{\alpha},\Psi,D\Psi\right).$	$L_{\mathrm{mat}}=L(g_{\alpha\beta},\vartheta^{\alpha},\Psi,D\Psi).$
mielke-geometrodynamics_100 EQ0291 (4.5.2)		■ 1.000 ● W -1	$\Sigma_{\alpha}:=\frac{\partial L}{\partial\vartheta^{\alpha}},\quad\sigma^{\alpha\beta}:=2\frac{\partial L}{\partial g_{\alpha\beta}},\quad\text{and}\quad\tau_{\alpha\beta}:=\frac{\partial L}{\partial\Gamma^{\alpha\beta}}=I_{\alpha\beta}\Psi\wedge\frac{\partial L}{\partial D\Psi}$	$\Sigma_{\alpha}:=\frac{\partial L}{\partial\vartheta^{\alpha}},\quad\sigma^{\alpha\beta}:=2\frac{\partial L}{\partial g_{\alpha\beta}},\quad\text{and}\quad\tau_{\alpha\beta}:=\frac{\partial L}{\partial\Gamma^{\alpha\beta}}=I_{\alpha\beta}\Psi\wedge\frac{\partial L}{\partial D\Psi}$	$\Sigma_{\alpha}:=\frac{\partial L}{\partial\vartheta^{\alpha}},\quad\sigma^{\alpha\beta}:=2\frac{\partial L}{\partial g_{\alpha\beta}},\quad\text{and}\quad\tau_{\alpha\beta}:=\frac{\partial L}{\partial\Gamma^{\alpha\beta}}=I_{\alpha\beta}\Psi\wedge\frac{\partial L}{\partial D\Psi}$
mielke-geometrodynamics_101 EQ0297 (8.10)		■ 1.000 ● N +0	$\delta\mathrm{L}=\delta\mathrm{g}_{\alpha}\,\beta\wedge\frac{\partial L}{\partial\mathrm{g}_{\alpha}\,\beta}+\delta\vartheta^{\alpha}\wedge\frac{\partial L}{\partial\vartheta^{\alpha}}+\delta\Psi\wedge\frac{\partial L}{\partial\Psi}+\delta(D\Psi)\wedge\frac{\partial L}{\partial D\Psi}.$	$\delta L=\delta g_{\alpha\beta}\wedge\frac{\partial L}{\partial g_{\alpha\beta}}+\delta\vartheta^{\alpha}\wedge\frac{\partial L}{\partial\vartheta^{\alpha}}+\delta\Psi\wedge\frac{\partial L}{\partial\Psi}+\delta(D\Psi)\wedge\frac{\partial L}{\partial D\Psi}.$	$\delta L=\delta g_{\alpha\beta}\wedge\frac{\partial L}{\partial g_{\alpha\beta}}+\delta\vartheta^{\alpha}\wedge\frac{\partial L}{\partial\vartheta^{\alpha}}+\delta\Psi\wedge\frac{\partial L}{\partial\Psi}+\delta(D\Psi)\wedge\frac{\partial L}{\partial D\Psi}.$
mielke-geometrodynamics_102 EQ0300 (4.5.11)		■ 1.000 ● N +0	$\delta\mathrm{L}=-\varepsilon_{\alpha}^{\beta}\left[-\sigma^{\alpha\gamma}g_{\gamma\beta}+\vartheta^{\alpha}\wedge\frac{\partial L}{\partial\vartheta^{\beta}}+D\frac{\partial L}{\partial\Gamma_{\alpha}^{\beta}}+I^{\alpha}_{\beta}\Psi\wedge\frac{\delta L}{\delta\Psi}\right].$	$\delta L=-\varepsilon_{\alpha}^{\beta}\left[-\sigma^{\alpha\gamma}g_{\gamma\beta}+\vartheta^{\alpha}\wedge\frac{\partial L}{\partial\vartheta^{\beta}}+D\frac{\partial L}{\partial\Gamma_{\alpha}^{\beta}}+I^{\alpha}_{\beta}\Psi\wedge\frac{\delta L}{\delta\Psi}\right].$	$\delta L=-\varepsilon_{\alpha}^{\beta}\left[-\sigma^{\alpha\gamma}g_{\gamma\beta}+\vartheta^{\alpha}\wedge\frac{\partial L}{\partial\vartheta^{\beta}}+D\frac{\partial L}{\partial\Gamma_{\alpha}^{\beta}}+I^{\alpha}_{\beta}\Psi\wedge\frac{\delta L}{\delta\Psi}\right].$
mielke-geometrodynamics_102 EQ0301 (4.5.12)		■ 1.000 ●	$\mathrm{D}\,\tau_{\alpha}\,\beta+\vartheta_{[\alpha}\wedge\Sigma_{\beta]}=-I_{\alpha\beta}\Psi\wedge\frac{\delta L}{\delta\Psi}\simeq 0.$	$D\tau_{\alpha\beta}+\vartheta_{[\alpha}\wedge\Sigma_{\beta]}=-I_{\alpha\beta}\Psi\wedge\frac{\delta L}{\delta\Psi}\simeq 0.$	$D\tau_{\alpha\beta}+\vartheta_{[\alpha}\wedge\Sigma_{\beta]}=-I_{\alpha\beta}\Psi\wedge\frac{\delta L}{\delta\Psi}\simeq 0.$
mielke-geometrodynamics_102 EQ0303 (4.5.13)		■ 1.000 ● W -1	$\begin{aligned}&\&R_{\alpha}\{\}^{\beta}\&R_{\alpha}\{\}^{\beta}=\mathrm{d}\,\Gamma_{\alpha}\{\}^{\beta}\\&\&\&\&- \Gamma_{\alpha}^{\gamma}\wedge\Gamma_{\gamma}^{\beta}\end{aligned}$	$R_{\alpha}^{\beta}=d\Gamma_{\alpha}^{\beta}-\Gamma_{\alpha}^{\gamma}\wedge\Gamma_{\gamma}^{\beta}$	$R_{\alpha}^{\beta}=d\Gamma_{\alpha}^{\beta}-\Gamma_{\alpha}^{\gamma}\wedge\Gamma_{\gamma}^{\beta}$
mielke-geometrodynamics_103 EQ0307 (4.5.17)		■ 1.000 ● N +0	$\hat{M}:=M^3/\left[M^2+\frac{\Lambda}{3}J^2\right],\quad\hat{J}:=JM^2/\left[M^2+\frac{\Lambda}{3}J^2\right],$	$\hat{M}:=M^3/\left[M^2+\frac{\Lambda}{3}J^2\right],\quad\hat{J}:=JM^2/\left[M^2+\frac{\Lambda}{3}J^2\right],$	$\hat{M}:=M^3/[M^2+\frac{\Lambda}{3}J^2],\quad\hat{J}:=JM^2/[M^2+\frac{\Lambda}{3}J^2],$
mielke-geometrodynamics_110 EQ0317 (5.2.1)		■ 1.000 ● W +0	$\mathrm{L}_{\mathrm{mat}}=\mathrm{L}(\gamma,\psi,\mathrm{D}\,\psi)=\frac{i}{2}\left\{\bar{\psi}^*\gamma\wedge D\psi+\overline{D\psi}\wedge^*\gamma\psi\right\}-m\bar{\psi}\psi\eta,$	$L_{\mathrm{D}}=L(\gamma,\psi,D\psi)=\frac{i}{2}\left\{\bar{\psi}^*\gamma\wedge D\psi+\overline{D\psi}\wedge^*\gamma\psi\right\}-m\bar{\psi}\psi\eta,$	$L_{\mathrm{D}}=L(\gamma,\psi,D\psi)=\frac{i}{2}\left\{\bar{\psi}^*\gamma\wedge D\psi+\overline{D\psi}\wedge^*\gamma\psi\right\}-m\bar{\psi}\psi\eta,$
mielke-geometrodynamics_111 EQ0322 (5.2.6)		■ 1.000 ● K +0	$\left.\mathrm{d}\,U=T^{\alpha}\wedge\mu_{\alpha}-\vartheta^{\alpha}\wedge D\mu_{\alpha}=\vartheta^{\alpha}\wedge\left[e_{\beta}\right]\left(T^{\beta}\wedge\mu_{\alpha}\right)-D\mu_{\alpha}\right]$	$dU=T^{\alpha}\wedge\mu_{\alpha}-\vartheta^{\alpha}\wedge D\mu_{\alpha}=\vartheta^{\alpha}\wedge\left[e_{\beta}\right]\left(T^{\beta}\wedge\mu_{\alpha}\right)-D\mu_{\alpha}]$	$dU=T^{\alpha}\wedge\mu_{\alpha}-\vartheta^{\alpha}\wedge D\mu_{\alpha}=\vartheta^{\alpha}\wedge\left[e_{\beta}\right]\left(T^{\beta}\wedge\mu_{\alpha}\right)-D\mu_{\alpha}]$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_E08324 (5.3.1)	111	■ 1.000 ● K +0	$L_{\mathrm{EC}}:=-\frac{1}{2}\kappa R^{\alpha\beta}\wedge\eta_{\alpha\beta}\wedge\eta_{\alpha\beta},$	$L_{\mathrm{EC}}:=-\frac{1}{2\kappa}R^{\alpha\beta}\wedge\eta_{\alpha\beta},$	$L_{\mathrm{EC}}:=-\frac{1}{2\kappa}R^{\alpha\beta}\wedge\eta_{\alpha\beta},$
mielke-geometrodynamics_E08325 (5.3.2)	112	■ 1.000 ● K +0	$G_{\alpha}:=\frac{1}{2}R^{\beta\gamma}\wedge\eta_{\alpha\beta\gamma}\wedge\eta_{\alpha\beta\gamma}=\kappa\Sigma_{\alpha},$	$G_{\alpha}:=\frac{1}{2}R^{\beta\gamma}\wedge\eta_{\alpha\beta\gamma}=\kappa\Sigma_{\alpha},$	$G_{\alpha}:=\frac{1}{2}R^{\beta\gamma}\wedge\eta_{\alpha\beta\gamma}=\kappa\Sigma_{\alpha},$
mielke-geometrodynamics_E08327 (5.3.4)	112	■ 1.000 ● N +0	$D^{\{\}\{\}}G_{\alpha}^{\{\}\{\}}\equiv 0$	$D^{\{\}\{\}}G_{\alpha}^{\{\}\{\}}\equiv 0$	$D^{\{\}\{\}}G_{\alpha}^{\{\}\{\}}\equiv 0$
mielke-geometrodynamics_E08331 (5.3.8)	112	■ 1.000 ● N +0	$D[\gamma,\Theta]\equiv 2i\left[\sigma,\Omega^g\right],\quad D\left[\gamma,\Omega^g\right]\equiv\left[\Theta,\Omega^g\right],$	$D[\gamma,\Theta]\equiv 2i\left[\sigma,\Omega^g\right],\quad D\left[\gamma,\Omega^g\right]\equiv\left[\Theta,\Omega^g\right],$	$D[\gamma,\Theta]\equiv 2i\left[\sigma,\Omega^g\right],\quad D\left[\gamma,\Omega^g\right]\equiv\left[\Theta,\Omega^g\right],$
mielke-geometrodynamics_E08332 (5.3.9)	113	■ 1.000 ● N +0	$D\,G\equiv i\left[\Theta,\gamma_5\Omega^g\right]$	$DG\equiv i\left[\Theta,\gamma_5\Omega^g\right]$	$DG\equiv i\left[\Theta,\gamma_5\Omega^g\right]$
mielke-geometrodynamics_E08333 (5.3.10)	113	■ 1.000 ● K +0	$\eta_{\alpha\beta\gamma}\wedge T^{\gamma}=2\kappa\tau_{\alpha\beta}$	$\eta_{\alpha\beta\gamma}\wedge T^{\gamma}=2\kappa\tau_{\alpha\beta}$	$\eta_{\alpha\beta\gamma}\wedge T^{\gamma}=2\kappa\tau_{\alpha\beta}$
mielke-geometrodynamics_E08334 (5.3.11)	113	■ 1.000 ● N +0	$\mathrm{T}_{\mu\nu}:=\frac{2}{\sqrt{g}}\frac{\delta L}{\delta g_{\mu\nu}}$	$T_{\mu\nu}:=\frac{2}{\sqrt{g}}\frac{\delta L}{\delta g_{\mu\nu}}$	$T_{\mu\nu}:=\frac{2}{\sqrt{g}}\frac{\delta L}{\delta g_{\mu\nu}}$
mielke-geometrodynamics_E08338 (5.3.15)	114	■ 1.000 ● N +0	$\{ \}^{\{3\}}T_{\alpha}=\frac{(-1)^s}{3}\ast\left(\vartheta_{\alpha}\wedge\mathscr{A}\right)$	$^{(3)}T_{\alpha}=\frac{(-1)^s}{3}\ast\left(\vartheta_{\alpha}\wedge\mathscr{A}\right)$	$^{(3)}T_{\alpha}=\frac{(-1)^s}{3}\ast\left(\vartheta_{\alpha}\wedge\mathscr{A}\right)$
mielke-geometrodynamics_E08342 (5.3.19)	115	■ 1.000 ● N -1	$\Delta\widetilde{L}:=\kappa\left(L_{\mathrm{EC}}-L_{\mathrm{HE}}\right)\simeq\frac{\kappa^2}{48}j_5\wedge^{\ast}j_5=4f^2j_5\wedge^{\ast}j_5,$	$\Delta\widetilde{L}:=\kappa\left(L_{\mathrm{EC}}-L_{\mathrm{HE}}\right)\simeq\frac{\kappa^2}{48}j_5\wedge^{\ast}j_5=4f^2j_5\wedge^{\ast}j_5,$	$\Delta\widetilde{L}:=\kappa\left(L_{\mathrm{EC}}-L_{\mathrm{HE}}\right)\simeq\frac{\kappa^2}{48}j_5\wedge^{\ast}j_5=4f^2j_5\wedge^{\ast}j_5,$
mielke-geometrodynamics_E08344 (5.4.2)	116	■ 1.000 ● N -1	$L_{\mathrm{HE}}:=L_{\mathrm{HE}}+\frac{\Lambda}{\kappa}\eta$	$L_{\mathrm{HE}}:=L_{\mathrm{HE}}+\frac{\Lambda}{\kappa}\eta$	$L_{\mathrm{HE}}:=L_{\mathrm{HE}}+\frac{\Lambda}{\kappa}\eta$
mielke-geometrodynamics_E08345 (5.4.3)	116	■ 1.000 ● N +0	$g_{\mathrm{N}}:=\kappa k^2,\quad\lambda:=\Lambda/k^2,$	$g_{\mathrm{N}}:=\kappa k^2,\quad\lambda:=\Lambda/k^2,$	$g_{\mathrm{N}}:=\kappa k^2,\quad\lambda:=\Lambda/k^2,$
mielke-geometrodynamics_E08346 (5.4.4)	116	■ 1.000 ● N +0	$k\frac{\partial}{\partial k}g_{\mathrm{N}}=\beta_1\left(g_{\mathrm{N}},\lambda\right)=\left(2+d_{\mathrm{N}}\right)g_{\mathrm{N}},\quad k\frac{\partial}{\partial k}\lambda=\beta_2\left(g_{\mathrm{N}},\lambda\right),$	$k\frac{\partial}{\partial k}g_{\mathrm{N}}=\beta_1\left(g_{\mathrm{N}},\lambda\right)=\left(2+d_{\mathrm{N}}\right)g_{\mathrm{N}},\quad k\frac{\partial}{\partial k}\lambda=\beta_2\left(g_{\mathrm{N}},\lambda\right),$	$k\frac{\partial}{\partial k}g_{\mathrm{N}}=\beta_1\left(g_{\mathrm{N}},\lambda\right)=\left(2+d_{\mathrm{N}}\right)g_{\mathrm{N}},\quad k\frac{\partial}{\partial k}\lambda=\beta_2\left(g_{\mathrm{N}},\lambda\right),$
mielke-geometrodynamics_E08348 (5.4.6)	117	■ 1.000 ●	$f^2=\frac{1}{192}\kappa^2=3\times 2^{-10}\left(\frac{\mu}{\Lambda}\right)^2=2^{-8}\frac{\mu}{\rho_{\Lambda}}$	$f^2=\frac{1}{192}\kappa^2=3\times 2^{-10}\left(\frac{\mu}{\Lambda}\right)^2=2^{-8}\frac{\mu}{\rho_{\Lambda}}$	$f^2=\frac{1}{192}\kappa^2=3\times 2^{-10}\left(\frac{\mu}{\Lambda}\right)^2=2^{-8}\frac{\mu}{\rho_{\Lambda}}$
mielke-geometrodynamics_E08350 (6.2.1)	123	■ 1.000 ● W -1	$C_{\mathrm{TT}}:=\frac{1}{2\ell^2}\left(\vartheta^{\alpha}\wedge T_{\alpha}\right)=-\frac{(-1)^{\mathrm{sig}}}{2\ell^2}\ast\mathscr{A},$	$C_{\mathrm{TT}}:=\frac{1}{2\ell^2}\left(\vartheta^{\alpha}\wedge T_{\alpha}\right)=-\frac{(-1)^{\mathrm{sig}}}{2\ell^2}\ast\mathscr{A},$	$C_{\mathrm{TT}}:=\frac{1}{2\ell^2}\left(\vartheta^{\alpha}\wedge T_{\alpha}\right)=-\frac{(-1)^{\mathrm{sig}}}{2\ell^2}\ast\mathscr{A},$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_124 EQ0351 (6.2.2)		■ 1.000 ● W -1	$\begin{aligned} C_{\mathrm{RR}} &:= -\operatorname{Tr} \left(\Gamma \wedge \Omega - \frac{1}{3} \Gamma \wedge \Gamma \wedge \Gamma \right) \\ &= -\frac{1}{2} \left(\Gamma_{\alpha}{}^{\beta} \wedge R_{\beta}{}^{\alpha} + \frac{1}{3} \Gamma_{\alpha}{}^{\beta} \wedge \Gamma_{\beta}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\alpha} \right). \end{aligned}$	$C_{\mathrm{RR}} := -\operatorname{Tr} \left(\Gamma \wedge \Omega - \frac{1}{3} \Gamma \wedge \Gamma \wedge \Gamma \right) \\ = -\frac{1}{2} \left(\Gamma_{\alpha}{}^{\beta} \wedge R_{\beta}{}^{\alpha} + \frac{1}{3} \Gamma_{\alpha}{}^{\beta} \wedge \Gamma_{\beta}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\alpha} \right).$	$C_{\mathrm{RR}} := -\operatorname{Tr} \left(\Gamma \wedge \Omega - \frac{1}{3} \Gamma \wedge \Gamma \wedge \Gamma \right) \\ = -\frac{1}{2} \left(\Gamma_{\alpha}{}^{\beta} \wedge R_{\beta}{}^{\alpha} + \frac{1}{3} \Gamma_{\alpha}{}^{\beta} \wedge \Gamma_{\beta}{}^{\gamma} \wedge \Gamma_{\gamma}{}^{\alpha} \right).$
mielke-geometrodynamics_124 EQ0352 (6.2.3)		■ 1.000 ● N -2	$\hat{C} = C_{\mathrm{RR}} - 2C_{\mathrm{TT}}$	$\hat{C} = C_{\mathrm{RR}} - 2C_{\mathrm{TT}}$	$\hat{C} = C_{\mathrm{RR}} - 2C_{\mathrm{TT}}$
mielke-geometrodynamics_124 EQ0353 (6.2.4)		■ 1.000 ● N -1	$d C_{\mathrm{TT}} \equiv \frac{1}{2\ell^2} \left(T^{\alpha} \wedge T_{\alpha} + R_{\alpha\beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta} \right),$	$dC_{\mathrm{TT}} \equiv \frac{1}{2\ell^2} \left(T^{\alpha} \wedge T_{\alpha} + R_{\alpha\beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta} \right),$	$dC_{\mathrm{TT}} \equiv \frac{1}{2\ell^2} \left(T^{\alpha} \wedge T_{\alpha} + R_{\alpha\beta} \wedge \vartheta^{\alpha} \wedge \vartheta^{\beta} \right),$
mielke-geometrodynamics_124 EQ0354 (6.2.5)		■ 1.000 ● N -1	$d C_{\mathrm{RR}} = -\operatorname{Tr}(\Omega \wedge \Omega) = \frac{1}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}$	$dC_{\mathrm{RR}} = -\operatorname{Tr}(\Omega \wedge \Omega) = \frac{1}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}$	$dC_{\mathrm{RR}} = -\operatorname{Tr}(\Omega \wedge \Omega) = \frac{1}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}$
mielke-geometrodynamics_124 EQ0355 (6.2.6)		■ 1.000 ● N +0	$D R_{\alpha\beta} \equiv 0,$	$DR_{\alpha\beta} \equiv 0,$	$DR_{\alpha\beta} \equiv 0,$
mielke-geometrodynamics_125 EQ0359 (6.2.10)		■ 1.000 ● N +0	$H_{\alpha}^{\parallel} := -\partial L_{\parallel} / \partial T^{\alpha} = (1/\kappa) \eta_{\alpha\beta\gamma} \wedge K^{\beta\gamma}$	$H_{\alpha}^{\parallel} := -\partial L_{\parallel} / \partial T^{\alpha} = (1/\kappa) \eta_{\alpha\beta\gamma} \wedge K^{\beta\gamma}$	$H_{\alpha}^{\parallel} := -\partial L_{\parallel} / \partial T^{\alpha} = (1/\kappa) \eta_{\alpha\beta\gamma} \wedge K^{\beta\gamma}$
mielke-geometrodynamics_126 EQ0361 (6.2.12)		■ 1.000 ● N +0	$\tilde{L}_{\parallel} = L_{\parallel} - R^{\alpha\beta} \wedge \lambda_{\alpha\beta}.$	$\tilde{L}_{\parallel} = L_{\parallel} - R^{\alpha\beta} \wedge \lambda_{\alpha\beta}.$	$\tilde{L}_{\parallel} = L_{\parallel} - R^{\alpha\beta} \wedge \lambda_{\alpha\beta}.$
mielke-geometrodynamics_126 EQ0362 (6.2.14)		■ 1.000 ● W +0	$\begin{gathered} D H_{\alpha}^{\parallel} - E_{\alpha}^{\parallel} = \Sigma_{\alpha}, \\ D \lambda_{\alpha\beta} + \vartheta_{[\alpha} \wedge H_{\beta]}^{\parallel} = \tau_{\alpha\beta}, \end{gathered}$	$\begin{gathered} DH_{\alpha}^{\parallel} - E_{\alpha}^{\parallel} = \Sigma_{\alpha}, \\ D\lambda_{\alpha\beta} + \vartheta_{[\alpha} \wedge H_{\beta]}^{\parallel} = \tau_{\alpha\beta}, \end{gathered}$	$\begin{gathered} DH_{\alpha}^{\parallel} - E_{\alpha}^{\parallel} = \Sigma_{\alpha}, \\ D\lambda_{\alpha\beta} + \vartheta_{[\alpha} \wedge H_{\beta]}^{\parallel} = \tau_{\alpha\beta}, \end{gathered}$
mielke-geometrodynamics_126 EQ0363 (6.2.15)		■ 1.000 ● N +1	$R^{\alpha\beta} = 0.$	$R^{\alpha\beta} = 0.$	$R^{\alpha\beta} = 0.$
mielke-geometrodynamics_126 EQ0365 (6.2.17)		■ 1.000 ● N +0	$D \left(\tau_{\alpha\beta} - \vartheta_{[\alpha} \wedge H_{\beta]}^{\parallel} \right) = 0$	$D \left(\tau_{\alpha\beta} - \vartheta_{[\alpha} \wedge H_{\beta]}^{\parallel} \right) = 0$	$D(\tau_{\alpha\beta} - \vartheta_{[\alpha} \wedge H_{\beta]}^{\parallel}) = 0$
mielke-geometrodynamics_127 EQ0369 (6.2.20)		■ 1.000 ● K +0	$\stackrel{(\pm)}{C}_{\mathrm{TT}} := \frac{1}{2\ell^2} \left(\vartheta^{\alpha} \wedge \stackrel{(\pm)}{T}_{\alpha} \right),$	$\stackrel{(\pm)}{C}_{\mathrm{TT}} := \frac{1}{2\ell^2} \left(\vartheta^{\alpha} \wedge \stackrel{(\pm)}{T}_{\alpha} \right),$	$\stackrel{(\pm)}{C}_{\mathrm{TT}} := \frac{1}{2\ell^2} \left(\vartheta^{\alpha} \wedge \stackrel{(\pm)}{T}_{\alpha} \right),$
mielke-geometrodynamics_127 EQ0373 (6.2.24)		■ 1.000 ● N +0	$\stackrel{(\pm)(\pm)}{D} \Pi_{\alpha} = \Sigma_{\alpha}.$	$\stackrel{(\pm)(\pm)}{D} \Pi_{\alpha} = \Sigma_{\alpha}.$	$\stackrel{(\pm)(\pm)}{D} \Pi_{\alpha} = \Sigma_{\alpha}.$
mielke-geometrodynamics_128 EQ0379 (6.2.30)		■ 1.000 ●	$\stackrel{(\pm)}{L}_{\mathrm{YM}} = \mp(1/4) \operatorname{Tr} \left(\stackrel{(\pm)}{F} \wedge \stackrel{(\pm)}{F} \right)$	$\stackrel{(\pm)}{L}_{\mathrm{YM}} = \mp(1/4) \operatorname{Tr} \left(\stackrel{(\pm)}{F} \wedge \stackrel{(\pm)}{F} \right)$	$\stackrel{(\pm)}{L}_{\mathrm{YM}} = \mp(1/4) \operatorname{Tr} \left(\stackrel{(\pm)}{F} \wedge \stackrel{(\pm)}{F} \right)$
mielke-geometrodynamics_129 EQ0383 (6.3.1)		■ 1.000 ● N +0	$\check{E}_{\alpha} := E_{\alpha} + \Gamma_{\alpha}{}^{\beta} \wedge H_{\beta} + \Sigma_{\alpha} \cong dH_{\alpha},$	$\check{E}_{\alpha} := E_{\alpha} + \Gamma_{\alpha}{}^{\beta} \wedge H_{\beta} + \Sigma_{\alpha} \cong dH_{\alpha},$	$\check{E}_{\alpha} := E_{\alpha} + \Gamma_{\alpha}{}^{\beta} \wedge H_{\beta} + \Sigma_{\alpha} \cong dH_{\alpha},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_130 EQ0386 (6.4.1)		■ 1.000 ● N +0	$n:=\partial_t-N^A\partial/\partial x^A,\quad A=1,2,3,$	$n:=\partial_t-N^A\partial/\partial x^A,\quad A=1,2,3,$	$n:=\partial_t-N^A\partial/\partial x^A,\quad A=1,2,3,$
mielke-geometrodynamics_132 EQ0395 (6.4.10)		■ 1.000 ● W +0	$\Sigma_{\alpha}:=\frac{\partial L}{\partial \vartheta^{\alpha}},\quad \tau_{\alpha\beta}:=\frac{\partial L}{\partial \Gamma^{\alpha\beta}}=\rho(I_{\alpha\beta})\Psi\wedge\frac{\partial L}{\partial D\Psi}$	$\Sigma_{\alpha}:=\frac{\partial L}{\partial \vartheta^{\alpha}},\quad \tau_{\alpha\beta}:=\frac{\partial L}{\partial \Gamma^{\alpha\beta}}=\rho(I_{\alpha\beta})\Psi\wedge\frac{\partial L}{\partial D\Psi}$	$\Sigma_{\alpha}:=\frac{\partial L}{\partial \vartheta^{\alpha}},\quad \tau_{\alpha\beta}:=\frac{\partial L}{\partial \Gamma^{\alpha\beta}}=\rho(I_{\alpha\beta})\Psi\wedge\frac{\partial L}{\partial D\Psi}$
mielke-geometrodynamics_136 EQ0408 (6.4.28)		■ 1.000 ● N +0	$d\,s^{\{2\}}=h^{\{2\}}\,d\,r^{\{2\}}+f^{\{2\}}\left[d\,\psi^{\{2\}}+\sin^{\{2\}}\,\psi\left(d\,\theta^{\{2\}}+\sin^{\{2\}}\,\theta\,d\phi^{\{2\}}\right)\right],$	$ds^2=h^2dr^2+f^2\left[d\psi^2+\sin^2\psi\left(d\theta^2+\sin^2\theta d\phi^2\right)\right],$	$ds^2=h^2dr^2+f^2\left[d\psi^2+\sin^2\psi\left(d\theta^2+\sin^2\theta d\phi^2\right)\right],$
mielke-geometrodynamics_136 EQ0410 (6.4.30)		■ 1.000 ● N -2	$\ell^2\underline{C}_{\mathrm{TT}}=\frac{1}{2}\underline{\vartheta}^{\alpha}\wedge d\underline{\vartheta}_{\alpha}=\frac{1}{2}\underline{\vartheta}^A\wedge d\underline{\vartheta}_A=3\underline{\vartheta}^{\hat{1}}\wedge\underline{\vartheta}^{\hat{2}}\wedge\underline{\vartheta}^{\hat{3}}$	$\ell^2\underline{C}_{\mathrm{TT}}=\frac{1}{2}\underline{\vartheta}^{\alpha}\wedge d\underline{\vartheta}_{\alpha}=\frac{1}{2}\underline{\vartheta}^A\wedge d\underline{\vartheta}_A=3\underline{\vartheta}^{\hat{1}}\wedge\underline{\vartheta}^{\hat{2}}\wedge\underline{\vartheta}^{\hat{3}}$	$\ell^2\underline{C}_{\mathrm{TT}}=\frac{1}{2}\underline{\vartheta}^{\alpha}\wedge d\underline{\vartheta}_{\alpha}=\frac{1}{2}\underline{\vartheta}^A\wedge d\underline{\vartheta}_A=3\underline{\vartheta}^{\hat{1}}\wedge\underline{\vartheta}^{\hat{2}}\wedge\underline{\vartheta}^{\hat{3}}$
mielke-geometrodynamics_136 EQ0412 (6.4.32)		■ 1.000 ● N +1	$f=\frac{a\,r^{\{2\}}}{r^{\{2\}}+c^{\{2\}}}$	$f=\frac{ar^2}{r^2+c^2}$	$f=\frac{ar^2}{r^2+c^2}$
mielke-geometrodynamics_137 EQ0414 (6.5.2)		■ 1.000 ● N +0	$\Psi(\gamma)=\int_{\mathscr{C}/\mathscr{G}}\mathscr{D}A\,W(A,\gamma)\Psi(A).$	$\Psi(\gamma)=\int_{\mathscr{C}/\mathscr{G}}\mathscr{D}A\,W(A,\gamma)\Psi(A).$	$\Psi(\gamma)=\int_{\mathscr{C}/\mathscr{G}}\mathscr{D}A\,W(A,\gamma)\Psi(A).$
mielke-geometrodynamics_137 EQ0416 (6.5.4)		■ 1.000 ● K +0	$\langle 0 W(\gamma) 0\rangle e^{\Lambda G(\gamma)}V(\gamma),$	$\langle 0 W(\gamma) 0\rangle e^{\Lambda G(\gamma)}V(\gamma),$	$\langle 0 W(\gamma) 0\rangle e^{\Lambda G(\gamma)}V(\gamma),$
mielke-geometrodynamics_138 EQ0417 (6.6.1)		■ 1.000 ● N +1	$\int T=\int d\,\sigma=2\pi\,N$	$\int T=\int d\sigma=2\pi N$	$\int T=\int d\sigma=2\pi N,$
mielke-geometrodynamics_139 EQ0418 (6.6.2)		■ 1.000 ● K +0	$W^{\{N\}}:=S^{\{3\}}\,\backslash\#^{\{N\}}\backslash\left(S^{\{1\}}\,\times S^{\{2\}}\right)\,.$	$W^N:=S^3\#^N\left(S^1\times S^2\right).$	$W^N:=S^3\#^N(S^1\times S^2).$
mielke-geometrodynamics_140 EQ0419 (6.7.1)		■ 1.000 ● K +0	$L:=L_{\mathrm{EC}}+L_{\theta}+L_{\mathrm{D}},$	$L:=L_{\mathrm{EC}}+L_{\theta}+L_{\mathrm{D}},$	$L:=L_{\mathrm{EC}}+L_{\theta}+L_{\mathrm{D}},$
mielke-geometrodynamics_140 EQ0420 (6.7.2)		■ 1.000 ● N -1	$L_{\theta}=\theta_{\mathrm{T}}dC_{\mathrm{TT}}+\theta_{\mathrm{L}}dC_{\mathrm{RR}}$	$L_{\theta}=\theta_{\mathrm{T}}dC_{\mathrm{TT}}+\theta_{\mathrm{L}}dC_{\mathrm{RR}}$	$L_{\theta}=\theta_{\mathrm{T}}dC_{\mathrm{TT}}+\theta_{\mathrm{L}}dC_{\mathrm{RR}}$
mielke-geometrodynamics_140 EQ0421 (6.7.3)		■ 1.000 ● N +0	$H_{\alpha}:= -\frac{\partial L}{\partial T^{\alpha}}=-\frac{\theta_{\mathrm{T}}}{\ell^2}T_{\alpha},$	$H_{\alpha}:= -\frac{\partial L}{\partial T^{\alpha}}=-\frac{\theta_{\mathrm{T}}}{\ell^2}T_{\alpha},$	$H_{\alpha}:= -\frac{\partial L}{\partial T^{\alpha}}=-\frac{\theta_{\mathrm{T}}}{\ell^2}T_{\alpha},$
mielke-geometrodynamics_140 EQ0422 (6.7.4)		■ 1.000 ● W +0	$H_{\alpha\beta}:= -\frac{\partial L}{\partial R^{\alpha\beta}}=\frac{1}{2\kappa}\eta_{\alpha\beta}-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}\wedge\vartheta_{\beta}-\theta_{\mathrm{L}}R_{\alpha\beta}$	$H_{\alpha\beta}:= -\frac{\partial L}{\partial R^{\alpha\beta}}=\frac{1}{2\kappa}\eta_{\alpha\beta}-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}\wedge\vartheta_{\beta}-\theta_{\mathrm{L}}R_{\alpha\beta}$	$H_{\alpha\beta}:= -\frac{\partial L}{\partial R^{\alpha\beta}}=\frac{1}{2\kappa}\eta_{\alpha\beta}-\frac{\theta_{\mathrm{T}}}{2\ell^2}\vartheta_{\alpha}\wedge\vartheta_{\beta}-\theta_{\mathrm{L}}R_{\alpha\beta}$
mielke-geometrodynamics_140 EQ0423 (6.7.5)		■ 1.000 ● N +0	$D\,\theta_{\mathrm{T}}\wedge T_{\alpha}+\theta_{\mathrm{T}}DT_{\alpha}+\ell^2E_{\alpha}=-\ell^2\Sigma_{\alpha}.$	$D\theta_{\mathrm{T}}\wedge T_{\alpha}+\theta_{\mathrm{T}}DT_{\alpha}+\ell^2E_{\alpha}=-\ell^2\Sigma_{\alpha}.$	$D\theta_{\mathrm{T}}\wedge T_{\alpha}+\theta_{\mathrm{T}}DT_{\alpha}+\ell^2E_{\alpha}=-\ell^2\Sigma_{\alpha}.$
mielke-geometrodynamics_140 EQ0425 (6.7.7)		■ 1.000 ●	$C_{\mathrm{TT}}\cong\frac{\kappa}{4\ell^2}j_5$	$C_{\mathrm{TT}}\cong\frac{\kappa}{4\ell^2}j_5$	$C_{\mathrm{TT}}\cong\frac{\kappa}{4\ell^2}j_5$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_141 EQ0427 (6.7.9)	141	■ 1.000 ● N +0	$T^{\gamma} \wedge \eta_{\gamma\beta} + \kappa D\theta_{\mathrm{L}} \wedge D T_{\beta} = 0$.	$T^{\gamma} \wedge \eta_{\gamma\beta} + \kappa D\theta_{\mathrm{L}} \wedge D T_{\beta} = 0$.	$T^{\gamma} \wedge \eta_{\gamma\beta} + \kappa D\theta_{\mathrm{L}} \wedge D T_{\beta} = 0$.
mielke-geometrodynamics_141 EQ0429 (6.7.11)	141	■ 1.000 ● N +0	$\kappa d\theta_{\mathrm{L}} \wedge T_{\beta} = -\eta_{\beta}$	$\kappa d\theta_{\mathrm{L}} \wedge T_{\beta} = -\eta_{\beta}$	$\kappa d\theta_{\mathrm{L}} \wedge T_{\beta} = -\eta_{\beta}$
mielke-geometrodynamics_141 EQ0430 (6.7.12)	141	■ 1.000 ● N -1	$\kappa \ell^2 d\theta_{\mathrm{L}} \wedge C_{\mathrm{TT}} = 2\eta$.	$\kappa \ell^2 d\theta_{\mathrm{L}} \wedge C_{\mathrm{TT}} = 2\eta$.	$\kappa \ell^2 d\theta_{\mathrm{L}} \wedge C_{\mathrm{TT}} = 2\eta$.
mielke-geometrodynamics_141 EQ0431 (6.7.13)	141	■ 1.000 ● N +0	$D T^{\gamma} \wedge \eta_{\alpha\beta\gamma} + \kappa D\theta_{\mathrm{T}} \wedge T_{[\alpha} \wedge \vartheta_{\beta]} / \ell^2 = 2\kappa \left(T_{[\alpha} \wedge \mu_{\beta]} - \vartheta_{[\alpha} \wedge D\mu_{\beta]} \right)$.	$D T^{\gamma} \wedge \eta_{\alpha\beta\gamma} + \kappa D\theta_{\mathrm{T}} \wedge T_{[\alpha} \wedge \vartheta_{\beta]} / \ell^2 = 2\kappa \left(T_{[\alpha} \wedge \mu_{\beta]} - \vartheta_{[\alpha} \wedge D\mu_{\beta]} \right)$.	$D T^{\gamma} \wedge \eta_{\alpha\beta\gamma} + \kappa D\theta_{\mathrm{T}} \wedge T_{[\alpha} \wedge \vartheta_{\beta]} / \ell^2 = 2\kappa \left(T_{[\alpha} \wedge \mu_{\beta]} - \vartheta_{[\alpha} \wedge D\mu_{\beta]} \right)$.
mielke-geometrodynamics_142 EQ0432 (6.8.1)	142	■ 1.000 ● N +0	$W = \int \left[L(\vartheta^{\alpha}, T^{\alpha}, R^{\alpha\beta}, \Psi, D\Psi) + L_g(\vartheta^{\alpha}, T^{\alpha}, R^{\alpha\beta}) \right]$.	$W = \int \left[L(\vartheta^{\alpha}, T^{\alpha}, R^{\alpha\beta}, \Psi, D\Psi) + L_g(\vartheta^{\alpha}, T^{\alpha}, R^{\alpha\beta}) \right]$.	$W = \int \left[L(\vartheta^{\alpha}, T^{\alpha}, R^{\alpha\beta}, \Psi, D\Psi) + L_g(\vartheta^{\alpha}, T^{\alpha}, R^{\alpha\beta}) \right]$.
mielke-geometrodynamics_142 EQ0433 (6.8.3)	142	■ 1.000 ● C -3	$\begin{gathered} \frac{\delta L}{\partial L} = \frac{\delta L}{\partial \Psi} - (-1)^p D \frac{\partial L}{\partial D\Psi} = 0, \quad (\text{MATTER}) \\ DH_{\alpha} - E_{\alpha} = \Sigma_{\alpha}, \quad (\text{FIRST}) \\ DH_{\alpha\beta} - E_{\alpha\beta} = \tau_{\alpha\beta}. \quad (\text{SECOND}) \end{gathered}$	$\frac{\delta L}{\partial \Psi} = \frac{\partial L}{\partial \Psi} - (-1)^p D \frac{\partial L}{\partial D\Psi} = 0, \quad (\text{MATTER}) \\ DH_{\alpha} - E_{\alpha} = \Sigma_{\alpha}, \quad (\text{FIRST}) \\ DH_{\alpha\beta} - E_{\alpha\beta} = \tau_{\alpha\beta}. \quad (\text{SECOND})$	$\frac{\delta L}{\delta \Psi} = \frac{\partial L}{\partial \Psi} - (-1)^p D \frac{\partial L}{\partial D\Psi} = 0, \quad (\text{MATTER}) \\ DH_{\alpha} - E_{\alpha} = \Sigma_{\alpha}, \quad (\text{FIRST}) \\ DH_{\alpha\beta} - E_{\alpha\beta} = \tau_{\alpha\beta}. \quad (\text{SECOND})$
mielke-geometrodynamics_142 EQ0435 (6.8.6)	142	■ 1.000 ● W +1	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = \frac{\partial L}{\partial \vartheta^{\alpha}} + D \frac{\partial L}{\partial T^{\alpha}}$	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = \frac{\partial L}{\partial \vartheta^{\alpha}} + D \frac{\partial L}{\partial T^{\alpha}}$	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = \frac{\partial L}{\partial \vartheta^{\alpha}} + D \frac{\partial L}{\partial T^{\alpha}}$
mielke-geometrodynamics_142 EQ0436 (6.8.7)	142	■ 1.000 ● W +0	$\tau_{\alpha\beta} := \frac{\delta L}{\delta \Gamma^{\alpha\beta}} = \rho(I_{\alpha\beta}) \Psi \wedge \frac{\partial L}{\partial (D\Psi)} + \vartheta_{[\alpha} \wedge \frac{\partial L}{\partial T^{\beta]} + D \frac{\partial L}{\partial R^{\alpha\beta}},$	$\tau_{\alpha\beta} := \frac{\delta L}{\delta \Gamma^{\alpha\beta}} = \rho(I_{\alpha\beta}) \Psi \wedge \frac{\partial L}{\partial (D\Psi)} + \vartheta_{[\alpha} \wedge \frac{\partial L}{\partial T^{\beta]} + D \frac{\partial L}{\partial R^{\alpha\beta}},$	$\tau_{\alpha\beta} := \frac{\delta L}{\delta \Gamma^{\alpha\beta}} = \rho(I_{\alpha\beta}) \Psi \wedge \frac{\partial L}{\partial (D\Psi)} + \vartheta_{[\alpha} \wedge \frac{\partial L}{\partial T^{\beta]} + D \frac{\partial L}{\partial R^{\alpha\beta}},$
mielke-geometrodynamics_142 EQ0438 (6.8.9)	142	■ 1.000 ● N +0	$E_{\alpha\beta} := -\vartheta_{[\alpha} \wedge H_{\beta]}$	$E_{\alpha\beta} := -\vartheta_{[\alpha} \wedge H_{\beta]}$	$E_{\alpha\beta} := -\vartheta_{[\alpha} \wedge H_{\beta]}$
mielke-geometrodynamics_143 EQ0439 (6.8.10)	143	■ 1.000 ● N +0	$\vartheta^{\alpha} \wedge E_{\alpha} = 4L_g + 2T^{\alpha} \wedge H_{\alpha} + 2R^{\alpha\beta} \wedge H_{\alpha\beta}$.	$\vartheta^{\alpha} \wedge E_{\alpha} = 4L_g + 2T^{\alpha} \wedge H_{\alpha} + 2R^{\alpha\beta} \wedge H_{\alpha\beta}$.	$\vartheta^{\alpha} \wedge E_{\alpha} = 4L_g + 2T^{\alpha} \wedge H_{\alpha} + 2R^{\alpha\beta} \wedge H_{\alpha\beta}$.
mielke-geometrodynamics_143 EQ0442 (6.8.13)	143	■ 1.000 ● N +0	$L_{\mathrm{qPG}} = -\frac{1}{2} T^{\alpha} \wedge H_{\alpha} - \frac{1}{2} R^{\alpha\beta} \wedge \left(H_{\alpha\beta} - \frac{a_0}{2\ell^2} \eta_{\alpha\beta} \right),$	$L_{\mathrm{qPG}} = -\frac{1}{2} T^{\alpha} \wedge H_{\alpha} - \frac{1}{2} R^{\alpha\beta} \wedge \left(H_{\alpha\beta} - \frac{a_0}{2\ell^2} \eta_{\alpha\beta} \right),$	$L_{\mathrm{qPG}} = -\frac{1}{2} T^{\alpha} \wedge H_{\alpha} - \frac{1}{2} R^{\alpha\beta} \wedge \left(H_{\alpha\beta} - \frac{a_0}{2\ell^2} \eta_{\alpha\beta} \right),$
mielke-geometrodynamics_143 EQ0444 (6.8.15)	143	■ 1.000 ● N +0	$H_{\alpha\beta} = -\frac{a_0}{2\ell^2} \eta_{\alpha\beta} - \frac{1}{\kappa} * \left(\sum_{j=1}^6 b_j^{(j)} R_{\alpha\beta} \right).$	$H_{\alpha\beta} = -\frac{a_0}{2\ell^2} \eta_{\alpha\beta} - \frac{1}{\kappa} * \left(\sum_{j=1}^6 b_j^{(j)} R_{\alpha\beta} \right).$	$H_{\alpha\beta} = -\frac{a_0}{2\ell^2} \eta_{\alpha\beta} - \frac{1}{\kappa} * \left(\sum_{j=1}^6 b_j^{(j)} R_{\alpha\beta} \right).$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_144 EQ0447 (6.9.3)	144	■ 1.000	$\eta=\{ \}^{\ast} 1=\frac{1}{4!} \eta_{\alpha \beta \gamma \delta} \vartheta^{\alpha} \wedge \vartheta^{\beta} \wedge \vartheta^{\gamma} \wedge \vartheta^{\delta},$ $\wedge \vartheta^{\alpha} \wedge \vartheta^{\beta} \wedge \vartheta^{\gamma} \wedge \vartheta^{\delta},$	$\eta=\ast 1=\frac{1}{4!} \eta_{\alpha \beta \gamma \delta} \vartheta^{\alpha} \wedge \vartheta^{\beta} \wedge \vartheta^{\gamma} \wedge \vartheta^{\delta},$	$\eta=\ast 1=\frac{1}{4!} \eta_{\alpha \beta \gamma \delta} \vartheta^{\alpha} \wedge \vartheta^{\beta} \wedge \vartheta^{\gamma} \wedge \vartheta^{\delta},$
mielke-geometrodynamics_145 EQ0448 (6.9.4)	145	■ 1.000 N +0	$p=\frac{1}{8 \pi^2} \int R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}.$ $R_{\beta}^{\alpha} \wedge R_{\alpha}^{\beta}.$	$p=\frac{1}{8 \pi^2} \int R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}.$	$p=\frac{1}{8 \pi^2} \int R_{\alpha}^{\beta} \wedge R_{\beta}^{\alpha}.$
mielke-geometrodynamics_151 EQ0453 (7.2.1)	151	■ 1.000 N +0	$\left[L_{a b}^{\left(\star\right)}:=\frac{1}{2} \varepsilon_{a b c d} L^{c d}=L_{a b}\right]$ $L^{c d}=L_{a b} \quad . \quad]$	$L_{a b}^{\left(\star\right)}:=\frac{1}{2} \varepsilon_{a b c d} L^{c d}=L_{a b}.$	$L_{a b}^{\left(\star\right)}:=\frac{1}{2} \varepsilon_{a b c d} L^{c d}=L_{a b}.$
mielke-geometrodynamics_152 EQ0454 (7.2.6)	152	■ 1.000 N +0	$\Omega^g=\Omega_{\mathrm{C}}+\bar{\Omega}^g-\frac{1}{6} \vartheta \wedge \vartheta R.$ $\frac{1}{6} \vartheta \wedge \vartheta R.$	$\Omega^g=\Omega_C+\bar{\Omega}^g-\frac{1}{6} \vartheta \wedge \vartheta R.$	$\Omega^g=\Omega_C+\bar{\Omega}^g-\frac{1}{6} \vartheta \wedge \vartheta R.$
mielke-geometrodynamics_152 EQ0455 (7.2.7)	152	■ 1.000 N +0	$\bar{\Omega}^g=\vartheta \wedge^{\ast}\left(\vartheta \wedge \Omega^g\right)-\frac{1}{4} \vartheta \wedge \vartheta R,$ $\vartheta \wedge^{\ast}\left(\vartheta \wedge \Omega^g\right)-\frac{1}{4} \vartheta \wedge \vartheta R,$	$\bar{\Omega}^g=\vartheta \wedge^{\ast}\left(\vartheta \wedge \Omega^g\right)-\frac{1}{4} \vartheta \wedge \vartheta R,$	$\bar{\Omega}^g=\vartheta \wedge^{\ast}\left(\vartheta \wedge \Omega^g\right)-\frac{1}{4} \vartheta \wedge \vartheta R,$
mielke-geometrodynamics_152 EQ0457 (7.2.9)	152	■ 1.000 N +0	$\operatorname{Tr}\left(\vartheta \wedge \vartheta \wedge \Omega_C\right)=\operatorname{Tr}\left(\vartheta \wedge \vartheta \wedge \bar{\Omega}^g\right)=0$ $\operatorname{Tr}\left(\vartheta \wedge \vartheta \wedge \Omega_C\right)=\operatorname{Tr}\left(\vartheta \wedge \vartheta \wedge \bar{\Omega}^g\right)=0$	$\operatorname{Tr}\left(\vartheta \wedge \vartheta \wedge \Omega_C\right)=\operatorname{Tr}\left(\vartheta \wedge \vartheta \wedge \bar{\Omega}^g\right)=0$	$\operatorname{Tr}\left(\vartheta \wedge \vartheta \wedge \Omega_C\right)=\operatorname{Tr}\left(\vartheta \wedge \vartheta \wedge \bar{\Omega}^g\right)=0$
mielke-geometrodynamics_153 EQ0463 (7.3.2)	153	■ 1.000 N +0	$D^{\ast} \Omega^g=0,$	$D^{\ast} \Omega^g=0,$	$D^{\ast} \Omega^g=0,$
mielke-geometrodynamics_153 EQ0464 (7.3.3)	153	■ 1.000 K +0	$D^{\ast} R_{\alpha \beta}=0,$	$D^{\ast} R_{\alpha \beta}=0,$	$D^{\ast} R_{\alpha \beta}=0,$
mielke-geometrodynamics_154 EQ0468 (7.3.7)	154	■ 1.000 N +0	$R_{\alpha \beta}=-\ast R_{\alpha \beta}^{\left(\star\right)},$	$R_{\alpha \beta}=-\ast R_{\alpha \beta}^{\left(\star\right)},$	$R_{\alpha \beta}=-\ast R_{\alpha \beta}^{\left(\star\right)},$
mielke-geometrodynamics_154 EQ0469 (7.3.8)	154	■ 1.000 N +0	$R_{\alpha \beta}=\ast R_{\alpha \beta}^{\left(\star\right)},$	$R_{\alpha \beta}=\ast R_{\alpha \beta}^{\left(\star\right)},$	$R_{\alpha \beta}=\ast R_{\alpha \beta}^{\left(\star\right)},$
mielke-geometrodynamics_157 EQ0479 (7.4.7)	157	■ 1.000 K +0	$G_{\alpha}-\Lambda \eta_{\alpha}=0$	$G_{\alpha}-\Lambda \eta_{\alpha}=0$	$G_{\alpha}-\Lambda \eta_{\alpha}=0$
mielke-geometrodynamics_157 EQ0480 (7.5.1)	157	■ 1.000 N -1	$\square^c:=\square-\frac{1}{6} R, \quad \square:=(1 / \sqrt{g}) \partial_i\left(\sqrt{g} g^{i j} \partial_j\right) .$ $\square^c:=\square-\frac{1}{6} R, \quad \square:=(1 / \sqrt{g}) \partial_i\left(\sqrt{g} g^{i j} \partial_j\right) .$	$\square^c:=\square-\frac{1}{6} R, \quad \square:=(1 / \sqrt{g}) \partial_i\left(\sqrt{g} g^{i j} \partial_j\right) .$	$\square^c:=\square-\frac{1}{6} R, \quad \square:=(1 / \sqrt{g}) \partial_i\left(\sqrt{g} g^{i j} \partial_j\right) .$
mielke-geometrodynamics_157 EQ0482 (7.5.3)	157	■ 1.000 N +0	$\left(\square-\frac{2}{3} \Lambda\right) h_{i j}=0$	$\left(\square-\frac{2}{3} \Lambda\right) h_{i j}=0$	$\left(\square-\frac{2}{3} \Lambda\right) h_{i j}=0$
mielke-geometrodynamics_158 EQ0483 (7.5.4)	158	■ 1.000 N +0	$\partial^i h_{i j}-\partial_j h=0$	$\partial^i h_{i j}-\partial_j h=0$	$\partial^i h_{i j}-\partial_j h=0$
mielke-geometrodynamics_158 EQ0484 (7.5.5)	158	■ 1.000 K +0	$\Delta m=\sqrt{2 \Lambda / 3}>0$	$\Delta m=\sqrt{2 \Lambda / 3}>0$	$\Delta m=\sqrt{2 \Lambda / 3}>0$
mielke-geometrodynamics_160 EQ0488 (7.7.5)	160	■ 1.000 K +0	$L_{\mathrm{EC}}=-\frac{1}{2 \kappa^2} R^{\alpha \beta} \wedge \eta_{\alpha \beta}$ $\wedge \eta_{\alpha \beta}$	$L_{\mathrm{EC}}=-\frac{1}{2 \kappa^2} R^{\alpha \beta} \wedge \eta_{\alpha \beta}$	$L_{\mathrm{EC}}=-\frac{1}{2 \kappa^2} R^{\alpha \beta} \wedge \eta_{\alpha \beta}$
mielke-geometrodynamics_160 EQ0489 (7.7.6)	160	■ 1.000 N +0	$\widetilde{L}=-\sum_{k=0}^K(1 / 2 k) R^{\alpha \beta} \wedge H_{\alpha \beta}^{(2 k)}+L_{\theta},$ $\wedge H_{\alpha \beta}^{(2 k)}+L_{\theta},$	$\widetilde{L}=-\sum_{k=0}^K(1 / 2 k) R^{\alpha \beta} \wedge H_{\alpha \beta}^{(2 k)}+L_{\theta},$	$\widetilde{L}=-\sum_{k=0}^K(1 / 2 k) R^{\alpha \beta} \wedge H_{\alpha \beta}^{(2 k)}+L_{\theta},$

Conf. MathPix confidence:

■

≥0.9

■

0.5-0.9

■

<0.5

—

no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E08491 (7.7.9)	161	■ 1.000 ● K +0	$\widetilde{L} \rightarrow L = -\frac{1}{2} \left(R^{\alpha\beta} \wedge \widetilde{H}_{\alpha\beta} - \widetilde{L} \right)$	$\widetilde{L} \rightarrow L = -\frac{1}{2} \left(R^{\alpha\beta} \wedge \widetilde{H}_{\alpha\beta} - \widetilde{L} \right)$	$\widetilde{L} \rightarrow L = -\frac{1}{2} \left(R^{\alpha\beta} \wedge \widetilde{H}_{\alpha\beta} - \widetilde{L} \right)$
mielke-geometrodynamics_E08493 (7.7.11)	161	■ 1.000 ●	$\widetilde{H}_{\alpha\beta\mu\nu} := \frac{\partial^2 \widetilde{L}}{\partial R^{\mu\nu} \partial R^{\alpha\beta}} = -\frac{\partial \widetilde{H}_{\mu\nu}}{\partial R^{\alpha\beta}}$	$\widetilde{H}_{\alpha\beta\mu\nu} := \frac{\partial^2 \widetilde{L}}{\partial R^{\mu\nu} \partial R^{\alpha\beta}} = -\frac{\partial \widetilde{H}_{\mu\nu}}{\partial R^{\alpha\beta}}$	$\widetilde{H}_{\alpha\beta\mu\nu} := \frac{\partial^2 \widetilde{L}}{\partial R^{\mu\nu} \partial R^{\alpha\beta}} = -\frac{\partial \widetilde{H}_{\mu\nu}}{\partial R^{\alpha\beta}}$
mielke-geometrodynamics_E08494 (7.7.12)	161	■ 1.000 ● W -1	$\widetilde{\eta}_{\alpha\beta} := \frac{\theta_{\mathrm{T}}^*}{2} \eta_{\alpha\beta} - \frac{\theta_{\mathrm{T}}}{2} \vartheta_{\alpha} \wedge \vartheta_{\beta} - \ell^2 \widetilde{H}_{\alpha\beta} = 0$	$\widetilde{\eta}_{\alpha\beta} := \frac{\theta_{\mathrm{T}}^*}{2} \eta_{\alpha\beta} - \frac{\theta_{\mathrm{T}}}{2} \vartheta_{\alpha} \wedge \vartheta_{\beta} - \ell^2 \widetilde{H}_{\alpha\beta} = 0$	$\widetilde{\eta}_{\alpha\beta} := \frac{\theta_{\mathrm{T}}^*}{2} \eta_{\alpha\beta} - \frac{\theta_{\mathrm{T}}}{2} \vartheta_{\alpha} \wedge \vartheta_{\beta} - \ell^2 \widetilde{H}_{\alpha\beta} = 0$
mielke-geometrodynamics_E08496 (7.7.14)	162	■ 1.000 ● K +0	$\eta := \eta_{\alpha\beta\gamma\delta} \vartheta^{\alpha} \wedge \vartheta^{\beta} \wedge \vartheta^{\gamma} \wedge \vartheta^{\delta} / 4!$	$\eta := \eta_{\alpha\beta\gamma\delta} \vartheta^{\alpha} \wedge \vartheta^{\beta} \wedge \vartheta^{\gamma} \wedge \vartheta^{\delta} / 4!$	$\eta := \eta_{\alpha\beta\gamma\delta} \vartheta^{\alpha} \wedge \vartheta^{\beta} \wedge \vartheta^{\gamma} \wedge \vartheta^{\delta} / 4!$
mielke-geometrodynamics_E08499 (7.7.17)	163	■ 1.000 ● K +0	$\gamma_{\alpha} \gamma_{\beta} + \gamma_{\beta} \gamma_{\alpha} = 2 o_{\alpha\beta} \mathbf{1}_4.$	$\gamma_{\alpha} \gamma_{\beta} + \gamma_{\beta} \gamma_{\alpha} = 2 o_{\alpha\beta} \mathbf{1}_4.$	$\gamma_{\alpha} \gamma_{\beta} + \gamma_{\beta} \gamma_{\alpha} = 2 o_{\alpha\beta} \mathbf{1}_4.$
mielke-geometrodynamics_E08501 (7.7.19)	163	■ 1.000 ● K +0	$\gamma := \gamma^{\alpha} \vartheta_{\alpha}$	$\gamma := \gamma^{\alpha} \vartheta_{\alpha}$	$\gamma := \gamma^{\alpha} \vartheta_{\alpha}$
mielke-geometrodynamics_E08502 (7.7.20)	163	■ 1.000 ● N +0	${}^* \gamma = \gamma^{\alpha} \eta_{\alpha} = \frac{i}{6} \gamma_5 \gamma \wedge \gamma \wedge \gamma,$	${}^* \gamma = \gamma^{\alpha} \eta_{\alpha} = \frac{i}{6} \gamma_5 \gamma \wedge \gamma \wedge \gamma,$	${}^* \gamma = \gamma^{\alpha} \eta_{\alpha} = \frac{i}{6} \gamma_5 \gamma \wedge \gamma \wedge \gamma,$
mielke-geometrodynamics_E08503 (7.7.21)	164	■ 1.000 ● N +1	$\sigma := \frac{1}{2} \sigma_{\alpha\beta} \vartheta^{\alpha} \wedge \vartheta^{\beta} = \frac{i}{2} \gamma \wedge \gamma, \quad {}^* \sigma = \frac{1}{2} \sigma_{\alpha\beta} \eta^{\alpha\beta} =: \sigma^{(*)} = i \gamma_5 \sigma.$	$\sigma := \frac{1}{2} \sigma_{\alpha\beta} \vartheta^{\alpha} \wedge \vartheta^{\beta} = \frac{i}{2} \gamma \wedge \gamma, \quad {}^* \sigma = \frac{1}{2} \sigma_{\alpha\beta} \eta^{\alpha\beta} =: \sigma^{(*)} = i \gamma_5 \sigma.$	$\sigma := \frac{1}{2} \sigma_{\alpha\beta} \vartheta^{\alpha} \wedge \vartheta^{\beta} = \frac{i}{2} \gamma \wedge \gamma, \quad {}^* \sigma = \frac{1}{2} \sigma_{\alpha\beta} \eta^{\alpha\beta} =: \sigma^{(*)} = i \gamma_5 \sigma.$
mielke-geometrodynamics_E08504 (7.7.22)	164	■ 1.000 ● N +0	$\sigma^{(*)} := i \gamma_5 \sigma, \quad \sigma^{(**)} = i^2 \gamma_5^2 \sigma = -\sigma.$	$\sigma^{(*)} := i \gamma_5 \sigma, \quad \sigma^{(**)} = i^2 \gamma_5^2 \sigma = -\sigma.$	$\sigma^{(*)} := i \gamma_5 \sigma, \quad \sigma^{(**)} = i^2 \gamma_5^2 \sigma = -\sigma.$
mielke-geometrodynamics_E08505 (7.7.23)	164	■ 1.000 ● N +0	$[\gamma, \sigma^{(*)}] = \gamma \wedge i \gamma_5 \sigma - i \gamma_5 \sigma \wedge \gamma = -i \gamma_5 (\gamma \wedge \sigma + \sigma \wedge \gamma) = 2 \gamma_5 \gamma \wedge \gamma \wedge \gamma$	$[\gamma, \sigma^{(*)}] = \gamma \wedge i \gamma_5 \sigma - i \gamma_5 \sigma \wedge \gamma = -i \gamma_5 (\gamma \wedge \sigma + \sigma \wedge \gamma) = 2 \gamma_5 \gamma \wedge \gamma \wedge \gamma$	$[\gamma, \sigma^{(*)}] = \gamma \wedge i \gamma_5 \sigma - i \gamma_5 \sigma \wedge \gamma = -i \gamma_5 (\gamma \wedge \sigma + \sigma \wedge \gamma) = 2 \gamma_5 \gamma \wedge \gamma \wedge \gamma$
mielke-geometrodynamics_E08508 (7.7.26)	165	■ 1.000 ● N +0	$\gamma \rightarrow \gamma^{\theta} = e^{i \gamma^5 \theta} \gamma e^{-i \gamma^5 \theta}$	$\gamma \rightarrow \gamma^{\theta} = e^{i \gamma^5 \theta} \gamma e^{-i \gamma^5 \theta}$	$\gamma \rightarrow \gamma^{\theta} = e^{i \gamma^5 \theta} \gamma e^{-i \gamma^5 \theta}$
mielke-geometrodynamics_E08510 (7.7.28)	165	■ 1.000 ● K +0	$\gamma := \vartheta^{\alpha} \gamma_{\alpha}, \quad \Gamma := \frac{i}{4} \Gamma^{\alpha\beta} \sigma_{\alpha\beta} = \Gamma^{\{\}} - K,$	$\gamma := \vartheta^{\alpha} \gamma_{\alpha}, \quad \Gamma := \frac{i}{4} \Gamma^{\alpha\beta} \sigma_{\alpha\beta} = \Gamma^{\{\}} - K,$	$\gamma := \vartheta^{\alpha} \gamma_{\alpha}, \quad \Gamma := \frac{i}{4} \Gamma^{\alpha\beta} \sigma_{\alpha\beta} = \Gamma^{\{\}} - K,$
mielke-geometrodynamics_E08513 (7.7.31)	166	■ 1.000 ● N +0	$\Omega^{(*)} = \frac{1}{4} R_{\alpha\beta}^{(*)} \sigma^{\alpha\beta}, \quad R_{\alpha\beta}^{(*)} := \frac{1}{2} \eta_{\alpha\beta\gamma\delta} R^{\gamma\delta}.$	$\Omega^{(*)} = \frac{1}{4} R_{\alpha\beta}^{(*)} \sigma^{\alpha\beta}, \quad R_{\alpha\beta}^{(*)} := \frac{1}{2} \eta_{\alpha\beta\gamma\delta} R^{\gamma\delta}.$	$\Omega^{(*)} = \frac{1}{4} R_{\alpha\beta}^{(*)} \sigma^{\alpha\beta}, \quad R_{\alpha\beta}^{(*)} := \frac{1}{2} \eta_{\alpha\beta\gamma\delta} R^{\gamma\delta}.$
mielke-geometrodynamics_E08514 (7.7.32)	166	■ 1.000 ● K +0	$D \sigma = \frac{i}{2} [\Theta, \gamma]$	$D \sigma = \frac{i}{2} [\Theta, \gamma]$	$D \sigma = \frac{i}{2} [\Theta, \gamma]$
mielke-geometrodynamics_E08517 (7.7.35)	166	■ 1.000 ● N +0	$d^* \gamma = 0, \quad d^* \Gamma = 0$	$d^* \gamma = 0, \quad d^* \Gamma = 0$	$d^* \gamma = 0, \quad d^* \Gamma = 0$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E09518 (7.7.36)	166	■ 1.000 ● N +0	$D \ D \ \Phi=\left[\Omega^g, \ \Phi\right],$	$DD\Phi = [\Omega^g, \Phi],$	$DD\Phi = [\Omega^g, \Phi],$
mielke-geometrodynamics_E09520 (7.7.38)	166	■ 1.000 ●	$G:=G_{\alpha}\gamma^{\alpha}\wedge\eta_{\mu\nu\lambda}\gamma^{\lambda}=-i\gamma_5\left(\Omega^g\wedge\gamma+\gamma\wedge\Omega^g\right)=i\left[\gamma,\gamma_5\Omega^g\right].$	$G:=G_{\alpha}\gamma^{\alpha}:=\frac{1}{2}R^{\mu\nu}\wedge\eta_{\mu\nu\lambda}\gamma^{\lambda}=-i\gamma_5\left(\Omega^g\wedge\gamma+\gamma\wedge\Omega^g\right)=i\left[\gamma,\gamma_5\Omega^g\right].$	$G:=G_{\alpha}\gamma^{\alpha}:=\frac{1}{2}R^{\mu\nu}\wedge\eta_{\mu\nu\lambda}\gamma^{\lambda}=-i\gamma_5(\Omega^g\wedge\gamma+\gamma\wedge\Omega^g)=i[\gamma,\gamma_5\Omega^g].$
mielke-geometrodynamics_E09521 (7.7.39)	167	■ 1.000 ● W +0	$D[\gamma,\Theta]\equiv 2i\left[\sigma,\Omega^g\right],\quad D\left[\gamma,\Omega^g\right]\equiv\left[\Theta,\Omega^g\right],\quad DG\equiv i\left[\Theta,\gamma_5\Omega^g\right],$	$D[\gamma,\Theta]\equiv 2i\left[\sigma,\Omega^g\right],\quad D\left[\gamma,\Omega^g\right]\equiv\left[\Theta,\Omega^g\right],\quad DG\equiv i\left[\Theta,\gamma_5\Omega^g\right],$	$D[\gamma,\Theta]\equiv 2i[\sigma,\Omega^g],\quad D[\gamma,\Omega^g]\equiv[\Theta,\Omega^g],\quad DG\equiv i[\Theta,\gamma_5\Omega^g],$
mielke-geometrodynamics_E09523 (7.7.42)	167	■ 1.000 ● N +0	$D\ R_{\alpha\beta}^{\left(\star\right)}\equiv 0,$	$DR_{\alpha\beta}^{(\star)}\equiv 0,$	$DR_{\alpha\beta}^{(\star)}\equiv 0,$
mielke-geometrodynamics_E09532 (7.7.52)	169	■ 1.000 ● K +0	$\left\{ \right\} ^{\left(6\right)}R^{\alpha\beta}=-\frac{1}{12}\ R\vartheta^{\alpha}\wedge\vartheta^{\beta}$	$^{(6)}R^{\alpha\beta}=-\frac{1}{12}R\vartheta^{\alpha}\wedge\vartheta^{\beta}$	$^{(6)}R^{\alpha\beta}=-\frac{1}{12}R\vartheta^{\alpha}\wedge\vartheta^{\beta}$
mielke-geometrodynamics_E09541 (8.3.2)	177	■ 1.000 ● N +0	$s\ \gamma=\psi-[C,\gamma]$	$s\gamma=\psi-[C,\gamma]$	$s\gamma=\psi-[C,\gamma]$
mielke-geometrodynamics_E09542 (8.3.3)	177	■ 1.000 ● K +0	$s\ \sigma=\frac{i}{2}\ s(\gamma\wedge\gamma)=\frac{i}{2}([\psi,\gamma]-[[C,\gamma],\gamma])=\frac{i}{2}[\psi,\gamma]-[C,\sigma].$	$s\sigma:=\frac{i}{2}s(\gamma\wedge\gamma)=\frac{i}{2}([\psi,\gamma]-[[C,\gamma],\gamma])=\frac{i}{2}[\psi,\gamma]-[C,\sigma].$	$s\sigma:=\frac{i}{2}s(\gamma\wedge\gamma)=\frac{i}{2}([\psi,\gamma]-[[C,\gamma],\gamma])=\frac{i}{2}[\psi,\gamma]-[C,\sigma].$
mielke-geometrodynamics_E09543 (8.3.4)	177	■ 1.000 ● N +0	$s\ \bar{c}=b,\quad s\ b=0\ .$	$s\bar{c}=b,\quad sb=0.$	$s\bar{c}=b,\quad sb=0.$
mielke-geometrodynamics_E09546 (8.3.7)	178	■ 1.000 ● N +0	$d\ s^2=\frac{1}{4}\operatorname{Tr}\{\gamma\otimes\gamma\}^*\circ_{\alpha\beta}D\xi^{\alpha}\otimes D\xi^{\beta}$	$ds^2=\frac{1}{4}\operatorname{Tr}\{\gamma\otimes\gamma\}^*\circ_{\alpha\beta}D\xi^{\alpha}\otimes D\xi^{\beta}$	$ds^2=\frac{1}{4}\operatorname{Tr}\{\gamma\otimes\gamma\}^*\circ_{\alpha\beta}D\xi^{\alpha}\otimes D\xi^{\beta}$
mielke-geometrodynamics_E09550 (8.4.4)	179	■ 1.000 ● N -1	$L_{\mathrm{TG}}=dC_{\mathrm{RR}}+L_{\mathrm{FP}}=dC_{\mathrm{RR}}+s\{\cdots\}$	$L_{\mathrm{TG}}=dC_{\mathrm{RR}}+L_{\mathrm{FP}}=dC_{\mathrm{RR}}+s\{\cdots\}$	$L_{\mathrm{TG}}=dC_{\mathrm{RR}}+L_{\mathrm{FP}}=dC_{\mathrm{RR}}+s\{\cdots\}$
mielke-geometrodynamics_E09559 (8.5.2)	181	■ 1.000 ● N +0	$\sigma:=\frac{i}{2}\gamma\wedge\gamma.$	$\sigma:=\frac{i}{2}\gamma\wedge\gamma.$	$\sigma:=\frac{i}{2}\gamma\wedge\gamma.$
mielke-geometrodynamics_E09563 (8.5.6)	182	■ 1.000 ● N +0	$\Omega^{g(\theta)}:=\sqrt{2}\left(\Omega^g\sin\theta+{}^*\Omega^{g(\star)}\cos\theta\right)$	$\Omega^{g(\theta)}:=\sqrt{2}\left(\Omega^g\sin\theta+{}^*\Omega^{g(\star)}\cos\theta\right)$	$\Omega^{g(\theta)}:=\sqrt{2}\left(\Omega^g\sin\theta+{}^*\Omega^{g(\star)}\cos\theta\right)$
mielke-geometrodynamics_E09564 (8.6.1)	183	■ 1.000 ● N +0	$L_{\mathrm{qPG}}=\frac{\Lambda}{\ell^2}\eta-\frac{a_0}{4\ell^2}R^{\alpha\beta}\wedge\eta_{\alpha\beta}-\frac{1}{2}T^{\alpha}\wedge H_{\alpha}-\frac{1}{2}R^{\alpha\beta}\wedge H_{\alpha\beta}.$	$L_{\mathrm{qPG}}=\frac{\Lambda}{\ell^2}\eta-\frac{a_0}{4\ell^2}R^{\alpha\beta}\wedge\eta_{\alpha\beta}-\frac{1}{2}T^{\alpha}\wedge H_{\alpha}-\frac{1}{2}R^{\alpha\beta}\wedge H_{\alpha\beta}.$	$L_{\mathrm{qPG}}=\frac{\Lambda}{\ell^2}\eta-\frac{a_0}{4\ell^2}R^{\alpha\beta}\wedge\eta_{\alpha\beta}-\frac{1}{2}T^{\alpha}\wedge H_{\alpha}-\frac{1}{2}R^{\alpha\beta}\wedge H_{\alpha\beta}.$
mielke-geometrodynamics_E09565 (8.6.2)	183	■ 1.000 ● N +1	$H_{\alpha}:= -\partial L_{\mathrm{qPG}}/\partial T^{\alpha}=\frac{1}{\ell^2}\ast\left(\sum_{M=1}^3a_{(M)}^{(M)}T_{\alpha}\right),$	$H_{\alpha}:= -\partial L_{\mathrm{qPG}}/\partial T^{\alpha}=\frac{1}{\ell^2}\ast\left(\sum_{M=1}^3a_{(M)}^{(M)}T_{\alpha}\right),$	$H_{\alpha}:= -\partial L_{\mathrm{qPG}}/\partial T^{\alpha}=\frac{1}{\ell^2}\ast\left(\sum_{M=1}^3a_{(M)}^{(M)}T_{\alpha}\right),$
mielke-geometrodynamics_E09569 (8.6.7)	184	■ 1.000 ● N +0	$\begin{aligned} D\,H_{\alpha}-E_{\alpha} \,\&= \Sigma_{\alpha} \\ D\,H_{\alpha\beta}+\vartheta_{[\alpha}\wedge H_{\beta]} \,\&= \tau_{\alpha\beta}, \end{aligned}$	$\begin{aligned} DH_{\alpha}-E_{\alpha} &= \Sigma_{\alpha}, \\ DH_{\alpha\beta}+\vartheta_{[\alpha}\wedge H_{\beta]} &= \tau_{\alpha\beta}, \end{aligned}$	$\begin{aligned} DH_{\alpha}-E_{\alpha} &= \Sigma_{\alpha}, \\ DH_{\alpha\beta}+\vartheta_{[\alpha}\wedge H_{\beta]} &= \tau_{\alpha\beta}, \end{aligned}$

Conf. MathPix confidence:

■

≥0.9

■

0.5–0.9

■

<0.5

—

no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ
Residual render vs scan (inkdrill):

●

C component

●

W weak

●

S stable

●

N noise

●

K clean

●

not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_184 EQ0571 (8.6.9)	184	■ 1.000 ● W -1	$\frac{\theta_{\mathrm{T}}^*}{2\ell^2}\eta_{\alpha\beta\gamma}T^\gamma+\frac{\theta_{\mathrm{T}}}{\ell^2}\vartheta_{[\alpha}\wedge T_{\beta]}+\vartheta_{[\alpha}\wedge H_{\beta]}(**)=\tau_{\alpha\beta}.$	$\frac{\theta_{\mathrm{T}}^*}{2\ell^2}\eta_{\alpha\beta\gamma}T^\gamma+\frac{\theta_{\mathrm{T}}}{\ell^2}\vartheta_{[\alpha}\wedge T_{\beta]}+\vartheta_{[\alpha}\wedge H_{\beta]}(**)=\tau_{\alpha\beta}.$	$\frac{\theta_{\mathrm{T}}^*}{2\ell^2}\eta_{\alpha\beta\gamma}T^\gamma+\frac{\theta_{\mathrm{T}}}{\ell^2}\vartheta_{[\alpha}\wedge T_{\beta]}+\vartheta_{[\alpha}\wedge H_{\beta]}(**)=\tau_{\alpha\beta}.$
mielke-geometrodynamics_185 EQ0576 (8.6.14)	185	■ 1.000 ●	$\Lambda_{\mathrm{eff}}=\frac{1}{\theta_{\mathrm{T}}^*}\left[\Lambda-\frac{1}{4}(\theta_{\mathrm{T}}^*-a_0)R\right]=\frac{\Lambda}{\theta_{\mathrm{T}}^*}-\Lambda_\theta$	$\Lambda_{\mathrm{eff}}=\frac{1}{\theta_{\mathrm{T}}^*}\left[\Lambda-\frac{1}{4}(\theta_{\mathrm{T}}^*-a_0)R\right]=\frac{\Lambda}{\theta_{\mathrm{T}}^*}-\Lambda_\theta$	$\Lambda_{\mathrm{eff}}=\frac{1}{\theta_{\mathrm{T}}^*}\left[\Lambda-\frac{1}{4}(\theta_{\mathrm{T}}^*-a_0)R\right]=\frac{\Lambda}{\theta_{\mathrm{T}}^*}-\Lambda_\theta$
mielke-geometrodynamics_186 EQ0578 (8.6.16)	186	■ 1.000 ● K +0	$\widetilde{L}_{\mathrm{qPG}}=L_{\mathrm{qPG}}+\lambda_\alpha\wedge T^\alpha.$	$\widetilde{L}_{\mathrm{qPG}}=L_{\mathrm{qPG}}+\lambda_\alpha\wedge T^\alpha.$	$\widetilde{L}_{\mathrm{qPG}}=L_{\mathrm{qPG}}+\lambda_\alpha\wedge T^\alpha.$
mielke-geometrodynamics_188 EQ0582 (8.6.20)	188	■ 1.000 ● S +0	$\Lambda_\theta=\frac{3(\theta_{\mathrm{T}}^*-a_0)^2}{2\ell^2(\theta_{\mathrm{L}}^*-b_6)\theta_{\mathrm{T}}^*}$	$\Lambda_\theta=\frac{3(\theta_{\mathrm{T}}^*-a_0)^2}{2\ell^2(\theta_{\mathrm{L}}^*-b_6)\theta_{\mathrm{T}}^*}$	$\Lambda_\theta=\frac{3(\theta_{\mathrm{T}}^*-a_0)^2}{2\ell^2(\theta_{\mathrm{L}}^*-b_6)\theta_{\mathrm{T}}^*}$
mielke-geometrodynamics_193 EQ0586 (9.1.4)	193	■ 1.000 ● N +0	$ds^2=\left(\frac{N(r\pm t)}{r}\right)^2(-dt^2+dr^2+r^2d\Omega^2),$	$ds^2=\left(\frac{N(r\pm t)}{r}\right)^2(-dt^2+dr^2+r^2d\Omega^2),$	$ds^2=\left(\frac{N(r\pm t)}{r}\right)^2(-dt^2+dr^2+r^2d\Omega^2),$
mielke-geometrodynamics_194 EQ0588 (9.1.6)	194	■ 1.000 ● N +0	$u^2-v^2=e^{r^*/4\mu}$	$u^2-v^2=e^{r^*/4\mu}$	$u^2-v^2=e^{r^*/4\mu}$
mielke-geometrodynamics_194 EQ0589 (9.1.7)	194	■ 1.000 ● N +0	$(u+v)/(u-v)=e^{t/2\mu},$	$(u+v)/(u-v)=e^{t/2\mu},$	$(u+v)/(u-v)=e^{t/2\mu},$
mielke-geometrodynamics_194 EQ0590 (9.1.8)	194	■ 1.000 ● N +0	$ds^2=16\mu^2e^{\nu-r^*/2\mu}(-dv^2+du^2)+r^2d\Omega^2,$	$ds^2=16\mu^2e^{\nu-r^*/2\mu}(-dv^2+du^2)+r^2d\Omega^2,$	$ds^2=16\mu^2e^{\nu-r^*/2\mu}(-dv^2+du^2)+r^2d\Omega^2,$
mielke-geometrodynamics_195 EQ0591 (9.1.9)	195	■ 1.000 ● N +0	$\Omega^{\{\}}=\Omega_{\mathrm{C}}^{\{\}}+\bar{\Omega}^{\{\}}-\frac{1}{6}R^{\{\}}\vartheta\wedge\vartheta,$	$\Omega^{\{\}}=\Omega_{\mathrm{C}}^{\{\}}+\bar{\Omega}^{\{\}}-\frac{1}{6}R^{\{\}}\vartheta\wedge\vartheta,$	$\Omega^{\{\}}=\Omega_{\mathrm{C}}^{\{\}}+\bar{\Omega}^{\{\}}-\frac{1}{6}R^{\{\}}\vartheta\wedge\vartheta,$
mielke-geometrodynamics_197 EQ0600 (9.1.18)	197	■ 1.000 ● N +1	$\begin{aligned}ds_{\mathrm{Sds}}^2&=\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{*2}}\right)d\tau^2\\&+\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{*2}}\right)^{-1}dr^2+r^2d\Omega^2\end{aligned}$	$ds_{\mathrm{Sds}}^2=\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{*2}}\right)d\tau^2+\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{*2}}\right)^{-1}dr^2+r^2d\Omega^2$	$ds_{\mathrm{Sds}}^2=\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{*2}}\right)d\tau^2+\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{*2}}\right)^{-1}dr^2+r^2d\Omega^2,$
mielke-geometrodynamics_197 EQ0601 (9.1.19)	197	■ 1.000 ● N +0	$\begin{aligned}&f=\frac{2\mu}{r^2}\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{*2}}\right)^{-1/2}\\&g=h=k=0\end{aligned}$	$f=\frac{2\mu}{r^2}\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{*2}}\right)^{-1/2}$ $g=h=k=0$	$f=\frac{2\mu}{r^2}\left(1-\frac{2\mu}{r}+\frac{\kappa}{4\ell^{*2}}\right)^{-1/2}$ $g=h=k=0$
mielke-geometrodynamics_198 EQ0603 (9.2.2)	198	■ 1.000 ● K +0	$p_1(M^4)=-3\tau$	$p_1(M^4)=-3\tau$	$p_1(M^4)=-3\tau$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_200 EQ0607 (9.2.8)	200	■ 1.000 ● W +0	$\begin{aligned} \chi(M \# N) &= \chi(M) + \chi(N) - \chi \\ \left(S^{\{n\}}\right), \quad \chi(M \times N) &= \chi(M) \cdot \chi(N), \\ \chi(M^2) &= 2(g-1), \quad \chi \\ \left(\mathbb{C}P^2\right) &= 3 \end{aligned}$	$\begin{aligned} \chi(M \# N) &= \chi(M) + \chi(N) - \chi(S^n), \\ \chi(M \times N) &= \chi(M) \cdot \chi(N), \\ \chi(M^2) &= 2(g-1), \\ \chi(\mathbb{C}P^2) &= 3 \end{aligned}$	$\begin{aligned} \chi(M \# N) &= \chi(M) + \chi(N) - \chi(S^n), \\ \chi(M \times N) &= \chi(M) \cdot \chi(N), \\ \chi(M^2) &= 2(g-1), \\ \chi(\mathbb{C}P^2) &= 3 \end{aligned}$
mielke-geometrodynamics_201 EQ0608 (9.2.10)	201	■ 1.000 ● K +0	$\chi\left(M^{\wedge 2\ k+1}\right)=0$	$\chi\left(M^{2k+1}\right)=0$	$\chi(M^{2k+1})=0$
mielke-geometrodynamics_208 EQ0612 (10.0.1)	208	■ 1.000 ● N +0	$L_{\mathrm{EC}}=-\frac{\chi}{\ell}\vartheta^{\alpha}\wedge R_{\alpha}^{\star}\wedge R_{\alpha}^{\star}$	$L_{\mathrm{EC}}=-\frac{\chi}{\ell}\vartheta^{\alpha}\wedge R_{\alpha}^{\star}$	$L_{\mathrm{EC}}=-\frac{\chi}{\ell}\vartheta^{\alpha}\wedge R_{\alpha}^{\star}$
mielke-geometrodynamics_209 EQ0614 (10.1.2)	209	■ 1.000 ● N +0	$T^{\alpha}:=d\vartheta^{\alpha}-(-1)^s\eta^{\alpha\beta}\wedge\Gamma_{\beta}^{\star}\wedge\Gamma_{\beta}^{\star}$	$T^{\alpha}:=d\vartheta^{\alpha}-(-1)^s\eta^{\alpha\beta}\wedge\Gamma_{\beta}^{\star}$	$T^{\alpha}:=d\vartheta^{\alpha}-(-1)^s\eta^{\alpha\beta}\wedge\Gamma_{\beta}^{\star}$
mielke-geometrodynamics_209 EQ0615 (10.1.3)	209	■ 1.000 ●	$R_{\alpha}^{\star}=\frac{1}{2}\eta_{\alpha\beta\gamma}R^{\beta\gamma}:=d\Gamma_{\alpha}^{\star}+\frac{(-1)^s}{2}\eta_{\alpha}^{\beta\gamma}\Gamma_{\beta}^{\star}\wedge\Gamma_{\gamma}^{\star},$	$R_{\alpha}^{\star}=\frac{1}{2}\eta_{\alpha\beta\gamma}R^{\beta\gamma}:=d\Gamma_{\alpha}^{\star}+\frac{(-1)^s}{2}\eta_{\alpha}^{\beta\gamma}\Gamma_{\beta}^{\star}\wedge\Gamma_{\gamma}^{\star},$	$R_{\alpha}^{\star}=\frac{1}{2}\eta_{\alpha\beta\gamma}R^{\beta\gamma}:=d\Gamma_{\alpha}^{\star}+\frac{(-1)^s}{2}\eta_{\alpha}^{\beta\gamma}\Gamma_{\beta}^{\star}\wedge\Gamma_{\gamma}^{\star},$
mielke-geometrodynamics_209 EQ0616 (10.1.4)	209	■ 1.000 ● N +0	$D\ T^{\alpha}\equiv(-1)^s\eta^{\alpha\beta}\wedge R_{\beta}^{\star}\wedge R_{\beta}^{\star}$	$DT^{\alpha}\equiv(-1)^s\eta^{\alpha\beta}\wedge R_{\beta}^{\star}$	$DT^{\alpha}\equiv(-1)^s\eta^{\alpha\beta}\wedge R_{\beta}^{\star}$
mielke-geometrodynamics_209 EQ0617 (10.1.5)	209	■ 1.000 ● N +0	$D\ R_{\alpha}^{\star}\equiv 0\ .$	$DR_{\alpha}^{\star}\equiv 0.$	$DR_{\alpha}^{\star}\equiv 0.$
mielke-geometrodynamics_209 EQ0618 (10.1.6)	209	■ 1.000 ● N +0	$C_{\mathrm{T}}:=\frac{1}{2\ell^2}\vartheta^{\alpha}\wedge T_{\alpha}=-\frac{(-1)^s}{\ell^2}\eta^{\alpha}\wedge K_{\alpha}^{\star},$	$C_{\mathrm{T}}:=\frac{1}{2\ell^2}\vartheta^{\alpha}\wedge T_{\alpha}=-\frac{(-1)^s}{\ell^2}\eta^{\alpha}\wedge K_{\alpha}^{\star},$	$C_{\mathrm{T}}:=\frac{1}{2\ell^2}\vartheta^{\alpha}\wedge T_{\alpha}=-\frac{(-1)^s}{\ell^2}\eta^{\alpha}\wedge K_{\alpha}^{\star},$
mielke-geometrodynamics_209 EQ0619 (10.1.7)	209	■ 1.000 ● N +0	$C_{\mathrm{L}}:=(-1)^s\Gamma^{\star\alpha}\wedge R_{\alpha}^{\star}-\frac{1}{3!}\eta_{\alpha\beta\gamma}\Gamma^{\star\alpha}\wedge\Gamma^{\star\beta}\wedge\Gamma^{\star\gamma}.$	$C_{\mathrm{L}}:=(-1)^s\Gamma^{\star\alpha}\wedge R_{\alpha}^{\star}-\frac{1}{3!}\eta_{\alpha\beta\gamma}\Gamma^{\star\alpha}\wedge\Gamma^{\star\beta}\wedge\Gamma^{\star\gamma}.$	$C_{\mathrm{L}}:=(-1)^s\Gamma^{\star\alpha}\wedge R_{\alpha}^{\star}-\frac{1}{3!}\eta_{\alpha\beta\gamma}\Gamma^{\star\alpha}\wedge\Gamma^{\star\beta}\wedge\Gamma^{\star\gamma}.$
mielke-geometrodynamics_209 EQ0621 (10.1.9)	209	■ 1.000 ● N -2	$\frac{\delta C_{\mathrm{L}}}{\delta\vartheta^{\alpha}}=0\quad\text{and}\quad\frac{\delta C_{\mathrm{L}}}{\delta\Gamma^{\star\alpha}}=(-1)^s2R_{\alpha}^{\star},$	$\frac{\delta C_{\mathrm{L}}}{\delta\vartheta^{\alpha}}=0\quad\text{and}\quad\frac{\delta C_{\mathrm{L}}}{\delta\Gamma^{\star\alpha}}=(-1)^s2R_{\alpha}^{\star},$	$\frac{\delta C_{\mathrm{L}}}{\delta\vartheta^{\alpha}}=0\quad\text{and}\quad\frac{\delta C_{\mathrm{L}}}{\delta\Gamma^{\star\alpha}}=(-1)^s2R_{\alpha}^{\star},$
mielke-geometrodynamics_210 EQ0622 (10.1.10)	210	■ 1.000 ● N -1	$C_{\mathrm{TL}}:=\frac{1}{\ell}\left(\Gamma^{\star\alpha}\wedge T_{\alpha}-\frac{(-1)^s}{2}\eta_{\alpha\beta\gamma}\Gamma^{\star\alpha}\wedge\Gamma^{\star\beta}\wedge\vartheta^{\gamma}\right).$	$C_{\mathrm{TL}}:=\frac{1}{\ell}\left(\Gamma^{\star\alpha}\wedge T_{\alpha}-\frac{(-1)^s}{2}\eta_{\alpha\beta\gamma}\Gamma^{\star\alpha}\wedge\Gamma^{\star\beta}\wedge\vartheta^{\gamma}\right).$	$C_{\mathrm{TL}}:=\frac{1}{\ell}\left(\Gamma^{\star\alpha}\wedge T_{\alpha}-\frac{(-1)^s}{2}\eta_{\alpha\beta\gamma}\Gamma^{\star\alpha}\wedge\Gamma^{\star\beta}\wedge\vartheta^{\gamma}\right).$
mielke-geometrodynamics_210 EQ0625 (10.1.13)	210	■ 1.000 ● N +0	$D\ H_{\alpha}^{\star}-S_{\alpha}=\tau_{\alpha}^{\star},$	$DH_{\alpha}^{\star}-S_{\alpha}=\tau_{\alpha}^{\star},$	$DH_{\alpha}^{\star}-S_{\alpha}=\tau_{\alpha}^{\star},$
mielke-geometrodynamics_211 EQ0626 (10.1.14)	211	■ 1.000 ● K +0	$H_{\alpha}:= -\frac{\partial L}{\partial T^{\alpha}},\quad H_{\alpha}^{\star}:= -\frac{(-1)^s}{2}\frac{\partial L}{\partial R^{\star\alpha}},$	$H_{\alpha}:= -\frac{\partial L}{\partial T^{\alpha}},\quad H_{\alpha}^{\star}:= -\frac{(-1)^s}{2}\frac{\partial L}{\partial R^{\star\alpha}},$	$H_{\alpha}:= -\frac{\partial L}{\partial T^{\alpha}},\quad H_{\alpha}^{\star}:= -\frac{(-1)^s}{2}\frac{\partial L}{\partial R^{\star\alpha}},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics E0628 (10.1.16)	211	■ 1.000 ● N +0	$S_{\alpha}:=\frac{\left(-1\right)^s}{2}\frac{\partial L}{\partial \Gamma^{\star\alpha}}=\frac{1}{2}\eta_{\alpha\beta}\wedge H^{\beta}.$		$S_{\alpha}:=\frac{\left(-1\right)^s}{2}\frac{\partial L}{\partial \Gamma^{\star\alpha}}=\frac{1}{2}\eta_{\alpha\beta}\wedge H^{\beta}.$
mielke-geometrodynamics E0629 (10.1.17)	211	■ 1.000 ● N +0	$\Sigma_{\alpha}:=\frac{\delta L_{\mathrm{MAT}}}{\delta \vartheta^{\alpha}},\quad \tau_{\alpha}^{\star}:=\frac{\left(-1\right)^s}{2}\frac{\delta L_{\mathrm{MAT}}}{\delta \Gamma^{\star\alpha}},$		$\Sigma_{\alpha}:=\frac{\delta L_{\mathrm{MAT}}}{\delta \vartheta^{\alpha}},\quad \tau_{\alpha}^{\star}:=\frac{\left(-1\right)^s}{2}\frac{\delta L_{\mathrm{MAT}}}{\delta \Gamma^{\star\alpha}},$
mielke-geometrodynamics E0630 (10.2.1)	211	■ 1.000 ● N -1	$L_{\mathrm{EC}}:=-\frac{\chi}{\ell}\vartheta^{\alpha}\wedge R_{\alpha}^{\star}\equiv-\chi C_{\mathrm{TL}}-\frac{\chi}{\ell}d\left(\Gamma_{\alpha}^{\star}\wedge\vartheta^{\alpha}\right)$		$L_{\mathrm{EC}}:=-\frac{\chi}{\ell}\vartheta^{\alpha}\wedge R_{\alpha}^{\star}\equiv-\chi C_{\mathrm{TL}}-\frac{\chi}{\ell}d\left(\Gamma_{\alpha}^{\star}\wedge\vartheta^{\alpha}\right)$
mielke-geometrodynamics E0631 (10.2.2)	211	■ 1.000 ● C -5	$L_{\mathrm{MB}}\left(\vartheta^{\alpha},\Gamma_{\alpha}^{\star}\right)=\theta_{\mathrm{T}}C_{\mathrm{T}}+\theta_{\mathrm{L}}C_{\mathrm{L}}+\theta_{\mathrm{TL}}C_{\mathrm{TL}},$		$L_{\mathrm{MB}}\left(\vartheta^{\alpha},\Gamma_{\alpha}^{\star}\right)=\theta_{\mathrm{T}}C_{\mathrm{T}}+\theta_{\mathrm{L}}C_{\mathrm{L}}+\theta_{\mathrm{TL}}C_{\mathrm{TL}},$
mielke-geometrodynamics E0632 (10.2.3)	212	■ 1.000 ● C -3	$-\theta_{\mathrm{TL}}R_{\alpha}^{\star}-\frac{1}{\ell}\theta_{\mathrm{T}}T_{\alpha}=\ell\Sigma_{\alpha}$		$-\theta_{\mathrm{TL}}R_{\alpha}^{\star}-\frac{1}{\ell}\theta_{\mathrm{T}}T_{\alpha}=\ell\Sigma_{\alpha}$
mielke-geometrodynamics E0634 (10.2.5)	212	■ 1.000 ●	$T_{\alpha}-\frac{\varepsilon}{\ell}\eta_{\alpha}=\frac{2}{A}\ell\left(\theta_{\mathrm{TL}}\tau_{\alpha}^{\star}-\theta_{\mathrm{L}}\ell\Sigma_{\alpha}\right)$		$T_{\alpha}-\frac{\varepsilon}{\ell}\eta_{\alpha}=\frac{2}{A}\ell\left(\theta_{\mathrm{TL}}\tau_{\alpha}^{\star}-\theta_{\mathrm{L}}\ell\Sigma_{\alpha}\right)$
mielke-geometrodynamics E0635 (10.2.6)	212	■ 1.000 ● C -3	$R_{\alpha}^{\star}-\frac{\rho}{\ell^2}\eta_{\alpha}=\frac{2}{A}\left(\theta_{\mathrm{TL}}\ell\Sigma_{\alpha}-\theta_{\mathrm{T}}\tau_{\alpha}^{\star}\right)$		$R_{\alpha}^{\star}-\frac{\rho}{\ell^2}\eta_{\alpha}=\frac{2}{A}\left(\theta_{\mathrm{TL}}\ell\Sigma_{\alpha}-\theta_{\mathrm{T}}\tau_{\alpha}^{\star}\right)$
mielke-geometrodynamics E0638 (10.2.9)	213	■ 1.000 ● N +0	$R_{\alpha}^{\star}\equiv R_{\alpha}^{\{\}\star}-DK_{\alpha}^{\star}+\frac{1}{2}\eta_{\alpha\beta\gamma}K^{\star\beta}\wedge K^{\star\gamma}$		$R_{\alpha}^{\star}\equiv R_{\alpha}^{\{\}\star}-DK_{\alpha}^{\star}+\frac{1}{2}\eta_{\alpha\beta\gamma}K^{\star\beta}\wedge K^{\star\gamma}$
mielke-geometrodynamics E0640 (10.2.11)	213	■ 1.000 ● C -3	$\Lambda_{\mathrm{eff}}=-\left[4\rho-\left(1+2\left(-1\right)^s\right)\varepsilon^2\right]/4\ell^2=\theta_{\mathrm{T}}^2\left[\left(9+2\left(-1\right)^s\right)\theta_{\mathrm{TL}}^2+8\theta_{\mathrm{T}}\theta_{\mathrm{L}}\right]/\left(2A\ell\right)^2$		$\Lambda_{\mathrm{eff}}=-\left[4\rho-\left(1+2\left(-1\right)^s\right)\varepsilon^2\right]/4\ell^2=\theta_{\mathrm{T}}^2\left[\left(9+2\left(-1\right)^s\right)\theta_{\mathrm{TL}}^2+8\theta_{\mathrm{T}}\theta_{\mathrm{L}}\right]/\left(2A\ell\right)^2$
mielke-geometrodynamics E0642 (10.2.13)	213	■ 1.000 ● N +0	$ds^2=-N^2(r)dt^2+N^{-2}(r)dr^2+r^2\left[d\phi+N^{\phi}(r)dt\right]^2$		$ds^2=-N^2(r)dt^2+N^{-2}(r)dr^2+r^2\left[d\phi+N^{\phi}(r)dt\right]^2$
mielke-geometrodynamics E0643 (10.2.14)	213	■ 1.000 ● N +0	$N^2(r)=-M-\Lambda_{\mathrm{eff}}r^2+\frac{J^2}{4r^2},\quad N^{\phi}(r)=-\frac{J}{2r^2},$		$N^2(r)=-M-\Lambda_{\mathrm{eff}}r^2+\frac{J^2}{4r^2},\quad N^{\phi}(r)=-\frac{J}{2r^2},$
mielke-geometrodynamics E0647 (10.3.4)	214	■ 1.000 ● K +0	$S_{\alpha}=\frac{\ell}{\varepsilon}E_{\alpha}$		$S_{\alpha}=\frac{\ell}{\varepsilon}E_{\alpha}$
mielke-geometrodynamics E0648 (10.3.5)	215	■ 1.000 ● N +0	$H_{\alpha}^{\star}=\frac{\ell}{\varepsilon}H_{\alpha}$		$H_{\alpha}^{\star}=\frac{\ell}{\varepsilon}H_{\alpha}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_215 EQ0649 (10.3.6)		■ 1.000 ● N +1	$T_{\{\alpha\}}=\frac{\{\varepsilon\}}{\{\ell\}}\ \eta_{\{\alpha\}},\ \text{quad}\ R_{\{\alpha\}}^{\{\star\}}=\frac{\{\rho\}}{\{\ell^2\}}\ \eta_{\{\alpha\}}$	$T_{\alpha}=\frac{\varepsilon}{\ell}\eta_{\alpha},\quad R_{\alpha}^{\star}=\frac{\rho}{\ell^2}\eta_{\alpha}$	$T_{\alpha}=\frac{\varepsilon}{\ell}\eta_{\alpha},\quad R_{\alpha}^{\star}=\frac{\rho}{\ell^2}\eta_{\alpha},$
mielke-geometrodynamics_215 EQ0650 (10.3.7)		■ 1.000 ● N +0	$H_{\{\alpha\}}^{\{\star\}}=\{\delta\ell\}\ H_{\{\alpha\}}+\frac{\{\gamma\}}{\{\ell\}}\ \{\vartheta_{\alpha}\}$	$H_{\alpha}^{\star}=\delta\ell H_{\alpha}+\frac{\gamma}{\ell}\vartheta_{\alpha}$	$H_{\alpha}^{\star}=\delta\ell H_{\alpha}+\frac{\gamma}{\ell}\vartheta_{\alpha}$
mielke-geometrodynamics_215 EQ0651 (10.3.8)		■ 1.000 ● N +1	$\{\delta\ell\}\ D\ H_{\{\alpha\}}+\frac{\{\gamma\}}{\{\ell\}}\ T_{\{\alpha\}}-\frac{\{1\}}{\{2\}}\ \eta_{\{\alpha\}\ \beta}\ \wedge\ H^{\{\beta\}}=\{\tau_{\alpha}^{\star}\}$	$\delta\ell D H_{\alpha}+\frac{\gamma}{\ell}T_{\alpha}-\frac{1}{2}\eta_{\alpha\beta}\wedge H^{\beta}=\tau_{\alpha}^{\star}$	$\delta\ell D H_{\alpha}+\frac{\gamma}{\ell}T_{\alpha}-\frac{1}{2}\eta_{\alpha\beta}\wedge H^{\beta}=\tau_{\alpha}^{\star}$
mielke-geometrodynamics_216 EQ0653 (10.3.10)		■ 1.000 ● K +0	$H_{\{\alpha\}}^{\{\star\}}=\frac{\{\gamma\}}{\{\ell(1+\{\delta\varepsilon\})}\ \{\vartheta_{\alpha}\}=-\frac{\{\ell\}}{\{\varepsilon\}}H_{\alpha},$	$H_{\alpha}^{\star}=\frac{\gamma}{\ell(1+\delta\varepsilon)}\vartheta_{\alpha}=-\frac{\ell}{\varepsilon}H_{\alpha},$	$H_{\alpha}^{\star}=\frac{\gamma}{\ell(1+\delta\varepsilon)}\vartheta_{\alpha}=-\frac{\ell}{\varepsilon}H_{\alpha},$
mielke-geometrodynamics_217 EQ0659 (10.3.16)		■ 1.000 ● K +0	$H=\{\}\ ^{\{\star\}}\{\left(\{\vartheta^{\{\alpha\}}\}\ \wedge\ H_{\{\alpha\}}\right)\},$	$H=\ ^{\star}(\vartheta^{\alpha}\wedge H_{\alpha}),$	$H=\ ^{\star}(\vartheta^{\alpha}\wedge H_{\alpha}),$
mielke-geometrodynamics_217 EQ0660 (10.3.17)		■ 1.000 ● N +0	$t=\{\}\ ^{\{\star\}}\{\left(T^{\{\alpha\}}\ \wedge\ H_{\{\alpha\}}\right)\},\ \text{quad}\ \{\mathscr{A}\}:=\{\}\ ^{\{\star\}}\{\left(\{\vartheta^{\{\alpha\}}\}\ \wedge\ T_{\{\alpha\}}\right)\},\ \text{quad}\ h=\{\}\ ^{\{\star\}}\{\left(\eta^{\{\alpha\}}\ \wedge\ H_{\alpha}\right)\},$	$t=\ ^{\star}(T^{\alpha}\wedge H_{\alpha}),\quad \mathscr{A}:=\ ^{\star}(\vartheta^{\alpha}\wedge T_{\alpha}),\quad h=\ ^{\star}(\eta^{\alpha}\wedge H_{\alpha}),$	$t=\ ^{\star}(T^{\alpha}\wedge H_{\alpha}),\quad \mathscr{A}:=\ ^{\star}(\vartheta^{\alpha}\wedge T_{\alpha}),\quad h=\ ^{\star}(\eta^{\alpha}\wedge H_{\alpha}),$
mielke-geometrodynamics_217 EQ0663 (10.3.20)		■ 1.000 ● N +2	$d\ s^{\{2\}}=(-1)^{\{s\}}\ d\ t^{\{2\}}+d\ \{\mathscr{A}\}^{\{2\}}+d\ h^{\{2\}}$	$ds^2=(-1)^s dt^2+d\mathscr{A}^2+dh^2$	$ds^2=(-1)^s dt^2+d\mathscr{A}^2+dh^2$
mielke-geometrodynamics_218 EQ0665 (10.3.22)		■ 1.000 ● N +0	$D^{\{\star\}}\ \{\vartheta_{\alpha}\}:=\frac{\{\ell\}}{\{\gamma\}}\ [D^{\star}H_{\alpha}^{\star}-\{\delta\ell\}D^{\star}H_{\alpha}]$	$D^{\star}\vartheta_{\alpha}:=\frac{\ell}{\gamma}\left[D^{\star}H_{\alpha}^{\star}-\delta\ell D^{\star}H_{\alpha}\right]$	$D^{\star}\vartheta_{\alpha}:=\frac{\ell}{\gamma}\left[D^{\star}H_{\alpha}^{\star}-\delta\ell D^{\star}H_{\alpha}\right]$
mielke-geometrodynamics_219 EQ0668 (10.3.25)		■ 1.000 ● N +0	$D^{\{\star\}}\ \{\vartheta^{\{\alpha\}}\}\ \cong\ 0,$	$D^{\star}\vartheta^{\alpha}\cong 0,$	$D^{\star}\vartheta^{\alpha}\cong 0,$
mielke-geometrodynamics_220 EQ0672 (10.3.29)		■ 1.000 ●	$P_{\{\alpha\}}=\int\ \{\check{\mathscr{E}}_{\alpha}\},\quad J_{\{\alpha\}}=\int\ \{\check{\mathscr{J}}_{\alpha}^{\star}\}.$	$P_{\alpha}=\int\ \check{\mathscr{E}}_{\alpha},\quad J_{\alpha}=\int\ \check{\mathscr{J}}_{\alpha}^{\star}.$	$P_{\alpha}=\int\ \check{\mathscr{E}}_{\alpha},\quad J_{\alpha}=\int\ \check{\mathscr{J}}_{\alpha}^{\star}.$
mielke-geometrodynamics_220 EQ0675 (10.3.32)		■ 1.000 ● N -1	$\{\check{\mathscr{J}}_{\alpha}^{\star}\}=-\frac{\{\theta_{\mathrm{L}}\}}{\{2\}}d\Gamma_{\alpha}^{\star}$	$\check{\mathscr{J}}_{\alpha}^{\star}=-\frac{\theta_{\mathrm{L}}}{2}d\Gamma_{\alpha}^{\star}$	$\check{\mathscr{J}}_{\alpha}^{\star}=-\frac{\theta_{\mathrm{L}}}{2}d\Gamma_{\alpha}^{\star}$
mielke-geometrodynamics_220 EQ0677 (10.3.34)		■ 1.000 ● N -1	$\{\check{\mathscr{J}}_{\alpha}^{\mathrm{MB}\star}\}=(-1)^s\frac{\{\chi\}}{\{2\ell\}}d\vartheta_{\alpha}-\frac{\{\theta_{\mathrm{L}}\}}{\{2\}}d\Gamma_{\alpha}^{\star}$	$\check{\mathscr{J}}_{\alpha}^{\mathrm{MB}\star}=(-1)^s\frac{\chi}{2\ell}d\vartheta_{\alpha}-\frac{\theta_{\mathrm{L}}}{2}d\Gamma_{\alpha}^{\star}$	$\check{\mathscr{J}}_{\alpha}^{\mathrm{MB}\star}=(-1)^s\frac{\chi}{2\ell}d\vartheta_{\alpha}-\frac{\theta_{\mathrm{L}}}{2}d\Gamma_{\alpha}^{\star}$
mielke-geometrodynamics_221 EQ0679 (10.3.36)		■ 1.000 ● N +0	$\{\widetilde{C}_{\{\mathrm{L}\}}\}=(-1)^{\{s\}}\ \{\widetilde{C}_{\{\alpha\}}^{\{\star\}}\}\ \wedge\ \{\widetilde{R}_{\alpha}^{\star}\}-\frac{\{1\}}{\{3!\}}\ \eta_{\{\alpha\}\ \beta}\ \{\widetilde{C}_{\{\alpha\}}^{\{\star\}}\}\ \wedge\ \{\widetilde{R}_{\beta}^{\star}\}\ \wedge\ \{\widetilde{C}_{\{\alpha\}}^{\{\star\}}\}\ \wedge\ \{\widetilde{R}_{\beta}^{\star}\},$	$\widetilde{C}_{\mathrm{L}}=(-1)^s\widetilde{\Gamma}^{\star\alpha}\wedge\widetilde{R}_{\alpha}^{\star}-\frac{1}{3!}\eta_{\alpha\beta\gamma}\widetilde{\Gamma}^{\star\alpha}\wedge\widetilde{\Gamma}^{\star\beta}\wedge\widetilde{\Gamma}^{\star\gamma},$	$\widetilde{C}_{\mathrm{L}}=(-1)^s\widetilde{\Gamma}^{\star\alpha}\wedge\widetilde{R}_{\alpha}^{\star}-\frac{1}{3!}\eta_{\alpha\beta\gamma}\widetilde{\Gamma}^{\star\alpha}\wedge\widetilde{\Gamma}^{\star\beta}\wedge\widetilde{\Gamma}^{\star\gamma},$
mielke-geometrodynamics_221 EQ0683 (10.3.41)		■ 1.000 ● N -1	$c=12\cdot 4\pi\ \{\theta_{\mathrm{L}}\},\quad \{\tilde{c}\}=12\cdot 4\pi\ \{\tilde{\theta}_{\mathrm{L}}\},$	$c=12\cdot 4\pi\theta_{\mathrm{L}},\quad \tilde{c}=12\cdot 4\pi\tilde{\theta}_{\mathrm{L}},$	$c=12\cdot 4\pi\theta_{\mathrm{L}},\quad \tilde{c}=12\cdot 4\pi\tilde{\theta}_{\mathrm{L}},$
mielke-geometrodynamics_222 EQ0684 (10.3.42)		■ 1.000 ● K +0	$\{\widetilde{L}\}=L+\lambda_{\alpha}\ \wedge\ T^{\alpha}.$	$\widetilde{L}=L+\lambda_{\alpha}\wedge T^{\alpha}.$	$\widetilde{L}=L+\lambda_{\alpha}\wedge T^{\alpha}.$
mielke-geometrodynamics_223 EQ0691 (10.5.2)		■ 1.000 ● K +0	$\{\left[\{\sigma^{\{\alpha\}}\}\ \{\sigma^{\{\beta\}}\}\right]\}=2\ i\ \eta^{\{\alpha\beta\gamma\}}\ \{\sigma_{\gamma}\}.$	$\left[\sigma^{\alpha},\sigma^{\beta}\right]=2i\eta^{\alpha\beta\gamma}\sigma_{\gamma}.$	$\left[\sigma^{\alpha},\sigma^{\beta}\right]=2i\eta^{\alpha\beta\gamma}\sigma_{\gamma}.$
mielke-geometrodynamics_224 EQ0692 (10.5.3)		■ 1.000 ● K +0	$\{\gamma_{\{0\}}\}=i\ \{\sigma^{\{2\}}\},\quad \{\gamma_{\{1\}}\}=\{\sigma^{\{1\}}\},\quad \{\gamma_{\{2\}}\}=\{\sigma^{\{3\}}\}.$	$\gamma_0=i\sigma^2,\quad \gamma_1=\sigma^1,\quad \gamma_2=\sigma^3.$	$\gamma_0=i\sigma^2,\quad \gamma_1=\sigma^1,\quad \gamma_2=\sigma^3.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_224 EQ0693 (10.5.4)		■ 1.000 ● N +0	$\gamma_{\alpha}\gamma_{\beta} = g_{\alpha\beta}\mathbf{1} + \eta_{\alpha\beta\nu}\gamma^{\nu}$, $\mathbf{1} + \eta_{\alpha\beta\nu}\gamma^{\nu}$,	$\gamma_{\alpha}\gamma_{\beta} = g_{\alpha\beta}\mathbf{1} + \eta_{\alpha\beta\nu}\gamma^{\nu}$,	$\gamma_{\alpha}\gamma_{\beta} = g_{\alpha\beta}\mathbf{1} + \eta_{\alpha\beta\nu}\gamma^{\nu}$,
mielke-geometrodynamics_224 EQ0695 (10.5.6)		■ 1.000 ● N +0	$i^{*}\gamma\wedge D\psi - m\psi\eta - \frac{i}{2}(D^{*}\gamma)\psi = 0$	$i^{*}\gamma\wedge D\psi - m\psi\eta - \frac{i}{2}(D^{*}\gamma)\psi = 0$	$i^{*}\gamma\wedge D\psi - m\psi\eta - \frac{i}{2}(D^{*}\gamma)\psi = 0$
mielke-geometrodynamics_224 EQ0697 (10.5.8)		■ 1.000 ● K +0	$\hat{D}:=D-\frac{1}{2}T$,	$\hat{D}:=D-\frac{1}{2}T$,	$\hat{D}:=D-\frac{1}{2}T$,
mielke-geometrodynamics_224 EQ0698 (10.5.9)		■ 1.000 ● N +0	$i^{*}\gamma\wedge D\psi - m\eta\psi = 0$,	$i^{*}\gamma\wedge \hat{D}\psi - m\eta\psi = 0$,	$i^{*}\gamma\wedge \hat{D}\psi - m\eta\psi = 0$,
mielke-geometrodynamics_224 EQ0699 (10.5.10)		■ 1.000 ● N +0	$\hat{R}=R-\frac{1}{2}dT=\frac{i}{4}R^{\alpha\beta}\sigma_{\alpha\beta}-\frac{1}{2}dT$	$\hat{R}=R-\frac{1}{2}dT=\frac{i}{4}R^{\alpha\beta}\sigma_{\alpha\beta}-\frac{1}{2}dT$	$\hat{R}=R-\frac{1}{2}dT=\frac{i}{4}R^{\alpha\beta}\sigma_{\alpha\beta}-\frac{1}{2}dT$
mielke-geometrodynamics_225 EQ0703 (10.6.2)		■ 1.000 ● N +0	$d^{*}F - m_{\text{photon}}F = 0$	$d^{*}F - m_{\text{photon}}F = 0$	$d^{*}F - m_{\text{photon}}F = 0$
mielke-geometrodynamics_226 EQ0705 (10.7.1)		■ 1.000 ● N +0	$X:=x_{\alpha}\vartheta^{\alpha}$, $Y:=y_{\alpha}\vartheta^{\alpha}$. $\vartheta^{\alpha}\wedge X=0$, $\vartheta^{\alpha}\wedge Y=0$.	$X:=x_{\alpha}\vartheta^{\alpha}$, $Y:=y_{\alpha}\vartheta^{\alpha}$.	$X:=x_{\alpha}\vartheta^{\alpha}$, $Y:=y_{\alpha}\vartheta^{\alpha}$.
mielke-geometrodynamics_226 EQ0706 (10.7.2)		■ 1.000 ● N +0	$\Sigma_{\alpha}=\varepsilon x_{\alpha}X\wedge Y$, $\tau_{\alpha}^{*}=\sigma y_{\alpha}X\wedge Y$,	$\Sigma_{\alpha}=\varepsilon x_{\alpha}X\wedge Y$, $\tau_{\alpha}^{*}=\sigma y_{\alpha}X\wedge Y$,	$\Sigma_{\alpha}=\varepsilon x_{\alpha}X\wedge Y$, $\tau_{\alpha}^{*}=\sigma y_{\alpha}X\wedge Y$,
mielke-geometrodynamics_226 EQ0707 (10.7.3)		■ 1.000 ● N +0	$\vartheta^{\alpha}\wedge\Sigma_{\alpha}=0$, $\vartheta^{\alpha}\wedge\tau_{\alpha}^{*}=0$	$\vartheta^{\alpha}\wedge\Sigma_{\alpha}=0$, $\vartheta^{\alpha}\wedge\tau_{\alpha}^{*}=0$	$\vartheta^{\alpha}\wedge\Sigma_{\alpha}=0$, $\vartheta^{\alpha}\wedge\tau_{\alpha}^{*}=0$
mielke-geometrodynamics_227 EQ0708 (10.7.4)		■ 1.000 ● N +0	$R^{\alpha\beta}=\varepsilon\ell^2x^{[\alpha}y^{\beta]}X\wedge Y$,	$R^{\alpha\beta}=\varepsilon\ell^2x^{[\alpha}y^{\beta]}X\wedge Y$,	$R^{\alpha\beta}=\varepsilon\ell^2x^{[\alpha}y^{\beta]}X\wedge Y$,
mielke-geometrodynamics_228 EQ0711 (10.7.7)		■ 1.000 ● N +0	$\mathscr{A}:=*(\vartheta^{\alpha}\wedge T_{\alpha})=-(-1)^s\frac{\varepsilon}{\ell^2}$	$\mathscr{A}:=*(\vartheta^{\alpha}\wedge T_{\alpha})=-(-1)^s\frac{\varepsilon}{\ell^2}$	$\mathscr{A}:=*(\vartheta^{\alpha}\wedge T_{\alpha})=-(-1)^s\frac{\varepsilon}{\ell^2}$
mielke-geometrodynamics_228 EQ0713 (10.8.2)		■ 1.000 ● K +0	$\gamma=\gamma_{\alpha}\vartheta^{\alpha}$.	$\gamma=\gamma_{\alpha}\vartheta^{\alpha}$.	$\gamma=\gamma_{\alpha}\vartheta^{\alpha}$.
mielke-geometrodynamics_228 EQ0714		■ 1.000 ●	$\bar{\Psi}\wedge\Psi=0$, $\bar{\Psi}\wedge\gamma_5\gamma^{\alpha}\Psi=0$, $\bar{\Psi}\wedge\gamma_5\Psi=0$.	$\bar{\Psi}\wedge\Psi=0$, $\bar{\Psi}\wedge\gamma_5\gamma^{\alpha}\Psi=0$, $\bar{\Psi}\wedge\gamma_5\Psi=0$.	$\bar{\Psi}\wedge\Psi=0$, $\bar{\Psi}\wedge\gamma_5\gamma^{\alpha}\Psi=0$, $\bar{\Psi}\wedge\gamma_5\Psi=0$.
mielke-geometrodynamics_229 EQ0715 (10.9.1)		■ 1.000 ● N -1	$L_{\text{RS}}=\frac{i}{4}(\bar{\Psi}\wedge D\Psi - \Psi\wedge\overline{D\Psi}) + \frac{i}{4}m\bar{\Psi}\wedge\gamma\wedge\Psi$,	$L_{\text{RS}}=\frac{i}{4}(\bar{\Psi}\wedge D\Psi - \Psi\wedge\overline{D\Psi}) + \frac{i}{4}m\bar{\Psi}\wedge\gamma\wedge\Psi$,	$L_{\text{RS}}=\frac{i}{4}(\bar{\Psi}\wedge D\Psi - \Psi\wedge\overline{D\Psi}) + \frac{i}{4}m\bar{\Psi}\wedge\gamma\wedge\Psi$,
mielke-geometrodynamics_229 EQ0719 (10.9.5)		■ 1.000 ● N +1	$\Sigma_{\alpha}=-\frac{i}{4}m\bar{\Psi}\wedge\gamma_{\alpha}\Psi=-(-1)^s2m\tau_{\alpha}^{*}$	$\Sigma_{\alpha}=-\frac{i}{4}m\bar{\Psi}\wedge\gamma_{\alpha}\Psi=-(-1)^s2m\tau_{\alpha}^{*}$	$\Sigma_{\alpha}=-\frac{i}{4}m\bar{\Psi}\wedge\gamma_{\alpha}\Psi=-(-1)^s2m\tau_{\alpha}^{*}$,
mielke-geometrodynamics_230 EQ0721 (10.10.1)		■ 1.000 ● N +0	$L=L_{\infty}(\vartheta^{\alpha},\Gamma_{\alpha}^{*},\Psi)=L_{\text{MB}}+L_{\Psi}$	$L=L_{\infty}(\vartheta^{\alpha},\Gamma_{\alpha}^{*},\Psi)=L_{\text{MB}}+L_{\Psi}$	$L=L_{\infty}(\vartheta^{\alpha},\Gamma_{\alpha}^{*},\Psi)=L_{\text{MB}}+L_{\Psi}$
mielke-geometrodynamics_230 EQ0726 (10.10.6)		■ 1.000 ● N +0	$\frac{2}{i}\frac{\delta L}{\delta\Psi}=D\Psi+\frac{1}{2}m\gamma\wedge\Psi\cong 0$	$\frac{2}{i}\frac{\delta L}{\delta\Psi}=D\Psi+\frac{1}{2}m\gamma\wedge\Psi\cong 0$	$\frac{2}{i}\frac{\delta L}{\delta\Psi}=D\Psi+\frac{1}{2}m\gamma\wedge\Psi\cong 0$
mielke-geometrodynamics_231 EQ0729 (10.10.9)		■ 1.000 ● N +0	$\gamma\wedge\gamma=-2\gamma^{\alpha}\eta_{\alpha}$	$\gamma\wedge\gamma=-2\gamma^{\alpha}\eta_{\alpha}$	$\gamma\wedge\gamma=-2\gamma^{\alpha}\eta_{\alpha}$
mielke-geometrodynamics_231 EQ0730 (10.10.10)		■ 1.000 ● W -2	$i\left[\left(\frac{\theta_{\text{TL}}}{\ell}-1\right)R_{\alpha}^{*}+\left(\frac{\theta_{\text{T}}}{\ell^2}-\frac{m}{2}\right)T_{\alpha}+\frac{m^2}{2}\eta_{\alpha}\right]\wedge\gamma^{\alpha}\Psi=0$,	$i\left[\left(\frac{\theta_{\text{TL}}}{\ell}-1\right)R_{\alpha}^{*}+\left(\frac{\theta_{\text{T}}}{\ell^2}-\frac{m}{2}\right)T_{\alpha}+\frac{m^2}{2}\eta_{\alpha}\right]\wedge\gamma^{\alpha}\Psi=0$,	$i\left[\left(\frac{\theta_{\text{TL}}}{\ell}-1\right)R_{\alpha}^{*}+\left(\frac{\theta_{\text{T}}}{\ell^2}-\frac{m}{2}\right)T_{\alpha}+\frac{m^2}{2}\eta_{\alpha}\right]\wedge\gamma^{\alpha}\Psi=0$,

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Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

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mielke-geometrodynamics_231 EQ0731 (10.10.11)		■ 1.000 ● W +0	$\begin{aligned} & i\left[\left(\theta_{\mathrm{L}}+\frac{\theta_{\mathrm{TL}}}{\ell}-1\right)R_{\alpha}^{\star}+\left((-1)^s\frac{\theta_{\mathrm{TL}}}{\ell}+\frac{\theta_{\mathrm{T}}}{\ell^2}-\frac{m}{2}\right)T_{\alpha}+\frac{1}{2}\left(\frac{\theta_{\mathrm{T}}}{\ell^2}+m^2\right)\eta_{\alpha}\right. \\ & \left.+\tau_{\alpha}^{\star}\right]\wedge\gamma^{\alpha}\Psi\simeq 0, \end{aligned}$		
mielke-geometrodynamics_232 EQ0732 (10.10.12)		■ 1.000 ● W +2	$\theta_{\mathrm{T}}\simeq -m^2\ell^2, \theta_{\mathrm{TL}}\simeq \frac{(-1)^s}{2}m(2m+1)\ell, \theta_{\mathrm{L}}\simeq 1-\frac{\theta_{\mathrm{TL}}}{\ell}=1-\frac{(-1)^s}{2}m(2m+1)$		
mielke-geometrodynamics_232 EQ0733 (10.10.13)		■ 1.000 ● N +0	$\vartheta_{\alpha}=(-1)^s\ell\Gamma_{\alpha}^{\star}+\bar{\sigma}\gamma_{\alpha}\Psi.$		
mielke-geometrodynamics_238 EQ0735 (11.1.2)		■ 1.000 ● N +0	$\gamma_{\alpha}\gamma_{\beta}+\gamma_{\beta}\gamma_{\alpha}=2\mathrm{o}_{\alpha\beta}\mathbb{1}$		
mielke-geometrodynamics_238 EQ0738 (11.1.5)		■ 1.000 ● N +0	$\mathrm{X}:=\imath x^{\alpha}\gamma_{\alpha}\longleftrightarrow x^{\alpha}=-\frac{l}{N}\operatorname{Tr}\left(\gamma^{\alpha}X\right).$		
mielke-geometrodynamics_238 EQ0739 (11.1.6)		■ 1.000 ● K +0	$\operatorname{det}\mathrm{X}=-\mathrm{o}_{\alpha\beta}x^{\alpha}x^{\beta},$		
mielke-geometrodynamics_238 EQ0741 (11.1.8)		■ 1.000 ● N +0	$\mathrm{X}-\mathrm{X}'=SXS^{-1},\quad S\in\widetilde{SO}_{\circ}(s,n-s).$		
mielke-geometrodynamics_239 EQ0743 (11.1.10)		■ 1.000 ● N +0	$\sigma^{\alpha\beta}:=\frac{i}{4}\left[\gamma^{\alpha},\gamma^{\beta}\right],\quad\Lambda\left(\sigma^{\alpha\beta}\right)=L^{\alpha\beta},$		
mielke-geometrodynamics_239 EQ0744 (11.1.11)		■ 1.000 ● N +0	$L_{\circ}^g(M):=P\left(M,SO_{\circ}(s,n-s),\pi,\delta\right)$		
mielke-geometrodynamics_239 EQ0745 (11.1.12)		■ 1.000 ●	$\widetilde{L}(M):=P\left(M,\widetilde{SO}_{\circ}(s,n-2),\tilde{\pi},\tilde{\delta}\right)\ni\tilde{p}$		
mielke-geometrodynamics_239 EQ0746 (11.1.13)		■ 1.000 ● N +0	$f(\tilde{p}S)=f(\tilde{p})\Lambda(S),S\in\widetilde{SO}_{\circ}(s,n-2).$		
mielke-geometrodynamics_240 EQ0748 (11.1.15)		■ 1.000 ● N +0	$\tilde{L}(M^4)=M^4\times SL(2,\mathbb{C})$		
mielke-geometrodynamics_241 EQ0749 (11.1.16)		■ 1.000 ● N +0	$V^{\bar{s}}:=V\left(M,\mathbb{C}^N,\rho^{\bar{s}}\left(\widetilde{SO}_{\circ}(s,n-s)\right)\subset GL(N,\mathbb{C}),\tilde{L}(M)\right)$		
mielke-geometrodynamics_241 EQ0751 (11.1.18)		■ 1.000 ● K +0	$D^{\frac{1}{2}}:=D^{(-\frac{1}{2},\frac{3}{2})}\oplus D^{(\frac{1}{2},\frac{3}{2})}\left(\widetilde{SO}_{\circ}(1,3)\right)$		
mielke-geometrodynamics_241 EQ0754 (11.1.21)		■ 1.000 ● N +0	$\psi\rightarrow^{S^{-1}}\psi:=S^{-1}\psi$		

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Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured
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mielke-geometrodynamics_242 EQ0755 (11.1.22)	242	■ 1.000 ● W +0	$\begin{gathered} S(p) \in \mathscr{G}_S \approx C^\infty \left(\tilde{L}(M) \times_{\text{Ad}} \widetilde{SO}_o(s, n-s) \right) \\ \Lambda \left(S_{(p)} \right) = G(p) \in \mathscr{G}_p \\ \end{gathered}$	$S(p) \in \mathscr{G}_S \approx C^\infty \left(\tilde{L}(M) \times_{\text{Ad}} \widetilde{SO}_o(s, n-s) \right) \\ \Lambda \left(S_{(p)} \right) = G(p) \in \mathscr{G}_p$	$S(p) \in \mathscr{G}_S \approx C^\infty (\tilde{L}(M) \times_{\text{Ad}} \widetilde{SO}_o(s, n-s)) \\ \Lambda(S_{(p)}) = G(p) \in \mathscr{G}_p$
mielke-geometrodynamics_242 EQ0757 (11.1.24)	242	■ 1.000 ● N +0	$S^{-1}(p) \gamma_\alpha S(p) = \gamma_\beta G_\alpha{}^\beta(p), \quad [G_\alpha{}^\beta(p)] \in \mathscr{G}_p.$	$S^{-1}(p) \gamma_\alpha S(p) = \gamma_\beta G_\alpha{}^\beta(p), \quad [G_\alpha{}^\beta(p)] \in \mathscr{G}_p.$	$S^{-1}(p) \gamma_\alpha S(p) = \gamma_\beta G_\alpha{}^\beta(p), \quad [G_\alpha{}^\beta(p)] \in \mathscr{G}_p.$
mielke-geometrodynamics_243 EQ0762 (11.1.29)	243	■ 1.000 ● N +0	$\bar{\psi} := \psi^+ \gamma^0,$	$\bar{\psi} := \psi^+ \gamma^0,$	$\bar{\psi} := \psi^+ \gamma^0,$
mielke-geometrodynamics_243 EQ0763 (11.1.30)	243	■ 1.000 ● W -2	$\bar{\sigma}^{\alpha\beta} := \gamma^0 \sigma^{\alpha\beta} + \gamma^0 = \sigma^{\alpha\beta}, \quad \bar{\gamma}^\alpha := \gamma^0 \gamma^{\alpha+} \gamma^0 = \gamma^\alpha.$	$\bar{\sigma}^{\alpha\beta} := \gamma^0 \sigma^{\alpha\beta} + \gamma^0 = \sigma^{\alpha\beta}, \quad \bar{\gamma}^\alpha := \gamma^0 \gamma^{\alpha+} \gamma^0 = \gamma^\alpha.$	$\bar{\sigma}^{\alpha\beta} := \gamma^0 \sigma^{\alpha\beta} + \gamma^0 = \sigma^{\alpha\beta}, \quad \bar{\gamma}^\alpha := \gamma^0 \gamma^{\alpha+} \gamma^0 = \gamma^\alpha.$
mielke-geometrodynamics_243 EQ0764 (11.1.31)	243	■ 1.000 ● K +0	$D \psi = d\psi + i\tilde{\omega}\psi,$	$D\psi = d\psi + i\tilde{\omega}\psi,$	$D\psi = d\psi + i\tilde{\omega}\psi,$
mielke-geometrodynamics_243 EQ0765 (11.1.32)	243	■ 1.000 ● K +0	$\Lambda(i\tilde{\omega}) = \omega^{\mathfrak{g}}$	$\Lambda(i\tilde{\omega}) = \omega^{\mathfrak{g}}$	$\Lambda(i\tilde{\omega}) = \omega^{\mathfrak{g}}$
mielke-geometrodynamics_244 EQ0771 (11.1.38)	244	■ 1.000 ● K +0	$\gamma \longrightarrow {}^{S^{-1}}\gamma = S^{-1}\gamma S$	$\gamma \longrightarrow {}^{S^{-1}}\gamma = S^{-1}\gamma S$	$\gamma \longrightarrow {}^{S^{-1}}\gamma = S^{-1}\gamma S$
mielke-geometrodynamics_246 EQ0775 (11.2.4)	246	■ 1.000 ● K +0	$\widetilde{\omega} = \gamma^0 \widetilde{\omega}^+ \gamma^0 = \widetilde{\omega}$	$\widetilde{\omega} = \gamma^0 \widetilde{\omega}^+ \gamma^0 = \widetilde{\omega}$	$\widetilde{\omega} = \gamma^0 \widetilde{\omega}^+ \gamma^0 = \widetilde{\omega}$
mielke-geometrodynamics_246 EQ0777 (11.2.6)	246	■ 1.000 ● N +0	$\tau_s = i \left(\bar{\psi}^* \gamma \psi \right).$	$\tau_s = i \left(\bar{\psi}^* \gamma \psi \right).$	$\tau_s = i (\bar{\psi}^* \gamma \psi).$
mielke-geometrodynamics_247 EQ0779 (11.2.7)	247	■ 1.000 ● K +0	$d\tau_s = 0$	$d\tau_s = 0$	$d\tau_s = 0$
mielke-geometrodynamics_247 EQ0781 (11.2.9)	247	■ 1.000 ● N +1	$(\psi, \chi) = i \int_{H^{n-1}} \psi^+ \chi d^{n-1}x$	$(\psi, \chi) = i \int_{H^{n-1}} \psi^+ \chi d^{n-1}x$	$(\psi, \chi) = i \int_{H^{n-1}} \psi^+ \chi d^{n-1}x.$
mielke-geometrodynamics_247 EQ0782 (11.3.1)	247	■ 1.000 ● N +0	$\widetilde{\omega} = \widetilde{\omega}^{\{\}} - \widetilde{K}, \quad \Lambda(i\widetilde{K}) = K$	$\widetilde{\omega} = \widetilde{\omega}^{\{\}} - \widetilde{K}, \quad \Lambda(i\widetilde{K}) = K$	$\widetilde{\omega} = \widetilde{\omega}^{\{\}} - \widetilde{K}, \quad \Lambda(i\widetilde{K}) = K$
mielke-geometrodynamics_248 EQ0785 (11.3.4)	248	■ 1.000 ● N +0	$[\Theta, \vartheta] = \ell^{*2} \tau_s.$	$[\Theta, \vartheta] = \ell^{*2} \tau_s.$	$[\Theta, \vartheta] = \ell^{*2} \tau_s.$
mielke-geometrodynamics_252 EQ0790 (11.4.2)	252	■ 1.000 ●	$\begin{aligned} \left(\bar{\psi} i \gamma^5 \gamma_\mu \psi \right) \left(\bar{\psi} i \gamma^5 \gamma^\mu \psi \right) &= \left(\bar{\psi} \gamma_\mu \psi \right) \left(\bar{\psi} \gamma^\mu \psi \right) \\ &= \left(\bar{\psi} \psi \right)^2 - \left(\bar{\psi} \gamma^5 \psi \right)^2. \end{aligned}$	$\begin{aligned} \left(\bar{\psi} i \gamma^5 \gamma_\mu \psi \right) \left(\bar{\psi} i \gamma^5 \gamma^\mu \psi \right) &= \left(\bar{\psi} \gamma_\mu \psi \right) \left(\bar{\psi} \gamma^\mu \psi \right) \\ &= \left(\bar{\psi} \psi \right)^2 - \left(\bar{\psi} \gamma^5 \psi \right)^2. \end{aligned}$	$\begin{aligned} \left(\bar{\psi} i \gamma^5 \gamma_\mu \psi \right) \left(\bar{\psi} i \gamma^5 \gamma^\mu \psi \right) &= \left(\bar{\psi} \gamma_\mu \psi \right) \left(\bar{\psi} \gamma^\mu \psi \right) \\ &= \left(\bar{\psi} \psi \right)^2 - \left(\bar{\psi} \gamma^5 \psi \right)^2. \end{aligned}$
mielke-geometrodynamics_253 EQ0793 (11.4.5)	253	■ 1.000 ● N +0	$\rho := \frac{\sqrt{2\pi}}{\ell^*} r,$	$\rho := \frac{\sqrt{2\pi}}{\ell^*} r,$	$\rho := \frac{\sqrt{2\pi}}{\ell^*} r,$
mielke-geometrodynamics_253 EQ0794 (11.4.6)	253	■ 1.000 ● N +0	$\bar{g}_{kl} \longleftarrow g_{kl} = e^\mu \bar{g}_{kl}.$	$\bar{g}_{kl} \longleftarrow g_{kl} = e^\mu \bar{g}_{kl}.$	$\bar{g}_{kl} \longleftarrow g_{kl} = e^\mu \bar{g}_{kl}.$
mielke-geometrodynamics_253 EQ0795 (11.4.7)	253	■ 1.000 ● W -2	$\bar{E}_{\cdot k}^\alpha \longleftarrow E_{\cdot k}^\alpha = e^{\mu/2} \bar{E}_{\cdot k}^\alpha$	$\bar{E}_{\cdot k}^\alpha \longleftarrow E_{\cdot k}^\alpha = e^{\mu/2} \bar{E}_{\cdot k}^\alpha$	$\bar{E}_{\cdot k}^\alpha \longleftarrow E_{\cdot k}^\alpha = e^{\mu/2} \bar{E}_{\cdot k}^\alpha$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_253 EQ0796 (11.4.8)	253	■ 1.000 ● N +0	$\overline{\left\{\begin{array}{c} \alpha \\ k l \end{array}\right\}}=\overline{\left\{\begin{array}{c} \alpha \\ k l \end{array}\right\}}+\frac{1}{2}\left(\delta_k^{\alpha} \partial_l+\delta_l^{\alpha} \partial_k-g_{k l} \partial^{\alpha}\right) \mu$	$\left\{\begin{array}{c} \alpha \\ k l \end{array}\right\}=\overline{\left\{\begin{array}{c} \alpha \\ k l \end{array}\right\}}+\frac{1}{2}\left(\delta_k^{\alpha} \partial_l+\delta_l^{\alpha} \partial_k-g_{k l} \partial^{\alpha}\right) \mu$	$\left\{\begin{array}{c} \alpha \\ k l \end{array}\right\}=\overline{\left\{\begin{array}{c} \alpha \\ k l \end{array}\right\}}+\frac{1}{2}\left(\delta_k^{\alpha} \partial_l+\delta_l^{\alpha} \partial_k-g_{k l} \partial^{\alpha}\right) \mu$
mielke-geometrodynamics_253 EQ0797 (11.4.9)	253	■ 1.000 ● N +0	$\Gamma_k=\bar{\Gamma}_k-\frac{1}{4} \delta_k^{[\alpha} \partial^{\beta]} \mu \sigma_{\alpha \beta} .$	$\Gamma_k=\bar{\Gamma}_k-\frac{1}{4} \delta_k^{[\alpha} \partial^{\beta]} \mu \sigma_{\alpha \beta} .$	$\Gamma_k=\bar{\Gamma}_k-\frac{1}{4} \delta_k^{[\alpha} \partial^{\beta]} \mu \sigma_{\alpha \beta} .$
mielke-geometrodynamics_254 EQ0801 (11.4.13)	254	■ 1.000 ● N +1	$\begin{aligned} & \& \bar{\Gamma}_0=\frac{i}{4} e^{(\nu-\mu) / 2} \partial_r(\nu-\mu) \gamma_0 \gamma_1 \\ & \bar{\Gamma}_1=0, \quad \bar{\Gamma}_2=\frac{i}{2} \gamma_2 \gamma_1 \\ & \bar{\Gamma}_3=\frac{i}{2}(\sin \vartheta \gamma_3 \gamma_1+\cos \vartheta \gamma_3 \gamma_2) \end{aligned}$	$\bar{\Gamma}_0=\frac{i}{4} e^{(\nu-\mu) / 2} \partial_r(\nu-\mu) \gamma_0 \gamma_1$ $\bar{\Gamma}_1=0, \quad \bar{\Gamma}_2=\frac{i}{2} \gamma_2 \gamma_1$ $\bar{\Gamma}_3=\frac{i}{2}(\sin \vartheta \gamma_3 \gamma_1+\cos \vartheta \gamma_3 \gamma_2)$	$\bar{\Gamma}_0=\frac{i}{4} e^{(v-\mu) / 2} \partial_r(v-\mu) \gamma_0 \gamma_1$ $\bar{\Gamma}_1=0, \quad \bar{\Gamma}_2=\frac{i}{2} \gamma_2 \gamma_1$ $\bar{\Gamma}_3=\frac{i}{2}(\sin \vartheta \gamma_3 \gamma_1+\cos \vartheta \gamma_3 \gamma_2)$
mielke-geometrodynamics_255 EQ0804 (11.4.16)	255	■ 1.000 ● N +0	$\chi_{\kappa}^m=\sum_{\bar{m}=\pm 1 / 2} C\left(l_{\frac{1}{2}} j ; m-\bar{m}, \bar{m}\right) Y_l^{m-\bar{m}}(\vartheta, \phi) \chi^{\bar{m}} .$	$\chi_{\kappa}^m=\sum_{\bar{m}=\pm 1 / 2} C\left(l_{\frac{1}{2}} j ; m-\bar{m}, \bar{m}\right) Y_l^{m-\bar{m}}(\vartheta, \phi) \chi^{\bar{m}} .$	$\chi_{\kappa}^m=\sum_{\bar{m}=\pm 1 / 2} C\left(l_{\frac{1}{2}} j ; m-\bar{m}, \bar{m}\right) Y_l^{m-\bar{m}}(\vartheta, \phi) \chi^{\bar{m}} .$
mielke-geometrodynamics_255 EQ0808 (11.4.20)	255	■ 1.000 ● W +0	$\frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } \chi_{\kappa}^m=-\chi_{-\kappa}^m$	$\frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } \chi_{\kappa}^m=-\chi_{-\kappa}^m$	$\frac{\vec{\sigma} \cdot \vec{x}}{ \vec{x} } \chi_{\kappa}^m=-\chi_{-\kappa}^m$
mielke-geometrodynamics_257 EQ0814 (11.4.27)	257	■ 1.000 ● W +1	$\beta=\frac{M^*}{2 m}, \quad M^*:=\frac{\sqrt{8 \pi} \hbar}{c \ell^*} .$	$\beta=\frac{M^*}{2 m}, \quad M^*:=\frac{\sqrt{8 \pi} \hbar}{c \ell^*} .$	$\beta=\frac{M^*}{2 m}, \quad M^*:=\frac{\sqrt{8 \pi} \hbar}{c \ell^*} .$
mielke-geometrodynamics_257 EQ0816 (11.4.30)	257	■ 1.000 ● K +0	$F^{\prime \prime}+\frac{2}{\rho} F^{\prime}-\frac{2}{\rho^2} F \simeq \frac{1-\omega^2}{\beta^2} F$	$F^{\prime \prime}+\frac{2}{\rho} F^{\prime}-\frac{2}{\rho^2} F \simeq \frac{1-\omega^2}{\beta^2} F$	$F^{\prime \prime}+\frac{2}{\rho} F^{\prime}-\frac{2}{\rho^2} F \simeq \frac{1-\omega^2}{\beta^2} F$
mielke-geometrodynamics_257 EQ0817 (11.4.31)	257	■ 1.000 ● N +0	$F \sim C_{\infty}\left(\frac{1}{\bar{\rho}}+\frac{1}{\bar{\rho}^2}\right) e^{-\bar{\rho}}$	$F \sim C_{\infty}\left(\frac{1}{\bar{\rho}}+\frac{1}{\bar{\rho}^2}\right) e^{-\bar{\rho}}$	$F \sim C_{\infty}\left(\frac{1}{\bar{\rho}}+\frac{1}{\bar{\rho}^2}\right) e^{-\bar{\rho}}$
mielke-geometrodynamics_257 EQ0818 (11.4.32)	257	■ 1.000 ● N +0	$H \sim C_{\infty} \sqrt{\frac{1+\omega}{1-\omega}} \frac{1}{\bar{\rho}} e^{-\bar{\rho}}$	$H \sim C_{\infty} \sqrt{\frac{1+\omega}{1-\omega}} \frac{1}{\bar{\rho}} e^{-\bar{\rho}}$	$H \sim C_{\infty} \sqrt{\frac{1+\omega}{1-\omega}} \frac{1}{\bar{\rho}} e^{-\bar{\rho}}$
mielke-geometrodynamics_257 EQ0819 (11.4.33)	257	■ 1.000 ●	$\bar{\rho}:=\frac{1}{\beta} \sqrt{1-\omega^2} \rho$	$\bar{\rho}:=\frac{1}{\beta} \sqrt{1-\omega^2} \rho$	$\bar{\rho}:=\frac{1}{\beta} \sqrt{1-\omega^2} \rho$
mielke-geometrodynamics_258 EQ0821 (11.4.35)	258	■ 1.000 ● K +0	$F(\rho) \simeq C_0 f^1(\bar{\rho}) \frac{1}{\cosh \bar{\rho}}$	$F(\rho) \simeq C_0 f^1(\bar{\rho}) \frac{1}{\cosh \bar{\rho}}$	$F(\rho) \simeq C_0 f^1(\bar{\rho}) \frac{1}{\cosh \bar{\rho}}$
mielke-geometrodynamics_258 EQ0822 (11.4.36)	258	■ 1.000 ● N +0	$H(\rho) \simeq C_0 \sqrt{\frac{1+\omega}{1-\omega}} f^0(\bar{\rho}) \frac{1}{\cosh \bar{\rho}}$	$H(\rho) \simeq C_0 \sqrt{\frac{1+\omega}{1-\omega}} f^0(\bar{\rho}) \frac{1}{\cosh \bar{\rho}}$	$H(\rho) \simeq C_0 \sqrt{\frac{1+\omega}{1-\omega}} f^0(\bar{\rho}) \frac{1}{\cosh \bar{\rho}}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E08823 (11.4.37)	258	■ 1.000 ● K +0	$C_{\{0\}}=(1-\omega)\sqrt{\frac{-2}{\varepsilon(\omega+1)}}$	$C_0 = (1 - \omega) \sqrt{\frac{-2}{\varepsilon(\omega + 1)}}$	$C_0 = (1 - \omega) \sqrt{\frac{-2}{\varepsilon(\omega + 1)}}$
mielke-geometrodynamics_E08826 (11.4.40)	259	■ 1.000 ● N +0	$m_{\{s\}}=\frac{1}{c}\int_{H^3}\Sigma_{\mathrm{D}}^0$	$m_s = \frac{1}{c} \int_{H^3} \Sigma_{\mathrm{D}}^0$	$m_s = \frac{1}{c} \int_{H^3} \Sigma_{\mathrm{D}}^0$
mielke-geometrodynamics_E08827 (11.5.1)	262	■ 1.000 ● N +1	$Q:=i\,e\int_{H^3}\bar{\psi}^*\gamma\psi=k\,e,\quad k=0,1,2,\ldots$	$Q := ie \int_{H^3} \bar{\psi}^* \gamma \psi = ke, \quad k = 0, 1, 2, \dots$	$Q := ie \int_{H^3} \bar{\psi}^* \gamma \psi = ke, \quad k = 0, 1, 2, \dots$
mielke-geometrodynamics_E08828 (11.5.2)	263	■ 1.000 ● N +0	$i\gamma\wedge^*d\psi(x)+e^2\int_{y\in M}\bar{\psi}(y)\gamma\psi(y)D_{\mathrm{F}}(y-x)\wedge^*\gamma\psi(x)=m\psi(x)\eta$	$i\gamma\wedge^*d\psi(x)+e^2\int_{y\in M}\bar{\psi}(y)\gamma\psi(y)D_{\mathrm{F}}(y-x)\wedge^*\gamma\psi(x)=m\psi(x)\eta$	$i\gamma\wedge^*d\psi(x)+e^2\int_{y\in M}\bar{\psi}(y)\gamma\psi(y)D_{\mathrm{F}}(y-x)\wedge^*\gamma\psi(x)=m\psi(x)\eta$
mielke-geometrodynamics_E08830 (12.1.2)	269	■ 1.000 ● N +0	$\left\langle d j_5(x)\right\rangle=2 i m\left\langle P\right\rangle-\left(1 / 96 \pi^2\right) F \wedge F$	$\langle dj_5(x)\rangle = 2im\langle P\rangle - (1/96\pi^2) F \wedge F$	$\langle dj_5(x)\rangle = 2im\langle P\rangle - (1/96\pi^2) F \wedge F$
mielke-geometrodynamics_E08838 (12.2.6)	271	■ 1.000 ● N +0	$d\,j_{\{5\}}=2\,i\,m\,P=2\,i\,m\,\bar{\psi}\gamma_5\psi\eta$	$dj_5 = 2imP = 2im\bar{\psi}\gamma_5\psi\eta$	$dj_5 = 2imP = 2im\bar{\psi}\gamma_5\psi\eta$
mielke-geometrodynamics_E08843 (12.2.11)	272	■ 1.000 ● N +0	$L_{a\psi\psi}=\frac{1}{2}d\theta\wedge j_5=\frac{1}{2f_a}da\wedge\bar{\psi}^*\gamma\gamma_5\psi,$	$L_{a\psi\psi} = \frac{1}{2}d\theta \wedge j_5 = \frac{1}{2f_a}da \wedge \bar{\psi}^* \gamma \gamma_5 \psi,$	$L_{a\psi\psi} = \frac{1}{2}d\theta \wedge j_5 = \frac{1}{2f_a}da \wedge \bar{\psi}^* \gamma \gamma_5 \psi,$
mielke-geometrodynamics_E08846 (12.2.14)	273	■ 1.000 ● N -1	$d\,j_{\{5\}}\cong 4\,d\,C_{\mathrm{TT}}=\frac{2}{\ell^2}\left(T^{\alpha}\wedge T_{\alpha}+R_{\alpha\beta}\wedge\vartheta^{\alpha}\wedge\vartheta^{\beta}\right)$	$dj_5 \cong 4dC_{\mathrm{TT}} = \frac{2}{\ell^2} (T^\alpha \wedge T_\alpha + R_{\alpha\beta} \wedge \vartheta^\alpha \wedge \vartheta^\beta)$	$dj_5 \cong 4dC_{\mathrm{TT}} = \frac{2}{\ell^2} (T^\alpha \wedge T_\alpha + R_{\alpha\beta} \wedge \vartheta^\alpha \wedge \vartheta^\beta)$
mielke-geometrodynamics_E08849 (12.3.2)	275	■ 1.000 ● N +0	$K\left(t,\,x,\,\not{D}^2\right)=\left(4\pi\right)^{-n/2}\sum_{k=0}^{\infty}t^{\left(k-n\right)/2}K_k\left(x,\,\not{D}^2\right)$	$K\left(t,x,\not{D}^2\right) = (4\pi)^{-n/2} \sum_{k=0}^{\infty} t^{(k-n)/2} K_k\left(x,\not{D}^2\right)$	$K(t,x,\not{D}^2) = (4\pi)^{-n/2} \sum_{k=0}^{\infty} t^{(k-n)/2} K_k(x,\not{D}^2)$
mielke-geometrodynamics_E08852 (12.3.5)	276	■ 1.000 ● N -1	$-K_{\{2\}}/\,t=\left(2\,\ell^2/t\right)dC_{\mathrm{TT}}\cong\left(\ell^2/2t\right)dj_5=\left(im\ell^2/t\right)\bar{\psi}\gamma_5\psi\sim\ell^2M^2m^4\rightarrow 0\,.$	$-K_2/t = (2\ell^2/t) dC_{\mathrm{TT}} \cong (\ell^2/2t) dj_5 = (im\ell^2/t) \bar{\psi}\gamma_5\psi \sim \ell^2 M^2 m^4 \rightarrow 0.$	$-K_2/t = (2\ell^2/t) dC_{\mathrm{TT}} \cong (\ell^2/2t) dj_5 = (im\ell^2/t) \bar{\psi}\gamma_5\psi \sim \ell^2 M^2 m^4 \rightarrow 0.$
mielke-geometrodynamics_E08856 (12.4.3)	277	■ 1.000 ● N +0	$\mathscr{G}^A\cong 0,\quad \text{A=1,2,3}$	$\mathscr{G}^A \cong 0, \quad A = 1, 2, 3$	$\mathscr{G}^A \cong 0, \quad A = 1, 2, 3$
mielke-geometrodynamics_E08858 (12.4.5)	278	■ 1.000 ● K +0	$\gamma:=\gamma_{\alpha}\vartheta^{\alpha},\quad \gamma^*=\gamma^{\alpha}\eta_{\alpha}.$	$\gamma := \gamma_{\alpha} \vartheta^{\alpha}, \quad \gamma^* = \gamma^{\alpha} \eta_{\alpha}.$	$\gamma := \gamma_{\alpha} \vartheta^{\alpha}, \quad \gamma^* = \gamma^{\alpha} \eta_{\alpha}.$
mielke-geometrodynamics_E08859 (12.4.6)	278	■ 1.000 ● N +0	$\Theta:=D\,\gamma=T^{\alpha}\gamma_{\alpha},\quad \Omega:=d\Gamma+\Gamma\wedge\Gamma=\frac{i}{4}R^{\alpha\beta}\sigma_{\alpha\beta}$	$\Theta := D\gamma = T^\alpha \gamma_\alpha, \quad \Omega := d\Gamma + \Gamma \wedge \Gamma = \frac{i}{4} R^{\alpha\beta} \sigma_{\alpha\beta}$	$\Theta := D\gamma = T^\alpha \gamma_\alpha, \quad \Omega := d\Gamma + \Gamma \wedge \Gamma = \frac{i}{4} R^{\alpha\beta} \sigma_{\alpha\beta}$
mielke-geometrodynamics_E08860 (12.4.7)	278	■ 1.000 ●	$C_{\mathrm{RR}}:=-\operatorname{Tr}\left(\Gamma\wedge\Omega-\frac{1}{3}\Gamma\wedge\Gamma\wedge\Gamma\right).$	$C_{\mathrm{RR}} := -\operatorname{Tr} \left(\Gamma \wedge \Omega - \frac{1}{3} \Gamma \wedge \Gamma \wedge \Gamma \right).$	$C_{\mathrm{RR}} := -\operatorname{Tr} \left(\Gamma \wedge \Omega - \frac{1}{3} \Gamma \wedge \Gamma \wedge \Gamma \right).$
mielke-geometrodynamics_E08862 (12.4.9)	279	■ 1.000 ● N +0	$C_{\mathrm{TT}}:=\frac{1}{8\ell^2}\operatorname{Tr}(\gamma\wedge\Theta)=\frac{1}{2\ell^2}\vartheta^{\alpha}\wedge T_{\alpha}.$	$C_{\mathrm{TT}} := \frac{1}{8\ell^2} \operatorname{Tr}(\gamma \wedge \Theta) = \frac{1}{2\ell^2} \vartheta^\alpha \wedge T_\alpha.$	$C_{\mathrm{TT}} := \frac{1}{8\ell^2} \operatorname{Tr}(\gamma \wedge \Theta) = \frac{1}{2\ell^2} \vartheta^\alpha \wedge T_\alpha.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E09863 (12.4.10)	279	■ 1.000 ● C +4	$d\,C_{\mathrm{TT}}=\frac{1}{8}\ell^2\operatorname{Tr}(\Theta\wedge\Theta-4i\Omega\wedge\sigma)=\frac{1}{2\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right).$	$dC_{\mathrm{TT}}=\frac{1}{8\ell^2}\operatorname{Tr}(\Theta\wedge\Theta-4i\Omega\wedge\sigma)=\frac{1}{2\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right).$	$dC_{\mathrm{TT}}=\frac{1}{8\ell^2}\operatorname{Tr}(\Theta\wedge\Theta-4i\Omega\wedge\sigma)=\frac{1}{2\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right).$
mielke-geometrodynamics_E09864 (12.4.11)	279	■ 1.000 ● C -3	$\hat{C}_{\mathrm{C}}_{\mathrm{RR}}=C_{\mathrm{RR}}-2\,C_{\mathrm{TT}}$	$\hat{C}_{\mathrm{RR}}=C_{\mathrm{RR}}-2C_{\mathrm{TT}}$	$\hat{C}_{\mathrm{RR}}=C_{\mathrm{RR}}-2C_{\mathrm{TT}}$
mielke-geometrodynamics_E09865 (13.2.1)	283	■ 1.000 ● N -1	$L_{\mathrm{BF}}=-B\wedge F=-B\wedge dA.$	$L_{\mathrm{BF}}=-B\wedge F=-B\wedge dA.$	$L_{\mathrm{BF}}=-B\wedge F=-B\wedge dA.$
mielke-geometrodynamics_E09866 (13.2.2)	283	■ 1.000 ● N +1	$\tilde{L}_{\mathrm{BF}}:=-B\wedge dA+\frac{1}{2}B\wedge B\cong-B\wedge dA+dC$	$\tilde{L}_{\mathrm{BF}}:=-B\wedge dA+\frac{1}{2}B\wedge B\cong-B\wedge dA+dC$	$\tilde{L}_{\mathrm{BF}}:=-B\wedge dA+\frac{1}{2}B\wedge B\cong-B\wedge dA+dC.$
mielke-geometrodynamics_E09868 (13.2.4)	283	■ 1.000 ● N +0	$C=\frac{1}{2}A\wedge F,\quad dC=\frac{1}{2}F\wedge F,$	$C=\frac{1}{2}A\wedge F,\quad dC=\frac{1}{2}F\wedge F,$	$C=\frac{1}{2}A\wedge F,\quad dC=\frac{1}{2}F\wedge F,$
mielke-geometrodynamics_E09869 (13.2.5)	283	■ 1.000 ● K +0	$A\rightarrow A'=A+d\theta(x),$	$A\rightarrow A'=A+d\theta(x),$	$A\rightarrow A'=A+d\theta(x),$
mielke-geometrodynamics_E09870 (13.2.6)	284	■ 1.000 ● N +0	$A\rightarrow\widetilde{A}=A+\sigma,\quad B\rightarrow\widetilde{B}=B+d\sigma,$	$A\rightarrow\widetilde{A}=A+\sigma,\quad B\rightarrow\widetilde{B}=B+d\sigma,$	$A\rightarrow\widetilde{A}=A+\sigma,\quad B\rightarrow\widetilde{B}=B+d\sigma,$
mielke-geometrodynamics_E09871 (13.2.7)	284	■ 1.000 ● K +0	$L_{\mathrm{Max}}=-B\wedge dA+\frac{1}{2}B\wedge{}^*B+L_{\mathrm{matter}},$	$L_{\mathrm{Max}}=-B\wedge dA+\frac{1}{2}B\wedge{}^*B+L_{\mathrm{matter}},$	$L_{\mathrm{Max}}=-B\wedge dA+\frac{1}{2}B\wedge{}^*B+L_{\mathrm{matter}},$
mielke-geometrodynamics_E09872 (13.2.8)	284	■ 1.000 ● N +1	$-dB=d{}^*F\cong j,$	$-dB=d{}^*F\cong j,$	$-dB=d{}^*F\cong j,$
mielke-geometrodynamics_E09875 (13.3.3)	284	■ 1.000 ● N +0	$[L^\alpha{}_\beta,L^\gamma{}_\delta]=\delta^\alpha_\delta L^\gamma{}_\beta-\delta^\gamma_\beta L^\alpha{}_\delta,$	$[L^\alpha{}_\beta,L^\gamma{}_\delta]=\delta^\alpha_\delta L^\gamma{}_\beta-\delta^\gamma_\beta L^\alpha{}_\delta,$	$[L^\alpha{}_\beta,L^\gamma{}_\delta]=\delta^\alpha_\delta L^\gamma{}_\beta-\delta^\gamma_\beta L^\alpha{}_\delta,$
mielke-geometrodynamics_E09880 (13.4.1)	285	■ 1.000 ● N +0	$A(x):=\left(\begin{array}{cc}\Lambda(x)&\tau(x)\\ \tau_*(x)&1\end{array}\right),$	$A(x):=\left\{\begin{array}{cc}\Lambda(x)&\tau(x)\\ \tau_*(x)&1\end{array}\right\},$	$A(x):=\left\{\begin{array}{cc}A(x)&\tau(x)\\ \tau_*(x)&1\end{array}\right\},$
mielke-geometrodynamics_E09881 (13.4.2)	285	■ 1.000 ● W +0	$\hat{\Gamma}=\left(\begin{array}{cc}\Gamma^{\mathrm{(L)}}&\Gamma^{\mathrm{(T)}}\\ \Gamma^{\mathrm{(T)}}_*&0\end{array}\right)=\left(\begin{array}{cc}\Gamma^{\mathrm{(L)}\beta}_\alpha L^\alpha_\beta & \Gamma^{\mathrm{(T)}\alpha}\ell P_\alpha\\ \Gamma^{*(\mathrm{T})}_\beta & \ell P^\beta_*\end{array}\right)$	$\hat{F}=\left(\begin{array}{cc}\Gamma^{(L)}&\Gamma^{(T)}\\ \Gamma^{(T)}_*&0\end{array}\right)=\left(\begin{array}{cc}\Gamma^{(L)\beta}_\alpha L^\alpha_\beta & \Gamma^{(T)\alpha}\ell P_\alpha\\ \Gamma^{*(T)}_\beta & \ell P^\beta_*\end{array}\right)$	$\hat{F}=\left(\begin{array}{cc}\Gamma^{(L)}&\Gamma^{(T)}\\ \Gamma^{(T)}_*&0\end{array}\right)=\left(\begin{array}{cc}\Gamma^{(L)\beta}_\alpha L^\alpha_\beta & \Gamma^{(T)\alpha}\ell P_\alpha\\ \Gamma^{*(T)}_\beta & \ell P^\beta_*\end{array}\right)$
mielke-geometrodynamics_E09882 (13.4.3)	286	■ 1.000 ● N +1	$\hat{\Gamma}\stackrel{A^{-1}(x)}{\longrightarrow}\hat{\Gamma}'=A^{-1}(x)\hat{\Gamma}A(x)+A^{-1}(x)dA(x).$	$\hat{F}\stackrel{A^{-1}(x)}{\longrightarrow}\hat{F}'=A^{-1}(x)\hat{F}A(x)+A^{-1}(x)dA(x).$	$\hat{F}\stackrel{A^{-1}(x)}{\longrightarrow}\hat{F}'=A^{-1}(x)\hat{F}A(x)+A^{-1}(x)dA(x).$
mielke-geometrodynamics_E09883	286	■ 1.000 ● N +0	$\tau\tau_*=-\Lambda\tau_*\Lambda^{-1}\tau.$	$\tau\tau_*=-\Lambda\tau_*\Lambda^{-1}\tau.$	$\tau\tau_*=-\Lambda\tau_*\Lambda^{-1}\tau.$
mielke-geometrodynamics_E09884 (13.4.4)	286	■ 1.000 ●	$\hat{R}:=d\hat{\Gamma}+\hat{\Gamma}\wedge\hat{\Gamma}=\left(\begin{array}{cc}R^{(L)}+\Gamma^{(L)}\wedge\Gamma^{(L)}_* & d\Gamma^{(T)}+\Gamma^{(L)}\wedge\Gamma^{(T)}\\ d\Gamma^{(T)}_*+\Gamma^{(T)}\wedge\Gamma^{(L)} & 0\end{array}\right)$	$\hat{R}:=d\hat{F}+\hat{F}\wedge\hat{F}=\left(\begin{array}{cc}R^{(L)}+\Gamma^{(L)}\wedge\Gamma^{(L)}_* & d\Gamma^{(T)}+\Gamma^{(L)}\wedge\Gamma^{(T)}\\ d\Gamma^{(T)}_*+\Gamma^{(T)}\wedge\Gamma^{(L)} & 0\end{array}\right),$	$\hat{R}:=d\hat{F}+\hat{F}\wedge\hat{F}=\left(\begin{array}{cc}R^{(L)}+\Gamma^{(L)}\wedge\Gamma^{(L)}_* & d\Gamma^{(T)}+\Gamma^{(L)}\wedge\Gamma^{(T)}\\ d\Gamma^{(T)}_*+\Gamma^{(T)}\wedge\Gamma^{(L)} & 0\end{array}\right),$
mielke-geometrodynamics_E09885	286	■ 1.000 ● K +0	$R^{(L)}:=d\Gamma^{(L)}+\Gamma^{(L)}\wedge\Gamma^{(L)}$	$R^{(L)}:=d\Gamma^{(L)}+\Gamma^{(L)}\wedge\Gamma^{(L)}$	$R^{(L)}:=d\Gamma^{(L)}+\Gamma^{(L)}\wedge\Gamma^{(L)}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C ● component ● W ● weak ● S ● stable ● N ● noise ● K ● clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_286 EQ0886 (13.4.5)	286	■ 1.000 ● N +0	$R^{\{T\}}:=d\,\Gamma^{\{T\}}+\Gamma^{\{L\}}\wedge\Gamma^{\{T\}}=\left(T^{\wedge\{\beta\}}-R_{\alpha}\,\xi^{\alpha}\right)\ell P_{\beta}\,\ell P_{\beta}$	$R^{(T)}:=d\Gamma^{(T)}+\Gamma^{(L)}\wedge\Gamma^{(T)}=\left(T^{\beta}-R_{\alpha}{}^{\beta}\xi^{\alpha}\right)\ell P_{\beta}$	$R^{(T)}:=d\Gamma^{(T)}+\Gamma^{(L)}\wedge\Gamma^{(T)}=\left(T^{\beta}-R_{\alpha}{}^{\beta}\xi^{\alpha}\right)\ell P_{\beta}$
mielke-geometrodynamics_286 EQ0887 (13.4.6)	286	■ 1.000 ● N +1	$\hat{R}\mapsto A^{-1}(x)\hat{R}A(x)\,.$	$\hat{R}\xrightarrow{A^{-1}(x)}\hat{R}'=A^{-1}(x)\hat{R}A(x)\,.$	$\hat{R}\xrightarrow{A^{-1}(x)}\hat{R}'=A^{-1}(x)\hat{R}A(x)\,.$
mielke-geometrodynamics_286 EQ0888 (13.4.7)	286	■ 1.000 ● N +0	$\Gamma^{\{L\}}\mapsto A^{-1}(x)\,\Gamma^{\{L\}}\,\Lambda(x)+\Lambda^{-1}(x)d\Lambda(x)$	$\Gamma^{(L)}\xrightarrow{A^{-1}(x)}\Gamma^{(L)'}=\Lambda^{-1}(x)\Gamma^{(L)}\Lambda(x)+\Lambda^{-1}(x)d\Lambda(x)$	$\Gamma^{(L)}\xrightarrow{A^{-1}(x)}\Gamma^{(L)'}=\Lambda^{-1}(x)\,\Gamma^{(L)}\,\Lambda(x)+\Lambda^{-1}(x)d\Lambda(x)$
mielke-geometrodynamics_287 EQ0891 (13.4.10)	287	■ 1.000 ● N +0	$\vartheta:=\Gamma^{\{T\}}+D\xi:=\vartheta^{\alpha}P_{\alpha},\quad\theta:=\Gamma^{\{T\}}_{*}+D\xi_{*}:=\theta_{\beta}P^{\beta}_{*},$	$\vartheta:=\Gamma^{(T)}+D\xi:=\vartheta^{\alpha}P_{\alpha},\quad\theta:=\Gamma^{(T)}_{*}+D\xi_{*}:=\theta_{\beta}P^{\beta}_{*},$	$\vartheta:=\Gamma^{(T)}+D\xi:=\vartheta^{\alpha}P_{\alpha},\quad\theta:=\Gamma^{(T)}_{*}+D\xi_{*}:=\theta_{\beta}P^{\beta}_{*},$
mielke-geometrodynamics_287 EQ0894 (13.4.13)	287	■ 1.000 ● N +0	$D\,\xi=0,\quad D\,\xi_{*}=0,$	$D\xi=0,\quad D\xi_{*}=0,$	$D\xi=0,\quad D\xi_{*}=0,$
mielke-geometrodynamics_287 EQ0895 (13.4.14)	287	■ 1.000 ● N +1	$\overline{\Gamma}=\left(\begin{array}{cc}\Gamma^{(L)}&\vartheta\\ \theta&0\end{array}\right),$	$\overline{\Gamma}=\left(\begin{array}{cc}\Gamma^{(L)}&\vartheta\\ \theta&0\end{array}\right),$	$\overline{\Gamma}=\left(\begin{array}{cc}\Gamma^{(L)}&\vartheta\\ \theta&0\end{array}\right),$
mielke-geometrodynamics_288 EQ0896 (13.5.1)	288	■ 1.000 ● N +0	$\hat{\Gamma}=\Gamma_{\alpha}{}^{\beta}L^{\alpha}_{\beta}+\vartheta^{\alpha}P_{\alpha}+\theta_{\beta}P^{\beta}_{*},$	$\hat{\Gamma}=\Gamma_{\alpha}{}^{\beta}L^{\alpha}_{\beta}+\vartheta^{\alpha}P_{\alpha}+\theta_{\beta}P^{\beta}_{*},$	$\hat{\Gamma}=\Gamma_{\alpha}{}^{\beta}L^{\alpha}_{\beta}+\vartheta^{\alpha}P_{\alpha}+\theta_{\beta}P^{\beta}_{*},$
mielke-geometrodynamics_288 EQ0899 (13.5.4)	288	■ 1.000 ● K +0	$\tilde{L}_{\mathrm{SL}(5,\mathbb{R})}=-\operatorname{Tr}\{\hat{B}\wedge\hat{R}\}-d\hat{C}\,.$	$\tilde{L}_{\mathrm{SL}(5,\mathbb{R})}=-\operatorname{Tr}\{\hat{B}\wedge\hat{R}\}-d\hat{C}\,.$	$\tilde{L}_{\mathrm{SL}(5,\mathbb{R})}=-\operatorname{Tr}\{\hat{B}\wedge\hat{R}\}-d\hat{C}\,.$
mielke-geometrodynamics_289 EQ0900 (13.5.5)	289	■ 1.000 ● N +0	$\hat{D}B_A{}^B\cong\hat{D}\hat{R}_A{}^B\equiv0\,.$	$\hat{D}B_A{}^B\cong\hat{D}\hat{R}_A{}^B\equiv0\,.$	$\hat{D}B_A{}^B\cong\hat{D}\hat{R}_A{}^B\equiv0\,.$
mielke-geometrodynamics_289 EQ0901 (13.5.6)	289	■ 1.000 ● N -1	$\tilde{L}_{\mathrm{SSB}}=\tilde{L}_{\mathrm{SL}(5,\mathbb{R})}+\frac{1}{2}\eta_{ABCDE}B^{AB}\wedge B^{CD}\Phi^E\,.$	$\tilde{L}_{\mathrm{SSB}}=\tilde{L}_{\mathrm{SL}(5,\mathbb{R})}+\frac{1}{2}\eta_{ABCDE}B^{AB}\wedge B^{CD}\Phi^E\,.$	$\tilde{L}_{\mathrm{SSB}}=\tilde{L}_{\mathrm{SL}(5,\mathbb{R})}+\frac{1}{2}\eta_{ABCDE}B^{AB}\wedge B^{CD}\Phi^E\,.$
mielke-geometrodynamics_289 EQ0902 (13.5.7)	289	■ 1.000 ● N +0	$\hat{g}_{AB}\Phi^A\Phi^B=\mu^2,\quad\Phi_E\hat{D}\Phi^E=0$	$\hat{g}_{AB}\Phi^A\Phi^B=\mu^2,\quad\Phi_E\hat{D}\Phi^E=0,$	$\hat{g}_{AB}\Phi^A\Phi^B=\mu^2,\quad\Phi_E\hat{D}\Phi^E=0,$
mielke-geometrodynamics_289 EQ0903 (13.5.8)	289	■ 1.000 ● K +0	$\langle\Phi^E\rangle=\Phi_0^E=(0,0,0,0,\mu)^T$	$\langle\Phi^E\rangle=\Phi_0^E=(0,0,0,0,\mu)^T$	$\langle\Phi^E\rangle=\Phi_0^E=(0,0,0,0,\mu)^T$
mielke-geometrodynamics_289 EQ0904 (13.5.9)	289	■ 1.000 ● N -1	$\tilde{L}_{\mathrm{SSB}}=-\operatorname{Tr}\{\hat{B}\wedge\hat{R}\}+\frac{\mu}{2}\eta_{\alpha\beta\gamma\delta}B^{\alpha\beta}\wedge B^{\gamma\delta}-dC_{\mathrm{RR}}+2dC_{\mathrm{TT}}$	$\tilde{L}_{\mathrm{SSB}}=-\operatorname{Tr}\{\hat{B}\wedge\hat{R}\}+\frac{\mu}{2}\eta_{\alpha\beta\gamma\delta}B^{\alpha\beta}\wedge B^{\gamma\delta}-dC_{\mathrm{RR}}+2dC_{\mathrm{TT}}$	$\tilde{L}_{\mathrm{SSB}}=-\operatorname{Tr}\{\hat{B}\wedge\hat{R}\}+\frac{\mu}{2}\eta_{\alpha\beta\gamma\delta}B^{\alpha\beta}\wedge B^{\gamma\delta}-dC_{\mathrm{RR}}+2dC_{\mathrm{TT}}$
mielke-geometrodynamics_290 EQ0905 (13.5.10)	290	■ 1.000 ● K +0	$d\,\hat{C}=d\,C_{\mathrm{RR}}-2d\,C_{\mathrm{TT}},$	$d\hat{C}=dC_{\mathrm{RR}}-2dC_{\mathrm{TT}},$	$d\hat{C}=dC_{\mathrm{RR}}-2dC_{\mathrm{TT}},$
mielke-geometrodynamics_290 EQ0906 (13.5.11)	290	■ 1.000 ● K +0	$B_{\alpha\beta}\cong\frac{1}{\mu}\eta_{\alpha\beta\gamma\delta}\left[R^{\gamma\delta}-\frac{\Lambda}{3}\theta^{\gamma}\wedge\vartheta^{\delta}\right]$	$B_{\alpha\beta}\cong\frac{1}{\mu}\eta_{\alpha\beta\gamma\delta}\left[R^{\gamma\delta}-\frac{\Lambda}{3}\theta^{\gamma}\wedge\vartheta^{\delta}\right]$	$B_{\alpha\beta}\cong\frac{1}{\mu}\eta_{\alpha\beta\gamma\delta}\left[R^{\gamma\delta}-\frac{\Lambda}{3}\theta^{\gamma}\wedge\vartheta^{\delta}\right]$
mielke-geometrodynamics_291 EQ0909 (13.5.14)	291	■ 1.000 ●	$\tilde{L}_{\mathrm{SSB}}\cong-\frac{1}{2\kappa}R^{\alpha\beta}\wedge\eta_{\alpha\beta}+\frac{\Lambda}{\kappa}\eta$	$\tilde{L}_{\mathrm{SSB}}\cong-\frac{1}{2\kappa}R^{\alpha\beta}\wedge\eta_{\alpha\beta}+\frac{\Lambda}{\kappa}\eta$	$\tilde{L}_{\mathrm{SSB}}\cong-\frac{1}{2\kappa}R^{\alpha\beta}\wedge\eta_{\alpha\beta}+\frac{\Lambda}{\kappa}\eta$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_EQ0910 (13.5.15)	291	■ 1.000 ● N +0	$\mu=\frac{4}{3}\kappa\Lambda=\frac{1}{3}(2\kappa)^2\rho_\Lambda$	$\mu=\frac{4}{3}\kappa\Lambda=\frac{1}{3}(2\kappa)^2\rho_\Lambda.$	$\mu=\frac{4}{3}\kappa\Lambda=\frac{1}{3}(2\kappa)^2\rho_\Lambda.$
mielke-geometrodynamics_EQ0912 (13.7.1)	293	■ 1.000 ● N +0	$g_{\mathrm{N}}:=\kappa k^2, \quad \lambda:=\Lambda/k^2,$	$g_{\mathrm{N}}:=\kappa k^2, \quad \lambda:=\Lambda/k^2,$	$g_{\mathrm{N}}:=\kappa k^2, \quad \lambda:=\Lambda/k^2,$
mielke-geometrodynamics_EQ0913 (13.7.2)	293	■ 1.000 ● S +0	$k\frac{\partial}{\partial k}g_{\mathrm{N}}=\beta_1(g_{\mathrm{N}},\lambda)=(2+d_{\mathrm{N}})g_{\mathrm{N}}, \quad k\frac{\partial}{\partial k}\lambda=\beta_2(g_{\mathrm{N}},\lambda),$	$k\frac{\partial}{\partial k}g_{\mathrm{N}}=\beta_1(g_{\mathrm{N}},\lambda)=(2+d_{\mathrm{N}})g_{\mathrm{N}}, \quad k\frac{\partial}{\partial k}\lambda=\beta_2(g_{\mathrm{N}},\lambda),$	$k\frac{\partial}{\partial k}g_{\mathrm{N}}=\beta_1(g_{\mathrm{N}},\lambda)=(2+d_{\mathrm{N}})g_{\mathrm{N}}, \quad k\frac{\partial}{\partial k}\lambda=\beta_2(g_{\mathrm{N}},\lambda),$
mielke-geometrodynamics_EQ0914 (13.7.3)	293	■ 1.000 ● N +0	$\mu_*\leq\frac{4}{3}g_{\mathrm{N}*}\lambda_*\simeq 0.2$	$\mu_*\leq\frac{4}{3}g_{\mathrm{N}*}\lambda_*\simeq 0.2$	$\mu_*\leq\frac{4}{3}g_{\mathrm{N}*}\lambda_*\simeq 0.2$
mielke-geometrodynamics_EQ0916 (13.8.2)	295	■ 1.000 ● N -1	$dC_{\mathrm{TT}}=\frac{1}{2\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right)$	$dC_{\mathrm{TT}}=\frac{1}{2\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right)$	$dC_{\mathrm{TT}}=\frac{1}{2\ell^2}\left(T^\alpha\wedge T_\alpha+R_{\alpha\beta}\wedge\vartheta^\alpha\wedge\vartheta^\beta\right)$
mielke-geometrodynamics_EQ0917 (13.8.3)	295	■ 1.000 ● N +0	$dC_{\mathrm{RR}}=-\frac{1}{2}R_\alpha{}^\beta\wedge R_\beta{}^\alpha$	$dC_{\mathrm{RR}}=-\frac{1}{2}R_\alpha{}^\beta\wedge R_\beta{}^\alpha$	$dC_{\mathrm{RR}}=-\frac{1}{2}R_\alpha{}^\beta\wedge R_\beta{}^\alpha$
mielke-geometrodynamics_EQ0918 (13.8.4)	295	■ 1.000 ● N +0	$D R_\alpha{}^\beta\equiv 0,$	$DR_\alpha{}^\beta\equiv 0,$	$DR_\alpha{}^\beta\equiv 0,$
mielke-geometrodynamics_EQ0921 (13.8.7)	296	■ 1.000 ● N +0	$L_{\mathrm{SKY}}:=-\frac{1}{2}R_{\alpha\beta}\wedge{}^*R^{\alpha\beta},$	$L_{\mathrm{SKY}}:=-\frac{1}{2}R_{\alpha\beta}\wedge{}^*R^{\alpha\beta},$	$L_{\mathrm{SKY}}:=-\frac{1}{2}R_{\alpha\beta}\wedge{}^*R^{\alpha\beta},$
mielke-geometrodynamics_EQ0922 (14.1.1)	302	■ 1.000 ● N +0	$L=L_g+L_\omega+L_{\mathrm{mat}}$	$L=L_g+L_\omega+L_{\mathrm{mat}}$	$L=L_g+L_\omega+L_{\mathrm{mat}}$
mielke-geometrodynamics_EQ0923 (14.1.2)	302	■ 1.000 ● N +1	$L_{\omega-g}=\mathrm{Tr}\left\{\wedge^*(\vartheta\wedge\Omega^g)\wedge^*(\vartheta\wedge\Omega)\right\}$	$L_{\omega-g}=\mathrm{Tr}\left\{\wedge^*(\vartheta\wedge\Omega^g)\wedge^*(\vartheta\wedge\Omega)\right\}$	$L_{\omega-g}=\mathrm{Tr}\left\{\wedge^*(\vartheta\wedge\Omega^g)\wedge^*(\vartheta\wedge\Omega)\right\}$
mielke-geometrodynamics_EQ0924 (14.1.3)	303	■ 1.000 ● N -1	$L_{\mathrm{GMD}.}=\frac{1}{\ell^{*2}}L_{\mathrm{W}.}+L_{\mathrm{YM}}+L_{\mathrm{mat}}$	$L_{\mathrm{GMD}.}=\frac{1}{\ell^{*2}}L_{\mathrm{W}.}+L_{\mathrm{YM}}+L_{\mathrm{mat}}$	$L_{\mathrm{GMD}.}=\frac{1}{\ell^{*2}}L_{\mathrm{W}.}+L_{\mathrm{YM}}+L_{\mathrm{mat}}$
mielke-geometrodynamics_EQ0933 (14.1.14)	305	■ 1.000 ● K +0	$e^{*\pi/2}=(-1)^{z*}, \quad z:=\frac{s-1}{2}.$	$e^{*\pi/2}=(-1)^{z*}, \quad z:=\frac{s-1}{2}.$	$e^{*\pi/2}=(-1)^{z*}, \quad z:=\frac{s-1}{2}.$
mielke-geometrodynamics_EQ0934 (14.1.15)	305	■ 1.000 ● K +0	$\xi:=e^{*(-\delta)}\Omega$	$\xi:=e^{*(-\delta)}\Omega$	$\xi:=e^{*(-\delta)}\Omega$
mielke-geometrodynamics_EQ0935 (14.1.16)	305	■ 1.000 ● N +0	$\Omega\wedge^*\Omega\rightarrow\xi\wedge^*\xi=\Omega\wedge^*\Omega\cos 2\delta-(-1)^z\Omega\wedge\Omega\sin 2\delta$	$\Omega\wedge^*\Omega\rightarrow\xi\wedge^*\xi=\Omega\wedge^*\Omega\cos 2\delta-(-1)^z\Omega\wedge\Omega\sin 2\delta$	$\Omega\wedge^*\Omega\rightarrow\xi\wedge^*\xi=\Omega\wedge^*\Omega\cos 2\delta-(-1)^z\Omega\wedge\Omega\sin 2\delta$
mielke-geometrodynamics_EQ0936 (14.1.17)	305	■ 1.000 ● N +0	$\Omega\wedge\Omega\rightarrow\xi\wedge\xi=\Omega\wedge\Omega\cos 2\delta-(-1)^z\Omega\wedge^*\Omega\sin 2\delta.$	$\Omega\wedge\Omega\rightarrow\xi\wedge\xi=\Omega\wedge\Omega\cos 2\delta-(-1)^z\Omega\wedge^*\Omega\sin 2\delta.$	$\Omega\wedge\Omega\rightarrow\xi\wedge\xi=\Omega\wedge\Omega\cos 2\delta-(-1)^z\Omega\wedge^*\Omega\sin 2\delta.$
mielke-geometrodynamics_EQ0937 (14.1.18)	305	■ 1.000 ● N +0	$\vec{E}=\{E_j:=F_{j0}\mid j=1,2,3\}$	$\vec{E}=\{E_j:=F_{j0}\mid j=1,2,3\}$	$\vec{E}=\{E_j:=F_{j0}\mid j=1,2,3\}$
mielke-geometrodynamics_EQ0938 (14.1.19)	305	■ 1.000 ● N +0	$\vec{B}=\left\{B_j:=\frac{1}{2}\epsilon_{ijk}F^{jk}\middle i,j,k=1,2,3\right\}$	$\vec{B}=\left\{B_j:=\frac{1}{2}\epsilon_{ijk}F^{jk}\middle i,j,k=1,2,3\right\}$	$\vec{B}=\{B_j:=\frac{1}{2}\epsilon_{ijk}F^{jk} i,j,k=1,2,3\}$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_306 EQ0943 (14.1.24)		■ 1.000 ●	$\Omega = e^{\ast \delta} \xi$.	$\Omega = e^{\ast \delta} \xi$.	$\Omega = e^{\ast \delta} \xi$.
mielke-geometrodynamics_306 EQ0944 (14.1.25)		■ 1.000 ● N -1	$\Sigma_{\mathrm{YM}}(\Omega) = \Sigma_{\mathrm{YM}}(\xi)$	$\Sigma_{\mathrm{YM}}(\Omega) = \Sigma_{\mathrm{YM}}(\xi)$	$\Sigma_{\mathrm{YM}}(\Omega) = \Sigma_{\mathrm{YM}}(\xi)$
mielke-geometrodynamics_307 EQ0950 (14.1.31)		■ 1.000 ● K +0	$\tau_e = \bar{\tau}_e \cos \delta - (-1)^z \bar{\tau}_p \sin \delta$	$\tau_e = \bar{\tau}_e \cos \delta - (-1)^z \bar{\tau}_p \sin \delta$	$\tau_e = \bar{\tau}_e \cos \delta - (-1)^z \bar{\tau}_p \sin \delta$
mielke-geometrodynamics_307 EQ0952 (14.1.34)		■ 1.000 ● N +0	$D \delta = d \delta = 0,$	$D \delta = d \delta = 0,$	$D \delta = d \delta = 0,$
mielke-geometrodynamics_308 EQ0955 (14.1.37)		■ 1.000 ● N +0	$(-1)^z \Sigma_{\mathrm{YM}}(\xi) \wedge d \delta = D \Sigma_{\mathrm{YM}}(\xi)$.	$(-1)^z \Sigma_{\mathrm{YM}}(\xi) \wedge d \delta = D \Sigma_{\mathrm{YM}}(\xi)$.	$(-1)^z \Sigma_{\mathrm{YM}}(\xi) \wedge d \delta = D \Sigma_{\mathrm{YM}}(\xi)$.
mielke-geometrodynamics_309 EQ0958 (14.1.40)		■ 1.000 ● N +1	$\delta = \int_{c_1} d \delta + \delta_0$	$\delta = \int_{c_1} d \delta + \delta_0$	$\delta = \int_{c_1} d \delta + \delta_0$.
mielke-geometrodynamics_314 EQ0969 (14.2.10)		■ 1.000 ● N +1	$\hat{\omega}^{AB} = -\hat{\omega}^{BA}, \quad A, B = 0, \dots, 3 + K$	$\hat{\omega}^{AB} = -\hat{\omega}^{BA}, \quad A, B = 0, \dots, 3 + K$	$\hat{\omega}^{AB} = -\hat{\omega}^{BA}, \quad A, B = 0, \dots, 3 + K$
mielke-geometrodynamics_315 EQ0975 (14.2.16)		■ 1.000 ● N +1	$\Omega^{\prime} := \pi \left(d \omega - \omega \wedge \overset{\circ}{\omega} - \omega^g \wedge \omega \right) = \Omega - \omega^g \wedge \omega$.	$\Omega^{\prime} := \pi \left(d \omega - \omega \wedge \overset{\circ}{\omega} - \omega^g \wedge \omega \right) = \Omega - \omega^g \wedge \omega$.	$\Omega^{\prime} := \pi (d \omega - \omega \wedge \overset{\circ}{\omega} - \omega^g \wedge \omega) = \Omega - \omega^g \wedge \omega$.
mielke-geometrodynamics_316 EQ0980 (14.2.21)		■ 1.000 ● N +0	$\hat{g}_{AB} \left(x, \xi^j \right) = \sum_{n=-\infty}^{+\infty} g_{AB}^{(n)}(x) Y_n \left(\xi^j \right)$	$\hat{g}_{AB} \left(x, \xi^j \right) = \sum_{n=-\infty}^{+\infty} g_{AB}^{(n)}(x) Y_n \left(\xi^j \right)$	$\hat{g}_{AB}(x, \xi^j) = \sum_{n=-\infty}^{+\infty} g_{AB}^{(n)}(x) Y_n(\xi^j)$
mielke-geometrodynamics_317 EQ0981 (14.2.22)		■ 1.000 ● N -1	$M^{\ast} = \sqrt{\hbar c / G_{\mathrm{N}}} \simeq 10^{19} \mathrm{GeV} / c^2$	$M^{\ast} = \sqrt{\hbar c / G_{\mathrm{N}}} \simeq 10^{19} \mathrm{GeV} / c^2$.	$M^{\ast} = \sqrt{\hbar c / G_{\mathrm{N}}} \simeq 10^{19} \mathrm{GeV} / c^2$.
mielke-geometrodynamics_318 EQ0983 (14.2.24)		■ 1.000 ● K +0	$G = S \, U(3) \, \times \, S \, U(2) \, \times \, U(1)$	$G = SU(3) \times SU(2) \times U(1)$	$G = SU(3) \times SU(2) \times U(1)$
mielke-geometrodynamics_318 EQ0984 (14.2.25)		■ 1.000 ● N +0	$B = \mathbb{C} P^2 \times S^2 \times S^1,$	$B = \mathbb{C} P^2 \times S^2 \times S^1,$	$B = \mathbb{C} P^2 \times S^2 \times S^1,$
mielke-geometrodynamics_318 EQ0985 (14.2.26)		■ 1.000 ● N +1	$G \, / \, H = \frac{S \, U(3) \, \times \, S \, U(2) \, \times \, U(1)}{S \, U(2) \, \times \, U(1) \, \times \, U(1)},$	$G/H = \frac{SU(3) \times SU(2) \times U(1)}{SU(2) \times U(1) \times U(1)},$	$G/H = \frac{SU(3) \times SU(2) \times U(1)}{SU(2) \times U(1) \times U(1)},$
mielke-geometrodynamics_318 EQ0986 (14.2.27)		■ 1.000 ● K +0	$\widetilde{S}^7 := \frac{S \, O(5) \, \times \, S \, O(3)}{S \, O(3) \, \times \, S \, O(3)}$.	$\widetilde{S}^7 := \frac{SO(5) \times SO(3)}{SO(3) \times SO(3)}$.	$\widetilde{S}^7 := \frac{SO(5) \times SO(3)}{SO(3) \times SO(3)}$.
mielke-geometrodynamics_319 EQ0988 (14.3.1)		■ 1.000 ● K +0	$\varphi \rightarrow \varphi^{\mathrm{C}} := \bar{\varphi}$	$\varphi \rightarrow \varphi^{\mathrm{C}} := \bar{\varphi}$	$\varphi \rightarrow \varphi^{\mathrm{C}} := \bar{\varphi}$
mielke-geometrodynamics_320 EQ0991 (14.3.4)		■ 1.000 ● N +0	$\hat{\psi} = \sum_{F=1}^{\dim \rho(G)} \left(\psi_{-}^F(x) \chi_{\rho F}^{-}(\xi) + \psi_{+}^F(x) \chi_{\rho F}^{+}(\xi) \right)$.	$\hat{\psi} = \sum_{F=1}^{\dim \rho(G)} \left(\psi_{-}^F(x) \chi_{\rho F}^{-}(\xi) + \psi_{+}^F(x) \chi_{\rho F}^{+}(\xi) \right)$.	$\hat{\psi} = \sum_{F=1}^{\dim \rho(G)} (\psi_{-}^F(x) \chi_{\rho F}^{-}(\xi) + \psi_{+}^F(x) \chi_{\rho F}^{+}(\xi)).$
mielke-geometrodynamics_321 EQ0994 (14.3.7)		■ 1.000 ● N +0	$D^{g+\omega} := d + i \widetilde{\omega}^g \otimes i \omega$	$D^{g+\omega} := d + i \widetilde{\omega}^g \otimes i \omega$	$D^{g+\omega} := d + i \widetilde{\omega}^g \otimes i \omega$
mielke-geometrodynamics_322 EQ0999 (14.3.12)		■ 1.000 ● N +0	$\left(\widehat{\gamma}^{n+1} \right)^2 = -1 \rightarrow \widehat{\gamma}^{n+1} \widehat{\psi} = \mp i \widehat{\psi}$.	$\left(\widehat{\gamma}^{n+1} \right)^2 = -1 \Rightarrow \widehat{\gamma}^{n+1} \widehat{\psi} = \mp i \widehat{\psi}$.	$(\widehat{\gamma}^{n+1})^2 = -1 \Rightarrow \widehat{\gamma}^{n+1} \widehat{\psi} = \mp i \widehat{\psi}$.
mielke-geometrodynamics_322 EQ1000 (14.3.13)		■ 1.000 ● N +0	$\left(\widehat{\gamma}^{n+1} \right)^2 = +1 \rightarrow \widehat{\gamma}^{n+1} \widehat{\psi} = \mp \widehat{\psi}$.	$\left(\widehat{\gamma}^{n+1} \right)^2 = +1 \Rightarrow \widehat{\gamma}^{n+1} \widehat{\psi} = \mp \widehat{\psi}$.	$(\widehat{\gamma}^{n+1})^2 = +1 \Rightarrow \widehat{\gamma}^{n+1} \widehat{\psi} = \mp \widehat{\psi}$.
mielke-geometrodynamics_324 EQ1006 (14.4.1)		■ 1.000 ● N +0	$\bar{Q}^{(i)} = Q^{(i)} \cos \delta + P^{(i)} \sin \delta, \quad Q^{(\circ)} \equiv Q_e$.	$\bar{Q}^{(i)} = Q^{(i)} \cos \delta + P^{(i)} \sin \delta, \quad Q^{(\circ)} \equiv Q_e$.	$\bar{Q}^{(i)} = Q^{(i)} \cos \delta + P^{(i)} \sin \delta, \quad Q^{(\circ)} \equiv Q_e$.

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_325 EQ1009 (14.4.4)		■ 1.000 ● N +0	$\rho^2:=r^2+\frac{J(J+1)}{M^2}\frac{\hbar^2}{c^2}\cos^2\vartheta$	$\rho^2:=r^2+\frac{J(J+1)\hbar^2}{M^2c^2}\cos^2\vartheta$	$\rho^2:=r^2+\frac{J(J+1)\hbar^2}{M^2c^2}\cos^2\vartheta$
mielke-geometrodynamics_325 EQ1010 (14.4.5)		■ 1.000 ●	$\Delta:=r^2-\frac{2M}{M^2}\frac{\hbar}{c}r+\frac{J(J+1)}{M^2}\frac{\hbar^2}{c^2}+\frac{\alpha_e}{M^2}\frac{\hbar^2}{c^2}\mathrm{Tr}\bar{Q}^2$	$\Delta:=r^2-\frac{2M\hbar}{M^{*2}c}r+\frac{J(J+1)\hbar^2}{M^2c^2}+\frac{\alpha_e\hbar^2}{M^{*2}c^2}\mathrm{Tr}\bar{Q}^2$	$\Delta:=r^2-\frac{2M\hbar}{M^{*2}c}r+\frac{J(J+1)\hbar^2}{M^2c^2}+\frac{\alpha_e\hbar^2}{M^{*2}c^2}\mathrm{Tr}\bar{Q}^2$
mielke-geometrodynamics_327 EQ1017 (14.4.12)		■ 1.000 ● N +0	$ds^2=-\frac{1}{\bar{\rho}^2}\left(\frac{4N}{Mc}\sin^2\left(\frac{\vartheta}{2}\right)d\phi+c\,dt\right)^2+\frac{\bar{\rho}^2}{\bar{\Delta}}(dr^2+\bar{\Delta}d\Omega^2),$	$ds^2=-\frac{\bar{\Delta}}{\bar{\rho}^2}\left(\frac{4N\hbar}{Mc}\sin^2\left(\frac{\vartheta}{2}\right)d\phi+c\,dt\right)^2+\frac{\bar{\rho}^2}{\bar{\Delta}}(dr^2+\bar{\Delta}d\Omega^2),$	$ds^2=-\frac{\bar{\Delta}}{\bar{\rho}^2}\left(\frac{4N\hbar}{Mc}\sin^2\left(\frac{\vartheta}{2}\right)d\phi+c\,dt\right)^2+\frac{\bar{\rho}^2}{\bar{\Delta}}(dr^2+\bar{\Delta}d\Omega^2),$
mielke-geometrodynamics_328 EQ1018 (14.4.13)		■ 1.000 ● N +0	$\bar{\Delta}:=r^2-\frac{2M}{M^2}\frac{\hbar}{c}r-\frac{N^2}{M^2}\frac{\hbar^2}{c^2}+\frac{\alpha_e}{M^2}\frac{\hbar^2}{c^2}\mathrm{Tr}\bar{Q}^2$	$\bar{\Delta}:=r^2-\frac{2M\hbar}{M^{*2}c}r-\frac{N^2\hbar^2}{M^2c^2}+\frac{\alpha_e\hbar^2}{M^{*2}c^2}\mathrm{Tr}\bar{Q}^2$	$\bar{\Delta}:=r^2-\frac{2M\hbar}{M^{*2}c}r-\frac{N^2\hbar^2}{M^2c^2}+\frac{\alpha_e\hbar^2}{M^{*2}c^2}\mathrm{Tr}\bar{Q}^2$
mielke-geometrodynamics_328 EQ1019 (14.4.14)		■ 1.000 ● N +0	$\bar{\rho}^2:=r^2+\frac{N^2}{M^2}\frac{\hbar^2}{c^2}$	$\bar{\rho}^2:=r^2+\frac{N^2\hbar^2}{M^2c^2}$	$\bar{\rho}^2:=r^2+\frac{N^2\hbar^2}{M^2c^2}$
mielke-geometrodynamics_328 EQ1020 (14.4.15)		■ 1.000 ● W +1	$\begin{aligned} A = & -\frac{1}{\bar{\rho}^2}(\bar{Q}\cos\delta+N\bar{Q}\sin\delta)c\,dt \\ & +\frac{1-\cos\vartheta}{\bar{\rho}^2}\bar{Q}\left[-\bar{\rho}^2\sin\delta+2r\frac{N\hbar}{Mc}\cos\delta\right]d\phi \end{aligned}$	$A = -\frac{1}{\bar{\rho}^2}(\bar{Q}\cos\delta+N\bar{Q}\sin\delta)c\,dt \\ +\frac{1-\cos\vartheta}{\bar{\rho}^2}\bar{Q}\left[-\bar{\rho}^2\sin\delta+2r\frac{N\hbar}{Mc}\cos\delta\right]d\phi$	$A = -\frac{1}{\bar{\rho}^2}(\bar{Q}\cos\delta+N\bar{Q}\sin\delta)c\,dt \\ +\frac{1-\cos\vartheta}{\bar{\rho}^2}\bar{Q}\left[-\bar{\rho}^2\sin\delta+2r\frac{N\hbar}{Mc}\cos\delta\right]d\phi.$
mielke-geometrodynamics_328 EQ1022 (14.4.17)		■ 1.000 ● W +0	$M^2/M_{\mathrm{ir}}^2=\frac{\left[1+\frac{1}{4}(\alpha_e^2\mathrm{Tr}\bar{Q}^2-2N^2)(M^*/M_{\mathrm{ir}})^4\right]^2}{1-\frac{1}{4}N^2(M^*/M_{\mathrm{ir}})^4},$	$M^2/M_{\mathrm{ir}}^2=\frac{\left[1+\frac{1}{4}(\alpha_e^2\mathrm{Tr}\bar{Q}^2-2N^2)(M^*/M_{\mathrm{ir}})^4\right]^2}{1-\frac{1}{4}N^2(M^*/M_{\mathrm{ir}})^4},$	$M^2/M_{\mathrm{ir}}^2=\frac{\left[1+\frac{1}{4}(\alpha_e^2\mathrm{Tr}\bar{Q}^2-2N^2)(M^*/M_{\mathrm{ir}})^4\right]^2}{1-\frac{1}{4}N^2(M^*/M_{\mathrm{ir}})^4},$
mielke-geometrodynamics_335 EQ1023 (15.0.1)		■ 1.000 ● N +0	$\ell:=2\pi\hbar/(M_Pc)=\ell^*/\sqrt{\kappa}\simeq1.3\times10^{-13}\,\mathrm{cm}$	$\ell:=2\pi\hbar/(M_Pc)=\ell^*/\sqrt{\kappa}\simeq1.3\times10^{-13}\,\mathrm{cm}$	$\ell:=2\pi\hbar/(M_Pc)=\ell^*/\sqrt{\kappa}\simeq1.3\times10^{-13}\,\mathrm{cm}$
mielke-geometrodynamics_335 EQ1024 (15.0.2)		■ 1.000 ● N +0	$V_{\mathrm{Yu}}(r)\sim\frac{1}{r}e^{-rm_\pi c/\hbar},$	$V_{\mathrm{Yu}}(r)\sim\frac{1}{r}e^{-rm_\pi c/\hbar},$	$V_{\mathrm{Yu}}(r)\sim\frac{1}{r}e^{-rm_\pi c/\hbar},$
mielke-geometrodynamics_336 EQ1026 (15.1.2)		■ 1.000 ● N +0	$L_{fg}=P(\vartheta^f,\vartheta^g)\eta,$	$L_{fg}=P(\vartheta^f,\vartheta^g)\eta,$	$L_{fg}=P(\vartheta^f,\vartheta^g)\eta,$
mielke-geometrodynamics_337 EQ1027 (15.1.3)		■ 1.000 ● N +0	$\delta L_{fg}=\delta\vartheta^f\wedge\Sigma^f+\delta\vartheta^g\wedge\Sigma^g.$	$\delta L_{fg}=\delta\vartheta^f\wedge\Sigma^f+\delta\vartheta^g\wedge\Sigma^g.$	$\delta L_{fg}=\delta\vartheta^f\wedge\Sigma^f+\delta\vartheta^g\wedge\Sigma^g.$
mielke-geometrodynamics_337 EQ1028 (15.1.4)		■ 1.000 ● N +0	$\frac{1}{a}\vartheta^f\wedge\Sigma^f+\frac{1}{b}\vartheta^g\wedge\Sigma^g=2L_{fg}.$	$\frac{1}{a}\vartheta^f\wedge\Sigma^f+\frac{1}{b}\vartheta^g\wedge\Sigma^g=2L_{fg}.$	$\frac{1}{a}\vartheta^f\wedge\Sigma^f+\frac{1}{b}\vartheta^g\wedge\Sigma^g=2L_{fg}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_337 EQ1029 (15.1.5)		■ 1.000 ● W +0	$L_{fg}^{\prime}=\frac{m_f^2}{\ell^2} \det g_{ij} ^u \det f_{ij} ^{1/2-u}\left(\vartheta^f-\vartheta^g\right)\wedge\left(\vartheta^f-\vartheta^g\right)\wedge\vartheta^g\wedge\vartheta^g$	$L_{fg}^{\prime}=\frac{m_f^2}{\ell^2} \det g_{ij} ^u \det f_{ij} ^{1/2-u}\left(\vartheta^f-\vartheta^g\right)\wedge\left(\vartheta^f-\vartheta^g\right)\wedge\vartheta^g\wedge\vartheta^g$	$L_{fg}^{\prime}=\frac{m_f^2}{\ell^2} \det g_{ij} ^u \det f_{ij} ^{1/2-u}\left(\vartheta^f-\vartheta^g\right)\wedge\left(\vartheta^f-\vartheta^g\right)\wedge\vartheta^g\wedge\vartheta^g$
mielke-geometrodynamics_338 EQ1035 (15.1.12)		■ 1.000 ● N +1	$\vartheta^{\mathrm{m}_{\{f\}}}:=\vartheta^{\mathrm{f}}-\vartheta^{\mathrm{g}}$	$\vartheta^{m_f}:=\vartheta^f-\vartheta^g$	$\vartheta^{m_f}:=\vartheta^f-\vartheta^g$
mielke-geometrodynamics_338 EQ1036 (15.1.13)		■ 1.000 ●	$\begin{aligned}ds^2&=\frac{1}{1+\kappa}g^{\alpha\beta}\left(\vartheta^g_{\alpha}\otimes_s\vartheta^g_{\beta}+\kappa\vartheta^f_{\alpha}\otimes\vartheta^f_{\beta}\right)\\&=\frac{1}{1+\kappa}(g_{ij}+\kappa f_{ij})dx^i\otimes_sdx^j\end{aligned}$	$ds^2=\frac{1}{1+\kappa}g^{\alpha\beta}\left(\vartheta^g_{\alpha}\otimes_s\vartheta^g_{\beta}+\kappa\vartheta^f_{\alpha}\otimes\vartheta^f_{\beta}\right)\\=\frac{1}{1+\kappa}(g_{ij}+\kappa f_{ij})dx^i\otimes_sdx^j$	$ds^2=\frac{1}{1+\kappa}g^{\alpha\beta}\left(\vartheta^g_{\alpha}\otimes_s\vartheta^g_{\beta}+\kappa\vartheta^f_{\alpha}\otimes\vartheta^f_{\beta}\right)\\=\frac{1}{1+\kappa}(g_{ij}+\kappa f_{ij})dx^i\otimes_sdx^j.$
mielke-geometrodynamics_338 EQ1037 (15.1.14)		■ 1.000 ● N +0	$\kappa=\ell^{*2}/\ell^2=G_{\mathrm{N}}/G_{\mathrm{S}}=0.38\times10^{-38}$	$\kappa=\ell^{*2}/\ell^2=G_{\mathrm{N}}/G_{\mathrm{S}}=0.38\times10^{-38}$	$\kappa=\ell^{*2}/\ell^2=G_{\mathrm{N}}/G_{\mathrm{S}}=0.38\times10^{-38}$
mielke-geometrodynamics_339 EQ1038 (15.1.15)		■ 1.000 ● W +1	$e^{-\lambda^{g,f}}:=1-\frac{2\mu^{g,f}}{r}-\frac{1}{3}\Lambda^{g,f}r^2,$	$e^{-\lambda^{g,f}}:=1-\frac{2\mu^{g,f}}{r}-\frac{1}{3}\Lambda^{g,f}r^2,$	$e^{-\lambda^{g,f}}:=1-\frac{2\mu^{g,f}}{r}-\frac{1}{3}\Lambda^{g,f}r^2,$
mielke-geometrodynamics_339 EQ1040 (15.1.17)		■ 1.000 ● N +0	$e^{-\lambda^f}\simeq1-\frac{2\mu^f}{r}\exp\left(-rm_fc/\hbar\right).$	$e^{-\lambda^f}\simeq1-\frac{2\mu^f}{r}\exp\left(-rm_fc/\hbar\right).$	$e^{-\lambda^f}\simeq1-\frac{2\mu^f}{r}\exp\left(-rm_fc/\hbar\right).$
mielke-geometrodynamics_340 EQ1041 (15.1.18)		■ 1.000 ● K +0	$E_{\alpha\beta}:(2^+)\oplus2(1^-)\oplus2(0^+)\oplus(1^+)$	$E_{\alpha\beta}:(2^+)\oplus2(1^-)\oplus2(0^+)\oplus(1^+)$	$E_{\alpha\beta}:(2^+)\oplus2(1^-)\oplus2(0^+)\oplus(1^+)$
mielke-geometrodynamics_342 EQ1043 (15.1.20)		■ 1.000 ● N +0	$\frac{M_{\mathrm{U}}}{M_{\mathrm{P}}}=\frac{\ell_{\mathrm{H}}}{\ell^*}\cdot\frac{\ell}{\ell^*}=\frac{1}{\kappa}\frac{\ell_{\mathrm{H}}}{\ell}\simeq10^{80}$	$\frac{M_{\mathrm{U}}}{M_{\mathrm{P}}}=\frac{\ell_{\mathrm{H}}}{\ell^*}\cdot\frac{\ell}{\ell^*}=\frac{1}{\kappa}\frac{\ell_{\mathrm{H}}}{\ell}\simeq10^{80}$	$\frac{M_{\mathrm{U}}}{M_{\mathrm{P}}}=\frac{\ell_{\mathrm{H}}}{\ell^*}\cdot\frac{\ell}{\ell^*}=\frac{1}{\kappa}\frac{\ell_{\mathrm{H}}}{\ell}\simeq10^{80}$
mielke-geometrodynamics_343 EQ1044 (15.2.1)		■ 1.000 ● W +0	$\begin{aligned}\bar{G}&=SL(2f,\mathbb{C})\otimes SL(2c,\mathbb{C})\\&\supset SU(f)_{\mathrm{L}}\otimes SU(f)_{\mathrm{R}}\otimes SU(c)_{\mathrm{L}}\otimes SU(c)_{\mathrm{R}}\end{aligned}$	$\bar{G}=SL(2f,\mathbb{C})\otimes SL(2c,\mathbb{C})\\ \supset SU(f)_{\mathrm{L}}\otimes SU(f)_{\mathrm{R}}\otimes SU(c)_{\mathrm{L}}\otimes SU(c)_{\mathrm{R}}$	$\overline{G}=SL(2f,\mathbb{C})\otimes SL(2c,\mathbb{C})\\ \supset SU(f)_{\mathrm{L}}\otimes SU(f)_{\mathrm{R}}\otimes SU(c)_{\mathrm{L}}\otimes SU(c)_{\mathrm{R}}$
mielke-geometrodynamics_343 EQ1045 (15.2.2)		■ 1.000 ● N +0	$\bar{L}(M):=M^4\times\bar{G}\supset M^4\times SL(2,\mathbb{C})$	$\bar{L}(M):=M^4\times\bar{G}\supset M^4\times SL(2,\mathbb{C})$	$\overline{L}(M):=M^4\times\overline{G}\supset M^4\times SL(2,\mathbb{C})$
mielke-geometrodynamics_344 EQ1046 (15.2.3)		■ 1.000 ● N -1	$\bar{\gamma}=\bar{E}_j^{\alpha}(m)\gamma_{\alpha}\otimes dx^j=\bar{E}_jdx^j$	$\bar{\gamma}=\bar{E}_j^{\alpha}(m)\gamma_{\alpha}\otimes dx^j=\bar{E}_jdx^j$	$\overline{\gamma}=\overline{E}_j^{\alpha}(m)\gamma_{\alpha}\otimes dx^j=\overline{E}_jdx^j$
mielke-geometrodynamics_344 EQ1048 (15.2.5)		■ 1.000 ● K +0	$\frac{1}{4}\mathrm{Tr}\left(\bar{\gamma}\otimes_s\bar{\gamma}\right)=f_{ij}dx^i\otimes_sdx^j$	$\frac{1}{4}\mathrm{Tr}\left(\bar{\gamma}\otimes_s\bar{\gamma}\right)=f_{ij}dx^i\otimes_sdx^j$	$\frac{1}{4}\mathrm{Tr}(\overline{\gamma}\otimes_s\overline{\gamma})=f_{ij}dx^i\otimes_sdx^j$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_345 EQ1052 (15.2.9)		■ 1.000 ● N +0	$\backslash\mathrm{frac}\{\backslash\omega\}_{i}=\backslash\mathrm{frac}\{1\}{2}\ \backslash\Gamma_{i}^{\{f\}}\ \backslash\alpha\ \backslash\mathrm{beta}\ j\ \backslash\mathrm{sigma}_{\{f\}}\{\backslash\alpha\ \backslash\mathrm{beta}\}\ \backslash\mathrm{lambda}_{\{j\}}^{\{f\}}\ \backslash\mathrm{oplus}\ \backslash\mathrm{frac}\{1\}{2}\ \backslash\Gamma_{i}^{\{c\}}\ \backslash\alpha\ \backslash\mathrm{beta}\ j\ \backslash\mathrm{sigma}_{\{c\}}\{\backslash\alpha\ \backslash\mathrm{beta}\}\ \backslash\mathrm{lambda}_{\{j\}}^{\{c\}}\ \backslash\mathrm{oplus}\ A_{i}\ .$	$\bar{\omega}_i=\frac{1}{2}\Gamma_i^{(f)\alpha\beta j}\sigma_{\alpha\beta}\lambda_j^{(f)}\oplus\frac{1}{2}\Gamma_i^{(c)\alpha\beta j}\sigma_{\alpha\beta}\lambda_j^{(c)}\oplus A_i.$	$\overline{\omega}_i=\frac{1}{2}\Gamma_i^{(f)\alpha\beta j}\sigma_{\alpha\beta}\lambda_j^{(f)}\oplus\frac{1}{2}\Gamma_i^{(c)\alpha\beta j}\sigma_{\alpha\beta}\lambda_j^{(c)}\oplus A_i.$
mielke-geometrodynamics_345 EQ1053 (15.2.10)		■ 1.000 ● N +0	$\backslash\mathrm{bar}\{\backslash\Theta\}:=\backslash\mathrm{bar}\{D\}\ \backslash\mathrm{bar}\{\backslash\gamma\}=d\ \backslash\mathrm{bar}\{\backslash\gamma\}-[\backslash\mathrm{bar}\{\backslash\omega\},\ \backslash\mathrm{bar}\{\backslash\gamma\}]$	$\bar{\Theta}:=\bar{D}\bar{\gamma}=d\bar{\gamma}-[\bar{\omega},\bar{\gamma}]$	$\bar{\Theta}:=\overline{D\gamma}=d\overline{\gamma}-[\bar{\omega},\overline{\gamma}]$
mielke-geometrodynamics_345 EQ1054 (15.2.11)		■ 1.000 ● W +1	$\backslash\mathrm{bar}\{0\omega\}=d\ \backslash\mathrm{bar}\{\backslash\omega\}-\backslash\mathrm{bar}\{\backslash\omega\}\ \backslash\mathrm{wedge}\ \backslash\mathrm{bar}\{\backslash\omega\}$	$\bar{\Omega}=d\bar{\omega}-\bar{\omega}\wedge\bar{\omega}$	$\overline{\Omega}=d\overline{\omega}-\overline{\omega}\wedge\overline{\omega}$
mielke-geometrodynamics_346 EQ1056 (15.2.13)		■ 1.000 ● N +0	$\backslash\mathrm{psi}=\backslash\mathrm{left}\{\backslash\mathrm{psi}^{\backslash\mathrm{left}\{q_f\},\ q_c\}\backslash\mathrm{right}\}\ \backslash\mathrm{mid}\ q_{\{f\}}=1,\ \backslash\mathrm{ldots},\ f\ ;\ q_{\{c\}}=1,\ \backslash\mathrm{ldots},\ c\backslash\mathrm{right}\}\ .$	$\psi=\left\{\psi^{(q_f,q_c)}\mid q_f=1,\ldots,f;q_c=1,\ldots,c\right\}.$	$\psi=\{\psi^{(q_f,q_c)} q_f=1,\ldots,f;q_c=1,\ldots,c\}.$
mielke-geometrodynamics_354 EQ1061 (16.1.2)		■ 1.000 ● W -2	$B=\mathrm{frac}\{2\ \mathrm{hbar}\ c\}{\mathrm{ell}^{\ast}\ 2}\}\ \backslash\mathrm{Lambda}_{\{\mathrm{mathrm}\{\mathrm{eff}\}\}}=-\mathrm{frac}\{3\ \mathrm{kappa}\}{2}\}\ \mathrm{frac}\{\mathrm{hbar}\ c\}{\mathrm{ell}^{\ast}\ 4}\}$	$B=\frac{2\hbar c}{\ell^{*2}}\Lambda_{\mathrm{eff}}=-\frac{3\kappa}{2}\frac{\hbar c}{\ell^{*4}}$	$B=\frac{2\hbar c}{\ell^{*2}}\Lambda_{\mathrm{eff}}=-\frac{3\kappa}{2}\frac{\hbar c}{\ell^{*4}}$
mielke-geometrodynamics_355 EQ1063 (16.2.2)		■ 1.000 ●	$\backslash\mathrm{square}=\mathrm{frac}\{1\}{\sqrt{\{f\}}}\ \backslash\mathrm{partial}_{\{i\}}\backslash\mathrm{left}(f^{\{i\}\ j}\ \backslash\mathrm{sqrt}\{\{f\}\}\ \backslash\mathrm{partial}_{\{j\}}\backslash\mathrm{right})$	$\square=\frac{1}{\sqrt{ f }}\partial_i\left(f^{ij}\sqrt{ f }\partial_j\right)$	$\square=\frac{1}{\sqrt{ f }}\partial_i(f^{ij}\sqrt{ f }\partial_j)$
mielke-geometrodynamics_355 EQ1064 (16.2.3)		■ 1.000 ● N +0	$\backslash\mathrm{varphi}=\mathrm{m}\ \mathrm{frac}\{c\}{\mathrm{hbar}}\ \mathrm{frac}\{1\}{\rho}\ F_{nm}^{\{i\}}\ F_{\{n\}\ m}^{\{i\}}\ \backslash\mathrm{left}(\rho^{\{i\}}\backslash\mathrm{right})\ Y_{\{i\}}^{\{m\}}(\backslash\mathrm{vartheta},\ \backslash\mathrm{phi})\ e^{\{-i\ t\ \omega\mathrm{e}\ m\ c^2\ /\ \mathrm{hbar}\}}\ .$	$\varphi=m\frac{c}{\hbar}\frac{1}{\rho}F_{nm}^l(\rho^*)Y_l^m(\vartheta,\phi)e^{-it\omega mc^2/\hbar}.$	$\varphi=m\frac{c}{\hbar}\frac{1}{\rho}F_{nm}^l(\rho^*)Y_l^m(\vartheta,\phi)e^{-it\omega mc^2/\hbar}.$
mielke-geometrodynamics_355 EQ1067 (16.2.6)		■ 1.000 ● K +0	$\backslash\mathrm{beta}=\mathrm{frac}\{\sqrt{2\ \pi}\ \mathrm{kappa}\}{\mathrm{ell}^{\ast}}\}\ /\ \mathrm{frac}\{m\ c\}{\mathrm{hbar}}=\mathrm{frac}\{\mathrm{ell}_{\{m\}}\}{\mathrm{ell}}\}\ .$	$\beta=\frac{\sqrt{2\pi\kappa}}{\ell^*}/\frac{mc}{\hbar}=\frac{\ell_m}{\ell}.$	$\beta=\frac{\sqrt{2\pi\kappa}}{\ell^*}\Big/\frac{mc}{\hbar}=\frac{\ell_m}{\ell}.$
mielke-geometrodynamics_358 EQ1072 (16.3.1)		■ 1.000 ● N +0	$\backslash\mathrm{operatorname}\{\mathrm{Tr}\}\ \backslash\mathrm{bar}\{0\}^{\{2\}}=-\mathrm{frac}\{4\ b\}{\backslash\alpha_{\{e\}}}\ Y+\mathrm{I}(\mathrm{I}+1)-\mathrm{frac}\{1\}{4}\}\ Y^{\{2\}}\ .$	$\mathrm{Tr}\,\bar{Q}^2=-\frac{4b}{\alpha_e}Y+I(I+1)-\frac{1}{4}Y^2.$	$\mathrm{Tr}\,\overline{Q}^2=-\frac{4b}{\alpha_e}Y+I(I+1)-\frac{1}{4}Y^2.$
mielke-geometrodynamics_358 EQ1073 (16.3.2)		■ 1.000 ● N +0	$Q_{\{e\}}=\mathrm{I}_{\{3\}}+\mathrm{frac}\{1\}{2}\}\ Y-\mathrm{frac}\{2\}{3}\}\ C\ .$	$Q_e=I_3+\frac{1}{2}Y-\frac{2}{3}C.$	$Q_e=I_3+\frac{1}{2}Y-\frac{2}{3}C.$
mielke-geometrodynamics_359 EQ1074 (16.3.3)		■ 1.000 ● N +0	$\backslash\mathrm{frac}\{M^{\{2\}}\}{M^{\{*\ 2\}}}\}=\backslash\mathrm{left}\{1-\mathrm{b}\ Y+\mathrm{frac}\{\backslash\alpha_{\{e\}}\}{4}\}\ \backslash\mathrm{left}[\mathrm{I}(\mathrm{I}+1)-\mathrm{frac}\{1\}{4}\}\ Y^{\{2\}}\backslash\mathrm{right}]\backslash\mathrm{right}\}^{\{2\}}+\mathrm{frac}\{1\}{4}\}\ J(J+1)$	$\frac{M^2}{M^{*2}}=\left\{1-bY+\frac{\alpha_e}{4}\left[I(I+1)-\frac{1}{4}Y^2\right]\right\}^2+\frac{1}{4}J(J+1)$	$\frac{M^2}{M^{*2}}=\left\{1-bY+\frac{\alpha_e}{4}[I(I+1)-\frac{1}{4}Y^2]\right\}^2+\frac{1}{4}J(J+1)$
mielke-geometrodynamics_360 EQ1075 (16.3.4)		■ 1.000 ● N +0	$\mathrm{T}_{\{\bullet\}}=\mathrm{frac}\{2\ \mathrm{hbar}\ c\}{S_{\{A\}}\ k_{\{B\}}}\backslash\mathrm{left}(r_{\{+\}}-\mathrm{frac}\{M\}{M^{\{*\ 2\}}}\}\ \mathrm{frac}\{\mathrm{hbar}\}{c}\backslash\mathrm{right})\ .$	$T_{\bullet}=\frac{2\hbar c}{S_Ak_B}\left(r_+-\frac{M}{M^{*2}}\frac{\hbar}{c}\right).$	$T_{\bullet}=\frac{2\hbar c}{S_Ak_B}\left(r_+-\frac{M}{M^{*2}}\frac{\hbar}{c}\right).$
mielke-geometrodynamics_364 EQ1076 (A.1)		■ 1.000 ● N +0	$x^{\{A\}}\ y_{\{A\}}:=\sum_{A=0}^n\ \sum_{B=0}^n\ x^{\{A\}}\ y^{\{B\}}\ g_{\{A\}\ B}\ .$	$x^Ay_A:=\sum_{A=0}^n\sum_{B=0}^nx^Ay^Bg_{AB}.$	$x^Ay_A:=\sum_{A=0}^n\sum_{B=0}^nx^Ay^Bg_{AB}.$
mielke-geometrodynamics_364 EQ1077 (A.2)		■ 1.000 ● N +0	$c=\mathrm{hbar}=1\ .$	$c=\hbar=1.$	$c=\hbar=1.$
mielke-geometrodynamics_365 EQ1080 (A.5)		■ 1.000 ● W +0	$\backslash\mathrm{left}\{\backslash\mathrm{sigma}^{\{i\}},\ \backslash\mathrm{sigma}^{\{j\}}\backslash\mathrm{right}\}=\mathrm{g}^{\{i\}\ j}$	$\{\sigma^i,\sigma^j\}=g^{ij}$	$\{\sigma^i,\sigma^j\}=g^{ij}$
mielke-geometrodynamics_365 EQ1081 (A.6)		■ 1.000 ● K +0	$\backslash\mathrm{left}(\backslash\mathrm{gamma}^{\{\backslash\alpha\}}\backslash\mathrm{right})^{\{\mathrm{mathrm}\{T\}}}\}:=\backslash\mathrm{left}(\backslash\mathrm{gamma}^{\{\emptyset\}},\ \backslash\mathrm{gamma}^{\{k\}}\backslash\mathrm{right})\},$	$(\gamma^\alpha)^{\mathrm{T}}:=(\gamma^0,\gamma^k),$	$(\gamma^\alpha)^{\mathrm{T}}:=(\gamma^0,\gamma^k),$
mielke-geometrodynamics_365 EQ1083 (A.8)		■ 1.000 ● N +0	$\backslash\mathrm{gamma}^{\{\backslash\alpha\}}\ \backslash\mathrm{gamma}^{\{\backslash\mathrm{beta}\}}+\backslash\mathrm{gamma}^{\{\backslash\mathrm{beta}\}}\ \backslash\mathrm{gamma}^{\{\backslash\alpha\}}=2\ \mathrm{g}^{\{\backslash\alpha\}\ \backslash\mathrm{beta}\}}$	$\gamma^\alpha\gamma^\beta+\gamma^\beta\gamma^\alpha=2g^{\alpha\beta}$	$\gamma^\alpha\gamma^\beta+\gamma^\beta\gamma^\alpha=2g^{\alpha\beta}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_365 EQ1084 (A.9)		■ 1.000 ● N +0	$\gamma^{\{0+\}}=\gamma^{\{0\}}, \quad \quad \gamma^{\{k+\}}=-\gamma^{\{k\}}, \quad \quad \gamma^{\{5+\}}=\gamma^{\{5\}} \quad .$	$\gamma^{0+} = \gamma^0, \quad \gamma^{k+} = -\gamma^k, \quad \gamma^{5+} = \gamma^5.$	$\gamma^{0+} = \gamma^0, \quad \gamma^{k+} = -\gamma^k, \quad \gamma^{5+} = \gamma^5.$
mielke-geometrodynamics_366 EQ1087 (A.14)		■ 1.000 ● K +0	$C(1,3)=C(0,2) \otimes C(1,1) \quad .$	$C(1,3) = C(0,2) \otimes C(1,1).$	$C(1,3) = C(0,2) \otimes C(1,1).$
mielke-geometrodynamics_368 EQ1091 (B.1)		■ 1.000 ● N +0	$\alpha^{\{p\}}=\frac{1}{p!} \, A_{\alpha_1\cdots \alpha_p} \, \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \in C^{\infty}(\wedge^p T^*(M)).$	$\alpha^{(p)} = \frac{1}{p!} A_{\alpha_1 \cdots \alpha_p} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \in C^{\infty}(\wedge^p T^*(M)).$	$\alpha^{(p)} = \frac{1}{p!} A_{\alpha_1 \cdots \alpha_p} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \in C^{\infty}(\wedge^p T^*(M)).$
mielke-geometrodynamics_368 EQ1092 (B.2)		■ 1.000 ● N +0	$\begin{gathered} \alpha^{\{p\}} \wedge \beta^{\{q\}} := \frac{1}{(p!) \cdot (q!)} A_{\alpha_1 \cdots \alpha_p} \cdot B_{\beta_1 \cdots \beta_q} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \\ \wedge \vartheta^{\beta_1} \wedge \cdots \wedge \vartheta^{\beta_q} \end{gathered}$	$\alpha^{(p)} \wedge \beta^{(q)} := \frac{1}{(p!) \cdot (q!)} A_{\alpha_1 \cdots \alpha_p} \cdot B_{\beta_1 \cdots \beta_q} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \\ \wedge \vartheta^{\beta_1} \wedge \cdots \wedge \vartheta^{\beta_q}$	$\alpha^{(p)} \wedge \beta^{(q)} := \frac{1}{(p!) \cdot (q!)} A_{\alpha_1 \cdots \alpha_p} \cdot B_{\beta_1 \cdots \beta_q} \vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_p} \\ \wedge \vartheta^{\beta_1} \wedge \cdots \wedge \vartheta^{\beta_q}$
mielke-geometrodynamics_368 EQ1093 (B.3)		■ 1.000 ● N +0	$\alpha^{\{p\}} \wedge \beta^{\{q\}}=(-1)^{pq} \, \beta^{\{q\}} \wedge \alpha^{\{p\}},$	$\alpha^{(p)} \wedge \beta^{(q)} = (-1)^{pq} \beta^{(q)} \wedge \alpha^{(p)},$	$\alpha^{(p)} \wedge \beta^{(q)} = (-1)^{pq} \beta^{(q)} \wedge \alpha^{(p)},$
mielke-geometrodynamics_368 EQ1094 (B.4)		■ 1.000 ●	$D(M):=\sum_{p=0}^{\infty} \wedge^p T^*(M)$	$D(M) := \sum_{p=0}^n \wedge^p T^*(M)$	$D(M) := \sum_{p=0}^n \wedge^p T^*(M)$
mielke-geometrodynamics_369 EQ1095 (B.5)		■ 1.000 ● N +1	$d\left(\alpha^{\{p\}} \wedge \beta^{\{q\}}\right)=d\alpha^{\{p\}} \wedge \beta^{\{q\}}+(-1)^p \alpha^{\{p\}} \wedge d\beta^{\{q\}}$	$d\left(\alpha^{(p)} \wedge \beta^{(q)}\right)=d\alpha^{(p)} \wedge \beta^{(q)}+(-1)^p \alpha^{(p)} \wedge d\beta^{(q)}$	$d(\alpha^{(p)} \wedge \beta^{(q)})=d\alpha^{(p)} \wedge \beta^{(q)}+(-1)^p \alpha^{(p)} \wedge d\beta^{(q)}$
mielke-geometrodynamics_369 EQ1097 (B.7)		■ 1.000 ● N +0	$\sigma: M \rightarrow N, \quad \sigma \in \mathscr{D}(M)$	$\sigma: M \rightarrow N, \quad \sigma \in \mathscr{D}(M)$	$\sigma: M \rightarrow N, \quad \sigma \in \mathscr{D}(M)$
mielke-geometrodynamics_369 EQ1098 (B.8)		■ 1.000 ● N +0	$\sigma_*: \wedge^p T^*(N) \rightarrow \wedge^p T^*(M)$	$\sigma_*: \wedge^p T^*(N) \rightarrow \wedge^p T^*(M)$	$\sigma_*: \wedge^p T^*(N) \rightarrow \wedge^p T^*(M)$
mielke-geometrodynamics_370 EQ1103 (B.15)		■ 1.000 ● K +0	$\{ \}^{\{ * \, * \}} \, \alpha^{\{ p \}}=(-1)^{p(n+1)+s} \, \alpha^{\{ p \}} \quad .$	$**\alpha^{(p)} = (-1)^{p(n+1)+s} \alpha^{(p)}.$	$**\alpha^{(p)} = (-1)^{p(n+1)+s} \alpha^{(p)}.$
mielke-geometrodynamics_370 EQ1105 (B.17)		■ 1.000 ● N +0	$(\alpha,\beta):=\int_M \alpha^{\{p\}} \wedge \beta^{\{p\}} = (\beta,\alpha)$	$(\alpha,\beta) := \int_M \alpha^{(p)} \wedge \beta^{(p)} = (\beta,\alpha)$	$(\alpha,\beta) := \int_M \alpha^{(p)} \wedge \beta^{(p)} = (\beta,\alpha)$
mielke-geometrodynamics_370 EQ1107 (B.19)		■ 1.000 ● N +0	$\square \alpha^{\{p\}}:=d\delta \alpha^{\{p\}}+\delta d \alpha^{\{p\}} \quad .$	$\square \alpha^{(p)} := d\delta \alpha^{(p)} + \delta d \alpha^{(p)}.$	$\square \alpha^{(p)} := d\delta \alpha^{(p)} + \delta d \alpha^{(p)}.$
mielke-geometrodynamics_371 EQ1108 (B.20)		■ 1.000 ● N +0	$\begin{aligned} (\alpha,\square \alpha) &= (\alpha,d\delta \alpha) + (\alpha,\delta d \alpha) \\ &= (\delta \alpha,\delta \alpha) + (d \alpha,d \alpha). \end{aligned}$	$\begin{aligned} (\alpha,\square \alpha) &= (\alpha,d\delta \alpha) + (\alpha,\delta d \alpha) \\ &= (\delta \alpha,\delta \alpha) + (d \alpha,d \alpha). \end{aligned}$	$\begin{aligned} (\alpha,\square \alpha) &= (\alpha,d\delta \alpha) + (\alpha,\delta d \alpha) \\ &= (\delta \alpha,\delta \alpha) + (d \alpha,d \alpha). \end{aligned}$
mielke-geometrodynamics_371 EQ1109 (B.21)		■ 1.000 ● K +0	$d \, \alpha=0 \quad .$	$d\alpha = 0.$	$d\alpha = 0.$
mielke-geometrodynamics_371 EQ1111 (B.23)		■ 1.000 ● N +0	$\beta^{\{p\}}=d \, \alpha^{\{p-1\}} \in C^{\infty}(B(M))$	$\beta^{(p)} = d\alpha^{(p-1)} \in C^{\infty}(B(M))$	$\beta^{(p)} = d\alpha^{(p-1)} \in C^{\infty}(B(M))$
mielke-geometrodynamics_371 EQ1112 (B.24)		■ 1.000 ● N +0	$H(M):=Z(M) \, / \, B(M)$	$H(M) := Z(M)/B(M)$	$H(M) := Z(M)/B(M)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_EQ1113 (B.25)	371	■ 1.000 ● N +0	$H(M)=\sum_{p=0}^{\infty}H^p(M):=\sum_{p=0}^{\infty}Z^p(M)/B^p(M)$.	$H(M)=\sum_{p=0}^nH^p(M):=\sum_{p=0}^nZ^p(M)/B^p(M).$	$H(M)=\sum_{p=0}^nH^p(M):=\sum_{p=0}^nZ^p(M)/B^p(M).$
mielke-geometrodynamics_EQ1115 (B.27)	371	■ 1.000 ● N +0	$H^n(M)=\wedge^nT^*(M)/B^n(M)$.	$H^n(M)=\wedge^nT^*(M)/B^n(M).$	$H^n(M)=\wedge^nT^*(M)/B^n(M).$
mielke-geometrodynamics_EQ1116 (B.28)	371	■ 1.000 ● N +0	$H^{\{0\}}(M)=Z^{\{0\}}(M)$	$H^0(M)=Z^0(M)$	$H^0(M)=Z^0(M)$
mielke-geometrodynamics_EQ1117 (B.29)	372	■ 1.000 ● N +0	$H^{\{0\}}(M)\approx\mathbb{R}$	$H^0(M)\approx\mathbb{R}$	$H^0(M)\approx\mathbb{R}$
mielke-geometrodynamics_EQ1118 (B.30)	372	■ 1.000 ● N -1	$b_p:=\operatorname{dim}H^p(M)$	$b_p:=\dim H^p(M)$	$b_p:=\dim H^p(M)$
mielke-geometrodynamics_EQ1119 (B.31)	372	■ 1.000 ● N +1	$\mathrm{f}_{\mathrm{M}}(t)=\sum_{p=0}^nb_p t^p$	$f_M(t)=\sum_{p=0}^nb_pt^p$	$f_M(t)=\sum_{p=0}^nb_pt^p,$
mielke-geometrodynamics_EQ1120 (B.32)	372	■ 1.000 ● N +0	$\chi(M):=\mathrm{f}_{\mathrm{M}}(-1)=\sum_{p=0}^n(-1)^pb_p,$	$\chi(M):=f_M(-1)=\sum_{p=0}^n(-1)^pb_p,$	$\chi(M):=f_M(-1)=\sum_{p=0}^n(-1)^pb_p,$
mielke-geometrodynamics_EQ1121 (B.33)	372	■ 1.000 ● N +0	$H^{\{0\}}\left(S^n\right)=H^n\left(S^n\right)\approx\mathbb{R}$	$H^0(S^n)=H^n(S^n)\approx\mathbb{R}$	$H^0(S^n)=H^n(S^n)\approx\mathbb{R}$
mielke-geometrodynamics_EQ1123 (B.35)	372	■ 1.000 ● N +0	$\mathrm{f}_{S^n}(t)=1+t^n,$	$f_{S^n}(t)=1+t^n,$	$f_{S^n}(t)=1+t^n,$
mielke-geometrodynamics_EQ1124 (B.36)	372	■ 1.000 ●	$\chi\left(S^n\right)=\left\{\begin{array}{l}0\text{ for }n=2k+1,\\2\text{ for }n=2k,\end{array}\right.$	$\chi(S^n)=\left\{\begin{array}{l}0\text{ for }n=2k+1,\\2\text{ for }n=2k,\end{array}\right.$	$\chi(S^n)=\left\{\begin{array}{l}0\text{ for }n=2k+1,\\2\text{ for }n=2k,\end{array}\right.$
mielke-geometrodynamics_EQ1125 (B.37)	372	■ 1.000 ● N +0	$\Delta_p:=\left\{x\in E^n\mid x=\sum_{i=0}^p\lambda^i\eta_i,\lambda^i\geq0,\sum_{i=0}^p\lambda^i=1\right\}.$	$\Delta_p:=\left\{x\in E^n\mid x=\sum_{i=0}^p\lambda^i\eta_i,\lambda^i\geq0,\sum_{i=0}^p\lambda^i=1\right\}.$	$\Delta_p:=\left\{x\in E^n x=\sum_{i=0}^p\lambda^i\eta_i,\lambda^i\geq0,\sum_{i=0}^p\lambda^i=1\right\}.$
mielke-geometrodynamics_EQ1127 (B.39)	373	■ 1.000 ● N +0	$\partial c=\sum_{i=0}^p(-1)^ic\left(\sum_{j=0}^{i-1}\lambda^j\eta_j+\sum_{j=i}^{p-1}\lambda^j\eta_{j+1}\right).$	$\partial c=\sum_{i=0}^p(-1)^ic\left(\sum_{j=0}^{i-1}\lambda^j\eta_j+\sum_{j=i}^{p-1}\lambda^j\eta_{j+1}\right).$	$\partial c=\sum_{i=0}^p(-1)^ic\left(\sum_{j=0}^{i-1}\lambda^j\eta_j+\sum_{j=i}^{p-1}\lambda^j\eta_{j+1}\right).$
mielke-geometrodynamics_EQ1129 (B.41)	373	■ 1.000 ● N +0	$H_{\mathrm{p}}(M,\mathbb{R}):=Z_{\mathrm{p}}(M,\mathbb{R})/B_{\mathrm{p}}(M,\mathbb{R})$	$H_p(M,\mathbb{R}):=Z_p(M,\mathbb{R})/B_p(M,\mathbb{R})$	$H_p(M,\mathbb{R}):=Z_p(M,\mathbb{R})/B_p(M,\mathbb{R})$
mielke-geometrodynamics_EQ1130 (B.42)	373	■ 1.000 ● N +1	$\int_{c_{p+1}}d\alpha^{(p)}=\int_{\partial c_{p+1}}\alpha^{(p)}$	$\int_{c_{p+1}}d\alpha^{(p)}=\int_{\partial c_{p+1}}\alpha^{(p)}$	$\int_{c_{p+1}}d\alpha^{(p)}=\int_{\partial c_{p+1}}\alpha^{(p)}.$
mielke-geometrodynamics_EQ1132 (B.44)	373	■ 1.000 ● K +0	$\langle d\alpha,c\rangle_M=\langle\alpha,\partial c\rangle_M$	$\langle d\alpha,c\rangle_M=\langle\alpha,\partial c\rangle_M$	$\langle d\alpha,c\rangle_M=\langle\alpha,\partial c\rangle_M$
mielke-geometrodynamics_EQ1133 (C.1)	375	■ 1.000 ● N +0	$I_j:=\left.\frac{\partial g(\xi)}{\partial \xi^j}\right _{g=e}\in T_e(G)\approx\mathfrak{g}$	$I_j:=\left.\frac{\partial g(\xi)}{\partial \xi^j}\right _{g=e}\in T_e(G)\approx\mathfrak{g}$	$I_j:=\left.\frac{\partial g(\xi)}{\partial \xi^j}\right _{g=e}\in T_e(G)\approx\mathfrak{g}$
mielke-geometrodynamics_EQ1138 (C.6)	376	■ 1.000 ● N +0	$[\mathrm{AB}]:=\mathrm{A}^{\mathrm{i}}\mathrm{B}^{\mathrm{j}}\mathrm{c}_{\mathrm{ij}}^{\mathrm{k}}\mathrm{I}_{\mathrm{k}}$	$[AB]:=A^iB^jc_{ij}^kI_k.$	$[AB]:=A^iB^jc_{ij}^kI_k.$

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

LaTeX source MathPix’s reading, unchanged; **Rendered** the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

[refined: *basis*] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — *basis* names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_376 EQ1139 (C.7)		■ 1.000 ● N +0	$\rho_{A}^{\mathrm{C}}(g_1)\rho_{C}^{\mathrm{B}}(g_2)=\rho_{A}^{\mathrm{B}}(g_1g_2)$	$\rho_A^C(g_1)\rho_C^B(g_2)=\rho_A^B(g_1g_2)$	$\rho_A^C(g_1)\rho_C^B(g_2)=\rho_A^B(g_1g_2)$
mielke-geometrodynamics_376 EQ1140 (C.8)		■ 1.000 ● K +0	$[A,\,B]\longrightarrow\rho([A,\,B])=[\rho(A),\,\rho(B)]\;.$	$[A,B]\longrightarrow\rho([A,B])=[\rho(A),\rho(B)].$	$[A,B]\longrightarrow\rho([A,B])=[\rho(A),\rho(B)].$
mielke-geometrodynamics_376 EQ1141 (C.9)		■ 1.000 ● K +0	$\operatorname{Ad}A(B):=[A,\,B]\; ; \quad A,\,B\in\mathfrak{g},$	$\operatorname{Ad}A(B):=[A,B];\quad A,B\in\mathfrak{g},$	$\operatorname{Ad}A(B):=[A,B];\quad A,B\in\mathfrak{g},$
mielke-geometrodynamics_376 EQ1142 (C.10)		■ 1.000 ● N +0	$(\operatorname{Ad}A)_{\{j\}\{ \}^{\{k\}}=c_{\{i\}\{ \}^{\{k\}}A^{\{i\}}\;.$	$(\operatorname{Ad}A)_j{}^k=c_{ij}{}^kA^i.$	$(\operatorname{Ad}A)_j{}^k=c_{ij}{}^kA^i.$
mielke-geometrodynamics_376 EQ1144 (C.12)		■ 1.000 ● K +0	$(A,\,B)=2\,N\operatorname{Tr}(A\,B)-2\operatorname{Tr}(A)\operatorname{Tr}(B)$	$(A,B)=2N\operatorname{Tr}(AB)-2\operatorname{Tr}(A)\operatorname{Tr}(B)$	$(A,B)=2N\operatorname{Tr}(AB)-2\operatorname{Tr}(A)\operatorname{Tr}(B)$
mielke-geometrodynamics_377 EQ1146 (C.14)		■ 1.000 ● N +1	$g=\exp^{\left(-\frac{i}{2}\xi^j\lambda_j\right)}\in S\,U(N),\,j=1,\,\ldots,\,N^2-1\;.$	$g=\exp^{\left(-\frac{i}{2}\xi^j\lambda_j\right)}\in SU(N),j=1,\ldots,N^2-1.$	$g=\exp^{\left(-\frac{i}{2}\xi^j\lambda_j\right)}\in SU(N),j=1,\ldots,N^2-1.$
mielke-geometrodynamics_377 EQ1148 (C.17)		■ 1.000 ● W +0	$\lambda_{\mathrm{i}}\lambda_{\mathrm{j}}=\frac{2}{N}\delta_{ij}+\left(d_{ij}^k+if_{ij}^k\right)\lambda_k\;.$	$\lambda_i\lambda_j=\frac{2}{N}\delta_{ij}+\left(d_{ij}^k+if_{ij}^k\right)\lambda_k.$	$\lambda_i\,\lambda_j=\frac{2}{N}\delta_{ij}+(d_{ij}{}^k+if_{ij}{}^k)\lambda_k.$
mielke-geometrodynamics_377 EQ1149 (C.18)		■ 1.000 ● N +0	$\operatorname{Tr}\left(\lambda_{\mathrm{i}}\lambda_{\mathrm{j}}\right)=2\delta_{ij}\;.$	$\operatorname{Tr}\left(\lambda_i\lambda_j\right)=2\delta_{ij}.$	$\operatorname{Tr}(\lambda_i\,\lambda_j)=2\delta_{ij}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Inline formulas that did not render (1 of 1794)

Identifier	Page	Conf.	LaTeX source	Rendered
mielke-geometrodynamics_F00431	100	—	$\mathrm{t}_{\xi}:=\xi \mathop{\mathrm{D}}\nolimits^{\mathrm{D}} \xi$	(not rendered)
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.				

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																												
mielke-geometrodynamics_06/05	075	—	\begin{tabular}[t]{l l } \hline Gauge Field Terminology & Fiber Bundle Terminology \\ \hline spacetime space of phase factors symmetry or gauge group gauge transformations gauge principle classical fields gauge potential gauge field strength & base space bundle space structure group inner automorphisms of the bundle G-equivalence principle cross section of a vector bundle connection 1-form curvature 2-form \\ \hline electromagnetism electromagnetism with monopoles Dirac's monopole quantization & dynamical theory with a connection in a U(1)-bundle connection in a nontrivial U(1)-bundle over \(\mathbb{R}^2 \times \mathrm{S}^2\) classification of U(1)-bundles according to the first Chern class \\ \hline Yang–Mills theory instanton number & dynamical theory with a connection in a SU(2)-bundle classification of SU(2)-bundles over \(\mathrm{S}^4\) according to the second Chern class \\ \hline \end{tabular}		<table><tr><th>Gauge Field Terminology</th><th>Fiber Bundle Terminology</th></tr><tr><td>spacetime</td><td>base space</td></tr><tr><td>space of phase factors</td><td>bundle space</td></tr><tr><td>symmetry or gauge group</td><td>structure group</td></tr><tr><td>gauge transformations</td><td>inner automorphisms of the bundle</td></tr><tr><td>gauge principle</td><td>G-equivalence principle</td></tr><tr><td>classical fields</td><td>cross section of a vector bundle</td></tr><tr><td>gauge potential</td><td>connection 1-form</td></tr><tr><td>gauge field strength</td><td>curvature 2-form</td></tr><tr><td>electromagnetism</td><td>dynamical theory with a connection in a U(1)-bundle</td></tr><tr><td>electromagnetism with monopoles</td><td>connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times \mathrm{S}^2$</td></tr><tr><td>Dirac’s monopole quantization</td><td>classification of U(1)-bundles according to the first Chern class</td></tr><tr><td>Yang–Mills theory</td><td>dynamical theory with a connection in a SU(2)-bundle</td></tr><tr><td>instanton number</td><td>classification of SU(2)-bundles over S^4 according to the second Chern class</td></tr></table>	Gauge Field Terminology	Fiber Bundle Terminology	spacetime	base space	space of phase factors	bundle space	symmetry or gauge group	structure group	gauge transformations	inner automorphisms of the bundle	gauge principle	G-equivalence principle	classical fields	cross section of a vector bundle	gauge potential	connection 1-form	gauge field strength	curvature 2-form	electromagnetism	dynamical theory with a connection in a U(1)-bundle	electromagnetism with monopoles	connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times \mathrm{S}^2$	Dirac’s monopole quantization	classification of U(1)-bundles according to the first Chern class	Yang–Mills theory	dynamical theory with a connection in a SU(2)-bundle	instanton number	classification of SU(2)-bundles over S^4 according to the second Chern class
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mielke-geometrodynamics_10/02	102	—	\begin{tabular}[t]{l l l l } \hline potential & field strength & Bianchi identity & Noether identity \\ \hline \(\vartheta^\alpha\) & \(T^\alpha = D\vartheta^\alpha\) & \(DT^\alpha \equiv R_\gamma{}^\alpha \wedge \vartheta^\gamma\) & \(\widehat{D} \Sigma_\alpha \cong (e_\alpha \rfloor R_{\beta\gamma}) \wedge \Delta^\beta{}_\gamma - \frac{1}{2}(e_\alpha \rfloor Q_{\beta\gamma}) \wedge \sigma^{\beta\gamma}\) \\ \hline \(\Gamma_\alpha{}^\beta\) & \(R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta - \Gamma_\alpha{}^\gamma \wedge \Gamma_\gamma{}^\beta\) & \(DR_\alpha{}^\beta \equiv 0\) & \(D\Delta^\alpha{}_\beta + \vartheta^\alpha \wedge \Sigma_\beta \cong \sigma^\alpha{}_\beta\) \\ \hline \end{tabular}	(not rendered)																													
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image																																																																																																																
mielke-geometrodynamics_2610	210	—	$\begin{tabular}{t}{ l l l l l l l } \hline & \text{Objects} & \\ \hline & \text{(\p\)-form} & \text{Components} & \text{(\mathrm{n}=4\)} & 3 & 2 \\ \hline & \text{(\vartheta^{\alpha}\)} & \text{Vector} & 1 & \text{(n^2\)} & \\ 16 & 9 & 1 \\ \hline & \text{(\Gamma_{\alpha}^{\star}\)} & \text{Vector} & 1 & \text{(n^2\)} & \\ & 1 & \text{(n^2\)} & 16 & 9 & 1 \\ \hline & \text{(T^{\alpha}\)} & \text{Vector} & 2 & \text{n^2(n-1)/2} & 24 & 9 & 2 \\ & \text{Vector} & 2 & \text{(n^2(n-1) / 2\)} & 24 & 9 & 2 \\ \hline & \text{(R^{\alpha} \beta\)} & \text{Bivector} & 2 & \text{n^2(n-1)^2/4} & 36 & 9 & 1 \\ & \text{(\Sigma_{\alpha}\)} & \text{Vector} & n-1 & \text{n^2} & 16 & 9 & 4 \\ & \text{(\tau_{\alpha\beta}\)} & \text{Bivector} & n-1 & \text{n^2(n-1)/2} & 24 & 9 & 2 \\ & \text{(\eta_{\alpha}\)} & \text{Vector} & n-1 & \text{n^2} & 16 & 9 & 4 \\ \hline \end{tabular}$	<table><tr><td></td><td>Objects</td><td><i>p</i>-form</td><td>Components</td><td>n = 4</td><td>3</td><td>2</td></tr><tr><td>ϑ^α</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma_\alpha^\star$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^α</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table>		Objects	<i>p</i> -form	Components	n = 4	3	2	ϑ^α	Vector	1	n^2	16	9	1	$\Gamma_\alpha^\star$	Vector	1	n^2	16	9	1	T^α	Vector	2	$n^2(n-1)/2$	24	9	2	$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1	Σ_α	Vector	$n-1$	n^2	16	9	4	$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2	η_α	Vector	$n-1$	n^2	16	9	4	<table><tr><td></td><td>Objects</td><td><i>p</i>-form</td><td>Components</td><td>n = 4</td><td>3</td><td>2</td></tr><tr><td>ϑ^α</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>$\Gamma_\alpha^\star$</td><td>Vector</td><td>1</td><td>n^2</td><td>16</td><td>9</td><td>1</td></tr><tr><td>T^α</td><td>Vector</td><td>2</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>$R^{\alpha\beta}$</td><td>Bivector</td><td>2</td><td>$n^2(n-1)^2/4$</td><td>36</td><td>9</td><td>1</td></tr><tr><td>Σ_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr><tr><td>$\tau_{\alpha\beta}$</td><td>Bivector</td><td>$n-1$</td><td>$n^2(n-1)/2$</td><td>24</td><td>9</td><td>2</td></tr><tr><td>η_α</td><td>Vector</td><td>$n-1$</td><td>n^2</td><td>16</td><td>9</td><td>4</td></tr></table>		Objects	<i>p</i> -form	Components	n = 4	3	2	ϑ^α	Vector	1	n^2	16	9	1	$\Gamma_\alpha^\star$	Vector	1	n^2	16	9	1	T^α	Vector	2	$n^2(n-1)/2$	24	9	2	$R^{\alpha\beta}$	Bivector	2	$n^2(n-1)^2/4$	36	9	1	Σ_α	Vector	$n-1$	n^2	16	9	4	$\tau_{\alpha\beta}$	Bivector	$n-1$	$n^2(n-1)/2$	24	9	2	η_α	Vector	$n-1$	n^2	16	9	4
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mielke-geometrodynamics_275	275	—	$\begin{tabular}{t}{ l l l } \hline & \text{Spin} & \text{Gravitational} & \text{YM anomaly} \\ \hline 1/2 & 1 & 1 \\ \hline 3/2 & -21 & 3 \\ \hline \end{tabular}$	<table><tr><td>Spin</td><td>Gravitational</td><td>YM anomaly</td></tr><tr><td>1/2</td><td>1</td><td>1</td></tr><tr><td>3/2</td><td>-21</td><td>3</td></tr></table>	Spin	Gravitational	YM anomaly	1/2	1	1	3/2	-21	3	<table><tr><td>Spin</td><td>Gravitational</td><td>YM anomaly</td></tr><tr><td>1/2</td><td>1</td><td>1</td></tr><tr><td>3/2</td><td>-21</td><td>3</td></tr></table>	Spin	Gravitational	YM anomaly	1/2	1	1	3/2	-21	3																																																																																														
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mielke-geometrodynamics_18096

364

—

$$\begin{tabular}{t}{|l|l|} \hline := & : \text{Equal by definition} \\ \hline \equiv & : \text{Identically equal} \\ \hline \approx & : \text{Isomorphic} \\ \hline \simeq & : \text{Isomorphic} \\ \hline \sim & : \text{Approximately equal} \\ \hline \sim & : \text{Asymptotic expansion} \\ \hline \times & : \text{Topological product} \\ \hline \otimes & : \text{Direct product} \\ \hline \oplus & : \text{Direct sum} \\ \hline \otimes & : \text{Semidirect product} \\ \hline G_{\circ} & : \text{Connected component of the group} \\ \hline \tilde{G} & : \text{Simply connected covering group} \\ \hline [A, B] := AB - BA & : \text{Commutator} \\ \hline \{A, B\} := AB + BA & : \text{Anticommutator} \\ \hline A^* = C A & : \text{Complex-conjugate matrix} \\ \hline A^+ = A^{*T} & : \text{Hermitian adjoint matrix} \\ \hline \pi^{-1}(N) & : \text{Original domain of N with respect to the mapping } \pi : M \rightarrow N \\ \hline T_{(\mu\nu)} := \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu}) & : \text{Symmetrized second-rank tensor} \\ \hline T_{[\mu\nu]} := \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu}) & : \text{Antisymmetrized second-rank tensor} \\ \hline \end{tabular}$$

:=	: Equal by definition
≡	: Identically equal
≈	: Isomorphic
≃	: Approximately equal
∼	: Asymptotic expansion
×	: Topological product
⊗	: Direct product
⊕	: Direct sum
⊗	: Semidirect product
G _◦	: Connected component of the group
$\tilde{G}$	: Simply connected covering group
$[A, B] := AB - BA$	: Commutator
$\{A, B\} := AB + BA$	: Anticommutator
$A^* = C A$	: Complex-conjugate matrix
$A^+ = A^{*T}$	: Hermitian adjoint matrix
$\pi^{-1}(N)$	: Original domain of N with respect to the mapping $\pi : M \rightarrow N$
$T_{(\mu\nu)} := \frac{1}{2}(T_{\mu\nu} + T_{\nu\mu})$	: Symmetrized second-rank tensor
$T_{[\mu\nu]} := \frac{1}{2}(T_{\mu\nu} - T_{\nu\mu})$	: Antisymmetrized second-rank tensor

Conf.

MathPix confidence:

■

 ≥0.9

■

 0.5-0.9

■

 <0.5

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 no value

LaTeX source

MathPix's reading, unchanged;

Rendered

the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ

Residual

render vs scan (inkdrill):

●

 C component

●

 W weak

●

 S stable

●

 N noise

●

 K clean

●

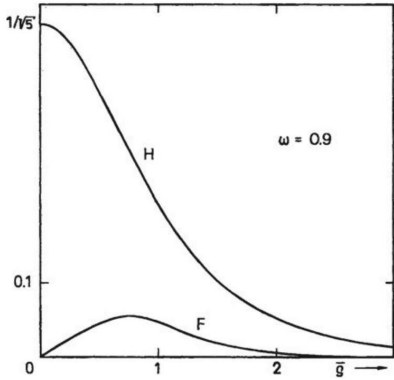
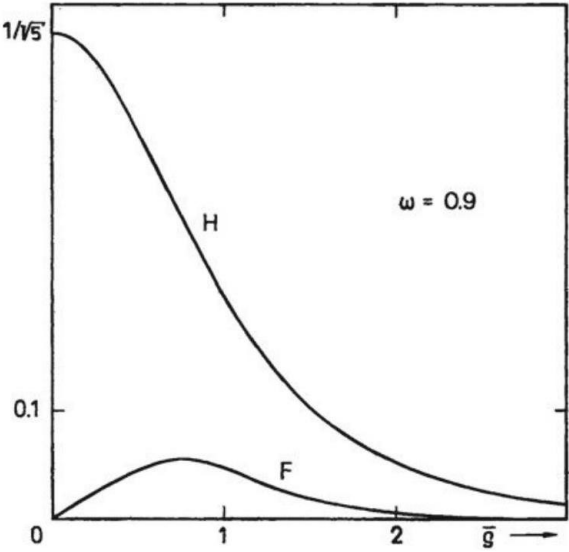
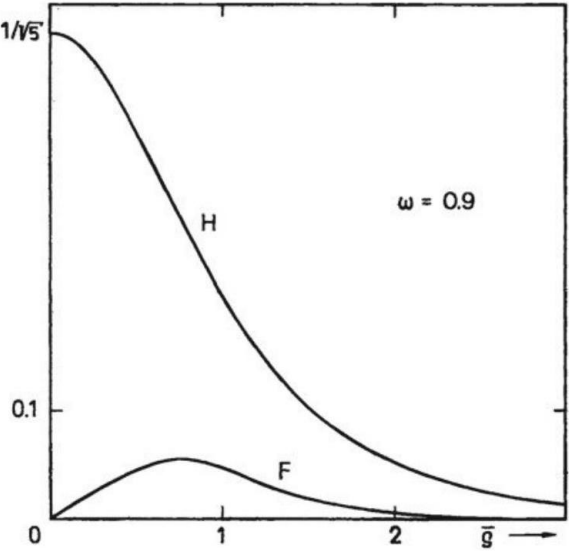
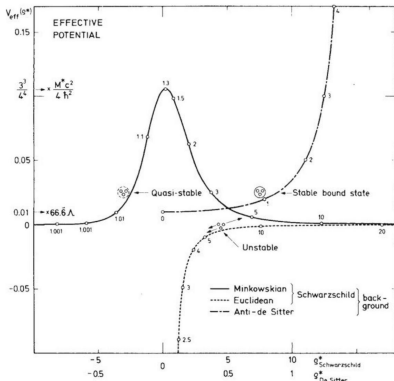
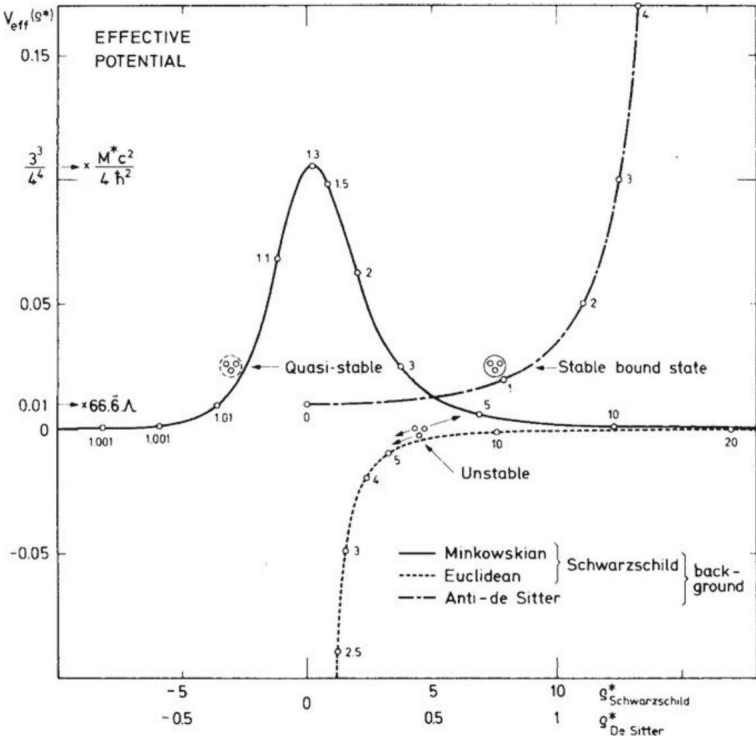
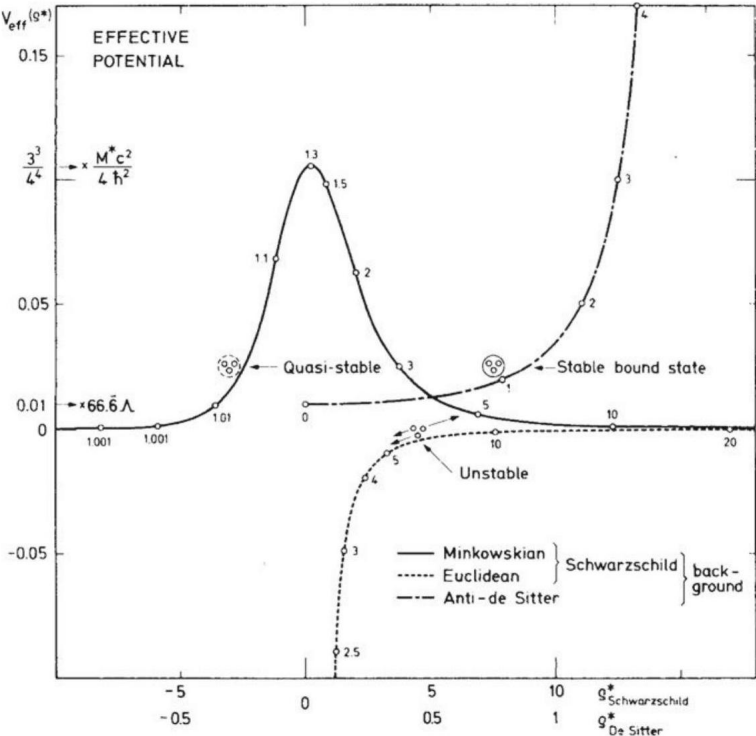
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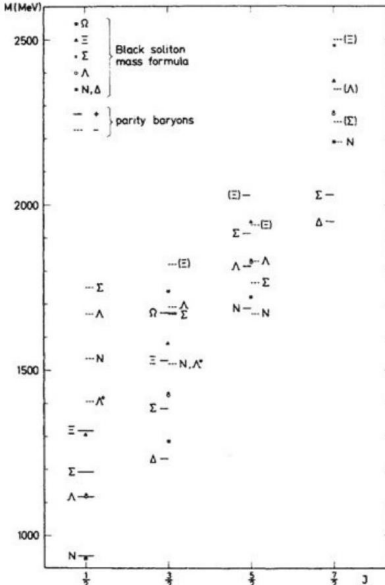
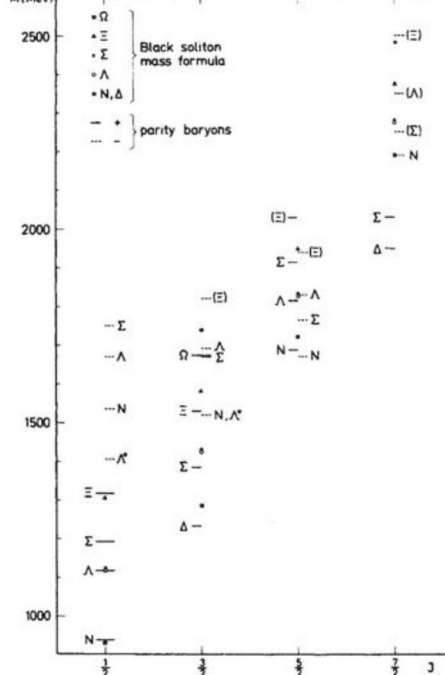
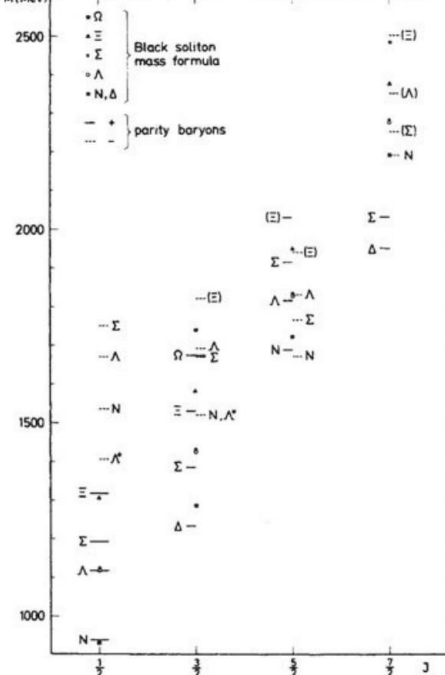
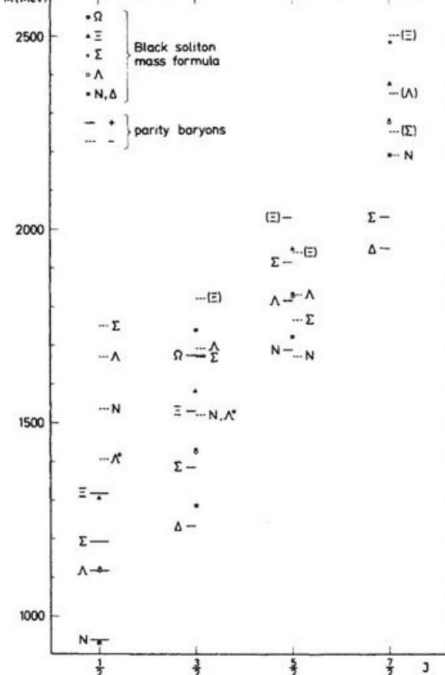
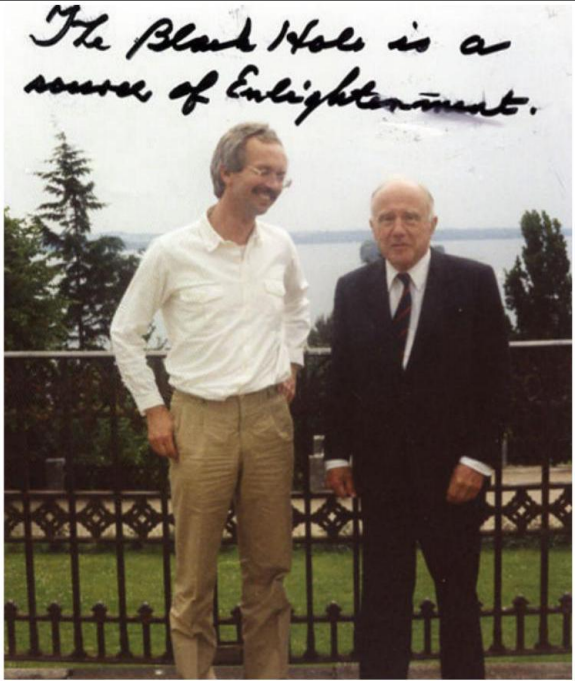
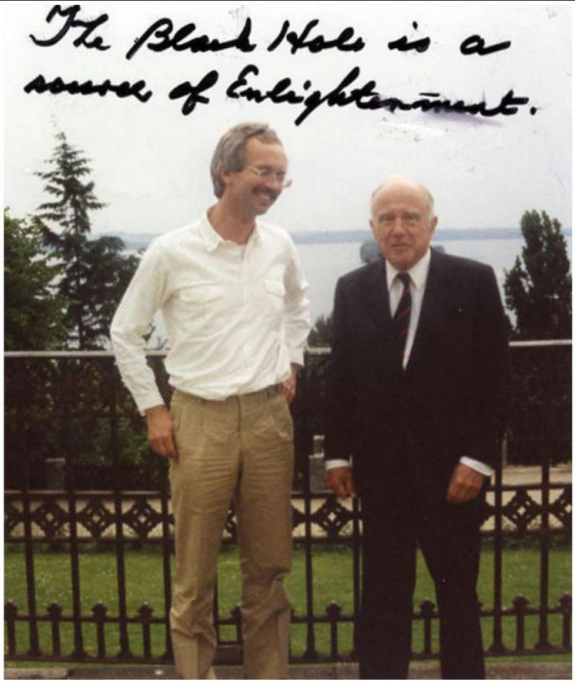
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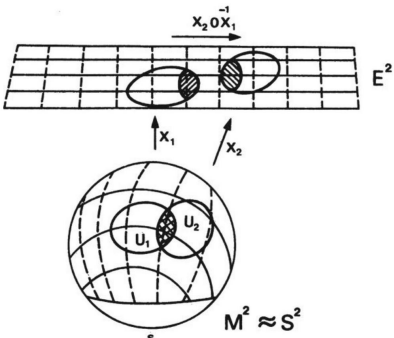
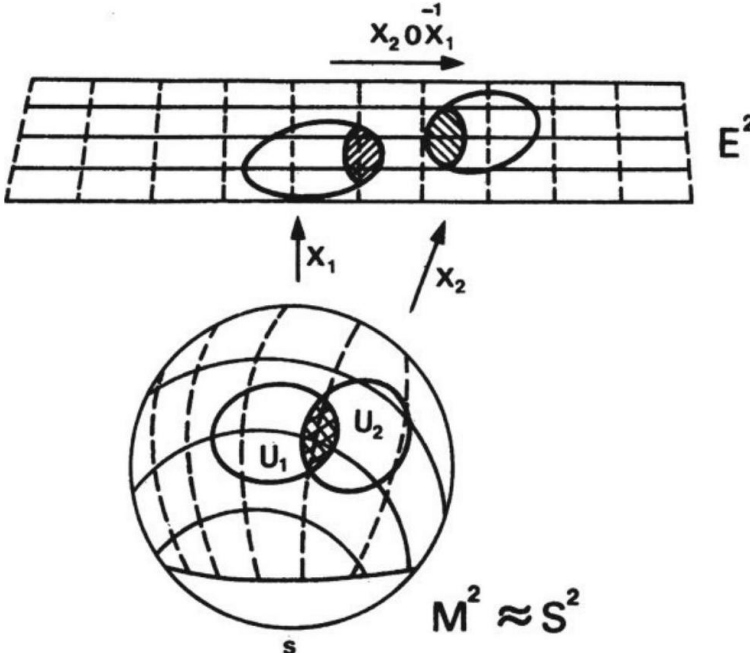
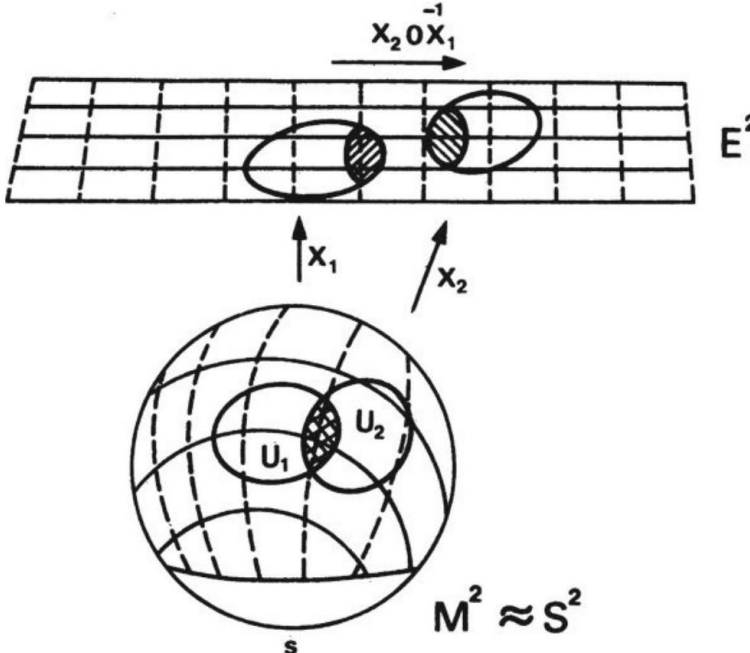
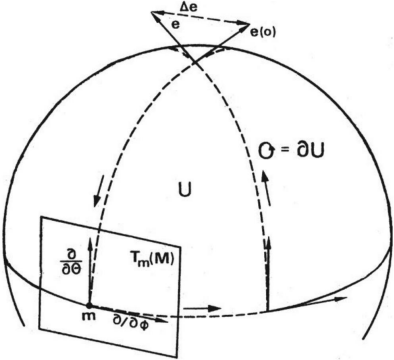
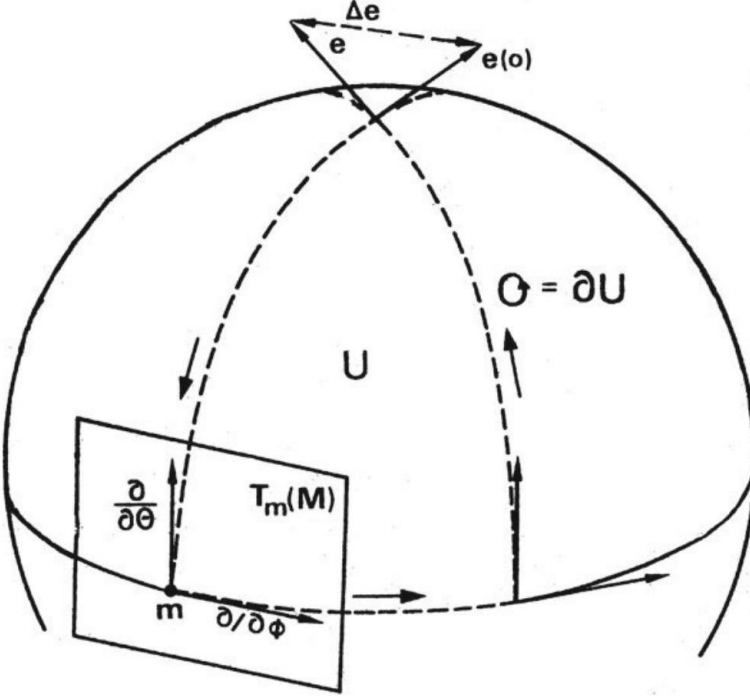
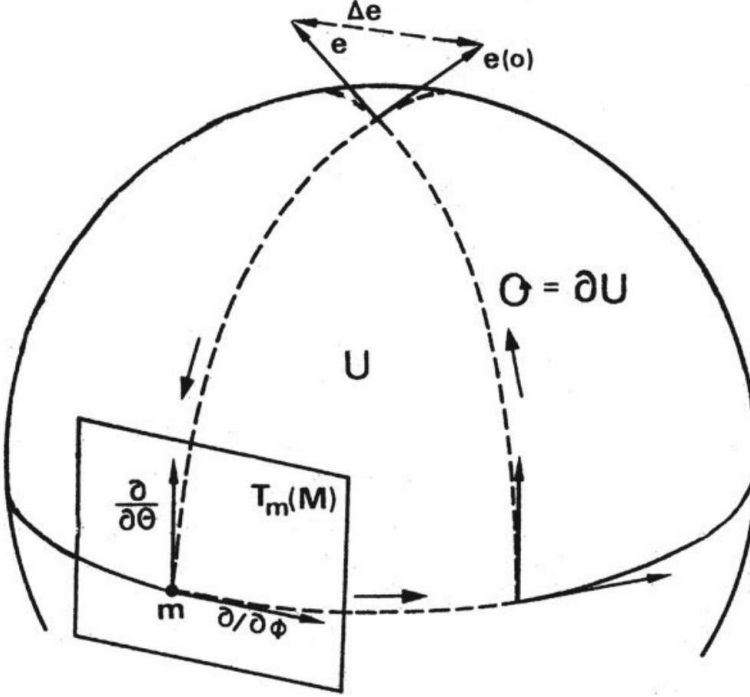
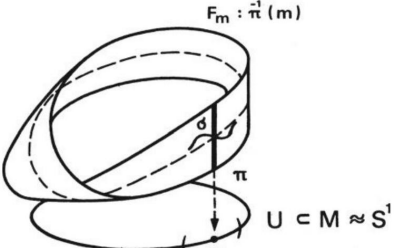
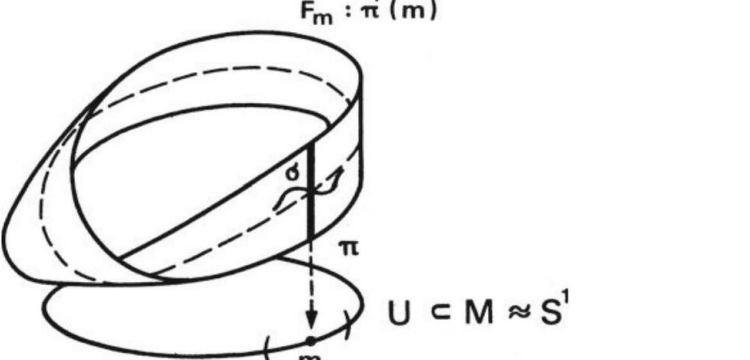
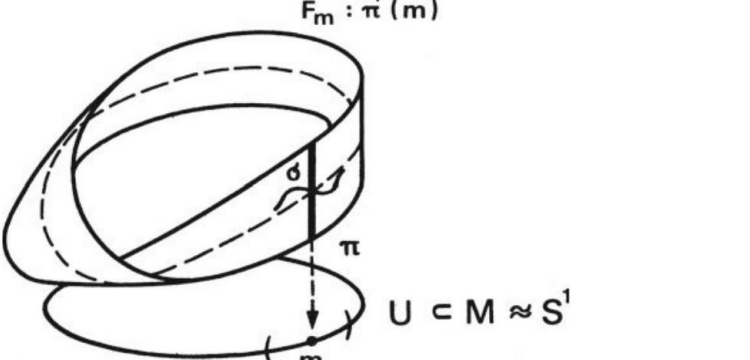
beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix's own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.

Image regions — rendered against scan

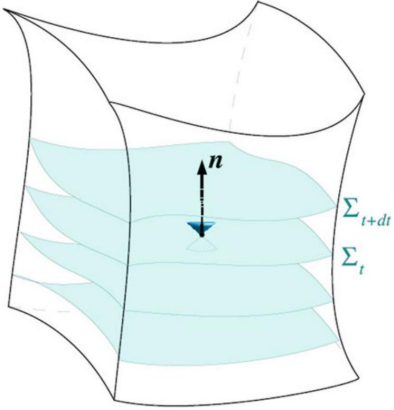
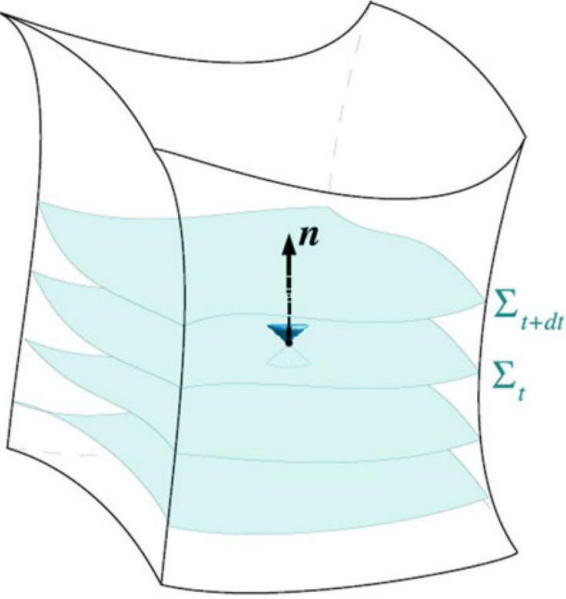
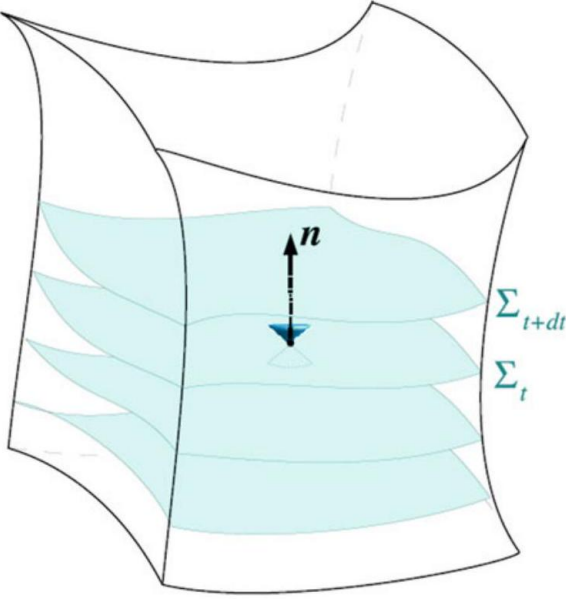
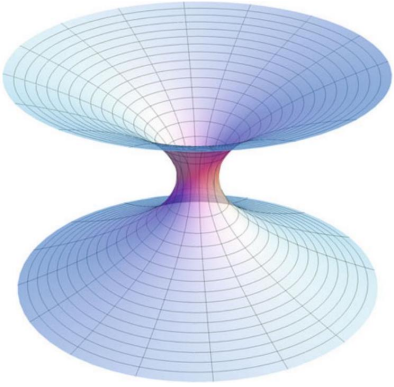
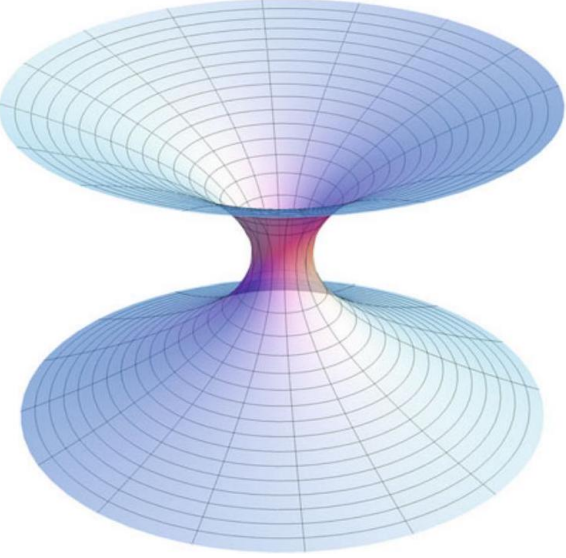
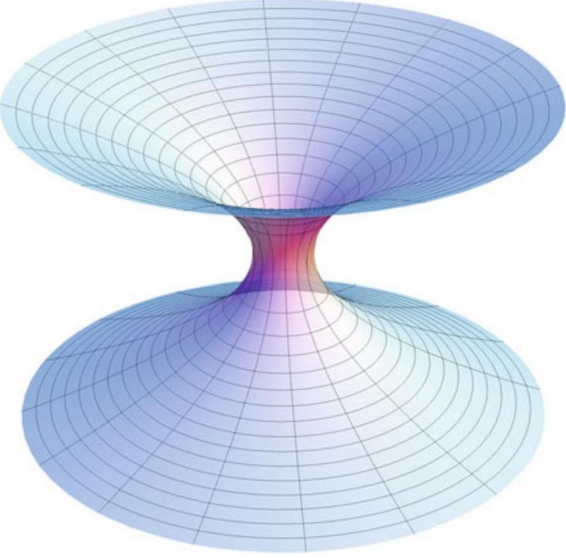
Two comparable cells per row. **Rendered:** the region’s own LaTeX compiled as its OWN document, or — where no LaTeX exists — the scan again, so every row has two cells and the ink difference reads as a floor rather than as missing-versus-present. A duplicated row is marked *(dup)* and its distance is a SELF-comparison, not agreement between two sources. **Scan:** the tex.zip image whose filename region 5-tuple matches this row, else the CDN crop. The Class column carries the residual class beside MathPix’s confidence, as the equation rows do. Setting a region’s LaTeX does NOT say it renders to what is on the page — that is what the two columns are for, and 281 is the open question they exist to let a reader answer. A region with no LaTeX can be reconstructed with pdfdrill vision; verify any LLM result against the real ink with inkdrill.

Identifier	Page	Class	Source	Author source	Rendered	Scan
						
mielke-geometrodynamics_R0C001	258	(dup)	b173365e-9184-4902-bd66-732270e80db9-258_719_748_1319_592.jpg			
						
mielke-geometrodynamics_R0C002	356	(dup)	b173365e-9184-4902-bd66-732270e80db9-356_1111_1146_196_186.jpg			

mielke-geometrodynamics_002003	359 (dup)	b173365e-9184-4902-bd66-732270e80db9-359_902_600_206_737.jpg				
mielke-geometrodynamics_002001	025 (dup)	b173365e-9184-4902-bd66-732270e80db9-025_885_748_204_592.jpg				

mielke-geometrodynamic_031_002	031	(dup)	b173365e-9184-4902-bd66-732270e80db9-031_864_999_194_261.jpg			
mielke-geometrodynamic_032_003	032	(dup)	b173365e-9184-4902-bd66-732270e80db9-032_915_995_1064_263.jpg			
mielke-geometrodynamic_036_004	036	(dup)	b173365e-9184-4902-bd66-732270e80db9-036_499_750_1538_594.jpg			

mielke-geotrodynamics_040	040	— (dup)	b173365e-9184-4902-bd66-732270e80db9-040_533_750_204_592.jpg			
mielke-geotrodynamics_057	057	— (dup)	b173365e-9184-4902-bd66-732270e80db9-057_449_744_206_596.jpg			
mielke-geotrodynamics_073	073	— (dup)	b173365e-9184-4902-bd66-732270e80db9-073_626_748_206_594.jpg			
mielke-geotrodynamics_084	084	— (dup)	b173365e-9184-4902-bd66-732270e80db9-084_573_748_207_592.jpg			

mielke-geotrodynamics_131_782_748_204_594.jpg	131	(dup)			
mielke-geotrodynamics_200_723_746_209_594.jpg	200	(dup)			
mielke-geotrodynamics_226_599_748_1382_592.jpg	226	(dup)	